Mashrabiya in passive cooling

Integrating Concepts of Mashrabiya into an Innovative Ceramic Evaporative Screen

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Abstract

Designing a multifunctional outdoor evaporative cooling screen in water-scarce hot-dry regions has posed a significant challenge. In areas where temperatures reach extreme levels during the summer and access to water is limited, the need for passive cooling strategies is urgent. While shading methods are commonly used to reduce sun exposure, existing systems rarely succeed in combining evaporative cooling and shading in a single component. Most approaches prioritise one function at the expense of the other or demand excessive water use, making them impractical for outdoor applications in such climates.

This research addresses this performance gap by exploring a design that balances cooling efficiency with conscious water consumption, considering its relevance for application in hot-dry regions. The research focuses on developing innovative ceramic screens that can function as both shading elements and efficient evaporative cooling systems. It integrates design principles from traditional Islamic screens, known as Mashrabiya. The design process builds on the environmental logic and spatial qualities of the Mashrabiya to propose a reinterpretation that connects cultural heritage with material innovation and functional designs, offering a passive cooling strategy that responds to the needs of contemporary hot-dry regions.

A hybrid methodology was adopted. Qualitative techniques, including literature review and interviews with craftsmen, were used to understand the essence of Mashrabiya's design. Quantitative methods, including environmental analysis and simulation tools, were employed to evaluate the performance. Traditional screen geometries were analysed and digitally documented to inform the design process. Alongside this, an investigation into ceramic manufacturing techniques supported the development of a screen that simultaneously achieves cooling, shading, and water efficiency.

The study also tested the screen's full-scale application under realistic conditions in Riyadh, monitoring its evaporative cooling, shading performance, water consumption, and design applicability. The results show that the proposed evaporative porous ceramic screen reduces surface and surrounding air temperature while acting as a fixed self-shading element that enhances comfort without reliance on mechanical systems. The screen offers an alternative passive solution rooted in cultural knowledge while addressing contemporary environmental demands.

Keywords: Mashrabiya, Maymooni screens, Evaporative cooling, Ventilation, mutual shading, shading, passive cooling, Ceramic.

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1. Introduction

Buildings consume more energy than any other sector, accounting for 30–40% of global energy use. In the United States, the building sector is responsible for about 41% of primary energy consumption (Ministerio de Industria, 2006) (Schianopahn, 2004). This trend is also evident in the Middle East, where buildings remain one of the highest energy-consuming sectors. In the Gulf region, the challenge is particularly acute: residential buildings account for around 43% of electricity consumption. (Almasri, 2022). Air conditioning dominates energy loads, often exceeding 70% of household consumption. However, studies confirm that targeted interventions such as thermal insulation, passive cooling, and energy-efficient systems can cut electricity use by up to 60–70% (Almasri, 2022), offering both environmental benefits and short payback periods.

In response to the growing demand for climate-appropriate passive cooling strategies, particularly in hot-dry regions, there is a need to develop low-carbon solutions that address thermal comfort while reducing dependency on mechanical systems (Brian Ford, 2020). Evaporative cooling emerges as a valuable bioclimatic response in such environments, especially when integrated within architectural elements that also provide shading. Within this context, the research aims to develop a multifunctional ceramic screen that can efficiently perform as an evaporative cooling element and a shading component. Therefore, it is essential to investigate the history of Mashrabiya implementation over time and how it evolved through the passive solutions of vernacular architecture. The research aims to critically evaluate the performance of Mashrabiya and how it can serve as a useful guiding tool in the design process, not by mimicking the Mashrabiya, but by integrating its concepts into innovative evaporative cooling screens.

Mashrabiya is a component that performs as a shading element, blocking multiple solar angles from penetrating the indoor space and regulating the light entering through the screen (Al-Shareef, 1996). Additionally, the screens enable efficient ventilation techniques that regulate airflow to an adequate depth within the space. As a result, Mashrabiya has embodied harmonious forms that combine both functionality and beauty (الزهران, 2007). Analysing Mashrabiya as a component and understanding its manufacturing history helps provide a better understanding of the applicable integration of its concepts in contemporary ceramic screens. Tracing the development of these patterns illustrates how they affect the shading and ventilation performance of the screens.

The efficiency of direct passive evaporative cooling depends on the surface area of the wet media, the airflow rate through it, and solar exposure (Brian Ford, 2020). Therefore, evaporative cooling requires sufficient ventilation strategies and controlled solar exposure to ensure efficiency while

maintaining balanced water consumption. Accordingly, merging the ventilation and shading in evaporative cooling systems is critical. The efficiency of evaporative cooling depends primarily on the evaporation rate, which is influenced by solar exposure that accelerates the evaporation process and increases the rate of evaporation, thereby affecting the microenvironment around the screen. (Brian Ford, 2020). The evaporation rate of the screens is also directly proportional to the water consumption of the evaporative cooling screen, where the water is consumed to contribute to the humidification of the air penetrating the wet media of the evaporative cooling system as well as cooling down the air temperature throughout releasing the energy in the high temperature of the air into the kinetic energy consumed in breaking the hydrogen bond in the water molecules (Zakinyan, 2022). Accordingly, the evaporation rate is not only motivated by solar exposure but also by the temperature of the air penetrating through the wet media, which will be an essential driver that the research will focus on during the evaporative cooling process achieved by the proposed system.

Evaporative cooling is a highly efficient passive cooling approach for hot-dry climates, as indicated by the overlapping Koppen climate classification scheme on the world map and the freshwater availability stress map; over 72% of hot-dry climatic zones are at greater risk of a freshwater crisis (Bloomberg, 2019). As a result, freshwater is a valuable resource, much like fossil fuels and other energy resources. In the deserts of the Middle East, water was seen as more precious than gold; its value was so deeply felt that in classical Arabic poetry, fresh water was often used as an adverb or metaphor to describe anything beautiful, rare, or deeply appreciated (lightfoot, 2024). As a result, it is valuable to develop a passive evaporative cooling strategy that prioritises the value of water while striking a balance between minimising water consumption and maximising the efficiency of the strategy. Controlling the solar exposure of the evaporative cooling screen can be achieved by adding external shading to the evaporative cooling system applied on the aperture or the main inlet of the space, where the shading devices control the solar accessibility to the system, such as shutters or overhangs that can be applied on the exterior of a building (Kreider, 2019). However, external shading devices may need to be designed and constructed to withstand strong winds, which can increase their cost and complexity, as well as obstruct airflow to the system to which they are applied (Kreider, 2019). On the other hand, external shading devices do not always provide consistent or sufficient solar control, particularly in areas exposed to high levels of direct sunlight for extended periods (Naguib, 2007). This is mainly because shading is a site-specific response that needs to adapt to the building's orientation and the shifting position of the sun throughout the day. A shading element that works well in blocking the low angles of the sun in the morning or late afternoon may become almost ineffective around midday when the sun sits high and casts light from a near-vertical angle, making it difficult for a single, fixed shading strategy to always perform equally efficiently at different times (Naguib, 2007).

The climatic applicability of the evaporative cooling strategy shaped the main objective of the research; accordingly, proposing an evaporative cooling screen designed to control undesired solar exposure and maximise the cooling performance of the screen applied on the aperture can be a new promising approach that can mitigate the contemporary design solutions of passive cooling. This research investigates the concepts of Mashrabiya, which can enhance the potential for improved performance when integrated into the proposed evaporative cooling ceramic screen.

Clay, as a material, portrays a unique set of features that encourage designers to incorporate ceramics into various design solutions, endowing them with a vital role in the development of passive environmental solutions (Chen, 2021). The use of ceramic in evaporative cooling is rooted in the porosity of the material that enables water absorption; therefore, the pores of the ceramic act like a sponge that keeps fluid in (Chen, 2021). The pores of the ceramic increase the internal surface area of the material, enabling the preservation of water inside the micro-pores of the element, which maintains the media moist for extended periods (Cabeza, 2020).

1.2 The primary aims of this research are:

The primary aim of the research is to develop a culturally and environmentally responsive ceramic evaporative cooling screen that integrates passive cooling and shading strategies inspired by Mashrabiya principles, for use in hot-dry climates. Accordingly, the research aims to:

1. To investigate the fundamental environmental, geometric, and cultural principles of Mashrabiya from both small-scale (details and patterns) and large-scale (façades and spatial performance) perspectives.
2. To develop design guidelines that quantify and evaluate the environmental performance of Mashrabiya components, aiding their integration into contemporary climate-responsive architectural strategies.
3. To integrate the design concepts of Mashrabiya elements into the design and fabrication of ceramic evaporative cooling screens, ensuring cultural continuity alongside environmental performance.
4. To design a ceramic evaporative cooling screen that optimises shading and cooling performance for hot, dry climates.
5. To explore and address the constraints and opportunities of manufacturing techniques such as 3D printing and ceramic casting for the contemporary application of cooling screens.

Incorporating historical Islamic design approaches into a suggested pattern design can construct structures that are not only practical and efficient for various environmental parameters but also

visually beautiful and culturally meaningful. It can also aid in preserving and recognising Mashrabiya's cultural legacy, promote the identity of the screens, and open a new field of experimentation towards better designs.

1.3 Objectives

The primary objective of this research is to develop an innovative evaporative cooling screen that effectively combines the functions of shading and evaporative cooling. In the face of increasing global energy consumption, there is an urgent need for innovative solutions in sustainable building design. Drawing inspiration from Mashrabiya, long recognised for its shading capabilities, the research aims to understand its design principles and extract key concepts to inform the development of a new ceramic screen. One of the research goals is to create a functional element that enhances cooling and shading efficacy and addresses the issue of water overconsumption in the cooling process. Achieving this requires striking a delicate balance between functionality, water conservation, suitability for outdoor use, and effectiveness in both shading and cooling. By incorporating the principles of mutual shading, where the geometry of the screen casts shade upon specific sections of its structure, the research aims to optimise the screen's performance while ensuring its adaptability to diverse environmental conditions.

Designing an evaporative cooling solution in hot-dry climates is commonly used indoors to avoid excessive solar exposure that can reduce the efficiency of the cooling process and increase the water consumption due to the rapid increase in the evaporation rate, which can be an issue due to the lack of water resources in the majority of hot-dry climatic regions that consider water as a valuable resource. On the other hand, outdoor evaporative cooling is rarely used, and when it is, crucial shading must be provided on top of the element (Musa, 2009). The fabrication of wet media for direct evaporative cooling requires consideration of several economic and technical factors (Schiano-Phan, 2001). Therefore, it is essential to design the wet media using the appropriate material and geometry to ensure the accurate criteria of water absorption and air friction with the wet media (Smith, 2011).

The proposed study aims to facilitate the equitable application of Mashrabiya concepts in contemporary evaporative cooling techniques and ceramic screens, considering the thermal comfort conditions of indoor spaces. Passive façade applications will be conducted in order to reduce the heat gains from excessive solar access and insufficient fresh air supply, translated into proper shading techniques and well-investigated ventilation techniques that increase the interaction between the indoor spaces and the outdoor environment. Ultimately, it will showcase the full potential of the cooling and shading effects of the intricate details of Mashrabiya patterns. This can help enhance the implementation of eco-friendly, sustainable, and energy-saving multifunctional ceramic evaporative

cooling screens. Incorporating principles from previous designs into a proposed design is a favoured strategy in architecture and design. To investigate the historical, environmental, and geometric principles of the Mashrabiya as a climatic and cultural architectural element, focusing on its application to passive cooling and shading. Accordingly, the research objectives of this study are as follows:

1. To investigate the historical, environmental, and geometric principles of the Mashrabiya as a climatic and cultural architectural element, focusing on its application to passive cooling and shading.
2. To develop a culturally responsive, ceramic-based evaporative cooling screen design, drawing inspiration from Mashrabiya typologies while adapting it for contemporary construction and environmental performance requirements.
3. To evaluate the proposed screen's thermal and evaporative cooling performance under hot and dry climatic conditions, with particular emphasis on temperature reduction, shading effectiveness, and airflow dynamics.
4. To test and validate the system at full scale through fieldwork and environmental monitoring, assess real-world performance, and address water consumption and airflow limitations.
5. To reflect on the process of reinterpretation, highlighting how traditional environmental design knowledge can inform innovative, sustainable architectural solutions suitable for water-scarce, high-heat regions.

1.4 Research Questions

1. Discuss the possibilities of successful ventilation and cooling techniques that provide proper shading when testing the small components of Mashrabiya. Can the design concepts of Mashrabiya help develop a more efficient responsive design?
2. It is possible to design a component as a passive solution for multiple environmental parameters and understand the matrix of relations between the different parameters and conditions, and how they affect each other.
3. What is the aspect of designing a multifunctional ceramic screen that can perform in multiple outdoor conditions simultaneously and overcome limitations, and how can reintroducing ceramic screens with integrated features from Mashrabiya be a revolutionary step in façade designs?

1.5 Methodology of the research:

This thesis is organised around its two primary goals:

• Designing and manufacturing an innovative porous ceramic evaporative cooling screen and an effective sizing method.

• Assessing the performance in advance to maximise its efficiency.

The three main research areas chosen for this study are:

1. Architectural research, particularly focusing on Environmental Design and Technology.
2. Experimentation with samples of traditional patterns, focusing on how the small details of the patterns affect the overall performance of the screens in terms of shading and ventilation.
3. Integrating the fundamentals of the screens in Mashrabiya into the proposed ceramic screens to maximise the efficiency of the screen performance in terms of shading and passive cooling.

Research Design and Methodological Approach:

The methodology followed a hybrid approach, combining both qualitative and quantitative analytical research in a validation process that uses qualitative methods to collect essential data and manufacturing details about Mashrabiya. This data was later used in the quantitative analytical stage to finalise the main performance criteria of Mashrabiya and to understand how the small details, specific manufacturing techniques, and turnery details managed to change the overall performance outcome of the screens and patterns.

This study builds on a previous MSc Final Thesis, "Evaluating the Environmental Performance of Mashrabiya" (University of Westminster, UK, 2019). In this previous research, various studies and documentation were utilised to understand Mashrabiya's total environmental performance and to evaluate it as a passive architectural solution rooted in vernacular architecture.

The research argument and methodology selection fall under qualitative studies using an inductive approach, with quantitative research used for performance measurement and testing.

Quantitative research

The first stage of the quantitative research involved continuous monitoring of the environmental performance of Mashrabiya screens. Samples of crafted traditional screens were collected for experimentation to quantify their physical environmental impact. This data was documented and underwent analytical reviews, and based on the outcome of the analysis, the main beneficial fundamentals of the screens were highlighted.

Therefore, the factors that improve the efficiency of the screens could be integrated into the proposed ceramic screens to enhance their efficiency. Understanding the impact of micro-level design details helped determine the desired functions that could be adjusted in the design of the ceramic screens.

This approach contributed to identifying the details that could be replicated in the proposed ceramic screens and recognising the features that may negatively impact the total performance of Mashrabiya.

A quantitative research method was therefore suitable for addressing the research questions in this study.

Additionally, the efficiency of the proposed ceramic screen was examined through quantitative research in the proposed fieldwork, which took place in a hot and dry climate. The screen was applied at a 1:1 scale to assess its performance and to optimise the clay properties used in the fabrication of the final screen.

In terms of sustainable performance and environmental design analysis, several programs were used to quantify the screens' environmental performance: Grasshopper, Ladybug, and Diva plug-ins for Rhino (used for radiation analysis, shadow studies, building energy, and daylighting simulation) (Ladybug Tools, 2025); Autodesk CFD (for fluid flow and thermal simulation) (Autodesk, 2025); and Climate Studio through Grasshopper and Rhino interfaces, among others (Solemma, 2025).

Qualitative research:

The research introduces a new multifunctional evaporative cooling screen, and designing a functional and efficient screen requires a conceptual design that could enhance environmental performance. Such functional screen designs can be found in the elements of Mashrabiya (Salem, 2019). Therefore, a qualitative research method was suitable for addressing the research questions in this study.

Qualitative research was also used to understand the design criteria of Islamic screens. This approach contributed to establishing design guidelines for ceramic screens that can efficiently function as both an evaporative cooling element and a shading component. This stage offered insights into the manufacturing details of the screens and the design stages of the patterns (محمد, 1997).

The first step in this qualitative research was a literature review of the history of Mashrabiya as a passive traditional solution. Relevant sources, including academic debates and journal articles, were used to understand its performance and fundamentals better.

The next step was to investigate manufacturing techniques by identifying experts in relevant fields and documenting the methods that craftsmen use in the wood lathing of the screen parts. This investigation aimed to determine the guidelines for designing and manufacturing Mashrabiya screens.

Previous studies have indicated that Mashrabiya is a successful passive solution (Allard, 1998). Its efficiency as an environmental solution is rooted in the screens' multifunctional performance. Therefore, Mashrabiya was a guiding stone in introducing an evaporative cooling element that can overcome challenging limitations.

A supplementary task stage of the qualitative study involved selecting the samples and defining the common traditional shapes of Maymooni screens. It also highlighted the most effective forms that incorporated the various design features that aid in corresponding changes in forms and small design details with the screen's function in terms of shading, mutual shading, and ventilation. Therefore, interviews were conducted with Egyptian craftsmen who specialise in refurbishing ancient screens of historical Mashrabiya to ensure that the screens were constructed using the same traditional elements featured in the Mashrabiya screens found in the chosen precedents.

After the screens were crafted, the samples were laboratory-scanned to ensure that all details were documented virtually and that all design features were included in the 3D virtual models to be tested using different analytical tools.

Next, ceramic screens and evaporative cooling elements were investigated. The outcome of this investigation was first-hand knowledge of the manufacturing techniques that will be followed in designing the screen. Moreover, they can also help identify the most applicable solution that can overcome limitations such as mass production and the environmental impact of the industrial stage.

Hybrid Validation Process

Throughout the design development stages, qualitative insights and quantitative findings were cross-referenced. This iterative validation ensured that the final screen design reflects cultural foundations and effectively performs as a passive, multifunctional evaporative cooling solution.

1.6 Structure of the research:

The fundamentals of shading and ventilation strategies in Mashrabiya and its common precedents are addressed in Chapters 2 and 3. On the other hand, Chapter 4 investigated the principles and fundamentals of evaporative cooling. Chapter 5 focused on understanding ceramic as a material—examining its properties and fabrication methods through various resources and a comprehensive literature review. Chapter 6 portrays the findings from the previous chapters and highlights the insights that shaped the analytical work. This collaborative effort aims to highlight the chosen samples and articulate the scale of designing and communicating with Mashrabiya craftsmen to facilitate the possession of screen samples for laboratory analysis, followed by the analytical work chapter that

investigates the performance of Mashrabiya and evaluates the screens in terms of shading and ventilation. These chapters shaped the findings and articulated the beneficial concepts found in the screens of Mashrabiya to integrate them into the design process of the proposed ceramic screen.

The first section of the research examined the roots of Mashrabiya, evaluated the intersection of local architecture and traditional lattice screens in Middle Eastern design, and investigated the approaches utilised by other studies in measuring the effectiveness of Mashrabiya in various contexts. Furthermore, the literature also addresses the current relevance of evaporative cooling, detailing how passive systems were utilised to define the design criteria for evaporative cooling screens that can be integrated into facades and the elements that hinder efficiency.

Furthermore, a comprehensive study of Mashrabiya principles regarding shading and ventilation formed the basis for selecting samples to be tested and analytically investigated for optimal scalability, pattern application, and design features that balance mutual shading, maximise effective surface area, and minimise airflow resistance. Samples were chosen from the Maymooni screen family, representing various traditional Mashrabiya screens, and laboratory scanning was conducted to document micro-details and design features into virtual models. These models were then evaluated using different analytical tools. A comparative study examined the technical qualities of the screens, correlating performance changes with design details and their impact on aerodynamics and shading performance, including mutual shading.

The findings of analysing the environmental performance of Mashrabiya guided the establishment of design guidelines to develop a multifunctional evaporative cooling screen. Subsequently, Chapter 7 explored the applicability of different ceramic shapes and colours as evaporators in architectural screens and façades, leveraging insights from testing the Mashrabiya typology to integrate relevant design concepts into the proposed screen. After finalising the performance evaluation of the Mashrabiya screen samples, the results highlighted how variations in design features, when fabricated in ceramic, can significantly influence the screen's environmental behaviour. Material properties such as emissivity and solar performance were critical in shaping the overall outcome.

In Chapter 8, Mashrabiya design concepts were fully integrated into the design workflow, with analytical testing conducted at each development stage to ensure the final prototype performed efficiently as a multifunctional element. An analytical workflow for designing an evaporative cooling screen was established, providing a structured approach to assess the benefits, challenges, and performance variables inherent in such systems. This workflow was applied to the proposed design, incorporating a climate study, ventilation strategy assessment, and evaporative cooling screen

performance testing. Through this process, the study identified how design geometry, material properties, and screen configuration influence thermal performance and water consumption, allowing for informed design decisions at each stage.

For fabricating the ceramic screen, the research intentionally focused on optimising its environmental performance through its designed geometry rather than altering ceramic chemical compositions or developing specialised manufacturing processes. To ensure the applicability of the proposed screen in various environments, traditional fabrication methods, specifically slip casting, were selected, along with commonly available clay materials with generic and widely used properties. The study evaluated the porosity levels and the water absorption capacity achievable through this conventional technique, demonstrating its suitability for evaporative cooling applications. This approach aligned with the project's aim to keep the design affordable, accessible, and easily replicable, ensuring that performance improvements were driven primarily by the geometry and form of the screen. Following the development and construction of the developed ceramic prototype, the hydraulic properties of the screen were carefully designed to allow efficient water flow through the hollow components, minimising the water loss factor and ensuring consistent surface wetting to enhance the evaporative cooling effect. Accordingly, chapter 8 aimed to finalise the prototype, covering both the design development and the performance evaluation of the proposed evaporative cooling screen. The ceramic screen underwent rigorous testing and validation to achieve a balanced multifunctional performance. Accordingly, different tests were carried out to determine the most efficient clay properties to employ during the fabrication of the finalised design to optimise its environmental response.

Chapter 9 presents the final reflection on the key results of this research, culminating in the real-world 1:1 application of the screen in a hot climatic region. This practical stage allowed the screen's efficiency and performance to be validated under actual environmental conditions, reinforcing the relevance and applicability of the research outcomes. The chapter also highlights how integrating Mashrabiya design concepts into the evaporative cooling strategy directly improved the performance and efficiency of the proposed screen standing as a testament to this research's innovative aspects, illustrating how Mashrabiya's applicable design principles were carefully utilised and integrated throughout the process, from early design thinking to final performance testing.

In line with the overall research progression, the work is divided into five main stages, which are:

Part 1. The Literature Review

This part is subdivided into four sub-sections, which are:

Section 1 – History of Mashrabiya:

1. Illustrating the passive solutions vernacular architecture gave birth to in dry-hot climatic regions.
2. Mashrabiya and its origin.
3. History of the traditional patterns.

Section 2 – Fundamentals of Mashrabiya:

1. Traditional perforated patterns and Shading.
2. Traditional perforated patterns and Ventilation.

Section 3 – Indoor Comfort and Evaporative Cooling:

1. Direct evaporative cooling and its boundaries.
2. The quantification of the cooling performance in Direct evaporative cooling
3. Evaporative cooling and solar exposure.
4. The Water Supply and Consumption in Evaporative Cooling.
5. The Wet Media.
6. Challenges in Evaporative Cooling Systems.

Section 4 – Modelling using Ceramic:

1. Ceramic.
2. The attractive features of applying Ceramic in architectural solutions.
3. The modelling techniques of ceramics.
4. How to control the porosity of Ceramic.
5. 3D printed Ceramic and different fabricating methods.

Part 2. Analytical Preliminaries and Environmental Performance Evaluation

The designs and geometry of the screens used in Mashrabiya vary from place to place, depending on aesthetic and architectural needs as well as regional preferences. Different examples of old Mashrabiya screens were selected and analysed to understand these ratios better. On the other hand, it was essential to understand and highlight the commonly implemented screens in Mashrabiya precedents and relate these designs to the analysed results. Therefore, to comprehend the performance of Mashrabiya in terms of shading, ventilation, and evaporative cooling, it was necessary first to grasp the etymology of the patterns and how they were formed. It was also crucial to understand the dominant characteristics and the factors that led to the dominance of particular forms and screens, which gained popularity and preference over time. Therefore, this stage focuses on choosing the samples that will be tested and analysed to understand the performance of Mashrabiya and highlight the design features that can be integrated into the proposed ceramic screen. This stage also includes the methodological approach of defining the design scale and the chosen samples' dimensions.

The analytical studies primarily focus on experimental investigations that evaluate the environmental performance of Mashrabiya (traditional perforated screens) using various analytical methods and software, such as Climate Studio, Ladybug, and Honeybee. This chapter is organised into three major subsequent phases, each of which adds to the growth of the following step. The first stage of the analytical work is to establish the objectives and determine what needs to be accomplished through the analytical studies to make them valuable and beneficial in the design process. This is followed by establishing the scope of the study, which will help us determine the outline of the approach and the work frame of the study.

Evaluating the environmental performance of Mashrabiya can be quantified by analysing the total performance of the entire screen or through micro-level studies that allow us to understand the effect of the elements that make up the screen and relate this performance to the design and manufacturing details that were passed down as an inherited science through generations of craftsmen in the field of manufacturing Mashrabiya.

The primary objective of this analytical study is to understand the design features of Mashrabiya screens and assess their impact on their environmental performance in terms of shading and ventilation.

The final stage of the analytical work will involve a comparative analysis of the performance of the material with different colours under direct and indirect solar exposure Accordingly, various colours of clay were used to fabricate the same geometry and were exposed to the same amount of solar incidence to help determine the materialistic properties and colour characteristics of the clay that will be used in the fabrication of the prototype. This analysis informed the decision-making process of choosing the clay colour for the prototype fabrication.

Part 3. Design development and fabricating the prototype

Chapter 8 summarises the conclusion of our design journey, based on knowledge gained from further experiments, such as the emissivity test, and the analytical analysis of Mashrabiya. This chapter examines the methodical development of the proposed screen design, documenting the rationale behind each decision made during the design process. Mashrabiya became a guiding influence, offering a wealth of concepts and ideas that influenced our design decisions. We aimed to validate our choices and ensure they aligned with the project's overall goals by thoroughly analysing each design aspect. Through this thorough investigation, the research successfully leveraged Mashrabiya's rich history to create a screen that effectively integrates innovation and tradition, addressing relevant cooling strategy concerns.

After finalising the design, we started fabricating the prototype, which was enhanced by valuable insights from numerous experiments. Some of these insights were meticulously documented within this chapter to guide the fabrication process. In contrast, others were recorded in our research journal, which aimed to document all experimentation undertaken throughout the research journey comprehensively. These efforts culminated in producing a 3D-printed prototype, which served as the basis for further research and development. Working closely with knowledgeable ceramicists and talented artisans, we leveraged their expertise to optimise the mould and complete the manufactured prototype. This cooperative effort was crucial to improving our prototype and ensuring it met its intended design goals.

Part 4. Testing the Full-Scale Screen

The finalised design of the screen, informed by thorough analytical studies of Mashrabiya's design features and performance, serves as a foundation for enhancing ventilation, shading, and ceramic production feasibility. Guided by Mashrabiya principles, decisions were made on pattern repetition, adaptability, and modular structure, enabling flexibility in design adjustments. However, further refinement can be made to optimise the screen's clay qualities and responsiveness for improved evaporative cooling effectiveness. Fieldwork in hot, dry climates will test the screens' performance at a 1:1 scale, connected to water supplies, to evaluate their real-world effectiveness. This hands-on approach will provide crucial insights for fine-tuning and ensuring that the screens meet performance standards, contributing effectively to cooling strategies.

Part 5. Design Journal

Along with the dissertation and the provided chapters of the study, a journal is included. This journal aims to document all the experiments carried out in the research and serve as a repository of all the experiments conducted throughout the research journey. It is designed to provide valuable insights into the background and evolution of the research process, offering guidance for fellow researchers interested in this domain. It also helps understand the successful trials and failures behind each stage and offers valuable insights regarding how the process was developed. The Journal also serves as an appendix, gathering images and process stages that were not essential to include in the main chapters, but are added here to provide additional insight into the intermediate stages and to further illustrate the design development.

2. History of Mashrabiya:

Understanding the environment in which we live is essential to engaging with it in a manner that conserves resources for future generations. Vernacular architectural practices in the Middle East are widely regarded as foundational to many of the ventilation, passive cooling, and shading techniques employed in contemporary sustainable design (Dabaieh, 2013). These traditional strategies offer valuable guidance in formulating design principles that enhance passive interaction between built structures and their surrounding outdoor environments.

Tracing the development of these patterns illustrates how they affect the screens' shading and ventilation performance. The Mashrabiya is a component that functions as a shading element, blocking multiple solar angles from penetrating the indoor space and regulating the light entering through the screens. In addition, the screens allow for efficient ventilation techniques that regulate the airflow to an adequate depth inside the space. As a result, the Mashrabiya has come to embody a harmonious integration of functionality and aesthetic appeal.

2.1 Mashrabiya in Vernacular Architecture:

Passive solutions in extreme climatic conditions were designed to respond to challenging circumstances at specific times of the year but may have negative impacts during other periods. All such factors fostered passive design solutions, such as suitable construction materials and various shading and ventilation techniques, which can mitigate the effects of extreme climatic conditions during particular seasons of the year (Weber, 2014).

The design solutions in vernacular architecture were meant to avoid discomfort in the users' surrounding environment and to improve the conditions of the building. In Middle Eastern vernacular architecture, buildings were designed to adapt to excessive solar radiation by using various shading techniques, as well as by employing the urban fabric of villages to provide mutual shading for the buildings by keeping them close to one another (Weber, 2014). Focusing on the micro-level and the small details embedded in the design techniques, the buildings used to have a rigid, coarse texture created using stones and bricks with a specific construction method. From a closer view, the craggy texture of the walls and bricks becomes noticeable. The concept behind these details is to create self-shaded areas on the walls themselves. In addition, variations in surface solar exposure can affect both the visual appearance of the material and its surface temperature (الزهراني, 2007).

Figure 1Vernacular buildings in Asser, Saudi Arabia and passive use of materials in the exterior walls

Hassan Fathy, known as the architect of the poor, was one of the pioneers in reintroducing and implementing vernacular passive solutions in his contemporary designs. He presented sustainable, environmentally friendly architecture by utilising local materials and employing low-cost methods to achieve thermal comfort—an approach influenced by the vernacular architecture of southern Egypt (EI-shorbagy, 2001).

Highlighting the Mashrabiya in Fathy's architectural works traces back to the cultural foundation of his designs. As a result, the Mashrabiya was incorporated into the design of the Abd al-Rahman Nassif House in Jeddah in 1973 (EI-shorbagy, 2001). Later, in the design of the Kazerouni House, the Mashrabiya appears implicitly on the main façade, alongside other passive solutions (EI-shorbagy, 2001). Fathy demonstrated how successful architectural designs can be achieved when cultural and heritage merge with environmental awareness. According to his principles of façade design, reducing the perforation of lattice screens at eye level during summer is essential to prevent direct sunlight. In winter, when the sun's angle is lower in the sky, the sunlight can pass through the same Mashrabiya and provide the desired light and warmth (fathi, 2000). The upper part of the Mashrabiya employed wider perforated screens to compensate for the reduction in airflow—an approach exemplified in the design of Jamal al-Din al-Zahabi's house in Cairo (fathi, 2000).

Figure 2 Mashrabiya at Jamal al-Din al-Zahabi's (Fathy, 1986)

2.2 The etymology and historical background:

Mashrabiya was one of the most well-known terms for lattice screens in Islamic architecture. These names varied depending on the region and the primary function of the screen. For instance, in some parts of the Middle East, the screens were called "Rawshan". Some sources attribute the common meaning and origin of this name to the Persian word "Rawshn," which means "twinkle" in English—referring to the shading performance of the screens that causes the light passing through them to twinkle and the screens themselves to shine in an artistically beautiful manner (حسنين, 1982)The term "Mashrabiya" is derived from the Arabic root "Sharab," meaning "to drink", and refers to the ceramic pot that used to be placed in the transitional space created by the extruded geometric form of the Mashrabiya (الوهاب, 1946) (Schiano-Phan, 2004). The primary purpose of placing a ceramic pot was passive environmental cooling. These pots were used to cool the water inside, and when positioned near the air inlet, they helped cool and humidify the air entering the space. The use of ceramic as a material for the water pot was based on its porosity, which allowed water droplets to form on its outer surface. Consequently, airflow over these micro-droplets produced what is known as the "evaporative cooling process"—a phenomenon this research will examine in detail (الوهاب, 1946) (Schiano-Phan, 2004).

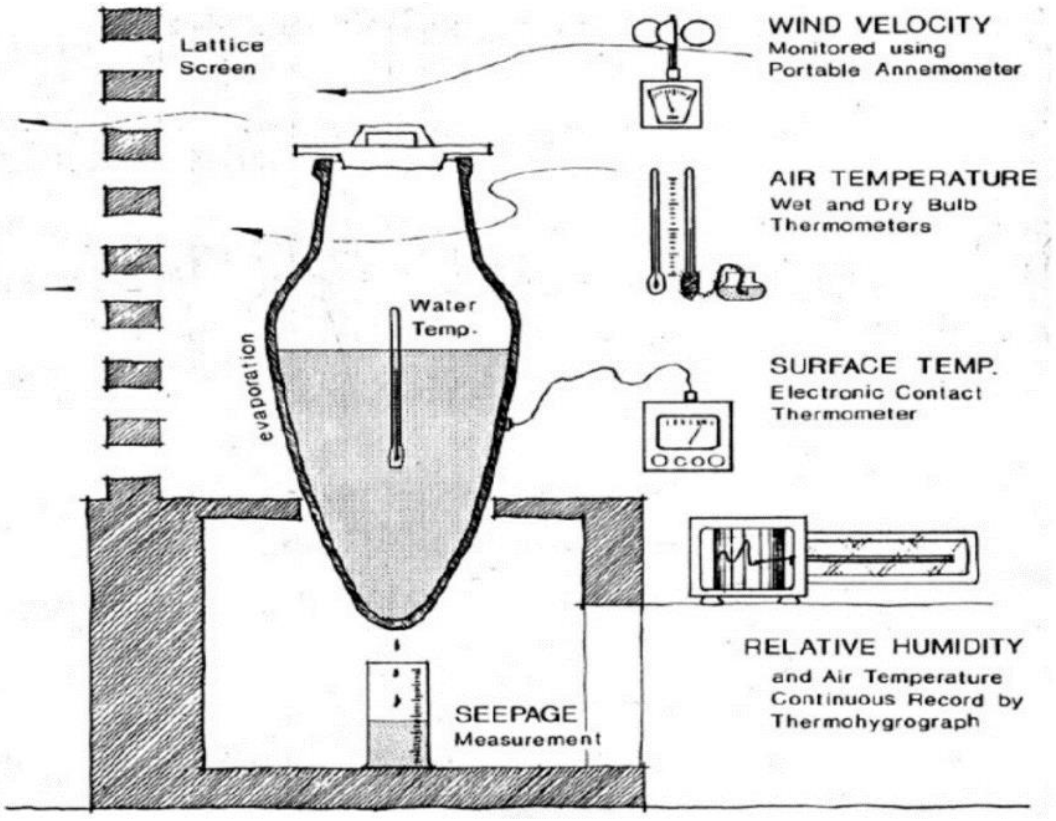


Figure 3windows or balconies containing a porous pottery or clay jar filled with water (Petersen, 1999)

From the various names of the Mashrabiya, it can be concluded that changes in shape and geometry were primarily linked to the function of the element and its dominant interaction with specific environmental conditions. Such insights can therefore be taken into consideration when determining the principal function behind implementing these traditional elements. The Mashrabiya is regarded by environmental researchers and designers as a passive design strategy developed to help buildings respond to environmental parameters that affect indoor comfort in hot climates (Schiano-Phan, 2004).

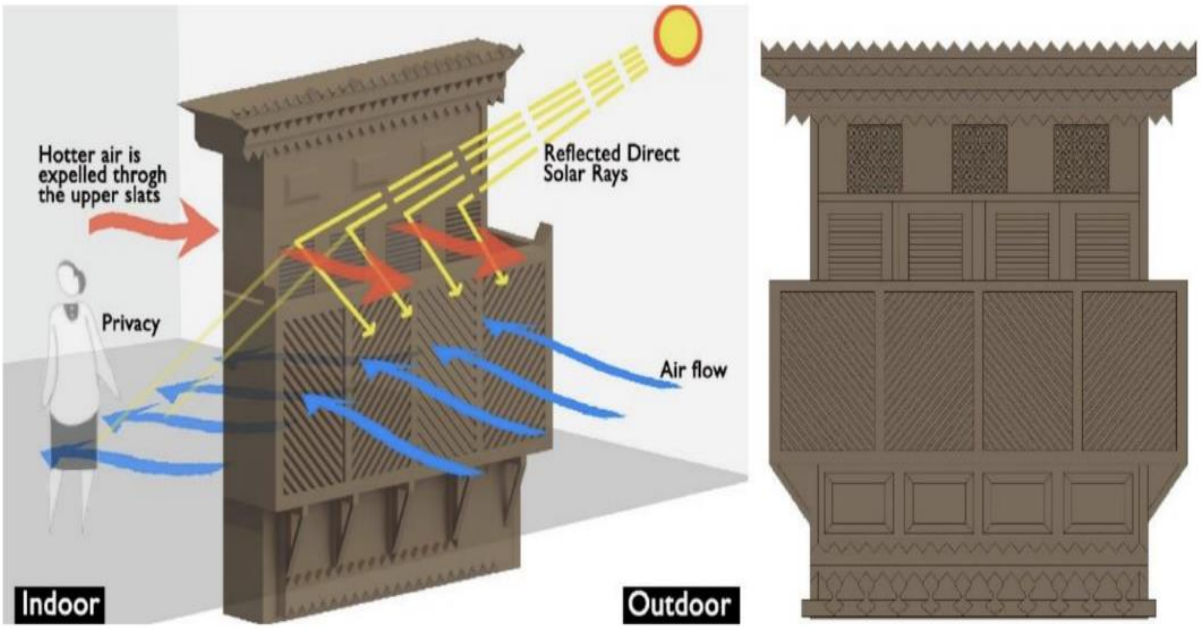


Figure 4Example of Mashrabiya demonstrating the principal functions. (Bagasi, 2020)

2.3 The screens and patterns:

Each screen had a specific name and a crafting technique that was commonly used among artisans. Sometimes, the name of the pattern was derived from a particular group of craftsmen or the family who originally designed the screen. For example, the Maymooni pattern was named after its place of origin, Maymooni village in Egypt. Understanding the design techniques behind these patterns, along with the historical context of their conceptual origins, can offer a clearer understanding of the craftsmanship principles behind these screens. Despite the evolution and development of the patterns, their names typically remained fixed in reference to the original screen, as illustrated in Figure 5 (Ashi,2010).

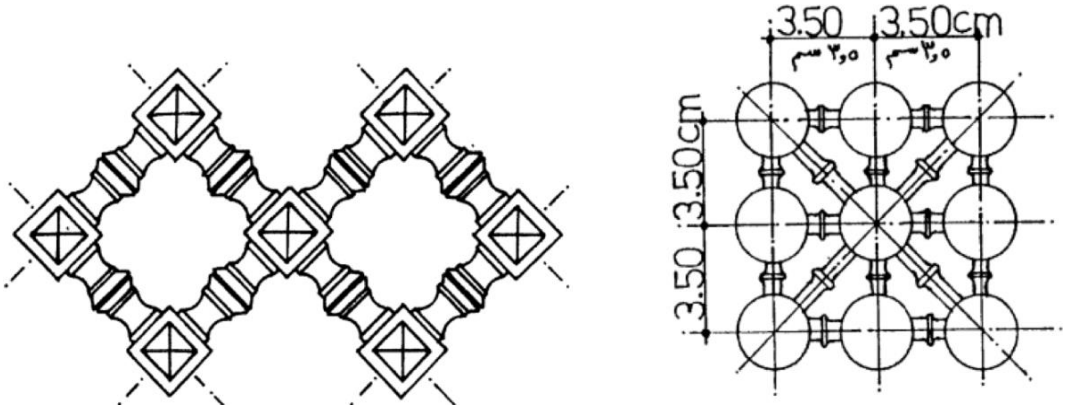


Figure 5 Types and shapes of Maymooni /Maymooni patterns (Salem, Evaluating the environmental performance of Mashrabiya, 2019)

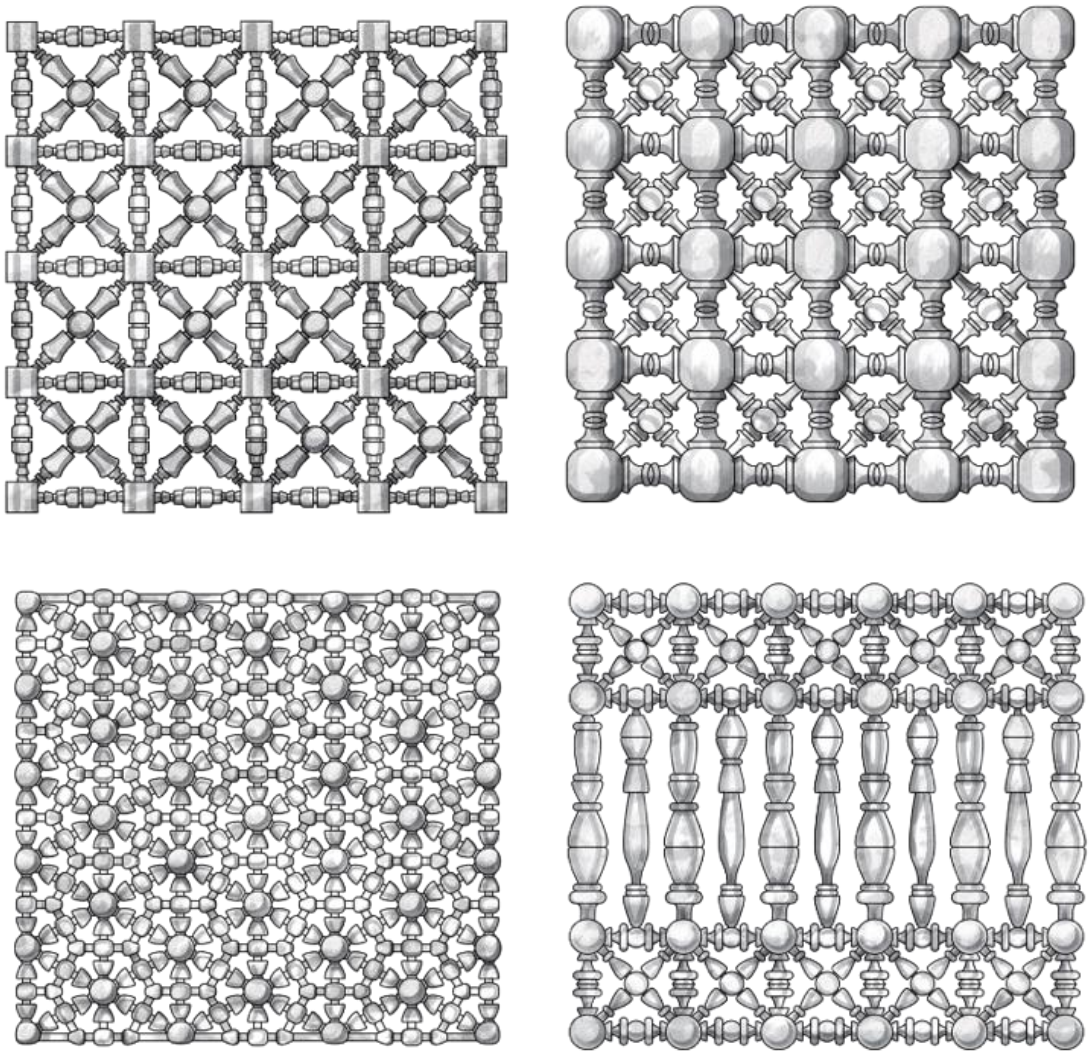


Figure 6 : Different shapes of the Maymooni pattern (Salem, Evaluating the environmental performance of Mashrabiya, 2019)

Looking back at Islamic history, art and geometry were considered high-priority fields in both knowledge and culture. Arabs emphasised calligraphy and the integration of Arabic letters with geometric design, which was initially used to decorate walls and later evolved into intricate decorative patterns. These patterns were originally applied to mosques and other high-value structures, but their popularity expanded over time, eventually making them common decorative elements in many Middle Eastern buildings of that era. However, certain patterns held greater value and incurred higher crafting costs due to their complexity and the challenges involved in the design process. Most patterns were developed using a specific ratio and a well-defined repetition module, which together created larger harmonious patterns to ensure consistency and visual appeal when applied on a larger scale. Repeating a pattern typically requires adherence to a technique that could be distilled into a formula or equation.

This formula was usually created by the designer or the craftsman who developed the pattern and required careful consideration of scale and material selection in the crafting process (محمد, 1997).

The primary material used in the manufacture of the Mashrabiya was wood, although some examples incorporated ivory or metallic components. In certain regions, Mashrabiyas were also crafted using alternative materials such as marble, stone, plaster, or bricks.

Greenlaw (1995) outlined the typical dimensions of the Mashrabiya: its width generally ranged between 2.4 and 2.8 meters, while its internal height was typically between 2.7 and 3.5 meters. The proportions of the Mashrabiya corresponded to the dimensions of the human body; it was broad enough to lie down comfortably—normally between 2.0 and 2.4 meters—and tall enough to stand within, approximately three meters in height, with a projection of around 60 centimetres into the street (Greenlaw, 1995).

The patterns used in Mashrabiya's design varied depending on the element that was repeated within a specific grid or module. For instance, wooden screens with identical perforation percentages could differ in overall appearance due to variations in the density and geometry of their microelements. These microelements are typically derived from the Islamic Geometric Patterns (IGP) A level, where the scale and repetition of a single shape on a larger grid are influenced by the original geometric form of the unit—illustrated in Figure 8 (Wilson, 1988).

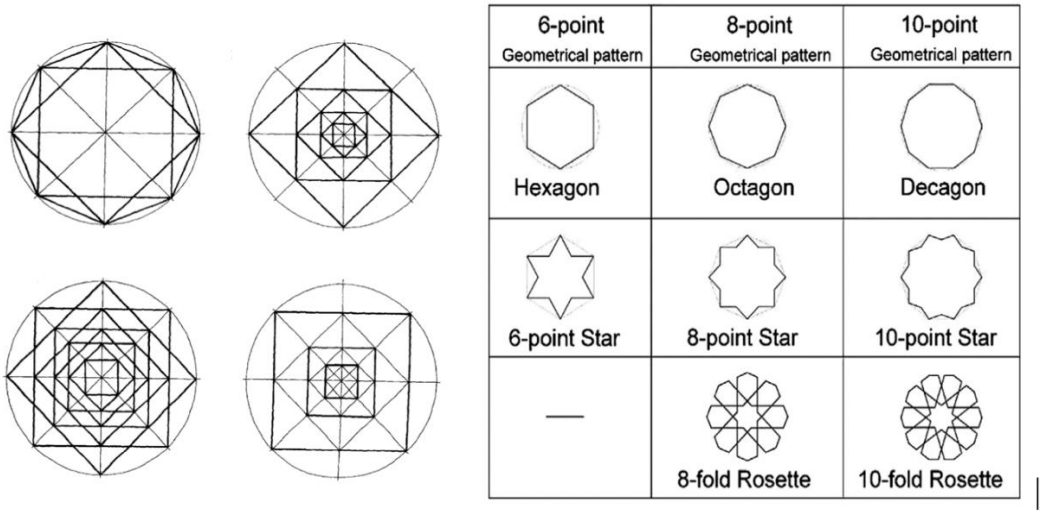


Figure 7 The First level of IGP design & designing stages of the 8-point star (Wilson, 1988).

The design of the patterns occurs on consecutive levels. These levels are:

- The first level concerns the overall meaning of the patterns and the concept the design represents. Typically, the concept begins with the intended meanings, followed by the cultural dimensions and human perspectives it aims to reflect, as well as the detailed semantics rooted in religious and social contexts. The design of patterns is frequently associated with values

embedded in the user's conscience, such as moderation, privacy, contrast, and connectivity (Wilson, 1988).

- The second level involves the roles that define the pattern's formation. At this stage, designers outline the unique structure of the pattern and delineate the boundaries of the space it occupies, which are primarily determined by the intended function of the pattern and the scale of its application (1997 ,محمد) (Wilson, 1988).
- The third level is the visual form of the component. This level focuses on the repetition of the shape and the methods used to preserve visual harmony across various forms and designs. It is essential for ensuring the delivery of well-executed patterns that maintain correct repetition ratios and scaling across different geometries. The repetition process largely depends on the geometry of the small elements that compose the screen. Thus, identifying the primary geometry that constitutes the pattern defines the repetition module and the axes of symmetry on a larger scale (1997 ,محمد) (1989 ,نظيف).

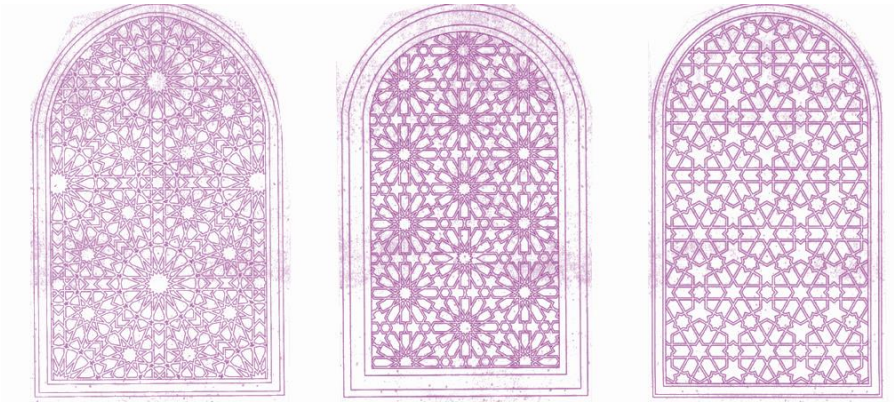


Figure 8out the development of the patterns formed on the same window by changing the diagonals of the star

- After generating designs on a 2D modular grid, the patterns that attracted the most attention and praise from viewers were more frequently requested for application on a three-dimensional scale, where designers crafted them in wood or stone, and sometimes carved them into the walls and columns of mosques (Wilson, 1988).

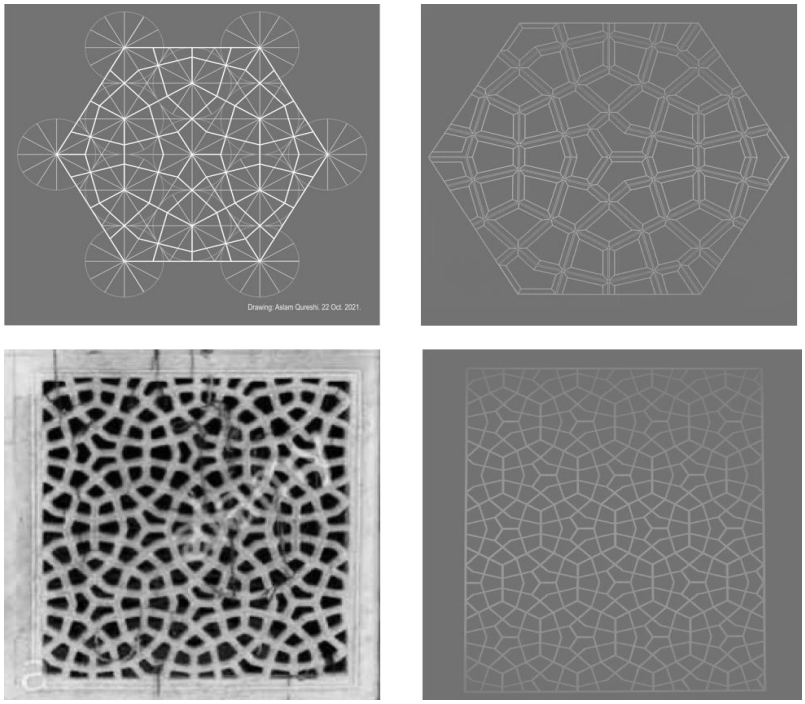


Figure 9 illustrating the evolution of a 2D pattern that is linked to the 3-dimensional ornaments by Aslam Qureshi

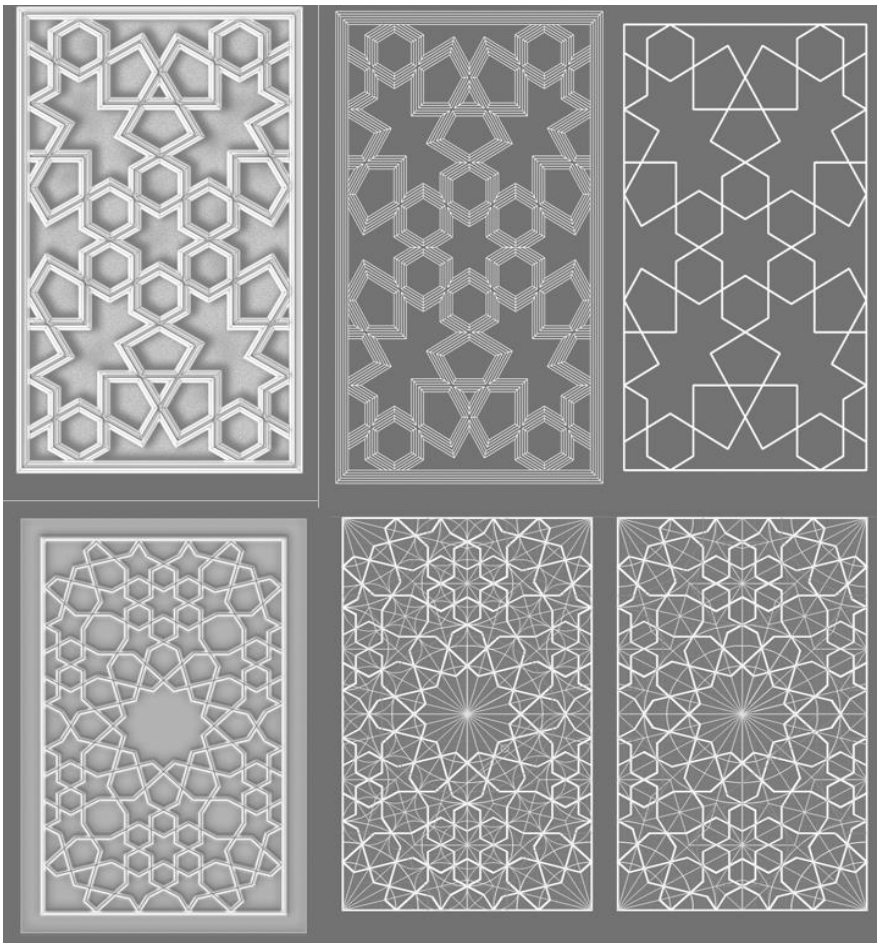


Figure 10 illustrating the evolution of a 2D pattern that is linked to the 3-dimensional ornaments by Aslam Qureshi

2.4 Manufacturing of the patterns:

The patterns in Islamic architecture varied in shape, structure, and design technique. Floral motifs, often found in Andalusian screens, differed from the geometric patterns used in Ottoman and Fatimid screens in terms of perforation percentage, drawing procedures, and manufacturing methods. The screen components were handcrafted using manual wood lathing tools. Despite the significant time and effort invested in manufacturing, the screens were expected to be rich in detail.

The principal tool was a piece of wood mounted on an axis, with a sharp blade or thread attached to facilitate carving the wood into the desired shape. The cutting thread, often bow-shaped, rotated the wood as the craftsman moved the bow back and forth. The first stage of manufacturing the screen is generating the pattern (الجواد, 1987). After producing the small latticework components, they were joined and dovetailed together without the use of nails or glue. The design intent was to create flexible forms that could regulate light and airflow, while also simplifying the repair of damaged sections without needing to replace the entire screen. Once assembled, the parts were secured within a structural frame to ensure the proper distribution of stress across the screen (الجواد, 1987). With advancements in crafting tools, it became more feasible to produce intricate components at lower costs, enabling mass production to meet market demand. Traditional Islamic lathing has historically demonstrated greater precision than electric lathing, which often struggles to replicate the fine modular beads that appear in diverse forms (الجواد, 1987).

Figure 11 The method of attaching the parts lathed together like Lego, the perforation, and the pattern are formed using the modular design—the image is from Nadeem craftsmen's workshop in Cairo.

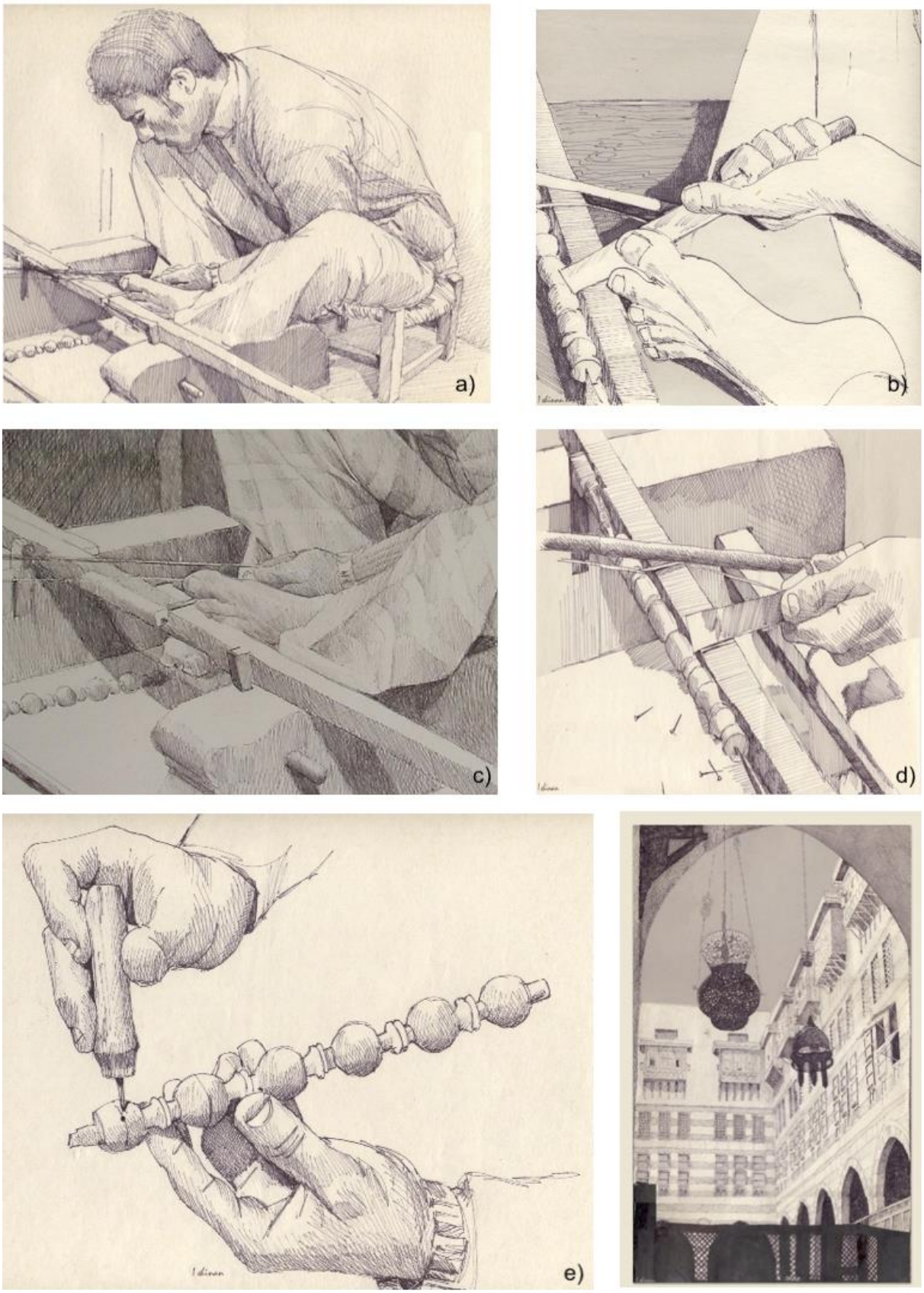


Figure 12 Steps of traditional crafting methods from the author's portfolio on the craft of khart in Egypt: a) The wood piece is fixed onto the axis of the tool. b) The wood is lathed using a sharp tool to achieve the initial form. c–d) The wood is twisted while the tool moves back and forth. e) The final shape is refined using fine lathing tools (Ishinan 1979).

2.5 Conclusion

The Mashrabiya demonstrated its function from its very name, which inherently reflects its environmental role within traditional architecture. Far from being a mere decorative element, the Mashrabiya served as a climatic device that enhanced the performance of buildings in hot arid environments. It generated dynamic interior patterns by manipulating light and shadow while simultaneously creating shaded and ventilated indoor spaces. Therefore, Inhabitants could enjoy thermal comfort without relying only on massive walls or small openings. Importantly, the Mashrabiya also maintained a degree of openness, enabling residents to remain visually and acoustically connected to the outside world. They could view the surroundings and hear the call to prayer from nearby mosques, experiences which would have been diminished had the screen functioned purely as a barrier. These observations support the argument that the Mashrabiya was never intended solely as a privacy device—its presence in courtyards, where privacy was already secured, highlights its environmental and social significance as an intermediary between inside and outside. In this way, the Mashrabiya allowed buildings to "breathe," mediating between external environmental challenges and the human need for comfort, while still engaging with the beauty of the external world.

Beyond its environmental performance, the Mashrabiya also pioneered design complexity and craftsmanship. It was not just a flat wooden panel with decorative perforations; it was a complex lattice formed of detailed lathed wooden elements, carefully assembled to achieve structural coherence and aesthetic richness. The dedication of this craftsmanship, achieved at a time when tools were relatively rudimentary, raises critical questions. Why did Mashrabiya craftsmen dedicate such time and precision to producing such elaborate details? Was it purely aesthetic reasons, or did the micro-scale of the components enable practical benefits such as flexibility and improved maintenance, or specific environmental functions that these elements could represent? The high degree of adjustability and the variation in lattice density suggest that the Mashrabiya may have offered control over airflow, daylight penetration, and privacy. Thus, its complexity was symbolic of artistic refinement and may indicate an advanced understanding of environmental performance that requires deeper investigation.

Accordingly, this review suggests that the Mashrabiya should be considered a multifunctional architectural element, one that integrates environmental, social, and aesthetic roles within a single device. The following chapter will, therefore, examine these environmental features in greater depth through a comprehensive literature review, while subsequent analytical studies (Chapter 7) will test the screens' environmental performance. Through this dual approach, the research seeks to uncover the historical intelligence embedded within the Mashrabiya and its potential relevance for contemporary sustainable design.

3. Environmental Fundamentals of Mashrabiya:

Mashrabiya provides a sensual division of interior space by creating a semi-transitional zone enriched with varying light patterns (الجواد, 1987). These light variations enhance the user's gradual interaction with both the space and the surrounding environment, helping to minimise abrupt shifts resulting from sudden transitions between exterior and interior conditions (waziri, 1999). Mashrabiya was commonly implemented to promote natural airflow into indoor spaces. The ventilation performance of the component was primarily influenced by the typology of the screen; the perforation percentage was often used as a key indicator of effectiveness in ventilation performance (waziri, 1999). Accordingly, the ventilation efficiency and the velocity of air penetrating the screens are directly proportional to the perforation ratio. Later research also revealed that ventilation is significantly affected by the shape and typology of the screens themselves. Floral patterns with smooth, organically shaped forms tend to enhance airflow through their perforations due to variations in the discharge coefficient at the micro level (الجواد, 1987).

3.1 Natural ventilation in Mashrabiya:

Buildings should be designed to adapt to the climatic conditions of their regions in order to achieve comfort while minimising energy consumption. Thus, a matrix of interrelations exists between the building, the user, and the environment (Brian Ford, 2020).). In this context, factors such as temperature, air velocity, air purity, and humidity play essential roles in determining spatial comfort and performance. Equally important, however, is the psycho-sociological context, which significantly influences user experience (Penn, 1990). Accordingly, developing an effective ventilation strategy requires a comprehensive understanding of all relevant factors. To design a successful ventilating component, designers must thoroughly consider the spatial characteristics, the surrounding context, climatic conditions, and any limitations that might negatively affect users within the space (Clements, 2002).

As Hassan Fathy famously observed: “When the modern architect replaced these decorative elements with air-conditioning equipment, he created a large vacuum in his culture. He has become like a football player playing football with a cannon. If the purpose of the game is scoring goals, then assuredly he can score a goal with every shot. But the game itself will

disappear, and so will any diversion for the spectators, except perhaps in the killing of the goalkeeper."(Fathy, 1986).

The natural ventilating component can be integrated into spatial openings—whether as windows or shading elements—and can be designed to adaptively respond to multiple environmental parameters simultaneously. This adaptability is enhanced through a systematic interior layout and the strategic placement of openings that facilitate effective air circulation (Dina Howeidy PhD., 2017). Moreover, numerous research studies and publications have explored common strategies for natural ventilation, emphasising the applicability of such methods in various design contexts based on the geometry of openings and the natural driving forces of wind and buoyancy (Elwan M. M., 2019).

Stack ventilation is one of the most effective strategies in natural ventilation. This approach has proven efficient in enhancing indoor airflow by promoting air circulation and delivering the required amount of fresh air into the space (Brian Ford, 2020). Researchers consider access to fresh air a fundamental element in creating a healthy indoor environment and in enhancing users' well-being and comfort (Allard, 1998).

Intermediate spaces represent one of the techniques that help users adapt to surrounding conditions. Therefore, architectural designs must establish mediation between different activities within the space (فكري, 1981). Such spaces encourage user interaction with the outdoor environment in a gradual, anticipatory manner that allows for a comfortable transition between contrasting interior and exterior conditions (نظيف, 1989). Designing an element that responds solely to a single environmental parameter introduces a limitation in the overall design strategy (Donald Watson, 1983). Given the interdependent nature of environmental parameters, addressing only one factor while disregarding others compromises both adaptability and performance efficiency. In other words, it is not possible to evaluate the performance of a building component in terms of ventilation without also considering its shading properties and its impact on heat gain (الجواد, 1987). Thus, implementing a strategy such as stack ventilation to provide adequate airflow throughout a space requires a sound understanding of the surrounding environmental context. Buildings should be designed to optimise the use of natural resources such as wind and sunlight in a passive and environmentally responsive manner (Clements, 2002).

The performance of the Mashrabiya can be analysed by comparing it to regular windows with the same effective aperture area, assessing how the screens influence the volume of penetrating airflow (Elwan M. M., 2019). Consequently, the efficiency of this passive strategy improves when it facilitates sufficient air circulation while simultaneously minimising solar heat gain through effective shading. This balance between airflow and solar control can be quantitatively assessed using the following equation (Srisamranrungruang, 2020):

$$Qv = p_{air}\ c_p\ Q_s\ \ (T_{odp,pp} - T_{zone,pp}) \quad (1)$$

Where:

- q_v: is the heat added to receiving zone air by airflow,
- p_{air}: is the air density [kg/m3],
- c_p : is the specific heat capacity [kJ/kg. K].
- $T_{odp,pp}$ & $T_{zone,pp}$: are the outdoor and the indoor temperatures prospectively.

While quantifying the efficiency, acknowledging the solar gains can be measured by the following equation (Srisamranrungruang, 2020):

$$\sum_{t=0}^{t=N} q_{pp} = p_{air}\ c_p\ \ Q_{s,pp}\ (T_{odp,pp} - T_{zone,pp}) \quad (2)$$

Where:

- (t=N) is the total solar gain over simulation time,
- q_{pp} is heat added to receiving zone air by airflow,
- p_{air} is air density [kg/m3],
- c_p : is specific heat capacity [kJ/kg. K].

Based on various analytical studies, the Mashrabiya has proven to be an efficient solution for single-sided ventilated spaces—a common scenario in many residential buildings (Srisamranrungruang, 2020). Most residential houses face limitations due to their surrounding context, which often restricts them to single-sided ventilation only (Elwan M. M., 2019). In such cases, providing fresh air and effective cooling necessitates a well-designed ventilation solution capable of delivering sufficient airflow across the full depth of the space (Givoni, 1976). Researchers broadly agree that by understanding and addressing the limitations of the Mashrabiya, its performance can be improved and adapted to meet users' comfort demands in contemporary architecture (Baarimah, 2019).

The strategies for single-sided ventilated spaces depend primarily on controlling the airflow rate at different parts of the opening and reducing obstructions to incoming air. Accordingly, the Mashrabiya, as a component for single-sided ventilation, can incorporate more than one strategy within the same element through the exploitation of pressure differences and the stack effect. As a primary air inlet, the Mashrabiya plays a vital role in directing airflow through the space. Therefore, the selection of the screens applied to the component was traditionally based on specific calculations to ensure adequate ventilation performance (Elwan M. M., 2019). When a building is naturally ventilated, the amount of heat added to or removed from the indoor space through airflow—denoted as Qv—is

calculated as the product of the airflow rate, the air density, the specific volumetric heat capacity of air, and the average temperature difference between indoor and outdoor conditions, as shown in Eq.*(1)*.

Single-sided openings in a space can also enable natural ventilation when wind-induced pressure differences occur across them. However, wind variability conditions are too complex to be expressed by a single general formula, due to the fluctuating nature of the airflow. These fluctuations are attributed both to the turbulent characteristics of the incoming wind and to turbulence generated by the building itself. Such turbulence contributes to variations in the pressure distribution around the openings, as demonstrated in Eq. (*3)* (Allard, 1998).

In 2019, a research paper by Elwan (2010) analysed the performance of the Mashrabiya in terms of ventilation, quantifying the behaviour of its components at the micro level *(Elwan M. , 2010)*. The study highlighted the functional behaviour of the individual elements that compose the screen, making it possible to assess its collective performance on a larger scale and evaluate the impact of combining multiple screens *(Elwan M. , 2010)*.). Accordingly, the central screen of the Mashrabiya, often with a lower perforation percentage, can function as the stack height in a single-sided stack ventilation strategy, while highly perforated screens are positioned at the top and bottom to serve as primary inlets and outlets (Salem, 2019). Moreover, in spaces with large openings—or dual openings located at upper and lower points on the same façade—the temperature difference between indoor and outdoor conditions generates an air density gradient: warmer air (lower density) rises, while cooler air (higher density) sinks. This gradient creates a pressure differential between interior and exterior zones. As a result, the elevated pressure at the upper interior causes air to exit, while fresh air is drawn in through the lower opening. This buoyancy-driven phenomenon initiates stack ventilation at a larger scale, working to equilibrate indoor and outdoor temperatures. Once this thermal equilibrium is reached, stack pressure approaches zero, and the driving force behind the ventilation diminishes gradually *(Givoni, 1976)*.

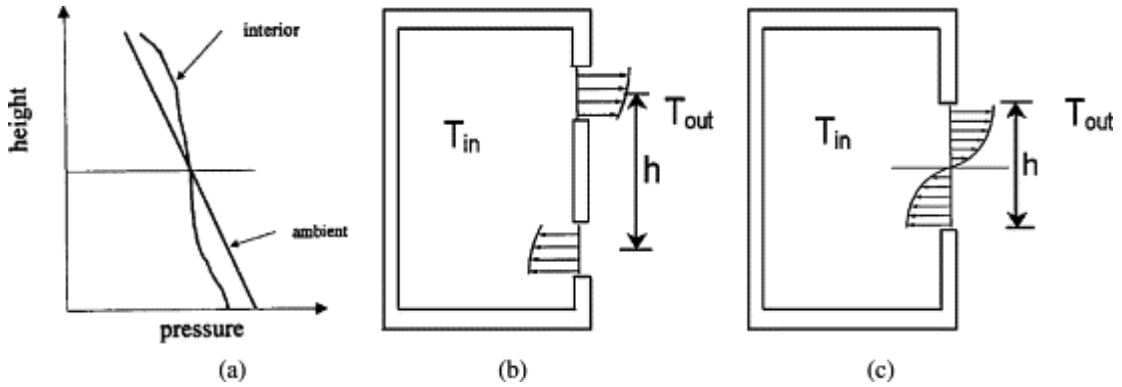


Figure 13. Indoor and outdoor pressure distribution for buoyancy-driven flow: (a) causes flow through an upper and lower opening; (b) or a single opening; (c) (Allocca, 2002).

Quantifying the efficiency of natural ventilation due to the stack effect is calculated using the following equation:

$$Q_s = C_D\ A_{oppening}\ F_{schedule} \sqrt{2\ g\ \Delta\ H_{NPL}(eclose - T_{odb}/T_{zone})} \quad (3)$$

Where:

- Q_s—Volumetric of airflow due to stack effect [m3/s],
- $\Delta\ H_{NPL}$—height from the midpoint of the lower opening to the neutral pressure level [m],
- $F_{schedule}$—open area fraction,
- $A_{oppening}$ —opening area [m2],
- T_{zone} —zone air dry-bulb temperature [K],
- T_{odb}—Outdoor air dry-bulb temperature [K],
- C_D—discharge coefficient.

Therefore, the efficiency of stack ventilation in single-sided ventilated spaces is directly proportional to the increase in the stack height, which can be represented by the less perforated screens used in the middle section and at the eye level of the Mashrabiya. While the relationship between stack ventilation and temperature difference is generally considered linear, studies have shown that screens with perforation percentages below 50% reduce the airflow rate due to a lower discharge coefficient and increased obstruction to air movement.

Some wooden lattice screens improve the ventilation process through the depth of the space and allow air circulation within the room. The buoyancy-driven airflow can be measured

by studying the air pressure difference at the upwind-facing window of common lattice screens such as *Al-Maymooni*, *Al-Salibi*, and *Abu-Sherwan*. A 2020 study visualised the ventilation performance of perforated screens using Computational Fluid Dynamics (CFD) simulations. The shape of the baluster and the curvature diameter of the components forming the screens have a significant impact on the airflow passing through, as they influence the pressure distribution across the screens—thereby enhancing performance in single-sided ventilation. The study further indicates that the shape of the screen elements, along with the crafting techniques, enhances the airflow rate by influencing the angle at which wind approaches the screen. Diagonal elements within the screens generate pressure-difference zones as a result of multiple round components facing the incoming airflow (Elwan M. M., 2019).

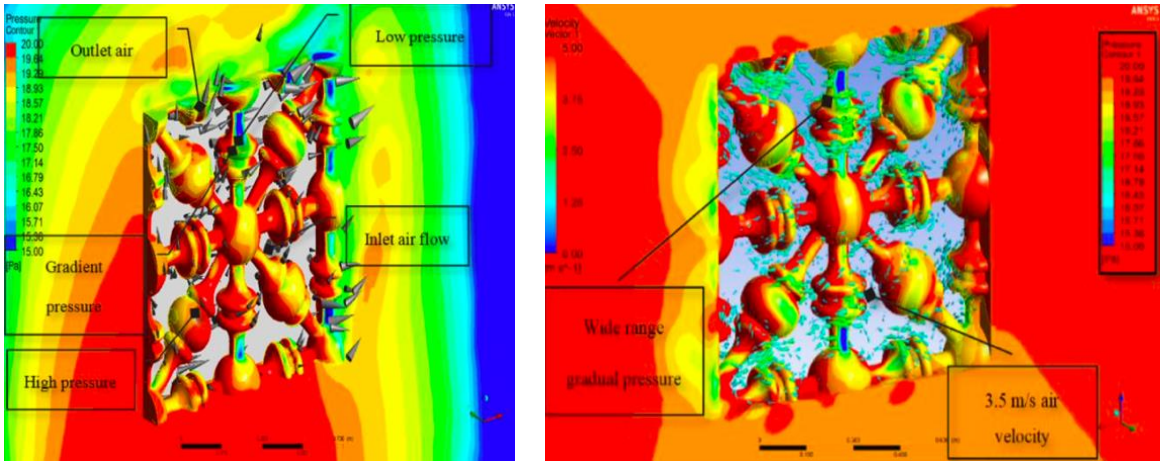


Figure 14. Pressure distribution and Coloured arrows representing air velocity through the Al-Salibi lattice window shape (Elwan M. M., 2019).

Given these points, the Mashrabiya can function as a versatile ventilating strategy that integrates multiple environmental solutions within a single component. The design of its screens allows for the regulation of airflow entering a space, which proves beneficial in contemporary responsive façades, offering a design approach that combines natural ventilation with the reduction of excessive solar gains. Natural ventilation has become a fundamental requirement for indoor well-being, and the demand for fresh air in gathering spaces is now considered crucial—even life-saving (GOV.UK, 2021). However, in much of contemporary architecture, the Mashrabiya is commonly used merely as a decorative perforated screen within a double façade system, thus depriving it of its original function as a natural ventilation device (بيومي, 2018).

3.2 Shading and Daylight in Mashrabiya

The primary function of shading element design varies across regions, depending on climatic conditions, building typology, context, and the specific function of the space requiring shading. Shading devices are typically designed to prevent glare and reduce visual discomfort. However, certain challenging environmental conditions require shading strategies that respond to solar angles and mitigate excessive solar gains. The design of such solutions is informed by calculations of the desired solar radiation incident on the openings, and by incorporating hourly variations in weather data and solar geometry. Common shading types range from horizontal to vertical fins, chosen based on façade orientation and solar path. Perforated screens, in particular, represent a hybrid strategy that integrates both horizontal and vertical shading principles. The effectiveness of a shading strategy can be quantified by evaluating daylight availability, daylight performance, and climate-based daylight metrics (CMBD).

The contemporary preference for modern architectural styles has limited the use of the Mashrabiya to perforated screens integrated within a double façade, often backed by glazed layers. As a result, the location and the form of these screens significantly influence indoor environment quality, particularly in terms of daylighting and thermal comfort. In hot climates, it becomes essential for the opaque elements to intercept direct solar radiation. This is achieved by positioning the shading layer on the outermost part of the façade to block excessive heat gain (الجواد, 1987). A wide range of passive shading solutions has been introduced in contemporary architecture by fusing traditional concepts with modern technological advancements. The Mashrabiya serves as a foundational idea for many of these strategies. With the aid of modern fabrication techniques, architects and designers have been able to overcome historical limitations in craft and scale — allowing for greater versatility and adaptability in the application of the Mashrabiya-inspired screens.

Daylight can be classified into three categories:

- Direct daylight
- Diffuse skylight
- Reflected glare

According to various studies, it is advisable to block direct sunlight from entering southern façade openings in order to reduce excessive solar gains and prevent glare that can lead to visual discomfort. Perforated screen serve as an effective strategy by integrating both horizontal and vertical shading principles, enabling them to obstruct undesirable sun angles while diffusing incoming solar radiation.

Accordingly, the amount of diffused light that enters the indoor space through the screes depends primarily on the perforation percentage, the shape of the screens, the reflectivity, and the materiality of the components composing the screen. The thin atmosphere characteristic of hot, arid climates presents challenging conditions that necessitate extensive shading—often resulting in dim indoor environments, further influenced by reflected light from the surroundings. This pronounced variation in light intensity typically causes visual discomfort. As a response, the Mashrabiya was traditionally employed to create a buffer zone functioning as a transitional space. The techniques embedded in the Mashrabiya are regarded as an adaptive shading and daylight system that regulates the quantity of light entering the space, effectively shading and obstructing lower-angle sunlight that often proves problematic in hot climate design scenarios (فكري, 1981).

“ .. And by incorporating the established D/L ratio, an accurate shading diagram can be produced, precisely mapping the periods throughout the year when complete shading is achieved by the Mashrabiya.” (Samuels, 2011).

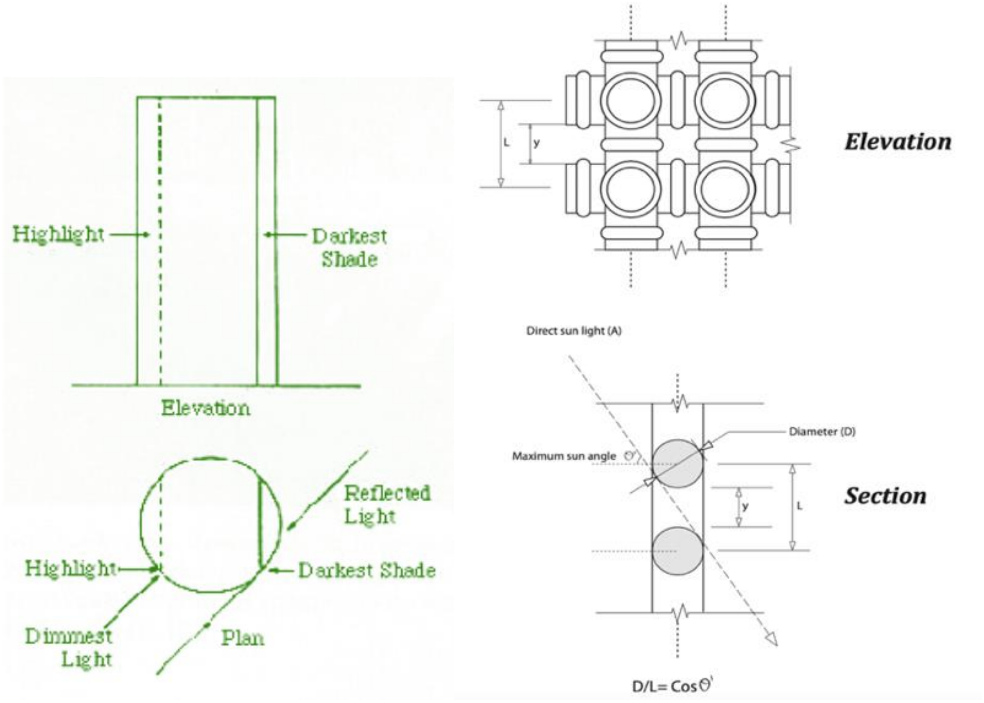


Figure 15. Plan and elevation of a Mashrabiya baluster illustrating the gradation of light across its rounded surface.

The distinction of Mashrabiya as a passive solution depends not only on its efficiency in shading and ventilation but also on its ability to provide sufficient visual comfort and appropriate daylight while mitigating excessive solar gains—all within the boundaries of environmental constraints, which it passively addresses. Accordingly, Mashrabiya designs varied across regions, adapting to specific functional requirements and responding to physical variables such as materiality, geometry, ornamentation, and pattern. Craftsmen traditionally selected Mashrabiya materials based on regional

availability and climatic performance. For instance, in the Middle East, commonly used materials included ebony, oak, and mahogany, whereas in India, clay and stone were used to create *Jali* screens. These material and formal choices serve as performance determinants. As such, the shading performance of traditional perforated screens can be assessed in terms of their perforation percentage and thickness. Most traditional patterns maintained consistent thickness due to structural considerations and the inherent properties of the materials used (Al-Shareef, 1996).

The design of the Mashrabiya is not limited to providing an optimal perforation percentage. For example, Ottoman Mashrabiyas often featured screens integrated into complex extruded geometries applied to window openings, frequently becoming an integral part of the building's structural composition. The geometry varied based on the function of the space and the user's interpretation of it—some users placed ceramic pots for evaporative cooling, while others used the Mashrabiya as a sitting area (الجواد, 1987). Typically, the Mashrabiya consists of three main sections: the upper, middle, and lower parts. Consequently, most Ottoman Mashrabiyas incorporated hybrid patterns or combined multiple screens, each with distinct ornamentation and perforation percentages. The criteria for selecting specific patterns generally depend on three principal factors, which are:

I. Achieving effective ventilation, as placing highly perforated screens at the top and bottom portions of the component, can promote stack-driven airflow.
II. Employing less perforated screens in the middle section allows for controlled airflow while prioritising shading and avoiding direct sunlight penetration.

The structure and outer form of the component reveal how it is designed to enhance shading performance. A structural crown, or "Tajj,"[1] often placed atop the component (and considered an integral part), functions as a canopy that casts shade over the highly perforated upper section. Simultaneously, it permits airflow, thereby optimising both shading and ventilation (Al-Shareef, 1996). The components of the Mashrabiya are divided into individual sashes; typically, the main (middle) portion is divided into three to five sashes. In some cases, each sash may incorporate a different pattern. This subdivision allows for design flexibility—enabling a mixture of ornamental patterns as well as the inclusion of opaque elements such as carved wooden decorations (Batterjee, 2010).

[1] Taj: the top part that is applied on top of the Mashrabiya to provide extra shading (Al-Shareef, 1996).

Figure 16. Gayer-Anderson Museum, Cairo, Egypt

The selection of the perforation percentage is based on identifying the extreme summer days of the year, when direct sunlight leads to excessive solar gains. Accordingly, diagonal elements were traditionally incorporated into the screen to block undesirable solar angles. The Mashrabiya screens were conceived as modular components—akin to pieces of Lego—enabling craftsmen to adapt and reconfigure them in response to varying environmental conditions (نظيف, 1989).

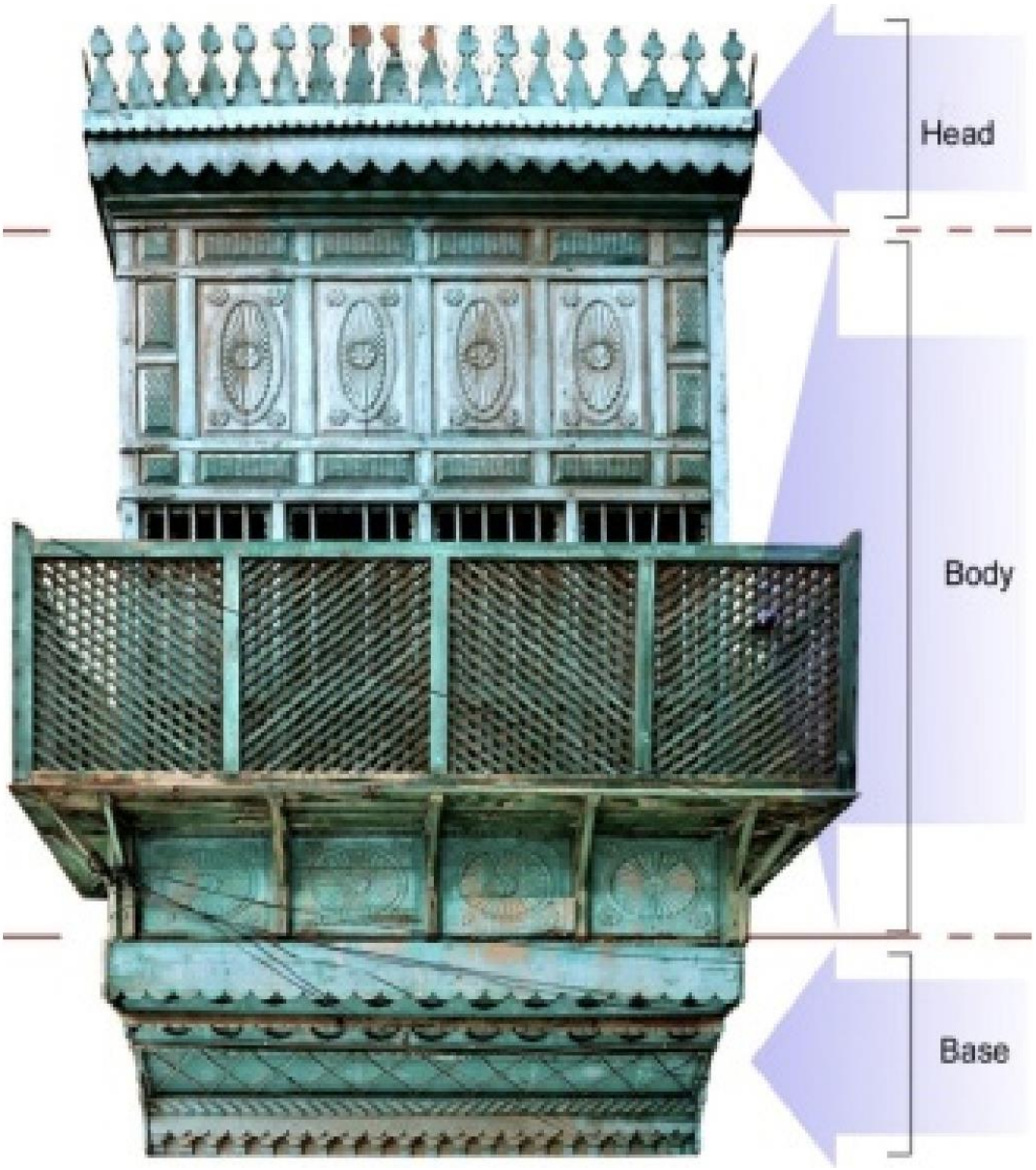


Figure 17. Main Part of the Mashrabiya (Alitani, 2011)

According to the spatial daylight anatomy criteria established by LEED v4 (2013), a study was conducted to evaluate the shading performance of perforated screens applied to office buildings compared to other shading systems. The analytical investigation, supported by Useful Daylight Illuminance (UDI) analysis, demonstrated that perforated screens not only block undesired glare but also deliver consistent daylighting throughout the full depth of the room. This results in a gradual reduction of light intensity as one moves deeper into the space—enhancing visual comfort and supporting energy efficiency (Alsharif, 2017).

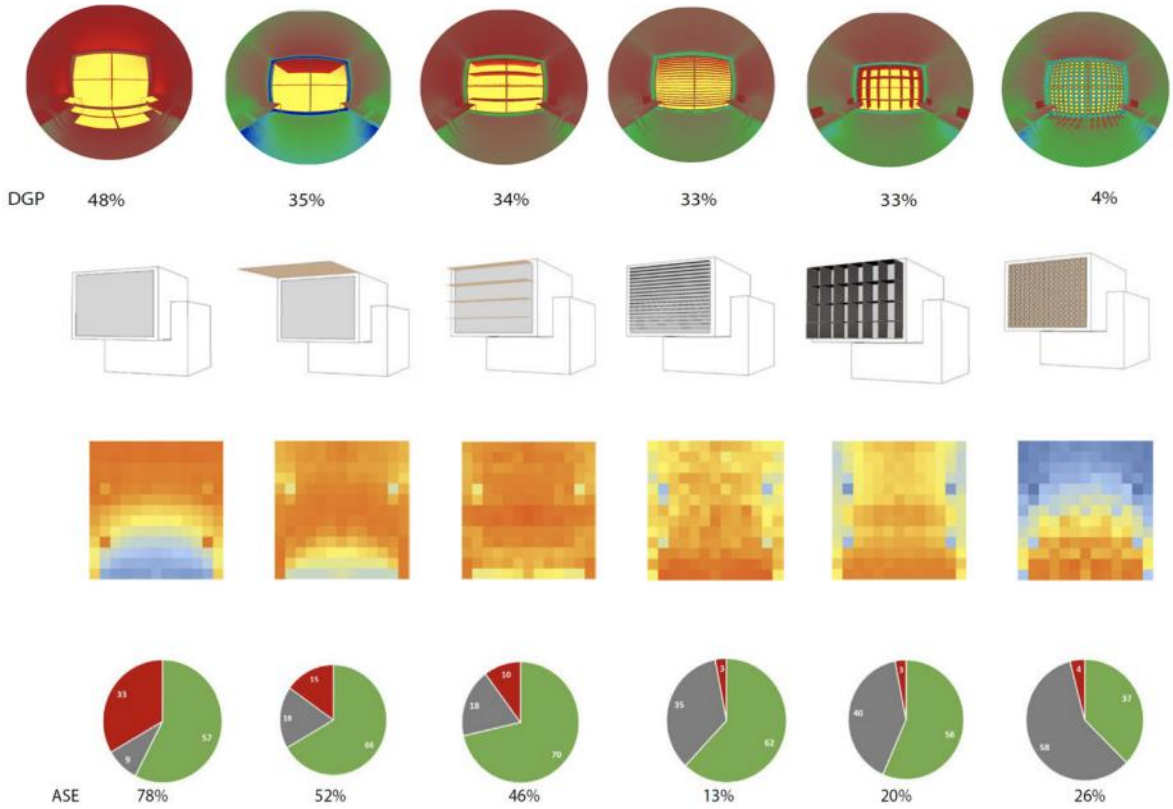


Figure 18 spatial daylight anatomy for percentages of the floor area that has a daylight autonomy value of 50% or more.

3.3 Conclusion:

The Mashrabiya has long stood as a layered environmental and cultural device, far beyond a decorative façade. It creates a sensual threshold between interior and exterior, filtering light, air, and views to form a gradual transition that softens the shift from outside to inside. Its performance lies in the precise orchestration of geometry, perforation, and materiality—choices made not for ornament alone, but to respond directly to climate, spatial use, and user comfort.

As a natural ventilation component, the Mashrabiya responds to prevailing winds, buoyancy forces, and the pressure differences it generates through its patterned apertures. Highly perforated upper and lower sections invite airflow, enabling stack-driven movement through the depth of the space, while less perforated middle sections moderate wind and glare at eye level. The shape of each baluster and the smoothness of its curves influence air velocity at the micro scale, demonstrating how small design decisions accumulate to shape overall performance. In single-sided ventilated spaces, these strategies achieve what mechanical systems often replicate at great energy cost.

Equally, its shading role is inseparable from its environmental value. Positioned on the outer façade, the Mashrabiya intercepts direct solar radiation before it strikes the building envelope, breaking

sunlight into diffused patterns that reduce glare and visual discomfort. The perforation percentage, thickness, and orientation of its elements all influence the balance between daylight penetration and solar control. By structuring its three parts—upper, middle, lower—each to serve a specific environmental purpose, it maintains comfort while preserving a visual and physical connection to the outside.

Traditionally, the Mashrabiya adapted to context. Materials varied from hardwoods in the Middle East to clay and stone in India, each chosen for both performance and availability. Patterns could be reconfigured like modular pieces to respond to seasonal light and ventilation needs. Yet in much of contemporary architecture, the Mashrabiya is reduced to a fixed, decorative layer within a double façade, stripped of its original adaptability and climatic intelligence. Modern fabrication now offers the opportunity to recover and enhance these functions, using precise control over geometry and materiality to achieve performance that is both environmentally responsive and culturally grounded.

The Mashrabiya's enduring relevance lies in this integration. It is at once a climatic mediator, a spatial filter, and a crafted artefact—uniting shading, ventilation, and daylighting into a single system. In revisiting its principles, contemporary design can move beyond surface imitation, reactivating the Mashrabiya as a living, adaptive strategy for buildings that must respond to the pressing environmental challenges of our time.

4. Evaporative Cooling

4.0. Keeping Cool with Passive Design

Designing a strategy for cooling the building requires consideration of different surrounding factors. Therefore, cooling the building with passive techniques does not only depend on supplying cool air; it also depends on avoiding heat gains and ensuring the proper airflow through a sufficient depth of space (Allocca, 2002). The following table illustrates the acceptable operative temperature ranges for different humidity levels and clothing insulation, showing how thermal comfort is influenced by both environmental and personal factors. It highlights that maintaining comfort is not defined by a fixed temperature but rather by a balance between air temperature, humidity, and seasonal conditions (Allocca, 2002).

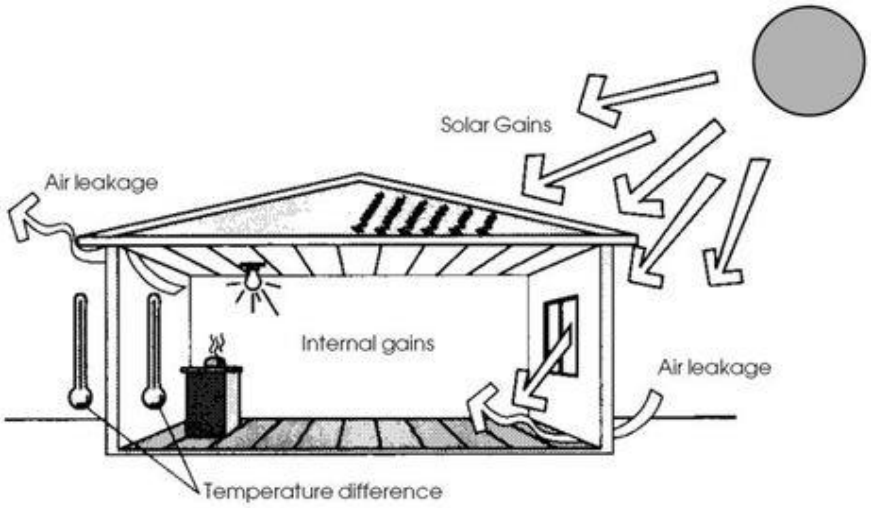


Figure 17 methods of heat gain in the building physics- source: (Al-Shareef, 1996)

Conditions	**Acceptable operative temperatures**	
	°C	° F
Summer (clothing insulation = 0.5 clo):		
Relative humidity 30%	24.5 – 28	76 – 82
Relative humidity 60%	23 – 25.5	74 – 78
Winter (clothing insulation = 1.0 clo):		
Relative humidity 30%	20.5 – 25.5	69 – 78
Relative humidity 60%	20 – 24	68 – 75

Table1. Examples of Acceptable Operative Temperature Ranges Based on Comfort Zone Diagrams in ASHRAE 55-2004

The prospects of passively achieved comfort in hot-arid climates depend on achieving a balance between different strategies. The climate of a typical hot, dry region requires substantial shading to ensure comfort and reduce excessive solar gains. On the other hand, the low moisture content of the air necessitates humidification to raise humidity levels to a comfortable range (Allard, 1998).

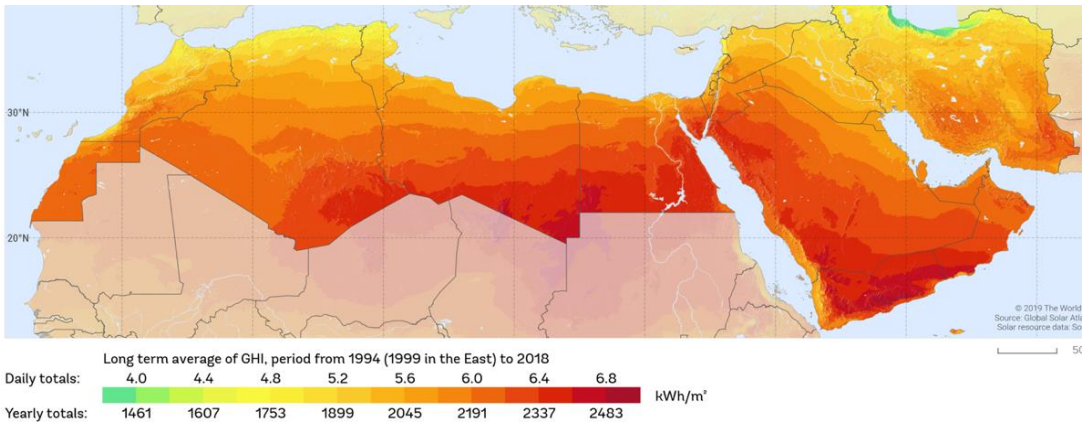


Figure 18 Average sunlight hours, Luxor, Egypt

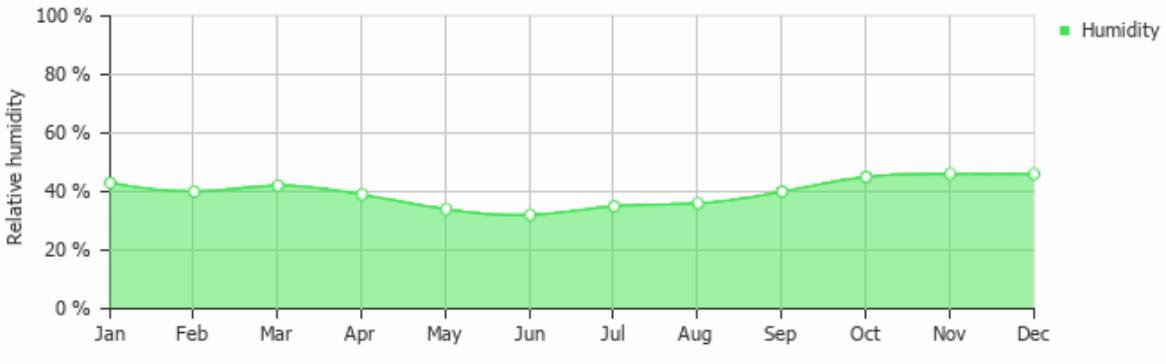


Figure 19 Average yearly humidity levels in Luxor, Egypt

Table 2. Number of Counties Classified by Level of Intervention within each Koppen Climate Zone

	Level of intervention							Level of intervention					
Köppen classification	VH	H	M	L	VL	-	Köppen classification	VH	H	M	L	VL	-
Af	0	0	0	0	0	1	Cwa	0	0	1	0	0	0
Am	0	0	0	0	0	2	Cwb	0	0	1	0	0	0
Aw	0	0	0	0	0	4	Dsa	0	0	2	2	0	0
BWh	9	2	0	0	0	0	Dsb	0	1	7	11	1	0
BWk	4	5	6	1	0	0	Dsc	0	1	3	1	1	3
BSh	2	2	5	6	1	2	Dwa	0	0	0	0	5	0
BSk	11	36	143	91	3	0	Dwb	0	0	0	3	1	0
Csa	3	6	10	3	1	0	Dwc	0	0	0	0	1	0
Csb	3	5	11	21	34	12	Dfa	0	0	15	12	410	0
Cfa	2	1	171	27	1414	101	Dfb	2	4	20	33	311	9
Cfb	0	0	1	5	62	7	Dfc	0	1	7	14	7	4
Cfc	0	0	0	0	0	2	ET	0	0	0	1	1	4

Energy consumption in residential buildings is primarily directed toward achieving indoor comfort standards. Approximately 70% of the energy used in these buildings is consumed by heating and cooling loads. Accordingly, the world has witnessed a rise in air-conditioning systems in hot climatic regions, leading to increased cooling demands in the residential sector. In Middle Eastern countries, where most of the land area is classified as a hot, dry climate, the affordability of energy combined with population growth has resulted in what is known as the "cheap energy phenomenon." Consequently, architectural designs have become reliant on non-passive cooling techniques to meet comfort requirements—ultimately contributing to increased carbon emissions and a greater environmental footprint (Dodoo, 2011).

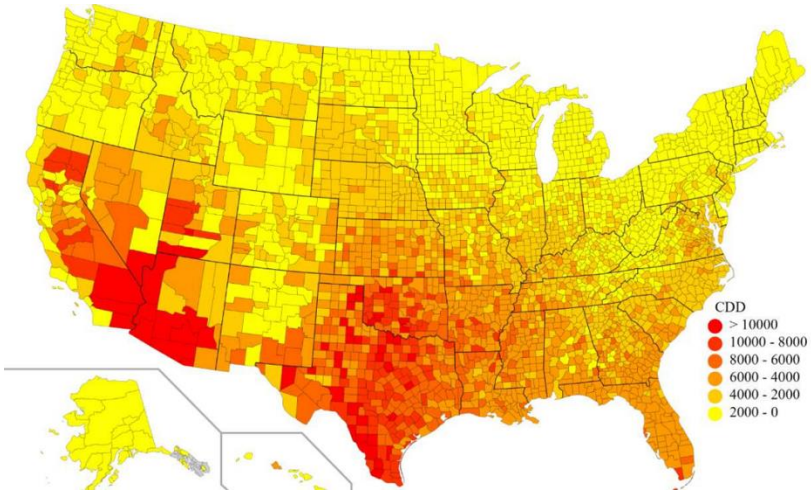


Figure 20. Degree of intervention and cooling degree days during summer in the USA

The difference between the dry-bulb temperature and the desired indoor comfort temperature primarily determines the need for cooling in buildings. The applicability of evaporative cooling, however, depends on the dry-bulb to wet-bulb temperature depression, which defines whether the strategy can effectively lower indoor temperatures. Hot-dry climates, therefore, provide the most suitable conditions for the application of passive evaporative cooling systems *(Moustafa, 2013)*.

In the Middle East, the rising cooling demands of residential buildings have significantly shaped energy consumption statistics over the past 15 years. The use of electric air conditioning peaked in 2015 and has continued to grow. In Cairo, for example, a typical two-storey residential building may consume up to 4000 kWh during the peak summer month of August, when cooling demand is at its highest. This

highlights the urgent need for effective passive cooling and humidification strategies in such climatic regions (Moustafa, 2013).

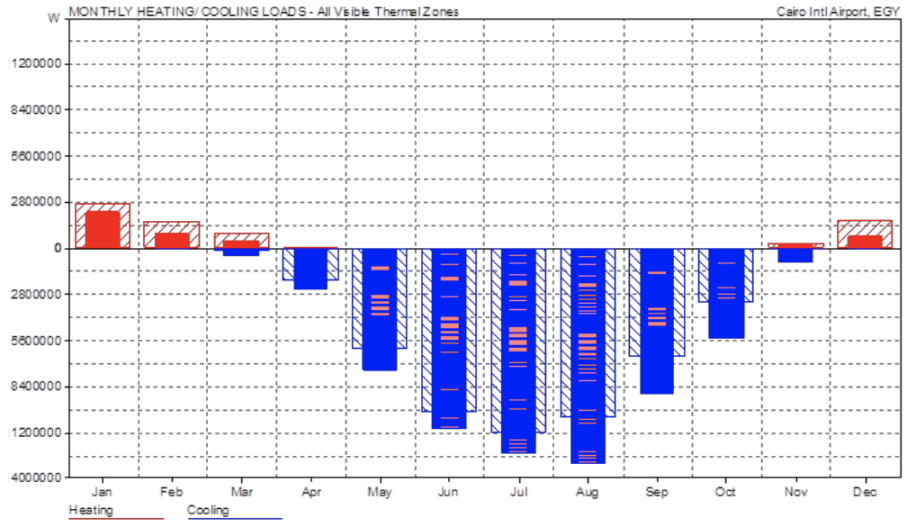


Figure 21. Monthly prototype heating/cooling loads in Cairo, Egypt. Hatched bars represent the loads simulated with Meteonorm, and solid bars represent the loads simulated with Energy Plus – Source: (Moustafa, 2013)

Evaporative cooling is the process of transferring heat and mass through the vaporisation of water. It depends primarily on the kinetic energy released through the breaking of hydrogen bonds in water molecules, driven by airflow and temperature differences. During this process, the air temperature and the water's surroundings temperature decrease gradually due to the energy transformation in the vaporisation process, with sensible heat being transformed into latent heat. EC can improve indoor conditions by reducing the average temperature and increasing the humidity of the air to a healthy, acceptable limit. However, when the relative humidity exceeds 80%, it negatively affects the building, causing warping, corrosion, and mildew in moisture-sensitive materials (Lechner, 2009).

Evaporative cooling can occur simply through the vaporisation of water. Still, for cooling indoor spaces, an airflow through a wet medium is desirable as it facilitates both cooling and humidification of the air. Humidity further helps reduce bacteria that accumulate in dry environments, while the wetted surface can filter out dust and particulate matter larger than PM2.5, contributing to cleaner indoor air (Lechner, 2009). However, the efficiency of evaporative cooling depends largely on effective air circulation to prevent an undesired rise in humidity levels and to ensure a consistent cooling effect throughout the entire space.

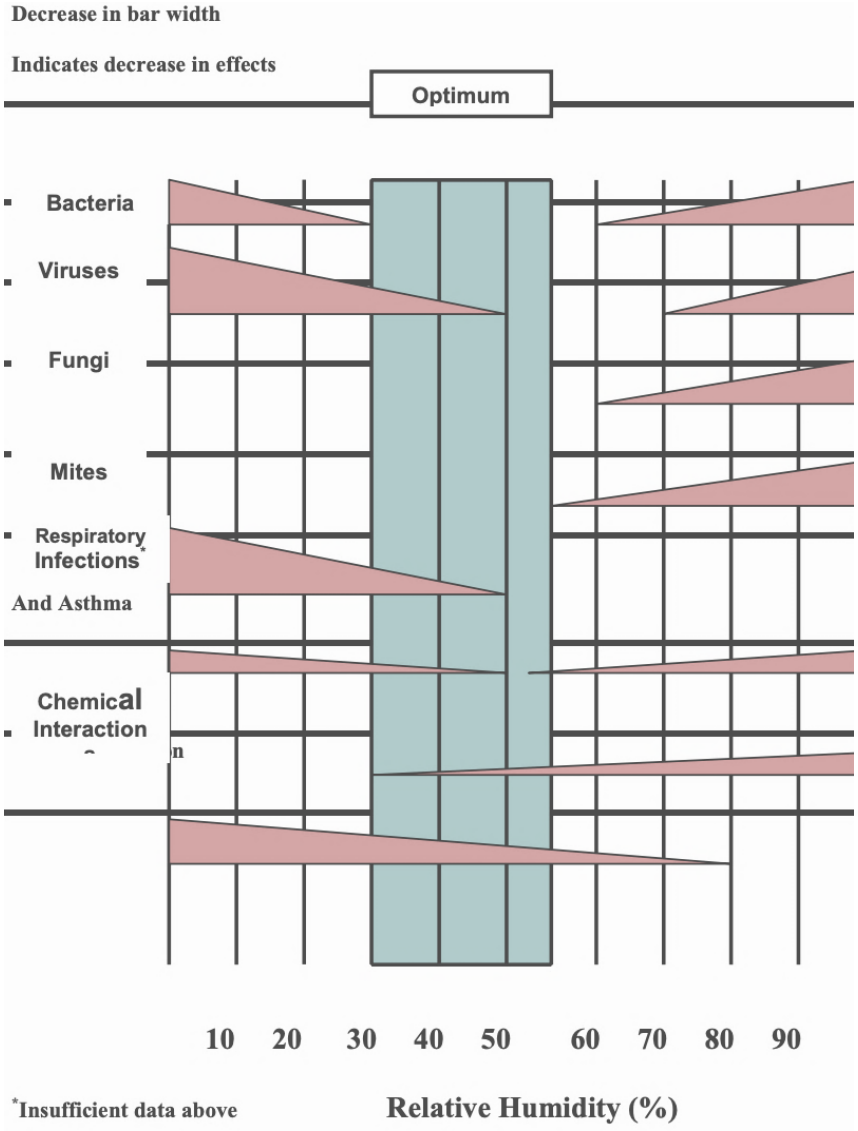


Figure 22. Effect of relative humidity on air quality

(Residential Energy Efficiency Database, 1990; Mark Gillot, 2000)

According to climatic regions, buildings should be designed using specific strategies to achieve indoor comfort, as illustrated in the psychrometric chart, which maps air properties and identifies appropriate design approaches. The theory of evaporative cooling relies on parameters such as dry-bulb temperature, wet-bulb temperature, relative humidity, and enthalpy, which are more accurately visualised through the psychrometric chart first developed by Dr. Willis in 1904. The comfort zone appears within this chart as a reference for

human thermal comfort, intersecting with the physical air properties that define the potential of different passive strategies, including evaporative cooling (Donald Watson, 1983).

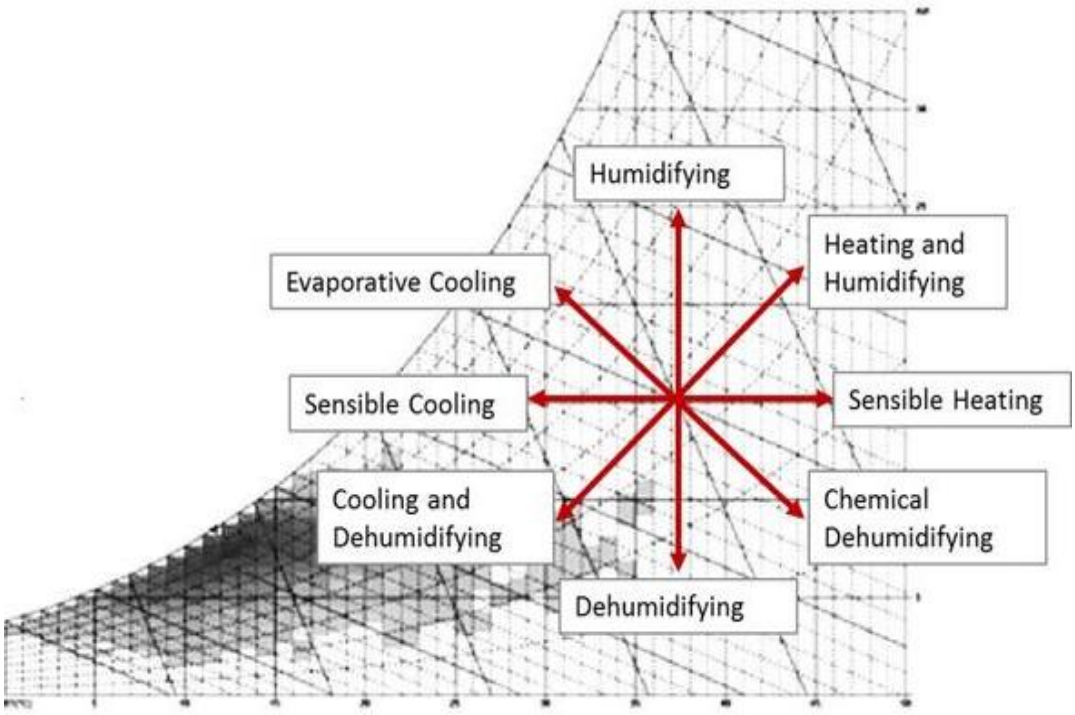


Figure 17. Design Strategies and Energy Requirements for Different Climatic Conditions

4.1 Direct Evaporative Cooling

Evaporative cooling became a common strategy applied alongside latticework and traditional perforated screens. It was also integrated into wind catchers and air towers to cool, humidify, and purify incoming air using wet mats. These mats functioned to filter the air and increase its moisture content. Evaporative cooling can be categorised into three main types:

I. Direct Evaporative Cooling Systems.
II. Indirect Evaporative Cooling Systems.
III. Combined systems that incorporate both direct and indirect EC.

In direct evaporative cooling, warm air becomes cooler as it passes through a wet medium (moisture content). This process is commonly used in hot, dry climates for greater efficiency, and it is frequently applied directly to the ventilation opening to provide the necessary airflow. Additionally, the quality of the water supplied to the wet medium has a significant impact on performance efficiency and long-term maintenance (Smith, 2011). The efficiency of direct evaporative cooling can range from 70% to 95%, which has led to its adoption in regions such as the USA, where it contributed to measurable energy savings in the building

sector—including reductions equivalent to 12 million barrels of fuel and 5 billion pounds of carbon emissions (Palmer, 2002). In the Middle East, however, cooling demand continues to rise with the growing reliance on air-conditioning units to cope with extreme exterior conditions. In contrast, the potential of evaporative cooling remains underexplored and largely unimplemented at scale.

Understanding the fundamental air properties is essential to explain the performance of evaporative cooling. Parameters such as dry-bulb temperature, wet-bulb temperature, and relative humidity define the behaviour of air in relation to moisture and heat and therefore determine the effectiveness of cooling strategies. The following section outlines these parameters and their relevance to evaluating evaporative cooling systems.

I. Dry-bulb temperature (DBT)

The temperature of the air, shielded from radiation, indicates the degree of sensible heat content without taking air humidity into consideration (Clements, 2002).

II. Radiant temperature (RT)

Thermal radiation emitted by a warm object significantly influences the heat exchange experienced by occupants, often more than the air temperature itself. Radiant heat sources can be primary, such as heaters, or secondary, such as surfaces warmed by solar exposure (STULL, 2011). When a surface absorbs heat from sunlight or other sources, it can transfer energy to the surrounding air through convection and, more importantly, directly affect the thermal comfort of nearby occupants by radiation (Donald Watson, 1983).

III. Wet-bulb temperature (WBT)

It indicates the latent heat content of the air. It corresponds to the lowest temperature that air can reach through evaporative cooling at constant pressure and is a key factor in processes such as evaporative cooling (STULL, 2011).

IV. Latent Heat

Latent heat is a critical factor in achieving thermal comfort and plays an essential role in the efficiency of passive evaporative cooling, as it is the amount of energy absorbed or released by a substance during a phase change, such as from liquid to vapour or solid to liquid, without causing a change in its temperature. In the context of air and evaporative processes, latent heat

represents the energy involved when water evaporates into the air, affecting humidity and thermal comfort (STULL, 2011).

V. Wet Bulb Depression (WBD)

Wet-bulb depression is the difference between the air's dry-bulb temperature and its wet-bulb temperature. It indicates the air's capacity for evaporative cooling, where a larger difference means the air can absorb more moisture and contribute to lowering temperatures (STULL, 2011).

VI. Cooling Efficiency (CE):

Cooling efficiency represents how effectively an evaporative cooling element lowers the air temperature. It is calculated as the ratio between the actual decrease in dry-bulb temperature across the cooling medium and the maximum possible temperature drop, which is determined by the wet-bulb depression. CE is often expressed as a percentage and is closely related to the saturation efficiency of the cooling medium (SE) (GEORGE, 2021).

VII. Relative Humidity (RH)

Relative humidity is the ratio of the actual water vapour pressure to the saturated vapour pressure at a fixed air temperature. It can be defined using the following equation:

Equation 2: is the formula used to determine the relative humidity – source: (STULL, 2011)

$$\phi = \frac{Ps}{Pss}$$

VIII. Moisture content

Defines the mass of water vapour contained in 1 kg of dry air within the space/ the vapour-specific mixture. It can be expressed as:

Equation 3 is the formula used to express the Moisture content– source: (STULL, 2011)

$$g = \frac{ms}{ma} = \frac{Kg.of\ water\ vapour}{Kg.of\ dry\ air}$$

IX. Specific Enthalpy

Specific enthalpy can be defined as the summation of the partial enthalpies present in the state of function, based on the heat loss or gain within the mixture, considering both water vapour and dry air.

Equation 4 defines the enthalpy in terms of internal energy, pressure and specific volume– source: (GEORGE, 2021)

$$H = E + Pv$$

Where:

"E" represents the internal energy of the system, "P" indicates the pressure and "v" is the specific volume. Specific volume refers to the volume occupied by 1 kg of dry air in 1m^3 of associated water vapour.

X. The Specific Heat

The required amount of energy to raise the temperature of one gram of a substance by one degree Celsius. It can be defined by the following formula (Bagasi, 2020):

$$\mathrm{CPm} = \mathrm{CP} + \omega\, \varrho \mathrm{CV}$$

Comfortable conditions in a typical Middle Eastern climate represent approximately 43% of the total annual hours, encouraging architectural approaches to adopt passive design solutions to enhance thermal comfort. About 13% of the uncomfortable conditions stem from hot, dry environments that require both cooling and humidification to counteract the effects of an arid desert climate. Accordingly, cooling efficiency (defined as the ratio of the actual temperature reduction achieved by an evaporative cooling system to the maximum possible reduction determined by the wet-bulb depression) in hot-arid climates ranges between 77% and 87%. In contrast, the average efficiency in hot-humid climates is approximately 72%. This indicates that passive cooling and humidification strategies are particularly effective in hot-dry regions, which constitute around 26% of the world's land area. Therefore, implementing strategies that reduce energy consumption in these regions holds the potential to significantly influence the future environmental footprint of the planet (Abdullah, 2022).

Table 3. Averages Cooling Efficiency and Temperature Drop in Different Climatic Regions – Source: (Abdullah, 2022)

Climatic conditions	Air entering the cooler			Air leaving the cooler			η (%)	D (°C)	q (W)
	T_{in}	RH_{in}	W_{in}	T_{out}	RH_{out}	W_{out}			
Arid	30.5	16	0.00432	18	70	0.00900	77	12.5	14595
	35.4	11	0.00391	20.1	63.1	0.00925	80.5	15.3	17581
	40	9	0.00411	22	59.4	0.00980	82.2	18	20379
	40	12	0.00549	22.6	66.4	0.01139	84	17.4	19700
	45	12	0.00715	24.5	69	0.01331	87	20.5	22845
	45	7.6	0.00451	24	59	0.01100	84.3	21	23402
	50	8.5	0.00651	26.5	65.4	0.01422	82	23.5	25783
Average	**40.8**	**10.9**	**0.00514**	**22.5**	**64.6**	**0.01114**	**82.4**	**18.2**	**20612**
Humid	30	42	0.01114	23.2	70	0.01247	71.6	6.8	8033
	31	67	0.01906	27.6	86	0.02014	67	3.4	3887
	30	71	0.01907	27.2	88	0.02013	65	2.8	3308
	35	58	0.02070	30	85	0.02297	68	5	5063
	38.8	46.2	0.02028	31	83	0.02378	75	7.8	7125
	40	45	0.02109	31.4	83	0.02435	74.5	8.6	7620
	40	56	0.02646	32.2	88	0.02713	84	7.8	6911
Average	**35**	**55**	**0.01969**	**28.9**	**83.3**	**0.02157**	**72.2**	**6**	**5992**
Moderate	28	19	0.00444	17	71	0.00856	80	11	13923
	29	43	0.01076	23	70	0.01232	63	6	7332
	30.5	16	0.00432	18	70	0.00900	77	12	14525
	30	25	0.00659	19.7	74	0.01061	78	10.3	12168
	30	42	0.01114	23.2	70	0.01247	71.6	6.8	8033
	35.4	11	0.00391	20.1	63.1	0.00925	80.5	15.3	15317
Average	**30.5**	**26**	**0.00686**	**20.2**	**69.7**	**0.01037**	**75**	**10.2**	**11883**

4.2 The Quantification of Cooling Performance in Direct Evaporative Cooling

The thermodynamics of evaporative cooling relies primarily on converting the sensible rise in temperature into latent heat, with the evaporation rate determining the amount of heat removed from the air. The vaporisation of 1Kg of water at 35 °C requires approximately 2257 kJ of internal energy. This energy is typically provided by the sensible heat of the air, which is transferred to the wet surface through convection, thereby lowering the air temperature. The breaking of hydrogen bonds in water molecules plays a key role in this process, contributing to the enthalpy change, energy release, and heat transformation into kinetic energy. This transformation simultaneously cools the air and increases its moisture content.

The cooling capacity of the evaporative cooler is typically defined as:

$$q = m_s + c_{pd}(T_{di} - T_{do})$$

Where,

- m_s : The rate of the airflow mass kg/s.
- c_{pd}: The specific heat capacity of the air.
- $(T_{di} - T_{do})$: Represents the temperature difference between the inlet and outlet air, indicating the degree of cooling achieved.

The efficiency of evaporative cooling performance can be expressed through the coefficient of performance (COP), defined as the ratio of heat absorption (i.e., the reduction in air temperature) to the total energy input. Additionally, water consumption in evaporative cooling must be considered to ensure a passive environmental solution with mindful use of resources. The amount of water consumption depends primarily on the following factors (GEORGE, 2021):

- The evaporation rate, which is influenced by airflow rate, air temperature, and relative humidity
- Solar exposure and its effect on water loss
- The properties of the material used in the evaporative cooling process, and its porosity.

Water consumption can be estimated using the following equation without considering the water loss (Abdullah, 2022):

$$m_w = m_a(g_{ai} - g_{ao})$$

Where,

- m_w : The water consumption rate in kg/s.
- m_a : The dry-air mass flow in kg/s can be calculated as: $m_a = \rho_{da} * Q_a$
  Where:
  ρ_{da} represents the air density in kg/m3 and Q_a represents the volumetric flow rate in m3/s
- $(g_{ai} - g_{ao})$: Represents the difference in humidity ratio of the air across the cooling medium

To evaluate the efficiency of evaporative cooling, the process is quantified using the differences between dry-bulb and wet-bulb temperatures, together with the corresponding humidity levels. These parameters define the actual cooling performance of the system. On

the other hand, to predict the potential efficiency of evaporative cooling, it is important to consider the factors that influence the rate of evaporation, such as air velocity, temperature, humidity, and the properties of the cooling medium, which in turn depend on the following key elements:

- The main radiant temperatures increase the Kinetic energy of water molecules.
- The porosity of the material determines how saturated the surface becomes and how much water it can absorb through its capillary structure.
- The airflow rate through the wet medium enables temperature exchange; increasing this rate enhances cooling efficiency. The surface area of the evaporative cooling element (wet medium): increasing the wet surface area exposes more water molecules to the airstream, facilitating and accelerating evaporation.
- The moisture content of the air passing through the wet medium influences performance, as evaporative cooling aims to both cool and humidify the airstream.
- The nature of the liquid affects the evaporation rate, depending on the strength of molecule bonds and the amount of kinetic energy required to break them.

The degree to which the outlet air temperature approaches the wet-bulb temperature of the inlet air—known as wet-bulb effectiveness—is a key technical indicator of evaporative cooling (EC) performance. This effectiveness typically ranges between 70% and 90%, depending on variation in the airflow rate through the wet medium. It can be defined by the following equation:

$$\varepsilon_{DEC} = \frac{T_{db,1} - T_{bd,2}}{T_{db,1} - T_{wb,2}}$$

Where:

- ε_{DEC} : wet-bulb effectiveness of the direct evaporative cooler
- $T_{db,1}$: inlet air dry-bulb temperature, o C
- $T_{db,2}$: outlet air dry-bulb temperature, o C
- $T_{wb,2}$: outlet air wet-bulb temperature, o C

The cooling capacity of the EC process can be defined using two different methods. The following method involves determining the intake air characteristics based on the indoor return air. Conversely,

when return air is recirculated and directed to the intake system, the cooling capacity can be quantified using the following equation (Abdullah, 2022).

$$Q_{cooling} = \frac{C_{p1f} p_f V^2 (T_{bdr} - T_{bd2})}{3.6}$$

Where:

- $Q_{cooling}$: Evaporative cooling capacity, W.
- C_{p1f}: specific heat of air at a certain pressure, kJ/ (kg k).
- p_f: air density of, kg/m3.
- V^2: Airflow rate, m3 /h.
- T_{bdr} : The reference dry-bulb temperature of the room, selected in accordance with ARI test standards (ARI, 2003).

The cooling efficiency of evaporative cooling depends on the system's ability to achieve thermal equilibrium and can be quantified by calculating the convective heat transfer coefficient. This coefficient is considered a process-based value that does not depend on a single material, but rather represents the rate of heat transfer from one medium to another (Musa, 2009):

$$\alpha = \frac{c_p \times d_m \times (T_2 - T_1)}{\mathrm{A} \times \rightarrow \Delta\, \mathrm{Tm}}$$

Where:

- α: Heat transfer coefficient
- $(T_2 - T_1)$: Temperature difference
- A: Heat transfer surface
- Δt: Examined time frame
- c_p: Heat capacity of the fluid
- d_m: Mass flow of the fluid

The evaporation rate of the fluid used in the EC significantly influences the overall cooling efficiency, directly impacting the estimated water consumption and the sensible cooling efficiency. It can be quantified using the following equation:

$$V_w = \frac{1000 V_3 \rho_{w,f}}{\rho_w} \times (w_3 - w_1)$$

Where:

V_w: water evaporation rate, litre/h

V_3: air flow rate, m3 /h

$\rho_{w,f}$: air density, kg/m3

ρ_w: Water film density, kg/m3

w_1: Inlet moisture content of secondary air

w_3: outlet moisture content of secondary air

The efficiency of the evaporative cooling depends primarily on the rate of evaporation and the amount of water vaporised to reduce the radiant temperature of the surrounding air. A higher evaporation rate indicates a greater consumption of kinetic energy required to break the molecular bonds of the liquid, thereby enhancing the cooling effect. The Evaporation rate governs the air's ability to absorb moisture and is influenced by several key factors:

I. Humidity of the Air

The surrounding relative humidity determines the vapour pressure in the environment. A decrease in relative humidity reduces the vapour pressure in the air, creating a greater potential for evaporation. As a result, the evaporation rate decreases when the entering air is more saturated (Brian Ford, 2020).

II. Solar Exposure

Designing an evaporative cooling strategy that is exposed to sunlight is very challenging. Direct solar radiation can increase the evaporation rate, but this does not necessarily mean the system becomes more efficient. Sunlight raises the surface temperature of the ceramic, which can reduce its ability to cool by radiation (Alsharif, 2017). Therefore, solar exposure introduces a balance: it may accelerate evaporation, but it also limits the ceramic's radiative heat loss, affecting the overall cooling performance (Musa, 2009).

III. The Airflow Rate at the Inlet.

The airflow rate at the inlet affects the evaporative cooling performance, but its impact depends on the ventilation strategy. Higher airflow does not necessarily lead to more cooling. It can improve evaporation only if the air passes through the full depth of the space and interacts with the cooled surface. In cases of single-sided or poorly directed ventilation, more air may simply bypass the cooled zones, limiting the system's efficiency (Allard, 1998).

IV. The Porosity of The Evaporative Cooling Element

The porosity of the wet medium allows water to be absorbed and maintains a wet surface, while permitting some air to pass through the material. Proper porosity ensures that the surface stays wet, which is essential for effective evaporative cooling. Beyond this, higher porosity does not automatically improve performance (Lechner, 2009). The porosity must also work in harmony with other factors, such as mesh density and the overall form, to achieve the desired efficiency while keeping water consumption under control (MCMATHIS, 2015).

V. **The Density of the Meshes in the Form of the Wet Media**

The density of the mesh and the final shape of the evaporative cooling element determine the resistance to airflow and influence turbulence and pressure variation.

VI. **The Surface Area of The Evaporative Cooling Element**

The more exposed the surface area occupied by the liquid, the faster the evaporation rate, ensuring a large contact interface between the air and the liquid.

VII. **The Thickness of The Wet Media**

The thickness of the wet medium affects water absorption by providing a greater volume of capillaries and increasing the amount of water the material can hold. Greater thickness also allows the development of self-supporting forms, as the material can maintain its structural integrity while remaining saturated.

Mass transfer occurs only when the wetted surface is exposed to a vapour partial pressure difference, which directly affects both the temperature and humidity of the flowing air. This phenomenon can be quantified by the following formula (Musa, 2009):

$$m\,wd\omega\ = hmA\,(\omega_{sat,w} - P\omega) \times dV$$

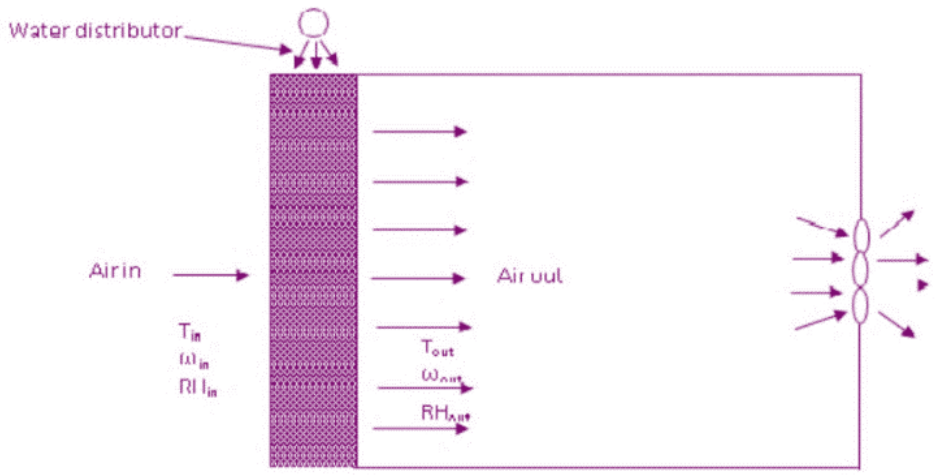


Figure 23. The Mechanism of Direct Evaporative Cooling (GEORGE, 2021)

4.3 Evaporative Cooling and Solar Exposure

The efficiency of the evaporative cooling is majorly affected by solar exposure. Direct sunlight on the wet media accelerates the evaporation rate, leading to excessive water consumption—an issue in regions with limited water supply. Consuming large amounts of water over a shorter period reduces the overall efficiency of the system. Therefore, designing an evaporative cooling element that also functions as a shading device presents a considerable

challenge. Solar exposure influences both the location and function of the evaporative cooling element, as well as whether the strategy targets a single environmental parameter or multiple surrounding conditions. Long-term sun exposure can degrade certain cooling media while also heating other parts of the evaporative cooler design.

4.4 The water supply and Consumption in Evaporative Cooling

Evaporative cooling depends on water as a fluid supply for the main process, unlike air-conditioning systems that depend mostly on the compression and expansion of fluorocarbon refrigerants. These refrigerants include chlorofluorocarbons and hydro-chlorofluorocarbons that can cause severe environmental damage and contribute to ozone layer depletion in the event of a leak (Musa, 2009). On the other hand, water is a valuable resource that must be consumed considerably. In the Middle East, water scarcity can become a pressing issue during certain parts of the year. Some Middle Eastern countries depend mainly on rainfall as the main water source. According to the following figure, most land areas experience a reduction in rainfall percentages during the rainy seasons, based on a vegetation analysis conducted in 2021 (analysis, 2021).

The water consumed in 2002 by the Zion National Park Visitor Center's passive cooling towers reached 420,938 litres (111,200 gals), despite the fact that the wetted media was shielded from direct solar exposure by being placed inside the wind-catching towers (Long, 2006). In arid climates, where water is scarce, such consumption levels raise concerns regarding the viability of passive evaporative cooling systems (Long, 2006). Proposing a successful cooling-humidifying passive strategy must address the issue of water consumption. An efficient evaporative cooling approach should maximise the building's cooling performance while utilising surrounding environmental factors judiciously and ensuring responsible water consumption.

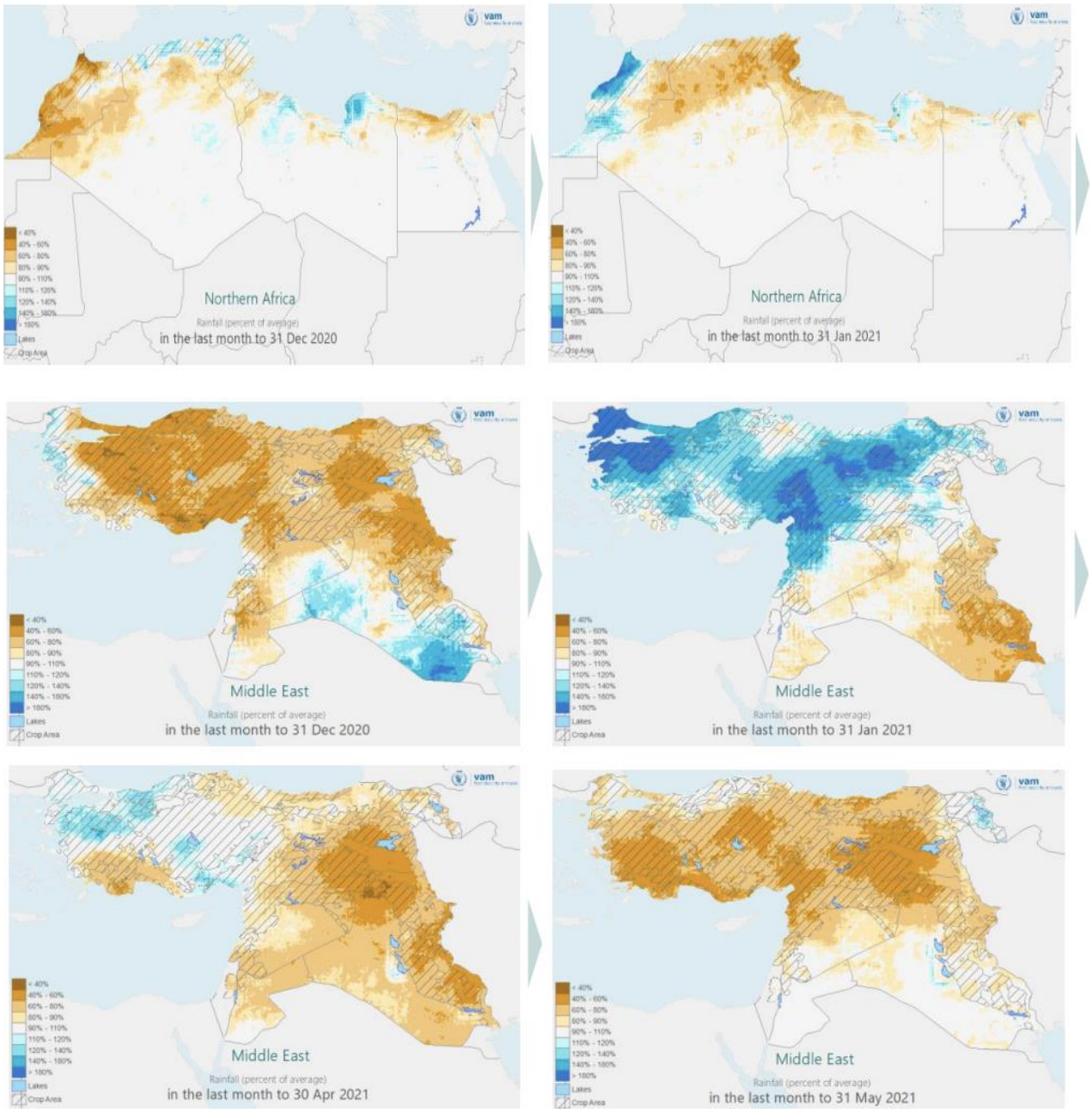


Figure 24. Rainfall Distribution in the Middle East Throughout the Year (analysis, 2021)

Water consumption in outdoor evaporative cooling designs presents a significant challenge. Therefore, some outdoor evaporative cooling elements are developed with additional shading components to reduce solar exposure and limit water usage—such as the *Cold Spot* project. *Cold Spot* is a design developed by Harvard graduate students, utilising Terracotta ceramic components for evaporative cooling that are shaded by horizontal shading fins positioned at the top of the structure (Schecter, Cold Spot: Evaporative Cooling through Ceramics, 2015).

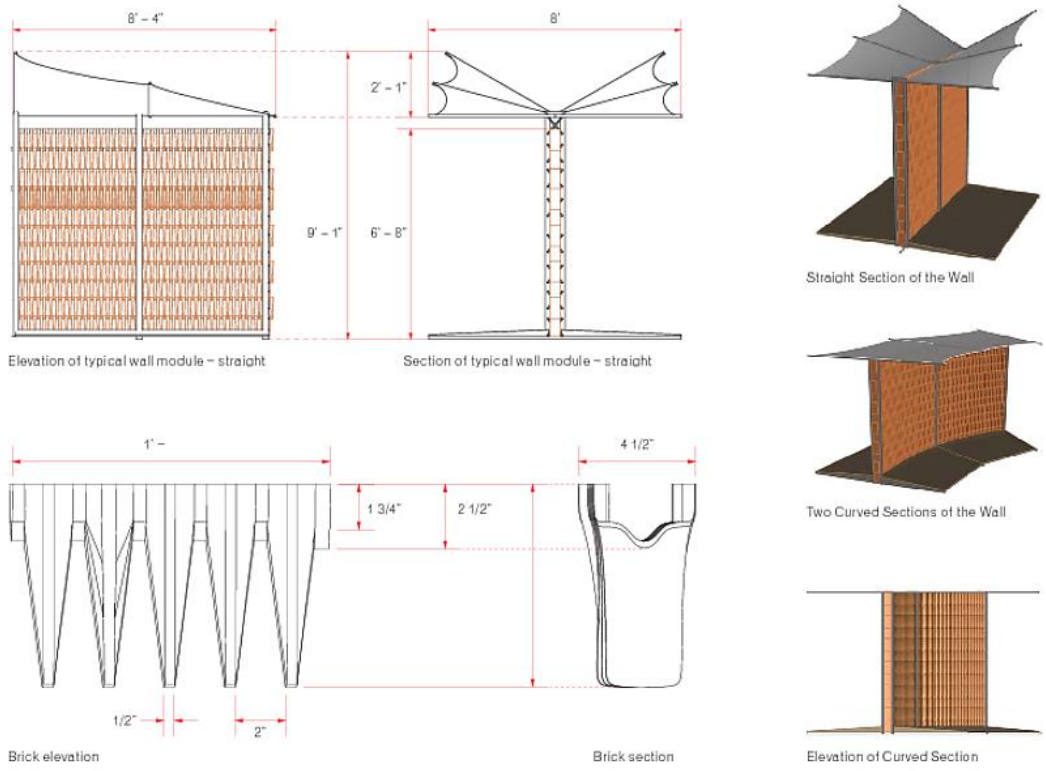


Figure 25. Schematic Diagram of the COLD SPOT Strategy (Schecter, 2015).

However, the *Cold Spot* strategy requires the ceramic components to be refilled with water once or twice a day, depending on the climatic conditions and the evaporative rate. (Schecter, Cold Spot: Evaporative Cooling through Ceramics, 2015) This can pose a challenge in hot, arid climates where water is a valuable resource.

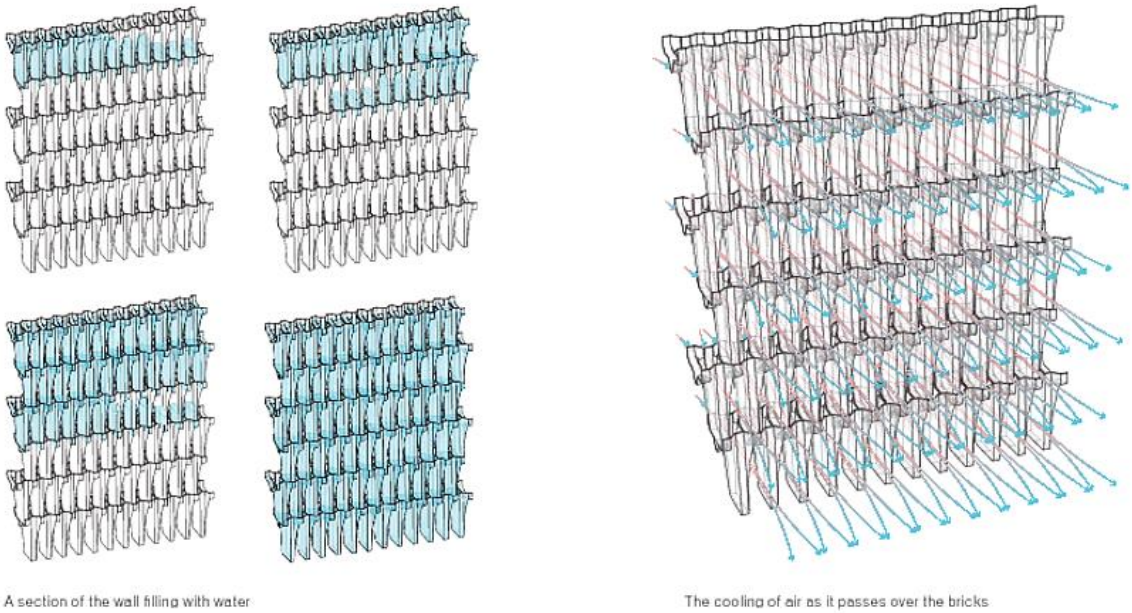


Figure 26. The Water Supply in the Cold Spot Strategy (Schecter, 2015)

Understanding the water distribution system is essential for designing an effective evaporative cooling setup—particularly in regions that demand responsible water consumption. Each evaporative cooling unit in direct evaporative cooling is divided into modules that distribute water from the source to the wet media through a continuous flow (al. K. e., 2006). Each module contains a water-spraying bar through a plurality of evenly spaced holes with area ranges between 15% and 25% of the cross-sectional area, as illustrated in Figure 10 (al. K. e., 2006).

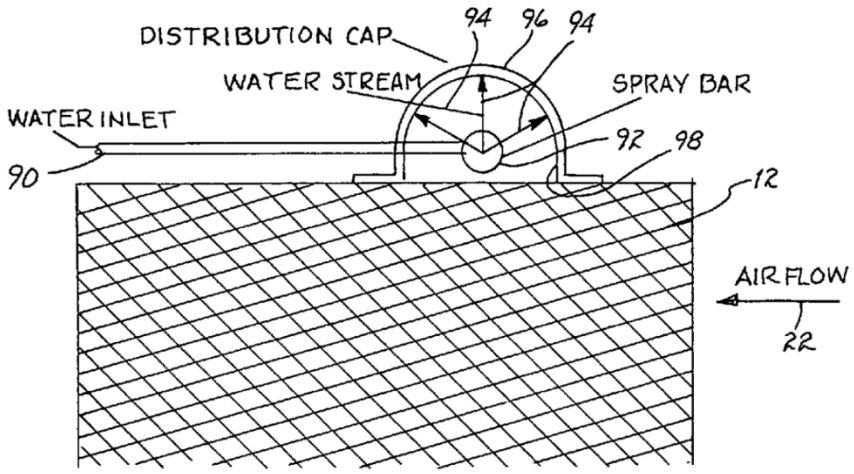


Figure 27 Water distribution system in Evaporative Cooler – Source: (al. K. e., 2006).

4.5 The Wet Media

The main component in the evaporative cooling system becomes saturated with water and allows the air to pass through it to facilitate the evaporation process. The typical thickness of the material ranges between 200 mm and 300 mm. It is usually made of a porous substance with a wide microscopic surface area, enabling it to retain water droplets. The criteria for selecting a suitable material for this medium depend on the following factors:

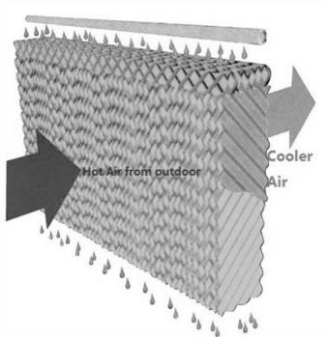
- Efficiency
- Cost
- Availability in the local environment
- Applicability
- Environmental factors
- The desired porosity

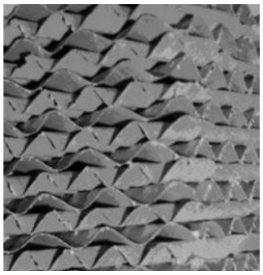
Multiple studies have been conducted to investigate suitable materials for evaporative cooling using both quantitative and qualitative analytical methods. The following table summarises the materials examined and their performance outcomes under specific conditions, based on findings from several research efforts (Kapilan, 2023):

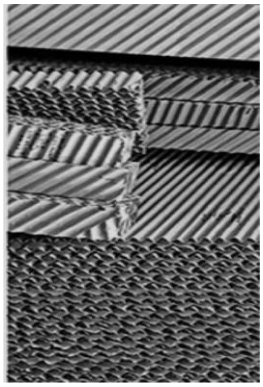
Table 4: The table illustrates the performance of the Synthetic wet pad - Source: (Kapilan, 2023)

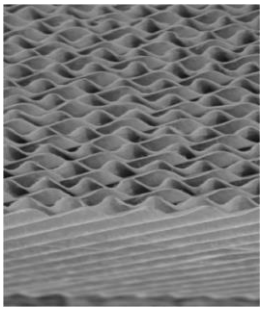
<table>
<tr><th>evaporative cooling material</th><th>Efficiency of Performance</th></tr>
<tr><td>

synthetic wet pad Ex.: GIASdek









</td><td>

Common Thermal characteristics of the fibres

Fibres	Thermal conductivity (W/m K)	Porosity (%)
Wood fibres	0.012–0.654	54.7–63.6 (very high water penetrability)
Latex with cement	0.28–0.52	1.10–2.32 (±0.02)
Methylcellulose with cement	0.32–0.42	2.07–2.12 (±0.02)
Silica fume	0.33–0.36	2.97–3.14 (±0.02)
Methylcellulose + fibre + cement	0.28–0.44	3.33–3.97 (±0.02)
Silica fume + methylcellulose	0.28–0.36	3.14–4.36 (±0.02)
Natural random fibrous materials (porosity = 10–58%)	0.02–0.13	Porosity = 10–58%
Cotton fibre (Sisal, Ramie and Jute)	0.176–0.351	40
Glass fibre	0.277	2.05–2.3 (±0.02)

Material

- Latex coated fibre, Fibreglass and Coated nonferrous metal
- Cellulose (Cardboard box/kraft paper) and Fibreglass
- Latex coated fibre, Fibreglass and Coated nonferrous metal

Performance

- Cellulose cooling pad is an alternative evaporative cooling pad material with a maximum effectiveness occurs at velocity 1.6 m/s for thickness 150 mm in pad 5090.
- Evaporated water is about 0.06 Lit/min for thickness 50 mm at 1.6 m/s air speed in pad 7090.
- Clogging resistance due to dust or sand. The water cleanses dirt and debris from the surface, and it is designed to distribute the water toward the air entering face of the pad, where it is most needed.
- Low pressure drop & Simple to maintain
- With a maximum saturation effectiveness of 90.37 percent, the cellulose pad with a thickness of 150mm is determined to be the most efficient of the cooling materials, while the aspen pad with a thickness of 36mm has a maximum saturation efficiency of 78.31 percent.
- In the case of cellulose and aspen pads, the efficiency rises as the pad thickness increases. However, when the thickness of the khus pad increases, the efficiency falls because the high density of the pad hinders air movement.
- As a result, the efficacy of the pads is Cellulose>Aspen>Khus pad in decreasing order of magnitude.

</td></tr>
</table>

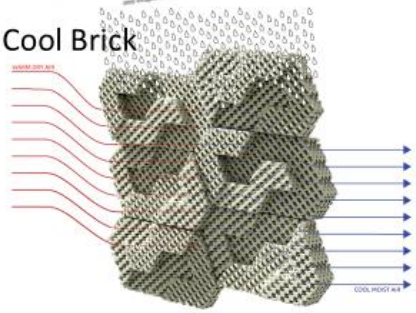
Table 5Table 4: The table illustrates the performance of the Synthetic wet pad - Source: (Kapilan, 2023), (al. P. e., 2021), (Yolasite, 2022)

evaporative cooling material	Efficiency of Performance

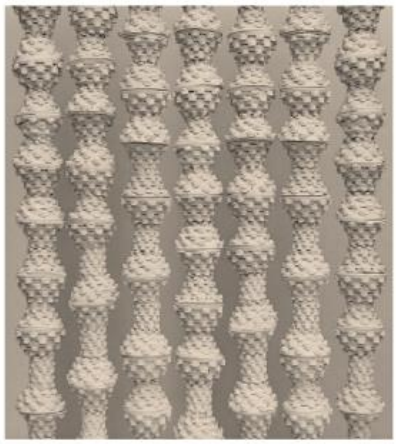
Porous ceramic materials

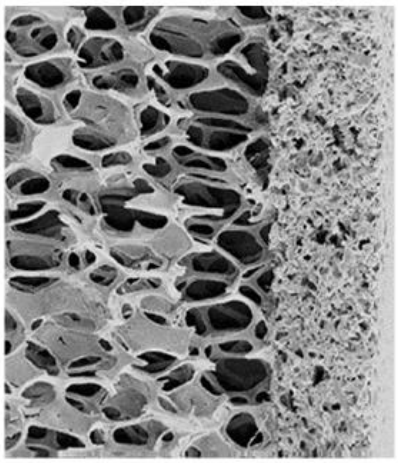
ECOOLER

Cool Brick



The Chronicle of Leslie Forehand





Common Thermal characteristics of Ceramic fibres

thermal conductivity [kW/m·K]	(400°C)	0.09×10^{-3}
	(600°C)	0.12×10^{-3}
	(800°C)	0.16×10^{-3}
	(1000°C)	0.21×10^{-3}
density [kg/m^3]		250
specific heat [kJ/kg·K]		1.0*
emissivity		0.9*

*approximated value, not guaranteed by manufacturer

No.	D (dry (g))	W (wetted (g))	I (immersed (g))	P_o (porosity (%))
1	1516	1646	901	17.4
4	1301	1415	775	17.8
5	1335	1456	796	18.3
6	1683	1829	990	17.4

*High porosity prototype-fired at 1110 C

Material

- ceramic structures combined with:
- alumina (Al2O3)
- nitrides (Si3N4)
- silicon oxides (SiO2)
- non-silicate glasses.

Performance

- ECOOLER operates on the same principle as human perspiration, where evaporation from the moist ceramic surface consumes heat and lowers the air temperature.
- Each ceramic unit holds about 550 ml of water, with approximately half of this amount evaporating over the course of a day.
- A set of 35 units stores around 19 litres of water, of which about 9.5 litres are consumed daily—comparable to the use of a single toilet flush.
- The airflow speed directly affects performance; stronger wind enhances evaporation and improves the delivery of cooled air into the surrounding space.
- The ceramic body itself functions as both the evaporative medium and the structural module, offering a combined role of cooling, shading, and spatial design flexibility.

4.6 Limitations of Evaporative Cooling Systems

Applying evaporative cooling has some associated problems that can be avoided when the system is well designed; however, such issues can be resolved and offset by the positive outcome of the process. Some of the disadvantages are:

- Adverse Effects of Elevated Humidity

  When the humidity level exceeds 55%, and the temperature is within the range of 20 °C to 35 °C, the excessive moisture in the air can promote the growth of bacteria and other microorganisms that may be harmful to the human body (GEORGE, 2021).

- Dust and Mites

  Some micro-organisms are common pollutant factors in indoor spaces with hot, humid conditions, which can trigger asthma attacks and cause inflammation of the linings. These pollutants affect the indoor air quality; humidified air is lighter, but without proper circulation, the lack of movement can still increase the concentration of dust, mites, and other allergens within the space (Stefaniak, 2025).

- Mould and Fungi

  Various types of Mould can exist in the indoor environment, and once they grow on wet media, they can cause a plethora of problems. The spread of the spores may lead to health issues and negatively affect air quality (Stefaniak, 2025).

Despite these factors, evaporative cooling offers significant advantages that can outweigh the aforementioned drawbacks. However, it is crucial to ensure adequate air circulation and implement proper ventilation techniques to prevent discomfort and maintain efficiency. Some of these disadvantages may be mitigated under conditions of solar exposure on different parts of the evaporative cooling element (Srisamranrungruang, 2020).

4.7 Conclusion

Evaporative cooling relies on water as the key element in its process, unlike conventional air-conditioning systems, which use refrigerants that have known environmental impacts, such as ozone depletion (Lechner, 2009). While evaporative cooling is often viewed as a sustainable alternative, the issue of water consumption cannot be overlooked, especially in dry regions where water scarcity poses a significant challenge during certain months of the year (Schecter, Cold Spot: Evaporative Cooling through Ceramics, 2015). Outdoor evaporative cooling systems that lack external shading are particularly vulnerable to excessive water loss. When wetted surfaces are exposed directly to sunlight, evaporation rates increase significantly, resulting in higher water demand to maintain effective cooling. This not only affects the system's efficiency but also raises important questions about its environmental viability. Even systems designed with shading, such as those inside wind-catching towers, can consume large quantities of water—as was the case at Zion National Park's passive cooling towers in 2002, which used over 400,000 litres annually. Therefore, when designing outdoor evaporative cooling systems, it is essential to prioritise shading strategies, whether architectural or natural, to reduce solar heat gain and minimise unnecessary water loss. Shading helps maintain cooler surface temperatures, improves the system's overall performance, and most importantly, ensures that water—a limited and valuable resource—is used responsibly. Without this balance, evaporative cooling risks becoming a water-intensive solution that contradicts its sustainable intentions.

5.0 Ceramic and Material Properties

The efficiency of ceramic applications largely depends on selecting the appropriate type of ceramic material that matches the desired structural features during manufacturing. Therefore, the production and fabrication stages require a comprehensive understanding of the process and the necessary characteristics. The pore configuration and geometry are critically determined during the synthesis process, which ultimately defines the final properties of the ceramic (chen, 2021).

5.1 Ceramic and Materialistic Properties

Ceramic materials are used across various industries and in different forms; accordingly, the typology of the material and the contents of the mixture vary depending on the primary function of the material and the features that need to be emphasised *(Groover, 2001)*.

a. Traditional Ceramic: This mixture is primarily composed of clay minerals (such as porcelain), along with various additives like cement, silica, or glass. This material is mainly used in the manufacture of pottery, bricks, and tiles *(Groover, 2001)*.

b. Advanced Ceramic: This material is made from synthetically produced raw materials and is primarily used in the manufacture of advanced ceramics, including bio-ceramics *(Groover, 2001)*.

The development of cellular ceramics began in the 1970s. These materials consist of numerous heat-resistant gaseous pores. The pore size ranges from the angstrom to the millimetre scale, with porosity levels spanning from 20% to 95%, depending on the properties and manufacturing process. Specifically, the pore size can vary between 3nm and 3mm *(Chen, 2014)*.

Porous ceramics are classified into two main categories:

1. Honeycomb Ceramics

   This type of cellular ceramic is considered an attractive alternative to conventional catalyst; however, honeycomb ceramics cannot be easily adapted for use in monolithic catalyst

structures (chen, 2021). The pore size and geometry of the modelled material can be controlled during the mould design phase, and the pore shapes are commonly triangular or square. The reliance on extrusion in the modelling process results in uniformity in both pore size and shape; therefore, producing a model with small pore sizes remains a significant challenge (Cabeza, 2020).

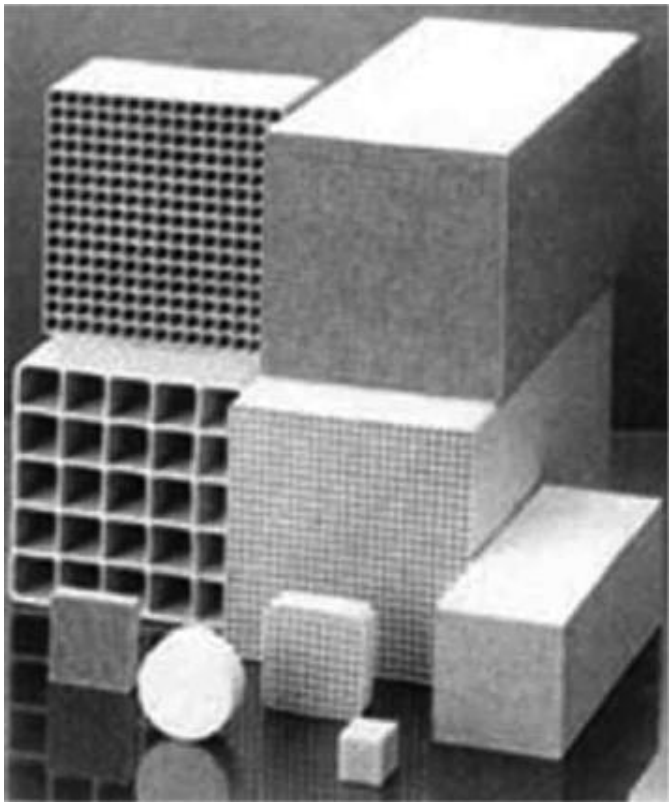


Figure 28. Honeycomb Ceramic Structure Featuring Regular Triangular Pores (Chen, 2021).

2. Ceramic Foam

Ceramic foam is commonly classified into two types:

a) Open-cell reticulated ceramic foam.

This type occurs when the pores within the ceramic structure are interconnected, forming a reticulated network (Cabeza, 2020).

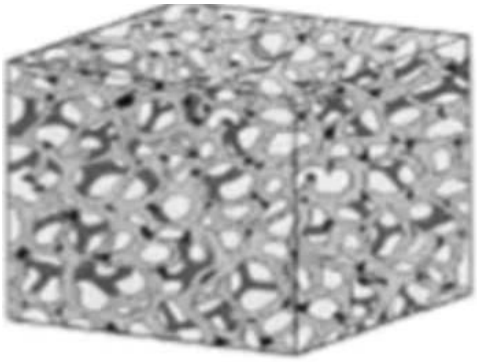


Figure 29. Open-cell Reticulated Ceramic Foam

b) Closed-cell bubble-like ceramic foam

This type occurs when a separating structure—such as solid cell walls—isolates the pores of the foam from one another (Cabeza, 2020).

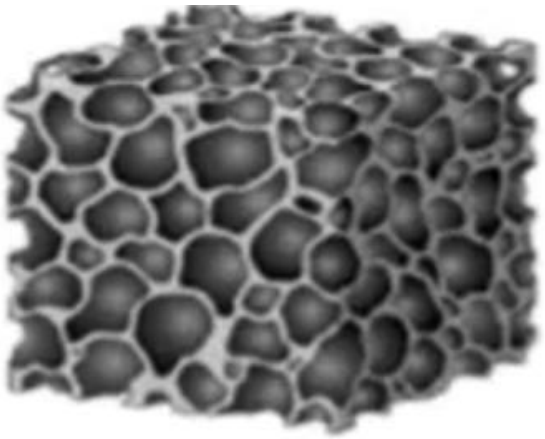


Figure 30. Closed-Cell Bubble-Like Ceramic Foam

- Such cases can be clearly identified by comparing the fluid penetrability through the ceramic foam. The degree of fluid saturation within the pore cells determines the presence of solid cell walls surrounding the pores. Some foams may exhibit hybrid configurations, where partially open-cell pores exist within the ceramic foam, and water retention in the pores is influenced by the specific type of each pore. The porous structures display varying levels of fluid permeability due to differences in bulk density and thermal conductivity. Consequently, open-cell ceramic foam typically has lower bulk density and reduced thermal conductivity, while the pore structural typologies enhance fluid penetrability (Cabeza, 2020). The porosity level of ceramic foam is controlled by adjusting the raw materials and preparation techniques. Accordingly, ceramic can also be classified based on the average pore size. According to the terminology of the International Union of Pure and Applied Chemistry (IUPAC), porous

materials are divided into three categories based on pore diameter, as follows: Micro-porous: The pore size is relatively small, less than 2 nm.

- Meso-porous: The pore size ranges between 2 and 50 nm.
- Macro-porous: The pore size is relatively large, typically greater than 50 nm.

Fluid permeability is influenced by the affinity between the solute and solvent and the porous material, particularly in processes such as ultrafiltration, nanofiltration, and reverse osmosis, where the pore size is relatively small. Filtration and microfiltration retain particles larger than the pore size, whereas ultrafiltration and nanofiltration are applied when finer separation is required (Chen, 2014).

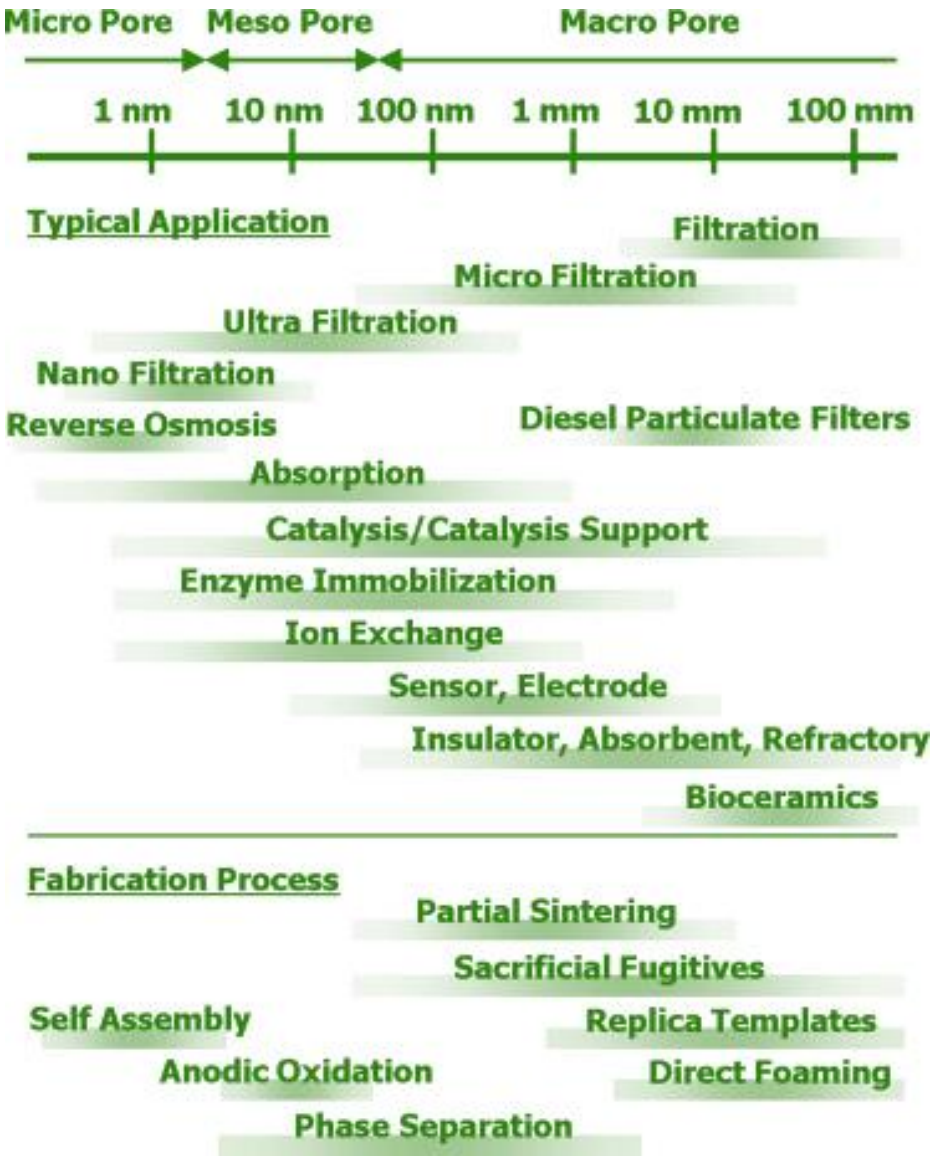


Figure 31. Classification of porous materials based on pore size, along with their typical applications and fabrication processes (Chen, 2014).

Ceramics can be classified into various types based on the material properties of the raw material used in the manufacturing process, including:

- Silicate
- Aluminosilicate
- Diatomite
- Corundum
- Silicon carbide

The open-cell type of ceramic foam, a novel form of highly porous ceramics, features a 3D reticulated structure with interconnecting pores. This configuration results in a high specific surface area, enhanced fluid contact capacity, and associated fluid pressure loss. Consequently, open-cell ceramic foams are used in various applications, including metallurgy, chemical engineering, environmental protection, and energy sectors (Cabeza, 2020). The porosity of ceramic in the evaporative cooling process acts as a fundamental factor in the general efficiency of the cooling strategy. Porosity is the key factor in controlling mass transfer per unit volume, which in turn directly affects heat transfer performance (Cabeza, 2020).

5.2 Attractive features of Ceramic: Key Advantages of Using Ceramics in the Fabrication of Wet Media

Porous ceramics offer high mechanical strength, abrasion resistance, chemical durability, and thermal stability, while the density, mass, and thermal conductivity are relatively low (H.V, 1979). These characteristics have attracted significant research interest, as porous ceramics present substantial economic and manufacturing advantages across various sectors. The advantages of Ceramics include:

- Low density
- Large specific surface area
- High toughness and strong thermal shock resistance
- Good thermal insulation capacity
- Excellent stability at high temperatures, with a low dielectric constant, in contrast to most metals, polymers, or dense counterparts (H.V, 1979).
- The manufacturing techniques of porous ceramics can create a foam with a high mechanical strength level, corrosion resistance, and stable reaction under high temperatures (Chen, 2014). Porous ceramics have several common characteristics: Relatively high chemical stability.

Porous items can be adapted for various corrosive environments in which they are intended to perform by selecting the appropriate material types and manufacturing processes.

- Particularly high specific strength and stiffness. Under gas pressure, liquid pressure, and other stress loadings, the form and size of the pores in porous ceramics remain unchanged.
- Excellent Thermal stability. High-temperature combustion gases can be filtered through porous components made of heat-resistant ceramics.

These remarkable properties point to a bright future for porous ceramics across a wide range of applications, and they offer porous ceramics' applicability in a variety of fields, including environmental design. As the fine pore structure of porous ceramics has become more precisely controlled, the resulting products have been increasingly applied in processes such as detachment, dispersion, and adsorption. These functions are now utilised in various industries, including chemical engineering, metal smelting, petroleum, textiles, pharmaceuticals, and food processing machinery (Chen, 2014).

5.3 Modelling Techniques of Ceramics

The modelling of ceramics begins with the preparation of the material to be used in the process. The composition of the mixture largely depends on the required strength and the specific material properties that need to be achieved (Chernyuk, 2015). Accordingly, the starting powders must meet formal requirements in terms of particle size and chemical composition, as these factors play a crucial role in determining the final strength of the ceramic. It is essential to control the grain size of the starting powder through both mechanical and chemical methods, given the direct relationship between the resultant strength and the grain size (Chernyuk, 2015). There are various modelling techniques used in the manufacturing of ceramics. Each modelling method has its advantages, disadvantages and limitations (Chen, 2014). Accordingly, the geometry and structural typology of the intended model are key factors in selecting the appropriate manufacturing method. Some of these modelling techniques are:

1- Ceramic Mould Casting
2- Ceramic Injection Moulded (CIM)
3- Extrusion
4- 3D Printing

Ceramic mould casting is a manufacturing method that primarily involves pouring the ceramic mixture onto a relief-patterned surface and allowing it to solidify over time, creating a cavity mould that can be used later in the casting process. The model formed in the portions of cope and drag can be physically enhanced through the firing. The resulting product from this technique is a hallow model

composed of separate parts that can be assembled later (Groover, 2001). The modelling pattern used in the casting process is made from various materials, such as metal, wood, resin, plaster, or rubber. The selection of the material depends on the cost and time considerations (Groover, 2001). The plaster method involves applying plaster to adjustable designs, which can be reinforced by the addition of fillers. Porous fillers, synthetic foams, and steam autoclaving can also enhance mould permeability (Groover, 2001). The patterns are robust structures made of high-quality refractory materials combined with an ethyl silicate binder and a gelling agent. These patterns pass through an elastic phase before solidifying, allowing the design to be transferred more easily, which also accommodates small undercuts and reduces the need for a taper (Chen, 2014). Therefore, Complex forms require manual mould fabrication and pattern stripping, leading to higher production costs and slower processing.

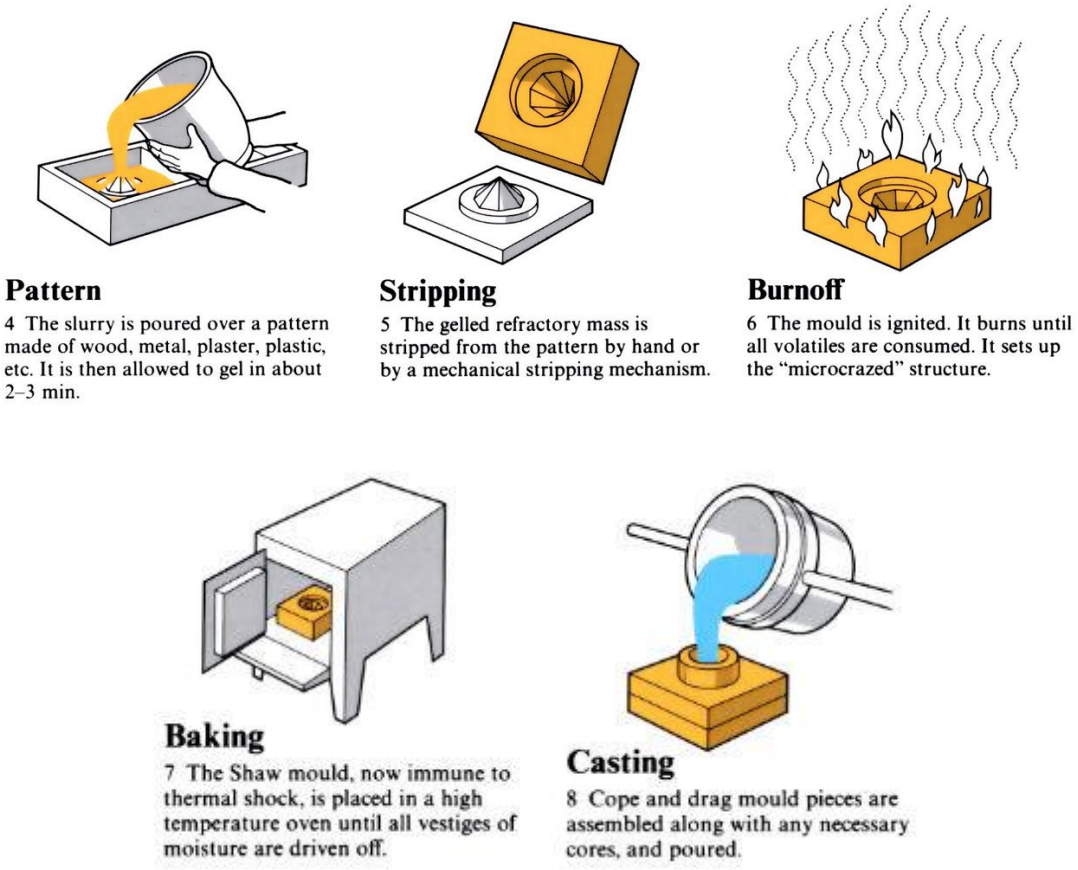


Figure 32. The Manufacturing Stages of Ceramic Mould Casting

In traditional ceramic, water is usually added to the powder for the temporary blinding of the particles to achieve the desired viscosity and consistency for shaping the intended form. In contrast, in advanced ceramics, other substances are introduced as binders during the modelling phase (Groover, 2001). In the initial stage of preparing the modelling mixture, the material undergoes grinding and milling using crushing tools to minimize particle size. After maintaining the desired grain size, water and other additives are incorporated to produce the required paste (Chen, 2014). Following the modelling stage,

the green parts are sintered or fired to enhance the structural properties and surface roughness of the model. The common fabrication techniques of ceramic models are:

I. Slip Casting: The main components of the mixture are water and clay, with a water content ranging from 25% to 40%. The method relies on drain casting, in which the mixture is poured into a pre-formed mould to develop a semi-solid layer, resulting in the formation of a hallow product (Groover, 2001).

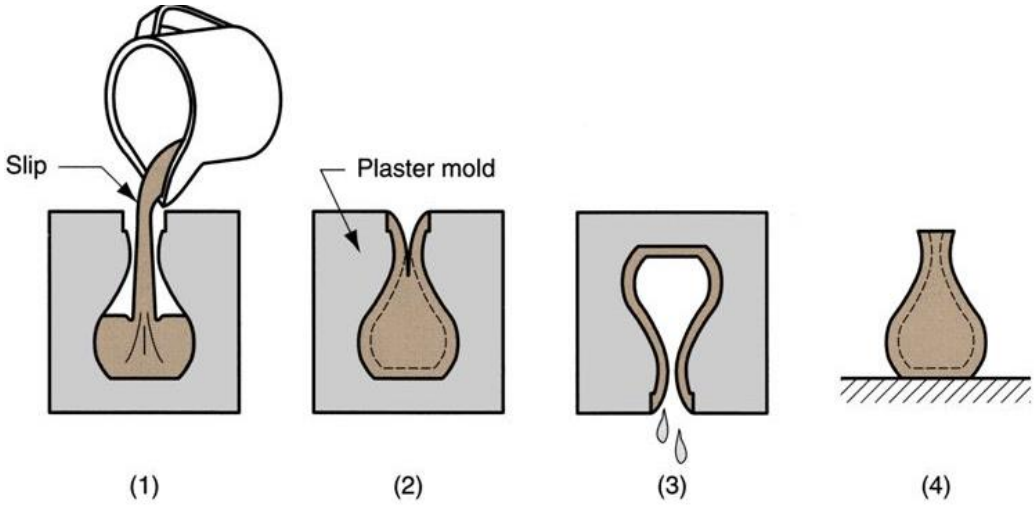


Figure 33. Steps of Drain Casting In the Modelling Of Ceramics Using a Fluid Mixture

II. Plastic Forming Method: The clay used is plastic with water content ranging between 15% and 25%. The process relies on adding water to the mixture to facilitate ceramic modelling, unless mechanical equipment is employed in the manufacturing process. This method includes various forming techniques such as:

a) Hand Modelling: This technique relies on the manipulation of the material into the desired form and geometry. In some cases, the modeller uses a spinning wheel that rotates 360° around a vertical axis to facilitate the shaping of all parts (Groover, 2001).

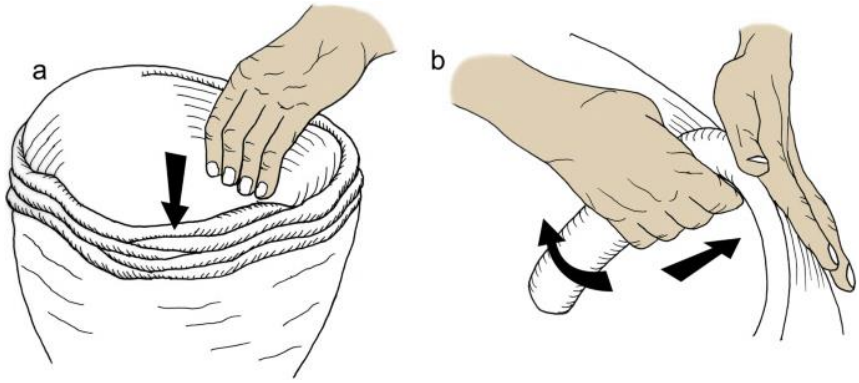


Figure 34. The Hand Modelling Technique for Ceramics

One well-known example of utilizing hand-modelled ceramic components in the design of efficient outdoor evaporative cooling systems is the use of terracotta cones in India. Munish Siripurapu of Ant studio, based in New Delhi, developed this concept by reimagining traditional terracotta cones—originally used in the introduction of yoghurt-based ice cream—as functional evaporative cooling screen (studio, 2020).

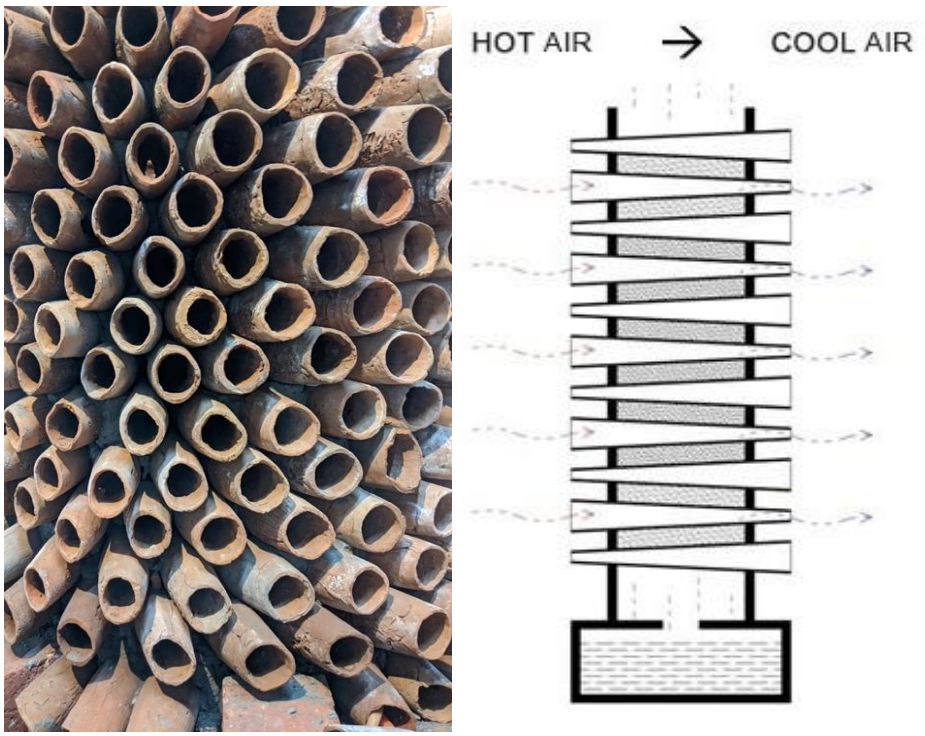


Figure 35. A Sustainable Cooling System Made with Wet Terracotta Cones (Studio, 2020)

Figure 36. A Sustainable Cooling System Made with Wet Terracotta Cones (Studio, 2020)

b) **Mechanical Method:** Also known as jiggering, this technique is similar to hand modelling but reduces human involvement by utilizing machines and equipment that minimize manual intervention (Groover, 2001).

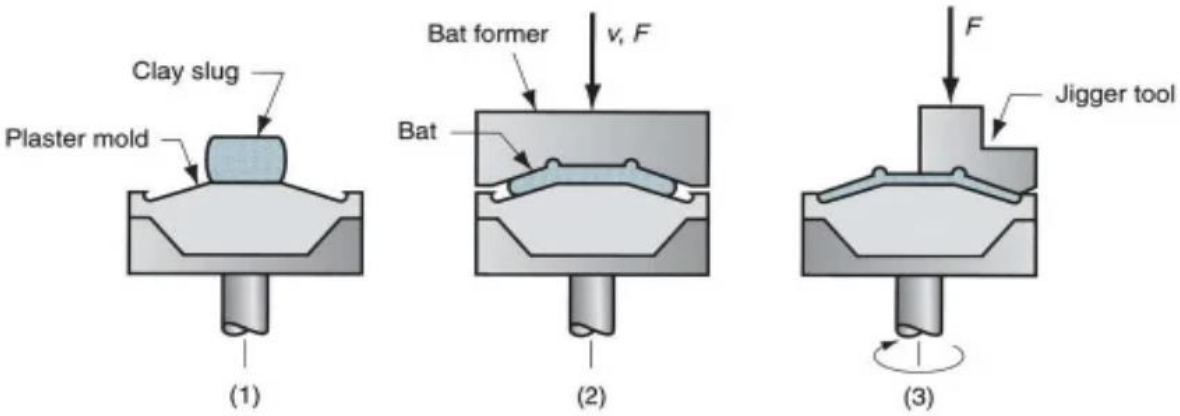


Figure 37. The Steps Followed In the Jiggering Method (Groover, 2001).

c) Plastic Pressing: This method involves pressing the material between two moulds held within metallic rings (H.V, 1979). The moulds are typically made from porous materials such as gypsum, which serve to reduce vacuum formation at the rear and facilitate the removal of excess moisture from the clay (Groover, 2001). Subsequently, positive air pressure is applied to open the moulds, preventing the sticking of the ceramic parts to the moulds. This technique overcomes limitations associated with jiggering, particularly in enabling mass production and overcoming the constraints of radial-symmetric shapes (Groover, 2001).

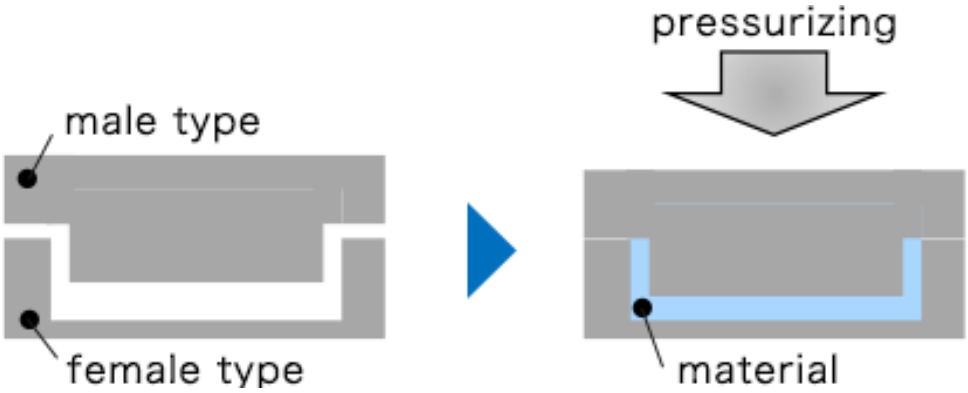


Figure 38.The Main Concept of the Plastic Pressing Method in Ceramic Manufacturing.

d) Extrusion: the clay mixture is compressed to be used in the production of uniform sections that can be cut into the desired geometry to be gathered in the required form. The method depends on utilizing different equipment that assists in the mixing and pushing of the clay mixture through the die opening (Groover, 2001). This technique

is also used in fabricating the starting clay slugs needed for other procedures such as jiggering and plastic pressing (Groover, 2001).

III. Semi-Dry Pressing: The main component of the mixture is moist clay with reduced plasticity. The process relies on the use of high pressure to avoid the low plasticity of the clay consequences (Groover, 2001).

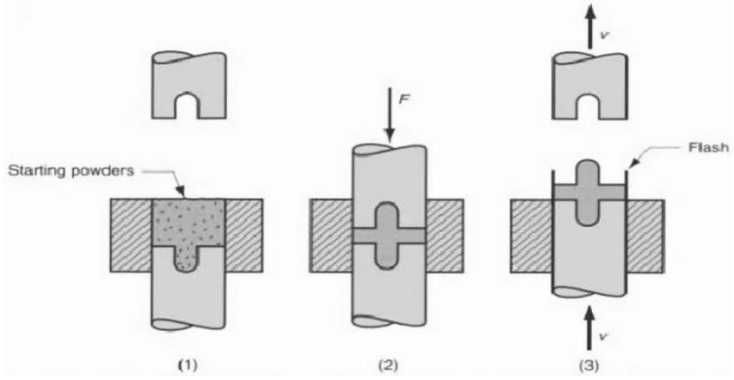


Figure 39. The Stages Followed in the Semi-Dry Pressing Method

IV. Dry Pressing: The main component of the mixture is dry clay with a water content of less than 5% and no plasticity. The dies used in this method must be made of hard materials, such as steel or cemented carbide, to minimise wear caused by the abrasiveness of the dry clay (Groover, 2001). This technique is commonly used in the manufacturing of bathroom tiles, refectory bricks and other simple forms (Chen, 2014).

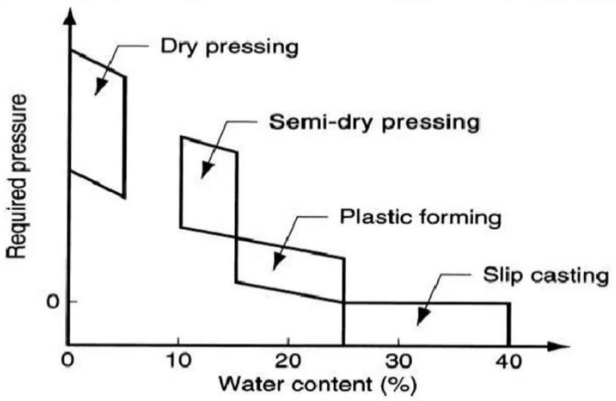


Figure 40. Categories of Shaping Mixtures Used in Traditional Ceramics

The water content in the ceramic mixture plays an important role in the shaping process of ceramics. Subsequently, this water must be eliminated from the model before the firing stage begins (Chen, 2014). Therefore, shrinking becomes a critical factor to consider during the drying process, as water contributes to the material's volume and its eventual reduction (Chernyuk, 2015). The drying process takes place in two consecutive stages. The first stage involves accelerating the drying rate until surface water begins to evaporate from the material. In the second stage, water content is gradually reduced until the ceramic grains come into contact with one another.

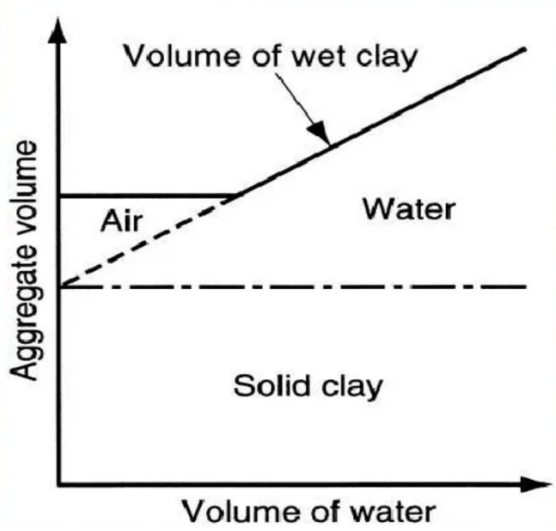


Figure 41. The Relationship Between the Volume of the Ceramic Model and the Water Content in the Used Mixture

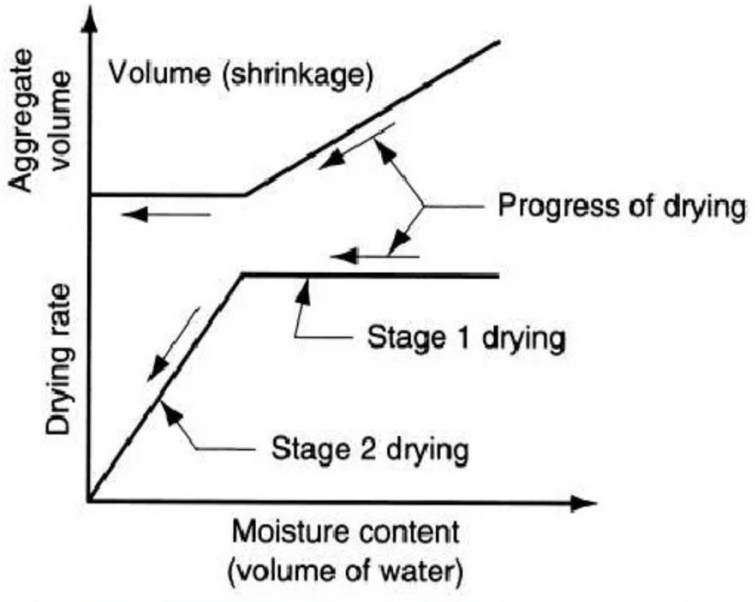


Figure 42. The Common Drying Rate Curve Associated with Volume and Water Content Reduction

Models of ceramic materials with diverse internal architectures are currently being developed on this foundation. One such approach is the fractal method, which represents similar porous structures (Chernyuk, 2015). However, this method does not account for non-periodic ceramic structures. An alternative approach describes ceramic pores based on their random number, size, and distribution. These pores are often modelled as spheres, although their actual shapes differ. Related computations are presented in a separate study that employs an analytical model with arbitrary subarea geometries. This framework provides an accurate representation of real grains, particulates, and pores.

Ceramic elements are commonly produced by mixing powder with water, while binders and other additives are occasionally incorporated.

Conventional technological equipment and tools, including injection moulding, tape casting, and gel casting, have enabled designers to overcome limitations and flexibly design various shapes and geometries (Zhangwei, 2019). However, managing densification remains essential; thus, the modelling stage requires sintering parts at higher temperatures (H.V, 1979). Consequently, such modelling techniques may entail certain limitations, including high cost and time-consuming processing (Zhangwei, 2019).

Modelling ceramic components with complex geometries, voids, or parametric shapes can be challenging when using the moulding method. Similarly, the machining method presents difficulties due to the delicate balance between hardness and brittleness. On the other hand, the cutting method is prone to manufacturing defects, such as cracks, poor surface quality, and imprecise dimensional modelling (Zhangwei, 2019).

3D printing in ceramic modelling, also known as additive manufacturing (AM), has become an attractive and efficient method for producing ceramic elements. The AM technique fabricates physical parts in a discrete manner—point by point, line by line, and layer by layer. It utilises a digital process in which three-dimensional models are adaptively sliced into multiple two-dimensional sections for sequential construction (Zhangwei, 2019).

After the modelling of the ceramic, the model undergoes the firing process, during which the ceramic model is exposed to heat to eliminate the remaining water content in the mixture (H.V, 1979). Firing can occur at high fusing (1290-1370℃), medium (1090-1260℃), or low (1065-870℃) temperatures (Zhangwei, 2019). Accordingly, additional shrinkage may occur in the polycrystalline material, compounding the shrinkage accumulated during the drying process (Chernyuk, 2015). Ceramic used in evaporative cooling typically does not undergo the glazing process to avoid obstructing the pores and capillaries of the component (Chernyuk, 2015). The sintering stage is a commonly required procedure in modelling ceramic components, helping to control the level of porosity, as well as the density and bonding of individual grains into the solid mass Chernyuk, 2015).

The final stage that certain models may require is the finishing procedure, which is applied to enhance dimensional accuracy, implement minor geometric adjustments, and improve surface finish (H.V, 1979).

5.4 Controlling the Porosity of Ceramic

The porosity of the ceramic refers to the proportion of all pores within the material, which may be open or closed, and either linear or nonlinear in shape. Open pores are typically interconnected through internal capillaries and channels, enhancing the ceramic fabric's ability

to absorb and retain various fluids. In contrast, closed pores generally form between enclosed particles as a result of residual volumes of decomposed substances gradually released during the firing process (Chen, 2014). Thus, the porosity level of the ceramic can be used to classify and describe different types of pores.

The porosity of the ceramic used in evaporative cooling plays a critical role in both process efficiency and the surface area of the wet medium. Therefore, it is essential to precisely control the porosity of the ceramic during the fabrication stage. One approach involves adding fugitive agents to the mixture, which dissolve during the sintering process and create pores and cavities. This method, however, requires additional heat during the firing stage to ensure the complete burning of these undesirable substances. Another technique is partial sintering, wherein powder compacts are sintered to a specific limit of densification (Dale Hardy, 1995). This method helps preserve the refined pore diameters within the grains after firing. Nevertheless, achieving precise control over porosity remains challenging due to various influencing factors, including the properties of the starting powder, the forming condition of the compacted mixture, and the proportion of added materials.

Sintering temperature and time are important factors in controlling porosity. However, it has been observed that the density of certain ceramics with poor sinterability is not significantly affected by sintering temperature (Dale Hardy, 1995). Therefore, regulating these factors helps adjust porosity by simultaneously influencing grain size, which plays a vital role in shaping the capillaries of the model and its mechanical properties (Yang, 2001).

It is challenging to infer a definitive link between mechanical characteristics and porosity in partly sintered fabricated models. It is crucial to identify an appropriate sintering method that enables the designer to control porosity without altering the microstructure and mechanical properties (Yang, 2001). In 2001, research was conducted to develop ceramics by hot pressing a powder combination to a predetermined dimension, beyond which the press is held to produce ceramics with varying porosity and high accuracy. As a result, the powder quantity may readily modify the density (Yang, 2001). The researchers relied on fixing the pressing force at a certain point upon reaching the required dimensions, employing a specific design in which rare-earth metal oxide can reduce the sinterability of the powder mixture while promoting fibrous grain production. The bulk density of the sintered materials is determined using distilled water and the Archimedes displacement technique. Both the bulk density and the theoretical density were used to compute the porosity (Yang, 2001).

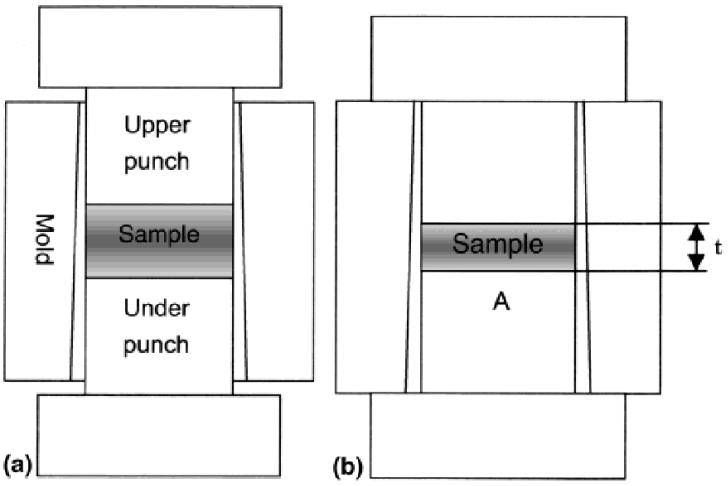


Figure 43. Hot Pressing Schematic Diagram during the Sintering Stage (Yang, 2001).

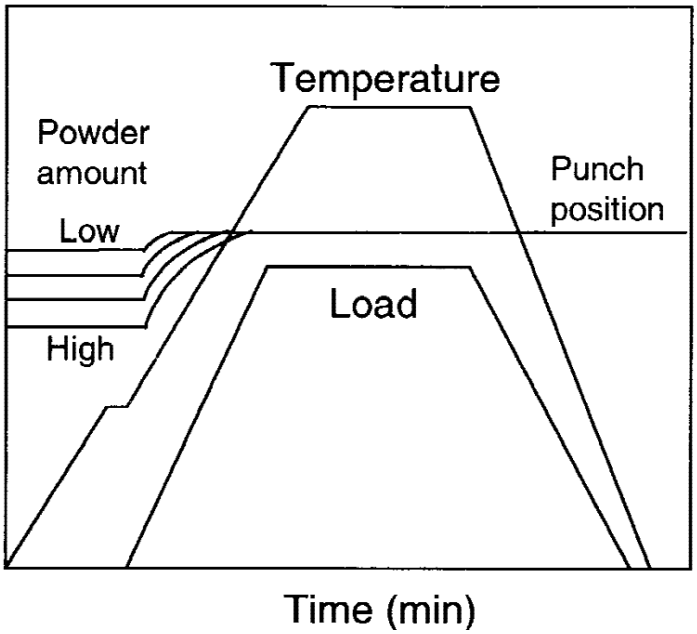


Figure 44 the time program of the sintering process of porous ceramic (Yang, 2001).

Accordingly, it was concluded that altering the temperature of the sintering process can regulate grain growth. Thus, the microstructure in partial hot pressing (PHP) can be adjusted through the firing temperature (Yang, 2001). This method enables the fabrication of various ceramic models with different porosities from the same powder composition by modifying the sintering conditions using the (PHP) technique (Yang, 2001). In ceramic components sintered under identical conditions, it was observed that the relationship between final mechanical properties and porosity was restricted to pores—specifically porosity and pore size—which holds significant importance from both scientific and industrial perspectives.

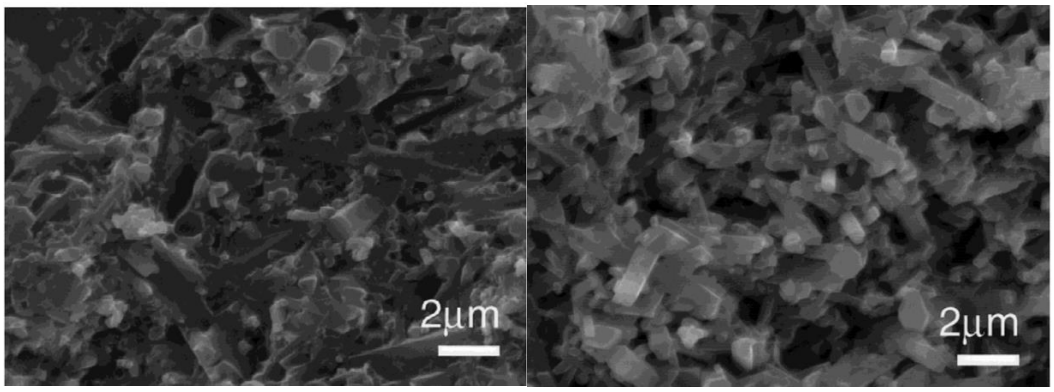


Figure 45. Ceramic Microstructure Produced Using the PHP Method; Porosities were (a) 0.004 and (b) 0.233.

Fracture energy (y) decreases as porosity (P) increases in single-phase ceramics with fine-to-medium grain size, following a trend similar to that observed in mechanical strength. This behaviour has been documented in various advanced ceramics with porosity levels reaching up to 50%.

Porosity also affects fatigue behaviour. Its presence contributes to the reduced fatigue strength of powder metallurgy (P/M) materials (Chen, 2014). Key porosity-related parameters—including porosity fraction, pore size, pore shape, and pore spacing—are critical factors influencing the fatigue performance of P/M materials (Yang, 2001).

The porosity of ceramic is not only associated with the material properties of the clay mixture used for moulding, but it may also change during the shrinkage process that occurs in the drying and firing stages. As the firing temperature gradually increases, the progressive formation of a liquid that fills the holes formed by chemical breakdown can manipulate the porosity level of ceramic. This fluid tends to solidify as the temperature decreases, producing glassy particles that cement within the interior microspores, thereby reducing average porosity and enhancing the strength and durability of the ceramic (Jastrzebski, 1959).

Furthermore, various processing procedures can influence the porosity, density, fluid resistance loss, and penetrability of these materials, with alumina and cordierite being the most commonly used material types (Chen, 2014). Alumina is employed to enhance the strength and thermal stability of the material, whereas cordierite is used as a raw component to improve resistance to thermal fluctuations. Additionally, porous silicon nitride and silicon carbide ceramics have been developed to enhance thermal stability further.

The suitability of porous ceramics for various applications is determined by their composition and structure. Initially, porous ceramics were used as filtration materials for removing bacteria and other microorganisms. Once the fine pores of porous ceramics were controlled to a higher degree, the resulting products became applicable to a growing number of applications, including detachment,

dispersion, and adsorption, which are now used in various industries: chemical engineering, metal smelting, petroleum, textiles, pharmaceuticals, and foodstuff machinery (Chen, 2014).

5.5 3D printed Ceramic

3D printing in ceramic modelling, also known as additive manufacturing (AM), became an attractive and efficient method for modelling ceramic elements. The AM manufacturing technique depends on fabricating physical parts using the point-by-point, line-by-line, and layer-by-layer discrete method. AM uses the adaptive manner from the three-dimensional software, which is digitally sliced into various 2D sections (Zhangwei, 2019).

3D printing is classified into different categories depending on the specific technological method used in the manufacturing of the ceramic model. Therefore, understanding the different aspects of each method helps designers to identify the needed method depending on the requirements of the modelling stage. Some of these methods are:

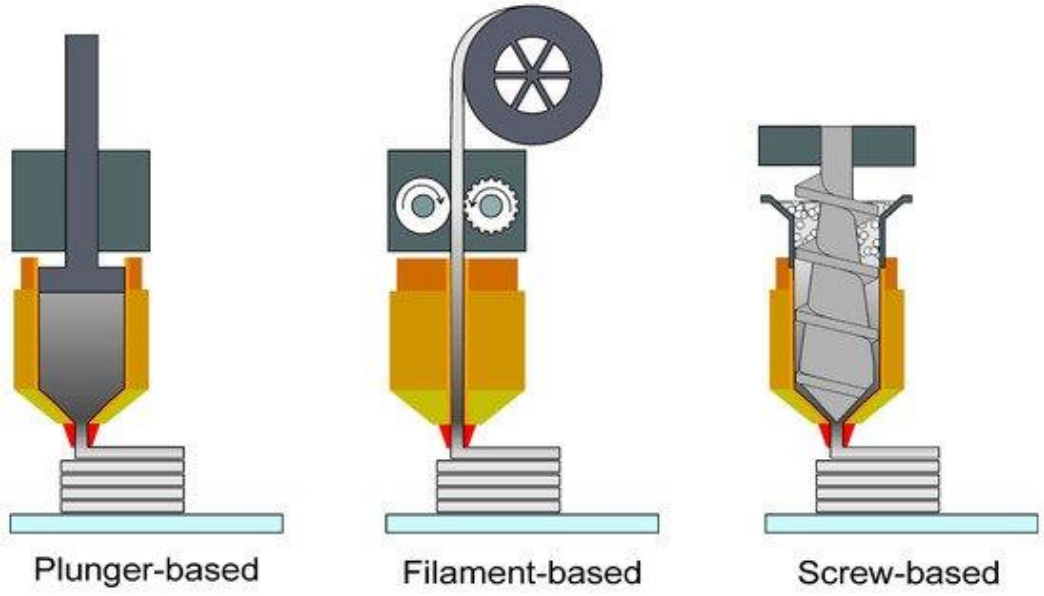


Figure 46. Extrusion-based Additive Manufacturing Encompasses Various Forms and Techniques.

The 3D-printed ceramic blocks represent an excellent example of applying 3D printing technology to develop an efficient ceramic screen for evaporative cooling. Each brick absorbs water like a sponge and is designed as a 3D lattice that allows air to pass through the structure, which consists of porous ceramic bricks arranged in a matrix (MCMATHIS, 2015).

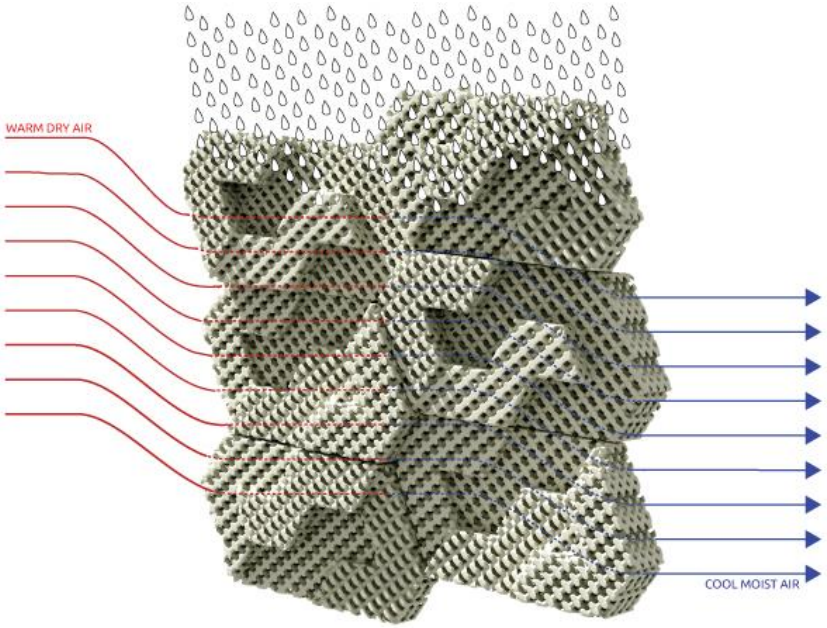


Figure 47. The working Technique of the 3D-Printed Compressed Ceramic Blocks (MCMATHIS, 2015).

The 3D-printed bricks were designed in a modular form to structurally support one other like interlocking bricks, while their geometry allows airflow through the cavities, cooling and humidifying the air to reduce the ambient temperature of the space (MCMATHIS, 2015).

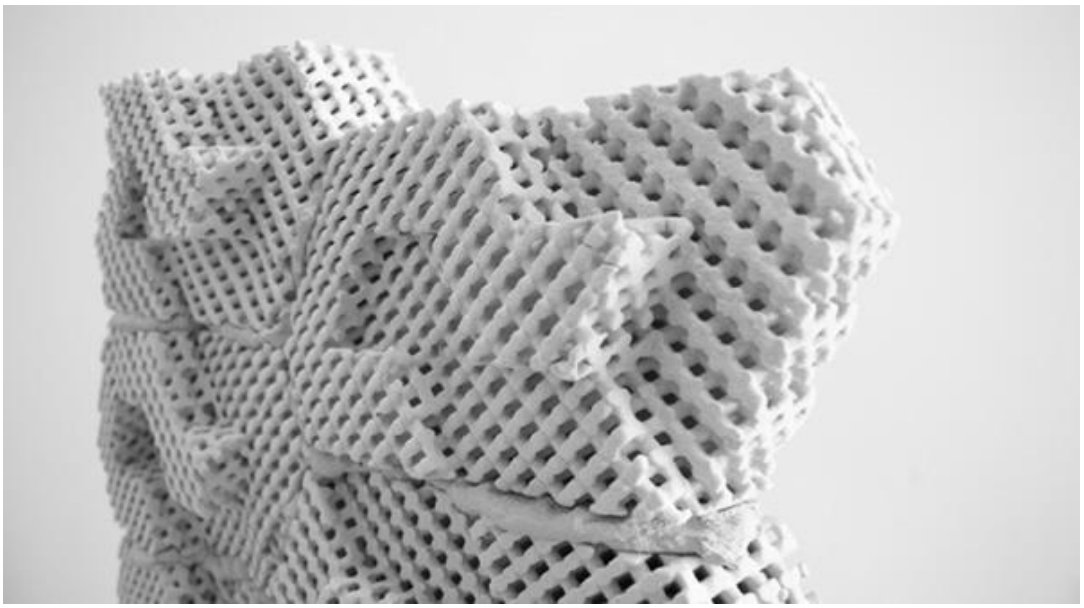


Figure 48. Close-up Image of the 3D-Printed Cooling Bricks (MCMATHIS, 2015)

Technologies Based on Slurry

This technology primarily depends on the viscosity and solid loading of the material used in the system. Accordingly, it often employs liquid or semi-liquid systems dispersed with small ceramic particles as

feedstock, either in the form of inks or pastes. Photopolymerization, inkjet printing, or extrusion can all be applied in slurry-based 3D printing (Zhangwei, 2019).

- Photopolymerization-based methods are categorised into different types, such as Stereolithography (SL) and its derivatives, including digital light processing (DLP).
- Inkjet-based inkjet printing (IJP).
- Extrusion-based direct ink writing (DIW) (Halloran, 1994).

1.1 SL Technique

Stereolithography, also referred to as the SL technique, is one of the most commonly used methods in the 3D printing of ceramics. It was introduced by Hull in 1986 and was later commercialised by several 3D printing pioneers, such as 3Dsystem Inc., which developed the 3D system SL technology (Wang, 2010). The method became commonly used in 1994, with the introduction of concentrated ceramic mixtures—up to 65% solids—which can include added substances such as silica, alumina and silicon nitride (Zhangwei, 2019).

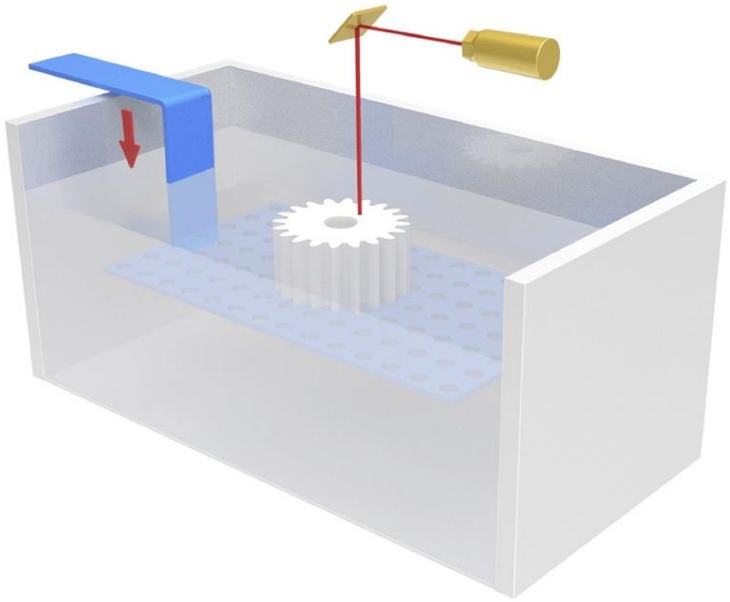


Figure 49. The Stereolithography (SL) Technique, as illustrated in this Figure (Zhongwei, 2019).

1.2- Digital Light Processing (DLP)

This method was first proposed by Nakamoto and Yamaguchi at the end of 1996, with further developments emerging in 1997 through the use of liquid crystal displays. The method continued to advance until 2001, when digital micromirror devices (DMD) were introduced (Hornbeck, 1997).

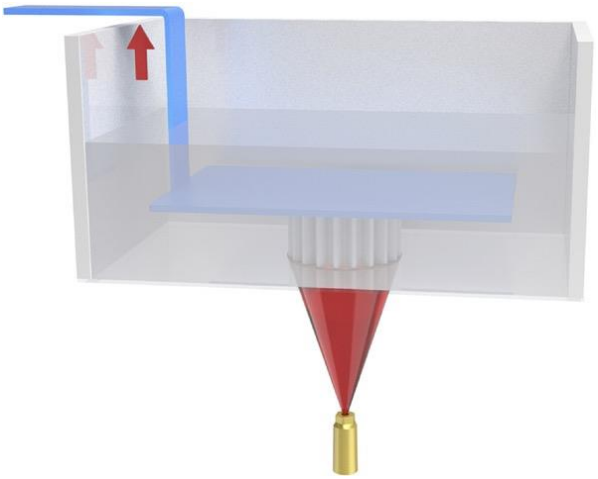


Figure 50. Schematic Diagram of the Digital Light Processing (DLP) Process (Zhongwei, 2019).

1.3- Inkjet Printing (IJP)

Inkjet printing (IJP) is a well-established technology used to produce 2D digital visuals by ejecting a fluid mixture—such as ink—in the form of droplets through printhead nozzles onto various surfaces. IJP technology was initially developed in the 1950s and, with the integration of computer assistance, was introduced to the commercial market after the 1970s by major manufacturers such as Epson, HP, and Canon (M. Singh, 2010).

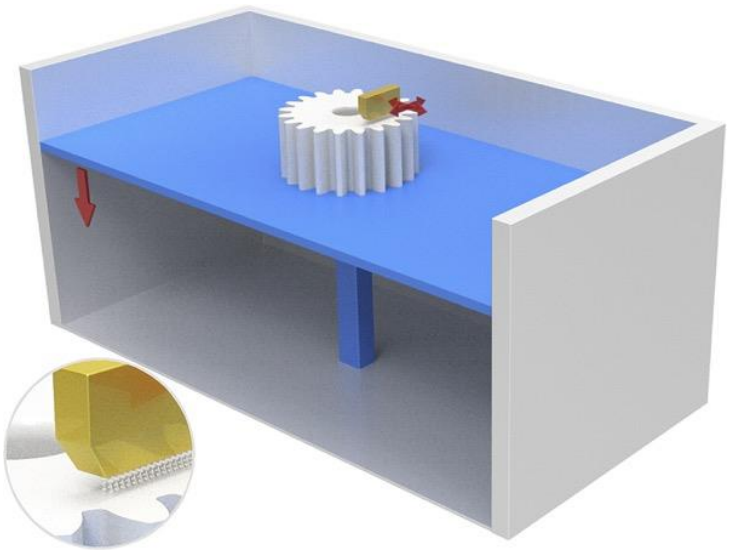


Figure 51. Schematic Diagram of Inkjet Printing (IJP) (M. Singh, 2010)

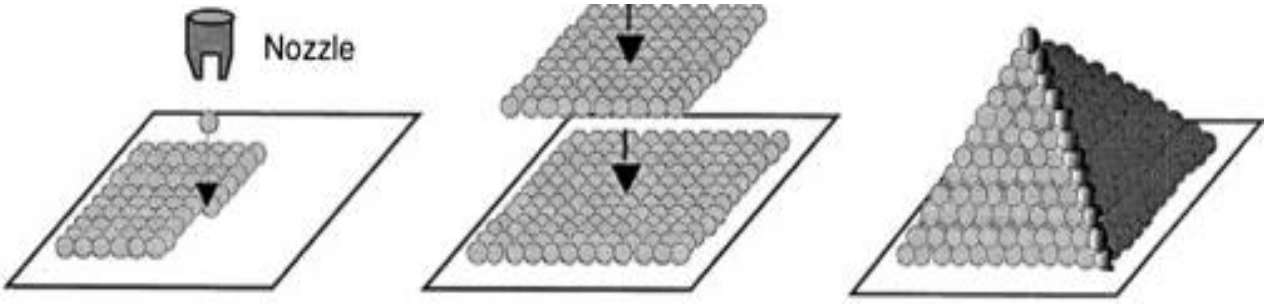


Figure 52. One of the Techniques Used in Inkjet Printing Involving a Jet Nozzle, Illustrating the Modelling Principle by Yamaguchi et al. (P.Tobler, 2006).

This method is used through two common techniques, one of which is described by P. Tobler (2006):

> Continuous or drop-on-demand (DOD). This method is preferred in the 3D printing of various models due to its accuracy in extrusion modelling, achieved by using small-sized material droplets that are produced by compressing ink through either thermal or piezoelectric stimulation (M. Singh, 2010). Due to their low solid loading, ceramic inks typically exhibit low viscosities, down to a few $\mu pas-$ which leads to extended drying durations and significant shrinkage, potentially compromising the final accuracy of the printed product.

One of the issues that can arise during the drying of 3D-printed ceramic patterns is the coffee stain effect, which occurs due to the segregation of solid particles from the centre to the edges of the printed pattern. This phenomenon is common when using inks with a low solid volume fraction and can lead to structural defects and faults. Therefore, it is important to mitigate this effect by adding substances such as polyethene glycol (PEG) to the ink (Rui Dou, 2011).

1.4- Direct Ink Writing (DIW)

This method was first employed by designers at Sandia National Laboratories in 1997 as part of efforts to develop a technique for processing highly concentrated materials, such as ceramic slurries with reduced organic content (Pedro Miranda, 2006).

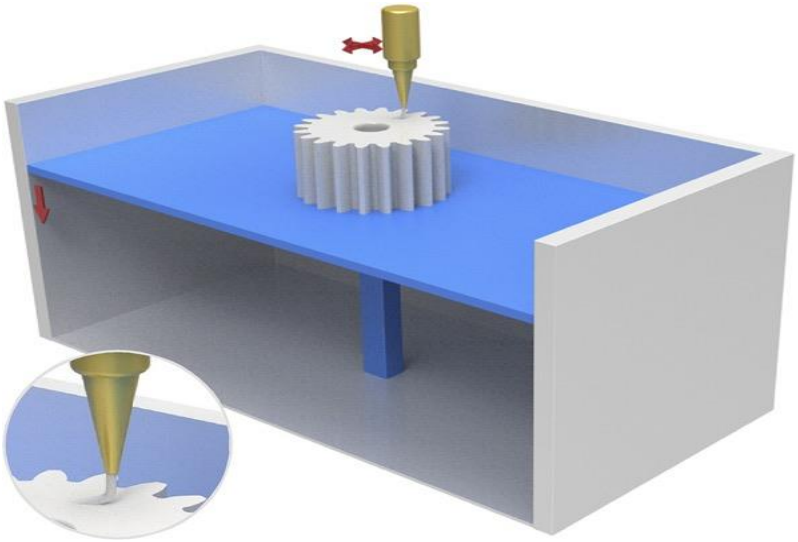


Figure 53. Direct Ink Writing (DIW) Method (Zhongwei, 2019).

This method relies on the coordinated movement of the nozzles to construct the designed geometry layer by layer, enabling a cost-and time-efficient process compared to photo-cutting methods (Pedro Miranda, 2006). The technology employs a semi-liquid paste that facilitates the formation of freestanding structures with a high aspect ratio, owing to its high solid loading and viscoelastic properties.

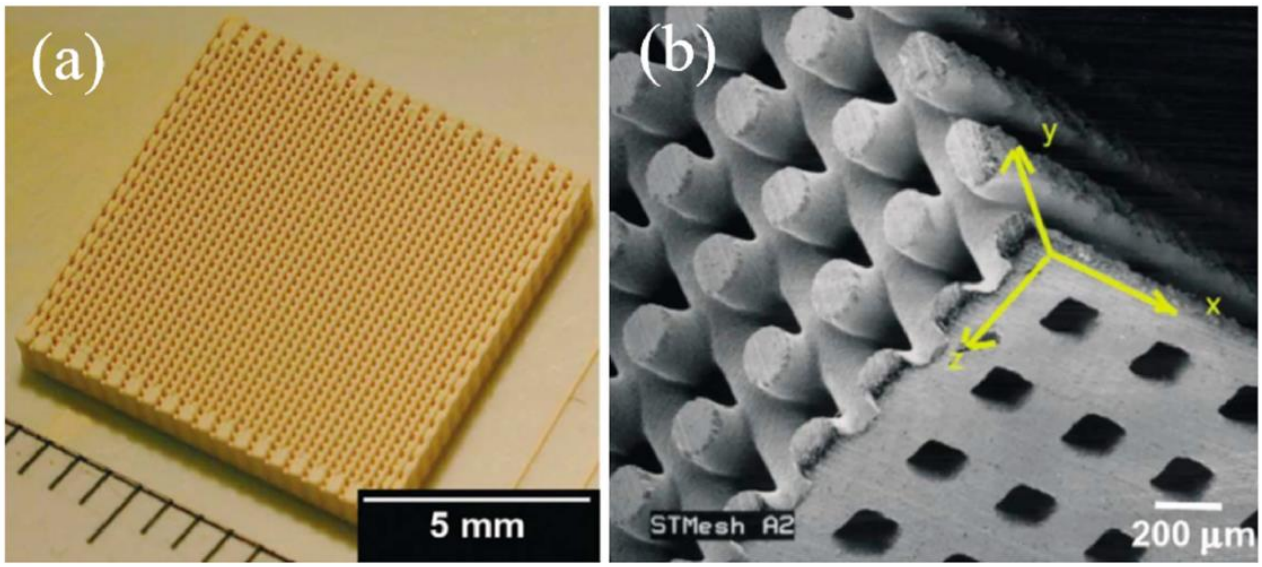


Figure 54. Concentrated Ceramic Paste Used in DIW and its Microstructure (Zhongwei, 2019

From the outcome of using 3D printing technology in the fabrication of ceramics, it can be concluded that ceramic 3d printing offers numerous attractive features that enable designers to fabricate complex geometries and overcome limitations inherent in traditional manufacturing methods (Pedro Miranda, 2006). The general advantages of ceramic adaptive fabricating methods are:

- This technology enabled designers to fabricate complex geometries and structures that are difficult to manufacture using traditional methods.
- It provides the desired qualities and properties needed in the model.
- The small-scale production of complex geometries can be more economically efficient than traditional methods.
- This technological method is advantageous in terms of cost and time.

While the disadvantages of the method are:

- AM necessitates an in-depth understanding of both 3D printing and the characteristic properties of the materials used in the fabrication.
- The sintering and firing procedure is an important stage in ceramic additive manufacturing, and it may be one of the primary roadblocks to the technology's broad acceptance.
- It is challenging to control the shrinkage of the model during the firing process (Pedro Miranda, 2006). Therefore, the designer should bear in mind the shrinkage factor during the design phase.
- This technology is not available to local manufacturers; therefore, mass production can be a limitation. However, Maggie J. W. George took the initiative of designing ceramic evaporative cooling screens using the 3D printing method to create prototype models, which were then employed in creating plaster casts for the remaining models (GEORGE, 2021).

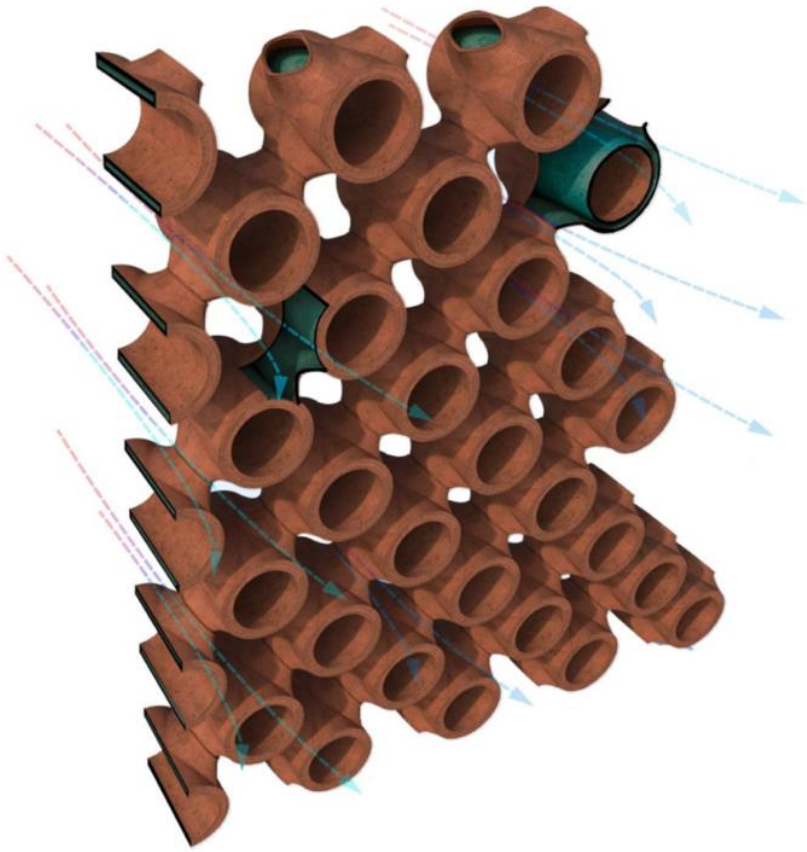


Figure 55 Digital Fabrication of Evaporative Cooling Blocks (Maggie J.W. George)

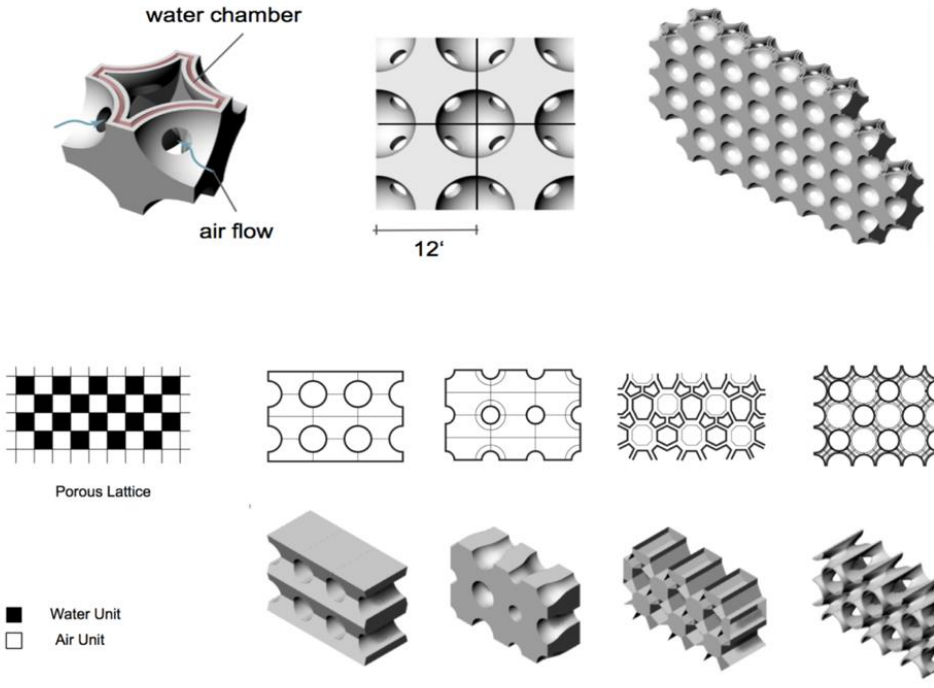


Figure 56. Digital Fabrication of Evaporative Cooling Blocks (Maggie J.W. George)

Figure 57. The Plaster Cast of the Evaporative Cooling Blocks (GEORGE, 2021)

The reason for using traditional methods in the final modelling stage—despite the availability of 3D printing technology—lies in the limitations of mass production. Accordingly, 3D ceramic printing technology is not accessible in some parts of the world, and the process is more time-consuming than slip casting. Maggie J.W. George also intended to create a system that could be manufactured within a commercial context and deployed rapidly across various sectors, particularly in impoverished regions, which would benefit most from this design feature and gain protection from excessive heat in hot, dry climates.

5.6 Porosity of Ceramic in Different Fabrications

Designing ceramic evaporative cooling systems requires understanding ceramic porosity and the factors that influence its level and linearity. The porosity of ceramic varies in terms of density, form, and linearity depending on the fabrication method and the firing conditions of the clay. Although some of these factors are not binary or fixed in their outcomes—and often rely on trial-and-error experimentation for full comprehension—it remains important to develop a generic understanding of how porosity can vary during the manufacturing and firing stages, based on the approaches adopted by previous researchers. However, the generic formula for quantifying porosity

Equation 4. The common formula for quantifying the porosity of Ceramic (Jastrzebski, 1959)

$$P = \frac{W - D}{W - S} \times 100$$

Where:

P is the apparent porosity of the ceramic.

D is the weight of the solid, dry clay or ceramic specimen.

S is the weight of the ceramic specimen when suspended in a fluid (typically water), with all internal pores fully saturated.

W is the weight of the soaked ceramic specimen after the excess surface water has been removed by dabbing.

The difference between the weight of the saturated ceramic (W) and the dry ceramic (D) represents the weight of the water that fills the pores of the ceramic, which can be used to identify the volume of the internal pores. However, the true porosity of ceramics is represented by the actual volumetric value of both the closed and open pores within the clay body. This can be precisely quantified using the following formula:

Equation 5. The true porosity of ceramic, including closed pores.

$$Percent\ of\ true\ porosity = \left(\frac{S_g - B_d}{S_g}\right) \times 100$$

Where:

S_g represents the specific gravity or the density of the solid material.

B_d is the bulk density of the ceramic.

The bulk density of a given mass can vary depending on changes in size, shape, surface area, and the degree of compacting, each of which is influenced by the specific manufacturing method used (Chen, 2014).

Conversely, the permeability of porous materials depends primarily on the interconnected open pores, as this characteristic determines the material's capacity to permit fluid penetration under a specific pressure gradient. These connected open pores, often referred to as the material's capillaries, influence both the effective porosity and the internal surface area. Therefore, having the same porosity level does not necessarily imply that the material will exhibit the same permeability (Jastrzebski, 1959).

However, Darcy's law, mentioned in the following equation, explains that the flow rate of the fluid through a material is directly proportionally to the specific pressure gradient of the fluid, assuming that the viscosity and permeability of the material remain constant.

Equation 6. Darcy's law of liquid pressure gradient (Jastrzebski, 1959)

$$\frac{\Delta P}{\Delta L} = \frac{1}{\propto} \times \eta . \upsilon$$

Where:

$\Delta P/\Delta l$ represents the pressure gradient across the porous material with thickness l.

υ is the specific flow rate of the liquid through the porous material.

η is the viscosity of the liquid (assumed constant).

$\propto$ is the permeability of the porous material.

The coefficient of permeability can also be quantified by measuring the fluid's flow volume through a given cross-sectional area. This area may be expressed in square inches or square centimetres, or in Darcy units, which represent a flow rate of 1 cm^3 across a cross-sectional area of 1 cm^2 (Jastrzebski, 1959).

The application of ceramic in façade design depends primarily on the material properties of the ceramic after firing. Firing is a fundamental process that enhances the durability of the material and transforms the designed clay from its humble, malleable state into a stronger, façade-appropriate substance, approaching the durability of stone.

Clays are primarily composed of hydrous aluminosilicates, which are the result of chemical weathering processes acting upon feldspar rocks. This transformation produces various clay minerals that share a characteristic sheet-like structure, facilitating the formation of flake-shaped particles ranging in size from 5 microns to less than 1 micron (Jastrzebski, 1959).

Accordingly, we can conclude that the characteristic properties of the clay used in fabrication can vary according to the following factors:

a) The geographical location of the area from which the clay is extracted.
b) The chemical composition of the raw clay.
c) The firing and drying procedures applied to the clay

The material of clay can vary significantly from one location to another due to these geographical differences. In Egypt, for instance, three distinct types of clay are commonly identified: White Aswan Clay, Red Aswan Clay, and Sinai Kaolin. Each type exhibits different percentages of SiO_2, AL_2O_3, and H_2O based on their molecular proportions, as shown in the following table (Gad, 1949).

Table 6. The Percentage of Molecular Proportions of Different Types of Egyptian Clay (Gad, 1949)

	Al_2O_3		SiO_2		H_2O
White Aswan clay	1	:	1·6	:	1·8
Red Aswan clay	1	:	2·2	:	2·1
Sinai kaolin	1	:	1	:	2·2

Such variations in the material composition of clay can affect the level of porosity and the properties of the ceramic after firing. However, these factors may also vary during the manufacturing stage, depending on the additives introduced to the clay to modify the molecular proportions of its components. Nevertheless, these characteristics can influence the behaviour of the material during the firing process, which will be discussed in the following pages.

The plasticity of clay can be described as the elastic property of raw clay when mixed with water, which facilitates its moulding into various shapes. Each manufacturing technique requires a different level of

plasticity that allows the clay to be shaped in a manner consistent with production demands, while minimizing the risk of drying cracks or deformation. During the manufacturing stage, water content forms a hydrophilic film around the flaky clay molecular particles, enhancing their parallel alignment. This alignment can subsequently improve the linearity of the pores within the ceramic component, as illustrated in the following images.

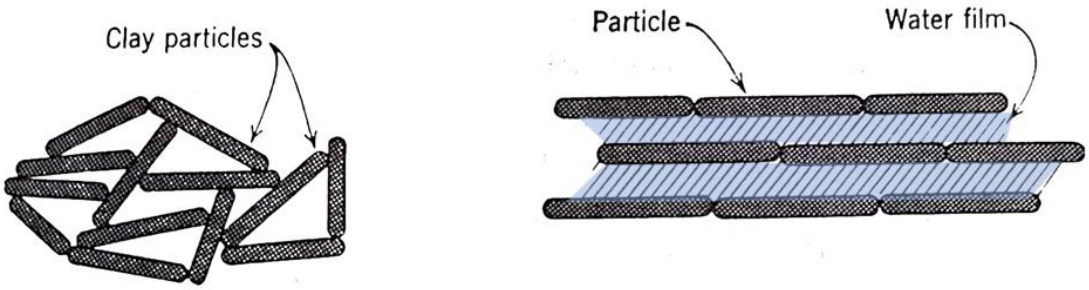


Figure 58. Plasticity of Clay and Changes in Molecular Structure after the Addition of Water to Form a Hydrophilic Film (Jastrzebski, 1959)

The amount of water added to the clay to manipulate its plasticity depends largely on the desired particle shape and the specific manufacturing technique employed. However, there are defined thresholds for the minimum and maximum water content. When the water content falls below 35%, H_2O makes the clay less plastic, crumbly, friable, and fixable in moulding—this is referred to as the *plastic limit* of the clay. Conversely, when H_2O (the water content) exceeds 50%, the clay becomes excessively wet and sticky, impeding the modelling process. This condition is known as the *liquid limit* of the clay, as illustrated in the following graph (Jastrzebski, 1959).

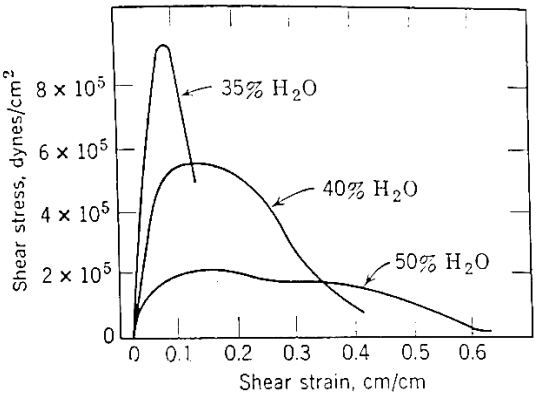


Figure 59. Stress-Strain Shear Force Diagram Illustrating Different Levels of Clay Plasticity (Jastrzebski, 1959)

Enhancing the plasticity of clay depends primarily on achieving a balanced ratio between water content and clay particles, which varies according to the specific requirements of the fabrication method. However, plasticity can also be improved during the mixing stage through the proper dispersion of clay particles, ensuring that each particle is surrounded by a water film to maximise the homogeneity of the clay mixture. Although water plays a critical role in shaping clay, this role is temporary; the water film

surrounding the clay particles gradually diminishes and ultimately disappears during the drying and firing stages.

The drying of clay models follows the modelling stage, during which the clay is left to air-dry in order to remove its water content prior to firing. As the water evaporates, the clay molecules are drawn closer together, resulting in a reduction in overall volume and an increase in the material's strength. This evaporation occurs gradually, depending on the location and nature of the water molecules within the clay. There are two types of water present in the clay body:

- Shrinkage Water refers to the water film surrounding the clay particles. It is the first portion of water to evaporate during the drying stage and is primarily responsible for the shrinkage of the clay body—hence the term "shrinkage water."
- Pore Water is the water held within the micropores of the ceramic. It evaporates later in the drying process, with a small percentage remaining that is typically released during the firing stage.

Accordingly, the drying stage is a critical phase that requires careful execution and proper guidance to prevent damage to the modelled ceramic during shrinkage and to avoid excessive shrinkage rates that could disrupt the movement of clay particles. Some resources recommend placing the clay in special environments with temperatures between 85°C and 96°C and high humidity levels to regulate the shrinkage rate by managing the evaporation rate of water content between the clay particles. Elevated air temperatures reduce the viscosity of the water, thereby accelerating its diffusion toward the surface (Jastrzebski, 1959). The greater the plasticity of the clay, the higher the potential shrinkage rate. For this reason, non-plastic materials—such as fine sand—are added to the clay to reduce water content and gradually minimise shrinkage during drying. In contrast, the removal of pore water does not affect the clay's volumetric shape and occurs gradually once the temperature exceeds 110°C (Jastrzebski, 1959).

To enhance the strength of moulded ceramic, the material undergoes a firing stage that improves its durability and refines its form to meet the desired specifications. The firing parameters required for ceramic production depend on the functional and structural properties sought, with temperatures ranging from 110°C to 1399°C. The transformation of the ceramic begins at approximately 427°C, at which the clay minerals start to decompose, initiating the dissolution of alumina and silica components. This decomposition is accompanied by the release of water, as represented by the following chemical formula:

$$AL_2O_3 \bullet 2SIO_2 \bullet 2H_2O \longrightarrow AL_2O_3 + 2SiO_2 + 2H_2O$$

Accordingly, the different stages of firing are accompanied by changes in the properties of the ceramic, including its strength and level of porosity, as well as the removal of organic substances and iron pyrites

attached to the clay particles (Jastrzebski, 1959). The transformation in shape and the shrinkage occurring during the firing process are typically associated with a reduction in porosity, which begins at a temperature of 689°C. The effect of firing on porosity level can be modulated by the gradual infiltration of fluids that saturate the pores. Once the temperature exceeds 1399°C, the porosity stabilises, and further firing has no additional effect on porosity levels (Jastrzebski, 1959).

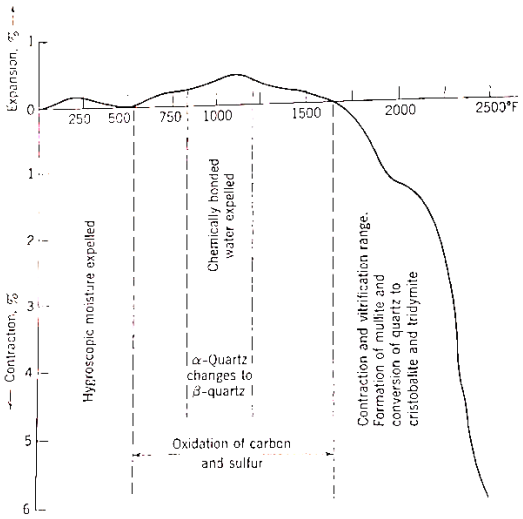


Figure 60. Gradual Changes in Clay during the Firing Process (Jastrzebski, 1959)

5.7 Conclusion:

Several fabrication methods are available for working with ceramics, each offering unique advantages and challenges depending on the design complexity and the level of craft expertise. Methods such as hand-building, wheel throwing, extrusion, and slip casting were explored throughout this chapter. While all can be used to produce ceramic components, slip casting emerged as the most suitable technique for the intended application.

Slip casting was selected due to its relative ease of use, accessibility, and adaptability across different regions and craft workshop levels. This method allows artisans to produce consistent, precise forms even in less industrialised settings, making it highly functional and widely available. Crucially, slip casting supports the creation of designs with hollow or cavitated interiors that are essential for water circulation in evaporative cooling systems. Its ability to reproduce intricate shapes efficiently while maintaining structural integrity makes slip casting the preferred fabrication method for developing ceramic cooling screens. The balance between craftsmanship and scalability offered by slip casting supports both traditional craft practices and contemporary production needs, thus positioning it as an ideal choice for implementing culturally rooted yet climate-responsive architectural elements.

6.0. Introduction to design studies

According to the literature and the adopted methodology, Islamic architecture in general—and the Mashrabiya in particular— relied heavily on specific geometric ratios commonly used in the design of traditional patterns. Craftsmen succeeded in preserving these details over time, treating them as defining features of each screen. Such data were reserved and transferred across generations of artisans through what is often referred to as "inherited science" (Gawad, 1986). Screens were also named and classified based on the shape, proportional ratios, and geographical origin of manufacture. Accordingly, the Maymooni pattern derived its name from the origin of its manufacturers (1997 ,محمد).

Accordingly, the details of the wooden lathed elements that constitute the Maymooni Mashrabiya retained their original forms and dimensions, despite the limited availability of resources and the absence of formal documentation concerning the geometric and orthographic details of these screens (waziri, 1999). The configuration of traditional wooden lathed screens can be defined by the shapes of the individual components that are assembled and joined to form the complete screen, as well as by the proportional relationships between the dimensions of these components (Hindi, 2011). To better understand these ratios, several examples of old Mashrabiya screens were selected and analysed. In parallel, it was essential to examine and highlight the commonly employed proportional systems in Islamic architecture and to relate these systems to the findings of the analytical study.

6.1 Insights into the Analytical Case Studies

Knowledge emerges from the act of asking the right questions. In this study, the literature review served as a critical starting point for framing those inquiries. The preceding chapters have explored Mashrabiya as a spatial and environmental element, examining the reasons behind its evolving design variations and its development into a functional passive cooling solution. Building on the previous chapter's discussion, we can begin laying the foundation for the case study and defining the analytical objectives that will guide the forthcoming exploration. These objectives are intended to unpack different dimensions of the study and to illuminate the key design features within Mashrabiya that may be interpreted, decoded, and reconfigured into a contemporary evaporative cooling screen. As introduced in Chapter One, Mashrabiya screens appear in diverse geometries and configurations. Nonetheless, the frameworks that form these screens often carry the same designation and share a common lineage—for example, the Maymooni screen.

The Mashrabiya screen is a complex architectural element that varies significantly in shape, form, and scale (Al-Shareef, 1996). Therefore, a clear understanding and definition of design scale is essential to identify the key functional features within the Mashrabiya that influence its overall

performance. Geometric ratios represent one of the most significant aspects of the design process in traditional Islamic patterns. These ratios were ultimately integrated into the design of the Mashrabiya, resulting in structured configurations that shape the visual quality of the final outcome. Each pattern is defined by its scale of application, which is determined by the axis scale and pattern ratio—factors that establish the spacing between the axes upon which the individual components are positioned and connected (Gawad, 1986). In traditional Islamic design, proportion serves as a self-guiding mechanism that ensures both aesthetic harmony and efficient application. Accordingly, in the second stage of this study, it was essential to examine these ratios and explore the underlying design concepts in order to understand the manufacturing logic of these screens and select the most appropriate samples to support the analytical objectives of the study.

The Maymooni screen was found to be an interesting design with complex details that still require thorough explanation and analysis. The intricacy of the small elements—assembled in a Lego-like fashion to form the main screen—preserved the original design details despite the limitations of fabrication methods, which were unable to keep pace with the mass production of these handcrafted components. This inherited complexity has not been addressed from a functional perspective in any research domain. However, certain parts of the screen may prove to be functionally effective in terms of shading if examined at the micro level. Nevertheless, this hypothesis must be validated through analytical work.

Although the Maymooni pattern is typically defined as one with minimal spacing between its constituent components, it has evolved over time to encompass a variety of shapes, forms, and perforation percentages (Gawad, 1986). The pattern gained widespread popularity across numerous historical examples, establishing itself as a distinctive family within the broader Mashrabiya typology, as illustrated in the following images. Consequently, the Maymooni pattern family was selected for analysis due to its prevalence in traditional Mashrabiya designs and the functional potential of its elements, characterised by their simple geometrical forms yet enriched with intricate manufacturing details.

Therefore, a deeper investigation was undertaken to examine the predominant shapes of the Maymooni screens and to analyse the distinctive characteristics and intricate details of these patterns. Accordingly, the following steps were carried out:

6.2 Choosing the Samples

According to Mustafa A. Mohamed's book *The Phenomena of Repetition in Islamic Arts*, the modular network structure of the pattern primarily relies on a grid mesh that serves as the fundamental load-bearing mechanism of the screen's components. The advantage of this assembly method lies in the limited number of distinct engineering pieces, whose dimensions must be adjusted based on spatial needs and installation requirements. This system supports the interface and relies on its framework to produce the required and regularly distributed openings and perforations within the pattern (محمد, 1997). Consequently, we may infer that each pattern is shaped by the application's scale, which in turn determines the proportional dimensions of the pattern—specifically, the spacing between the axes to which the elements are affixed and arranged.

The common ratios in Islamic pattern design and architecture are (Hindi, 2011):

a) 1:1

b) $1{:}1\frac{1}{2}$

c) $1{:}1\frac{1}{3}$

d) $1{:}1\frac{1}{4}$

e) $1{:}1\frac{1}{8}$

Islamic pattern design is a combination of integral elements that have been manufactured and designed using a specific method to maximise the functionality of these elements. Such a method proves to be creatively functional, as it acknowledges the matrix of relations between the designed elements and the application process (Gawad, 1986). Middle Eastern artists prioritised ratios and proportions, since using them in their artwork was considered a process of adapting to aspects of nature within the designs. Therefore, artists and craftsmen divided the ratios of Islamic pattern designs into three main categories:

- Quantitative: which depends on the numerical quantification of the ratio. (Gawad, 1986)
- Qualitative, which depends on the geometrical explanation of the ratio.
- Quantitative and qualitative are integrated.

Therefore, it is important to understand the common examples of both quantitative and qualitative design ratios. As noted in the literature, several research studies have identified the geometric ratios employed for the initial design of traditional Islamic patterns, which are highly useful for understanding the design phases of Mashrabiya screens. However, these studies would have been more informative had they concurrently addressed these ratios in terms of both numerical quantity and algebraic formulation. Furthermore, some scientific publications indicate the specific numerical values of the

ratios. Accordingly, this chapter will provide more detailed mathematical explanations of these ratios, along with references to their most prominent applications in Mashrabiya design.

The golden ratio is one of history's most well-known mathematical and geometric proportions. It is commonly observed in nature and has been incorporated into various designs across architectural schools. This ratio contributes aesthetic value to works of art and can be quantified as the proportion between two-line segments—or the distance between two dimensions at a certain level—where the larger segment equals the sum of both segments. It is represented mathematically by the following equation:

$$\frac{a}{b} = \frac{b}{a+b}$$

Equation 7: The Mathematical Formula of the Golden Ratio

Accordingly, the golden ratio can represent both summative and geometric sequences, which may explain its prevalence in various architectural and natural examples. Many researchers attribute the application of the golden ratio in the designs of the internal components of Mashrabiya and traditional Middle Eastern bay windows to the predominance of proportional patterns in Islamic art—patterns initially grounded in both mathematical and engineering principles (Hammoda, 1981). This notion is evidently supported by the observable ratios in the distribution of lathed wooden elements within Mashrabiya structure and the dimensions of these components, some of which correspond to what is known as the "golden rectangle" (Hammoda, 1981).

The Fibonacci sequence is an algebraic series that resembles the golden ratio in numerical and quantitative applications. The sequence begins with 0, 1, and progresses as 1, 2, 3, 5, 8, 13, 21, 34, 55, and continues infinitely, where each term equals the sum of the two preceding terms. As the sequence extends toward infinity, the ratio between consecutive numbers converges toward the golden ratio (Sautoy, 2022).

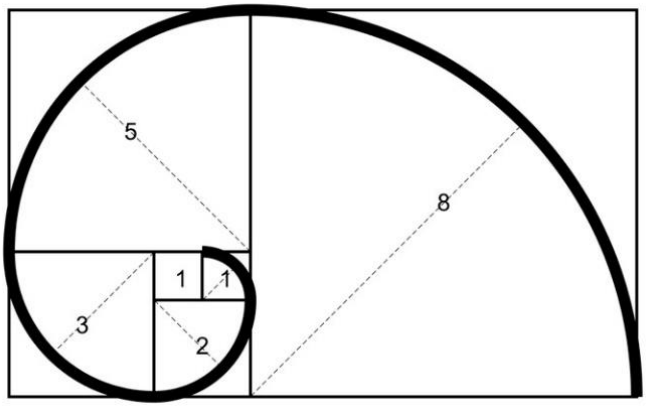


Figure 61 A: Fibonacci spiral composed of quarter-circle arcs inscribed within a series of squares, where the side lengths follow the Fibonacci sequence (Sautoy, 2022)

In the figure below, this Mashrabiya is one of the largest and most artistically complex screens in Bayt Al-Suhaymi, distinguished by its variable internal partitions. Each rectangular frame contains a series of distinct subdivisions, creating a sense of visual movement through elongation. This dynamic quality emerges from the variation in sizes and proportional spacing between individual elements, imparting a layered rhythm and spatial dynamism to the overall screen composition.

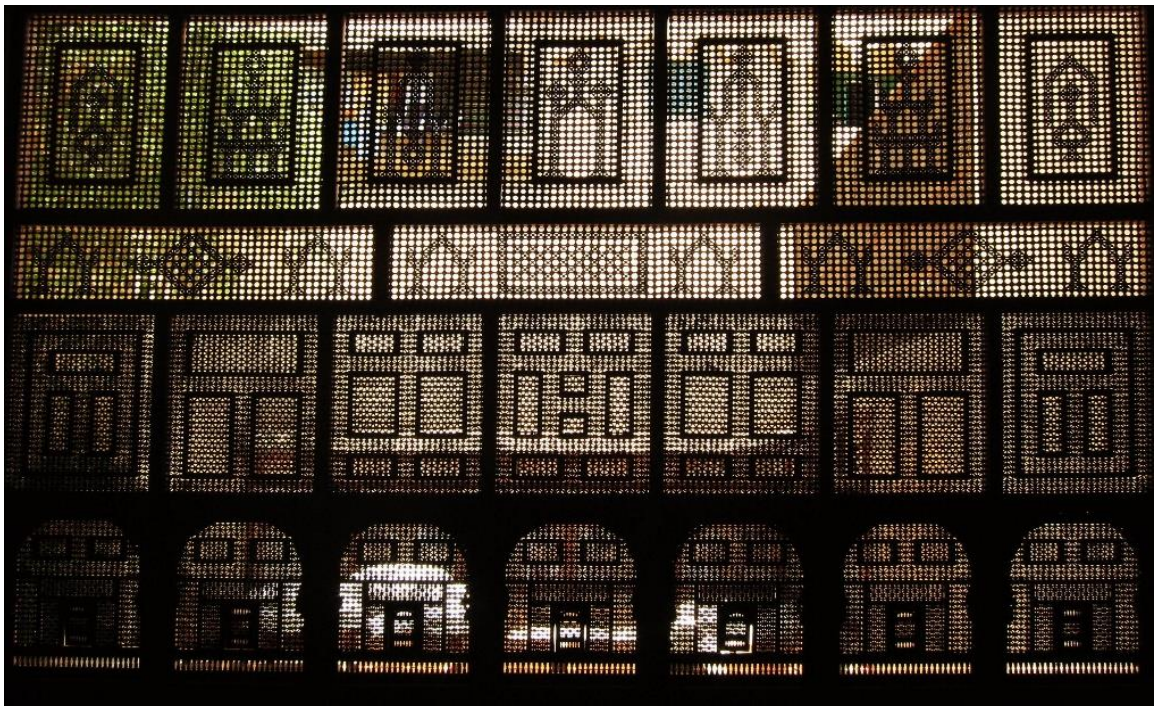


Figure 62. Main Mashrabiya in Bayt Al-Suhaymi (Hindi, 2011)

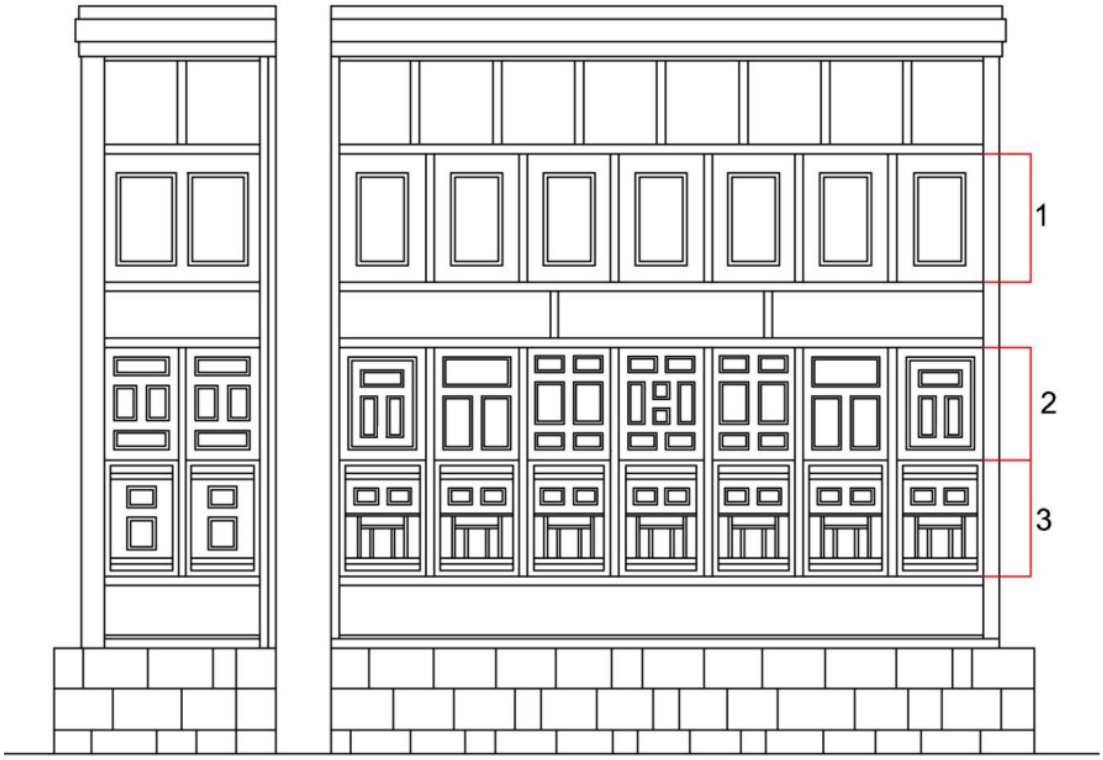


Figure 63: The Application of the Golden Ratio in Mashrabiya in Bayt Al-Suhaymi (Hindi, 2011)

Dr. Amani M. Hindi conducted research in the Faculty of Applied Arts at Damietta University, which provided insights into analysing this Mashrabiya. It concluded that this Mashrabiya example had the following ratios: 1.5, 1.6, 1.4, 1.7, 1.3, 1.8, and 1.1 between its parts and divisions. It was also found that in specific partitions, the golden ratio was achieved three times with minor deviations (Hindi, 2011). Mashrabiya is usually manufactured as a square aperture element, with a length roughly equal to its width. However, the above example did not preserve these ratios as a whole (Hindi, 2011). However, the elements chosen to be attached to form the generic figure of the screen have a closer 1:1 ratio in length and width (Hindi, 2011). The craftsmen resorted to setting up the Lego-lathed elements inside frames made of different-sized squares or rectangles, stacking them to create decorative shapes, and then manipulating the perforation percentage and the density of the mesh by adding various artistic forms—some of which now include shapes resembling niches, abstract geometric or botanical drawings. Some also featured inscriptions in square Kufic script, whose geometric nature complements the shapes of the wooden elements (Hindi, 2011).

It is also worth mentioning that this example was endangered due to being in a neglected historical site in the middle of medieval Cairo. According to the Supreme Council of Antiques' site documentation of Bayt Al-Suhaymi, although this building was not that old compared to the historical buildings in the area, it was in significantly poorer condition than the neighbouring structures, which might be attributed to the degeneration of construction skills in the Ottoman period to which this building

belongs (Antiques, 1997). However, in early 1997, the Supreme Council of Antiques of the Egyptian Ministry of Culture launched a renovation project for the site, collaborating with Egyptian researchers in different fields of expertise. One of the leading researchers who was in charge of renovating the traditional perforated screens in a way that maintains the same character and shape was Dr. Adham Nadeem, an expert researcher in traditional wooden lathed screens and methods of lathing wooden ornaments. He was also the CEO and founder of the Nadeem Foundation, which was in charge of renovating the majority of Cairo's historical Mashrabiya screens. (Antiques, 1997). Later, a detailed collaboration with the Nadeem Foundation will be explained in order to obtain the research samples for the analytical work of this research.

Figure 64: An archived image of Bayt Al-Suhaymi in 1931, the year the government acquired the essence of the house from private owners to declare it public property (Antiques, 1997)

Figure 65: Image taken inside the workshops during the renovation process documented by the council (Antiques, 1997)

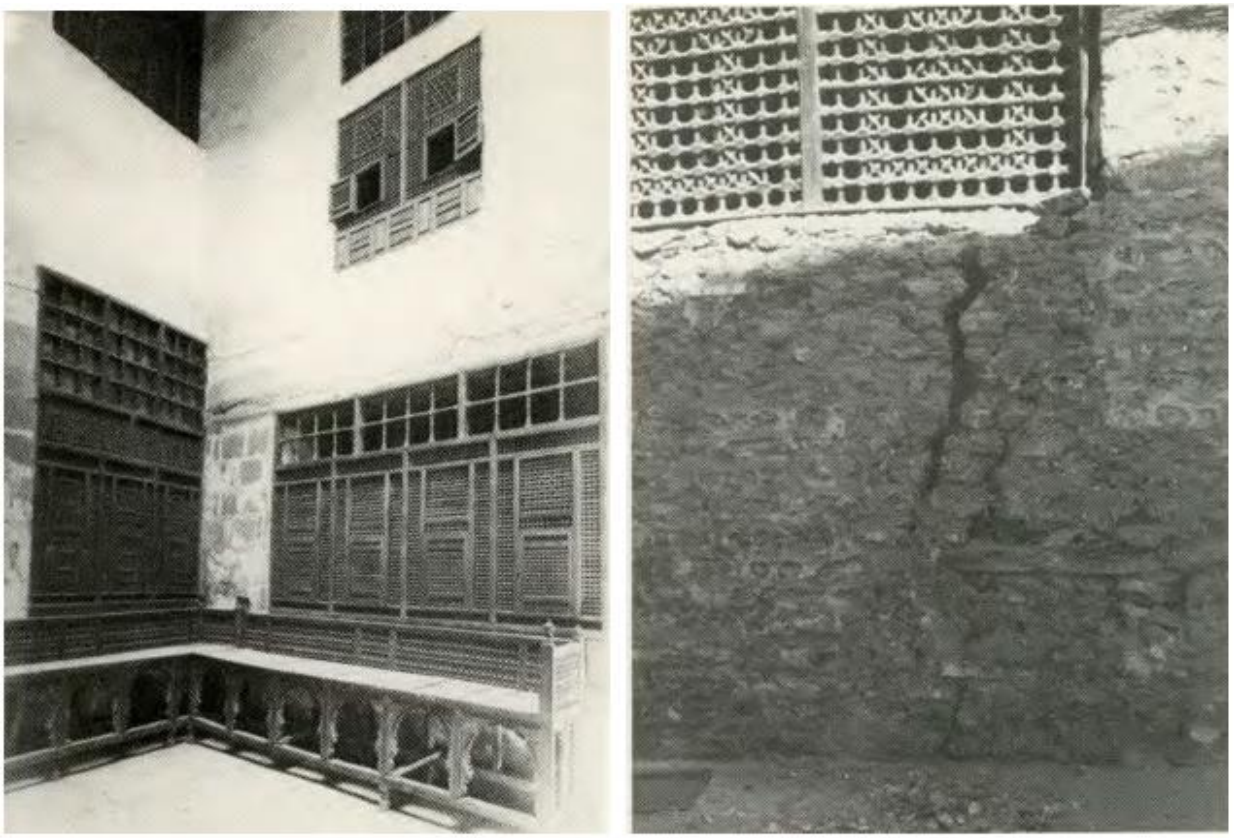


Figure 66: Archived image of the old screens inside the house before renovation (Antiques, 1997)

Mashrabiya is distinguished between two varieties of wood lathing: the first is small in size, with little space between the components and parts, and is called *Maymooni*; the second is distinguished by huge

barrels and considerable space between them and is called *Cistern lathed wood* (Gawad, 1986). Due to the fact that craftsmen used the interlocking technique rather than metal nails or glue, it was easier to renovate the screens by taking the parts to the craftsmen and replicating them, since there was no sufficient resource to help the researchers understand the manufacturing criteria, where the criteria for combining the elements and the components are assigned by favouring the efficient type.

From the previous example, we can conclude that the typology of the screens possesses its own categorisation methods, and by understanding these methods, we can select our samples based on what needs to be clarified through their analysis and what these samples will be useful for. In the next stage, the context and analysis matrix will be constructed to define the future steps and establish the outlines of the analytical procedure. Therefore, to comprehend the performance of Mashrabiya in terms of shading, ventilation, and evaporative cooling, it was necessary to first grasp the etymology of the patterns and the process of their formation. It was also critical to understand the dominant characteristics and the factors that initiated the prevalence of particular forms and screens, which became more popular and preferred over time.

B) Defining the Scale of Design

The regular size of windows in the Middle East (which is a typical case for hot, dry climatic regions) ranges between 120 – 300 cm, with 150 cm being the commonly used standard in windows for the application of Mashrabiya (Gawad, 1986). When a screen is applied to such windows, they are divided into 3-5 partitions; based on the size of each partition, the scale of the pattern is determined. This, in turn, helps define the size of the elements and the spacing between them, in order to maintain the perforation percentage, which ranges between 25% and 47 % (1997 ,محمد).

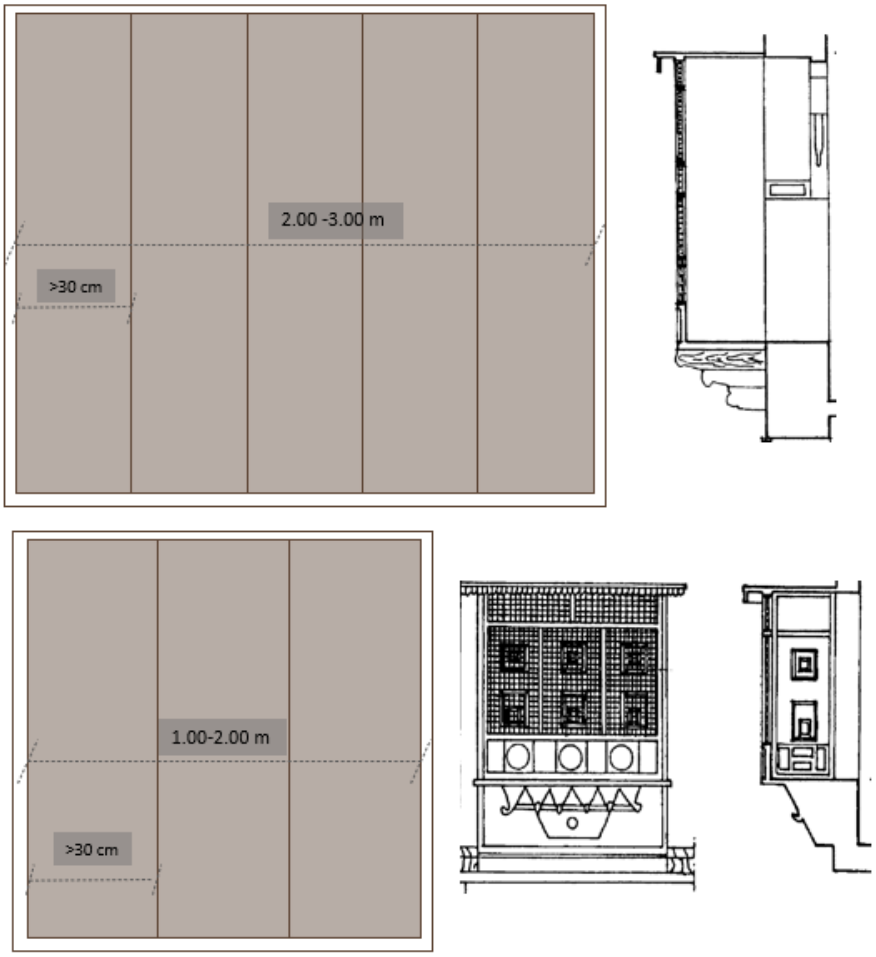


Figure 67: An illustration of the standard dimensions of the domestic aperture area and how it was patronised in the case of applying Mashrabiya – (Author)

Designing a perforated screen for shading an aperture, as in the case of Mashrabiya, requires dividing the design stage into different levels and phases, where the designer analyses the area to which the screen will be applied and the type of screen to be implemented. Consequently, the typologies of the screen will indicate the ratio between the element and the percentage of perforation in the screens. Some screens in the Maymooni family have different components, which may include five or more distinct elements arranged to form a unit that is repeated across the modular grid system of the screens, as shown in the following example.

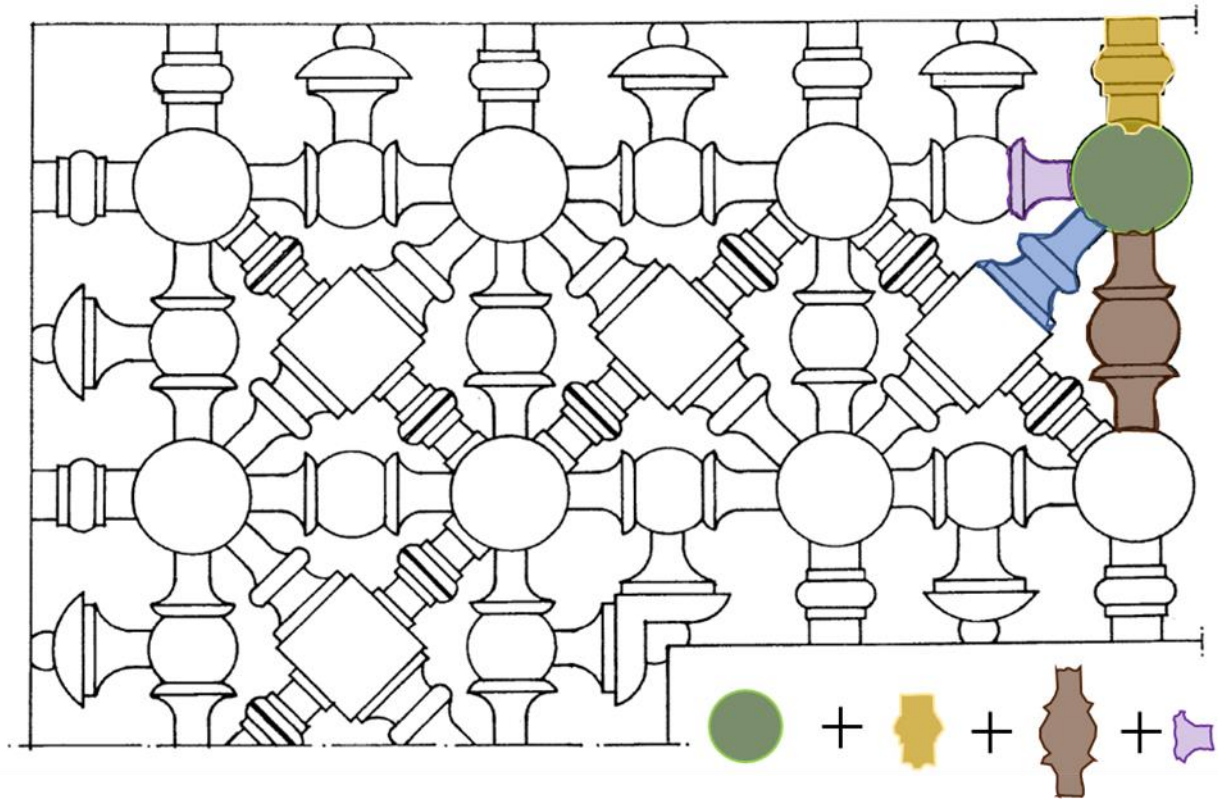


Figure 68: Illustration that explains how the Maymooni pattern can be composed of more than four elements that contribute to the total shape of the screen- (Author)

Accordingly, the components are distributed and positioned based on the central axis of the screen; these axes help define the elements' rotation angle and the spacing distances between the components. The efficiency of these stages is rooted in the fact that each stage complements the others, contributing to the outcome of a perfect interlocked screen, where the size of each component is related to the size of the elements forming it, which in turn corresponds to a fixed ratio that ultimately determines the spacing between the axis based of the aperture dimension, as shown in the following illustration.

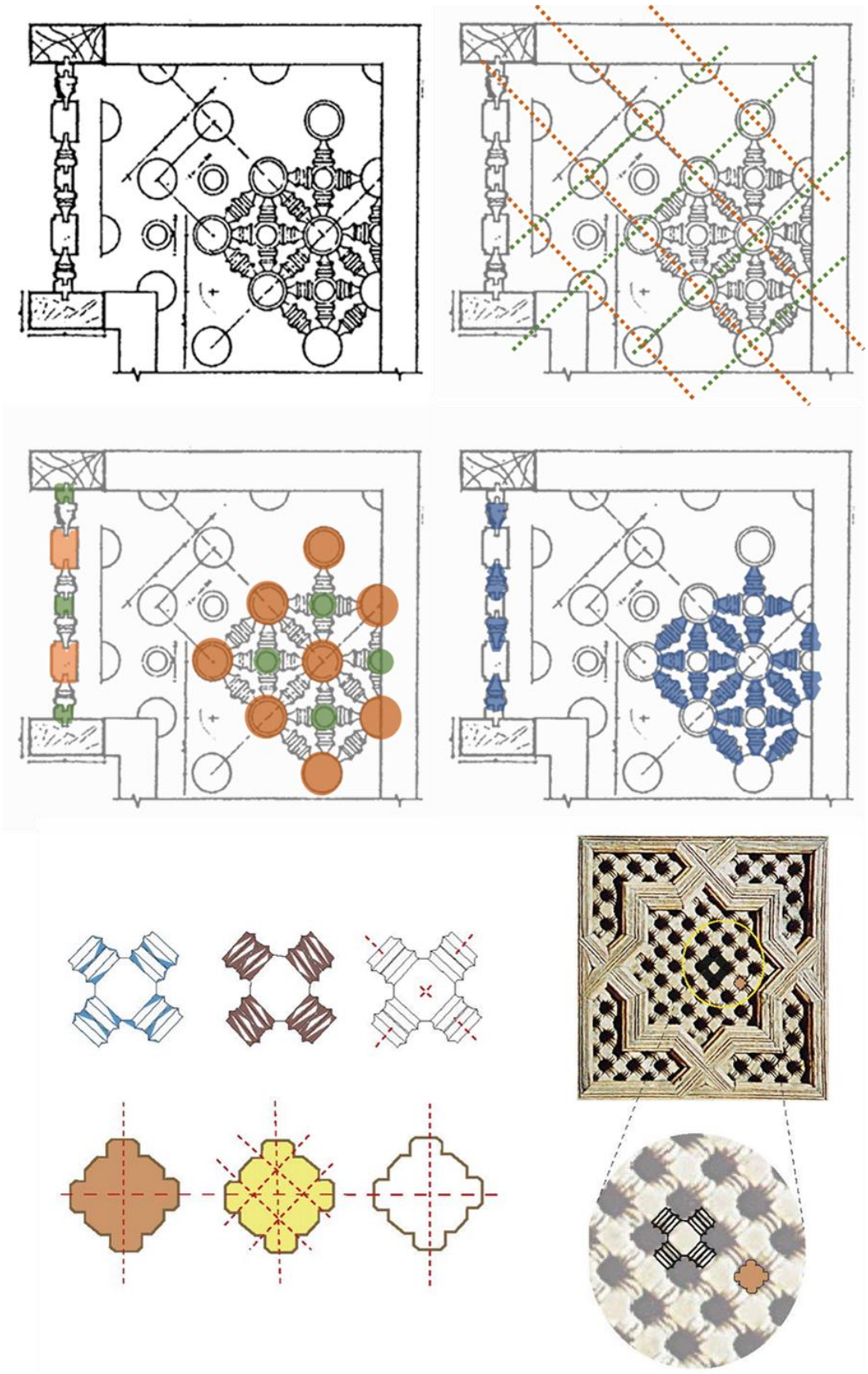


Figure 69: Illustration of how the small components are attached and combined along the drawn axis to form the perforated screens - (Author)

The simplicity of regular polygons facilitates the use of diagonal axes in the design process, which forms the foundation of the beauty of proportions. However, this theory does not fully explain why the elements that form these components are carved and lathed with such particular detail. Theoretically, for efficiency and simple manufacturing procedures, the elements could have retained a basic shape, and craftsmen would have avoided such complex detailing to accommodate mass production. On the other hand, the absence of sources covering this subject indicates a knowledge gap and introduces certain restrictions to the research. Nevertheless, by addressing these gaps, this research holds the potential to significantly advance our understanding of traditional Mashrabiya architecture.

Finally, we can conclude that the size and scale of the screen are determined by the number of partitions into which the aperture is divided, where the shading system is applied. This practical understanding allows for the determination of the preferred testing scale, which must closely reflect the scale of the design proposed by the research. As a result, a square with 30 cm dimensions was selected to accommodate various aperture sizes and partitioning methods. This scale was also determined using the following formula:

Considering that the common aperture length ranges between 120 cm and 300 cm, and that this aperture was commonly divided into 3-5 sections in the design of traditional Mashrabiya:

The Equation to Determine the Scale of Analysis and Design – Source: (Author)

$$\frac{l}{y} = x + f$$

$$length\ of\ aperture\ window(l) \div the\ number\ of\ divisions(y)$$

$$= length\ of\ the\ screen\ that\ will\ be\ applied(x) + frame\ factor\ (0 \sim 2.5\ cm)$$

Where:

- l = length of aperture window
- y = number of divisions
- x = length of the screen that will be applied
- f = frame factor, ranging from 0 to 2.5 cm

The framing factor considers different case scenarios. A frame is usually added to surround the screen, eventually contributing to its overall length, which usually does not exceed the dimension of a regular wooden frame—2.5 cm (Hammoda, 1981). Several taxonomies for the framing factor have been developed in different calculations, where the framing factor can obstruct light or air coming through the aperture (Hammoda, 1981). If this equation is followed with different lengths, as in the following calculations, we can find that the common range that can be used is a 30 cm-dimensioned screen.

Equation 8: Calculations made to determine the scale of analysis and design

$$120\ cm\ window\ \div 3\ parts = 40\ cm + frame\ factor$$
$$200\ cm\ window\ \div 5\ parts = 40\ cm + frame\ factor$$
$$200\ cm\ window\ \div 3\ parts = 66\ (33 + 33)\ cm + frame\ factor$$
$$150\ cm\ window\ \div 3\ parts = 50\ cm + frame\ factor$$
$$300\ cm\ window\ \div 5\ parts = 60\ (30 + 30)\ cm + frame\ factor$$
$$150\ cm\ window\ \div 5\ parts = 30\ cm + frame\ factor$$

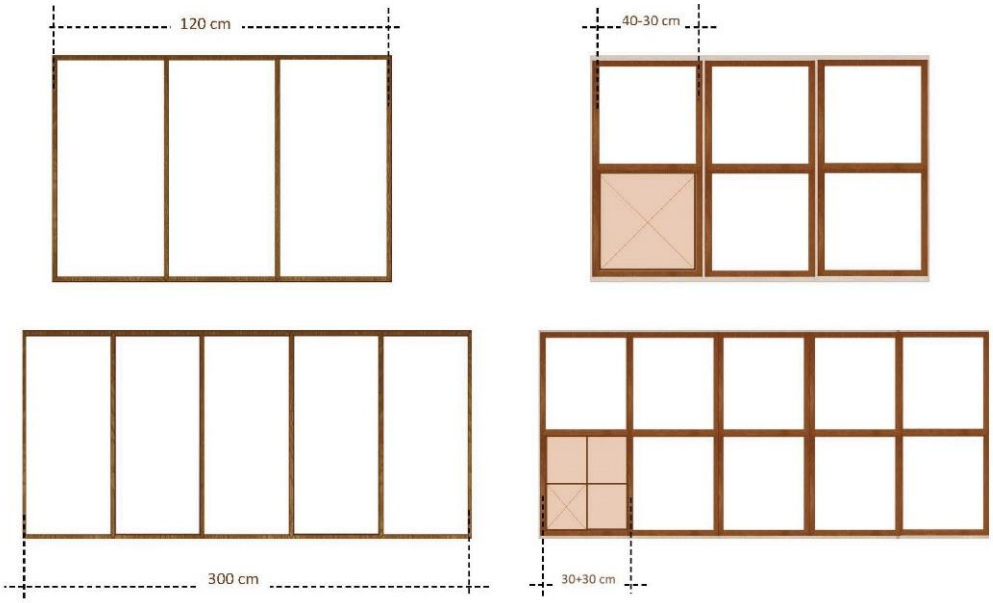


Figure 70: Illustration showing how the windows are divided using 30cm screens – Source: (Author)

C) Setting the Context

To understand the typologies of the patterns and the design features of the wooden-lathed screens, it was important to highlight a fixed context in which the screens predominantly exist and respond to the climatic conditions of that context. Consequently, it was necessary to locate the precedence of Maymooni screens across different Mashrabiya window designs worldwide.

The *Maymooni* screens are named after the Maymooni district in Cairo, Egypt, referring to the craftsmen of this area who were renowned for crafting these screens using traditional wood-lathing methods and techniques (Gawad, 1986). Therefore, choosing Cairo as the fixed context was more logical, since the majority of documented *Maymooni* examples are found there.

D) Choosing Sites in the Context

Choosing the context is a crucial stage in this research; it is important to highlight the location of the screens to facilitate the selection of the samples that will be analysed. It is also necessary to specify the

context in order to examine the environmental performance of the screen within the climatic conditions specific to that region, allowing for a realistic correlation between the performance of traditional screens and their geographic setting.

Cairo, the capital of Egypt and the cradle of Maymooni screens, was selected as the context due to its status as the origin of traditionally manufactured Mashrabiya screens. An even more compelling reason for selecting Cairo as the fixed frame of reference for this analytical study is its Hot-Dry climate, which served as a primary incentive for implementing passive cooling strategies in Middle Eastern vernacular architecture. This climatic condition presents an empowering opportunity for recommending passive evaporative cooling strategies with promising efficiency. Consequently, it is critical to comprehend the climatic conditions in Cairo throughout the year and to highlight the key features that can influence the environmental analytical studies of the screens.

The latitude of Cairo is 30°–31.5° N, placing it at a transitional point between Africa and Asia, within the geographical bounds of both continents. Cairo occupies a crucial location in Egypt, where the Delta region converges with Upper Egypt— the southern portion of the country. The general climate of Egypt is relatively wet and cool in winter (October to March), and dry and hot in summer (April to September) (حمدان, 1996).

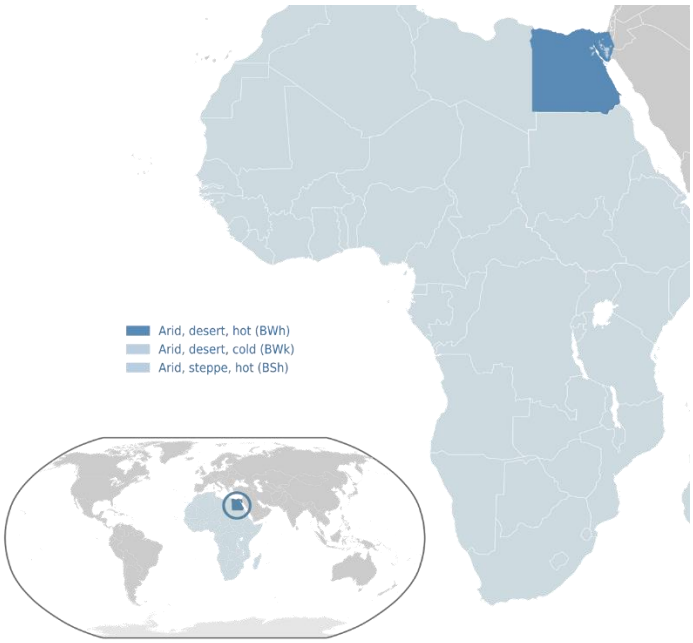
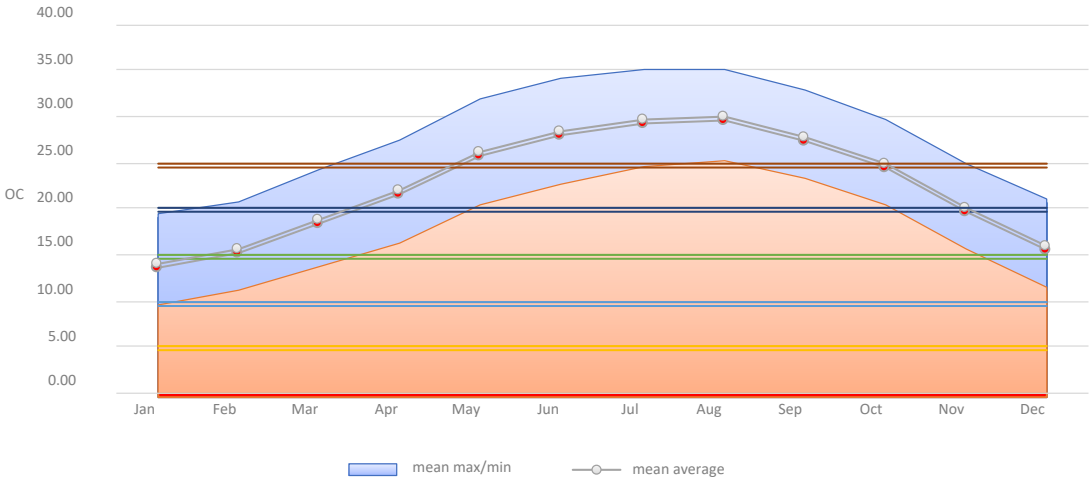


Figure 71: *Köppen-Geiger climate classification map for Egypt (1980-2016), overlaid on the country's geographical location – Source:* (Author)

Meteonorm is software that provides climatic data based on irradiation and historically monitored records from 8,320 weather stations worldwide. The software was used to extract the most recent monitored weather data for Cairo, which dates back to 2020. This data was later analysed and transformed into accessible graphs that illustrate the main climatic features in Cairo through the year's seasonal transitions (Meteonorm, 2022).

Figure 2: Monthly Average Dry Bulb Temperature for Cairo, Based on Meteonorm Data



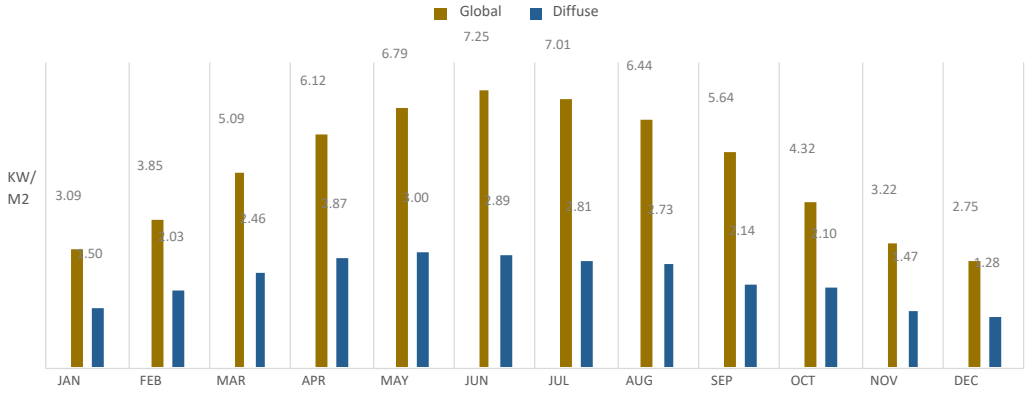


Figure 2: Monthly Average Global Vertical Radiation for Cairo, Based on Meteonorm Data

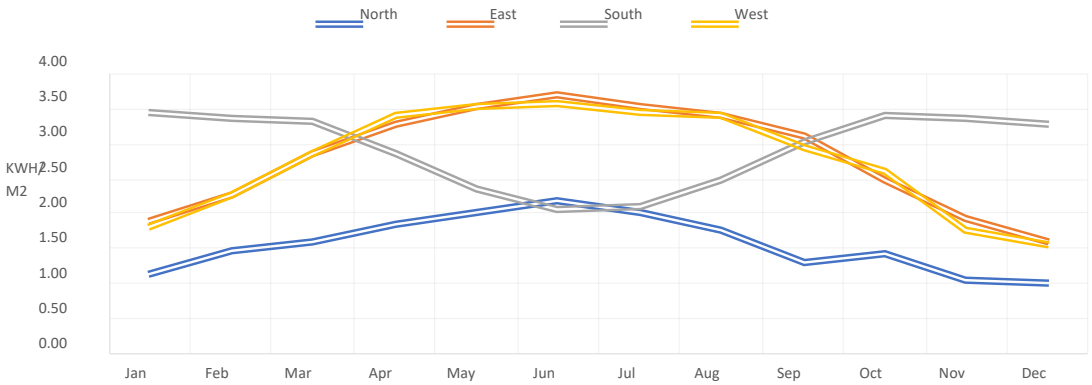


Figure 72: Monthly Average Global Horizontal Radiation for Cairo, Based on Meteonorm 2020 Data (Author)

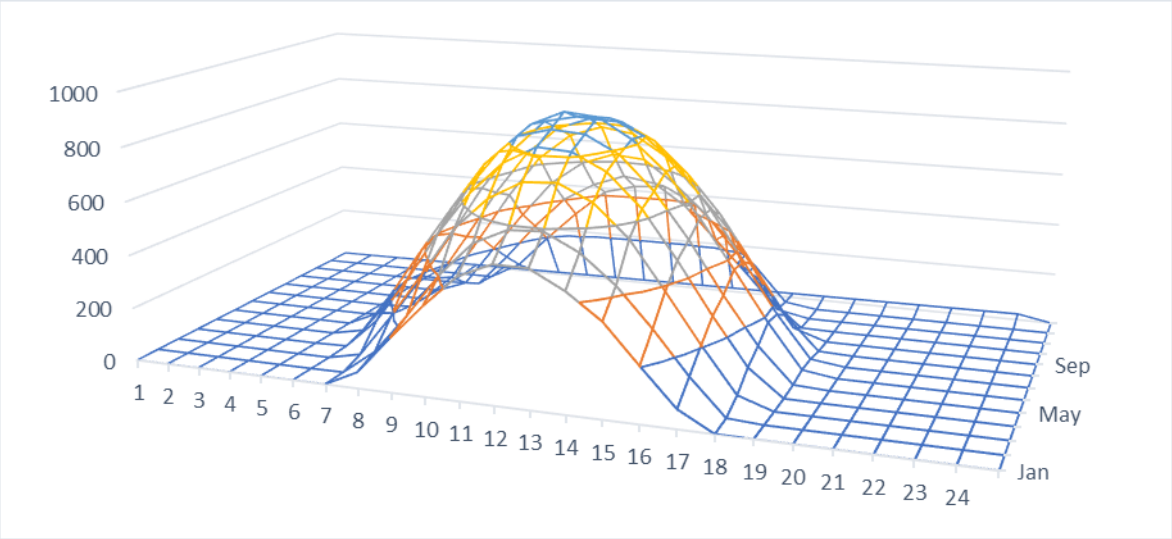


Figure 73: *Daily Average Global Horizontal Radiation for Cairo, Based on Meteonorm Data*

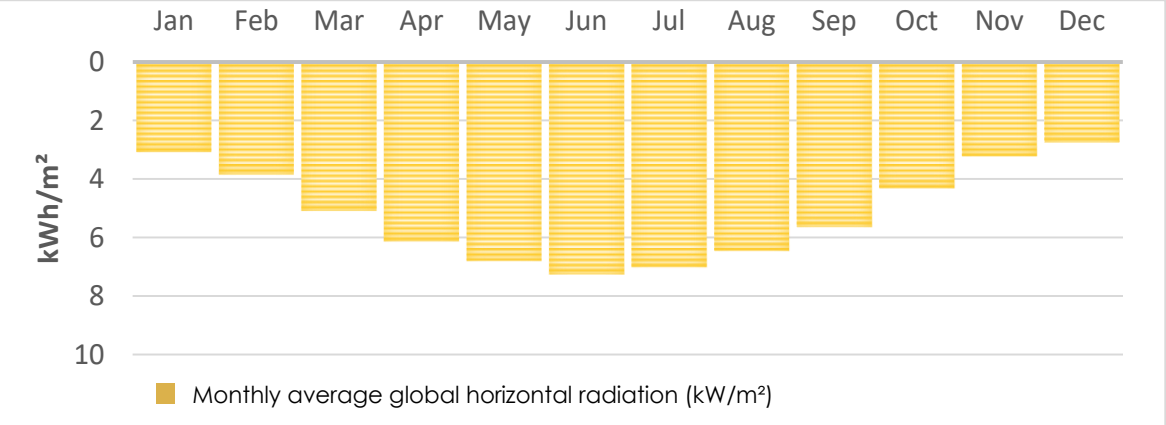
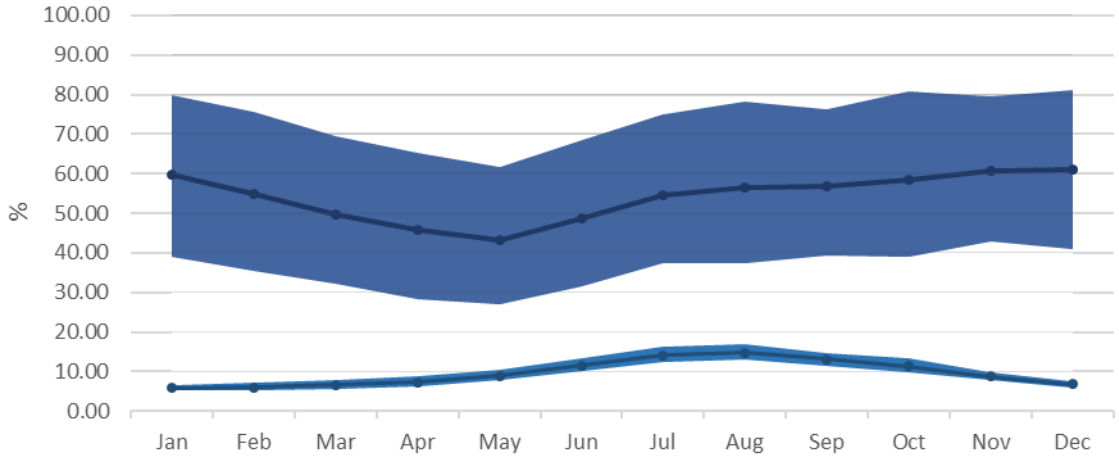


Figure 74i: *Monthly Average Global* Horizontal *Radiation for Cairo, Based on Meteonorm Data*

Figure 75: *Monthly Average Relative and A*



bsolute Humidity for Cairo, Based on Meteonorm Data – Source: (Author)

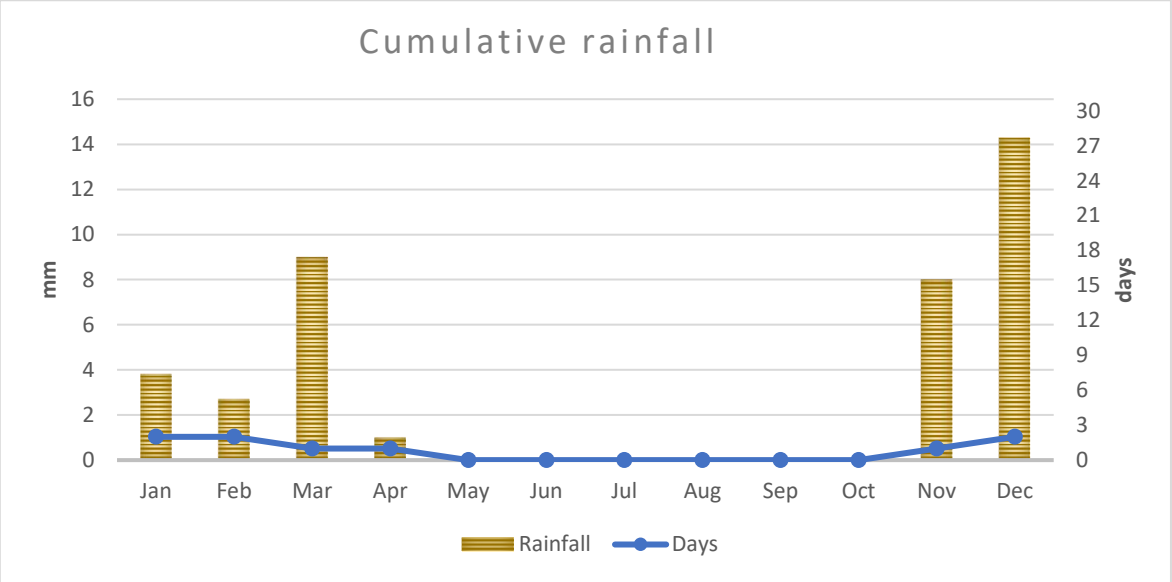


Figure 76: *Cumulative Monthly Rainfall Data for Egypt, Based on Meteonorm Records– Source:* (Author)

The above graphs demonstrate that a building façade in Cairo requires shading strategies for most of the year, particularly on the eastern and western façade sides, where solar exposure exceeds 3,000 kWh/m^2. This insight helps identify the optimal orientation of the screen to be tested in the solar radiation analysis.

The efficiency of shading strategies depends on maximising their success rate in blocking undesired solar angles. In hot-dry climates, solar energy—being the primary drive behind the implementation of shading systems—varies in orientation throughout the year. This seasonal fluctuation complicates the design of a shading solution that can perform efficiently across multiple orientations and solar angles. Therefore, the main objective of developing the methodology for analysing the shading performance of the screens in different orientations is to enhance the level of experimentation and exploration, ultimately maximising the effective application of the proposed strategy under various climatic conditions.

Accordingly, the screens will be tested across eight different orientations to estimate and quantify their shading performance under various case scenarios.

The average relative humidity in Cairo remains below 60% for most of the year and drops below 50% from March to June. This reduction is primarily due to decreased cumulative rainfall during these months, which typically contributes to the air's moisture content.

E) Choosing Screens from the Site

There are various shapes and geometries of the screens used in Mashrabiya, which vary from one place to another according to the artistic and architectural requirements, as well as regional

preferences. This diversity allowed artists and craftsmen to cultivate a distinctive identity in crafting each pattern, enabling them to build the artistic and scientific knowledge required to design these patterns across different scales. As a result, this engineering technique developed various ratios that promoted the implementation of these patterns at multiple scales. This correlation has been clarified in the texts of Yahya Bin Ismail bin Abbas al-Bouzjani, whose book explains the relationship between the manufacturing process and the engineering principles that underpin (IQBAL, 2007).

The patterns will be selected from the Maymooni screen family to assess how their detailed craftsmanship influences overall environmental performance in terms of shading and ventilation. Accordingly, the patterns will be chosen from Cairo, the cradle of the Maymooni style, where numerous traditional examples remain accessible. The components of the Maymooni screens were manufactured separately and assembled in a way that creates a sculpted three-dimensional texture. This technique produces varied levels of extrusion and depth across the screen patterns, as illustrated in the following diagram (1997 ,محمد).

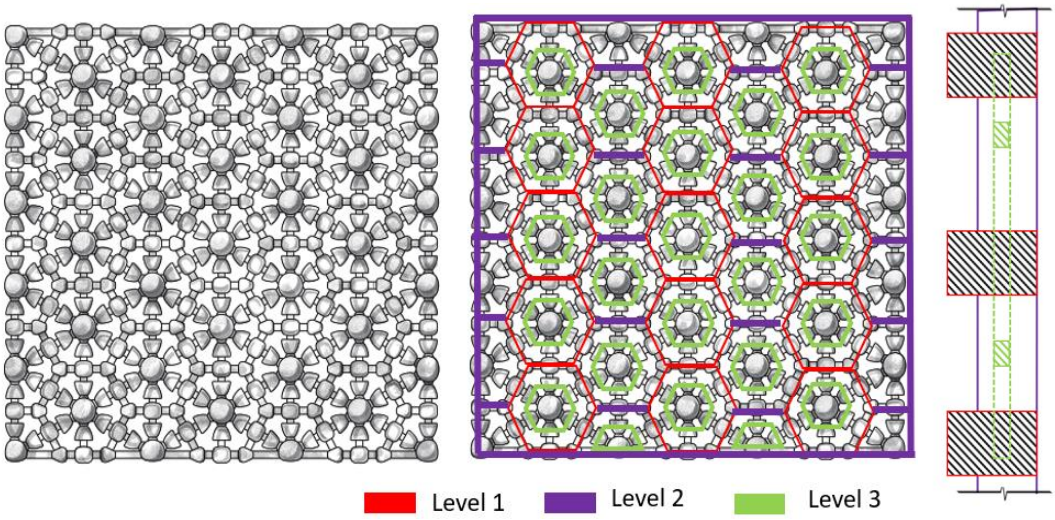


Figure 77. Diagram illustrating the sequential layering of elements after application, with a sectional view showing the variation in extrusion depth (Author)

Another valid reason for selecting samples from traditional examples in Cairo is the established connection with the Nadeem foundation, which was responsible for the restoration of the traditional perforated screens in Bayt Al-Suhaymi, previously referenced in this study. The foundation will support the research by providing access to data and facilitating meetings with local craftsmen to better understand the methodologies behind traditional manufacturing techniques. Additionally, the foundation will supply the research with the necessary physical samples.

Based on the afore-mentioned factors, 7 to 10 screens from the Maymooni pattern family were selected to represent the full range of traditional shapes and ratios. Since the screen size was previously standardised at 30 cm x 30 cm, all samples will conform to this dimension. Each pattern will be carefully

verified to maintain its original ratio, as commonly observed in the traditional examples from which it was derived. Even if the actual application size of the screens differs or exceeds the 30 cm format, the proportions and element dimensions must remain consistent with the original configurations.

The ratio was one of the primary factors influencing the selection of the samples. While it was relatively straightforward to identify commonly used ratios in the design of Islamic patterns and screens through various sources, it proved challenging to locate actual screen examples that applied these ratios. Therefore, it is essential to identify traditional, existing examples from well-documented sites so that this information can serve as a reference for the craftsmen preparing the samples. Such referential data will be crucial for ensuring the accuracy of the samples and their alignment with historical precedents.

Accordingly, the following ratios were examined in common examples of traditional Mashrabiya screens found across the following sites:

1. Bayt[2] Al-Suhaymi, Cairo, Egypt – An Ottoman-era house originally built in 1648 and expanded in 1796. The building underwent full restoration in early 1997, supported by the Nadeem foundation, to rehabilitate the traditional Mashrabiya screens originally installed on site (Antiques, 1997).

2. Bayt Al-Razzaz Palace, Cairo, Egypt – the palace was first built in the 15th century during the Mamluk era, featuring multiple Mashrabiya components with highly decorative handcrafted screens.

3. Bayt Zeinab Khatun, Cairo, Egypt—Originally constructed in 1468 and expanded in 1713, this Mamluk-era residence is located in the heart of medieval Cairo, near Al-Azhar Mosque.

[2] "Bayt" is a common term in Egyptian Arabic that refers to a personal house or home—specifically, the place where one spends the night (house, 2001).

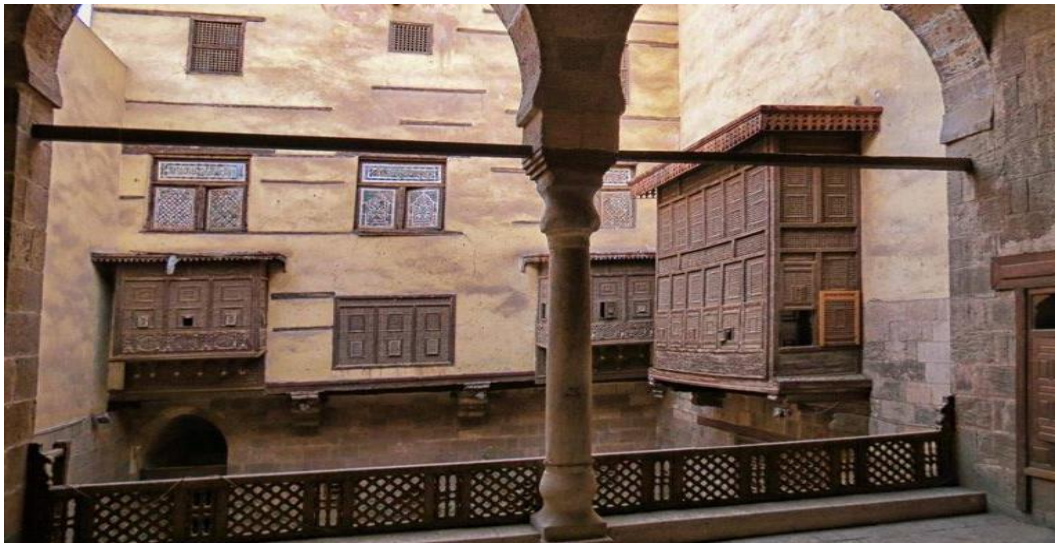


Figure 78. View inside the courtyard of Bayt Zaynab Khaton (eats, 2022)

4. Wikala Al-Ghuri—A caravanserai constructed during the Mamluk era for Sultan Qansuh Al Ghuri, serving as an international centre for commerce. Located in the heart of medieval Cairo on Tablita Street, near al-Azhar Street.

Figure 79: Interior View of Wikala Al- Ghuri (Aga Khan Trust for Culture, 2013)

5. Manzil[3] al-Kiridliya (Gayer-Anderson House, "front"), located adjacent to the Mosque of Ibn Tulun on Ibn Tulun Street.

[3] "Manzil" is an Arabic term that refers to a group of personal residences within the same building or a multi-family house (house, 2001)

6. House of Ali Effendi Labib in Cairo – originally built in the late 12th century and located in the Al Darb Al Ahmar district in medieval Cairo. The house is renovated through time from the remnants of Tekiyat Taqy El-Deen El-Bistami, a late Mamluk[4]-era structure.

Figure 80: Interior view of the Mashrabiya inside Manzil Ali Labib. (house, 2001)

7. Bayt al-Kritliya in Cairo – Originally constructed in 1631, now named "Gayer-Anderson Museum," within the historic Tulunid[5] capital of Al-Qata'I. It was built using materials salvaged from the surrounding area, including tiles from the Mokattam cliffs. Located in the south-eastern side of the Mosque of Ibn Tulun in the Sayyida Zeinab district, the building exemplifies 17th-century domestic architecture of the medieval Tulunid Cairo.

[4] The Mamluk period refers to a significant era in Islamic civilization in Cairo, named after the Mamluk military caste who ruled Egypt from the mid-13 century until the early 16th century, when the Ottoman Empire assumed control (1996 ,حمدان).

[5] The Tulunid period refers to a dynastic era in Islamic Egypt named after Ahmad ibn Tulun, who declared independence from the Abbasid Caliphate in 868 CE. The Tulunids maintained autonomous rule until 905 CE, establishing Egypt's first independent state under Ahmad ibn Tulun's leadership. (1996 ,حمدان).

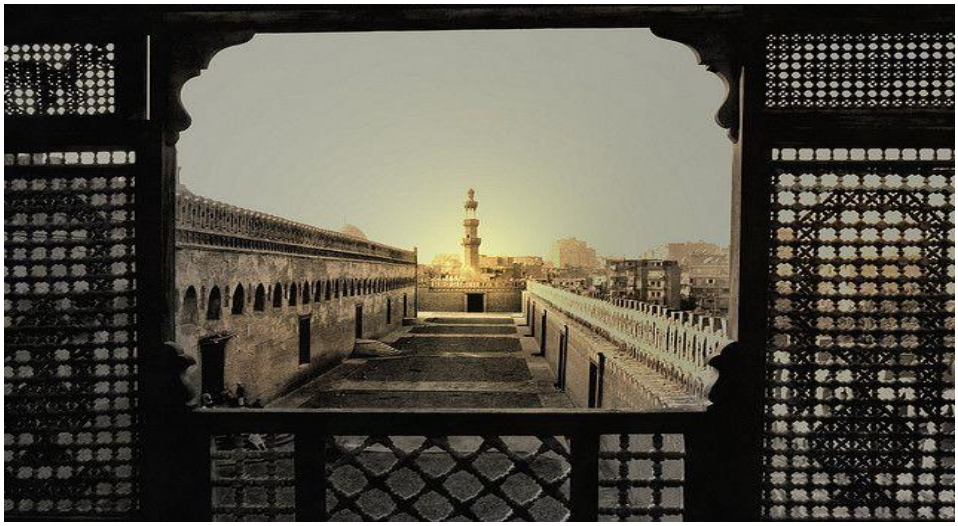


Figure 81: Interior view of *the Gayer-Anderson Museum- Source: (tours, 2021)*

8. Prince Mohammed Ali Tewfik Palace in Cairo – Constructed in the early 20th century, this relatively modern residence does not belong to the historic periods during which Mashrabiya screens were widely developed or implemented. However, the palace features refined lathed wooden Mashrabiya screens, primarily installed in the women's guest quarters. These screens were crafted based on historical Ottoman architectural drawings and assembled from salvaged elements, as documented in the Egyptian Ministry of Monuments' governmental report (tourism, 2022). The palace, now known as the Manial Palace Museum, reflects a synthesis of Islamic architectural styles and serves as a repository of cultural heritage.

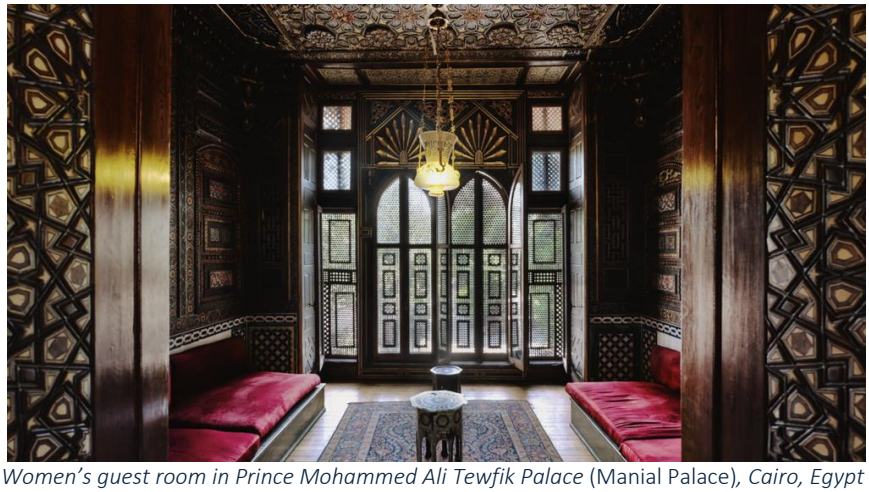


Figure 82: *The Women's guest room in Prince Mohammed Ali Tewfik Palace* (Manial Palace)*, Cairo, Egypt (1899 - 1929) – Source: (tours, 2021)*

F) Illustrating the Ratios of Each Screen

Following the selection of sites, the patterns identified at each location were examined in terms of the ratios they tend to manifest in the dimensional configuration of their constituent elements. These

ratios, which govern the geometric logic of the components forming the screens, were systematically documented. The rationale for investigating these ratios lies in their pivotal role in the construction of Islamic patterns and conventional Mashrabiya screens. Understanding and categorising the ratios aids in filtering out redundant findings, thereby streamlining the analytical process and minimising delays caused by unnecessary repetition.

Each of the previously mentioned sites may contain more than ten distinct Mashrabiya screens, varying in shape and form. While some screens may appear identical to the naked eye, subtle differences in detailing and variations in their small components often distinguish them from one another. These nuanced distinctions may result in different nomenclatures for screens that are otherwise visually similar, as illustrated in the following example.

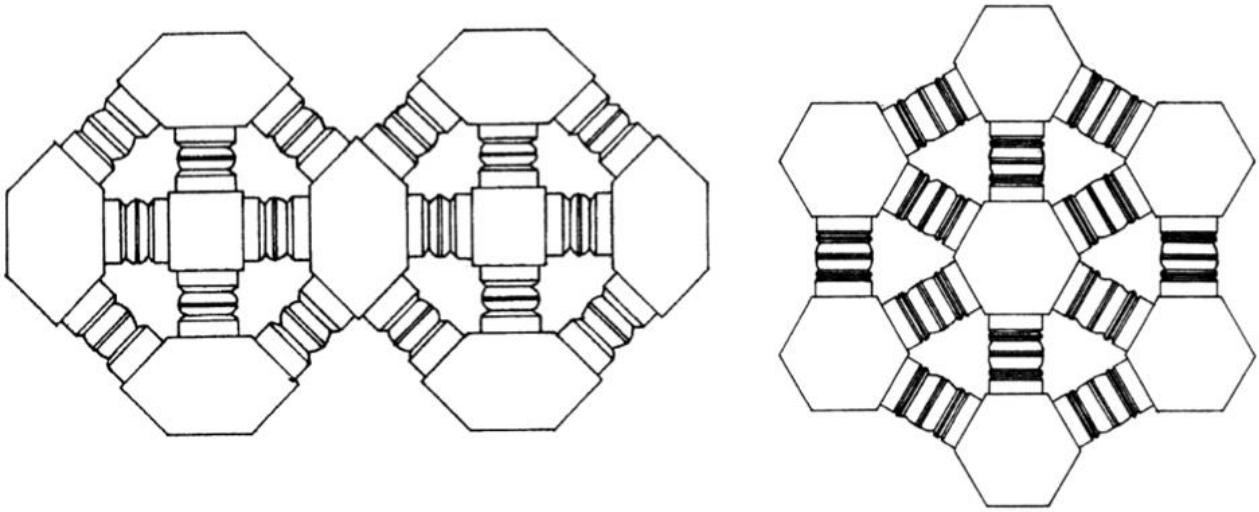


Figure 83: Examples of different designs from the Maymooni screen family, featuring similarly styled small elements with subtle variations (محمد, 1997)

Accordingly, each site will be investigated through targeted research, and virtual images of the existing screens will be analysed to determine the proportional ratios of each pattern. Although the traditional screen examples will be examined through imagery rather than physical measurement, the focus of the analyses is on understanding the underlying ratios. This approach does not require precisely scaled images, as it relies on relative proportions. The primary tools for this analytical process will be Adobe Photoshop and Illustrator, as demonstrated in the following example.

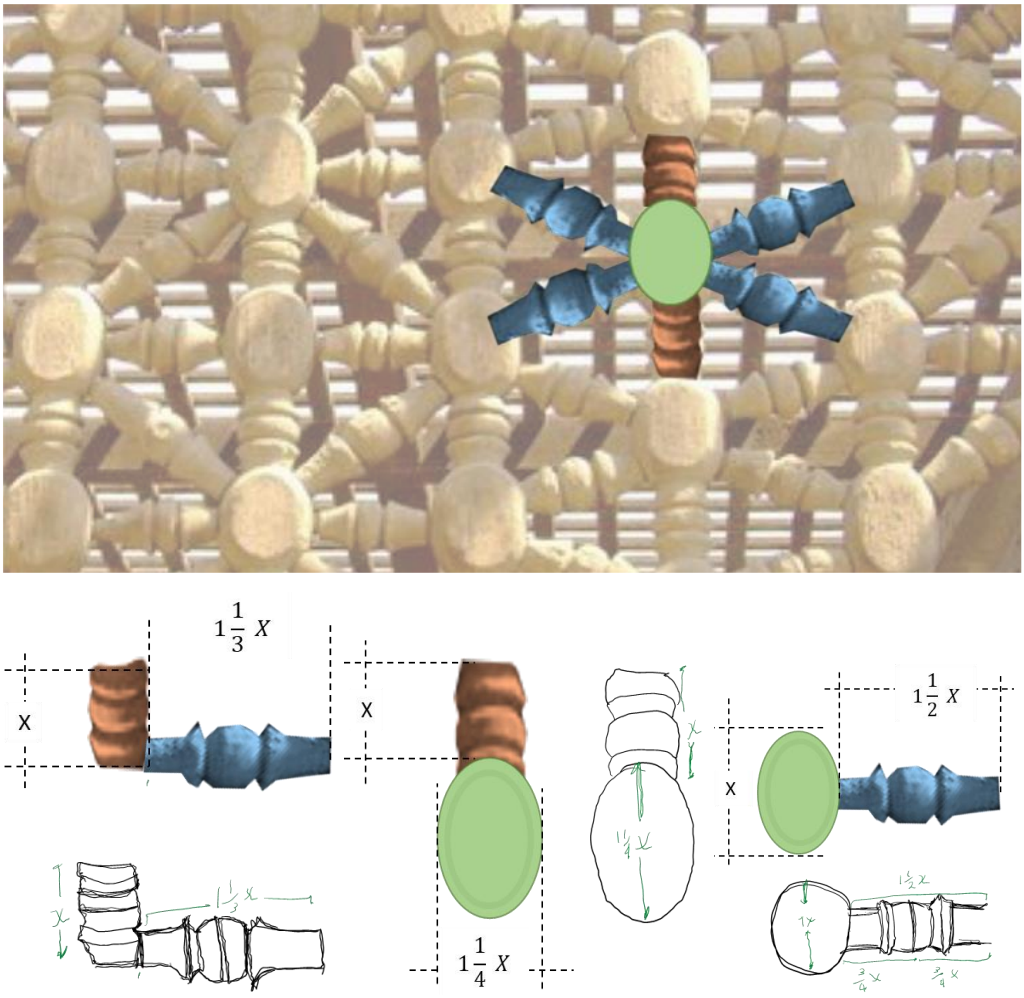


Figure 84: *Illustration showing how the dimensions of individual screen components can be expressed as proportional ratios-* (Author)

G) Finalise the Chosen Samples Based on Ratio Variation

The selected samples will be fabricated in workshops specialising in the renovation of traditional perforated screens in Cairo. Each workshop will be instructed to produce wooden samples measuring 30 cm x 30 cm, with particular attention to the fine details of each screen's components.

Once the physical samples are completed, the patterns will undergo three-dimensional scanning to capture and digitally model their intricate features. These 3D virtual models will subsequently be used within analytical platforms to evaluate the environmental performance of each screen design.

The 3D virtual model of the screens will be tested according to different environmental matrices in order to understand the variation of performance and the impact of the small details on the total performance.

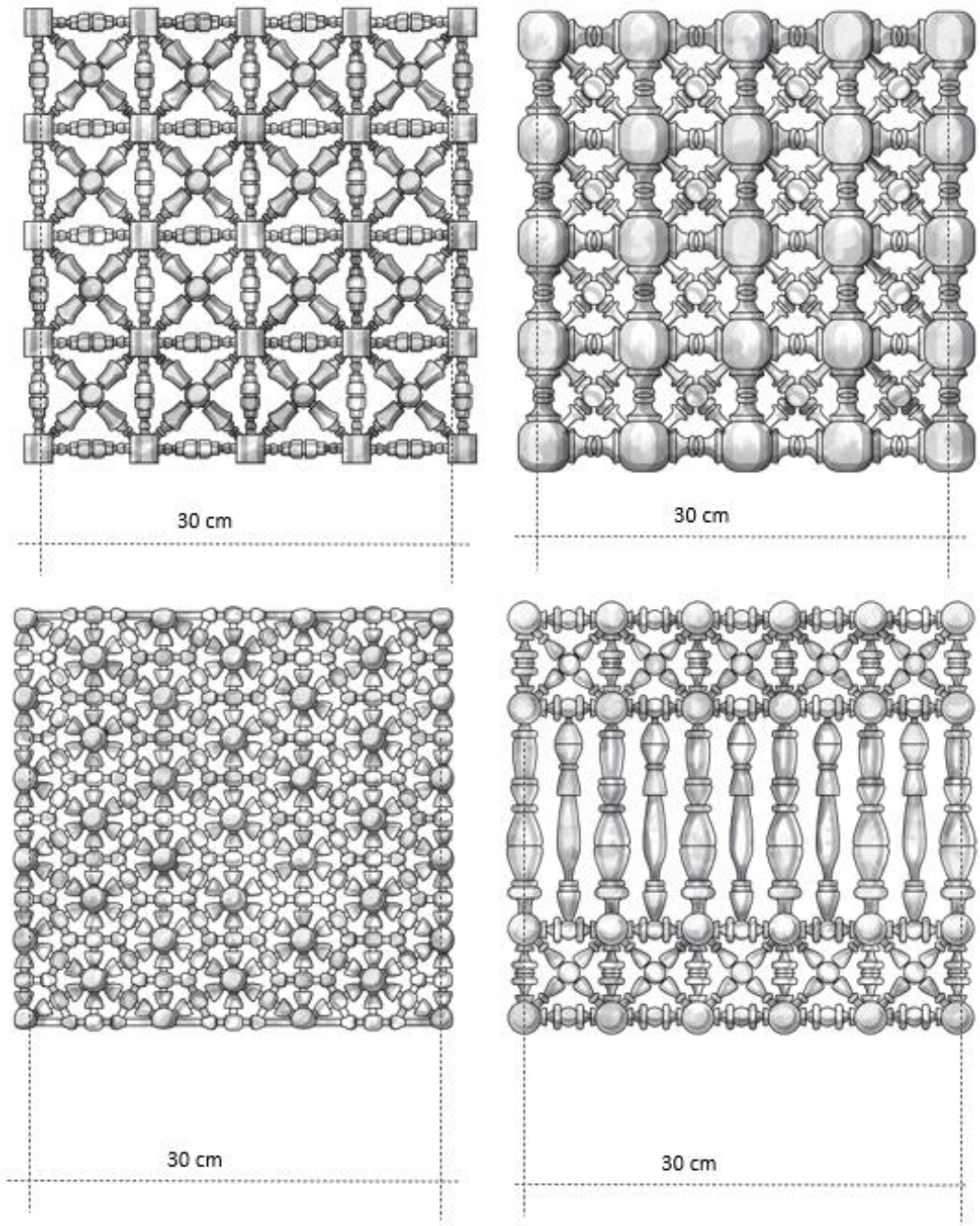


Figure 85: Illustration of the dimensional standards established for the screen samples, showing expected proportions and structural detailing (Author)

6.3 Samples Portfolio

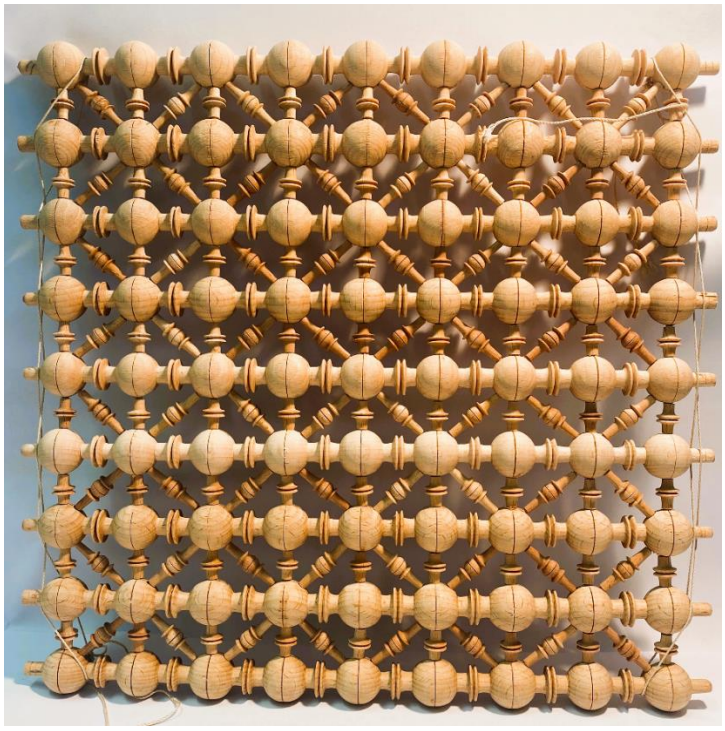


Figure 86: *Close-up image of sample 3-* (Author)

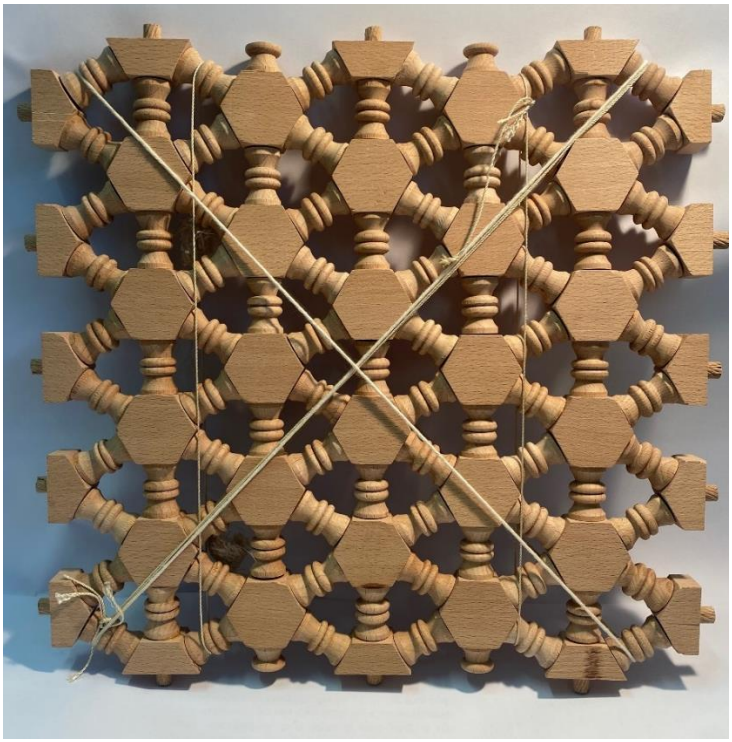


Figure 87: *Close-up image of sample 9-Source:* (Author)

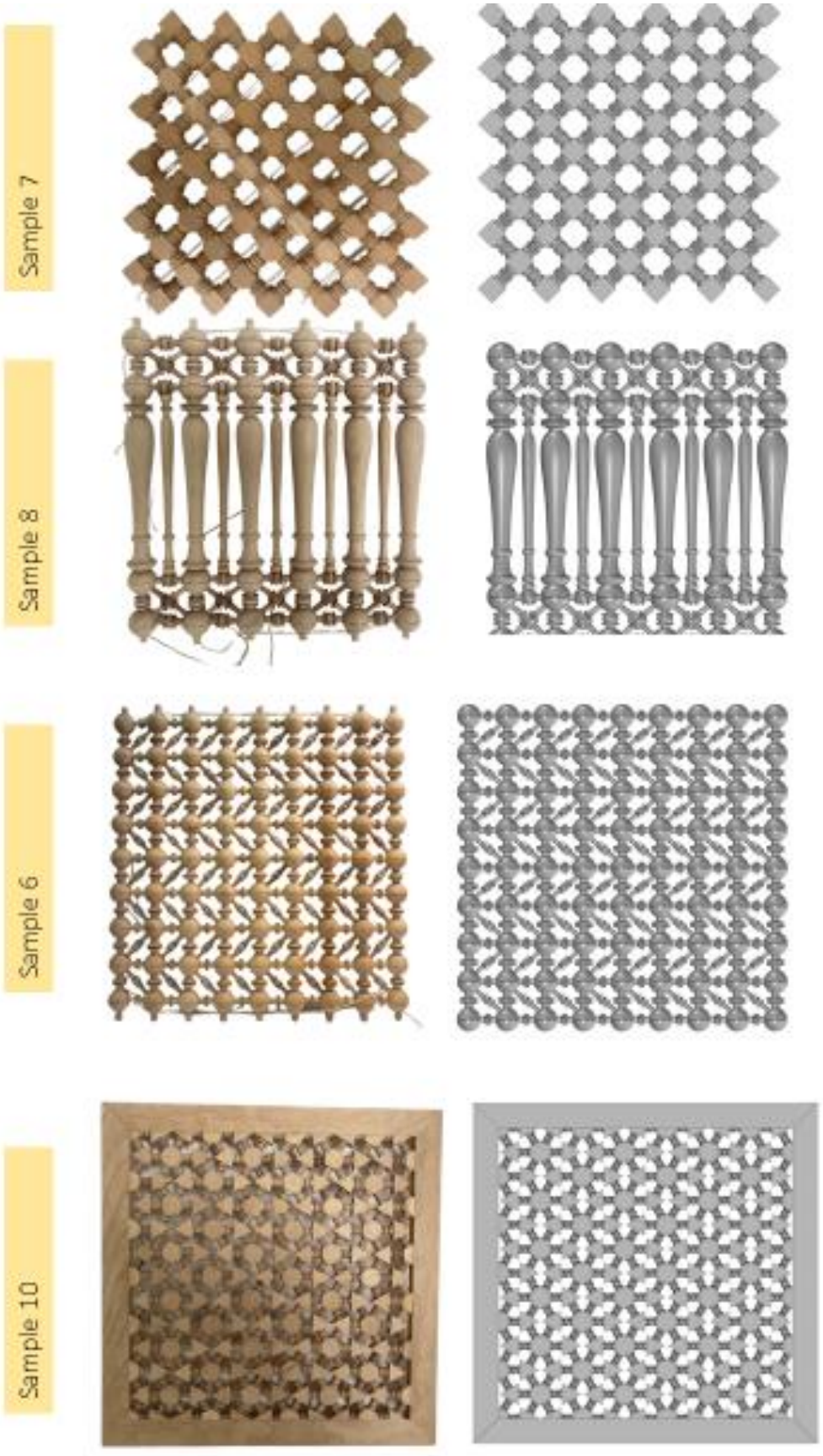


Figure 88: *Catalogue of sample images and corresponding virtual geometries following 3d scanning -Source:*(Author)

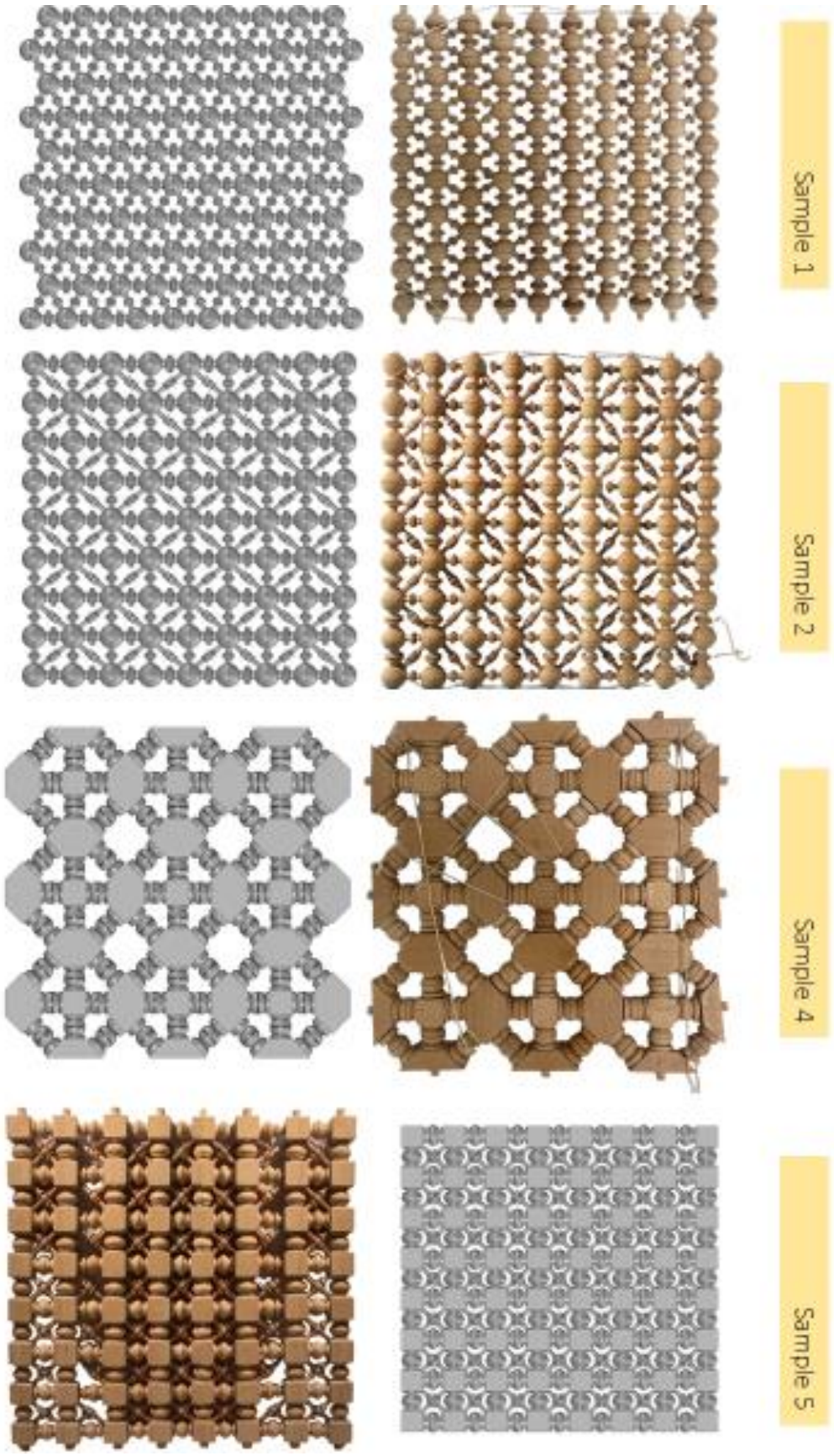


Figure 89: *Catalogue of sample images and corresponding virtual geometries following the 3d scanning Source:(Author)*

6.3.1 3D Scanning of the Samples (virtual documentation of Mashrabiya details)

This part of the study requires focusing on the small details manufactured in the screen that give the screen its sculpted texture and manipulate the effect of light and shadow on the screen, which can impact the mutual shading of the screen and how the screen shades itself. Such a feature can be critical as a concept to be integrated into a multi-functional ceramic evaporative cooling system and a shading screen, which is one of the main objectives of the study.

To understand the impact of such small-scale manufactured details on mutual shading and aerodynamics, it is important to document these details to the highest level of applicable accuracy. Accordingly, a part of the analytical work will be dedicated to the three-dimensional scanning of the screen samples provided for the study. This documentation will not only be beneficial to the study's analytical work but also serve as a useful approach for the community by preserving such endangered artistic details for future generations of designers and craftsmen.

The University of Westminster facilities provide three-dimensional screening of models and prototypes. However, the 3D scanning technology's maximum dimensional allowance should not exceed 15cm in length and width, which allows for the scanning of partial areas of the provided samples. Understanding the limitations of the tools and technologies used will indicate some of the constraints the study will need to overcome; therefore, it was important to highlight this limitation and discuss the outcome with the craftsmen who will be manufacturing the samples for the analytical work.

From the literature, Mashrabiya is mentioned in various resources as a smartly designed shading component. The elegance and creativity of the design are not only rooted in the shading and ventilation performance of the screens, but also in their efficient application. The elements of Mashrabiya are lathed separately in the required quantity, and then the parts are gathered and attached to each other without the need for nails or carpentry glue. After attaching the elements to form the screens, the pattern is usually not strong enough to be transported to the site. The frame of the screens, which is also used as a partitioning method for the overall component, serves as the main support for the elements, as it is attached to the exact size of the components to apply pressure that holds them together. The stress pressure resulting from the frame is the primary support that keeps the components composed and intact.

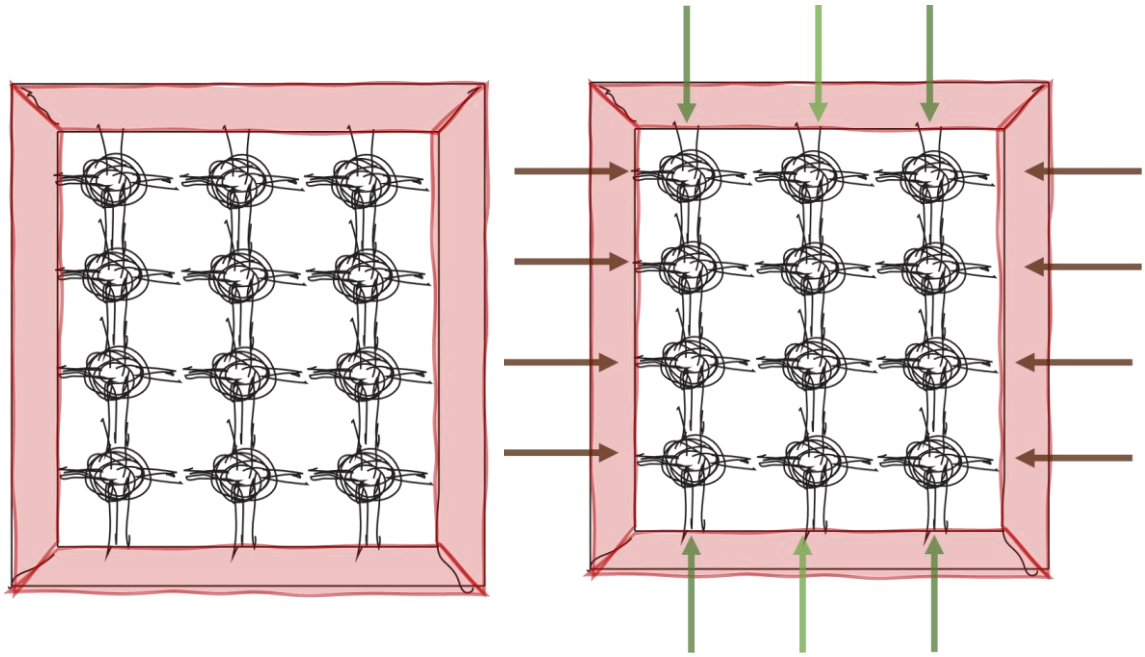


*Figure 90. Illustration explaining how the frame contributes to the stress pressure that helps the components remain combined and attached- (*Author)

Accordingly, the samples were selected, and a note was highlighted to the craftsmen responsible for manufacturing the samples. The note indicated that the samples should come attached but without a frame, so the pattern parts can be easily detached and sampled again. The unhitched screens will allow for overcoming the scale limitation in the process of 3D scanning.

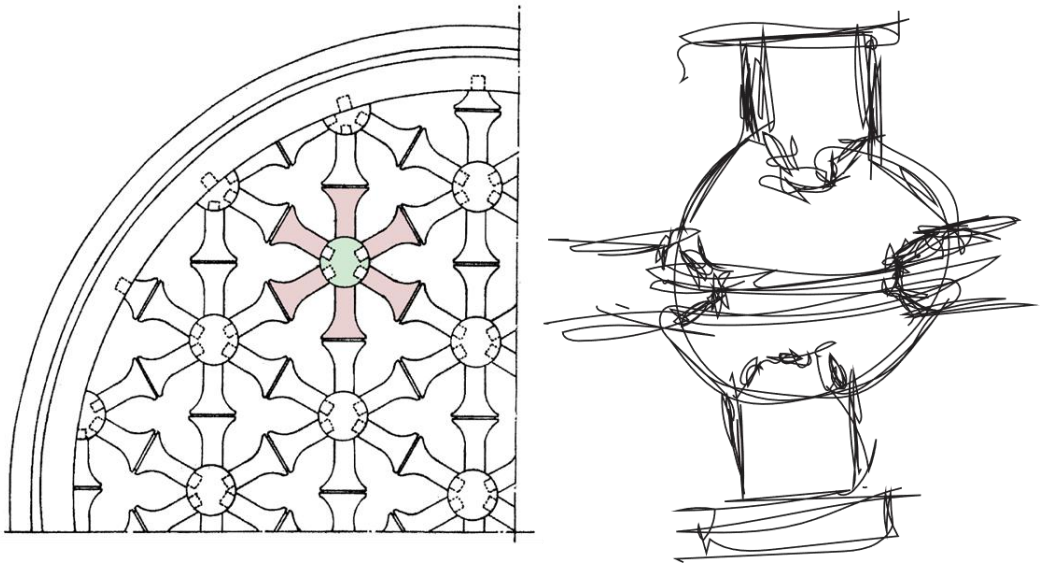


Figure 91: Illustration of the detaching method used for wooden components without glue or nails, and how this technique is repeated at the larger scale of the complete assembly-Source:(Author)

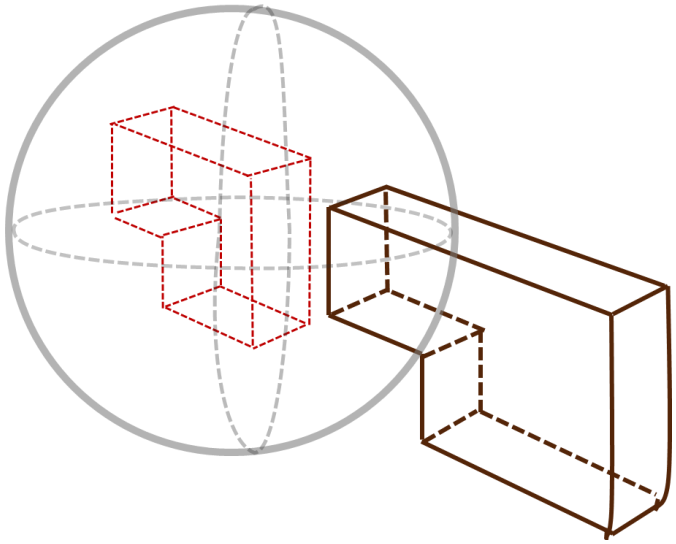


Figure 92: *Illustration of the detaching method used for wooden components without glue or nails- Source:*((Author)

The scanning of the samples was a crucial stage and required many experiments to ensure all the details of the patterns were correctly scanned. The scanners available in the surrounding facilities could not scan objects with two-dimensional lengths exceeding 15 cm, as shown in the following image. Therefore, the samples were dismantled and taken apart to fit the dimensional standards required for the scanners.

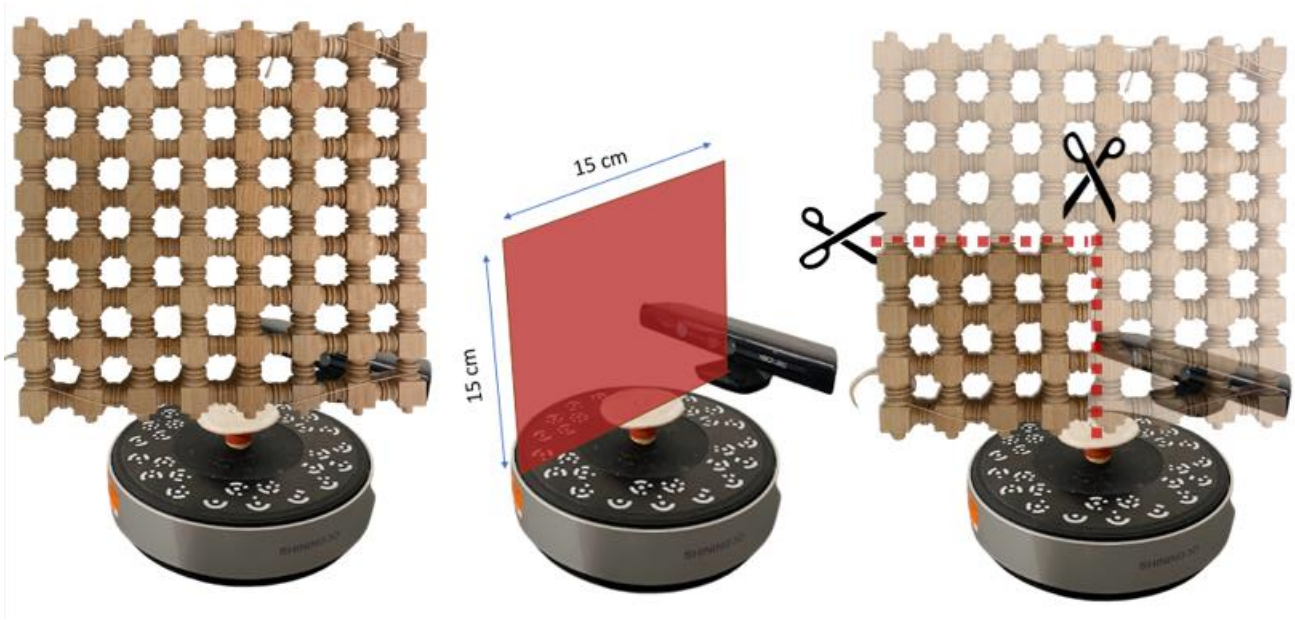


Figure 93: *Overcoming the divisional challenges in 3D scanning- Source:* (Author)

Samples were dismantled and taken apart to fit the dimensional standards required for the scanners. After scanning a quarter of the screen area, the model is exported from the scanner as complex NURBS and geometries and then simplified into meshes and closed preps in order to facilitate simulating the patterns in different analytical tools, as shown in the following image. Simplifying the geometry of the

scanned screens is followed by repeating the geometry and joining the parts to complete the screen in its final dimensions.

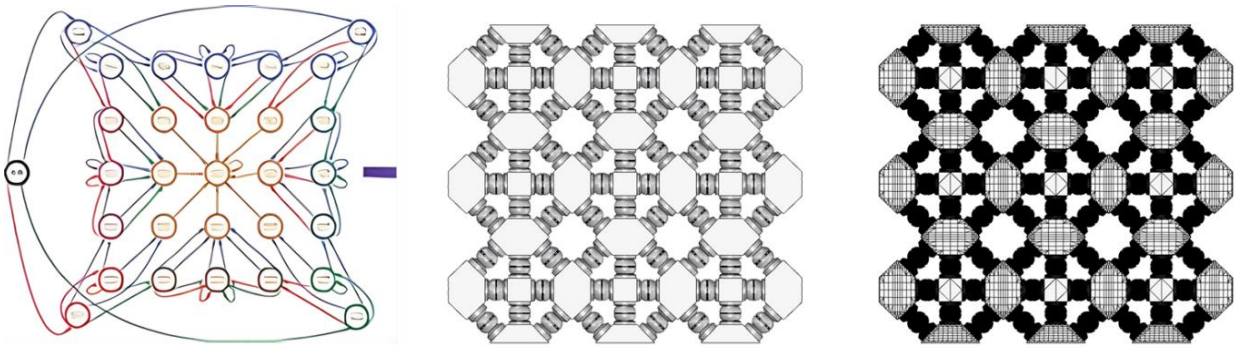


Figure 94: *The process of simplifying the scanned samples into meshes suitable for simulation workflows-* (Author)

6.3.2 Methodological Challenges

1. The limitations that this part of the study can face are mostly in the scanning procedure of the screens, where the screens come as 30 cm squares, and the available scanning technology can scan only 15 x 15 cm dimensional boundaries.
2. Another limitation was faced during the preparation of the samples and their transportation from the workshops in Cairo to the fabrication laboratory in London. It was challenging to transport these samples without the frame support that structurally holds the elements together, preventing them from detaching into multiple parts in a way that could complicate the rebuilding of the screens.
3. Maximising the accuracy of the results from the analytical tools.

A. Overcoming the Challenges

Overcoming the limitation of the dimensional boundaries in the 3D scanning procedure depends on highlighting the manufacturing of the samples without a frame, allowing them to be easily detached to fit the scanning boundaries. Accordingly, the samples will be fragile and breakable under any stress, which is challenging to avoid during the transportation of these samples from the workshops in Cairo to the fabrication laboratory in London. Therefore, the samples will be supported by knitting wool threads tied around the corners of the screen to partially support the parts together, as shown in the following image.

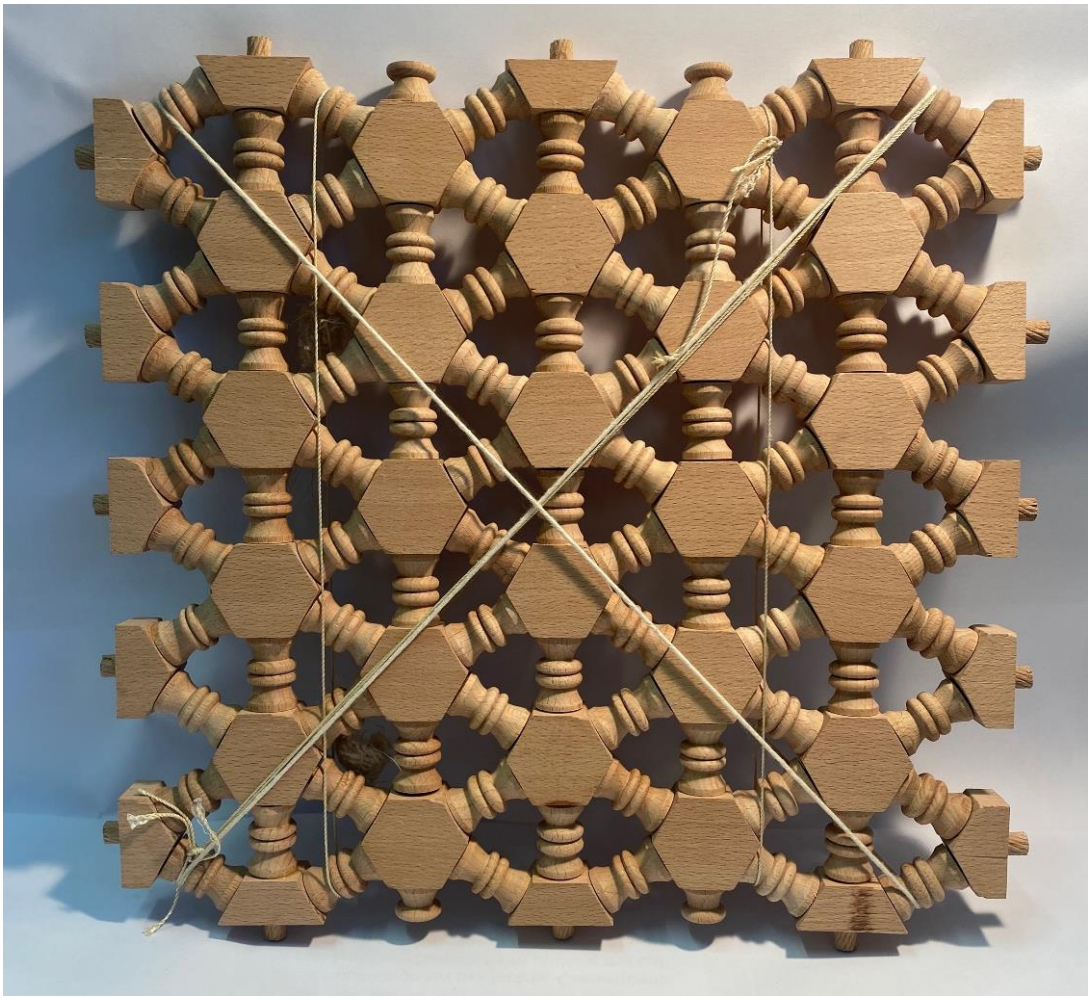


Figure 95: *Sample 9 tied with threads to maintain the structural integrity without the supporting frame- Source:* (*Author*)

6.4 Conclusion

Selecting the samples and collaborating with craftsmen who have more than thirty years of experience in fabricating these patterns brought the Mashrabiya details to life. Documenting them virtually and addressing the limitations gave this chapter a dual role: it provided a foundation for the analytical studies and served as a resource for evaluating the environmental performance of the screens. At the same time, it added another layer of value by recording and preserving the intricate details of the Mashrabiya, an aspect rarely addressed in previous research. This documentation establishes a library of Maymooni screens that enriches the study and supports the analytical work with greater depth and precision, while remaining aligned with the aims of the research. In doing so, the chapter not only prepares for the evaluation of performance but also contributes to a design process that is efficient, well-rooted, and culturally meaningful.

7. Performance of the wooden predominant examples of Mashrabiya

7.1. Introduction

This chapter focuses on experimental analytical investigations that aim to evaluate the environmental performance of Mashrabiya (traditional perforated screens) using various analytical methods and software such as Climate Studio, Ladybug, and Honeybee. This chapter is organised into three major subsequent phases, each of which contributes to the development of the following step. The first stage of the analytical work is to establish the objectives and determine what needs to be accomplished from the analytical studies to ensure they are valuable and beneficial to the design process. This is followed by establishing the scope of the study, which will help determine the outline of the approach and the framework of the study.

Evaluating the environmental performance of Mashrabiya can be quantified by analysing the total performance of the entire screen or through micro-level studies that allow for understanding the effect of the elements that make up the screen and relating this performance to the design and manufacturing details passed down as an inherited science through generations of craftsmen in the field of Mashrabiya manufacturing.

The form of the screens used to change depending on the size of the application; thus, if the traditional screens were chosen to be analysed and used as a cradle for concepts that would be merged into a ceramic screen, the pattern will need to meet the scale of the proposed element as well as the commonly used scale of the traditional screen.

7.1. 1 Scope

The study goals aided in establishing the scope as well as the scale of the analytical work, in order to portray the productivity of the analytical process and support the development of the design methodology. According to the objectives of the study and the necessity of understanding the effect of the manufacturing details of the elements that are attached to form the screen on its environmental performance, a review was carried out to identify the predominant screens used in the implementation of traditional Mashrabiya screens, in order to correlate the factors that led to this predominance while maintaining such details. Another review was conducted to understand the process required to comprehend the environmental performance of these screens.

Given the scale of the design and the scope of the analysis, the focus of the study shifted from on-site fieldwork to an experimental investigation of the screens using virtual analytical tools and software. As a result, the patterns were selected to be tested and analysed using environmental design analytical software packages, including Climate Studio, Autodesk CFD, and Grasshopper Ladybug and Honeybee tools.

Since the manufacturing features and designed shapes of traditional screens were critical to understanding the analytical investigations, it was paramount to visualise and model these features to be virtually identical to their current counterparts with a high degree of accuracy. Accordingly, the study needs to be organised into developmental stages in order to finalise the scale of design before initiating the analytical testing process, enabling the issuance and organisation of the required samples for the analytical work. These steps can also run concurrently with gaining deeper insight into ceramic fabrication processes and overcoming any limitations in modelling detailed screens with ceramic.

7.1. 2. Objectives

The overall objective of this study is to develop a multifunctional ceramic perforated screen that efficiently performs as a passive evaporative cooling system while overcoming the limitations of façade application and undesired solar exposure, which can obstruct the screen's efficiency in terms of operational cooling with considerable water consumption.

As part of a double-glazed façade, ceramic components have been introduced independently as shade components. The material was also included in the façade design via solar control systems installed on the façade, and the materiality of the screens is edited in terms of glazing to alter the reflectivity of the components. Ceramics, on the other hand, is often employed in evaporative cooling components because of the materialistic qualities it features, as discussed in the preceding chapter. Despite the multiple uses of ceramic, it is challenging to design a multifunctional ceramic screen that can act as a shading system and perform efficiently in terms of ventilation and evaporative cooling. However, the study tries to incorporate elements that might be useful in Mashrabiya designs with the proposed ceramic screen.

The main objective of the analytical study is to understand the design features in the Mashrabiya screens and evaluate the impact of these features on the environmental performance of the screens in terms of shading and ventilation. Mashrabiya has been introduced into modern architecture as a traditional façade component that performs efficiently as a shading solution.

Researchers have also related the efficiency of the screens in environmental performance to the mutual shading they feature in various traditional examples, as mentioned in the literature review.

The patterns of traditional Maymooni screens used to consist of between 2-10 elements that are crafted in detail depending on the scale of the screen and the repetition frequency of each element. Accordingly, the main objectives of the study can be simplified into three main points, which are:

1. Define the impact of Maymooni screens and the manufacturing details featured in the traditional patterns on:

    a) The shading performance of the screens on a micro level, where direct, diffuse, and reflected solar radiation will be assessed at different areas of the elements that compose the screen to clarify mutual shading and fluctuation in solar exposure and surface temperature.

    b) The ventilation performance of the screen by determining the discharge coefficient around the screens, as well as the various aerodynamic attributes surrounding the tested screens.

2. Examine the results of analytical studies that explored the mutual shading performance in terms of solar exposure fluctuations and the change in discharge coefficient surrounding each element, to understand the correlation between the different parameters and identify the matrix of relations.
3. Emphasise the concepts that can improve the effectiveness of the suggested ceramic evaporative cooling screen by designing the screen based on the proposed ideas that can be integrated.

As a consequence of the aforementioned goals, this chapter intends to analyse Mashrabiya screens in terms of shading and ventilation on a micro level, which will shape the context of the analytical study.

7.1.3. Methodology

The method of study in this stage depends on analysing the performance of the screens based on evidence obtained from virtual testing using various analytical tools. Analysing the shading performance of the screen involves quantifying the amount of solar radiation that projects onto the screen, penetrates it, and reflects from it. This study seeks to develop a systematic approach to disaggregate and quantify the influence of mutual shading and mutual reflection, and to examine how the manufactured geometry of the pattern affects the fluctuation in solar exposure across the screen.

The study methodology must encompass all the fundamentals of the research in order to explain and answer the research questions. As a result, the methodology will adopt a hybrid approach that collects both qualitative and quantitative analytical data through a validation process. The qualitative approach will be used to gather important manufacturing details regarding Mashrabiya, and this data will later inform the quantitative analytical stage to finalise the focal performance of Mashrabiya and assess how small turnery details and distinctive manufacturing techniques have influenced the eventual outcome of the screens.

7.2. Analysing the Shading Performance of the Samples

Conventionally, the objective of a shading device is to shield a structure from undesired solar radiation, particularly under extreme weather circumstances (Al-Obaidi, 2015). A shading system's mechanism relies on modulating and transforming direct solar energy into diffuse light, as well as controlling the amount of light and solar radiation penetrating the screen (Al-Obaidi, 2015). Evaluating the efficiency of the shading element depends on quantitative studies of two primary objectives:

a) *Visual (optical) performance*, which focuses on avoiding glare while maintaining adequate levels of indoor illumination.
b) *Thermal comfort and solar radiation control*, aimed at ensuring comfortable temperatures by regulating solar heat gains from both direct and diffuse solar radiation.

The efficiency of the shading element gradually increases according to a matrix of relations that balances the desired set of interactions with various factors. As a result, the configurations of the shading element can be achieved through innovative design that maximises the beneficial performance of the fixed shading component, by basing the design on calculations that account for changes in solar angle and solar radiation exposure throughout different times of the year.

Analysing the shading performance of perforated shading screens depends primarily on the amount of solar radiation blocked by the screen and how the pattern performs in relation to direct, indirect, and diffused solar radiation across different seasons, orientations, and times of day, as the sun’s angle and position change.

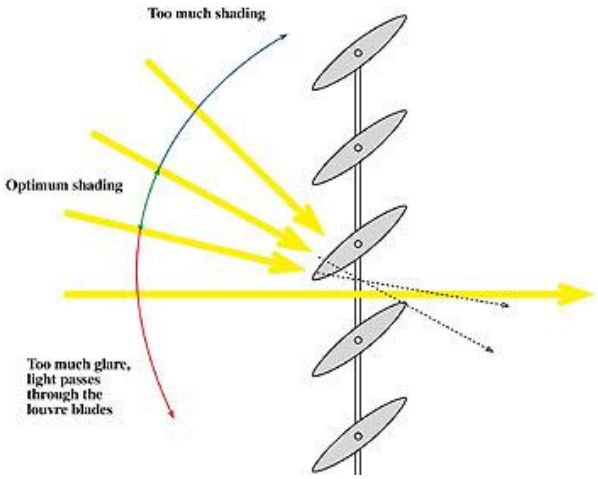


Figure 96: *Solar shading systems provide the best possible shading angle at all times and lessen the probability of over- or under-shading* (Colt, 2020).

To perform the analytical testing of the screens through solar radiation analysis, it is important to define the methods of analysis and review the most accurate means of quantifying variations in solar incidence across different parts of the screen. This analysis can be conducted using two distinct methods, which are:

a) Analysing the shading performance of the screens on a designated show box, where the analysis surface used to quantify variation in solar radiation incidence is positioned behind the screen.

b) Testing the screen itself, where the 3D-scanned model of the screen is refined to preserve the manufactured details captured during scanning, and the screen geometry is subsequently assigned as the analysis surface.

The two aforementioned methods are commonly used in analysing the shading performance of screens. Both will be tested to understand their differences and to determine which method provides a clearer explanation of the mutual shading performance. In parallel, different orientations, colours, and shapes will be investigated at various times of the year to develop a comprehensive understanding of shading performance.

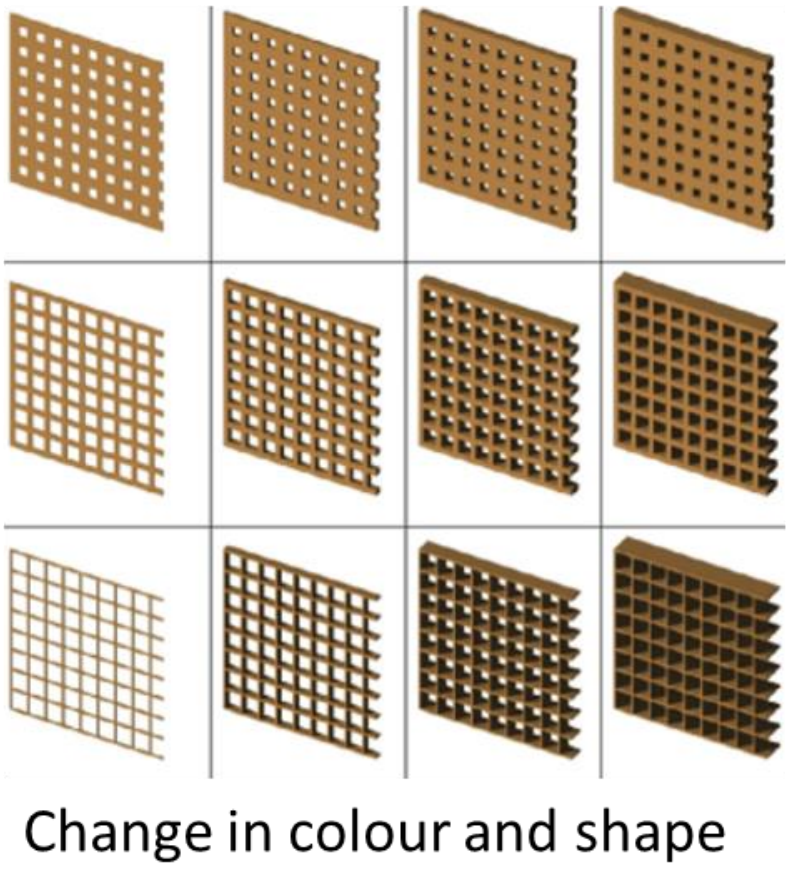


Figure 97: *The testing factors of the screens according to Colour- Source:*(Author)

According to the aforementioned factors and the data presented in the previous chapter regarding the key features of the selected climatic context, a matrix was developed to establish the conditions under which the patterns will be analysed. These conditions depend on the following factors, each of which the screens will be subjected to:

1. The Colour of the Pattern

Although all the pattern samples are made of lathed wood to correlate with realistic examples, the finishing of the patterns used to vary in tone, as the wood employed different colours. The colour of the pattern can influence the amount of solar radiation reflected to the exterior, interior, and the pattern itself, which in turn can affect the mutual shading performance of the patterns.

2. The Timing of the Simulations

To understand the shading performance of the screens, it is essential to evaluate their behaviour across different times of the year. Accordingly, Metronome weather data was analysed to identify representative days that reflect varying sun positions for each façade.

3. The Location of the Site

Cairo will serve as the fixed location for analysing and composing the screens under consistent environmental conditions.

4. The Amount of Direct Solar Radiation Each Part of the Screen Receives

Climate Studio will be used to calculate the variations in direct solar radiation across the same screen.

5. The Amount of Diffuse and Reflected Solar Radiation Each Part of the Pattern Receives

The solar thermal experimentation of the patterns focused on analysing the performance of the screens in relation to both direct and indirect solar radiation. Accordingly, various analytical tools supported the experimental setup used to examine the influence of partial shadowing across different parts of the screens.

The following figure depicts the experimental setup and the two methods used to quantify the shading performance and mutual shading of the screens.

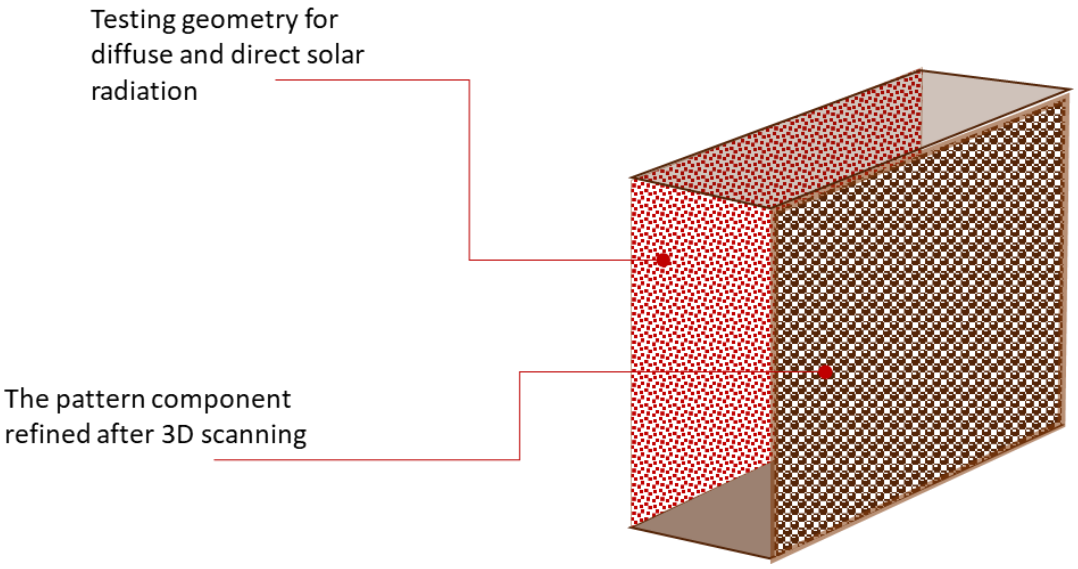


Figure 98: Method A in analysing the shading performance of the screens- Source: (Author)

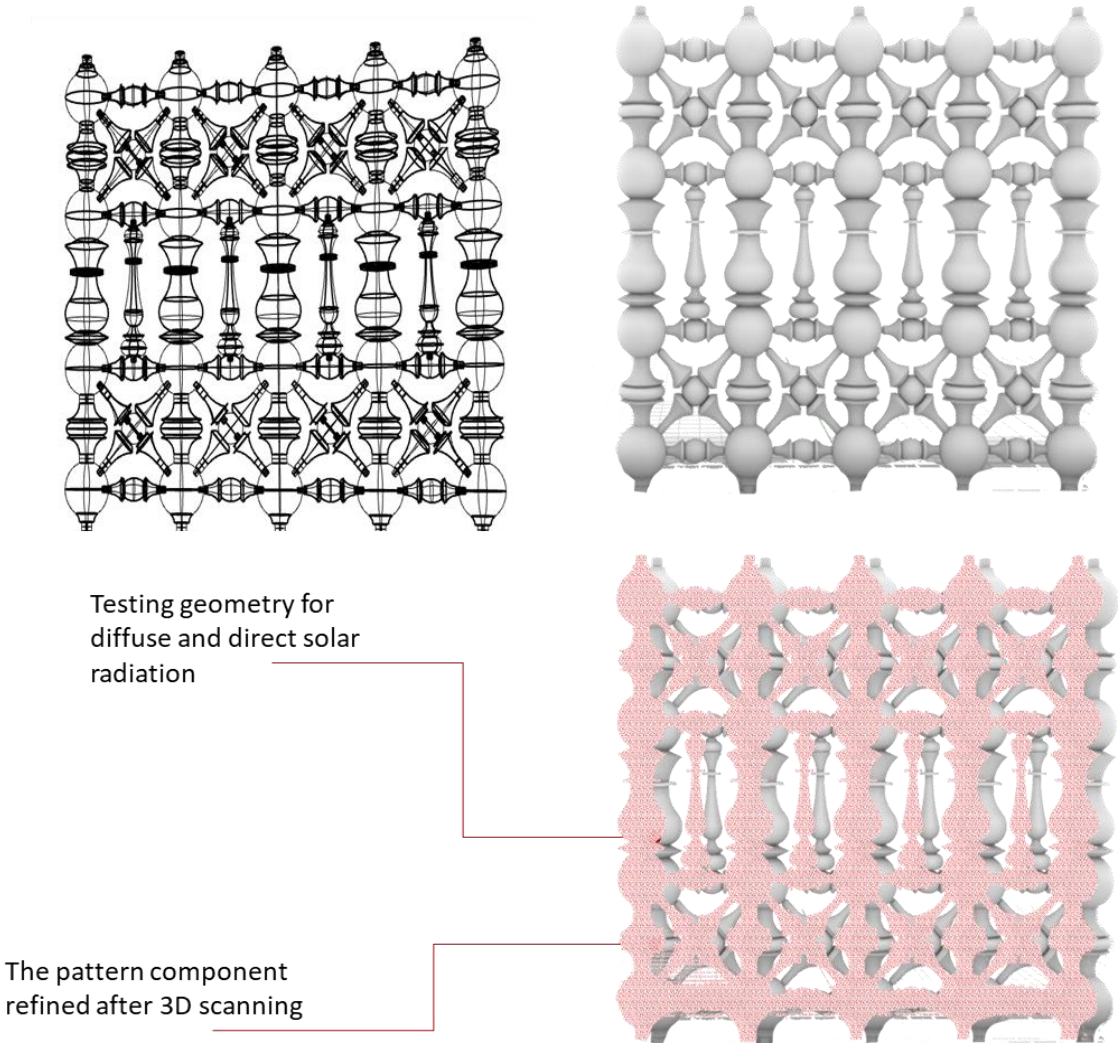


Figure 99: Method B in analysing the shading performance of the screens - Source: (Author)

7.3 Analysing the Ventilation Performance of the Screens

Understanding how the screens perform in terms of ventilation depends on various aspects and factors that can contribute to manipulating the overall performance of a given environmental strategy. The aerodynamics around perforated screens can be quantified and visualised using different analytical methods, including laboratory experimental testing and virtual analytical tools. A perforated screen's aerodynamics are determined by its geometry and application. Engineers can optimise the screen's design for specific purposes and evaluate its impact on the aerodynamic performance of the item to which it is connected, using computational fluid dynamics (CFD) simulations and experimental observations.

Many industries that rely on aerodynamics studies, such as aircraft design environments, have employed various analytical methods involving physical experimentation. Among these is the smoke

tunnel technique, documented by NASA as a method for visualising smoke flow and tuffs around a given object (Hall, 2022). However, the engineering design process has witnessed a transformation over the past 25 years, marked by the widespread adoption of computational fluid simulations for accurate quantification of aerodynamic performance (Jameson, 2002). Although CFD and other virtual analytical tools are still evolving and have yet to reach full maturity, extensive efforts are being made to advance these tools (Jameson, 2002). This underscores the reality that fluid flow is typically more complex and difficult to predict than the behaviour of solid structures.

Mashrabiya screens have different forms and shapes that behave differently in terms of ventilation, due to variations in the discharge coefficient around the screens, which is typically defined when air pressure drops through the aperture area. The air passing through the pattern, driven by natural airflow, can be considered incompressible flow, derived from the Bernoulli equation, which depends on the pressure change across the aperture area, the airflow rate, and the discharge coefficient. Accordingly, quantifying the environmental performance of the screens depends on understanding the pressure variation around the screens, which occurs at a small scale. As a result, it was more justifiable to use analytical tools such as Autodesk CFD for this procedure, in order to compare different patterns in terms of performance at a small scale.

$$\Delta p_{window} = \xi \frac{1}{2} \rho \frac{V^2{}_{window}}{A^2{}_{window}}$$

$$V_{window} = c_d A_{window} \sqrt{\frac{2}{p} \Delta p_{window}}$$

Equation 9: Discharge Coefficient

Where:

Δp_{window}: Pressure difference across the window.

V_{window} : Airflow rate for a defined air pressure loss coefficient ξ.

A_{window}: Area of the window.

ρ: Density of the flow.

c_d: Discharge coefficient, calculated in this equation as $1/\sqrt{\xi}$.

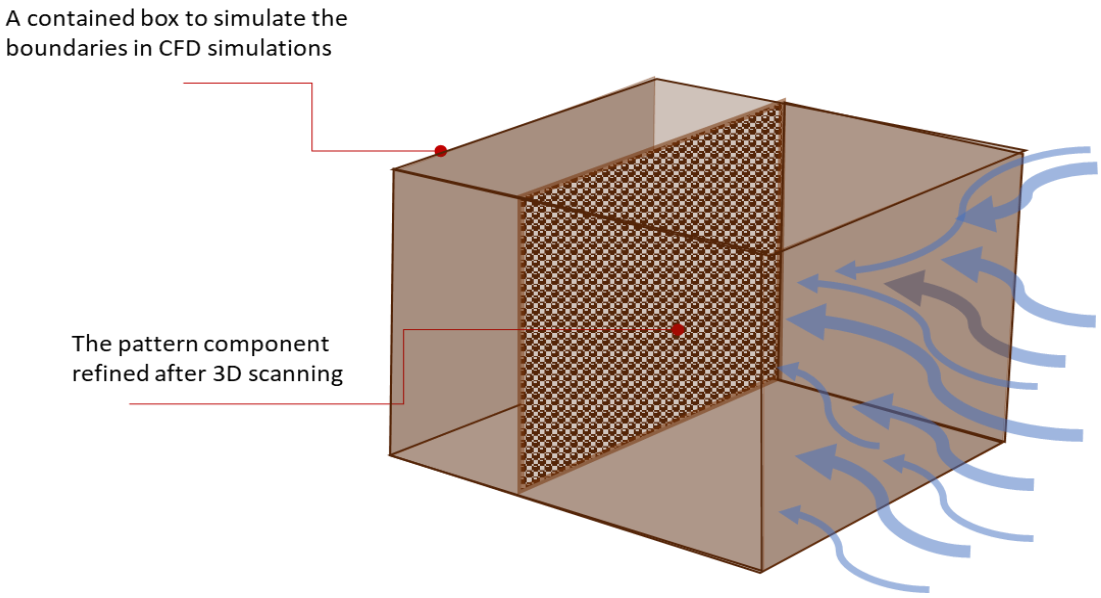


Figure 100: Illustration of quantifying the ventilation performance of the screens and the preparation of the testing environment for CFD pattern analysis

Overall, testing aerodynamics is a complex and multifaceted procedure that necessitates a combination of physical tests and computer simulations, along with a thorough understanding of fluid dynamics and aerodynamics concepts. Physical testing and experimentation will be required at later stages to examine the correlation between the aerodynamics of the designed screen and the process of evaporative cooling occurring through the designed evaporative cooling ceramic screen. Accordingly, this chapter will focus on computational fluid dynamics (CFD) simulations, which involve utilising computer software to model the flow of air over an object, depending on its shape and other physical properties.

CFD (Computational Fluid Dynamics) screen testing involves utilising simulation software to analyse the behaviour of fluids as they flow through screens or filters. The goal of this study is to understand flow patterns, pressure drops, and other fluid characteristics as they interact with the screen. Accordingly, the following steps were followed:

1. Create a 3D model of the screen using Rhino software.
2. Import the model into CFD software and configure the simulation parameters, such as fluid type, flow rate, and boundary conditions.
3. Run the simulation and examine the results, which may include information such as pressure drops, flow velocities, and particle dispersion.
4. Analyse the data to evaluate the screen's performance, including its capacity to allow airflow through the perforations or to maintain acceptable values for drag and discharge coefficient.

5. Modify the screen design as necessary, then repeat the simulation and analysis until the required performance is obtained.

7.4 Testing the samples

This study focuses on designing a multifunctional ceramic screen that efficiently performs as both a shading component and an evaporative cooling strategy. The main objective of the study shaped the methodology for analysing the screens and their performance in relation to specific environmental parameters, with the aim of understanding how certain concepts derived from the Mashrabiya can be beneficial when integrated into the design stages of a multifunctional evaporative cooling ceramic screen.

To ensure the accuracy of the results from testing the environmental performance of the screens, different analytical tools will be used to assess the reliability and precision of the results. Accordingly, analysing the shading performance of the screens was developed into a two-stage process, wherein the screens were tested to quantify the average solar exposure and the ratio of mutual shading. The concept of mutual shading can be further elaborated by testing the daily average sunlight hours at different times, in order to estimate the percentage of the area shaded by the screen compared to a simple extruded screen. The latter is defined as having 50% of its area exposed to the sun, while the remaining 50% is shaded—consisting of the backside, including the top, front and bottom surfaces of the screen itself.

B. Shading Performance of the Samples

Understanding the shading performance of the screens is essential for identifying key features that can be integrated into the ceramic screen to maximise its multifunctional efficiency. It was important to quantify both the overall shading performance and the mutual shading that occurs as a result of the screen's own geometry.

The shading performance was initially examined by studying the fluctuation in solar penetration onto a defined plan beneath each screen, where the plan served as the principal geometry and the screen represented the contextual element influencing solar exposure on that surface. This analytical methodology was useful in evaluating the shading performance of a shading system and its potential to enhance indoor visual and thermal comfort by identifying the most effective shading devices as part of façade system design. However, this methodology does not provide an accurate approach for understanding the mutual shading produced by complex extruded geometries, nor does it account for how variations in extrusion affect the shadows cast on the screen itself.

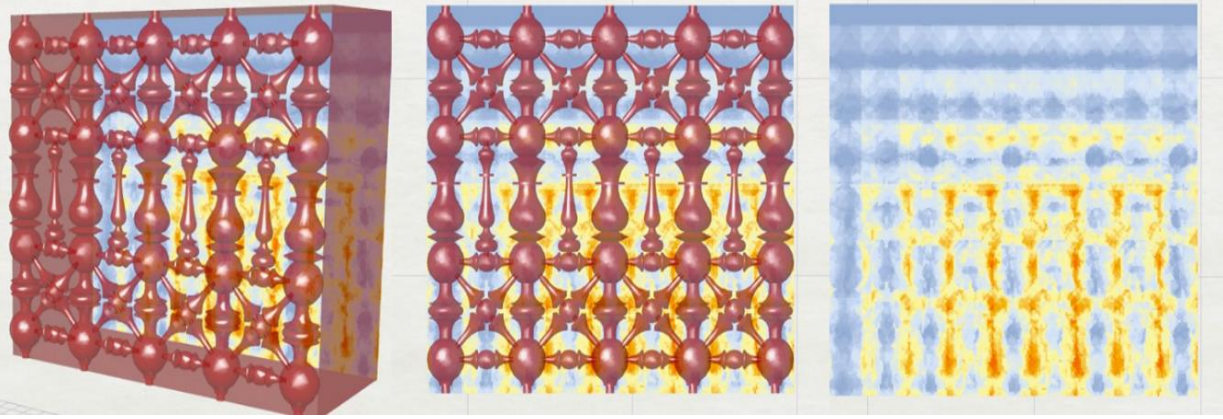


Figure 101: Experimental approach for analysing the solar performance of the screens by testing the amount of solar radiation allowed onto the adjacent surface - Source: (Author)

Accordingly, the analytical methodology was adapted to align with the main objective of the study. The first part of the investigation focused on understanding how each screen performs when exposed to direct sunlight during the peak hour of each season—specifically in summer, when solar exposure necessitates effective shading that directly addresses the primary objective of the analysis. The daily average sunlight exposure onto a regular plan during the peak hour of a typical summer day, oriented towards the east or west, corresponds to a mathematically predictable distribution (50% to 50%), where 50 percent of the surface is exposed to direct sunlight and the other 50% is fully shaded due to the opaque nature of the plan obstructing solar access. This indicates that a simple planar surface blocks zero percent of the sunlight received on its exposed side; such results are inherently predictable and do not require analytical tools to substantiate.

When exposing a screen under the same orientation and during identical direct sunlight hours, it is predicted that the results will remain consistent. The screen in question is a perforated surface initially designed in a two-dimensional scale, then extruded to a specific thickness. Although the average exposed area may vary, the ratio of sunlight to shaded areas is expected to remain the same, as reflected in the comparative analysis.

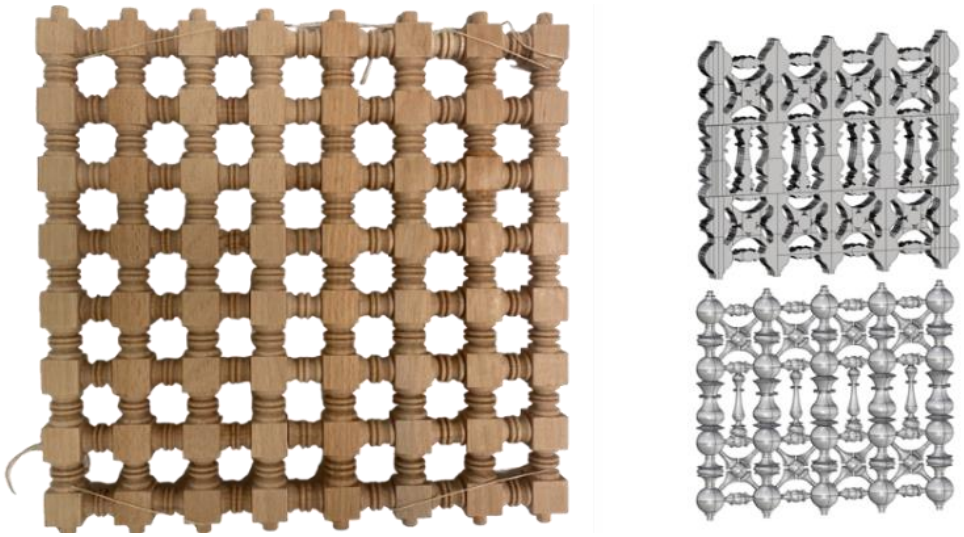


Figure 102: *Illustration of how scanning the details in the screens alters the final geometry compared to a simple extruded screen- Source:* (Author)

Having the screens in their realistic form and sculpted geometry, with varying thicknesses across the same screen, it is predicted that the results will differ. These variations are expected to help identify the efficiency of mutual shading for each screen. Accordingly, the solar analysis of the screens was divided into two main parts that contribute to a general understanding of shading performance across different scales, which are:

1. Mutual Shading of the Screen

Where the screen serves as the primary geometric element and the key parameter to be tested under direct sunlight during peak hours across various times of the year, with the results primarily dependent on the screen's orientation.

The orientation of the shading device determines the peak hours during which it receives the maximum amount of direct solar incidence, and the shadow range is at its minimum. These peak hours vary according to orientation, due to the continuous change in the sun's position throughout the day, accompanied by fluctuations in solar azimuth and altitude.

To understand the shading performance of the screens—and the extent to which the screen shades itself during peak solar hours when sun angles are most direct and shadow lengths are minimised—it is difficult to define a fixed time of day across different solar solstices for all orientations. Accordingly, a brief solar study was conducted to determine the peak hour for each orientation in each season. It was found that, in both summer and equinox months, the hours with minimum shadow length are nearly identical. The following images illustrate the stages of defining the main hour during which each screen would be exposed to direct solar radiation, varying by orientation due to changes in shadow length resulting from solar angles

throughout the day. For each orientation, the hour was selected based on the peak day of the year and the peak hour, which was calculated by determining the minimum shadow length for each orientation in square meters.

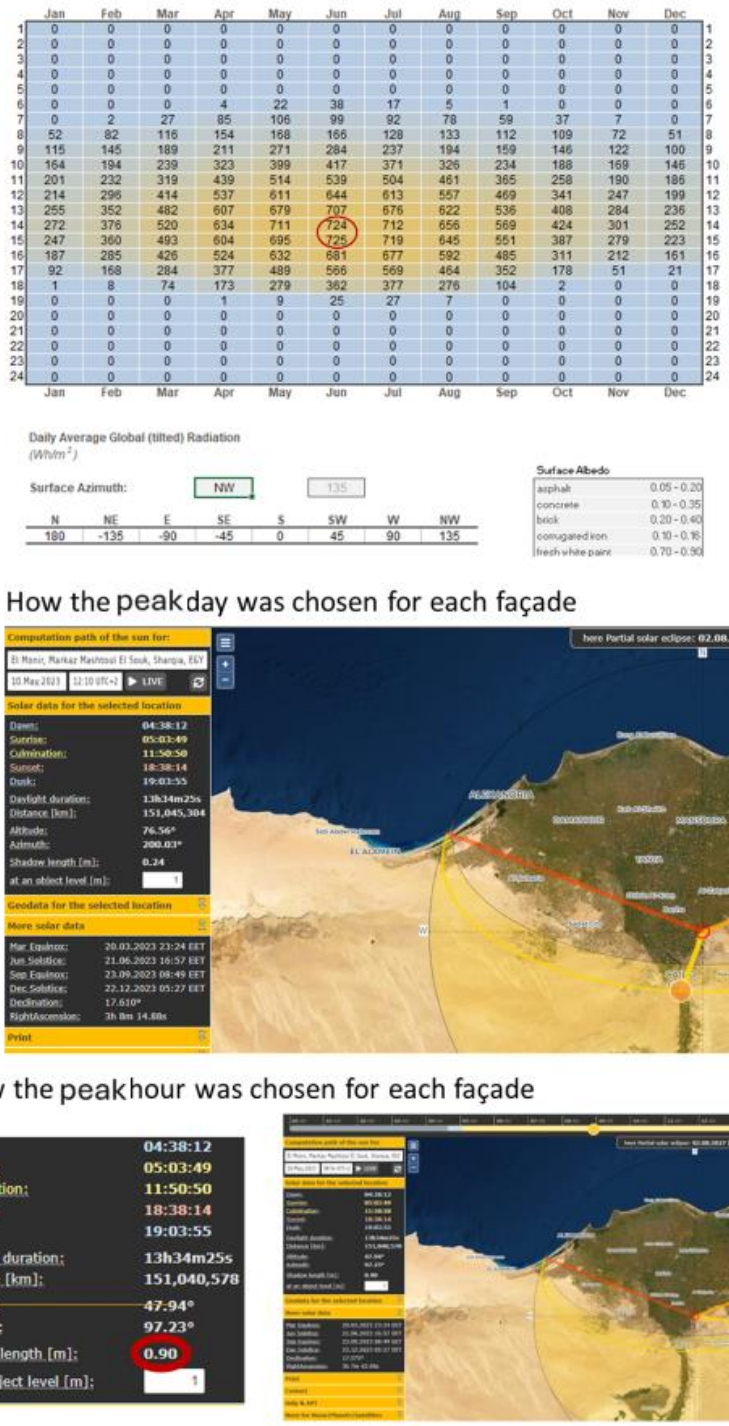


Figure 103: *Methodological stages for defining the solar angle and the hour of each orientation, using Meteonorm weather data for Cairo (2020) and an online sun calculator to determine solar angles- Source:*(Author)

The methodology illustrated in the previous page explains how the simulation context for the analytical work was selected in terms of time for each orientation. There are multiple approaches to determining the peak hour, when the sun is directed toward each orientation and the area of overcasting shadow is minimised. To clarify the terminology of peak hour, it is important to define the main objective of this part of the simulation and why identifying the peak hour is necessary for each orientation.

The primary objective of analysing the mutual shading of the screen is to quantify how much of the screen shades itself due to its sculpted features and complex extrusions. To understand these characteristics, it was essential to simulate the screens in their exact dimensions and design complexity by 3D scanning the screens and modelling the percentage of the area that does not receive direct solar radiation relative to the area exposed to direct sunlight—when the solar angle is perpendicular to the screen for orientation.

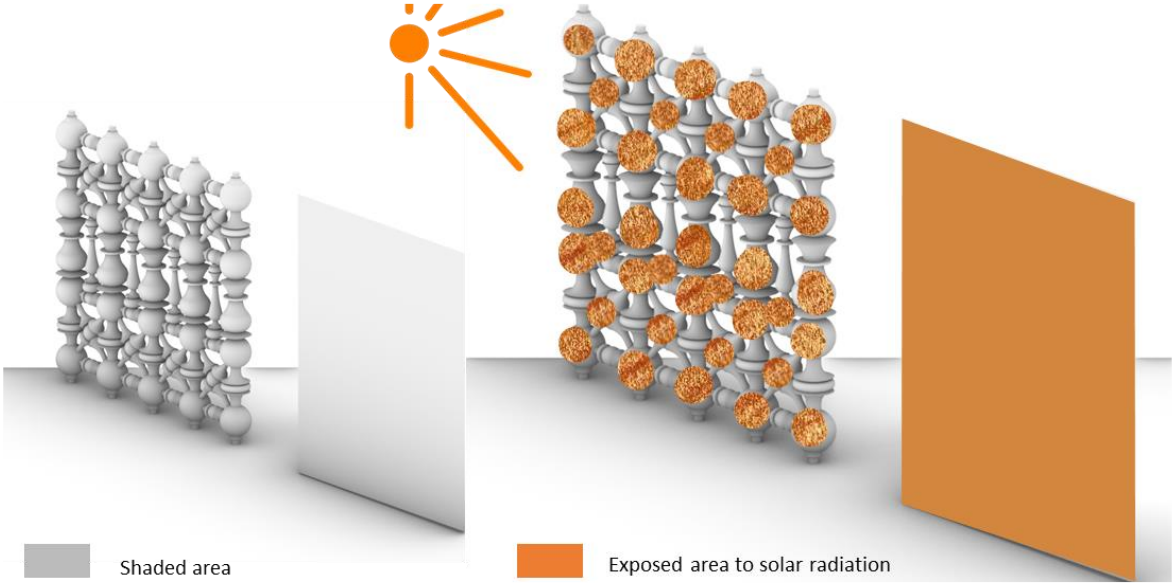


Figure 104: *Illustration explaining the variation in direct solar exposure between a complex extruded Mashrabiya screen and a standard vertical surface- Source:* (Author)

Quantifying the mutual shading of the screens relies on comparing analytical results of daily average sunlight hours during the peak hour with those of a standard vertical surface. A typical vertical surface on a façade, oriented in a specific direction, receives direct solar radiation such that approximately 50% of its area is exposed, while the remaining 50% is unexposed. The shadow portion is calculated from the rear side of the surface, which does not receive solar radiation due to the opacity, effectively blocking solar penetration. This is shown in the following image.

Accordingly, these hypotheses must be tested using different time variables to understand the correlation between changes in solar angles and the ratio of the area exposed to direct sunlight versus the area shaded by the screen's extruded geometry. The analysis conducted on Sample

3 compares its performance with that of a simple vertical surface, using Ladybug and Honeybee plugins within the Grasshopper software interface.

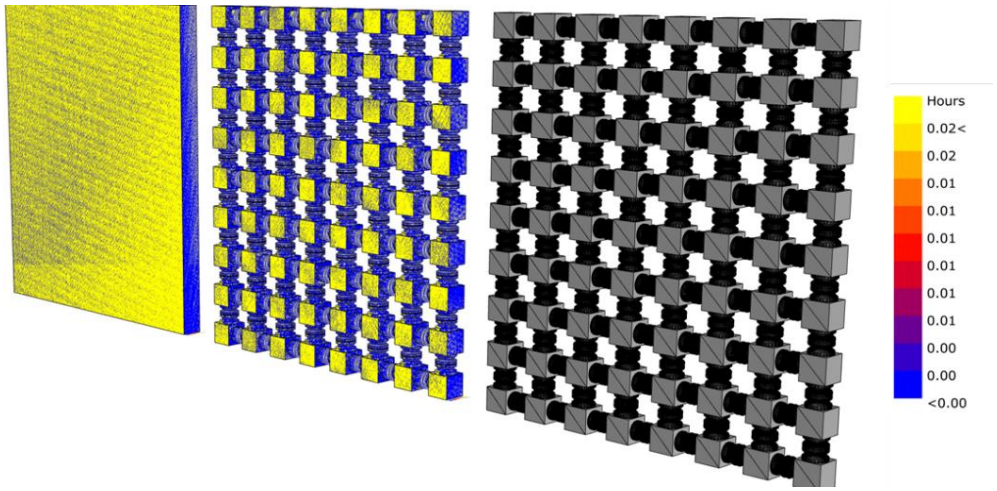


Figure 105: Results of daily average sunlight hours during the peak hour in the south orientation, comparing a standard vertical surface with the scanned geometry of sample 3- Source: (Author)

As a consequence, the initial results of the analytical simulation on daily average sunlight hours for the screens revealed remarkable outcomes and a wide variation in performance compared to a standard vertical surface facing the same orientation. This variation is demonstrated through comparative analysis, where the first pie chart represents the percentage of shaded area relative to exposed area on a standard vertical surface, followed by a second pie chart illustrating the same ratio for the scanned geometry of sample 3.

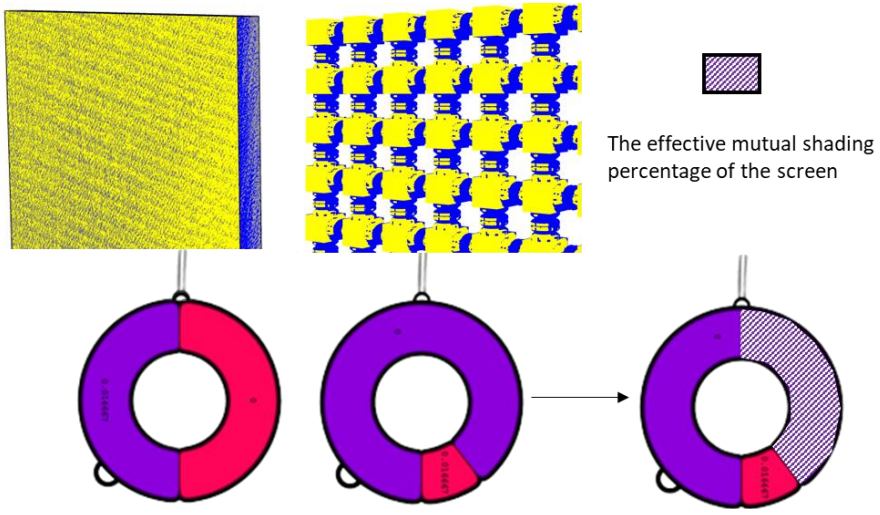


Figure 106: Pie chart illustrating the results of each case scenario and the corresponding percentage of effective mutual shading- Source: (Author)

2. Average Solar Radiation of the Screen

The screens were tested in various orientations at different times of the year to examine the variation in solar incidence according to average, maximum, and minimum values. This analysis aimed to correlate the fluctuations observed on each screen with the influence of surface colour—realistically represented through the use of different wood types, each exhibiting distinct tonal variations.

Mashrabiya screens were traditionally manufactured from wood, although precedents exist worldwide of similar elements fabricated from stone—such as the Jaali screens in India. However, the Maymooni screens selected as case studies in this study are based in Egypt and were exclusively crafted from wood. Therefore, given that the primary objective of the analytical study is to assess the environmental performance of these screens, it is appropriate to simulate them virtually using the same material.

Following research and investigation into the types of wood predominantly used for Mashrabiya in Egypt, it was found that when customer preferences require varying wood tones to suit the façade design and building typology, craftsmen typically alternate between several types of wood:

- Tudor Oak
- Pine
- Walnut

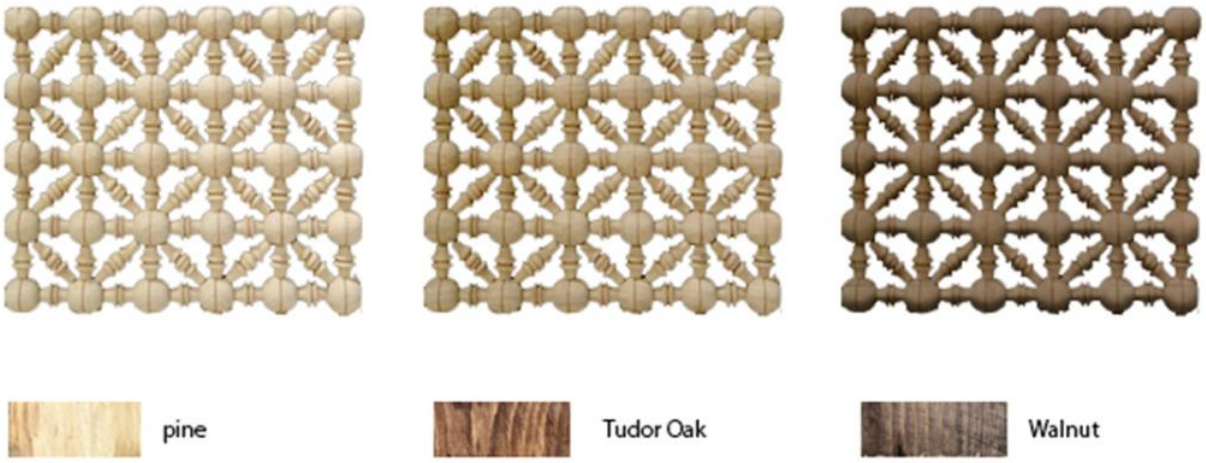


Figure 107: Different wood tones used in Maymooni screens in Egypt- (Author)

According to Mr. Mohsen, the head craftsman at the Nadeem Foundation, these types of wood were preferred by artisans due to the ease of carving pine wood, which can be shaped into small components that are assembled to form the overall screen. He also noted that the tone of the wood is selected based on user preferences and the façade design to which the screen will be applied. However, most of the existing historical examples feature medium-toned oak wood, chosen for its superior durability and greater availability.

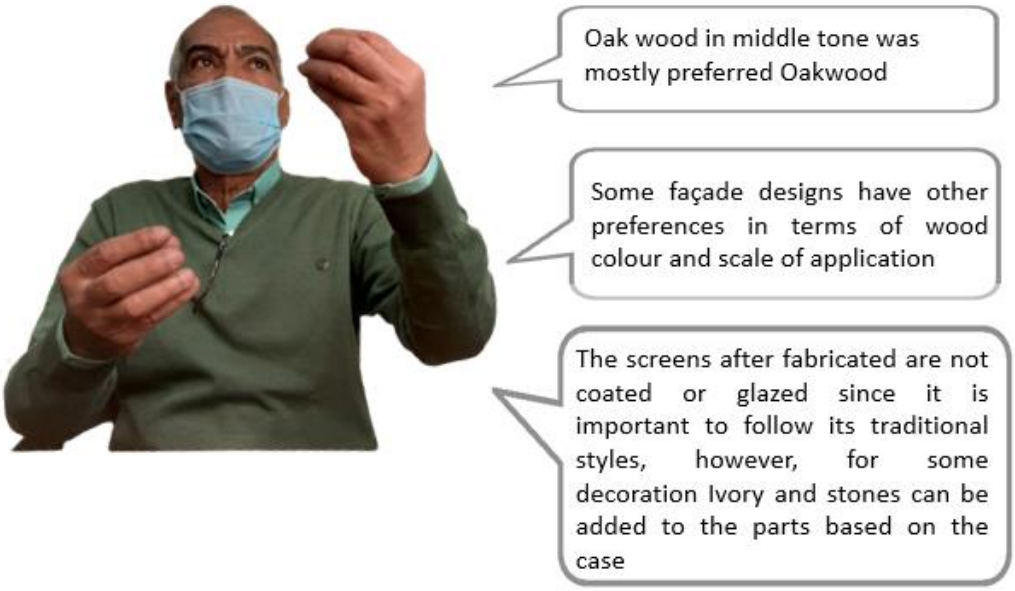


Figure 108: *Highlighted points from a recorded, consented interview with Mr. Mohsen at the Nadeem Foundation, regarding the materiality and design criteria of Traditional Mashrabiya —Source:* (Author)

Based on interviews with the craftsmen, it was found that once the small components are manufactured and assembled, the screen is applied without any coating or glazing. According to Mr. Mohsen, this practice preserves the traditional aesthetic of the screen and enhances its durability when installed on façade. However, ivory and polished stones may be incorporated into the individual parts to introduce decorative features. This is shown in the following image:

Figure 109: Addition of *Ivory and stone to screen components, as observed in the workshops of the Nadeem Foundation — Source:* (Author)

Accordingly, it was essential to assess the solar performance of these different tones and correlate them with the existing colours of clay, as the screens were analysed to integrate these concepts into a multifunctional evaporative cooling ceramic screen. Evaluating the shading performance—specifically

the variation in direct and indirect solar radiation received by the screen across different times of the year and various orientations—depends on two key factors:

a) The context

The orientation, time of day, and solar angles play a critical role in determining the amount of direct and indirect solar radiation received by the screen.

b) The screen

The geometry of the screen influences the distribution of solar radiation across its surface. However, the screen's colour and material significantly affect the reception of indirect solar radiation. Therefore, it was necessary to examine the correlation between the simulation results and the material characteristics.

These features are instrumental in refining the conceptual framework for the proposed ceramic prototype and in establishing design guidelines applicable to any perforated shading screen.

Consequently, the analytical scope of the study was redirected to investigate the solar performance of various wood colours and correlate these characteristics with the material properties of clay. The primary analytical tool employed to assess variations in direct and indirect solar exposure was Climate Studio accessed through the Rhino interface for result visualization. Subsequently, the Grasshopper interface was utilised to extract and process the simulation data.

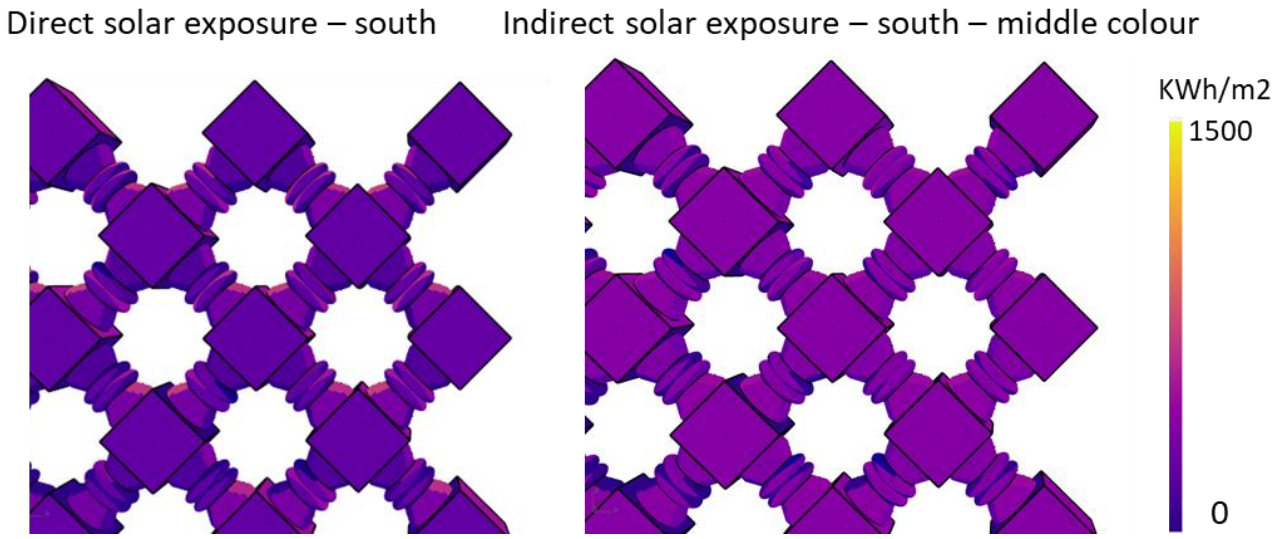


Figure 110: Variation in direct and indirect solar exposure on sample 7, oriented southward during the summer solstice, using the Climate Studio interface in Rhino—Source: (Author)

3. Aerodynamics Around the Samples

Following the scanning of the 3D patterns, the virtual geometries were simplified into closed Breps to facilitate export into fluid dynamics analysis software. Autodesk CFD was primarily

employed to evaluate volumetric airflow through the screens and to extract input parameters for calculating the drag and discharge coefficient of each tested sample.

Due to the geometric complexity of the screens and the intricacy of the aerodynamic simulations, the screens were not exported in their entirety. Instead, each screen was divided into quarters, and the representative repeated segment was selected for analysis, as illustrated in the following image.

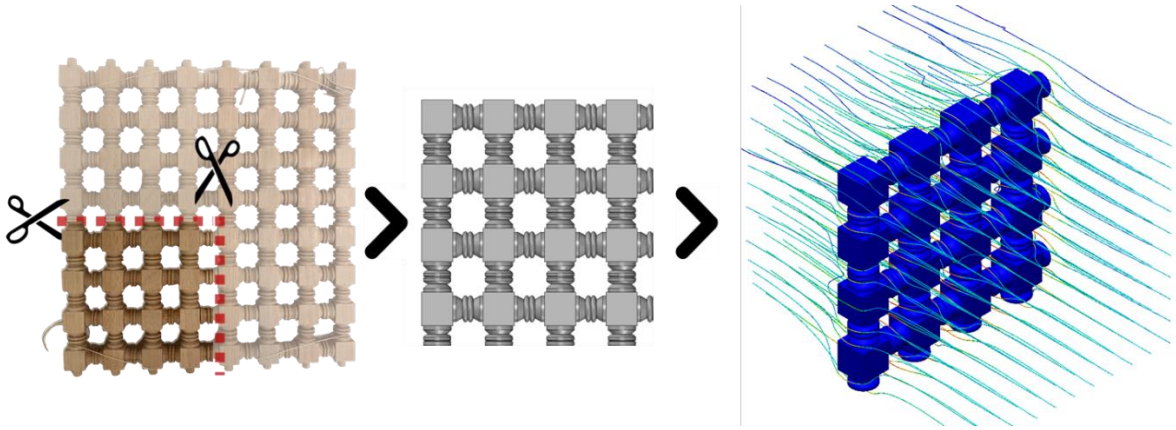


Figure 111: Simplification of screen geometry to facilitate aerodynamic simulations and ensure analytical accuracy — Source: (Author)

The patterns were analysed under consistent conditions to compare their performance and identify the key features of the screens that influence ventilation efficiency and aerodynamic behaviour. Aerodynamic testing involves monitoring airflow over and through the screens to evaluate sample performance and highlight critical design attributes. The primary objective of such testing is to optimise the proposed screen design by minimising drag and enhancing overall aerodynamic performance. Accordingly, Computational Fluid Dynamics (CFD) was employed to simulate wind tunnel testing, which entails placing a scale model or full-size replica of the object within a wind tunnel and measuring the forces and pressures exerted by the airflow, as illustrated in the following image.

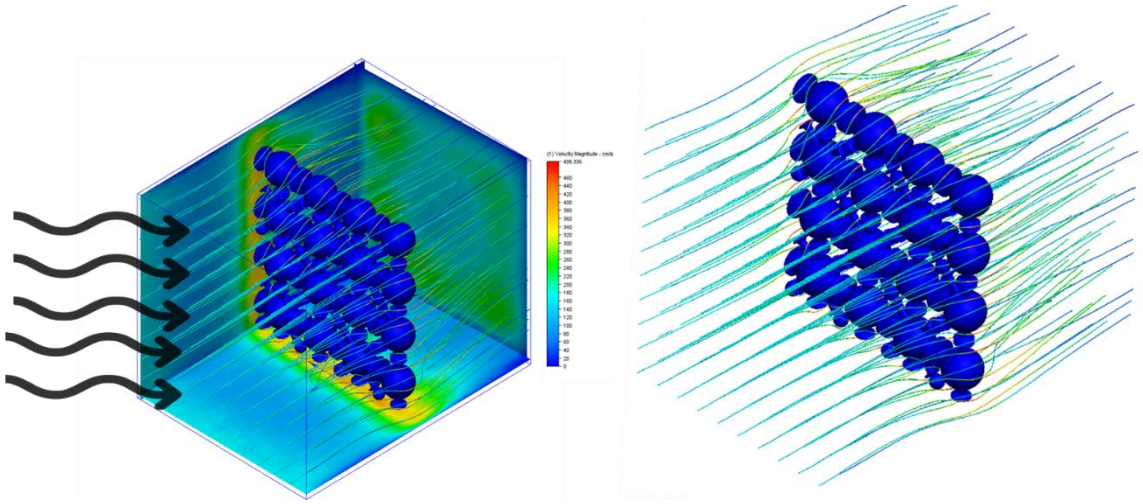
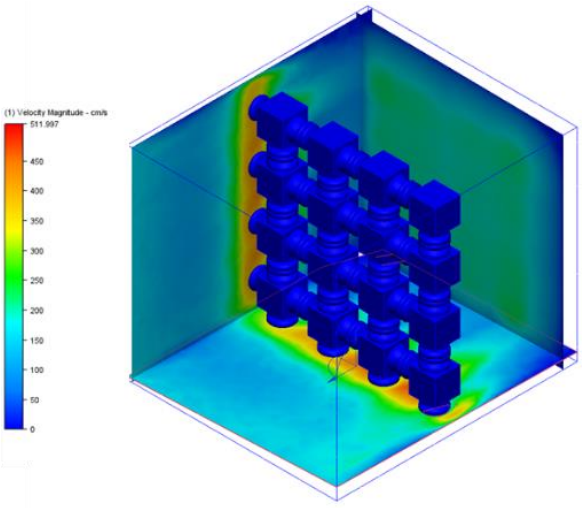


Figure 112: *Wind tunnel simulation using the test samples as geometric input within Autodesk CFD — Source:* (Author)

The wind tunnel was simulated in CFD using fixed dimensions for all samples, which were positioned at equal distances from the inlet. The inlet air velocity was uniformly set at 0.6 m/s across all tests. This consistent setup enabled comparative analysis of the CFD results for each sample, allowing conclusions to be drawn based on observed variations in aerodynamic performance.

Understanding the outcomes of Computational Fluid Dynamics (CFD) simulations involves analysing the output data generated by the simulation software. This data may include velocity vectors, pressure contours, temperature profiles, and other metrics that provide insight into fluid behaviour and its interaction with the screens examined in the analytical investigation. For instance, the following results were extracted from the CFD analysis of Sample 3.



Point 1			
Velocity Magnitude	-0.0050431	N/A	cm/s
Vx-Velocity	5.3867e-05	N/A	cm/s
Vy-Velocity	-0.00503358	N/A	cm/s
Vz-Velocity	-0.000255077	N/A	cm/s
Static Pressure	-40.672	N/A	dyne/cm2
Temperature	0	N/A	Celsius
Scalar variable	0	N/A	None
Density	0.00120473	N/A	g/cm3
Viscosity	0.0001817	N/A	poise
Shear Stress XY (Tau-XY)	-3.7226	N/A	dyne/cm2
Shear Stress YZ (Tau-YZ)	15.0474	N/A	dyne/cm2
Shear Stress ZX (Tau-ZX)	-0.554348	N/A	dyne/cm2
Shear Stress (Tau) Magnitude	16.0003	N/A	dyne/cm2
Wall Film Coefficients	0.00110212	N/A	W/cm2/K
Wall Heat Flux	0	N/A	W/cm2

Figure 113: *Extracted aerodynamic results for Sample 3from the Autodesk CFD simulation study — Source:* (Author)

Accordingly, the patterns were compared based on the extracted results from the CFD simulations, wherein the data served as input for quantifying and calculating the drag and discharge coefficients of each screen. This analysis aimed to elucidate the aerodynamic behaviour around the screen components and assess their ventilation performance.

The drag coefficient is a dimensionless parameter that characterizes the degree of aerodynamic resistance, correlating the interaction between the fluid and the screen. It is defined as the ratio of the drag force acting on the object to the product of the fluid density, the object's cross-sectional area, and the square of the fluid velocity. The drag coefficient can be calculated using the following formula:

Equation 10: Drag Force Calculation—NASA Aerodynamic Guide, 2027

$$F_d = \frac{1}{2} \times \rho \times V^2 \times A \times C_d$$

The drag coefficient is calculated by dividing the drag force (F_d) by the dynamic pressure (which is 1/2 air density (ρ) multiplied by the square of the speed (v) and multiplying it by the shape's reference area A of the object. Accordingly, the drag force for each sample was extracted directly from the CFD simulations and used to evaluate aerodynamic performance.

Given that the simulation inputs—such as wind tunnel geometry, inlet air velocity, and outlet pressure—are fixed, the analytical process enables meaningful insights into the fundamental physics of fluid flow by comparing the drag coefficients of different geometries. Objects with streamlined profiles, for instance, typically exhibit lower drag coefficients than those with blunt or angular forms, underscoring the significance of design in minimising aerodynamic resistance.

Overall, comparing results across varied forms under consistent conditions provides valuable information for optimising object design with respect to aerodynamic performance and deepening the understanding of fluid flow behaviour.

The discharge coefficient is a critical parameter in the design and optimisation of fluid flow devices, as it directly influences their efficiency and effectiveness in facilitating fluid movement. By understanding the factors that govern the discharge coefficient, engineers can develop systems that minimise energy loss and maximise fluid transmission.

In the context of ventilated screens, calculating the discharge coefficient is essential for evaluating and enhancing their performance. It determines the screen’s efficiency in promoting airflow and influences the volumetric flow rate, thereby serving as a key metric in the aerodynamic optimisation of the design.

The discharge coefficient is calculated by dividing the actual mass flow rate by the theoretical mass flow rate, both of which are derived from CFD outputs and extracted simulation data. The theoretical mass flow rate is calculated by using the following formula:

Equation 11: Theoretical Mass Flow Rate Calculation— NASA Aerodynamic Guide, 2027.

$$m = \rho \times A \times \sqrt{\frac{2 \times \Delta p}{\rho}}$$

The discharge coefficient for a single Venturi calibration point was computed using the data and equations previously outlined. Since the simulation inputs were fixed across all screens, the resulting values allow for comparative analysis. Although the results are related to the drag coefficients calculated for each screen, it was also essential to examine the correlation between the discharge coefficient and the distribution of screen components along different axes, which define the geometric variation among samples. This relationship directly influences the overall mass airflow rate.

Accordingly, the analytical investigation provides insights into how screen geometries can be optimised to minimise airflow obstruction and maximise volumetric flow. Such optimisation is particularly critical for perforated ceramic screens intended for evaporative cooling, where performance depends primarily on airflow and the interaction between hot, dry air and the wetted ceramic medium.

7.5 Results of Sample Testing

This section focuses on visualising the variation in results across different samples tested under identical conditions and inputs. The time variable, geographical location, climatic context, and orientation were held constant for each sample to ensure consistency. The study also emphasised the separation of direct and indirect solar exposure data to investigate the influence of colour materiality on changes in indirect solar intensity across specific regions of the screens.

The simulation of mutual shading conditions for each screen yielded the following results. In this phase of the analysis, wood colour was not considered, as the focus was solely on evaluating direct solar exposure across the screen geometries.

A perforated or mesh screen can significantly influence the aerodynamic behaviour around the aperture to which it is attached. As fluid—such as air—passes through a perforated screen, it encounters resistance due to both the presence of the screen and the configuration of its perforations. This resistance induces a pressure drop across the screen, which may affect the overall aerodynamic performance of the object.

7.5.1 Mutual Shading Analytical Study of the Screens

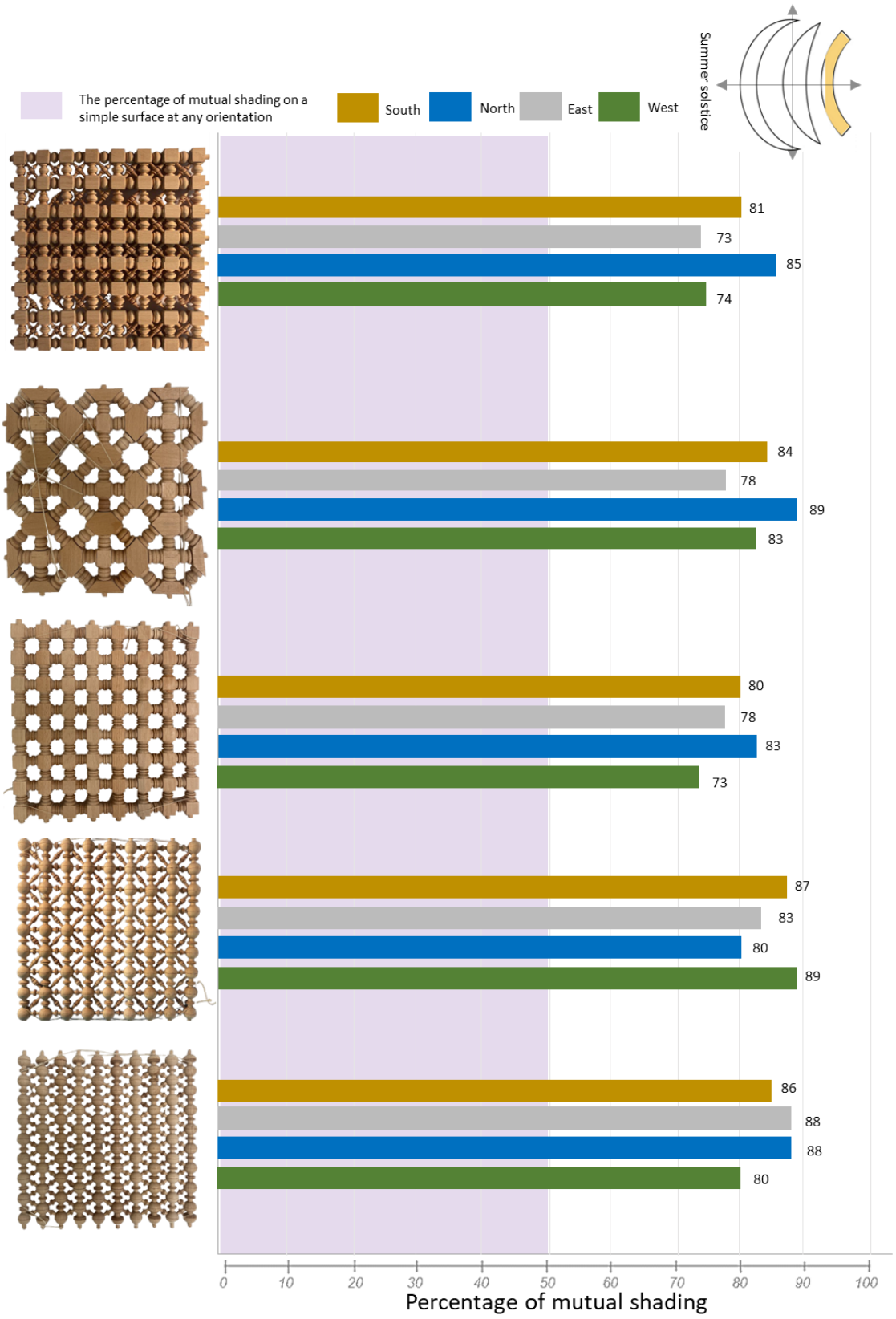


Figure 114: Percentage of mutual shading for screens—1~5 during the summer solstice across different orientations, extracted from CS analytical studies—(Author)

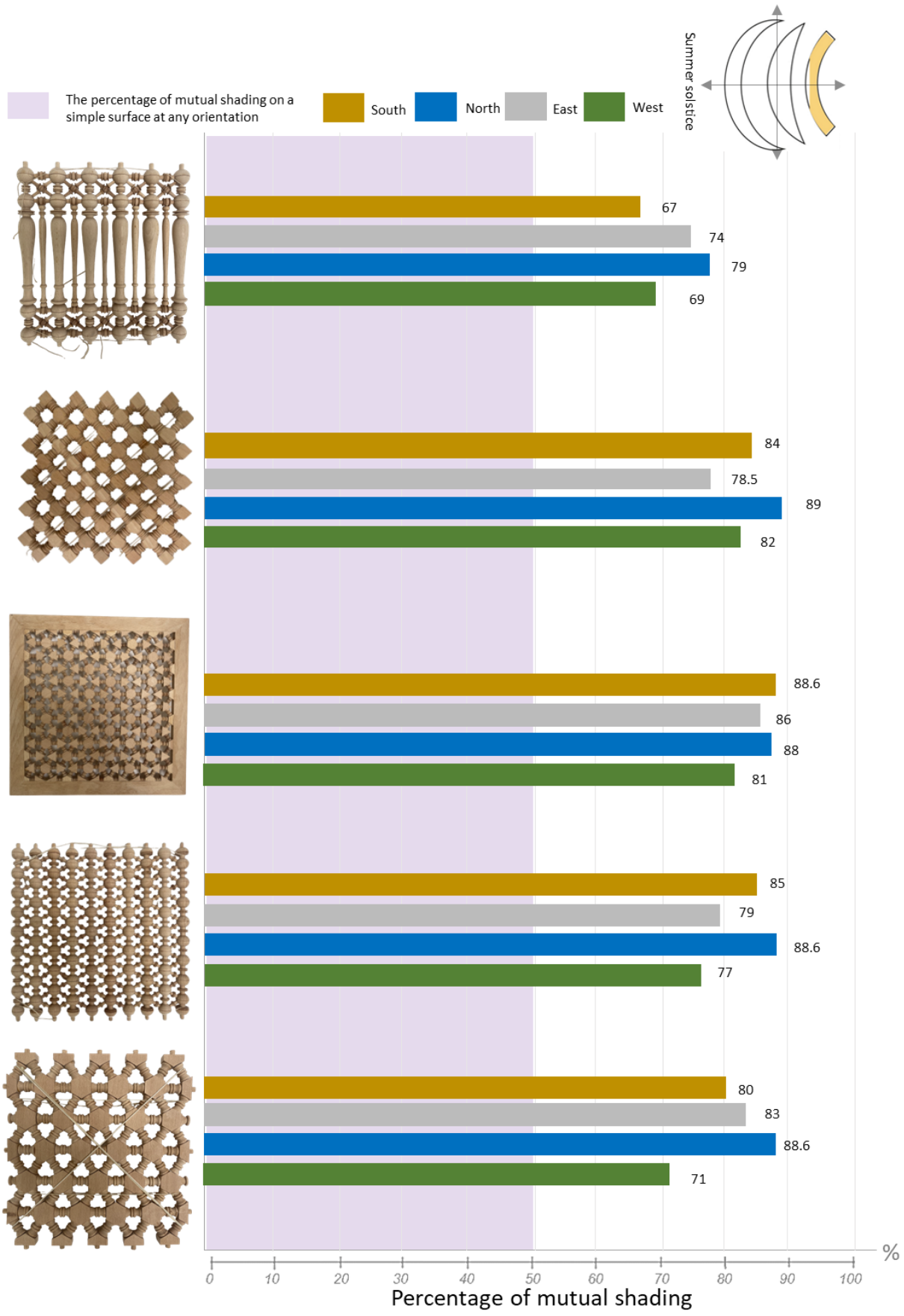


Figure 115: Percentage of mutual shading for screens 6~10 during the summer solstice across different orientations, extracted from CS analytical studies—(Author)

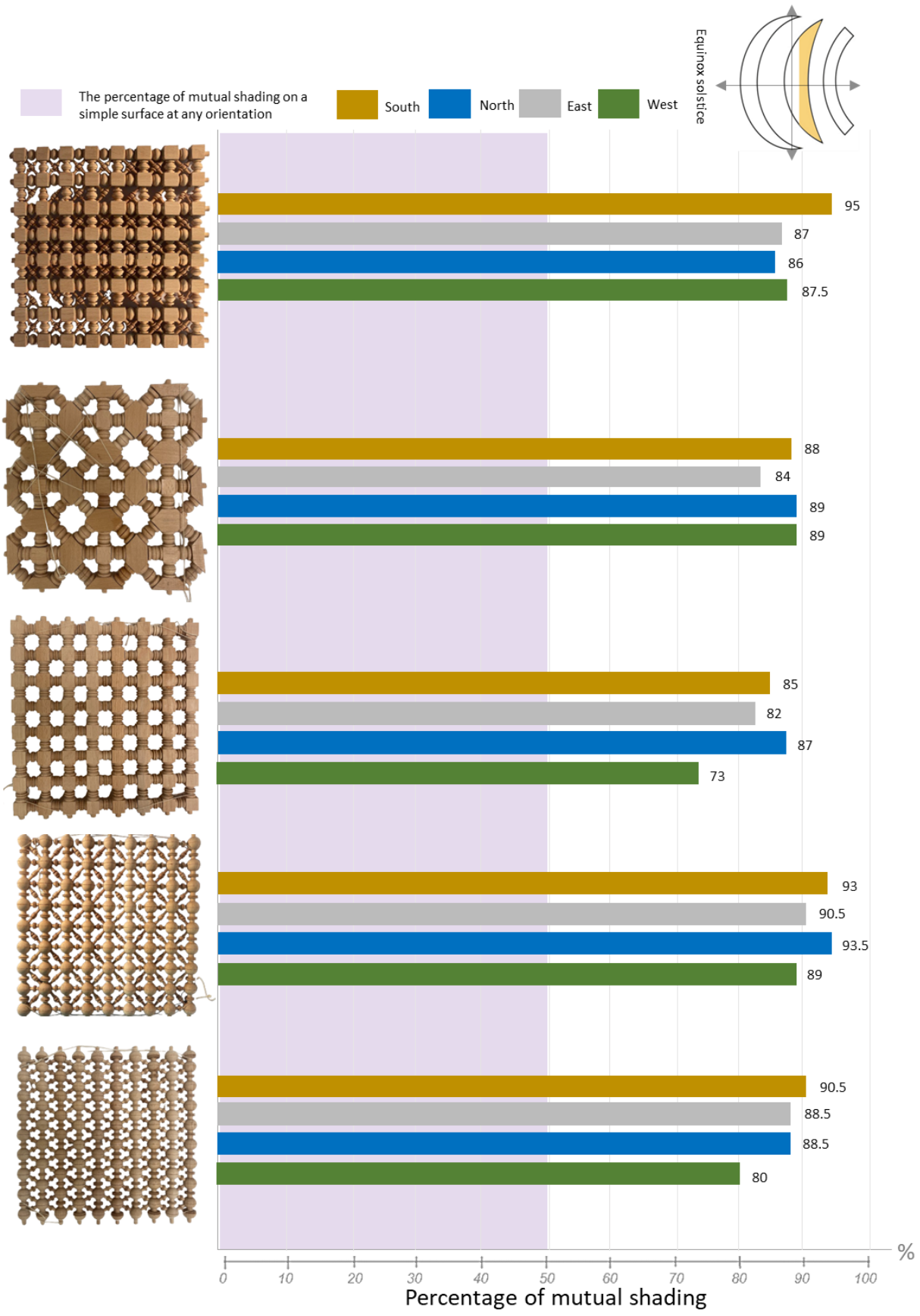


Figure 116: *Percentage of mutual shading for screens 1~5 during the equinox solstice across different orientations, extracted from CS analytical studies*—(Author)

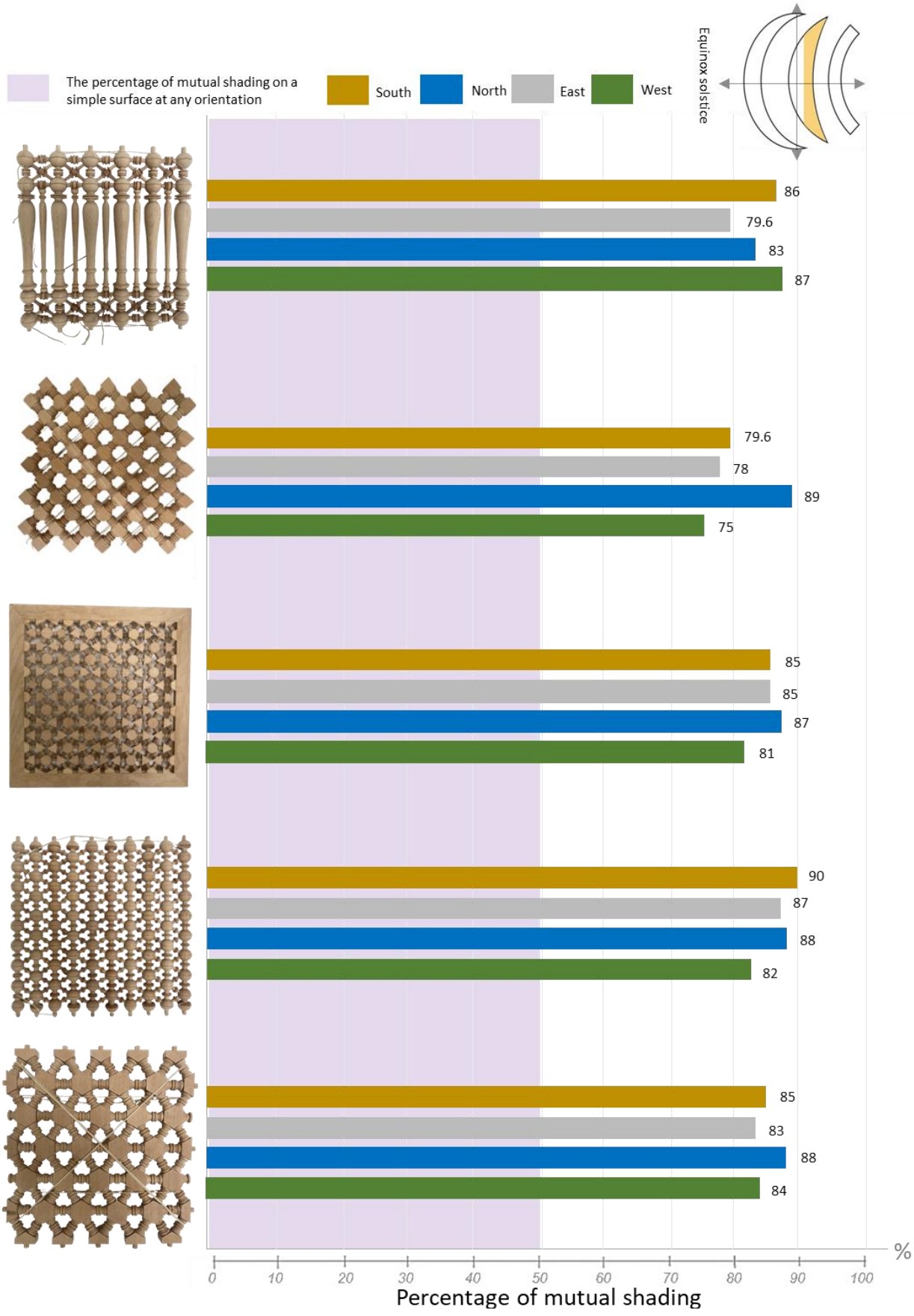


Figure 117: Percentage of mutual shading for screens 6~10 during the equinox solstice across different orientations, extracted from CS analytical studies— (Author)

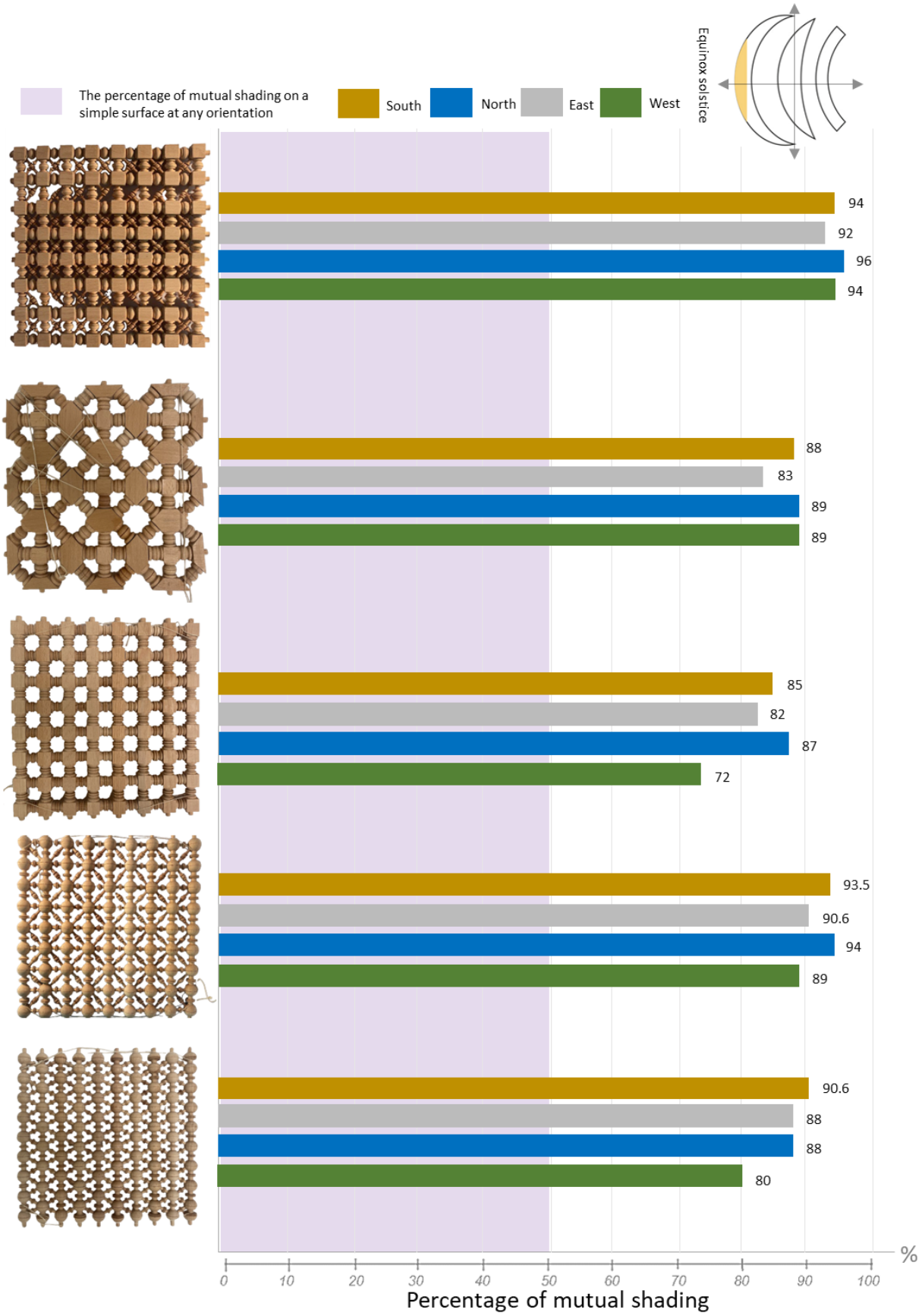


Figure 118: Percentage of mutual shading for screens 1~5 during the winter solstice across different orientations, extracted from CS analytical studies—(Author)

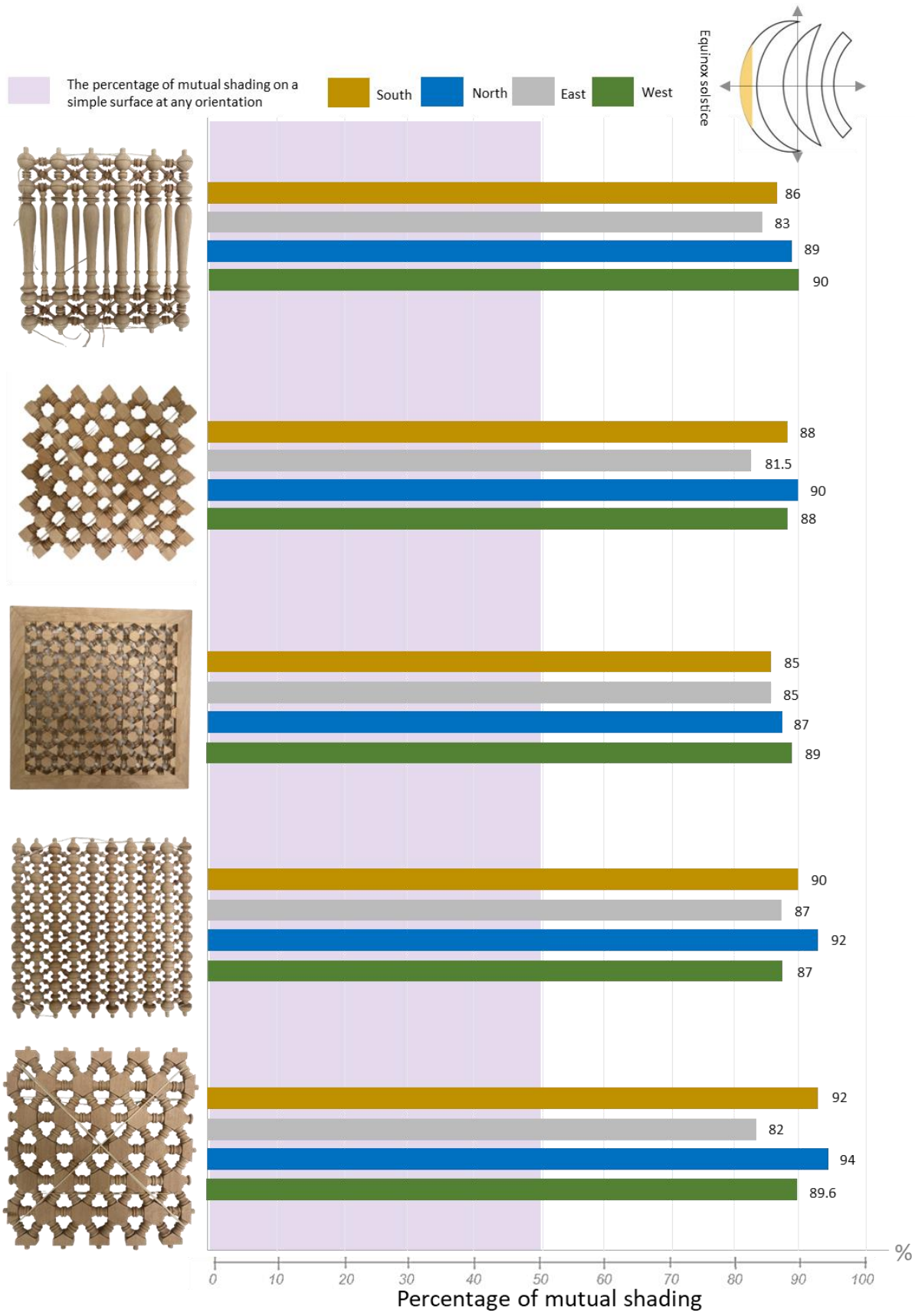


Figure 119: Percentage of mutual shading for screens 6~10 during the winter solstice across different orientations, extracted from CS analytical studies—(Author)

7.5.2 Solar Radiation Analytical Study of the Screens

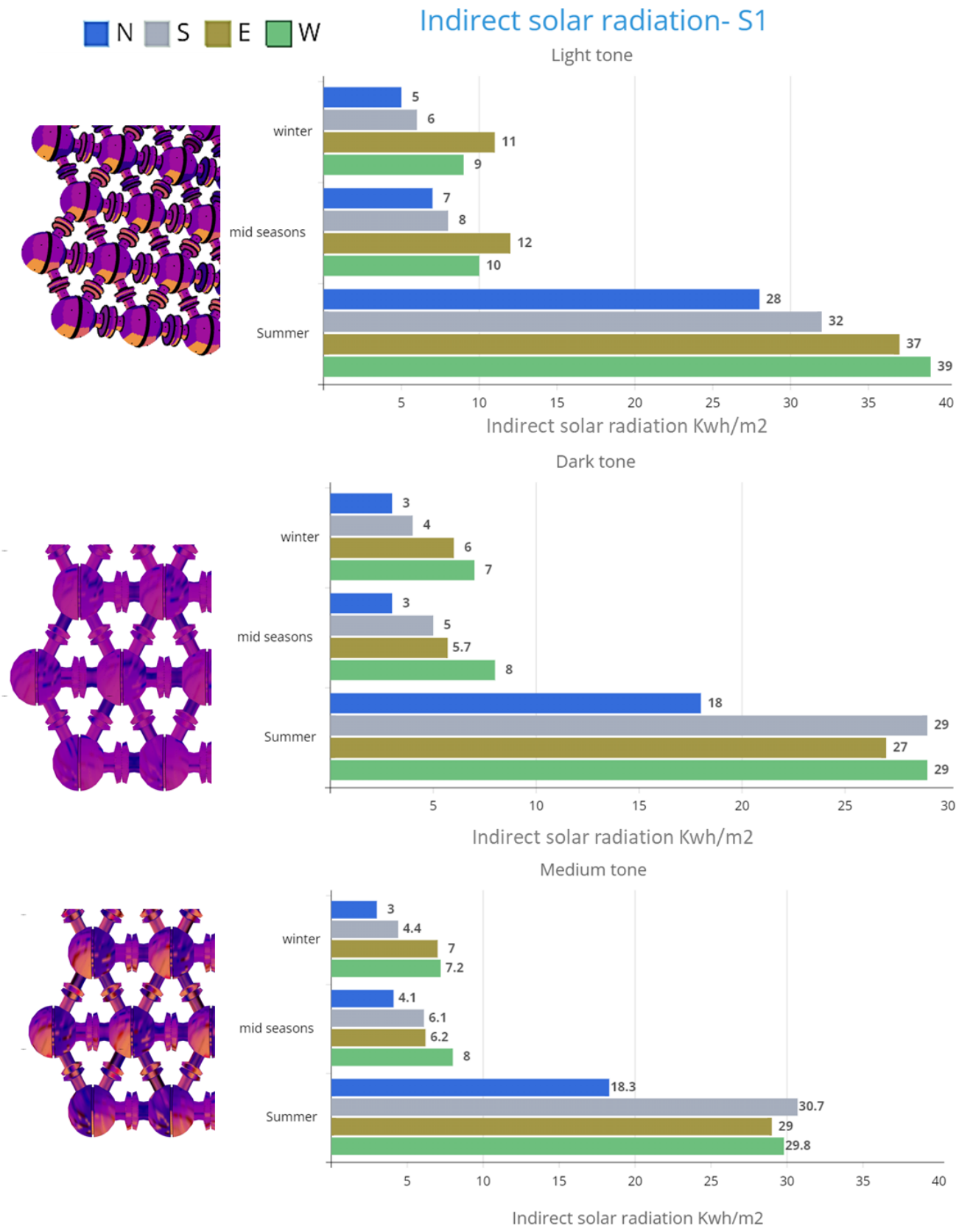


Figure 120: Indirect solar radiation results extracted from CS for sample 1, comparing dark, medium and light screen colours across different orientations to illustrate the impact of colour variation on final solar exposure outcomes — (Author)

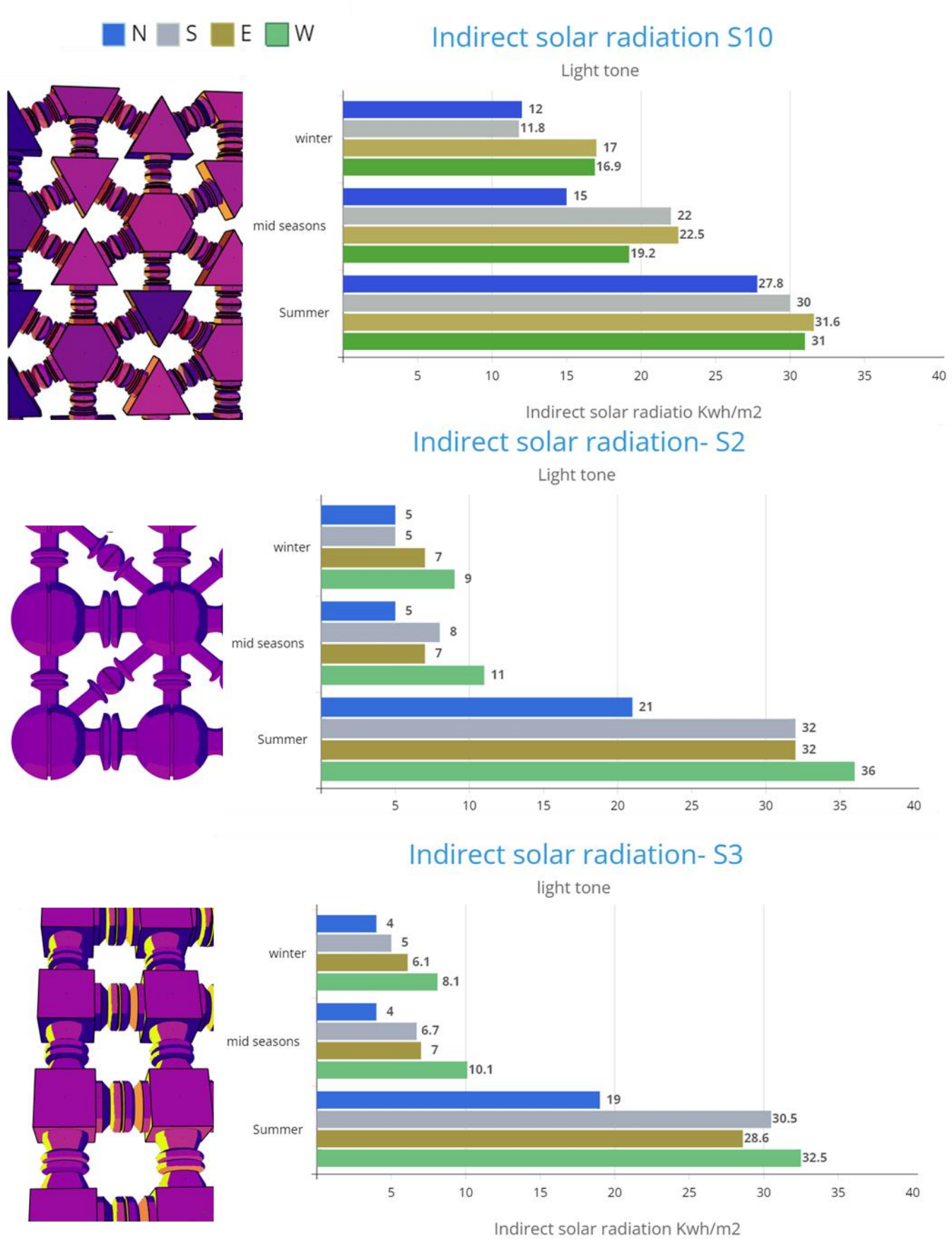


Figure 121: Indirect solar radiation results for each orientation for samples 10, 2, 3 — (Author)

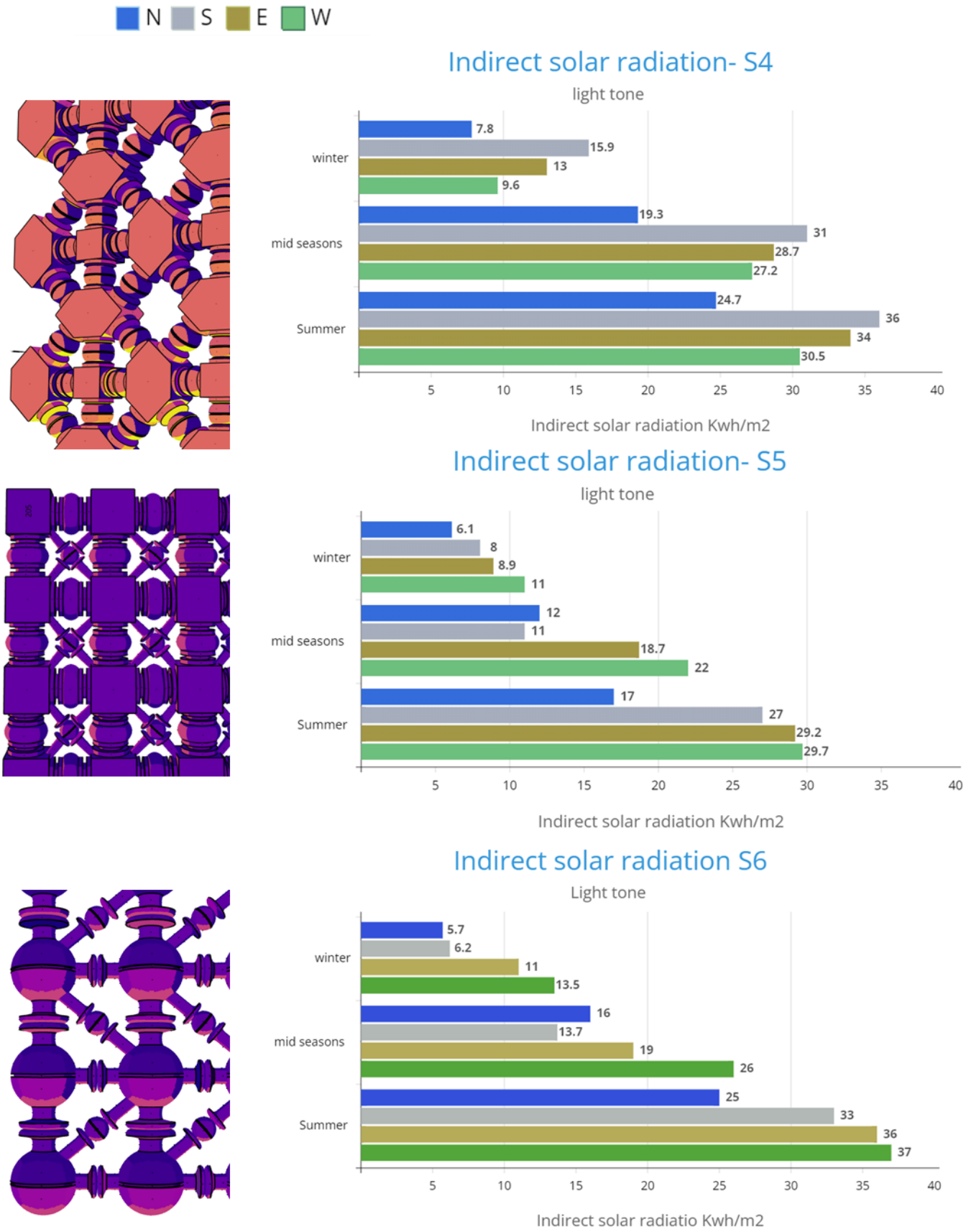


Figure 122: Indirect solar radiation results for each orientation for samples 4, 5, 6 —(Author)

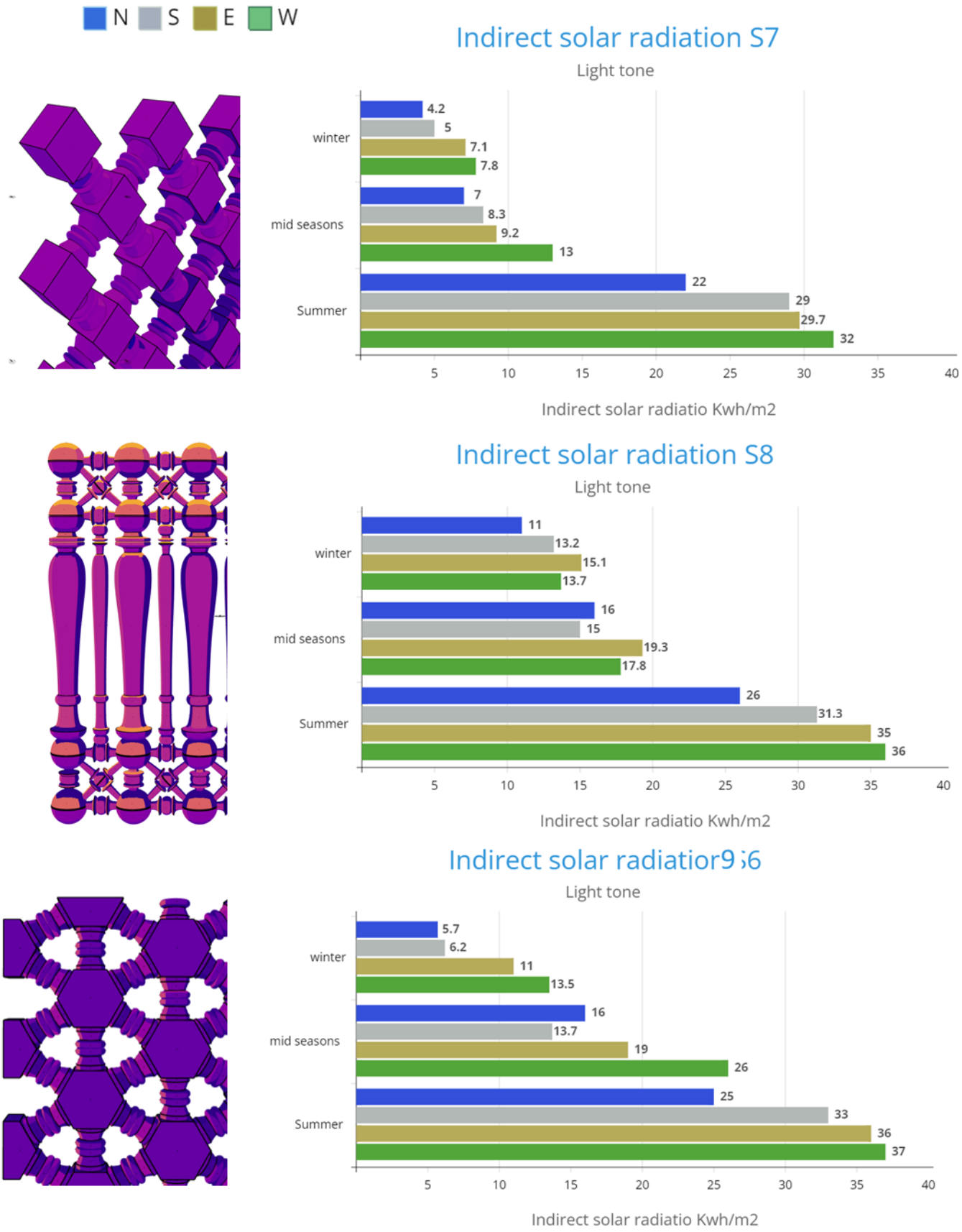


Figure 123: Indirect solar radiation results for each orientation for samples 7, 8, 9 — Source: (Author)

7.5.3 Aerodynamics Analytical Study of the Screens

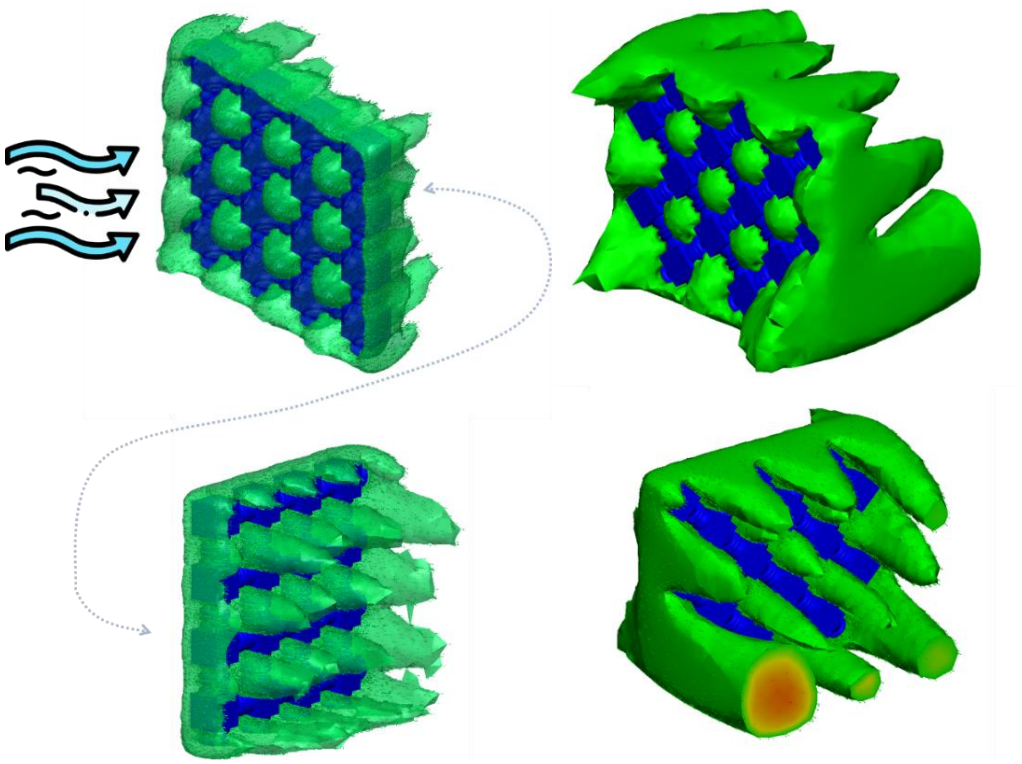


Figure 124: Volumetric airflow in CFD simulations through samples 3 and 7, illustrating changes in airflow and aerodynamic behaviour when components are attached diagonally rather than vertically— Source: (Author)

The value of the discharge coefficient may vary depending on the geometry of the device and the properties of the transported fluid. It can also be influenced by changes in fluid flow conditions. However, in this study, the fluid inputs and wind tunnel parameters were held constant across all screen samples. Therefore, the only variables affecting the resulting drag and discharge coefficient values were the geometry and configuration of the screen's forming components.

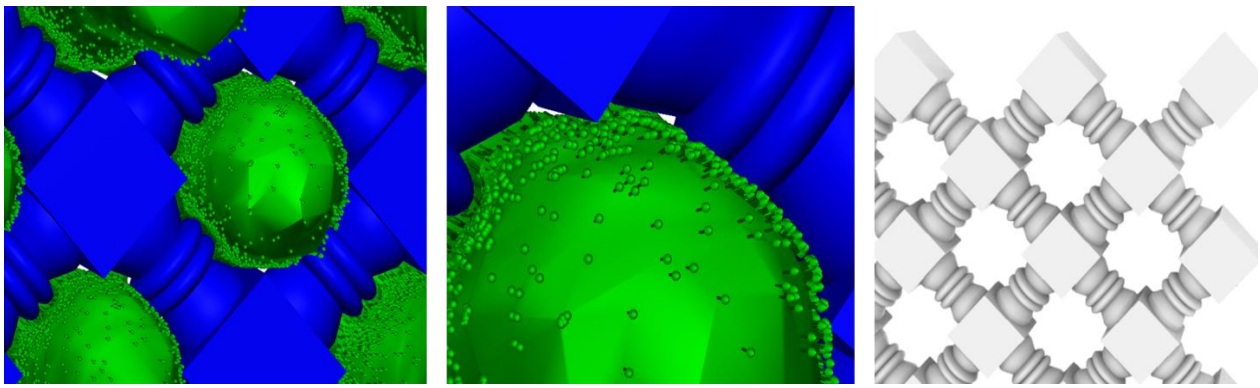


Figure 125: Volumetric airflow for the diagonal distribution of sample particles in sample 7, illustrating how the sharp edges of the geometry obstruct airflow across the pattern— Source: (Author)

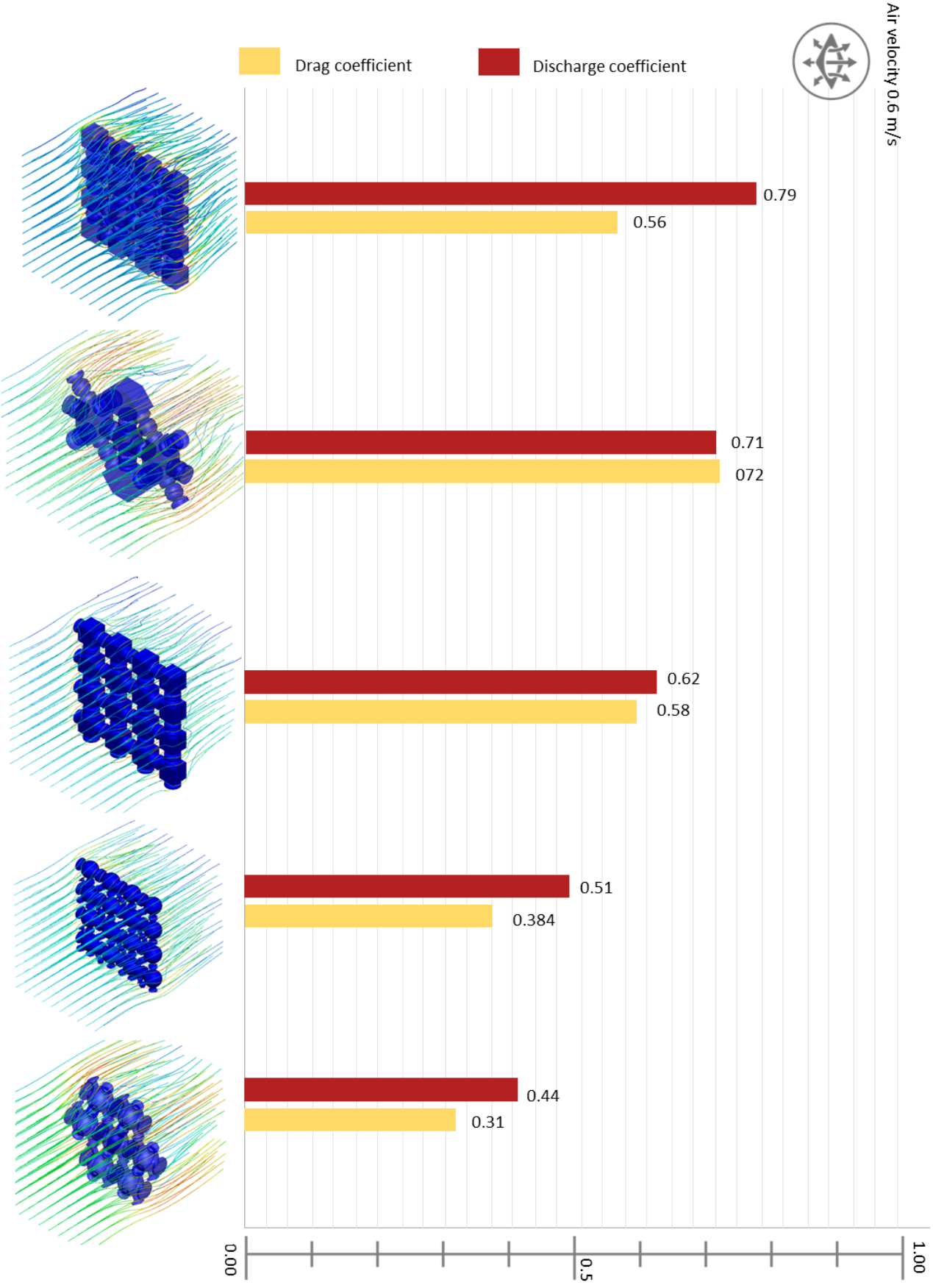


Figure 126: Calculation of drag and discharge coefficients from extracted CFD results for samples 1-5 — Source: (Author)

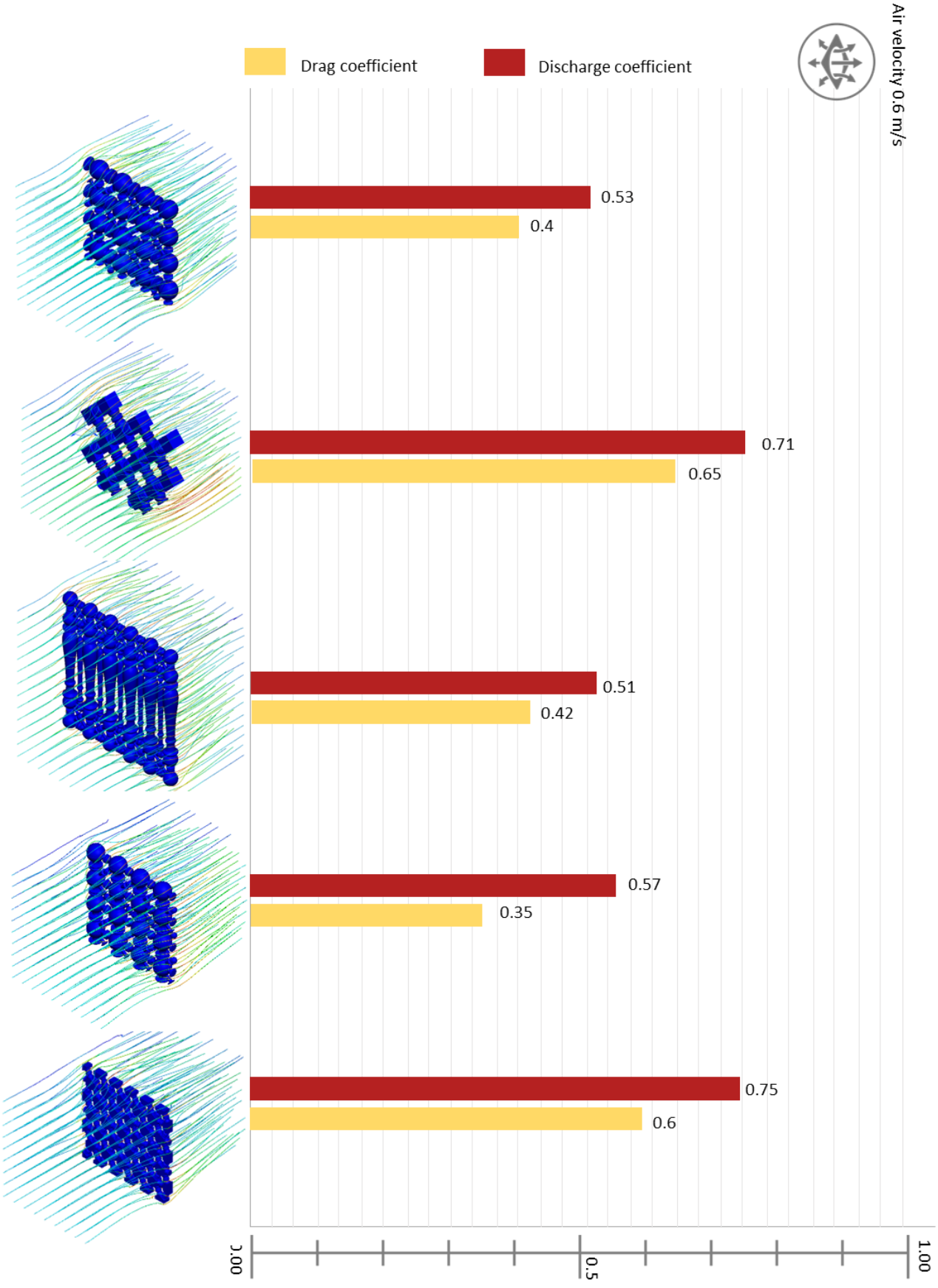


Figure 127: Calculation of drag and discharge coefficients from extracted CFD results for samples 6-10 — Source:(Author)

7.7. Emissivity Test

Analytical studies have provided insights that expand the foundation knowledge required to optimise the efficiency of the proposed screen designs. One key area of exploration involves examining changes in the emissivity based on the screen's colour, particularly when clay or ceramic is used as the primary material instead of wood. This investigation aims to assess how material and chromatic variations influence the thermal performance and overall results.

Another important approach to explore involves examining changes in ceramic porosity resulting from different fabrication methods and geometric configurations. This investigation will be conducted by experimenting with one of the clay-modelled samples and applying the following methods:

- Image analysis: This entails capturing microscopic photographs of the ceramic material and calculating pore size distribution and overall porosity using image analysis software. It also enables visualisation of the pore network, allowing assessment of capillary linearity.
- Water absorption: A commonly used test that involves weighing the ceramic sample before and after submersion in water for a specified duration. Porosity is then calculated based on the quantity of water absorbed by the material (Jastrzebski, 1959).

Although the insights gained from the analytical work were highly valuable in shaping the design concepts for Mashrabiya-inspired screens, further experimentation and analysis remain essential to ensure the efficiency of the proposed multi-functional evaporative cooling system. The screen design must balance multiple functions while maintaining optimal performance. Accordingly, it can be customised to provide shade, seclusion, or aesthetic impact by adjusting the size, shape, and spacing of the perforations to achieve the desired outcome.

In addition to these factors, close collaboration with clay specialists and a thorough understanding of design regulations for evaporative cooling systems will be essential to ensure that the perforated screen complies with all safety requirements, building codes, and regulatory standards. Further investigations will address water supply, application methods, structure support, and material durability. Multiple approaches will be tested to maximise overall system efficiency.

The analysis of selected Mashrabiya patterns provided essential insights into how form, colour, and the geometry of individual components influence screen performance in terms of shading and airflow. Subsequent analytical experiments were conducted to validate these correlations and assess their impact on overall efficiency, with the aim of fine-tuning the design specifications for ceramic screens. Accordingly, the following tests were implemented:

1- Emissivity test
2- Aerodynamics (as a part of the design development iterations)

3- Mutual shading (as a part of the design development iterations)

Understanding the correlation between the emissivity and reflectivity of wood and ceramics

This experiment adheres to the standard testing procedures employed in numerous research publications on the subject of "Experimental Studies Concerning the Impact of Solar Radiation on the Properties of Various Materials" (Nowo´swiat, 2019).

From the analytical studies of Mashrabiya, it was found that using lighter-coloured wood as the assigned material for the screen during solar radiation analysis in Climate Studio can behave unpredictably, and unexpected outcomes were observed. Mashrabiya was commonly found in mid-colour wood tones, yet some precedents were identified with lighter colours depending on regional application. Accordingly, it was important to test and analyse all the identified wood colours.

Some of the samples are characterised by lower perforation percentages (e.g., patterns 4, 6, and 10) when lighter material is assigned to the screen. When running the solar radiation analysis in Climate Studio, the screens behave in contradiction to the expected outcome. Lighter colours are generally preferred as construction materials in hot, dry climatic contexts due to their thermal characteristics and behaviour towards solar radiation, where brighter colours are more reflective and their ability to absorb heat is lower than that of darker colours, in relation to the emissivity of each material and colour. In the case of Mashrabiya, some screens have parts that formalise the screen closer to each other, which makes it tricky when lighter colours are used, since the parts are more likely to reflect solar radiation onto adjacent components instead of reflecting it outward to the external context—leading to the opposite of the anticipated result and contradicting the main function of the Mashrabiya. Understanding the contradictory behaviour of the screens in relation to the colour of the material initiated significant concern regarding the colour of the clay that will be used for the evaporative cooling screen.

The lighter colours of the wooden screen's performance initiated a question regarding the material used and its colour, since it was found that it can have a major impact on performance.

7.7.1 Objectives and Methods

Objectives

Designing a pattern used for shading and evaporative cooling can be challenging in terms of materiality. Although it is clear that clay will be used, clay comes in different colours and tones that may affect the probability of reflecting some of the solar incident rays within the screen itself. Accordingly, different colours of clay were used to fabricate the same geometry and were exposed

to the same amount of solar incidence to help determine the material properties and colour characteristics of the clay that will be used in the fabrication of the prototype.

Methods

To understand the performance of the material with different colours under direct and indirect solar exposure, it was more reliable to use a typical summer sunny day and natural sun exposure as the main tool in the experiment. Since the study is a comparative one that relies mainly on comparing how the different colours of the material vary in reflectivity and emissivity, it was not crucial to focus on the climatic context or the location of the experiment, but only on the availability of sunlight and the intensity of solar radiation. Accordingly, a typical summer sunny day was chosen during the peak hour of the day (1 p. m.) at a site with no surrounding obstructions (the centre of Primrose Hill Park), and the tools used were:

i- **Thermal camera** to detect the gradual variation in the surface temperature of the samples, based on the continuous detection of infrared wave variations emitted from the samples.

ii- **Aftab application** to create HDR images, where high dynamic range imaging captures a wider range of illuminance levels and helps define the reflectivity of the material colours, correlating the relationship between reflectivity and the emissivity.

7.7.2 Materials and Geometry

Material	Reflectivity
a. Wood - Pine	0.52
b. Wood – Tudor oak	0.31
c. Wood – Walnut	0.24
d. Grey Clay	0.51
e. Red Clay	0.30

Figure 128: *The materials used in the emissivity test experiment were a, b and c, representing different colours of wood, while d and e represented different colours of clay* — Source: (Author)

Geometry

Designing the geometry started with developing a two-dimensional representation for each material and testing how these surfaces can successfully facilitate a comprehensive understanding of emissivity variations induced by solar exposure, as shown in the following image.

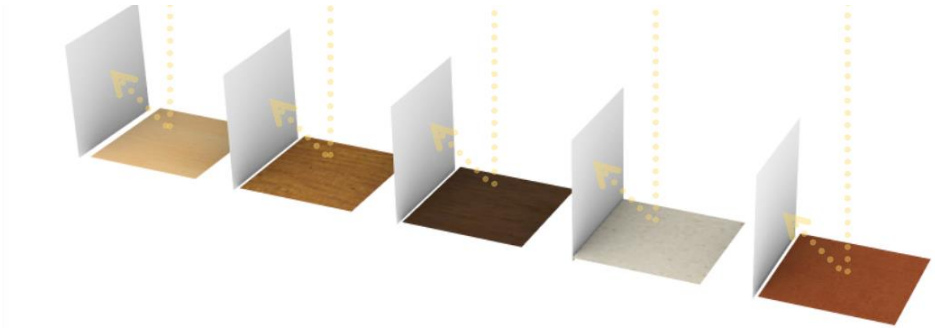


Figure 129: *Illustrates how different materials can be modelled on a two-dimensional surface to test variations in emissivity due to solar exposure— Source:* (Author)

Accordingly, the geometry was developed to maximise the exposure of different solar angles in order to understand how emissivity varies from one material to another. A spherical or cylindrical form proved more efficient than a planar surface, since spheres and round surfaces are three-dimensional objects and are more exposed to varying sun angles than flat or planar surfaces. This is due to the nature of the geometry, which presents different angles of solar incidence. A flat surface, on the other hand, has a constant angle of exposure. Additionally, the spherical shape maximises solar exposure by allowing it to absorb sunlight from a greater range of angles. In contrast, a flat surface has a limited capacity to absorb sunlight, particularly when it is incident at an acute angle, as shown in the following image, where Climate Studio solar radiation analysis visualises the variation between the two geometries when exposed to direct and indirect sunlight.

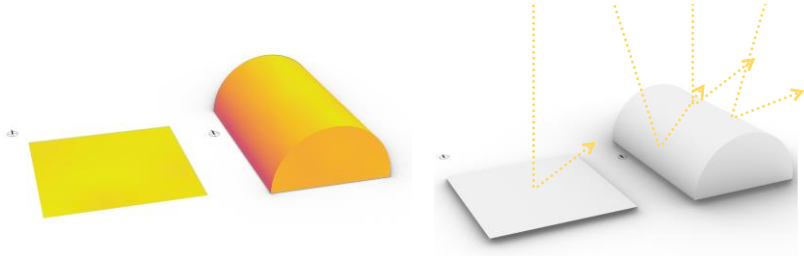


Figure 130: *Illustration of how solar exposure results can vary between a planar surface and a cylindrical form*— (Author)

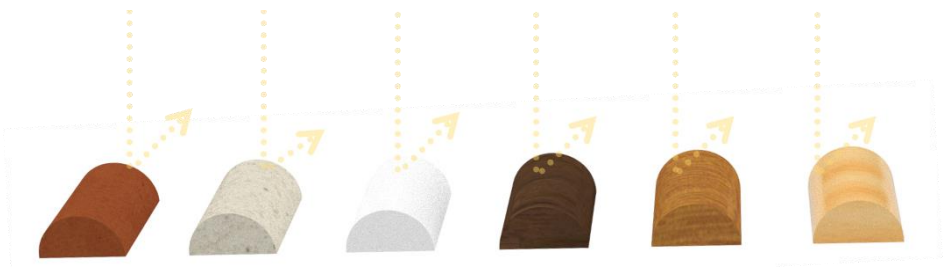


Figure 131: *The finalised geometry of each material with the different colours for the emissivity test*— (Author)

Figure 132: *Images of the realistic finalised geometry of each material with the different colours for the emissivity test*— (Author)

7.7.3 Results

When ceramics of different colours were examined, the results varied and did not follow the patterns observed in samples of wood, except in the case of dark-coloured clay. As expected, dark-coloured clay exhibited an emissivity that was comparatively greater than that of lighter colours, which was consistent with the findings from the wood samples.

As a result, lighter ceramic tones were found to have lower emissivity ratings than their darker substitutes. Lighter ceramic colours also exhibited better reflection than darker colours, according to a more thorough investigation of reflectivity. The lighter-coloured ceramics sustained lower reflectivity than that of wood; however, this is notable, as it shows a 32% reduction in reflected solar radiation when compared to the same-coloured wood. These results, which came from thorough research in Climate Studio, highlight the complex relationship between material colours and performance under solar exposure.

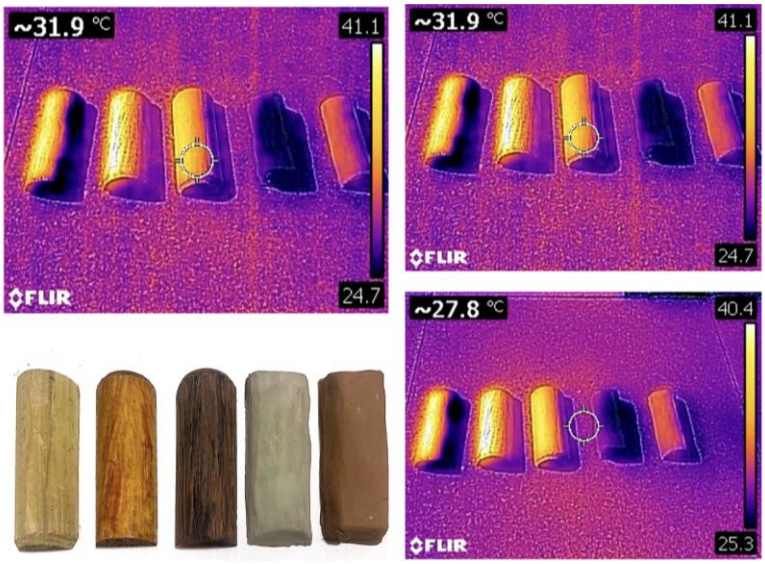


Figure 133: *Infrared thermal images and realistic images of the samples for three different colours of wood and two different colours of ceramic*—source: (Author)

To better understand how colour variation affects emissivity, three different ceramic colours were investigated in a rerun of the experiment. The objective of this experimental phase was to methodically examine the ways in which the emissivity of the resulting geometric formations is affected by the blackness of the ceramic colour. Although it was feasible to predict certain results based on previous data, the main goal of this experiment extended beyond investigating the correlation between emissivity and the colour of the material. Instead, the aim was to explore the subtle colour and physical qualities that the clay has to offer. This investigation serves two purposes: it adds to our knowledge of emissivity variations and provides important information on the unique properties of the clay that will be crucial for building the prototype. Therefore, it becomes easier to determine the material properties and colour characteristics of the clay that will be used in the fabrication of the prototype.

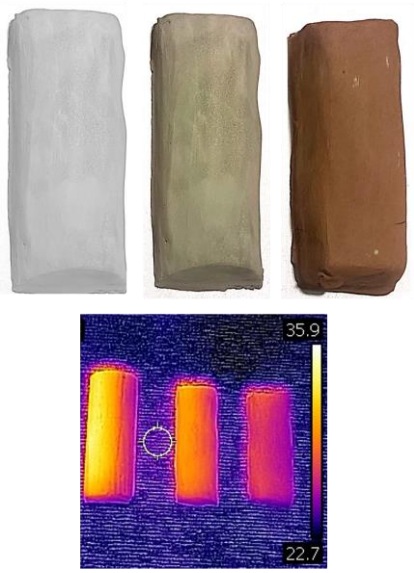


Figure 134: Infra*red thermal images and realistic images of the samples for three different colours of ceramic* — (Author)

Findings

It was important to understand how the change in colour in the wooden screens could impact performance due to the change in emissivity, and to correlate this with the use of clay as the fabricating material. The test was not directly concerned with the design of the screens, but it added to the analytical process by providing a clearer understanding of material behaviour, which in turn supports an efficient design process. The natural characteristics of ceramics are crucial in establishing emissivity. The results indicate that, unlike wood, the emissivity of ceramic material is not significantly affected by colour. The experiment showed that dark-coloured clay, such as the red clay used here, exhibited relatively lower emissivity and reflectivity compared to wood. This realisation opens possibilities for adaptability in developing the ceramic prototype, since darker hues of clay may be used without compromising performance, while lighter colours can still be prioritised for maximum efficiency. Interestingly, even lighter ceramic colours show lower reflectance than wood, partly due to the matte quality of the surface. These empirical findings differ from the expectations of computer-based modelling programmes such as Climate Studio, which tend to classify wood and clay within the same reflectance range per colour. The results therefore highlight the importance of understanding physical properties through direct testing, as real-world behaviour can contradict projected outcomes from virtual simulations.

7.8 Conclusion

This chapter marks a pivotal advancement in leveraging comparative data on drag coefficients across various geometries under identical conditions, aiming to assess the efficacy of each form in reducing drag and to identify the geometry with the lowest coefficient. Analytical studies yielded numerous observations that inform the design guidelines for a multifunctional evaporative cooling screen, integrating principles derived from the Mashrabiya. Accordingly, the analytical work proved instrumental in defining the key features and generic parameters of the proposed ceramic screen. The following features were concluded and incorporated, alongside additional inputs intended to enhance the overall performance of the ceramic prototypes:

1. Material Selection: The choice of material for the perforated screen significantly influences its durability, maintenance requirements, and overall performance in terms of shade, ventilation, and daylighting. Colour also plays a critical role in the absorption and reflection of solar radiation, thereby affecting the building's thermal behaviour and energy efficiency. Below are key considerations related to colour and solar radiation:

- **Reflection:** A material's colour significantly influences the amount of solar energy it reflects. Lighter colours tend to reflect more solar radiation than darker ones, thereby reducing heat gain within the structure. Reflective coatings may also be applied to enhance the material's solar reflectivity. This phenomenon was observed in samples 1, 2 and 6, where the inner parts of the screens—when simulated in lighter colours—negatively impacted overall performance by reflecting solar angles back onto themselves. This behaviour contradicts the principle of mutual shading inherent to Mashrabiya screens, as illustrated in the following image:

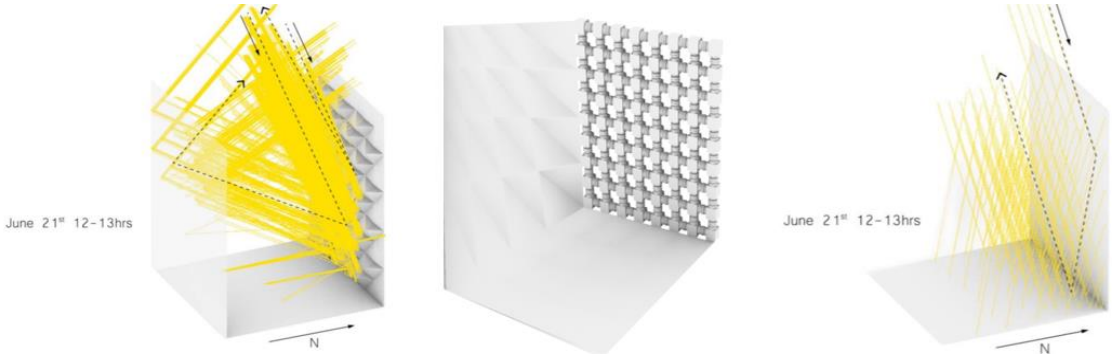


Figure 135: *Illustration showing the reflection of solar radiation across different parts of the screen, based on solar vectors recorded over a 13-hour period on June 21st*— (Author)

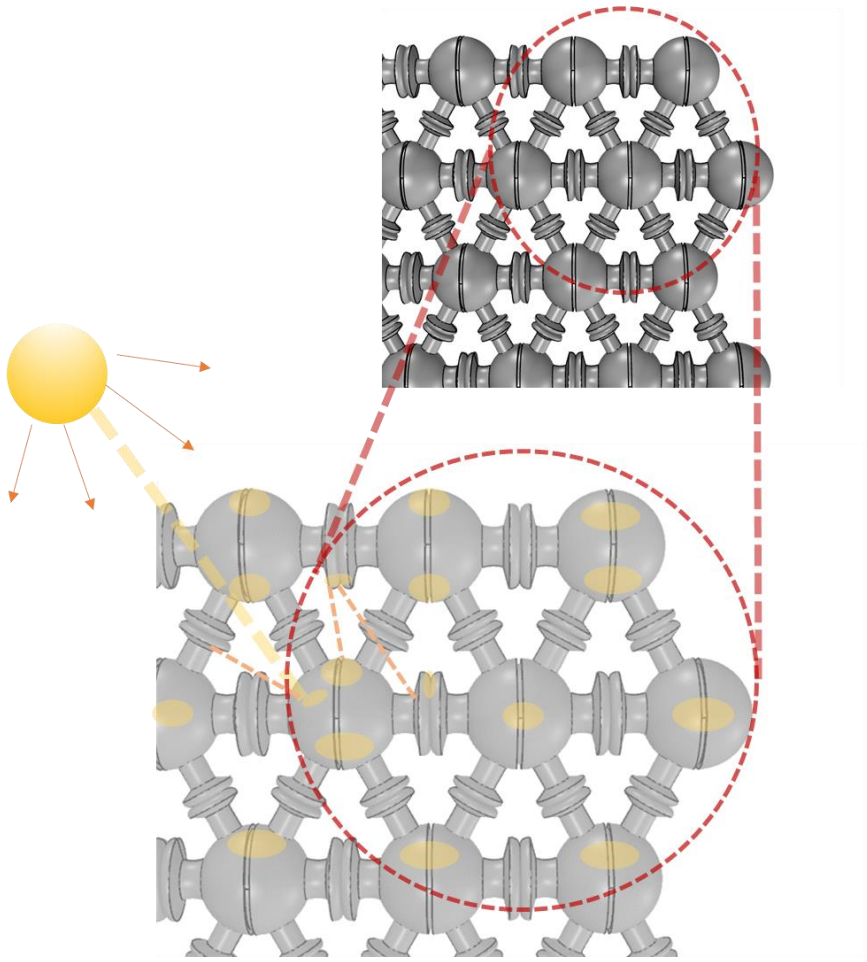


Figure 136: Illustration showing how different parts of the same screen receive varying levels of solar radiation and how mutual shading occurs between its components — (Author)

- **Absorption:** The amount of solar energy absorbed by the screen is influenced by its colour. Darker tones, such as black or dark blue, absorb significantly more solar radiation than lighter hues like white or pale yellow. This increased absorption leads to higher surface temperatures and greater heat uptake within the structure.

The analytical shading studies focused exclusively on quantifying direct and indirect solar exposure on the screens, measured in WH/m^2. The average values of indirect solar exposure varied according to the materiality and colour of the screens. It was found that indirect solar exposure was directly proportional to screen colour, with lighter tones exhibiting higher average values. These results are logically consistent to a certain extent, as lighter-coloured screens—with higher reflectivity—tend to receive greater solar reflection across their surfaces. However, these findings only establish a correlation between screen colour and solar exposure, specifically in terms of solar radiations reflected between screen components (solar reflection), without accounting for the emissivity of the materials. While the external colour of the screen contributes to its aesthetic and visual impact, it is essential to strike a balance between beauty and functionality. The following

diagrams illustrate the correlation between solar reflection and the colour tone of the wood.

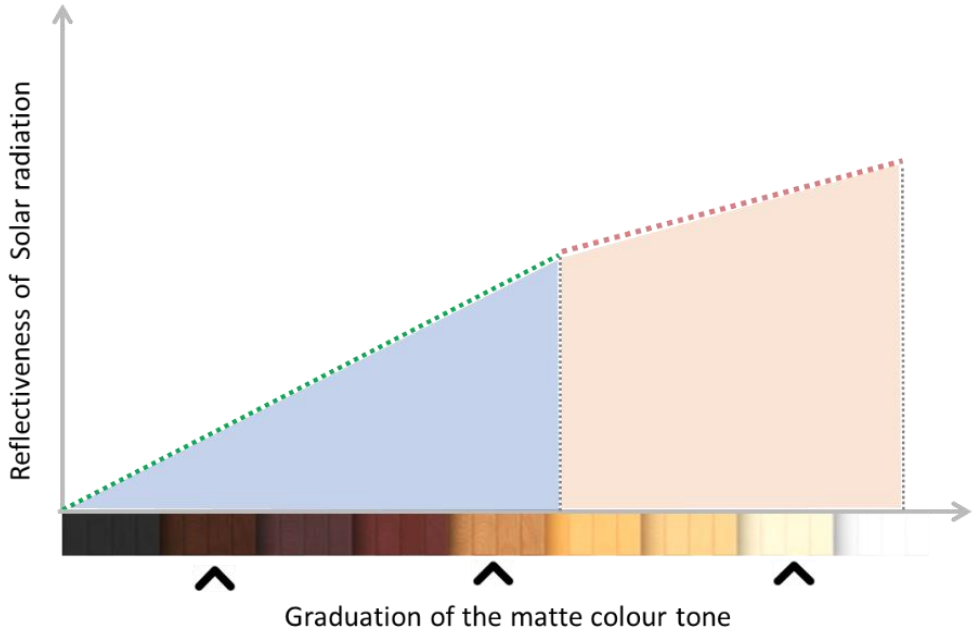


Figure 137: *Graph illustrating the correlation between the colour tone of the screens and solar reflection, based on extracted inputs form Climate Studio analytical simulations results* — (Author)

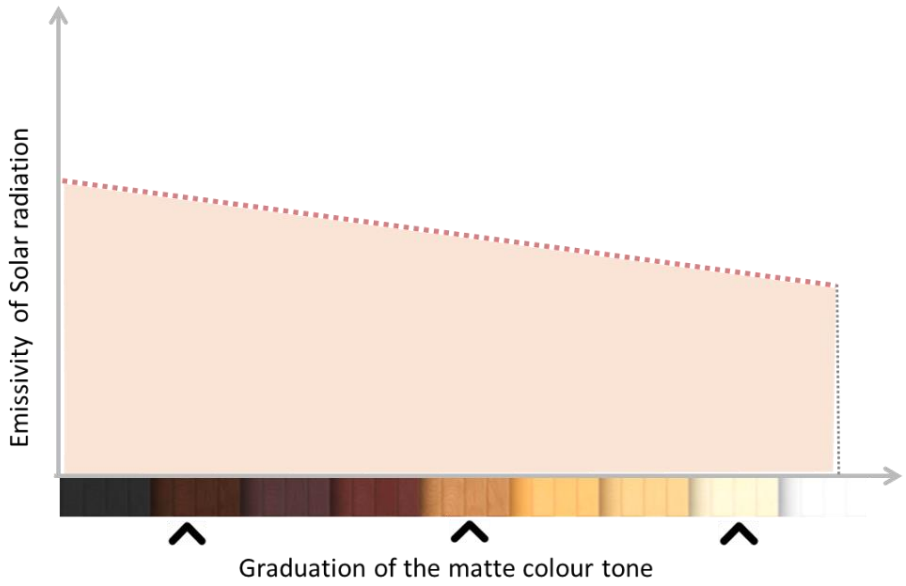


Figure 138: *Graph illustrating the theoretical correlation between the colour tone of the screens and emissivity, based on extracted inputs from the table below* — (Author)

Table 7: Comparison *between emissivity and solar reflection—ASHREA*

Sample	Emissivity	Solar reflectance
Cool Yellow	0.92	0.52
Standard Yellow	0.87	0.31
Cool Black	0.90	0.06
Standard Black	0.91	0.05
Cool Brown 1	0.90	0.29
Standard Brown 1	0.92	0.85
Cool Brown 2	0.89	0.24
Standard Brown 2	0.89	0.18

Emissivity can be described as the material's capacity to emit thermal radiation, and it is influenced by colour, intrinsic material properties, and climatic context. Materials with higher emissivity release more thermal radiation, thereby reducing heat gain on the designed component. This is particularly critical for building envelopes and façade elements exposed to solar radiation, such as roofs and walls (Colt, 2020). Consequently, understanding the impact of colour and materiality on emissivity is essential when designing a passive evaporative cooling screen, where solar exposure, airflow rate, and heat exchange are the primary influencing factors.

It is widely accepted that the most suitable colour for a structure is determined by its climatic context and geographical location. In hot regions, lighter colours are generally more effective in reducing heat gain, while darker tones may prove beneficial in cooler climates by enhancing thermal absorption.

2. Considerations for Design: The design of the perforated screen—particularly the size and arrangement of its holes or slots—significantly affects its performance in terms of shading, ventilation, and daylighting.

   The design of the screens, along with variations in the levels of extrusion across different components, has a significant impact on shading performance and the distribution of solar exposure, as illustrated in the following images. These design features can be strategically employed to reduce solar heat gain and glare, while simultaneously enhancing natural daylighting and ventilation. Moreover, they serve to maximise mutual shading—where screen elements shade one another—thereby improving thermal efficiency. This mutual shading effect is particularly advantageous in the development of a multifunctional ceramic evaporative cooling screen.

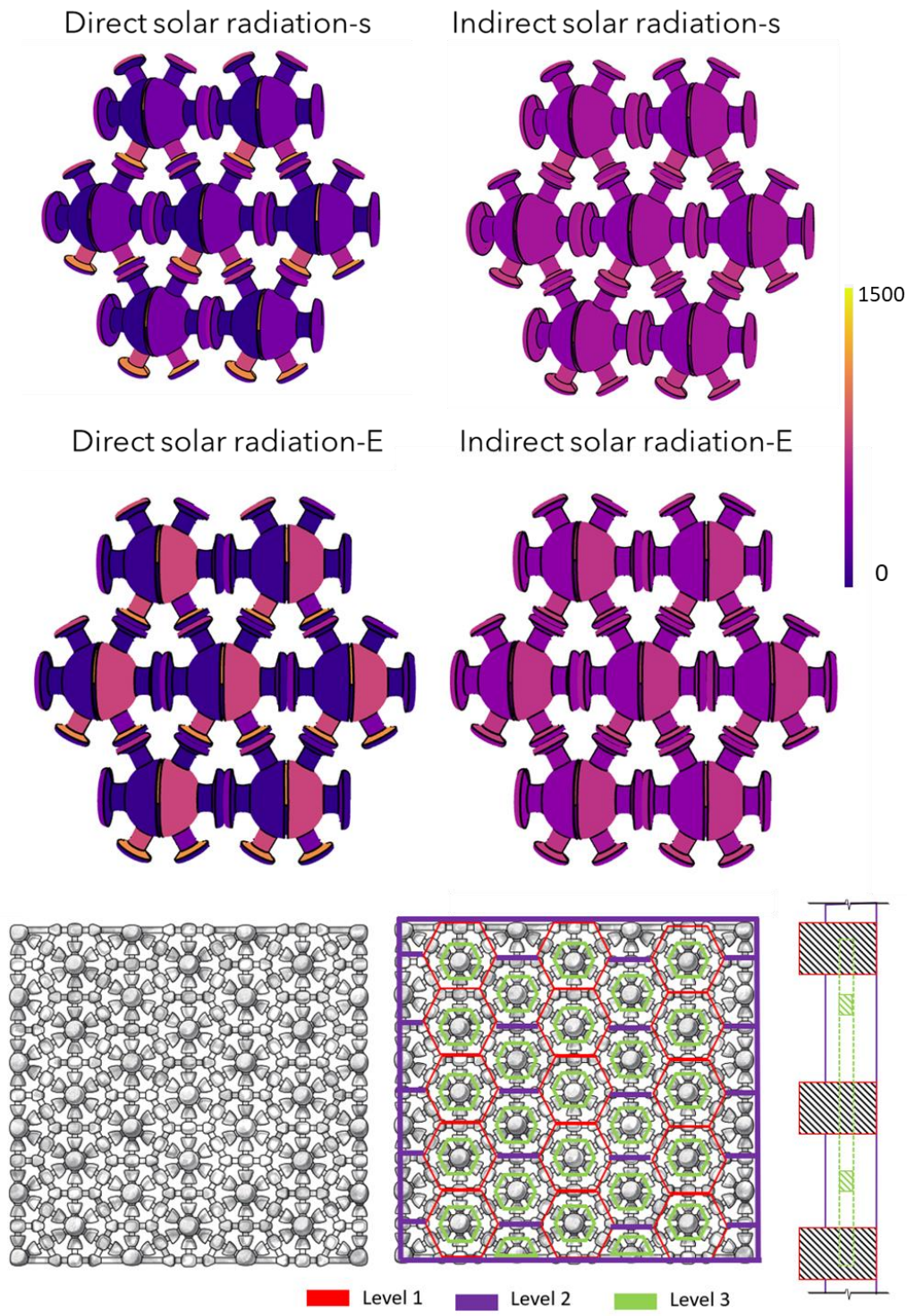


Figure 139: *Illustration showing how different extrusions across various screen components influence solar exposure outcomes. The top images present results from the Climate Studio solar radiation analytical study for a sample-oriented eastward* – (Author)

In summary, perforated screens offer a versatile and effective shading solution, providing both functional and aesthetic advantages to building façades. To ensure optimal efficiency and cost-effectiveness, their design and installation must be carefully considered. Colour plays a pivotal role in influencing both energy performance and visual impact and should be addressed thoughtfully during the design and construction phases. The selection of colour and material properties should be guided by multiple factors, including climate conditions, energy efficiency objectives, and desired performance outcomes. Moreover, mutual shading among extruded parametric screens presents a promising strategy for reducing solar heat gain and enhancing overall screen efficiency. To achieve the best balance between performance and aesthetics, the design must account for variables such as solar angles, material selection, structural integrity, and visual composition.

However, the Mashrabiya demonstrated remarkable performance in mutual shading across varying solar angles and orientations throughout the year in hot, dry climates. The screen successfully shaded over 35% of the primary exterior surface exposed to direct and indirect solar radiation, revealing valuable insights for the design of multifunctional evaporative cooling screens that eliminate the need for external shading devices. This outcome also serves as a compelling precedent for façade design strategies that address multiple objectives and complex challenges faced by contemporary designers. Moreover, it confirms that such subtle design interventions at a micro scale can significantly influence the overall of the screen, ultimately contributing to the development of efficient, multifunctional passive environmental solutions.

To comprehend the aerodynamics of a perforated screen, it is essential to consider the geometric variations that influence its performance—namely, the size and shape of the apertures, the spacing between perforations, and the thickness of the screen. These variables can affect the fluid's flow behaviour, and the thickness of the screen. The parameters affect fluid flow behaviour and contribute to pressure loss across screen, which can diminish its ventilation efficiency. To mitigate pressure drop, it is advisable to employ screens with wider apertures or reduced thickness, though this may increase solar access to the interior spaces where the screens are installed. In general, screens with more streamlined geometries exhibit lower drag coefficients than those with more blunt or angular forms. Additionally, increasing fluid flow velocity tends to raise the drag coefficient, while greater surface roughness of the object can also contribute to elevated drag values, as illustrated in the following images.

The selected samples featured a rich array of details that yielded diverse analytical data and valuable insights. Some screens incorporated sharply carved, angular components, while others included

spherical or smoothly contoured elements. Although certain samples shared similar carved features, their distribution and final assembly varied, as illustrated in the following image. Softened and sharp geometries represent two distinct formal typologies that can significantly influence the airflow characteristics within fluid dynamic systems.

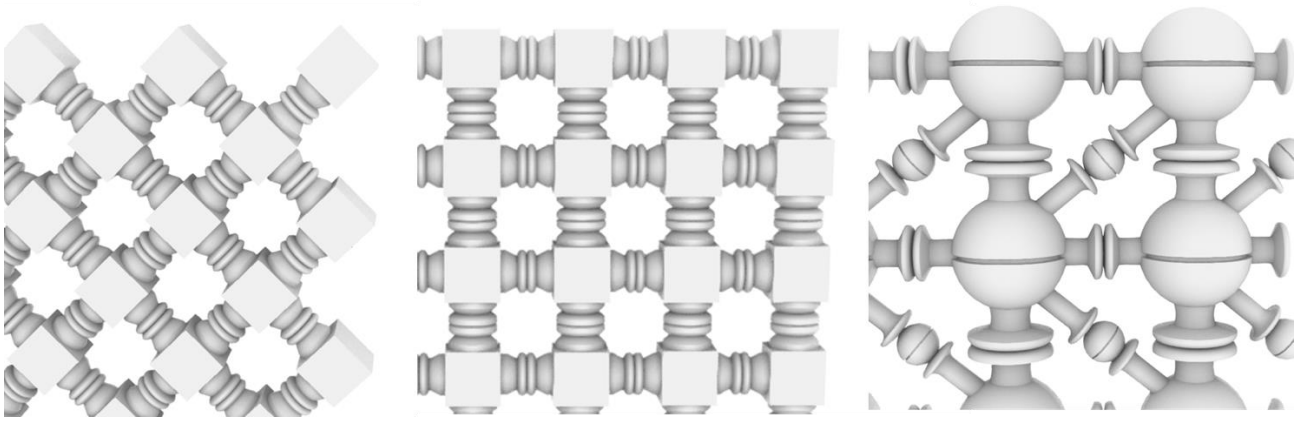


Figure 140: images of the scanned samples illustrating shape variations and how similar components can be assembled differently to produce distinct outcomes, as demonstrated in the first two images— (Author)

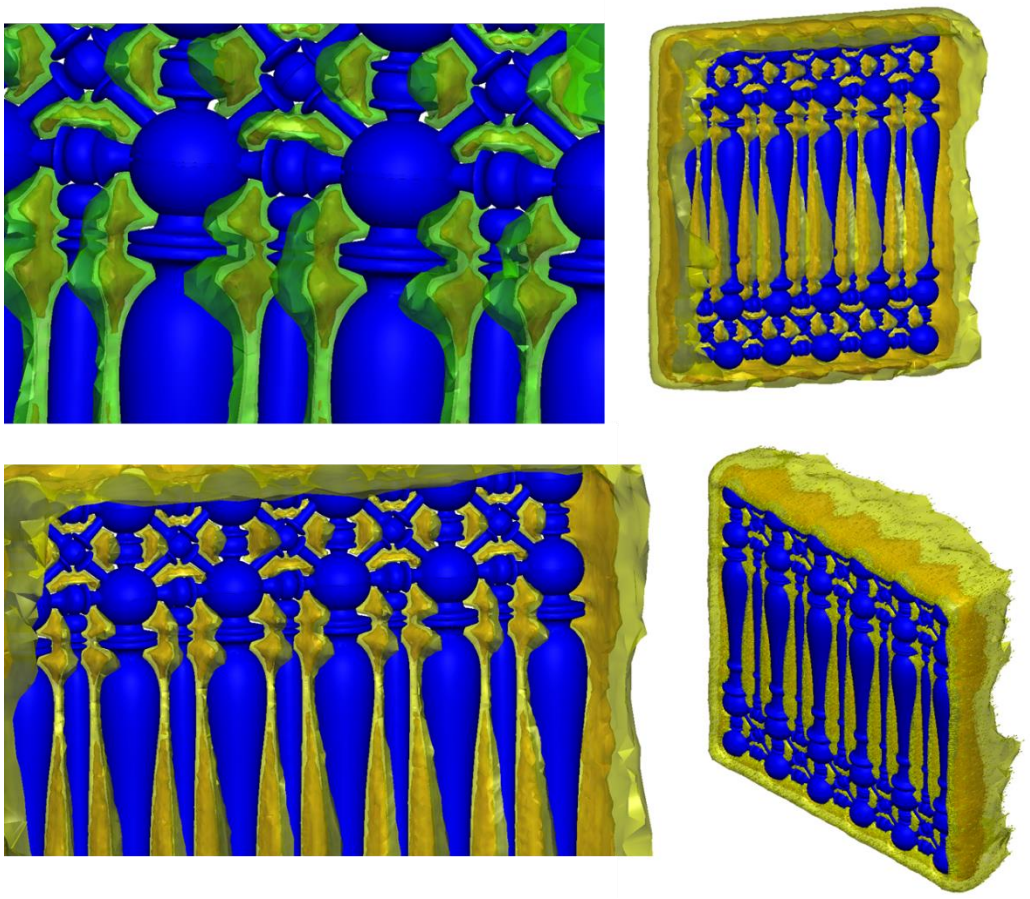


Figure 141: Volumetric airflow results for sample 8 from CFD simulations, demonstrating how the smooth geometries of the screen components enhanced airflow performance— (Author)

An example of smooth geometries is presented in the top image, where CFD simulation results were used to visualize volumetric airflow across the screen. The objective was to understand how the streamlined shapes of the screen components enhance airflow by reducing drag and turbulence around the structure. Consequently, fluid systems perform more efficiently when smooth geometries minimise turbulence and flow separation.

Sharp geometries include items with pronounced edges—such as cubes, edged polygons, and blunt forms with flat rear faces. These configurations tend to generate turbulent flow patterns, which can elevate drag and diminish the efficiency and overall performance of fluid systems.

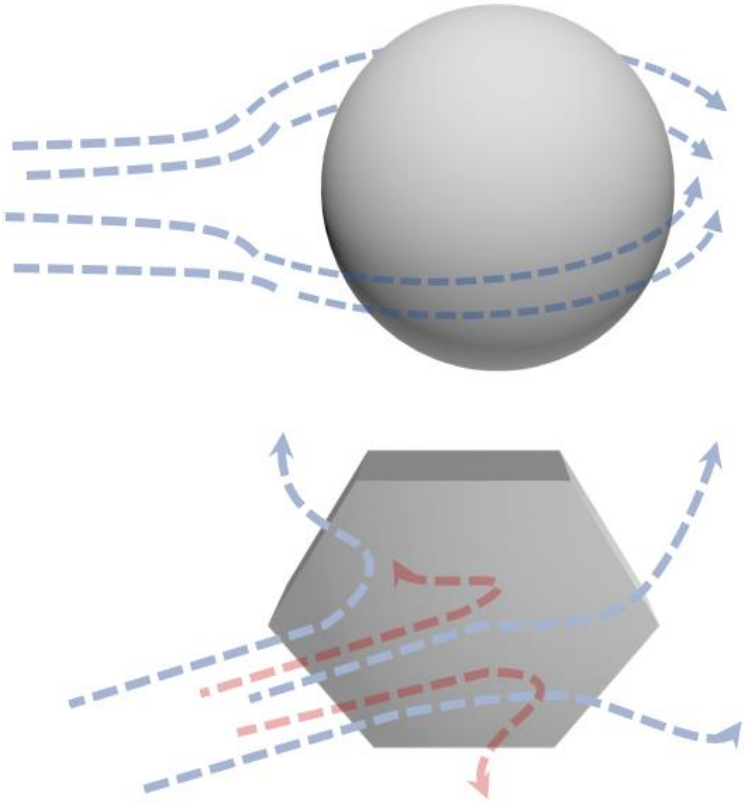


Figure 142: Illustration showing how airflow is enhanced around smooth geometries and obstructed by sharp geometries — (Author)

Another key insight from the aerodynamic analysis of the screens: diagonal components within perforated screens can obstruct airflow. However, this obstruction can be mitigated through symmetrical design, wherein diagonal elements are evenly distributed to form balanced angles that promote uniform air distribution. This principle was validated through comparing aerodynamic studies of Sample 1 and Sample 2, as illustrated in the following images.

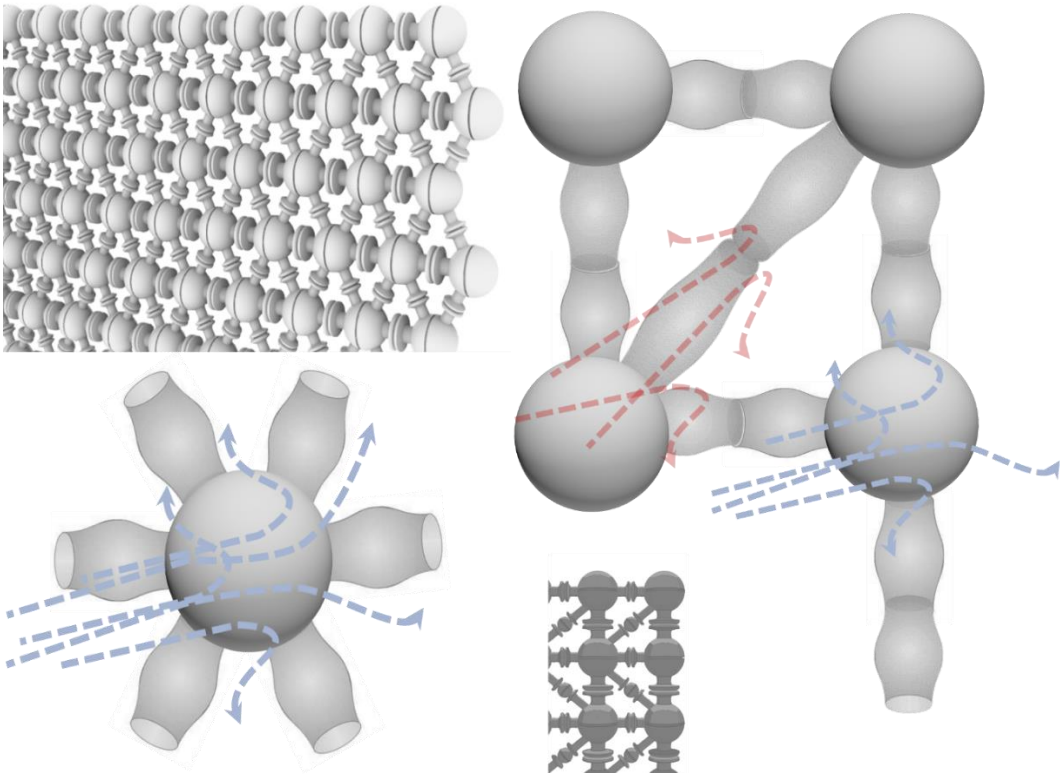


Figure 143: *Illustration showing changes in aerodynamic performance when diagonal obstructions within the screen lose their symmetry, based on extracted results from CFD simulations of Samples 1 and 2* —(Author)

Analytical studies of the aerodynamics surrounding the screens yielded valuable insights into the key factors that must be considered when designing a perforated screen for efficient passive cooling. Notably, phenomena observed at the micro level—around the screen's smaller components—were found to significantly influence the overall aerodynamic and shading performance. This was particularly evident when calculating the discharge coefficient for both the entire screen and its individual parts, where the results showed strong correlation. These findings reflect a "butterfly effect," in which micro-level variations in screen geometry contribute meaningfully to macro-level outcomes. The subtle differences in the shapes and configurations of screen elements were shown to produce substantial changes in performance from one case to another.

Ultimately, the screens demonstrated a successful example of effective mutual shading, offering a promising strategy for integration into ceramic evaporative cooling systems that require precise control over solar exposure. The aerodynamic analyses further provided valuable insights into optimising the proposed geometries, helping to mitigate design constraints and enhance airflow performance. These findings contribute meaningfully to the development of ceramic evaporative cooling screens with multifunctional capabilities.

8. Design stage.

Traditionally, generating the pattern is the first step in the design of a screen, as it establishes the underlying grid from which the whole composition emerges, as mentioned in Chapter 2 (Manufacturing the Samples). In this research, the objective is to design a pattern that serves as the main grid for the prototype. Islamic architectural patterns—some of which are as intricate as a thirty-one-point star—are often based on tessellation grids and may be observed in *muqarnas* screens from the thirteenth century (Moradzadeh, 2020). Several academic studies have systematically explored the mathematical structure of these patterns. The findings reveal a strong geometric basis for their construction, some of which have been addressed in the first chapter of this study.

Nevertheless, current approaches to creating these patterns often rely on a trial-and-error methodology, adjusting pattern edges, angles, and repetition axes to approximate the desired visual outcome. This process frequently requires the expertise of skilled craftsmen (Moradzadeh, 2020). However, an alternative method exists for describing and generating Islamic geometric patterns using digital tools, such as those explored by Eric Broug (Broug, 2013).

One such method is the introduction of a parametric system, which defines the fundamental isotropic repeat unit, followed by a basic paneling procedure used in the design process. Another approach involves aiding the formation of the pattern through mathematical reasoning, as discussed by Sir Roger Penrose in his book *The Islamic Geometric Patterns*, which focuses on designing three-dimensional structures rather than two-dimensional patterns (Penrose, 2017).

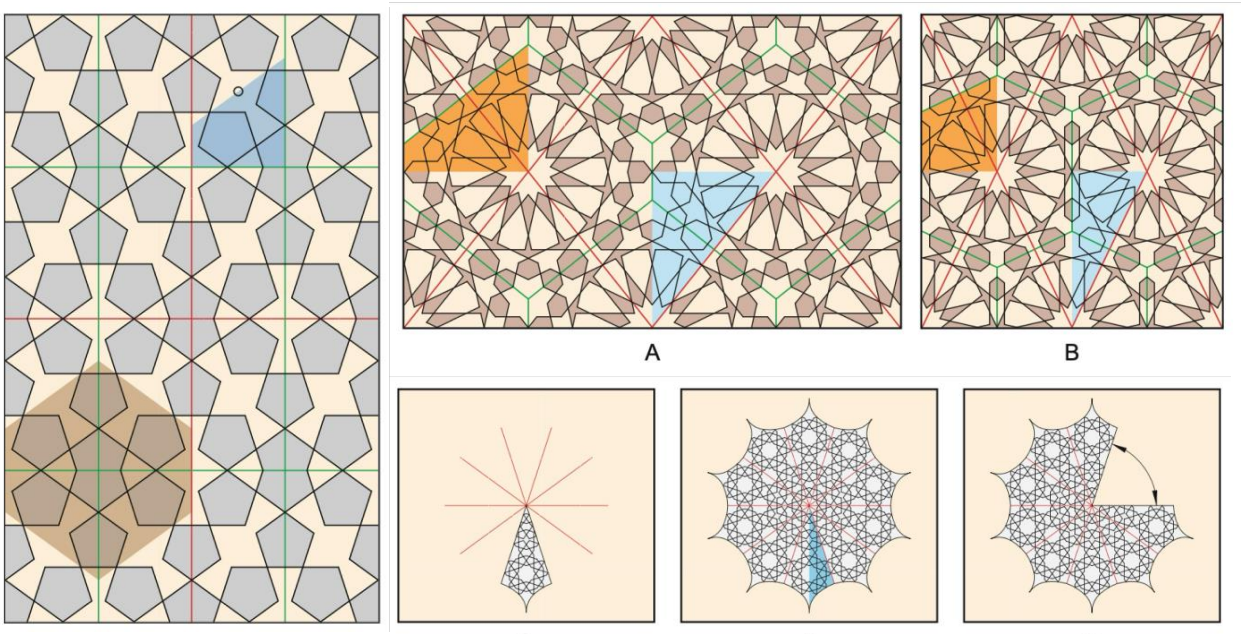


Figure 144 Methodology employed in crafting the gore segments for the four domes in Fatehpur Sikri and how the pattern can be formed when repeated on a specific axis with a chosen rotation angle, India (Penrose, 2017)

It has been reported that Islamic culture assigns complex meanings to geometric symbols, with each shape conveying a distinct message. The hexagon, representing Hebron, is crafted by intersecting two triangles, symbolising divine mercy and prayer (Fares, 2017). Simple geometric shapes like the hexagon have a long history of being used to beautify walls and inscriptions, as seen in the windows of the Al-Aqsa Mosque. Its elaborate presence is also evident in the Alhambra Palace, and it gained popularity during the Mughal era in India, where it was used to adorn buildings such as the Taj Mahal (Fares, 2017). In Islamic culture, the triangle is a well-known spiritual symbol, representing the devout connection with the divine through ascending prayers—elegantly depicted by the downward-pointing triangle. These geometrical figures form the foundation of designs that extend far beyond their spiritual significance. Such patterns are most prominently seen in Islamic decoration, particularly in Salah al-Din's pulpit, which features extensive ornamentation. In this instance, interlocking triangles function as highly sophisticated decorative elements, illustrating how geometric forms are employed to convey both spiritual and aesthetic dimensions in Islamic art (Hassan, 2007).

In the context of art, Islamic philosophical principles emphasise visual engagement and regard geometry as a source of intellectual stimulation and spiritual representation. Numerous connections exist between Islamic art and mathematics. In Diploma Studio 10 at the University Of Westminster School Of Architecture, Puimun Lee explored a parametric Penrose design inspired by Islamic geometry. The project focused primarily on the aesthetics of wooden architecture and the parametric approach. The principles guiding this design development were informed by existing literature and provided valuable insights for this stage of the study (Puimun Lee, 2019).

As shown in the following illustration, the resultant pattern is derived from an examination of Sir Roger Penrose's research. His exploration focused primarily on the intersection of six-pointed stars with hexagonal and pentagonal geometric forms. When this combination is parametrically repeated on a specific grid, it generates a complementary pattern. Penrose tiling may be categorised into three categories, all of which adhere to the golden ratio. These various configurations exhibit five-fold rotational symmetry as well as reflective symmetry (Puimun Lee, 2019)

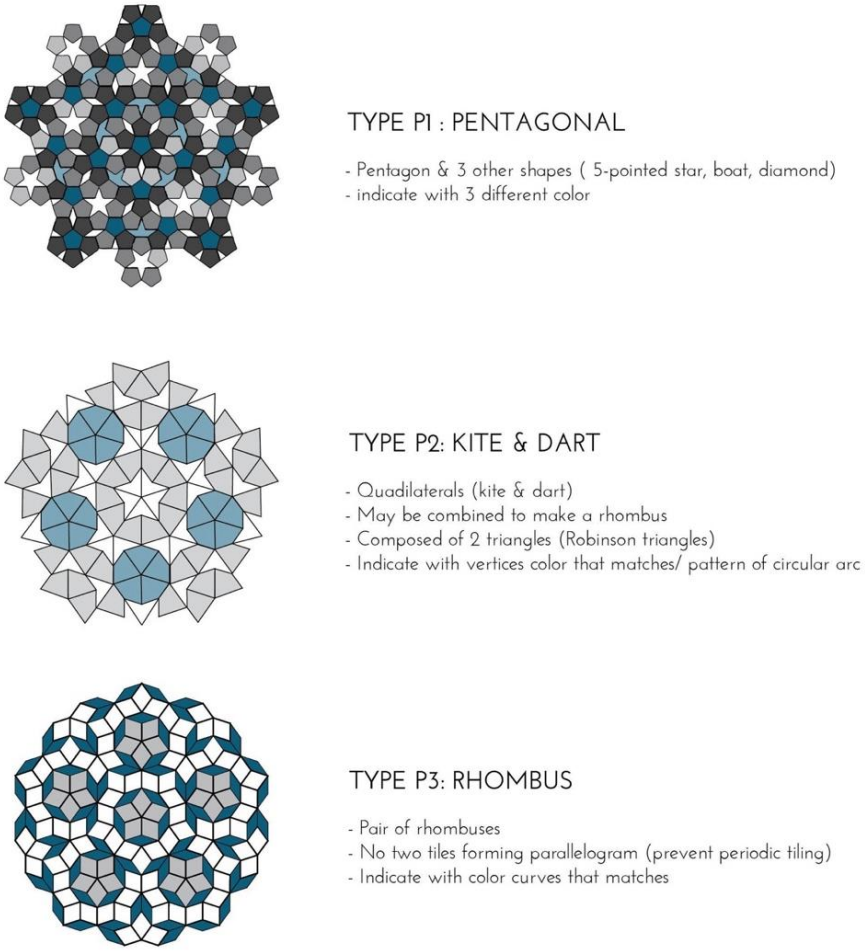


Figure 145 The Penrose tiling, discovered by Sir Roger Penrose, bears a notable resemblance to Islamic tiling patterns found in shrines (puimunlee, 2019).

8.1.a. Pattern development

Islamic art is renowned for its wide range of patterns, which often rely on regular formations and designs that repeat rhythmically across surfaces. When tasked with designing a screen inspired by Islamic pattern principles, it can be challenging to select one specific pattern over another. Therefore, a generic pattern is often preferred to accommodate such diversity. A project focused on Islamic patterns and geometry nearly always requires a particular interest in—and sensitivity to—spiritual and religious art. This is closely related to the idea that the researcher becomes more deeply connected to the beauty embedded within these works. The purpose of this stage of the study is to propose a method for mapping the development of an Islamic pattern in relation to screen design—specifically, by calculating the shape's center and vertices as it evolves. Islamic surface decoration employs a limited set of geometric forms, applied across surfaces in varied and inventive configurations.

Accordingly, numerous textbooks and academic publications on the history of Islamic art were analysed to investigate the various ways in which Islamic patterns are applied in practice (Fares, 2017). Real-world examples were studied to provide inspiration and conceptual grounding for designing the initial

pattern to be used in the design process (Hassan, 2007). This inquiry has contributed to the emergence of a new field of study focused on producing artwork inspired by classical and historical Islamic patterns, using digital tools such as Adobe Illustrator (Fares, 2017) (Hassan, 2007).

Designing an innovative and widely applicable ceramic model presents a significant challenge, given the abundance of historical icons from diverse Islamic cultures dispersed across regions. The vast variation in Islamic patterns—sometimes exceeding one hundred distinct designs within a single locality—makes it difficult to formulate a universal or generic pattern.

To meet the challenge of designing a culturally responsive generic screen, many scientific and design-oriented projects seek inspiration in the history of Islamic art, as well as in the construction guidelines and production procedures provided by manufacturers and reference manuals. Through extensive engagement with academic articles and books on the evolution of Islamic art—and the many ways Islamic patterns were applied in daily life—numerous designs were identified that reflect symbolic values and cultural priorities specific to each historical period. As a result, a total of 132 patterns were collated from Islamic pattern reference materials, including *Arabic Geometrical Pattern and Design* by J. Bourgoin. These patterns were then used to develop a design framework that is culturally responsive across various Islamic screen applications (Bourgoin, 2000). Nevertheless, creating a pattern that can meaningfully respond to such a wide range of historical designs remains a complex task, requiring the development of an efficient and adaptable methodology.

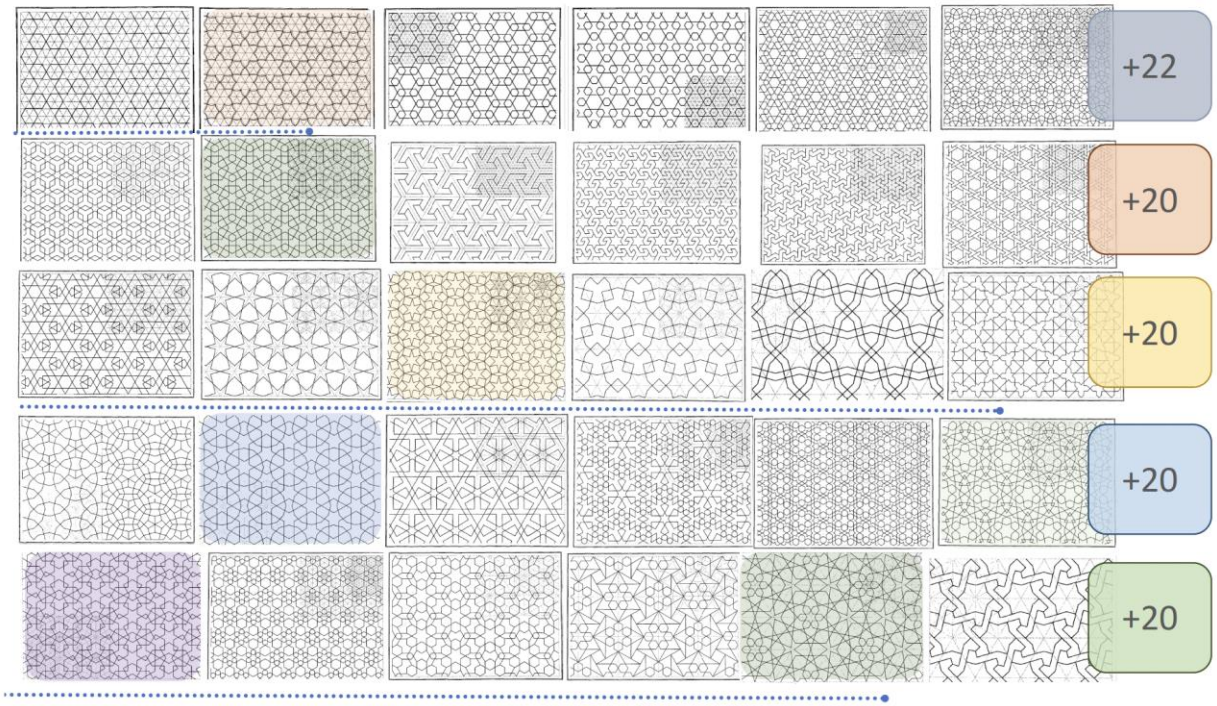


Figure 146 The selected 132 patterns from Islamic architecture across various regions were sourced from the Arabic Geometric Pattern by J. Bourgoin (2000) (Bourgoin, 2000) - source: Author.

To simplify the algorithm for pattern generation, a preliminary analysis was conducted by examining the frequent use of geometrical shapes in the development of Mashrabiya screens (Hassan, 2007). This process was facilitated through the use of AI-based generative design tools, including *MidJourney* and *Illuminarty*, alongside Python scripting (Illuminarty, 2023) (Midjourney, 2023). These tools enabled the shortlisting of patterns from a database of 132 historical designs sourced from different Islamic regions (Illuminarty, 2023). The method was designed to identify similarities while simultaneously preserving the uniqueness of geometrical patterns historically used in screen design. AI screening played an effective role in revealing recurring motifs and structural affinities across the selected patterns. This methodological approach stemmed from a desire to uncover common design principles as they evolved under the influence of different historical eras, architectural contexts, and the artistic philosophies embedded within each pattern (Awad, 2018).

"Feature detection" can be performed using the Python programming language, with popular methods focusing on identifying similarities, changes, or representations across image collections relevant to the task. Feature extraction refers to the process of identifying specific attributes or characteristics within an image. This mechanism is widely used in applications such as object recognition, image classification, and scene interpretation. As a result, visual data processing and recognition algorithms become increasingly efficient, enabling computer vision systems to perform more complex functions. These systems can locate similarities between different patterns and shortlist input data to highlight the most frequently recurring designs. In this research, AI was employed through Python to perform feature detection and extraction, where the tools were provided with a dataset of Islamic pattern images and guided the process by defining prompts and parameters. The system then analysed the collection to detect recurring features, highlight similarities, and identify common attributes across the images. This guided use of AI-assisted methods ensured that the outcomes were not fully automated but shaped by researcher input, allowing the system to shortlist the most frequently recurring designs for further development.

Thus, the inclusion of this feature, empowered by AI technology, enabled this analysis stage to streamline the input process, compressing it into six primary features that effectively reflected the majority of the observed patterns. This process facilitated the handling of complex data by simplifying the analysis, thereby making it more manageable to assess. The visualisation of the selected patterns, which is further compared to the input dataset in the accompanying image, provides a summary of the dataset's diversity by displaying all present features. As shown in the image below, the key selected patterns are highlighted through repetition, emphasising the common elements. These subsets, when mirrored onto the grid, produce the initial sketch that will evolve into the planned screen and serve as the foundation pattern for the design. This stage indeed simplified the data, as a smaller number of

patterns were shortlisted for examination, making the search for common patterns easier. The usage of AI in the earlier stage proved extremely beneficial, orchestrating the process that ultimately led to the crucial identification of key patterns.

Upon closer examination of the preceding figure, it became evident that, although the six patterns appeared visually distinct from the others, they shared a common reliance on the repetition of basic geometric components as their main identifying elements. Notably, the study revealed that a recurring motif—comprising six-pointed, star-like figures—was consistently present across these patterns.

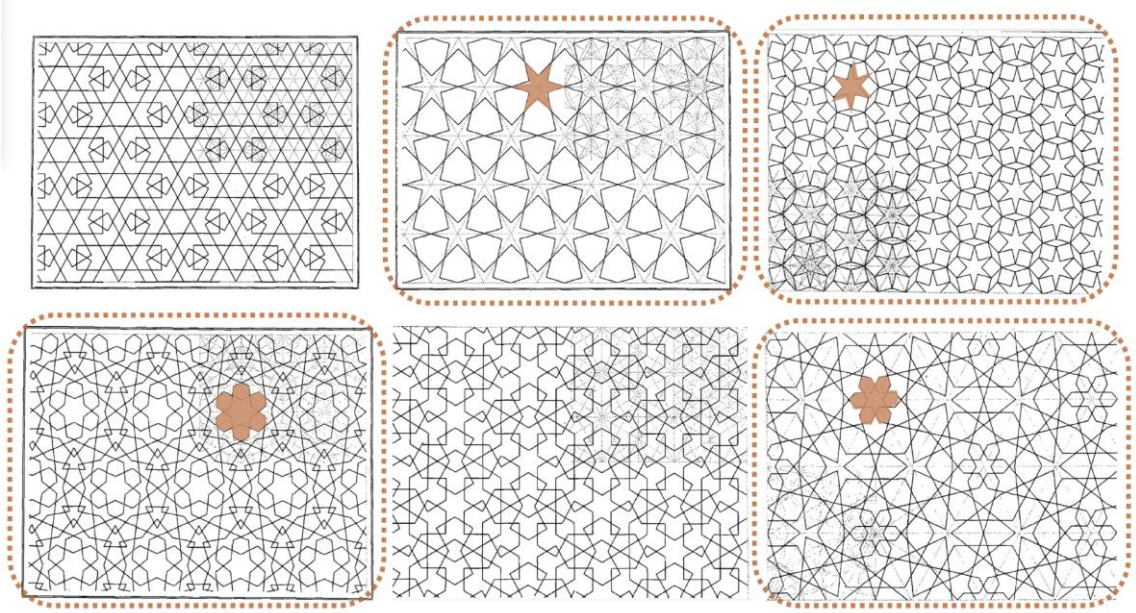


Figure 147 The six highlighted patterns were identified through AI feature detection on the input database. Among these, four patterns (outlined in orange) were shortlisted for further development. – Source: Author

An additional observation revealed that, even as the constituent elements grew increasingly complex, they retained a discernible connection to the foundational six-pointed star. These intricate variations often emerged through the introduction of supplementary nodal points and the strategic rotation of axis relative to the primary structure, as illustrated in the following figure. This phenomenon brings forth the remarkable adaptability and structural compatibility of the six-pointed star motif, which consistently serves as a generative core for the construction of diverse and elaborate pattern systems.

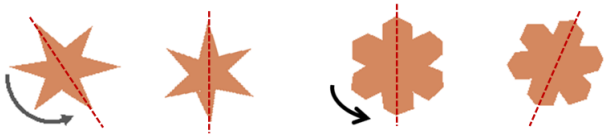


Figure 148 The commonly used star component in the 6 highlighted patterns and how it changes from one pattern to another based on the rotation of the symmetrical axis. -source: Author

Distinguishing between the geometric configurations of star patterns requires attention to several structural factors. The following key elements reinforce this differentiation (Broug, 2013):

1. Number of Points: One essential feature for distinction is that stars may vary in the number of points they possess.

2. Symmetry: A crucial aspect to consider when observing stars is symmetry. Some stars exhibit bilateral symmetry, while others display radial symmetry. Accordingly, distinguishing the stars' geometrical form becomes easier when the axis of symmetry is defined—where the shape is mirrored or symmetrical around that axis.

3. Angle Measures or number of edges: Like other mathematical polygons, the angles influence the geometric form of a star. Differentiation may benefit from measuring these angles and understanding how they relate to one another.

4. Line Segment Arrangement: The overall geometric form is shaped by how the line segments are connected to form the star and by the specific arrangement of these lines.

By combining these elements and methodically analyzing the distinctive characteristics of each star, one can effectively differentiate between geometric configurations and understand how each pattern is uniquely constructed—especially in relation to other motifs within the same cultural context. Accordingly, when examining the six highlighted screens prioritized by the AI as the most generically distinctive among all inputted patterns, we observe that each selected shape is driven from the six-pointed star. The variations across these patterns stem from two primary factors:

- The rotation of the main axis of the star (the axis that symmetrically divides the star).
- The number of points taken from the main axis (diagonal).

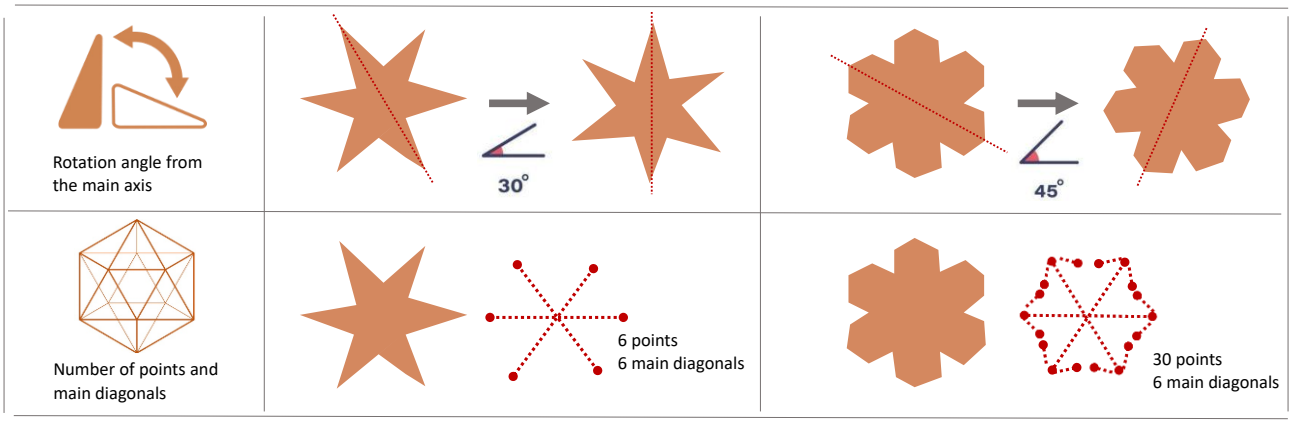


Figure 149 Illustration showing how the selected forms were derived from the six-pointed star shape, with pattern variations based on the primary axis and the number of points. – Source: Author.

When applying these types of patterns in ceramic fabrication, two distinct approaches can be employed. The first involves translating the Islamic Screen pattern into a three-dimensional ceramic form through a computational framework and digital modelling. By defining a custom algorithm, the planar pattern can be manipulated and warped into a three-dimensional surface or solid. Once the digital form is finalized, a suitable fabrication technique is used to automatically reproduce the pattern on the physical ceramic piece.

"Foundational generic pattern development", also known as base pattern (Azari, 2023), is understood as a stepwise methodology in which both the articulation of the design problem and its resolution are mediated through visualisation and mathematics. Each stage in the creation of a digital Islamic design pattern poster—intended as a base grid for the eventual ceramic piece—serves to demonstrate the effectiveness of these design samples in bridging conceptual abstraction with material realisation.

The distribution of this type of stellate pattern may be attributed to its flexibility and versatility, which make star-shaped figures an ideal pattern type for parametric repetition (Azari, 2023). Those with an even number of vertices are most likely to be parametrically periodic, as they exhibit greater symmetry and uniformity. The sides, characterised by uniform spacing between equally distributed vertices, confirm that the shape is polygonal and possesses an even number of diagonals, further indicating an even number of sides. Parametric geometry inherently supports symmetry, which not only enhances architectural beauty but also simplifies and clarifies the underlying calculations (Azari, 2023).

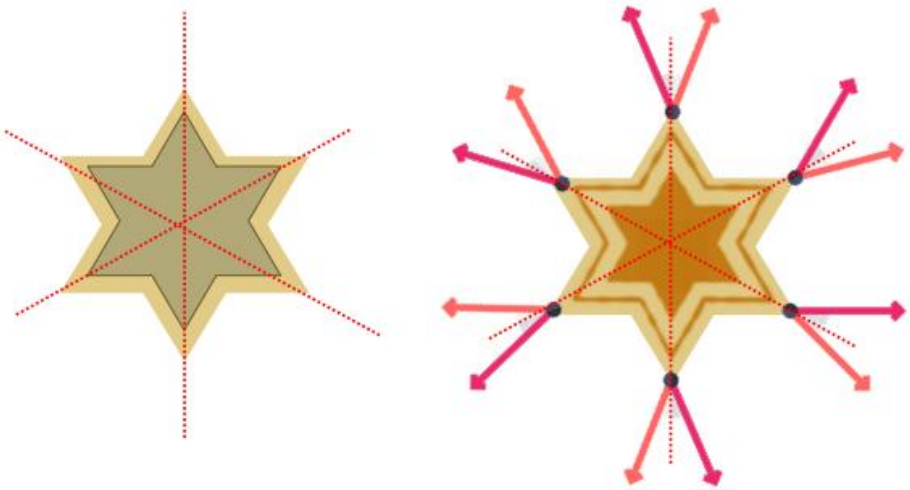


Figure 150 Image illustrating the adaptability and plasticity that make star-shaped figures the ideal pattern type for parametric repetition, possibly explaining the distribution of this kind of stellate pattern. – Source: Author.

Consequently, when working with polygons, their diagonals can be used to create quadrilaterals if the number of vertices is even. The four-sided figure offers a manageable set of parameters compared to triangles, which have an odd number of diagonals. This is because, unlike triangles—whose three sides and angles are known—quadrilaterals tend to be more stable and, with fewer angles and sides, may be easier to define parametrically. However, the number of diagonals may not be a significant factor in the

simplicity of parametric repetition for structures that are more complex or asymmetrical in form. It was easy to begin with a simple design, as forming the regular hexagram pattern was our initial goal. The composition began with six circles superimposed over two principal axes, along which the basic diagonal lines of the star were inscribed. These lines were spaced at 45-degree intervals, repeatedly, to evoke the classical effect of perspective. The central point within the star was emphasised through the intersection of its outlines, clearly revealing its intended purpose. Subsequently, three primary diagonal incisions were introduced to calibrate the proportions of the illustration with greater accuracy. To ensure that the six-pointed star appeared as intended, necessary modifications and amendments were made, entailing careful adjustments to both lines and angles. Precision was embedded into the design at this stage, and the form gradually evolved as the process advanced. Throughout this phase, the design underwent iterative refinement, and the final image representation captured the stages of its development, culminating in a cohesive and accurate visual outcome.

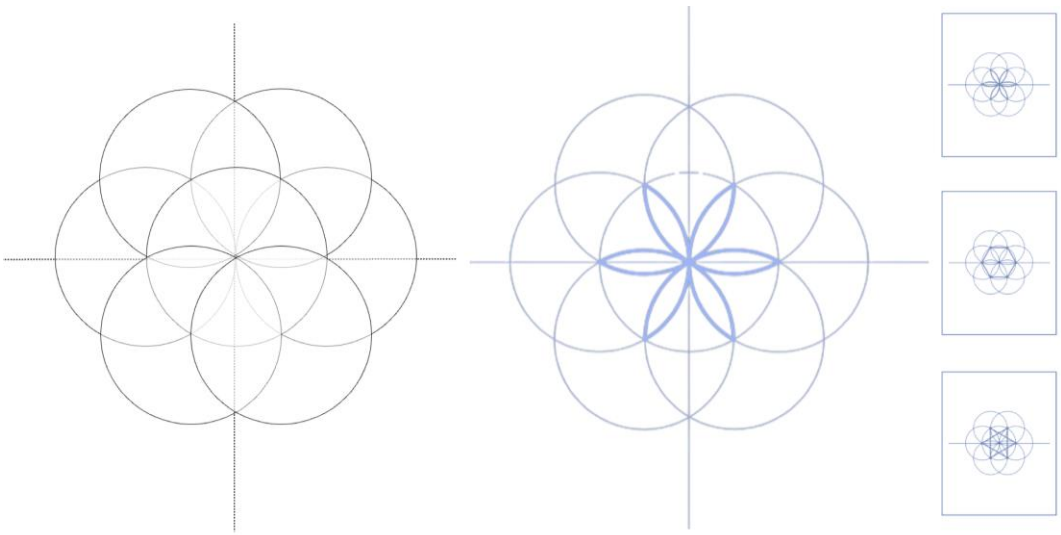


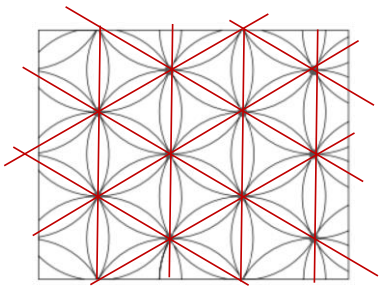
Figure 151 Initial phase in the development of the base pattern, serving as the foundation for the prototype's design evolution - Source: Author

Then, from the first point of the circle's edge, the following points were successively connected to form three diagonal lines. Each oblique line connected a point located two steps clockwise or two steps anticlockwise from another point. Another part of the analysis involved carefully researching and noting the geometric traits the design might possess, which were later tested for suitability in parametric repetition. To ensure replicability across various locations and conditions, this process required close examination and regulation of design parameters to allow for necessary adjustments. The stage-by-stage design method added value through strategic dimension changes, deliberate spacing, and perspective analysis, which supported the evaluation of the design's scalability and flexibility. Surveys conducted through the lens of this research confirm the model's success and its strong potential for widespread application in multiple scenarios.

From that moment onward, production processes were integrated into the design workflow by gradually repeating the created form to verify its applicability and sustained symmetry throughout the parametrically repeated sequence demonstrated on a predefined grid. Through the method of precisely staging the rows, a close analysis of the grid and its structural logic was conducted—examining whether a diagonal or orthogonal configuration would best suit the repetition pattern. The discussions held during these meetings played a pivotal role in shaping decisions regarding symmetry and spatial integration among the various shape elements that comprise the partial grid, as illustrated below.

Analyzing the geometry of the image AI mid-journey generated:

1- Applying the axes on the seven overlapping circles grid



2- To create a triangle grid, place the tracing paper over the circle grid and join the dots using horizontal and diagonal lines.

3- Overlapping the pattern and the diagonal grid created a third pattern that can highlight a hexagon around the outside of six adjacent triangles to indicate the hexagon grid, as indicated.

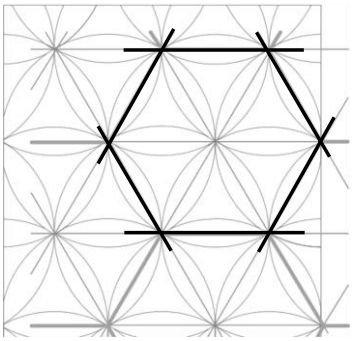


Figure 152 The second phase of the design development of the base pattern, used for the design development of the prototype - Source: Author

However, despite the complexity of the procedure, getting the right outlines of the six-pointed star, as well as the ideal geometric polygon that could contain the star without overlapping or crossing along the repeat grid, was among the most critical stages. This detailed vision was essential, considering the prospect of later refactoring or architectural upgrades. Applying lessons from Mashrabiya, the main extrusions and the overlaps of basic geometries—executed in the original form of the building to create some spots of shade—indicated the rationale for starting with such an initial step, which led to the differentiation of the proposed design in terms of flexibility and sustainability. This strategic way of decomposing the initial conceptions formed a basis for all further design iterations, allowing for the consideration of future corrections or add-ons, and thereby enhancing the strength of the design concept as well as its capability to accommodate adjustments to the maximum.

The intersection of the primary diagonal line axes acted as both the pivot point for the star's rotation and the focal point of the geometry, which gravitated from lower to higher level nodes, exemplified by

the size of stars and the number of lines they contained. The complex star's appearance and meaning were enriched by its purposeful placement, which helped establish it as both a structural element and an aesthetic requirement. Finally, the task was executed professionally by removing all construction marks and guidelines related to the star's shape, ensuring a flawless image. This particular attention to every minor detail reveals the designer's commitment to purity and precision. Ultimately, the final geometric pattern blends all elements and transforms into a beautiful version, as shown in the following illustration.

Degrading the image of the generated pattern into a simple pattern that can be used as a tracing layer for the designed prototype. Source: Rhino

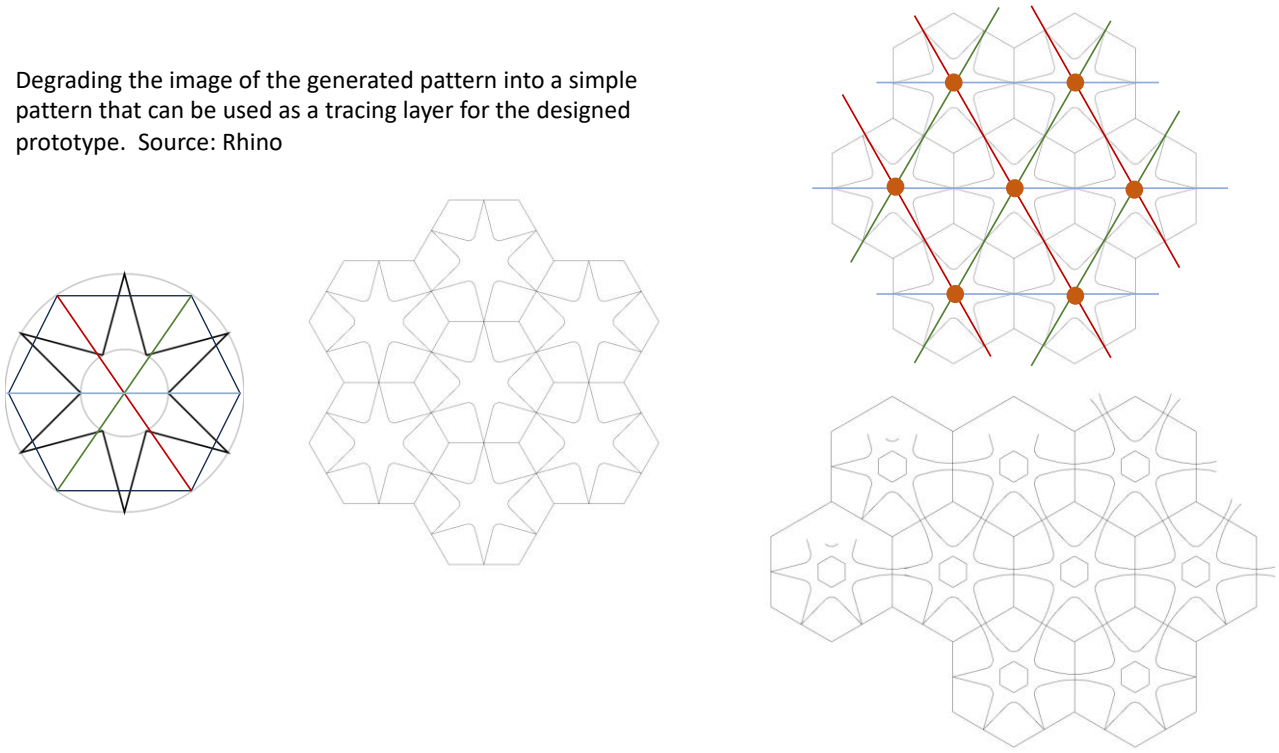


Figure 153 The third phase in the design development of the base pattern used for the prototype's design development- Source: Author

The developed fundamental pattern was generated through an AI algorithm that imitates the rendering procedure, aiming for reliable replications of standard patterns. This stage was important for understanding how the use of a more refined visualisation could enhance the perception of the design's aesthetic appeal and allow for adequate assessment. Next, digital design tools such as Photoshop were employed to iteratively apply the same modified pattern, simulating parametric repetition. This included a traditional approach using continuous application, under which the pattern's replicability and the use of different combinations were tested—yielding valuable insights into its overall visual effect and spatial arrangement, as demonstrated in the following illustration.

Figure 154 The image generated by AI and its visualization when repeated parametrically - Source: Author

8.2.b. The three-dimensional design of the pattern

Building on the previously developed prototypes, the design was further transformed into a three-dimensional format to simulate its appearance within a physical setting. The primary objective at this stage was to achieve a more realistic representation by converting the embossed foundations into volumetric forms. This transition revealed the spatial structure and enhanced the design's dimensional qualities. Using specialised 3D software, it became possible to conduct a detailed assessment of both the aesthetic contours and structural integrity of the pattern. The digital environment also facilitated an analysis of light and shadow interactions, offering deeper insight into the visual impact of the design in real-world contexts. In parallel, the exploration of contemporary ceramic preferences and the slip-casting technique proved essential, as both significantly influence the screen's functional, environmental and aesthetic performance. Production constraints— particularly those associated with slip-casting—play a decisive role in shaping the design process, especially when intricate geometric forms are involved. Moreover, the integration of a water supply system requires careful analysis to ensure seamless functionality and efficient management. This becomes especially critical when the screen is parametrically repeated across a larger surface. Accordingly, a strategic approach to water

distribution must be developed to guarantee consistent performance and effective operation of each screen unit within the system.

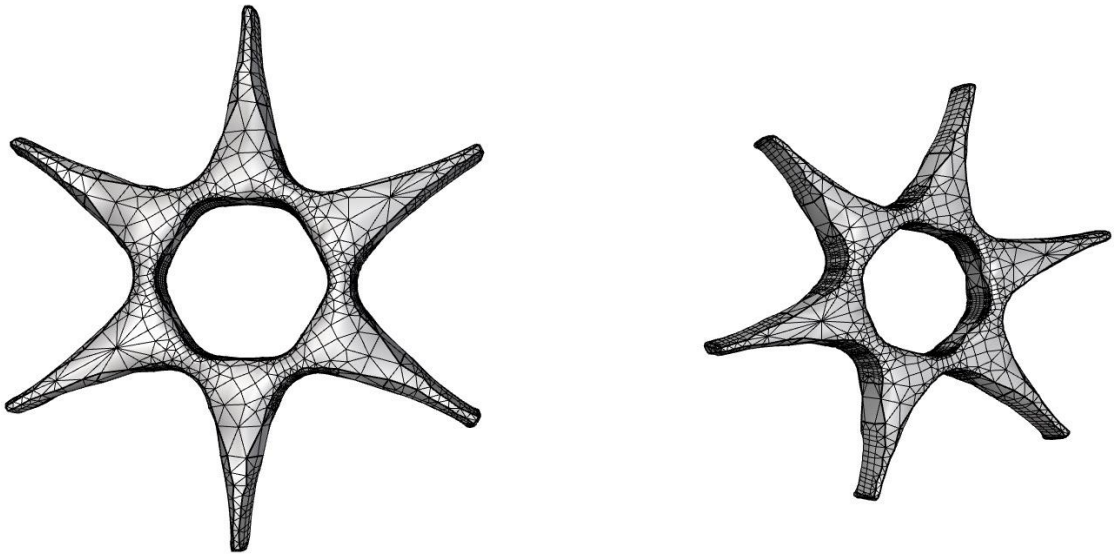


Figure 155 Early stages of modelling the star geometry in Rhino - Source: Author

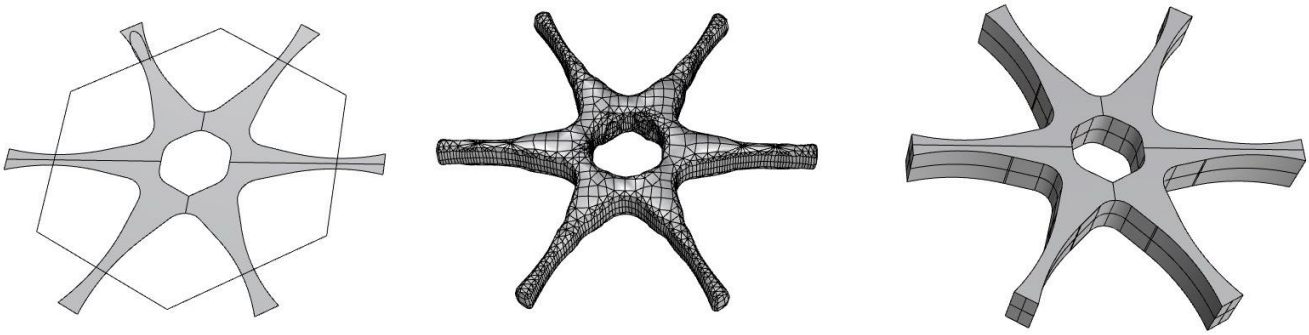


Figure 156 Modelling iterations of the star geometry in Rhino - Source: Author

In the manufacturing context under consideration, the slip-casting method—despite its general utility—reveals certain limitations, particularly when replicating intricate patterns found in architectural elements such as the Mashrabiya. The Mashrabiya is a traditional construction form characterised by its dual function: it provides shade while maintaining an efficient design profile. This interplay of functions is often referred to as "mutual shading," highlighting its capacity to balance environmental control with aesthetic and structural finesse.

On the contrary, the complexity of the Mashrabiya's design has drawn the attention of clay artists, as slip casting can exhibit certain limitations. Although the technique is not the most difficult to perform, casting such structures remains a labour-intensive process with notable setbacks. The inherent constraints of the modern method make it somewhat challenging to replicate the intricate details of the Mashrabiya latticework with precision.

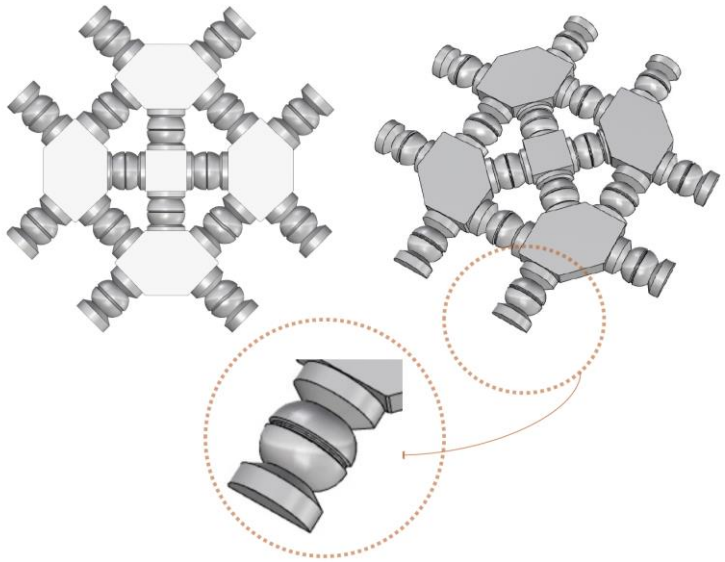


Figure 157 Visualization of the intricate micro-details captured in the scanned geometries of Mashrabiya samples utilized in the analytical study - Source: Author.

The idea of producing a model that is dependent on how the capillaries are used to flow the water inside the cavities created in the component straightforwardly clarifies all the constraints facing the slip-casting fabrication process. Therefore, the need for much caution and the avoidance of unnecessary complexity, particularly in the first phase of development, has dictated the production approach, as illustrated in the picture below. However, the new prototypes shown in the figure below demonstrate that this design approach will fill in the gaps, as they provide a platform for refining the first phase of the design and evaluating other elements of complexity that will overcome the constraints of the slip-casting fabrication process and thus realise the intended function.

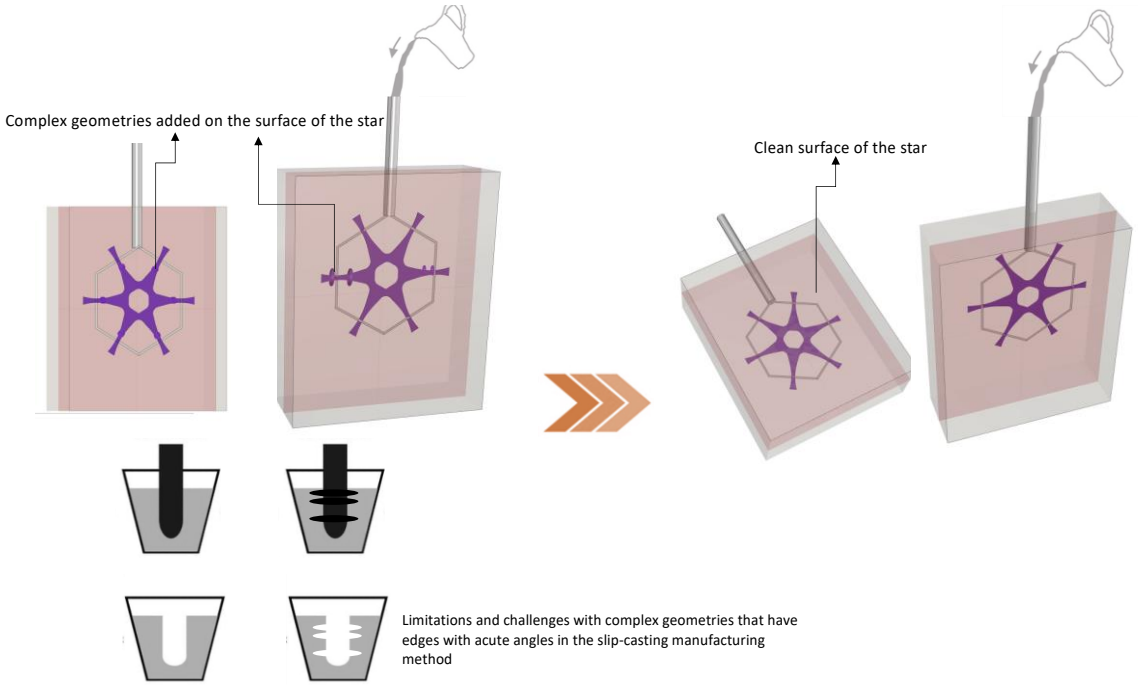


Figure 158 Illustration showing how the proposed geometry overcomes the limitations of slip casting, avoiding unnecessary complexity as seen on the left side of the image. Source: Author.

Consequently, a different strategy was tested, which introduced geometric complexity at the very beginning of the production process. This complexity was designed to remain compatible with slip casting, even though it required greater depth in the mould. Such an approach not only facilitates handling and transportation but also generates shading effects within the star geometry itself. As a result, the amount of incident solar radiation absorbed by the surface is reduced. Furthermore, the intentionally indented geometry of the screen element enhances surface articulation, allowing for more effective interaction with solar radiation.

Taking these challenges into account, an alternative approach emerged: the introduction of additional geometry considered from the outset of the manufacturing process. This added geometry is designed for ease of production via the slip-casting method and can be extruded to a greater depth. Functioning as a multifunctional element, it casts a shadow over the star geometry, thereby reducing direct solar exposure while also providing structural support to the system.

The strategic design of the over-extruded geometry will play a vital role in blocking incident solar rays, preventing them from reaching the star geometry, where the water channels will be flowing inside. This innovative approach not only overcomes the limitations of the slip-casting manufacturing method but also enhances the shading efficiency of the final product, keeping the proposed design committed to the features of Mashrabiya's mutual shading concept.

It was quite evident in the early stages of design what type of polygon the star geometry would fit, and it became clear which polygon would be apt to capture the intricate star geometry. The most poignant concern that emerged during these early stages was choosing the best polygon to simplify a complicated star geometry. The built base pattern provided an important guide in determining the exact polygon to use. It became evident that the most common outline of a six-pointed star might fit within the border of a six-sided polygon, since the common outline of a six-pointed star could fit into the border of six equals. Therefore, the hexagon became the best form upon which to base my design; it offered the right symmetry to balance the complex star geometry. The star feeding on the polygon is a perfect example of the design process’s obsession with detail and perfection. Such careful decision-making highlighted the rigorous method used to establish unity and balance in the final design iteration. At the same time, the use of AI technology served as a supporting tool to validate the logic behind this decision and to determine the prevalence of hexagonal configurations in fitting the delicate six-pointed star geometry. A comprehensive computer study was conducted to analyse the frequency and ubiquity of hexagonal polygons in the context of encapsulating the six-pointed star geometry. This analytical endeavour aimed to demonstrate the intrinsic appropriateness of hexagonal frameworks for effectively accommodating the complexities of star geometry. The study sought to identify patterns and trends indicative of the widespread adoption of hexagonal configurations in similar design contexts by

examining various historical and contemporary examples. This thorough investigation resulted in a deeper understanding of the geometric links between the hexagon and the six-pointed star, highlighting their inherent compatibility and synergistic potential within architectural and design discourse, as illustrated in the following images created by the AI Copilot Microsoft platform (Copilot, 2023).

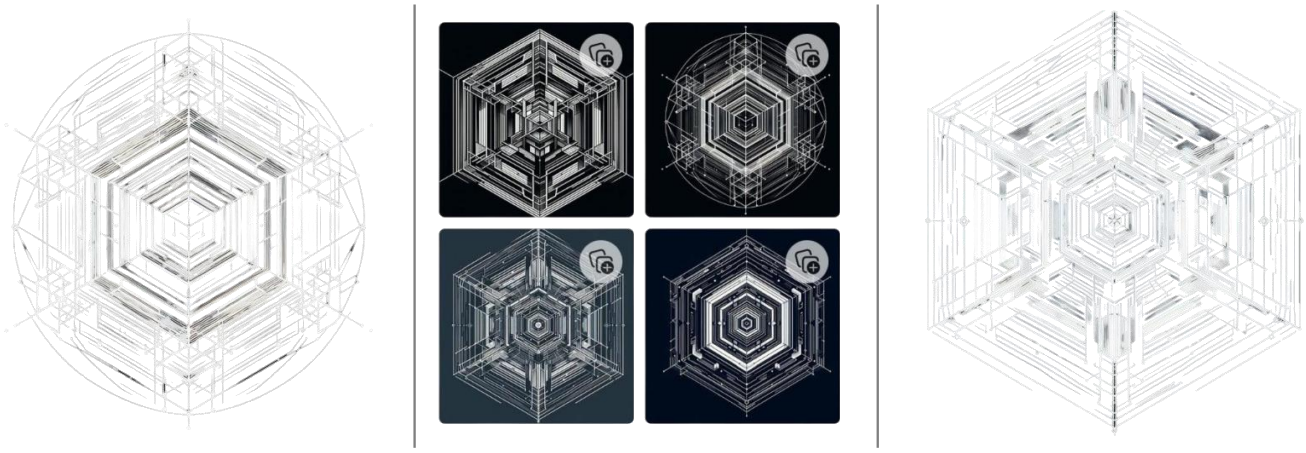


Figure 159 Collage of images created by the AI Copilot Microsoft platform, illustrating which polygon can be used in a repetitive pattern to accommodate a six-pointed star- Source: Author, (Copilot, 2023)

8. 1. C. i. Mutual shading optimisation of the component:

To determine the efficacy and legitimacy of the suggested strategy, analyse its performance and quantify the efficiency of shading with respect to solar radiation, which is critical for understanding its practical application. The shading dynamics were carefully analysed across multiple orientations using virtual simulation techniques such as Climate Studio simulations, with particular focus on areas with high sun exposure. Cairo, located in Egypt, serves as a realistic example of a hot and dry climate and provides the principal context for experimentation due to its accurate environmental characteristics. Using such advanced analytical approaches, comprehensive knowledge of shading efficacy might be obtained, guiding and confirming the suggested design strategy. This rigorous approach demonstrates a dedication to empirical validation and evidence-based design principles, which are critical for assuring the usefulness and performance of architectural interventions in real-world settings.

Accordingly, it was shown that the extruded hexagon geometry around the star limits the exposure of the newly cut star base to sunlight by 40%, and this percentage can be manipulated by increasing the length of the extrusion of the hexagon frame in relation to the sun angles in the application context. Nevertheless, it demands careful balancing so that solar radiation is not entirely obscured, given the fact that shade is fundamental while also contributing to the desirable evaporation rate. In addition, it was important to achieve a delicately balanced level of optimal evaporation performance while reducing excessive water consumption. The application of these implications is a subtle approach that

focuses on the richness of shade to regulate ambient temperatures and thermal splitting, while at the same time forming a perfected environment that catalyses the evaporation process at the desired rate. Consequently, the extrusion depth was adjusted to match the shade variation, as illustrated in this photo, which shows different versions after reiterating.

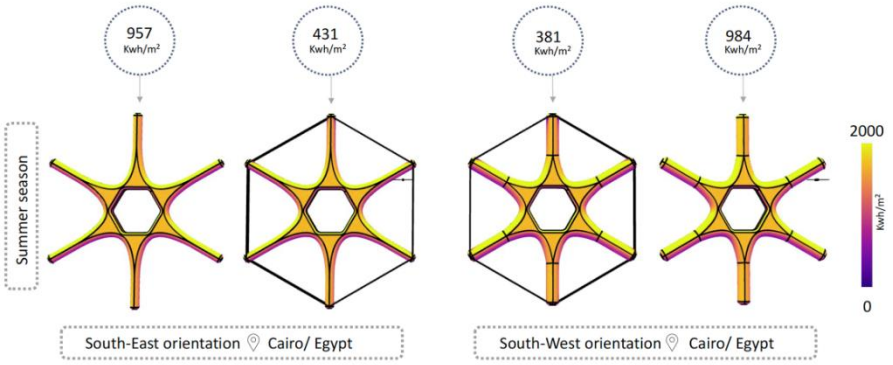


Figure 160 Final shading performance of the designed component, with and without the optimised extruded geometry, for south-east and south-west orientations respectfully in Cairo, Egypt, as simulated using Climate Studio. - Source: Author.

To conclude, although the slip-casting method presents certain limitations, the novel integration of over-extruded cap geometry offers effective solutions to several of these challenges. It opens new possibilities for producing low-energy structures that resonate with the distinctive architectural qualities of the Mashrabiya and the design principles it embodies.

8.1.b. ii. Aerodynamics optimisation of the component:

With regard to the subsequent design phase of the iteration method, the next step involved adjusting the styli's cross-sectional shape to allow air to flow over the screen in the most efficient manner. To highlight the significant impact of airflow on the evaporative cooling performance of the proposed screen, the modelled star was shaped to ensure unobstructed and free air movement. This outcome was based on a thorough study of fluid dynamics principles, which formed the foundation for selecting a cross-sectional shape that minimized airflow resistance and maximized air exchange.

The overall aim of the designed star's cross-section characteristic was to offer a relatively open channel through which airflow could be accelerated through the screen, allowing for the heat exchange and evaporation processes to occur. The hypothesis and methodology for design evolution and form optimisation were summarised in an informative diagram, with an illustration of the general roadmap and the following phases of analysis. This visual guide served a dual purpose: providing a bird's eye view of the sequential advancement of design processes, while also highlighting the iterative nature of the refinement process that determined which design configuration was the most efficient.

Every stage of the investigation was planned exactly as it had been designed, to include each respective intermediate criterion such as structural integrity, thermal performance, aerodynamic efficiency, and aesthetic appeal. The plan highlighted all the stages of the method used in order to track the route by

which designs improved after several assessment phases, providing insight into the iterative process of design and diverse elements that accelerated the process to optimise the shape.

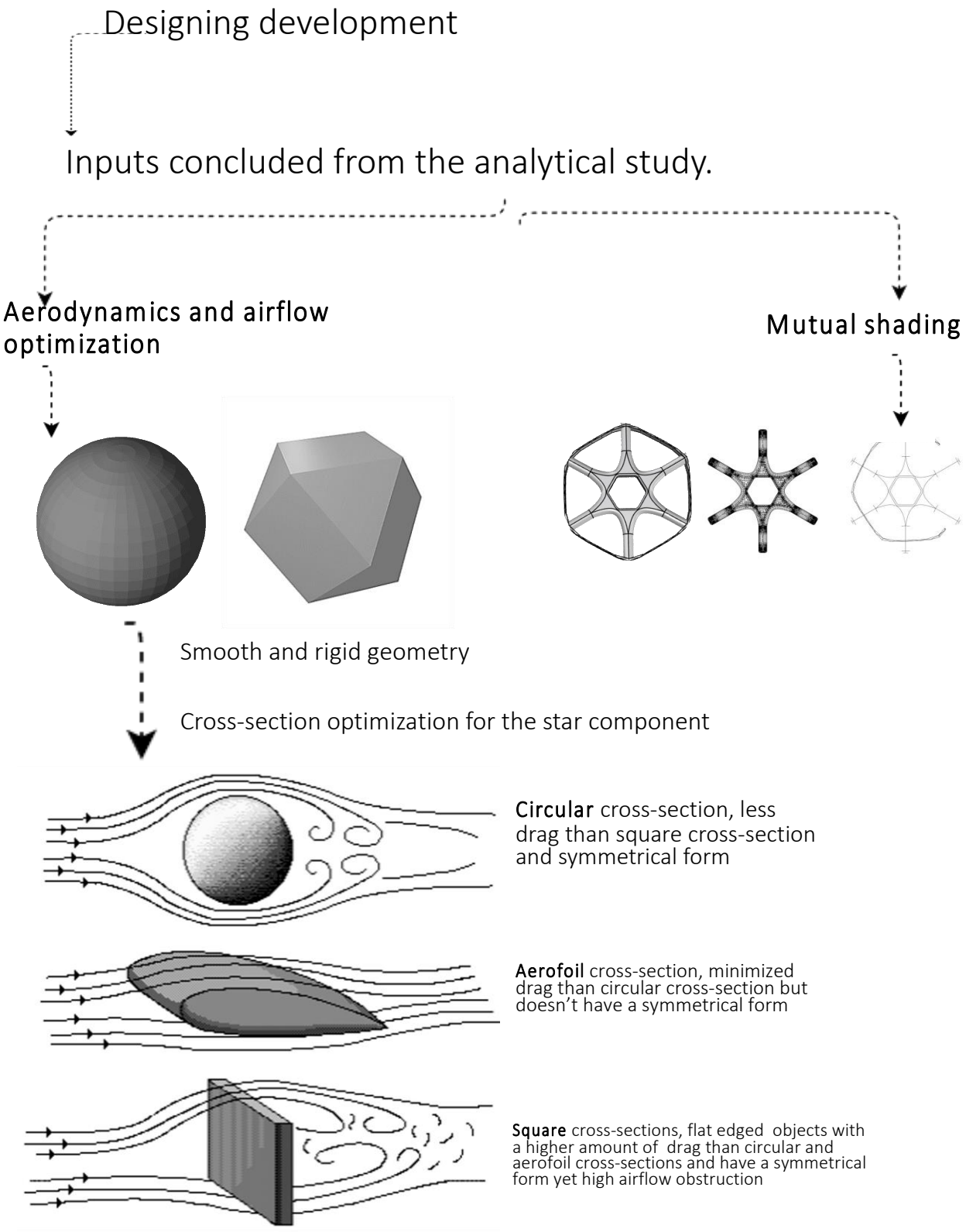


Figure 161 A diagram that illustrates the overall roadmap and the subsequent stages of analysis, providing a concise summary of the design evolution, form optimisation process, and hypothesis. – Source: Author.

Data from aerodynamic tests consistently demonstrate that rounded objects advance airflow more than straight forms with sharp edges. This occurs because the aerodynamic nature of the rounded shapes, which leads to a decrease in air resistance and turbulence, causes the airflow to be streamlined. The intrinsic curvature of a rounded shape generates a streamlined flow of air particles around the object, thereby reducing drag and improving aerodynamics. On the other hand, sharp edges and irregular curves disrupt airflow; as a result, turbulence is created, and the efficiency of the aerodynamic profile is decreased. This distinct difference in performance can be seen in the following visualisation.

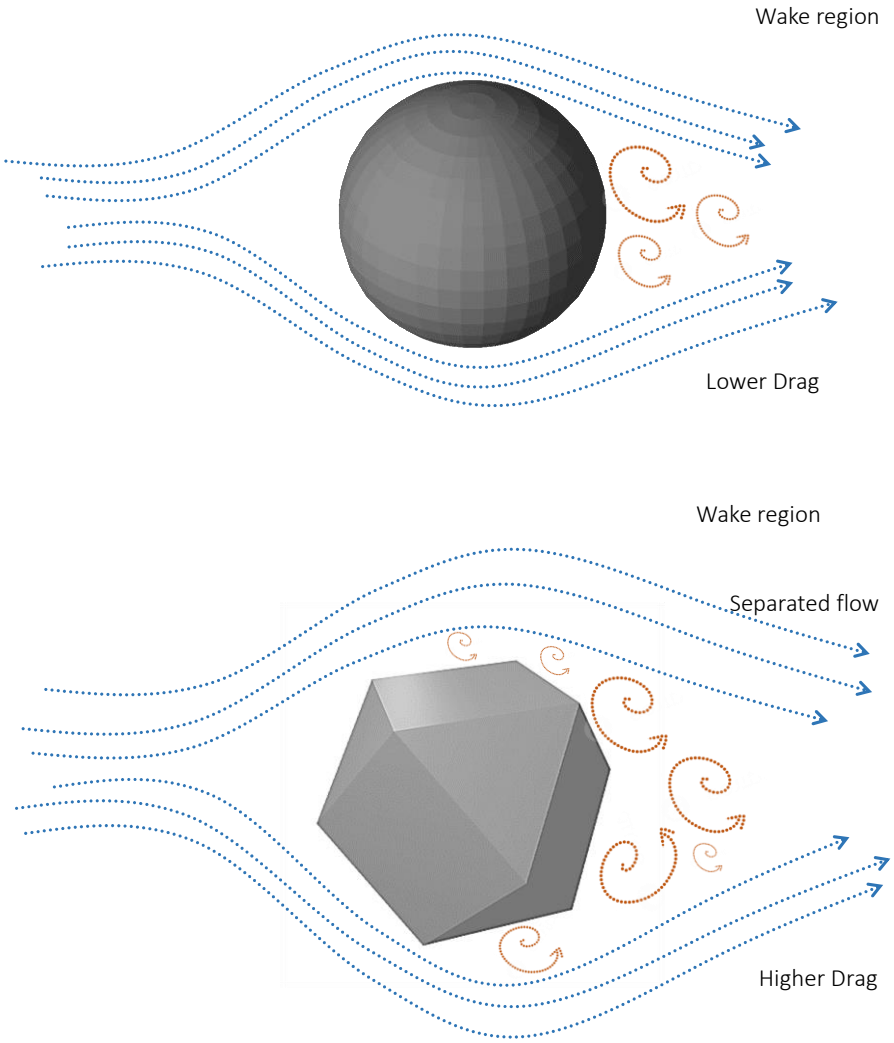


Figure 162Illustration explaining how drag is affected by the nature of geometry and the disruption of airflow around specific geometrical forms. Source: Author.

More streamlined forms appeal more to engineers and designers, for they capitalise on the aerodynamic benefits that smoother curves provide, hence gaining better efficiencies in performance in various other aerodynamic applications. In addition, diverse configurations in the shape of closed forms and aerofoils were computationally designed and tested by making use of the CFD techniques. Determining the drag coefficient for each model was used to evaluate uniformity in aerodynamic efficiency, to facilitate the understanding of the bond between geometry and aerodynamic efficiency,

which could be successfully unveiled by increasing the understanding of different configurations under the same conditions, and finally having a clear and accurate comparison between ridged section and a more streamlined and aerodynamic cross-section for the star component. The system was modified to perform testing for different cross-section examples using CFD tools, after which the models were compared. This included a comparison of three shapes:

:

- Circular cross-section (Shape A): exhibits less drag than the square cross-section and maintains a symmetrical form, with a resulting drag coefficient of 0.43 under the fixed context, typical of various spherical forms.
- Aerofoil NASA23015 cross-section (Shape B): minimised drag compared to the circular cross-section, but does not have a symmetrical form, with a resulting drag coefficient of 0.07 under the fixed context.
- Aerofoil Reflex Cambered Cross-section (Shape C): This cross-section minimises drag compared to the circular cross-section but does not have a symmetrical form. Due to the flat top and bottom surfaces, it has a resulting drag coefficient of 0.31 under the fixed context.

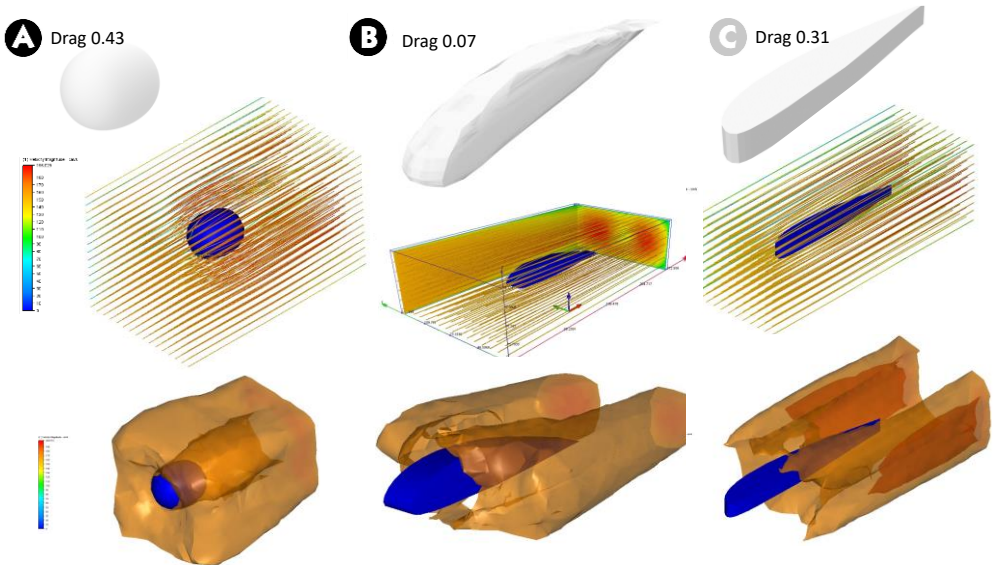


Figure 163 Examination of the drag coefficients for the NASA23015 aerofoils, Reflex Cambered aerofoils, and circular cross-sections using CFD simulations - Source: Author.

The aim was to compare and articulate the aerodynamic performance values of the different forms, thereby revealing the specific strengths and limitations inherent to each. These meticulous studies, in turn, equip decision-makers with the insights needed to pursue more effective optimization strategies and ultimately arrive at impactful, performance-driven design solutions.

Consequently, the star component was designed in accordance with the selected cross-sectional shape, and a CFD simulation was performed, which compared the results for the NASA23015 and Circular

cross-sections. This study attempts to demonstrate that the airflow characteristics and aerodynamic efficiency of each section differ substantially and consequently offer additional information about them relative to each other, namely their positive and negative effects. It is a compulsory step for the optimisation of the performance indicators, such as drag reduction, which is evaluated to determine the trailing edge geometry, as shown in the following images that capture some of the testing stages in CFD simulations.

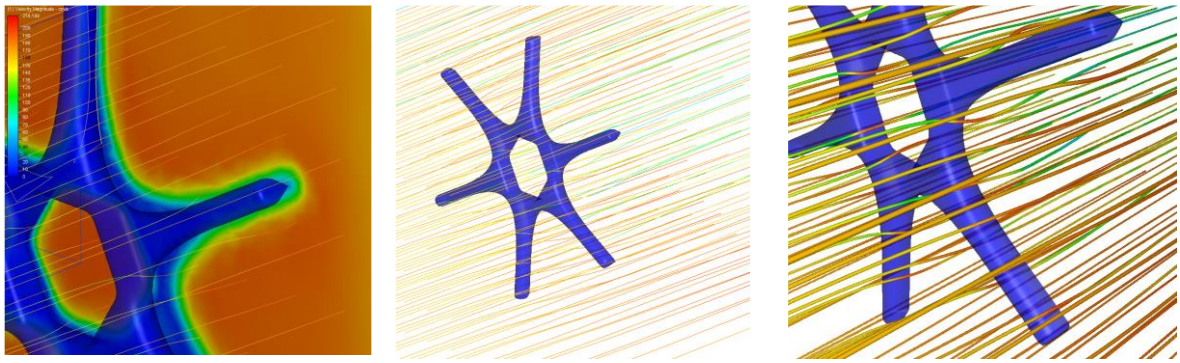


Figure 164 Detailed examination of the aerodynamic performance of the aerofoil cross-section applied to the star component, and its influence on surrounding airflow behaviour as analysed through CFD simulations. – Source: Author.

Examining the drag coefficients of the NASA23015 aerofoil and circular cross-sections—0.04 and 0.43, respectively, when modelled in isolation—revealed a considerable disparity in performance. However, when these profiles were implemented as cross-sectional geometries within the star component, an unexpected discovery emerged. Despite the significant differences in their standalone aerodynamic behaviour, both designs yielded comparable drag results when applied to the proposed six-point star configuration, as illustrated in the following images. This outcome suggests that, when simulated under real-world conditions, there is no discernible distinction in airflow behaviour or aerodynamic optimisation between the two cross-sectional profiles.

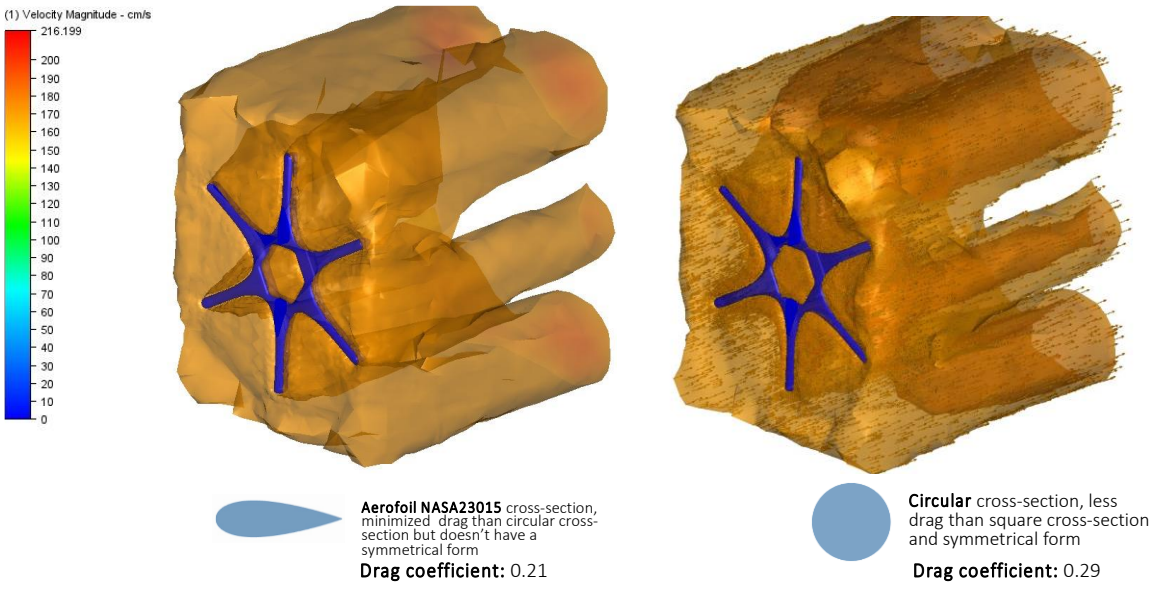


Figure 165 Analysis of the drag coefficients of the NASA23015 aerofoil and circular cross-sections when applied to the star component, as examined in detail through CFD simulations. – Source: Author.

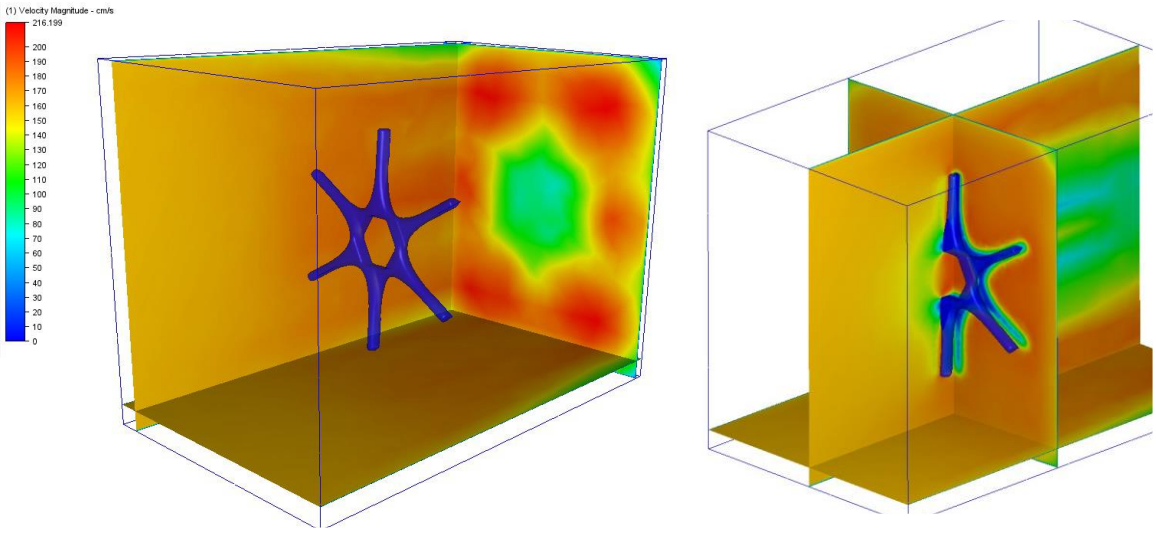


Figure 166 Analysis of the drag coefficient of the NASA23015 aerofoil when applied to the star component, as examined in detail through CFD simulations. – Source: Author.

Aerodynamic testing and CFD simulations revealed that the drag coefficients of the different cross-sectional shapes were initially quite distinct, with the NASA23015 aerofoil registering at 0.07 and the circular cross-section at 0.43. However, when these shapes were incorporated into the star component, the drag values converged significantly. This outcome illustrates how subtle design adjustments can lead to substantial improvements in overall drag performance, underscoring the impact of micro-level changes on optimising system aerodynamics. Based on these results, the circular cross-section was identified as the most appropriate configuration for the constituent element of the star, considering both aerodynamic enhancement and broader design factors. The circular profile offers a stable, symmetrical form that is convenient to model and reliably retains its shape during slip-casting production. In contrast, the aerofoil configuration may require complex modifications to the pattern sections and can result in inconsistencies in the produced mould.

The circular structure simplifies the production process and mitigates potential issues, such as the presence of sharp edges that increase the risk of internal cracking during the slip-casting process—an issue observed with the aerofoil scenario. This simplification reduces limitations in production speed and improves material integrity, while enhancing the repeatability and structural strength of the manufactured components. By prioritising manufacturing practicality, the circular cross-section aligns with the overarching objective of streamlining processes while maintaining design integrity and performance consistency. It also provides a validated design iteration at each stage of development.

8.1. c. Water supply and structural support

The integration of the water supply system and the structural support structure was also taken into consideration and played an important role in the design process. This factor prompted a detailed assessment of how the star components would work together effectively when applied on a larger

scale. Addressing the logistical needs and the structure's strength in parallel was essential to achieving a coherent and robust design solution that can effectively fulfil the required function while maintaining structural stability and durability.

8.1. c. a. Water supply

Looking at the Ecooler screen, as discussed in the literature review chapter, it presents a remarkable approach to designing a screen with an integrated internal water supply. This system is essential for enabling the cooling process and ensuring the screen's effective operation. Evaporative cooling relies on the circulation of water, which is typically incorporated into the design to move water fluid around the ceramic components. Water tanks, pumps, distribution pipelines, channels, and nozzles are among the common mechanisms used to facilitate this circulation, while reservoirs store the water required for cooling.

Figure 167 The detailed water connection system of the Ecooler screen. - Source: (Kahn, 2010)

The water supply in Ecooler is designed to provide consistent wetting for efficient cooling through channels that distribute water evenly across the ceramic components. Water is released via pumps onto the ceramic, which contains internal cavities that function as capillaries within the system. The porosity of the ceramic plays a crucial role in wetting its external surface, where evaporation occurs and cools the surrounding air by absorbing heat. Additionally, the water supply system may include components such as filters to remove contaminants, sensors to monitor water levels and flow rates, and controllers to regulate system performance based on ambient conditions and cooling requirements (Kahn, 2010).

Principally, the water distribution system in the Ecooler mesh pot aims to ensure a continuous and consistent flow of water to facilitate the evaporative cooling process, thereby enhancing the overall efficiency of the system (Kahn, 2010). At the same time, this approach supports water conservation and mindful consumption during cooling, minimising the loss typically associated with direct evaporative cooling. Since the system is designed to circulate water internally within the screen, only the necessary amount is used for cooling. This technique reduces waste by directing water precisely to the ceramic

components, where it evaporates and cools the surrounding air—unlike conventional cooling systems, which may consume excessive amounts of water.

In parallel, a similar method was integrated into the design phase of the star component, incorporating interior chambers that render it hollow from within—a feature that can naturally be achieved with uniformity through slip-casting, provided the model is designed with this aspect in mind. Accordingly, the star component was designed to enable the circulation of water through capillaries (internal water channels). The inclusion of these internal cavities contributed to reducing water loss and promoting responsible water usage throughout the system. This aspect reflects a conscientious effort to optimise water resources while adhering to sustainability principles and eco-friendly design practices.

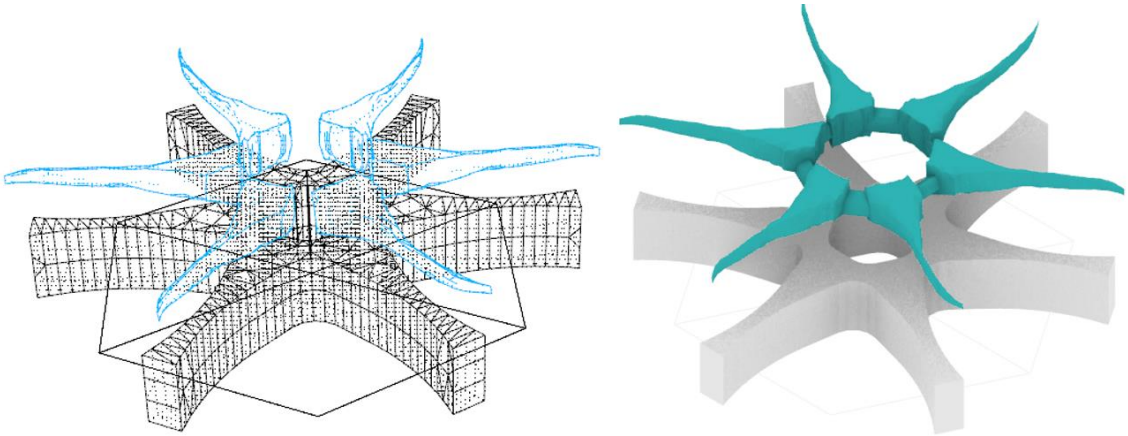


Figure 168 Illustration showing the intended design of the internal water capillaries within the star component of the screen. - Source: Author

To ensure a hollow cavity inside the star component, its thickness must be maintained within a specific limitation. First, the appropriate thickness enhances both structural stability and functional performance, while also allowing for water circulation. It further facilitates the integration of essential features such as internal water channels. This approach stems from the need to achieve a well-balanced relationship between frame stability and design practicality. Through this meticulous attention to detail, it can be concluded that the structure was conceived holistically—not merely as a form—while maintaining a deliberate balance between form and function.

Hence, time was dedicated to producing initial models with specific dimensions and weight to anticipate both the opportunities and challenges of the project. A key factor in this process was close collaboration with experienced craftsmen who understood the complexities of the slip-casting production procedure, an effort that proved invaluable in avoiding potential mistakes. Through collaborative engagement and iterative design methods, the intricacies of the manufacturing process were carefully navigated, fostering a dynamic interaction between theoretical design considerations and practical production

constraints. This approach ultimately led to the development of a comprehensive design, as illustrated in the following sketches that outline the proposed design modifications.

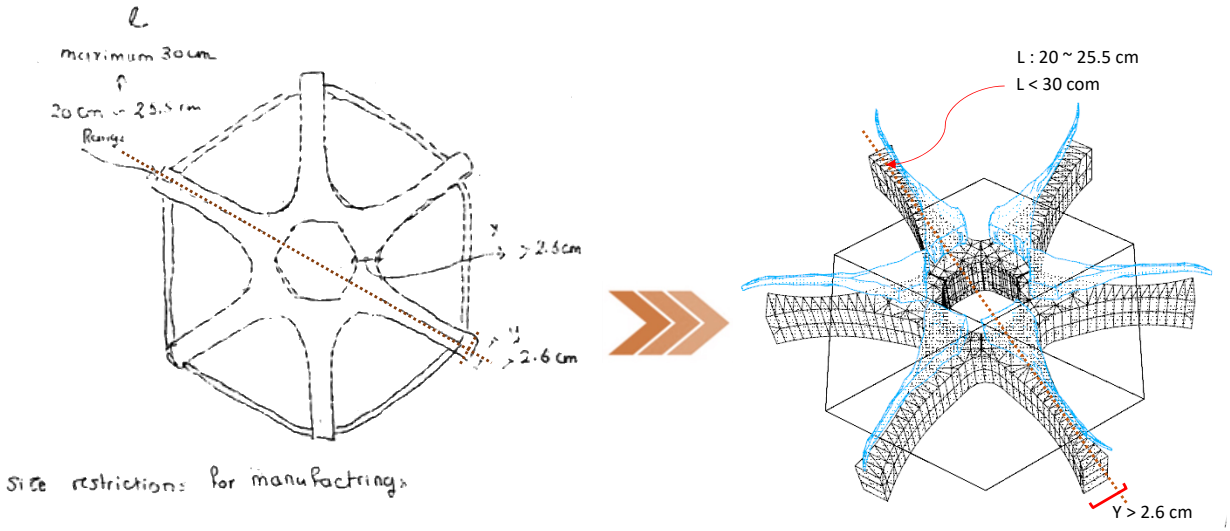


Figure 169 Illustration detailing the dimensional specifications devised to address the limitations inherent in slip-casting, thereby ensuring the precise formation of the screen's internal water capillaries. - Source: Author.

Accordingly, as the component undergoes parametric repetition, the water supply system evolves into a network of interconnected capillaries spanning the entire screen. This intricate system enables the continuous circulation of water, ensuring uniform distribution and optimal performance across the full surface. The design strategy not only facilitates efficient hydration of all segments but also preserves the structural integrity of the screen, as illustrated in the following image.

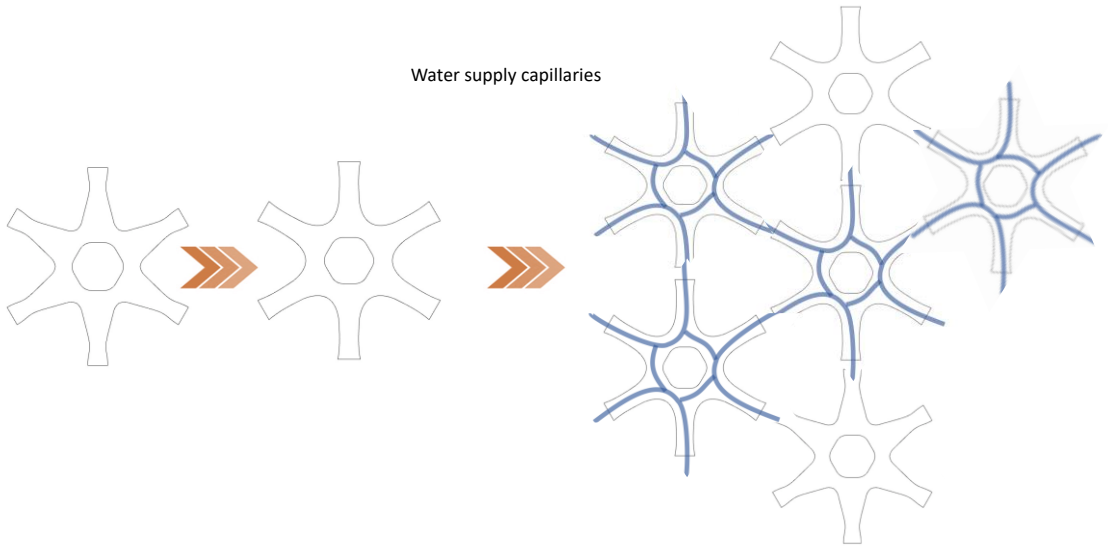


Figure 170 Visualisation of the water supply system as it emerges through parametric repetition of the screen's pattern, illustrating the integration of capillary networks across the full surface. - Source: Author.

8.1. c. b. Structural support

The integration of the water supply system, while incorporating the potential to maintain the structural integrity of the system, also played a role in its design. As a result, it was necessary to evaluate how these pieces would effectively work at a larger scale, given this essential concern. The simultaneous examination of the design requirements and the strength of the structure necessitated the development of a coherent and robust design solution that could support its purpose. Thus, it was equally essential to determine the integrity of the extruded hexagon and its capacity to support loads, which are part of the design validation. This also required a detailed analysis of how the cap would dissipate loads and sustain forces while being employed for its purposes. Consequently, debates were raised about whether water capillaries should be placed inside the hexagonal extruded form or not, and whether they would improve performance while maintaining structural stability, as shown in the following image.

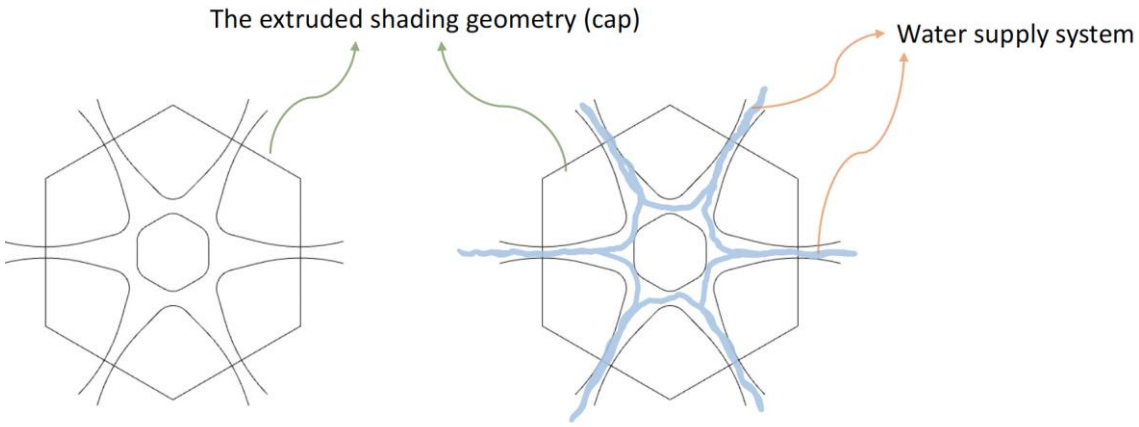


Figure 171 Image illustrating how the water supply system of the screen is confined to the star component, excluding the hexagonal extruded structure. - Source: Author.

Accordingly, a series of evaluations was conducted to identify the optimal configuration that would maximise the balance between structural integrity and functional efficiency. These assessments considered the following factors:

- The internal water channels, including cavities and capillaries, through which the water system circulates within each component.
- The method of component connection ensures watertight seals that prevent leakage throughout the system.
- The adaptability of the screen system, examining how it can be integrated across various apparatuses and design configurations, including mechanisms of attachment and suspension.

In the fine line of rating the ability of the stacked hexagonal structure vertically against numerous loading parameters, all these parameters sum up the system's total load endurance capacity, including its ability to transmit loads by virtue of its design. Next, these surfaces contribute significantly by

distributing forces consistently across every single point of their overall surface level. As a result, load stress concentrations are reduced. Besides that, the hexagonal structure creates the interlaced forms that allow smooth unions between adjoining units, resulting in maximum transfer of loads. This joining system ensures durability and reduces the probability of structural failure due to external causes, as evident from the following figure.

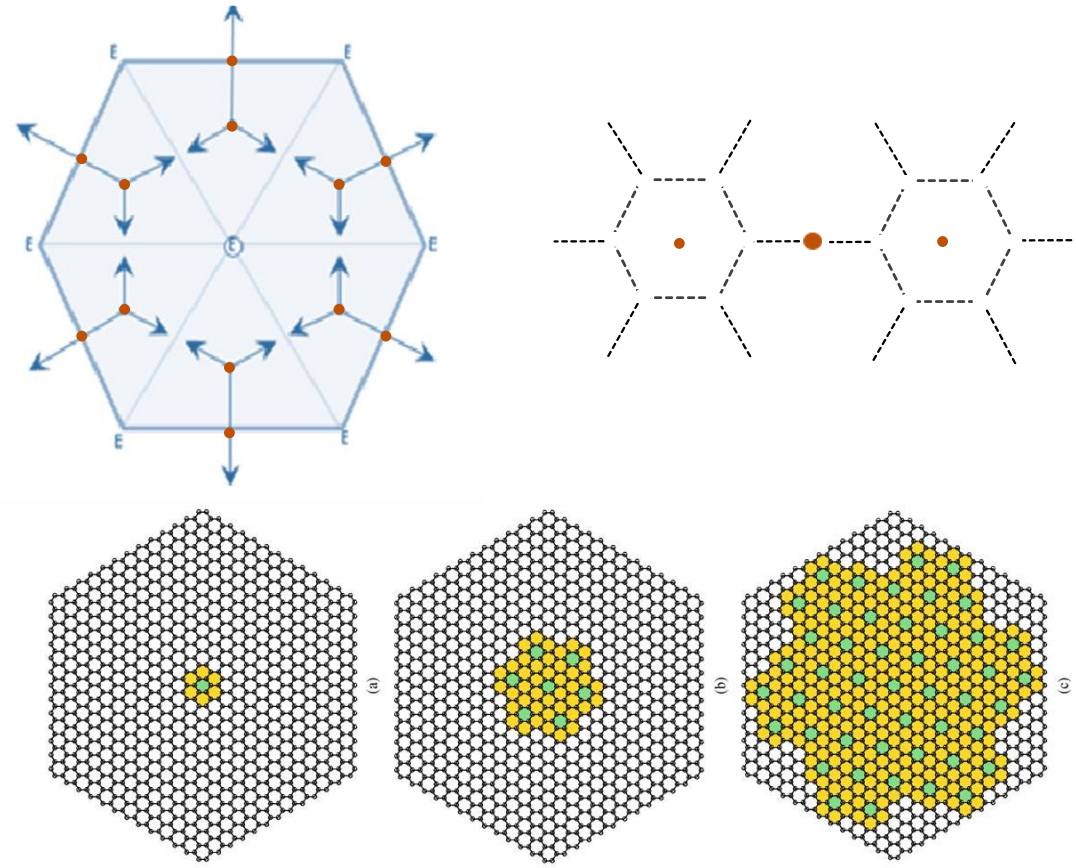


Figure 172 This joining system ensures durability and reduces the probability of structural failure caused by external factors, as evidenced by the repetition of the hexagonal structure shown in the figure - Source: Author

Additionally, hexagons show the best arrangement efficiency, which is useful for covering a surface using a minimum amount of wasted space. Consequently, material is used with maximum efficiency. The use of such a method of coordination in construction minimises voids and gaps, resulting in a heavy and durable structure. Furthermore, the highly symmetrical nature of hexagonally arranged structures enables balanced loading in all directions, regardless of the direction of the forces. On the other side, when the hexagonal cover is removed and the star patterns are joined, there is a reduced connection between the nodes of the structure due to the smaller number of node connections. Changing the cap geometry can result in a lack of connection structures, creating a localised stressed area and leading to a decrease in stability. This figure portrays the various load transfer systems found in two types of structures. The purple arrows indicate the paths through which loads are transferred into each structure, showing shear stress passing through. The picture accentuates the fact that the nodes of the

star component suffer the most from shear stress. Such nodes, where shear stress and load are exerted, are identified as areas more prone to developing structural failures due to the impact of shear stress. The instability that accompanies this, conversely, highlights the potential efficiency of distributing or loads through the addition of the hexagonal shape.

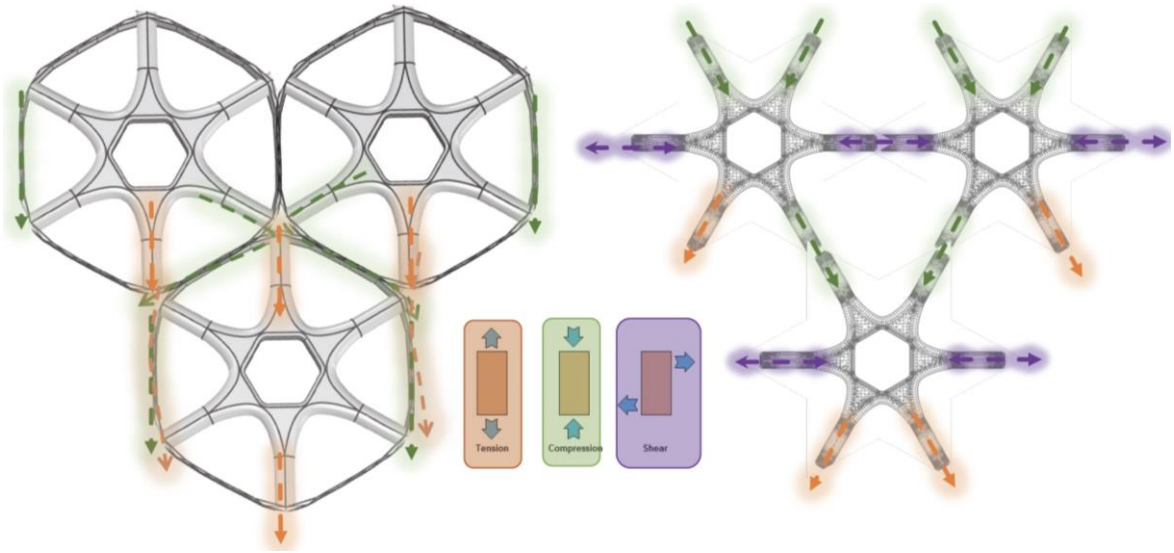


Figure 173 The distribution of loads in the proposed pattern with and without the hexagonal geometry - Source: Author

In sum, the hexagonal shell formed by the linked design reduces load concentration and consequently mitigates stress distribution, an effect that validates the engineering preference for shared bearing methods to enhance structural safety and integrity. This exemplifies the inherent symmetry of the hexagonal form, which contributes to greater stability and robustness, rendering hexagonal configurations ideal for screen construction and a wide range of architectural applications.

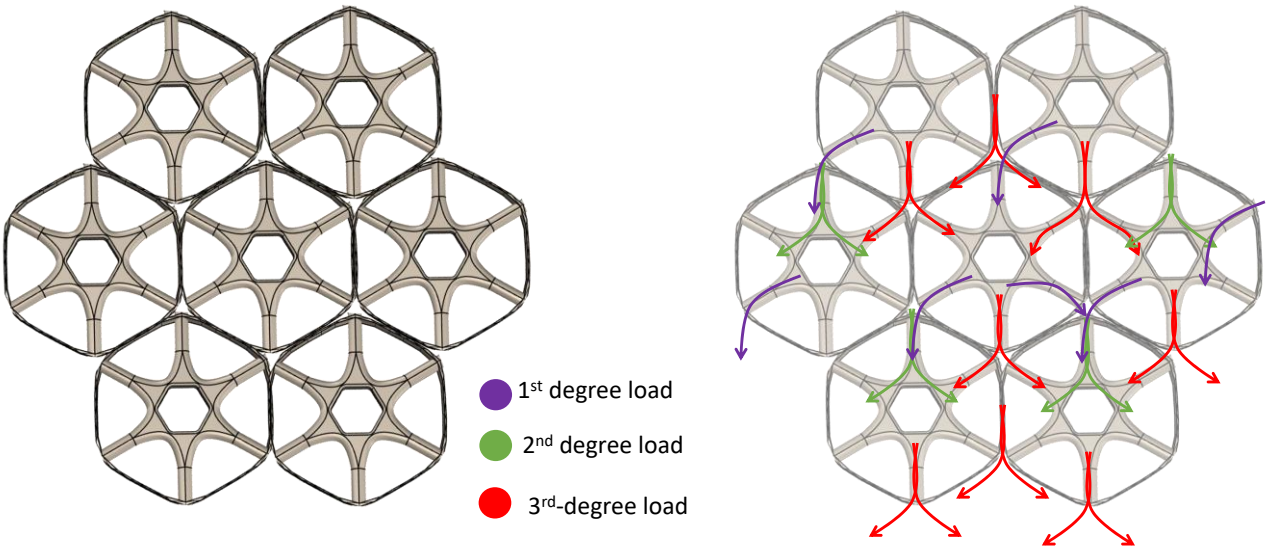


Figure 174 Illustration explains the distribution of loads in the proposed pattern, the hexagonal geometry, and how the hexagonal form helps in efficient load distribution among the structure of the screen- Source: Author

In conclusion, the combination of these variables leads to the fact that vertically stacked hexagonal constructions are highly efficient in carrying loads and maintaining structural integrity. However, this structural integrity accounts only for the load of the screen itself when the components are placed vertically, without considering the impact of wind force on the screen. Hence, the applications of the screen may vary in real-world scenarios, and further structural analysis will be needed to maximise structural integrity and ensure resilience against any unforeseen conditions.

Another aspect considered in the structural support is the planned method for attaching the components to the window, ensuring that this method remains both durable and flexible for current and future applications. Accordingly, the structure holder and the attachment of the screen were addressed during the design stage.

1. c. c. Structure holder.

The hexagonal shape added to the screen is considered more durable than the star component itself, as it lacks cavities for water circulation and is not hollow on the inside. This characteristic facilitates the attachment of the components to the window and ensures that the method remains durable and versatile in 1:1 future applications. As a result, the structure holder and attachment of the screen were planned throughout the design phase, as shown in the following image.

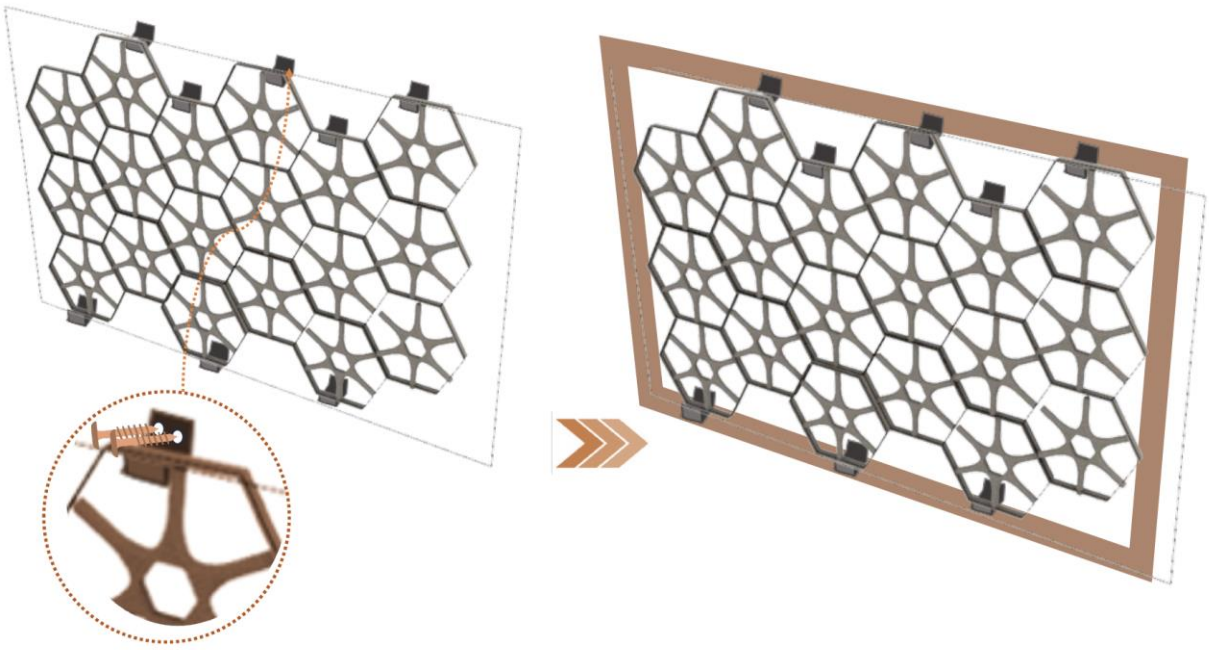


Figure 175 Illustration explains how the hexagonal geometry facilitates the structural attachment of the screen to the aperture using steel hooks and screws - Source: Author.

This method further validates the inclusion of the hexagonal form within the star component and its integration as part of the star in the slip-casting fabrication, as shown in the following image. It also

supports the decision to exclude the hexagonal form from adding the water supply system, thereby enhancing the structural durability of the component when applied at a 1:1 scale.

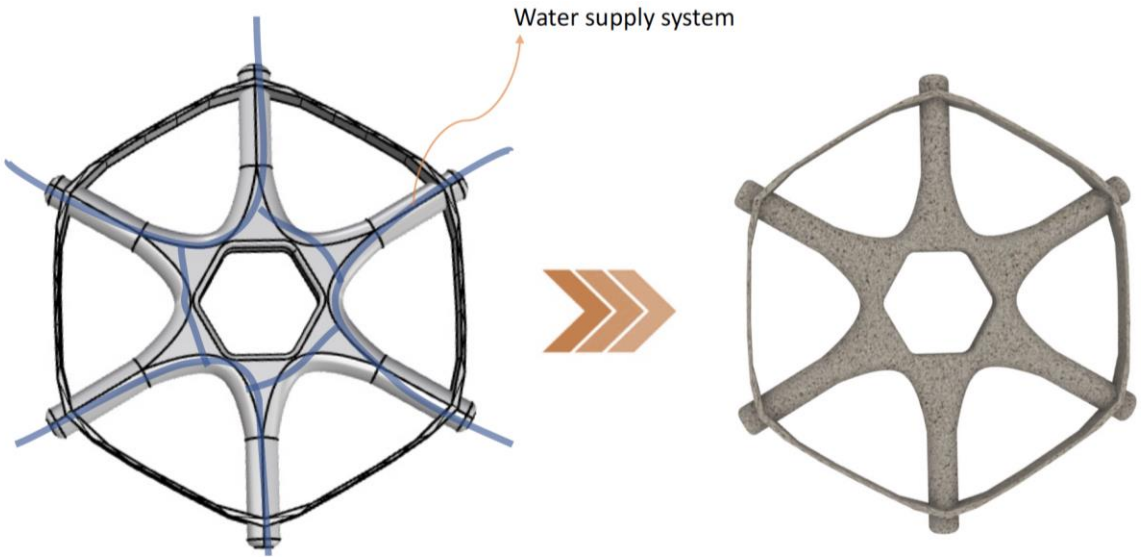


Figure 176 Image explains how the water circulation system is limited to the star component, excluding the hexagonal geometry, which enhances the structural durability of the hexagonal form - Source: Author.

The acceptability and efficacy of design decisions are systematically evaluated, resulting in maximum performance and durability in real-world applications. Therefore, it was important to also consider the appropriate connection method between the components — one that is durable enough to maintain structural stability and support the water-flowing system inside the screens without being affected by overexposure to water content. Another aspect to consider when selecting the connection material is its characteristic properties and the necessity of the lightweight composition, so that the connections between the screens do not add to the overall weight of the screen when applied. On the other hand, the material should be water-resistant to prevent any leakage between components when water is ducted inside the screens, thereby maintaining reasonable water consumption limited to the evaporative cooling process without any loss factor.

Cork is a natural material readily available in the local environment, as it is derived from oak trees commonly found in Mediterranean regions such as North Africa. It is distinguished by its lightweight composition, water resistance, watertightness, and effective thermal insulation properties. These characteristics enable the maintenance of water temperature within the channel that will be intuitively carved into the cork to facilitate water circulation between the components. Accordingly, cork was selected as the connection medium between each component. The application of this connection element is illustrated in the following image, which demonstrates how the cork will be drilled to accommodate a channel that allows the water system to flow across the screen.

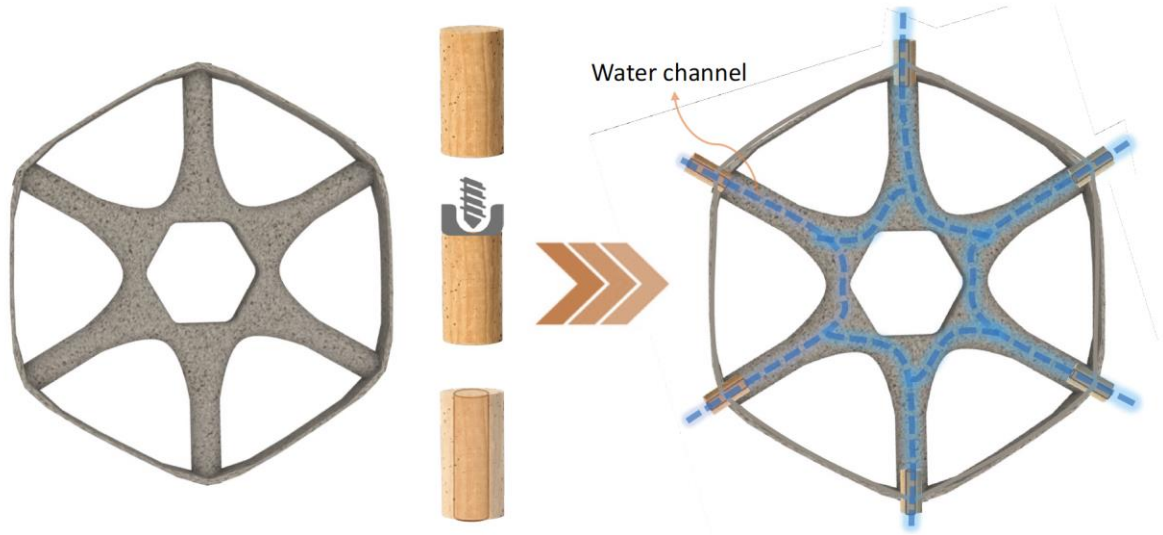


Figure 177 Illustration showing how the cork is drilled to accommodate an internal channel, enabling water circulation across the screen components - Source: Author.

As part of the effort to identify the most effective solution for the screen connection mechanism—specifically the use of cork—a series of targeted experiments was conducted to evaluate its material properties. These tests assessed cork's durability and its resistance to water and other environmental factors that could compromise its performance. The rigorous experimental procedures, as documented in the academic journal, encompassed a range of scenarios simulating real-world conditions. In addition to confirming cork's suitability, the testing process revealed potential challenges and limitations that may arise during implementation. This comprehensive evaluation enabled the research team to fully characterise cork's behaviour as a connecting material. Beyond material validation, the experiments also informed the precise design of the connection mechanism and the overall screen system. The following graphic illustrates the selected material and reflects the outcome of the various cork-based trials, which are discussed in detail in the forthcoming scholarly publication.

Figure 178 Picture illustrates the final material selection and shows the results of many cork experiments that will be shown in the academic journal of the research – Source: Author.

8.2. Size and scale guidelines

The preceding stages' debates and analyses demonstrated a commitment to a thorough design examination in order to improve and optimise the proposed solution. As a result, it was critical to understand the scale of the application and the appropriate screen size to minimise waste area when the screen is applied to the aperture, while also maximising the screen's sustainable approach when it is applied, considering application efficiency and managing resources.

The screen was designed to be highly flexible and expandable, anchored in a base pattern that maximises adaptability for future modifications. This foundational geometry allows for seamless alterations, ensuring the screen can evolve in response to emerging design needs. Accordingly, the component's scale was carefully developed to maintain specific dimensions that overcome the limitations of slip-casting, while also accommodating internal water cavities. These cavities played a key role in determining the minimum viable dimensions of the screen. Conversely, identifying the maximum effective size was equally critical. In this regard, responding to the apparatus's dimensions served as a strategic starting point for addressing the challenges of scale optimisation.

Most commonly, window manufacturers produce windows in standard sizes, which are determined by architectural norms and resource availability. Single-hung windows, for instance, usually measure up to 24x36 inches, while double-hung sizes include up to 30x60 inches. These dimensions are popular, as they require only common construction practices and readily available building materials. Nonetheless, typical differences in window sizes appear between regions due to the diversity of building codes, architectural preferences, and the specific needs of the building project.

Considering the typical sizes of windows is necessary to propose the screen for different building typologies that usually come with standard-designed apertures. Adjusting the screen dimensions to be commonly used for different window sizes promotes the implementation of the screen in various scenarios.

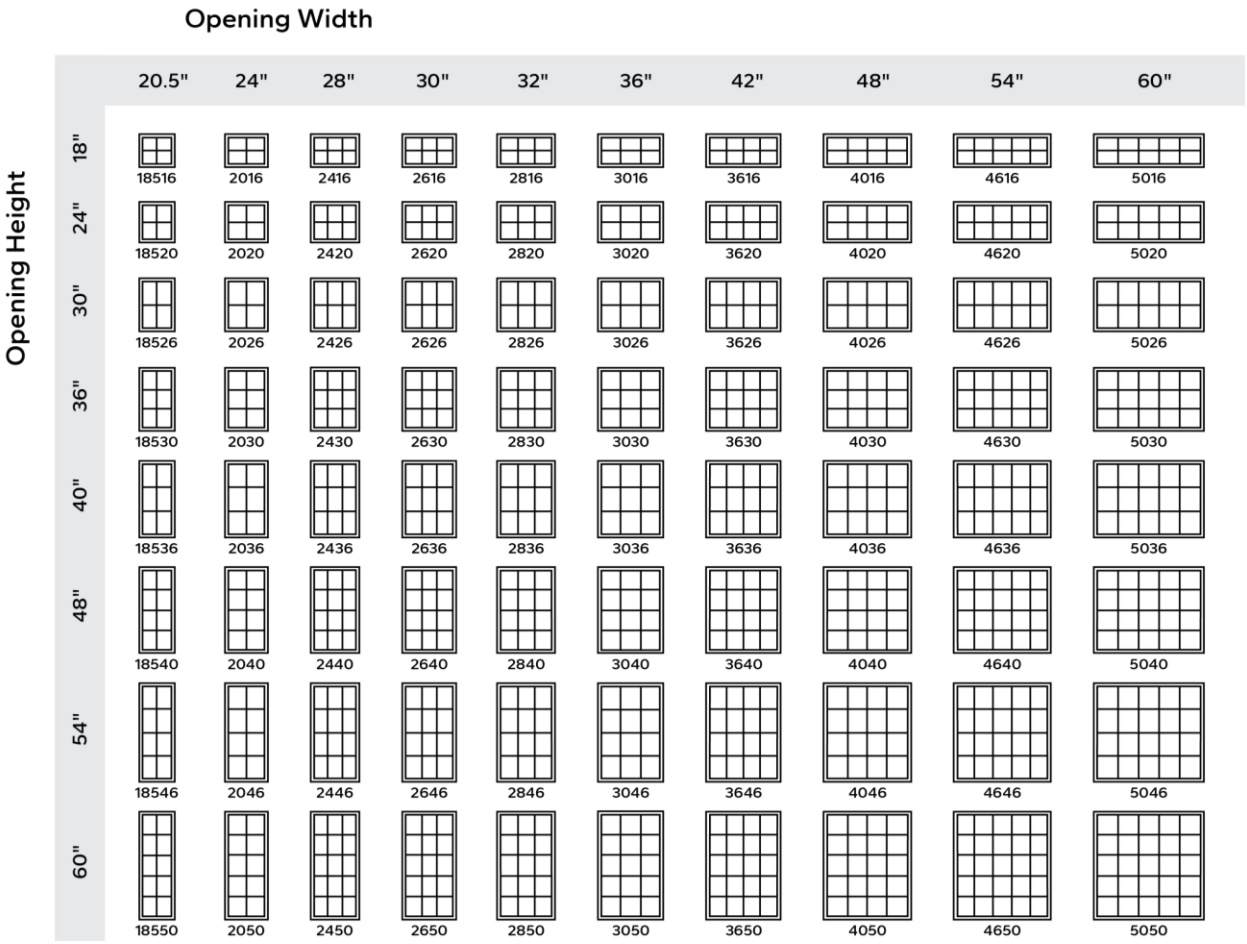


Figure 179 Standard window sizes globally for awning windows - source: (Vries, 2023)

On the other hand, regarding compatibility with standard sizes, the screen integrates very smoothly into existing architectural designs, thereby avoiding cost-intensive and time-consuming adaptation. In this way, the screen becomes easier to fit, and the labour costs associated with individualisation can be reduced.

In addition, screens manufactured to the appropriate size of window frames are readily available in the market; hence, they offer customers greater convenience and flexibility, while also overcoming challenges in the manufacturing process, such as the need for multiple prototypes to fit different window sizes. Holistically, by following the same sizing principles, the visual appeal of the screen is also enhanced, as shown in the following image. This provides evenness with the surrounding structures and ensures the preservation of architectural integrity. From a pragmatic perspective, screens optimised to standard window sizes yield efficient use of materials, reduce wastage, and encourage sustainable practices. Moreover, commonly sized windows facilitate the easy implementation of standard industry protocols and guidelines related to window installation, assuring safety and adherence to regulations and standards.

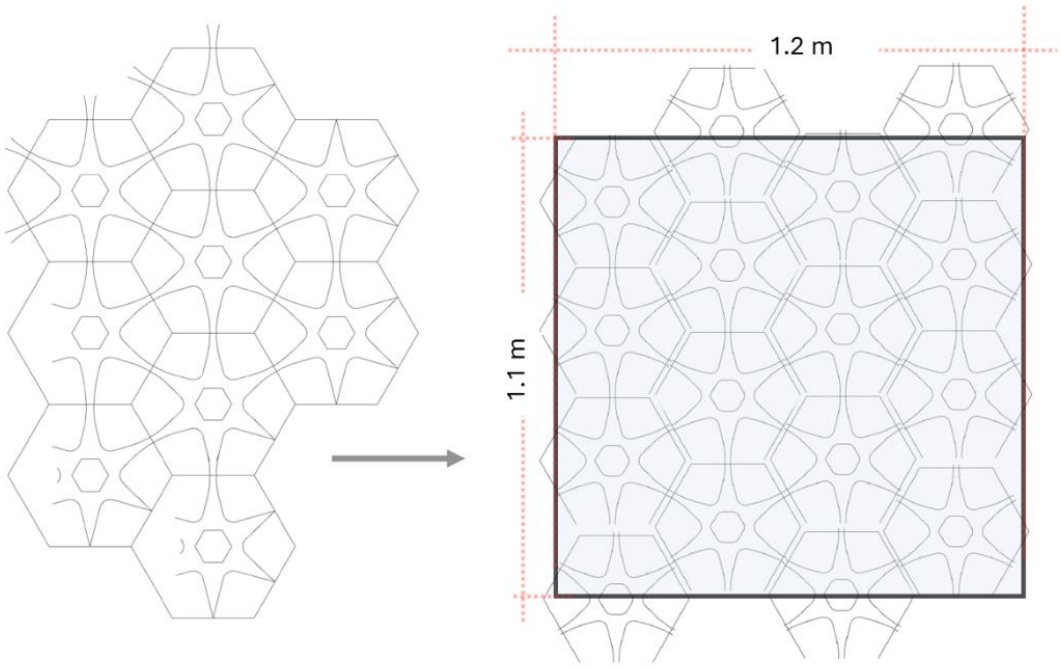


Figure 180 The image illustrates how the component is applied repeatedly to a standard-sized window – Source: Author.

Keeping a standard model also functioned to solve both the limited-resources and aesthetic-related challenges, which are:

- **Manufacturing:** On the one hand, it has a simple structure, avoiding the corresponding complication of producing successive prototypes and time-wasting. This measure guaranteed effectiveness and affordability—the resources were distributed well, with inevitable overspending cut down.

- **Application:** Due to unified prototyping, all modifications did not affect the screen's initial build integrity. That was accomplished in such a way that the view of the patterns remained sustainable after application, so as not to confuse the core design elements.

- **Aesthetic vision of the pattern:** The visual statements, having a uniform (consistent) appearance, brought about a tone that was cohesive and coordinated, hence improving the overall visual presentation (screen as a whole). Consequently, this reduction in prototyping complexity consisted not only of standardisation of the prototype but also of safeguarding the production's artistic and aesthetic line, resulting in a design that perfectly fitted its intended purpose and lay elegantly on the aperture.

In conclusion, this method enabled the flexibility of suitable apprehension, providing for the addition or removal of the components of the whole screen system in response to the disparity in design specifications and customer preferences, making the screen accommodate a Lego-like structure inspired by the designs and fundamentals of Mashrabiya, as shown in the following image. In addition

to making assembly and component modifications easier, the methodology established the framework for user-centric design, which involves involving users in the design process. This approach brings the design perspective to address the individualised requirements by making it compatible with various architectural styles, as well as quickly adapting to shifting design trends. As a result, it is more adaptable and modular, making it more appealing and applicable to various scenarios.

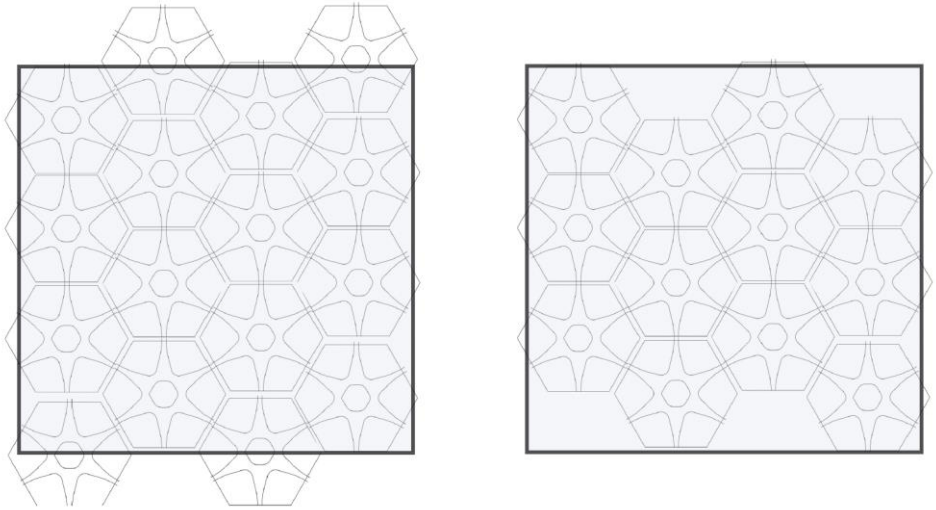


Figure 181 The image illustrates how the entire screen system's components respond to differing design requirements and client preferences, resulting in a structure analogous to Legos. - Source: Author

8.3. Fabrication of the prototype

The construction of the prototype marked the successful culmination of the design development process. It also served as a tangible response to all prior challenges and the realisation of the envisioned design. Although the measurable version evolved over time, it offered a much clearer representation of the potential framework for concept development, which was subsequently refined in the final prototype. Consequently, the creation of the prototype acts as a bridge between theoretical perspectives and practical outcomes. These initial stages embody the transition from abstract ideas to a concrete opportunity for evaluating the technical capabilities and feasibility of the proposed design in a real-world context.

Consequently, the emergence of the initial prototype through the application of innovative 3D printing technologies represented a significant stage in the fabrication process. This preliminary method laid the foundation for subsequent phases, particularly slip casting and moulding, which constituted the final steps in fabricating the prototype. Through the capabilities of 3D digitisation, precision and efficiency were maximised in translating digital designs into physical forms. Moreover, 3D printing significantly accelerated rapid prototyping and enabled iterative refinement and experimentation, resulting in minor adjustments and modifications aligned with the desired design and performance requirements. The initial prototype produced using 3D printing technology, primarily from recycled

plastic coils available within university facilities, played a crucial role in realising the prototype's initial physical form.

8.3.1. Prototype preparation

3D-printing technology relies on recycled plastic coils available in the university's laboratories. The 3D printer is connected to a computer system, where the digital STL Rhino model serves as the primary input. Additional software is used to translate this geometry for the 3D printer. OrcaSlicer is employed to slice the STL digital geometry and generate structure supports that may be required to maintain the stability of the geometry during the printing process. The slicing stage in this software determines the refinement and accuracy of the printed geometry's details, as well as the duration of the printing process.

The available 3D printers on the university premises present size limitations, as the maximum working canvas is 15x15 cm, while the prototype measures 28x28 cm. Overcoming these dimensional constraints relies primarily on overlaying the canvas size onto the digital geometry to minimise the number of cut sections and reduce the visibility of manufacturing lines on the final prototype, as illustrated in the following image.

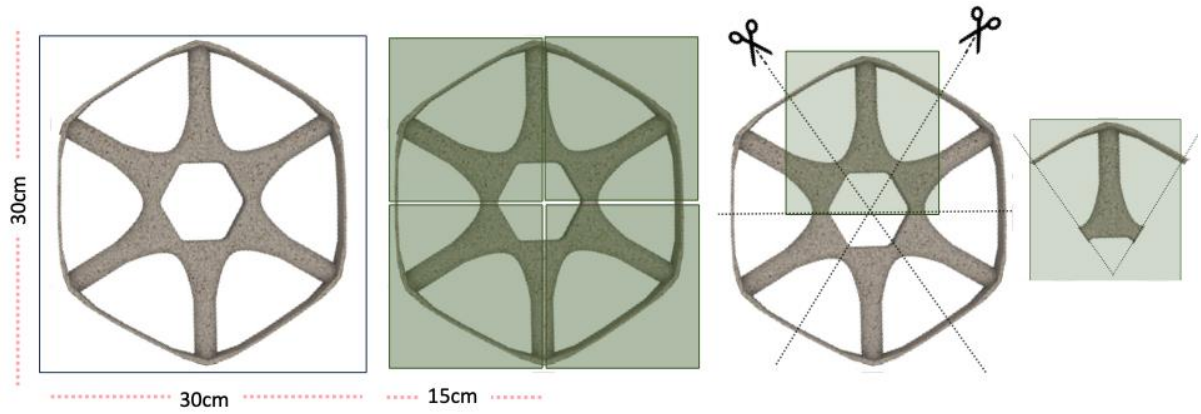


Figure 182 Overcoming size restrictions in the 3D printing canvas and minimising the cutting areas - Source: Author.

Structure support for the geometry is also one of the enabling features in OrcaSlicer software, and it is essential to ensure the successful printing of the input geometry, despite presenting challenges during the printing stage. Support lines can leave marks on the geometry, and the smooth texture of the prototype may not be preserved, even if the geometry is sliced as thinly as possible to enhance accuracy and detail refinement. Nevertheless, structural support was necessary to ensure the success of the process. Consequently, the geometry was positioned horizontally rather than vertically to reduce the number of support lines. This approach also confined the support lines to one side of the geometry, which is symmetrical to the opposite face, taking into account the geometry's symmetrical axes, as shown in the following images.

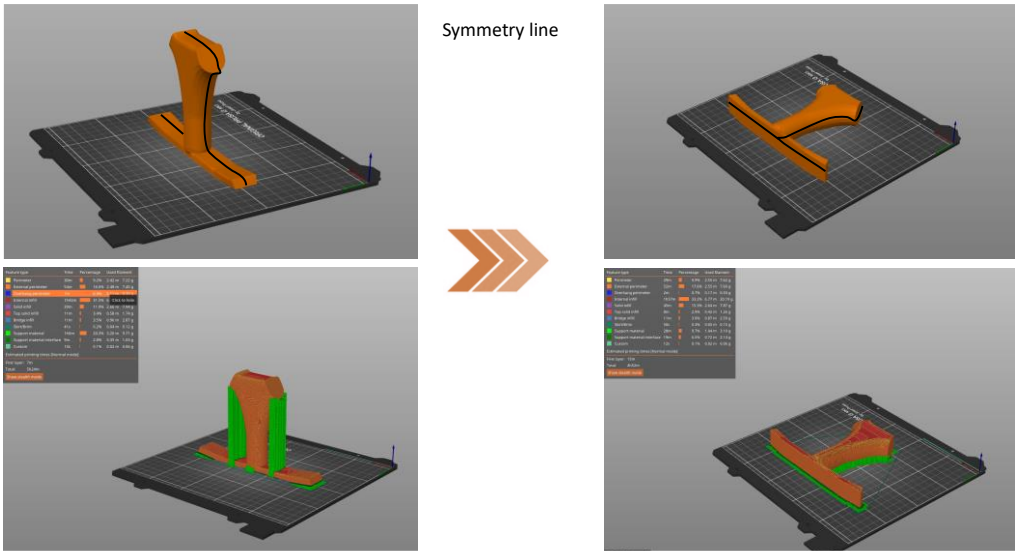


Figure 183 Screenshots from Orcaslicer illustrating the variation in the number of structure support lines when the geometry is oriented horizontally versus vertically, relative to its symmetrical axis. This repositioning minimizes support interference and concentrates structural traces on one symmetrical face of the geometry. Source: Author.

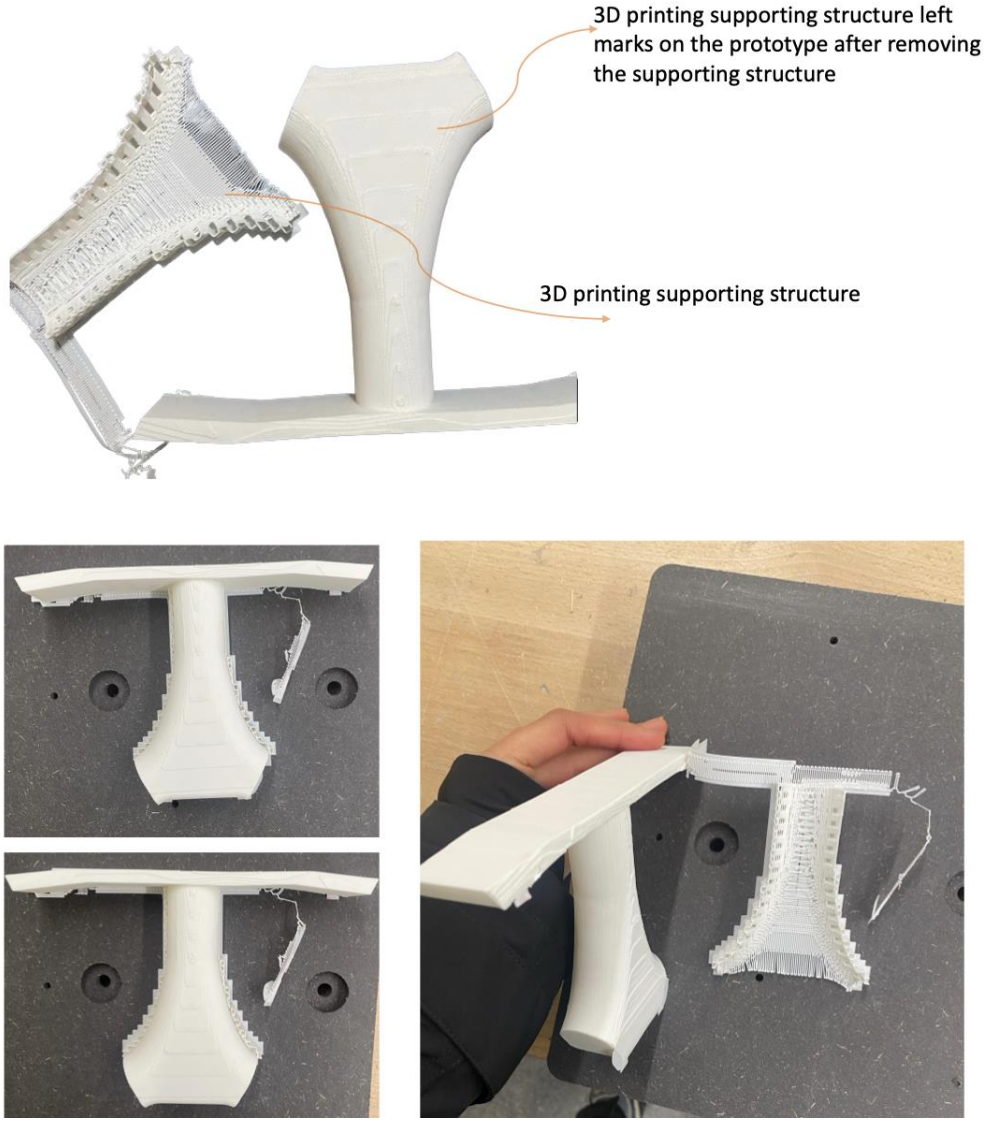


Figure 184 Close-up images showing the residual marks left on the prototype surface after the removal of the 3D printing supporting structures. These traces highlight the impact of structural supports on surface finish and texture - Source: Author.

This method allowed one side of the geometry to remain free of structural support lines, mirroring the opposite side that accommodates them. Such an approach is particularly effective for slip casting, as it enables the creation of two mould halves from the same face—ensuring a refined surface finish without visible traces of support structures. Moreover, by strategically placing the cutting line along a specific portion of the star-shaped geometry, the final form achieved a more seamless and subdued appearance. The cut line was effectively concealed once the sections were joined, as demonstrated in the following image.

Figure 185 Image showing the arrangement of 3D printed components after the printing process, assembled to form the final prototype. This step illustrates the transition from individual printed parts to a unified structure. - Source: Author.

8.3.2. Fabrication of the mould

The production of a mould that the prototyping process involved came at a milestone in the creation of the screen model. When it came to ceramic slip casting, an integral step of this process, partnering with the master craftsmen in the vicinity has been a priority to guarantee the desired result of this stage. The partnership with the local ceramic slip casters who are admired for their skills in the casting process was an important decision which was aimed at guaranteeing accuracy and efficiency in this vital phase. Being a part of this teamwork not only benefited from the traditional craftsmanship and the specialised knowledge but also accelerated the manufacturing and testing stage, making it faster to test different types of clay and different mould creation methods to optimise the mould that will be used. On the other hand, the collaboration brought the old and the new together, celebrating traditional craftsmanship and specialised knowledge while also developing the artisan's understanding of the local ceramic manufacturing industry. It was precisely this approach that enabled such a mould to be copied accurately and applied in different geographical realities as long as the given instructions were followed. The adaptation and application of ceramic industry resources was an efficient way to make the mould

preparation easily available and adaptable, hence leading the way to the global applicability of the proposed screen design. Gorbon Factory for Ceramics, a highly reputable and long-operating establishment since 1957 in Istanbul, Turkey, is commonly known for Islamic ceramic tiles and their earthenware products, which are composed of extremely valuable ingredients from the centre of Turkey, including glazes, clay, colours, and minerals (tiles, 2017). Additionally, the noticeable ambition displayed by Mr Orhan, the owner of the factory, contributed to a passionate endorsement of this artistic approach of utilising such ceramic innovation with an Islamic art background.

Consequently, the printed prototype of the designed screen was taken to Turkey, where a slip-casting expert at the factory assessed its potential for mould production. The design demonstrated a deep understanding of the challenges inherent in slip-casting, with most manufacturing considerations addressed during the early design stages. Rather than receiving criticism from the craftsmen, the prototype earned their admiration, as it acknowledged both the difficulties and opportunities of slip-casting and introduced a novel strategy that inspired them to actively participate in the creation of this innovation.

Figure 186 Documenting images during the discussion with the slip-casting expert in Gorbon Ceramic Factory to discuss the preparation of the mould - Source: Author.

The transportation of the 3D-printed prototype of the designed screen to Turkey was more feasible due to its lightweight feature, which is a crucial advantage offered by the 3D printing process. Additionally, the prototype was intentionally divided into six identically symmetrical parts, as explained in the 3D printing process. These six symmetrical pieces were easy to store and transport, allowing for convenient assembly during the mould-making process, as shown in the following images.

Figure 187 Documented steps of preparing the mould using liquid plaster poured over the 3D-printed sample. - Source: Author.

8.3.4. Finalising the sample

The creation of an evaporative cooling screen extends beyond the geometric structure design. It also involves the type of clay used, as well as the drying and firing processes it undergoes—both of which significantly influence the screen's absorption capacity and pore rate. These factors are critical in determining whether the screen performs effectively for evaporative cooling purposes. Therefore, careful attention must be given to both the design and the material properties of the clay in order to achieve the intended objective and ensure the desired functionality.

The level of clay porosity emerges as a central point of contention, as it determines whether clay-based materials exhibit high performance or suffer from poor durability—particularly when intended for evaporative cooling applications. Within this context, where moisture management is the primary objective, the degree of porosity directly influences the clay's ability to absorb, retain, and release water for effective evaporation cooling.

To begin with, clay's porosity has a significant effect on its water absorption capacity. Naturally porous clay can absorb water easily, allowing uptake from surrounding sources—an essential feature for supporting aquatic ecosystems and biodiversity. This dynamic enables clay to draw moisture from beneath or from direct contact with water. The pores in ceramic materials regulate the rate at which water evaporates from the clay surface. Air passes through these micro-cracks rather than flowing over the surface, resulting in a cooler ambient temperature due to increased evaporation from the highly porous clay surface.

Moreover, the tortuous structure of the pores facilitates effective water movement through the clay, promoting optimal evaporation and enabling natural air conditioning (Jastrzebski, 1959). The porous

nature of clay also contributes to its internal structure, which interacts with the mechanical forces and influences its varying degrees of resistance (Jastrzebski, 1959). Higher porosity increases the potential for inter-grain absorption and evaporation. Therefore, achieving the right balance of porosity is crucial to combine the benefits of efficient performance and material durability, ensuring the screen functions effectively without compromising its structural integrity.

Accordingly, when designing and selecting materials for evaporative cooling devices, one of the most important factors is the amount of water evaporation (Jastrzebski, 1959). Since the screen will be connected to the main water supply system, and porosity will be crucial in preventing leaks, the design was tested at various porosity levels to balance the most effective porosity level while maintaining the screen's durability.

8.3.5 . Manufacturing the sample

After the mould was fabricated, as documented in the previous parts of this chapter and thoroughly documented in the accompanying journal, it was finalised and cleaned to be used in the slip-casting process to prepare the sample screen. Therefore, it was important to ensure the correctness of the classification of the mould. Working with the locally available clay in the Gorbon factory, which is commonly used in the slip-casting manufacturing method, represents the beginning of a chain of events aimed at acquiring the final clay model of our visionary screen. The following table explains the chemical characteristics of the local clay used, as provided by the factory experts.

Table 8 The chemical properties of the local clay that was used, as supplied by industry specialists. - source: (tiles, 2017)

Chemical analysis:		
	SiO_2	72.2 %
	Al_2O_3	22.4 %
	TiO_2	1.3 %
	Fe_2O_3	0.9 %
	CaO	0.0 %
	MgO	0.3 %
	K_2O	2.8 %
	Na_2O	0.0 %

The visual story depicted in the following images aims to track the early manufacturing phases in a linear order. Every step is logically mapped out; thus, the entire process can be understood in great depth through the accompanying journal. It was also important to highlight the stage in which the liquefied clay was prepared and combined for use in the slip-casting process. This phase involves a blend of artistry and technical precision, as manufacturers and researchers collaborate to achieve optimal clay mixture consistency. This stage includes several experimental procedures and trial-and-error iterations, all of which were documented in the journal. Through these visualisations, viewers gain comprehensive insight into the specific requirements of each step, as well as the intricacies of

calibration. In addition, the following image may be utilised as recorded evidence of the accomplished efforts. This form of documentation serves as a knowledge base for future research and provides primary material for discussion and investigation within the relevant field.

Figure 188 Tracking the early manufacturing phases of the sample using the slip-casting method, and illustrating how the clay was prepared and poured into the mould. - Source: Author.

Since some parts of the modelled ceramic screen were hollow on the interior to accommodate the water channels, making them more delicate before firing, it was crucial to ensure that the sample was not shattered inside the mould when the model was removed. Therefore, it was advised to use the hexagonal structure as the holding grip of the model during its removal from the mould, considering it the structurally strongest part of the screen, as shown in the following image.

Figure 189 Tracking the early manufacturing phases of the sample using the slip-casting method, and showing how the sample was carefully removed from the mould. - Source: Author.

8.3.6. Testing the sample

The initial manufacturing of the sample depended mainly on the use of the available clay in Gorbon, which manufacturers use for slip-casting. The clay powder properties of the used sample consisted of 72.2% of silica, and water was then added gradually to the powder. The mixture of liquid clay goes through several staged actions involving precise measurements to ensure the quality and consistency of the final product. To start with, clay powder is fined and grounded to ensure that no solid, clotted particles remain within the powder. Sifting is a critical measure applied during the preparation process to guarantee the absence of any solid, clotted particles, which, if retained, would adversely affect the quality of the final product.

This is carried out by knocking the clay powder through a fine mesh screen or sieve to remove any lumps and impurities. Sieving of the powdered clay facilitates its breakdown and the elimination of any lumps or pellets, resulting in a more even and fluffier texture. In addition, it can also introduce froth into the clay powder to aerate it, improving flow characteristics and making it easier to mix with water to form a slurry. Through very careful sieving of the clay powder, ceramicists can achieve the desired fluidity and uniformity of the material, which will be of great importance in the subsequent processes of mixing the powder with water to obtain a thick, smooth fluid pulp with uniform consistency.

Table 9 The chemical properties of the local clay that was used, as supplied by industry specialists. - source: (tiles, 2017)

Body in powder form no. 209

Chemical analysis:		
	SiO_2	72.2 %
	Al_2O_3	22.4 %
	TiO_2	1.3 %
	Fe_2O_3	0.9 %
	CaO	0.0 %
	MgO	0.3 %
	K_2O	2.8 %
	Na_2O	0.0 %

During the water addition stage, the mixture was subjected to thorough mechanical stirring, as documented in the preceding images. This stirring ensured a uniform consistency and eliminated air bubbles. Both sifting the clay powder and employing mechanical agitation were essential to the success of the process, yielding a homogeneous mixture without the need for flocculant agents such as sodium silicate or sodium carbonate—commonly injected into the slip to enhance fluidity and facilitate casting. Instead, the process relied on natural manufacturing methods, prioritising applicability across diverse

production contexts using locally available materials, and deliberately avoiding dependence on chemical additives.

On the other hand, the required porosity of the clay was a variable that had to be determined through testing. Since the screen will be connected to the main water supply system, which governs how much water the surface can absorb, the process will rely on a 1:1 application of the screen. This ratio will define how the screen must function and what evaporation rate is necessary. The application context will also influence the amount of water that evaporates, which must be balanced against the volume of water supplied from the main system to the screen, in order to prevent leaks and deterioration. This iterative approach highlights the dynamic interplay between material properties, design parameters, and environmental conditions in achieving optimal performance.

However, the initial model adhered to the same porosity level achieved in comparable design approaches, such as Ecooler and Cold Spot ceramic screens (Schecter, Cold Spot: Evaporative Cooling through Ceramics, 2015). An initial absorption rate of 13% was selected and applied across several experimental stages by adjusting the firing procedure and the drying time after the sample was removed from the mould and left to dry at ambient temperature. The following table presents the final input parameters used to obtain the initial sample. This material experimentation was meticulously documented and will be made available in the journal.

Table 10 Highlights the final input parameters used to obtain the initial sample. Source: (tiles, 2017)

Ceramic data:

Dry shrinkage in %:	5	3	
Firing temperature:	1070 °C	1140 °C	1240 °C
Firing colour:	white	light	light
Firing shrinkage %:	2	4	6
Water absorption %:	13	12	1
Coefficient of Thermal Expansion (CTE):			
20 °C - 400 °C	59	64	62
20° C - 500 °C	60	67	65
20 °C - 600 °C	73	80	74
CTE * 10^{-7}/K			
Moisture expansion in%:	0.185	0.08	

8.3.7 Testing the clay

After the sample was dried and fired, it was weighed, and it was found that one component weighed around 0.57 Kg. Referring back to the scale and the size of the screen, when the research examined how many screens would be needed to cover common apparatus sizes, it was found that the aperture could accommodate 4 to 8 components vertically. This means that one component could have 8 similar components stacked on top of it vertically, resulting in a total weight of around 6 kg, in addition to the extra loads caused by wind pressure, saturated capillaries in the ceramic pores, and the water in the internal water channels.

Following the completion of the initial sample, additional samples were prepared with varying absorption rates. The screen's durability was evaluated by applying a 20 kg load to test the performance of these samples. The maximum absorption rate that could be sustained under this load was 13%; beyond this threshold, the screen began to crack.

Figure 190 The image illustrates the conceptual framework employed in the durability testing of the screen- Source: Author.

After the initial sample was completed, additional samples were prepared with varying absorption rates. To assess the structural integrity of each sample, the durability of the screen was tested by applying a 20 kg load. The maximum absorption rate that could be sustained under this load was 13%; once this threshold was exceeded, the screen began to crack.

The provided images depict the first iteration of the sample screen following the firing process, alongside its initial durability testing. Although the geometry of both the screen and the mould has

been finalised, this iteration is designated as "initial" due to the ongoing experimental process. Subsequent phases aim to further refine the clay composition of the screen to enhance its responsiveness and performance in evaporative cooling applications. Additionally, the refinement process seeks to optimise the screen's efficiency when deployed at 1:1 scale. Through continued iteration and experimentation, the objective is to develop a final product that fulfils the desired criteria for functionality and effectiveness in real-world conditions.

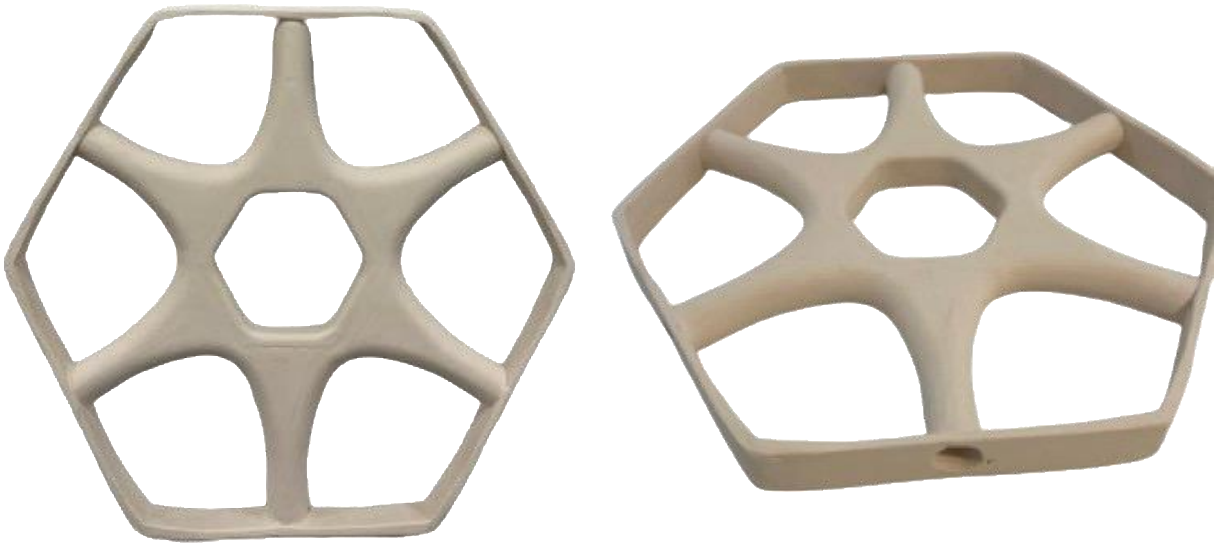


Figure 191 Images of the initial sample following the firing process and subsequent durability testing. - Source: Author

8.4 Conclusion:

Having valuable inputs from analysing the Maymooni screens of Mashrabiya, the design of the proposed screen was organised based on environmental performance optimisation, with the knowledge formulated about Islamic patterns. This developed a very efficient approach towards creating a multifunctional Islamic ceramic screen. The design evolved from a base pattern that was developed by blending various Islamic screens, resulting in a generic Islamic screen that can be culturally applicable in different regions due to its roots in diverse Islamic patterns. The base pattern was simple and less complex, as it depended only on two components merged together (a star and a hexagon), and then repeated modularly on a grid to formulate the final pattern, a pattern that can be found in a large number of Islamic designs as one layer among many layers that form different patterns when overlaid. The design was brought to life through the knowledge and expertise of craftsmen and ceramic artisans, overcoming many of the limitations that could arise during fabrication. Their input highlighted techniques that could be integrated into the design process, enabling the development of a screen inspired by the Mashrabiya. At the same time, the design remained practical and feasible for production in simple ceramic workshops.

9. Testing the Full-Scale Screen

9.1. Introduction

At this stage of the study, it is important to emphasise that the design geometry has been finalised following an extensive analytical study, as presented in the previous chapter. The design is the outcome of a comprehensive analysis of Mashrabiya's features and its performance in relation to solar exposure and airflow. This study provided crucial insights that guided the development of the module pattern, which forms the basis of the proposed design. As a result, the pattern is not only visually striking but also environmentally and culturally adaptive. Mashrabiya's design principles played a central role in shaping decisions throughout the design process, including the screen's application over the aperture. They also allowed for modifications to the system's components to accommodate varying design needs and client preferences, ultimately creating a modular structure inspired by the Mashrabiya.

As the study progressed, attention shifted towards testing various types of clay to assess porosity and evaluate the screen's performance under real-world conditions, where climatic factors remain dynamic and uncontrollable. The fieldwork was structured to examine the full-scale screen's performance in these realistic conditions, with a focus on a hot and dry climate. During this phase, the samples were applied at a 1:1 scale in a real context, connected to the main water supply, and subjected to environmental testing to determine their effectiveness in evaporative and radiant cooling, as well as shading. This practical approach provided essential insights into how the screens perform under dynamic environmental conditions, enabling necessary refinements and adjustments.

The fieldwork was strategically planned to take place in Riyadh, Saudi Arabia (24.7136° N, 46.6753° E), a location characterised by a hot and arid climate. A detailed climate analysis of the city was conducted to identify the optimal time for the fieldwork and assess the site's suitability. This careful selection was key to ensuring the effective implementation of the evaporative cooling strategy and to evaluating its performance under extreme climatic conditions. By choosing Riyadh as the fieldwork site, the study aimed to collect valuable data on how the proposed cooling strategy functions in a real-world setting, thereby strengthening the reliability and relevance of the research findings.

This chapter details the process of applying the screen, the results obtained, and the conclusions drawn from this phase. It is important to outline the objectives and scope of the fieldwork, while acknowledging that the research journey involved numerous experiments, tests, and failed trials, as discussed in the previous chapter. Only the key stages that contributed to the research outcome are covered here, while the remaining trials are included in the journal.

In this chapter, experimental investigations were conducted on porous ceramic evaporators. The experimental tests focused on high-porosity ceramic evaporators, which were found to consistently perform better than the low-and medium-porosity prototypes (Ibrahim et al, 2003). These investigations allowed for an in-depth comparison of the real-world performance of the porous ceramic-based evaporators with the theoretical model predictions discussed earlier, providing further validation for the proposed screen design and its performance in extreme climatic conditions. Additionally, this work confirmed the impact of the developed design and the integration of Mashrabiya's concepts on the screen's water consumption under realistic conditions, highlighting reduced water usage and increased design flexibility.

9.1.a. Objective:

The primary objective of this fieldwork is to evaluate the screen's performance in a hot, dry climate, testing its applicability beyond the theoretical concepts discussed in previous chapters. The results aim to validate the proposed design by assessing how its features influence performance and practical implementation, providing performance ranges and demonstrating its impact on indoor conditions. This will also illustrate how the water system can be effectively connected, and how the screen's assembly and attachment to the façade comply with structural safety standards, ensuring no friction between components.

To achieve a realistic assessment, the screen was tested in a dynamic environment, measuring its performance under real-life conditions rather than in controlled settings. Since the screen operates within a complex environment influenced by thermal mass, natural ventilation, infiltration, and external weather conditions, evaluating its interaction with these factors was crucial.

Another key objective of this application is to examine how the screen's self-shading performance and solar exposure affect water consumption. Additionally, the study explores how individual screen components can be configured to achieve various blockage ratios and shading coefficients, thereby offering greater flexibility in both design and performance.

9.1.b. Scope:

Since it is important to test the realistic evaporative cooling performance of the screen in a hot, dry climate that impacts water consumption, it is essential to consider the defining characteristics of such climates. A hot, dry climate is marked by high temperatures, low humidity, intense solar radiation, and minimal rainfall, leading to rapid heat loss at night and high evaporation rates. These conditions emphasise the necessity of shading and passive cooling strategies to enhance thermal comfort (Alshahrani, 2019).

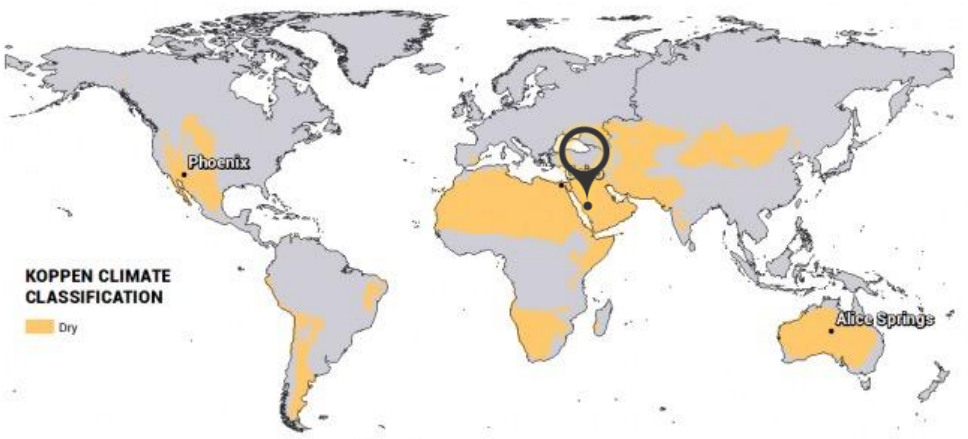


Figure 192 Map illustrating the chosen location for the fieldwork in a hot, dry climatic region according to the Köppen climate classification - Source: Author & (Mind, 2019)

In this regard, the fieldwork plan was to utilise a suitable location within a hot and arid climate—south Riyadh, Saudi Arabia (24.7136° N, 46.6753° E). A detailed analysis of climate data for the city was performed to identify the optimal period for conducting the fieldwork. This was a necessary step to ensure the successful implementation of the evaporative cooling strategy.

Limitations in the Fieldwork

The chosen site and indoor space introduced some limitations—some of which were anticipated, and pre-analytical approaches were taken to address them. However, additional limitations emerged during the testing process.

Pre-existing Limitations

Airflow Patterns and Cooling Efficiency

The designated site for the comprehensive screen testing featured a single-sided ventilated area, with an opening on the west-facing façade and no additional openings present in the room.

- Restricted Air Exchange: In single-sided natural ventilation strategies, the airflow depends significantly on forces such as thermal drafts and minor wind currents, which affect the volume of fresh air penetrating the screen, thereby reducing its overall cooling capacity. If air does not move consistently across the screen, a crucial catalyst for evaporative cooling is diminished, resulting in reduced cooling. The consistency of airflow depends on the temperature difference between indoor and outdoor environments. Since outdoor temperatures can sometimes drop during the night, this temperature difference decreases, which in turn affects airflow consistency.

- Potential for Stagnation: Air moves downward around the screen, then becomes stagnant or recirculates over the area that was just cooled, and thus evaporation is not as efficient as cooling through the full depth of the space. Air can become stagnant or recirculate around the

screen when airflow is insufficient to move the cooled air away or bring in fresh air. This typically occurs when the velocity of incoming air is very low or when cooling is too localised, reducing the effectiveness of evaporation and overall cooling.

- The effective aperture area of the inlet decreases: Since there is only one window in the room, part of this window functions as an inlet, while another part serves as an outlet. This division ultimately reduces the effective aperture area of the inlet, impacting the flow rate and, subsequently, the cooling effect of the screen.

Solution

The limitation of stagnant or recirculating air due to the room's depth and limited ventilation affected the cooling efficiency of the screen. While these factors were beyond control, they reflect the real-world challenges that natural ventilation systems face (Camille Allocca, 2001). In dense urban contexts, most residential buildings rely solely on single-sided ventilation, influenced by temperature fluctuations and wind variations (Camille Allocca, 2001). Future studies could investigate ways to improve airflow consistency by integrating ventilation systems to address this limitation.

Number of Data Loggers

Two Data loggers were used to document the indoor and outdoor dry-bulb temperature and relative humidity. Greater value could have been achieved by using additional data loggers to record changes in temperature and humidity across different distances within the room.

Solution

One data logger was used indoors, positioned at the centre of the room, where the depth was considered relatively small compared to the window size, which in this case was 1.5 m. The placement proved highly functional, as it allowed for assessment of how cooled air spread into the space and its impact on indoor conditions at 50% of the room’s depth.

Even with minimal spatial data, significant patterns can be obtained from a single data logger and from performance measurement near the screen. Continuous monitoring through data loggers was supplemented by regular hourly measurements of the surface temperature of the screen and surrounding surfaces using the UNI-T UTi160S Infrared Thermal Imager.

Emerging Limitations

Hard Water

During the testing period of the screen, a noticeable drop in performance occurred around midday on July 17th. This decline was observed through surface temperature monitoring, spot-measured humidity, and a reduction in water consumption, which was continuously monitored and checked daily (and hourly on the 3rd, 4th, 5th, and 7th of August). The issue became more apparent when the screen

began dripping water from its connection parts, indicating a decrease in evaporation processes and an increase in water pressure from the main supply, which led to leakage at the connection points.

Figure 193 Hard water deposits in the screen cavities and cleaning procedure - Source: Author

Solution

The screen was carefully disassembled after the water supply was stopped, left to drain for three hours, and then thoroughly inspected. Upon examination, it was found that the hard water from the main supply had caused mineral deposition, resulting in scaling on the internal surfaces of the top fur rows of screen components. This buildup clogged the internal pores of the components, thereby obstructing the evaporation process occurring on the external surfaces of the screen.

Defining the issue is always the key to finding the solution. Accordingly, similar approaches were followed by researchers, alongside the craftsmen's inherited knowledge. The screen components were left to soak for two days in a solution of white vinegar (acetic acid, CH_3COOH) and baking soda (Sodium bicarbonate, $NaHCO_3$), diluted with warm water. After the two-day soaking period, the screen was rinsed once with the same solution, then rinsed again with fresh starting water, and flushed with high pressure (5000 PSI) through the external surfaces and internal cavities to ensure that all deposited minerals had been removed. Effectiveness was verified by examining the visible parts of the cavities near the holes in the screen. Following that, the screen was reapplied and reconnected to the main water supply after the water tank had been cleaned, drained, and refilled.

Rinse with fresh streaming water with high pressure 5000 PSI

Soak for 2 days

500 ml - white vinegar

300 grams Baking soda

Figure 194 Summary of steps followed to resolve hard water mineral deposition inside the screen- Source: Author

9.1.c. Climate suitability:

Climate analysis is a fundamental stage in the environmental design process, essential for evaluating the feasibility of any passive strategy based on weather data interrogation. Most weather datasets provide hourly records of key environmental parameters (DBT, WBT, RH, v, etc.), representing a typical design summer year. For this study, climate data for Riyadh were sourced from Energy Plus, specifically from the Riyadh International Airport weather station, located five miles from the chosen site. DBT and RH were the primary parameters analysed to assess the viability of the evaporative cooling system, in alignment with the applicability methods outlined in Chapter 1.2.

In the context of evaporative cooling, dry bulb and wet bulb temperatures are key indicators of cooling performance. The dry-bulb temperature represents the actual thermal energy of the air, measured with a standard thermometer, while the wet-bulb temperature reflects the cooling effect of evaporation (Schianopahn, 2004).

The difference between these two values, known as wet-bulb depression, indicates the potential for evaporative cooling. A higher wet-bulb depression suggests drier air, which enhances evaporation and improves cooling efficiency (Schianopahn, 2004).

Consequently, an evaporative cooling screen operating under conditions of greater wet-bulb depression is likely to achieve superior cooling performance by absorbing more heat from the surrounding air, leading to a greater reduction in dry-bulb temperature and increased thermal comfort (Schianopahn, 2004).

To determine the suitability of evaporative cooling systems, DBT and RH were the primary parameters examined, as outlined in the literature review. Therefore, it was crucial to acknowledge two main factors:

1. The first indicator is the ratio of hours during which the air temperature outdoors is high enough to cause thermal stress. The estimation of thermal stress in this study follows an empirical approach, using fixed temperature thresholds defined by ISO 7730, set by the Saudi Arabia Building and Urban Development Institute and ASHRAE 55 (DBT > 27.7°C) as an indicator of thermal stress (Elnaklah, August 2021).

The following graph is sourced from EnergyPlus for the Riyadh International Airport weather station. The graph shows the ASHRAE 55 adaptable comfort zones and the monthly average dry-bulb temperature (DBT) in Riyadh. May to September sees the most hours exceeding this critical threshold, with July and August registering peak DBT. ASHRAE 55 indicates the week of August around the 24th as typically the hottest, providing sufficient time to conduct the testing and offering flexibility in scheduling fieldwork within this broader high-temperature period. This extended window facilitates thorough testing of the screen's evaporative cooling efficiency, specifically under thermally stressful conditions. Therefore, fieldwork was prioritised from July to September to maximise data collection during these crucial periods of thermal stress.

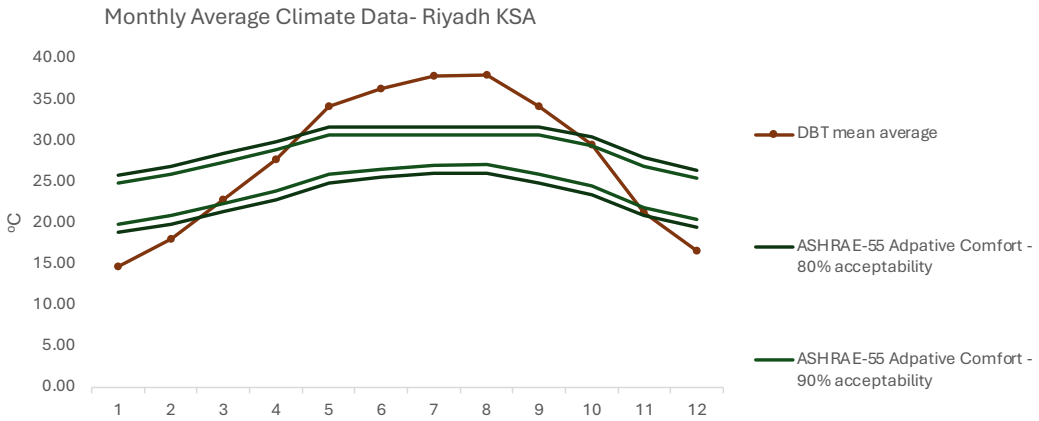


Figure 195 Monthly average dry-bulb temperature in Riyadh, KSA, combined with ASHRAE 55 Comfort band - source: Meteonorm

Heat stress in Riyadh reaches a peak from mid-July to mid-August as temperatures exceed 45°C, with minimal rain and intense sun. The desert climate, driven by high-pressure systems, limits moisture. With little shade, buildings and roads re-radiate heat, which negatively impacts human comfort (Elnaklah, August 2021). On the other hand, the high temperatures restrict the implementation of natural cooling strategies, increasing discomfort and widening the gap between outdoor and comfortable indoor conditions (Elnaklah, August 2021). Accordingly, the combination of extreme heat, arid conditions, and urban structures results in significant thermal discomfort and

heat stress, highlighting the demand for the evaporative cooling screen. This demand is generally based on the frequency of hot hours or days during the period of application, where a greater number of hot hours signifies a higher need for passive cooling, as shown in the following UTCI heatmap of Riyadh generated by Climate Studio.

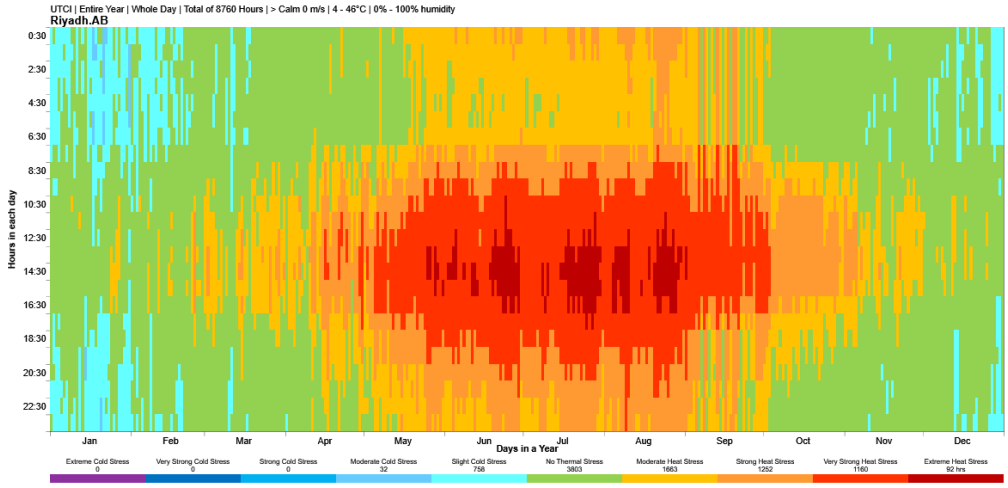


Figure 196 Annual UTCI heat map for Riyadh, KSA, highlighting the peak months with increased hours of heat stress - Source: Climate Studio

- The second index develops the potential of evaporative cooling systems, and this potential is mainly dependent on wet-bulb depression, given by the difference between Dry-Bulb Temperature, DBT (°C), and Wet-Bulb Temperature, WBT (°C), responding to the question: *How dry is the climate to apply an evaporative cooling strategy?* (Vallejo, 2018). The wet-bulb temperature may be estimated from air temperature, air pressure, and relative humidity; however, the wet-bulb value was obtained during the research process using the data monitored through the data loggers

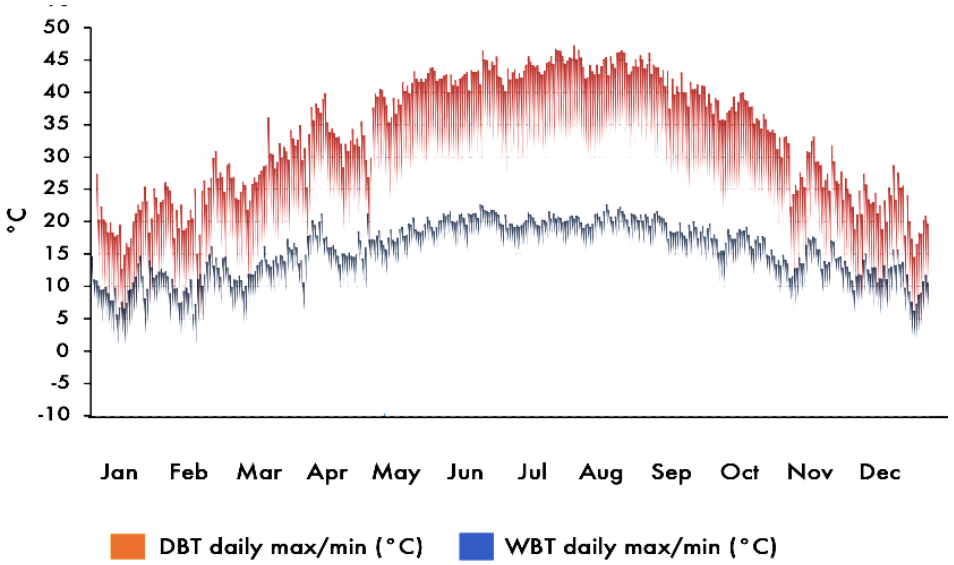


Figure 197 Chart illustrating the annual difference between wet-bulb and dry-bulb temperatures, along with daily global horizontal radiation of Riyadh, Saudi Arabia - source: Metronome.

9.2 System Description (Experimental Setup Outline)

Evaporative cooling systems depend on a wider range of factors than those typically considered in studies of natural ventilation, which focus on DBT, RH, and v. In contrast, strong design characteristics of the ceramic elements, the blockage ratio of the screen, and the nature of airflow through the screens play a significant role. Therefore, the ambient air temperature and absolute humidity for a normal operating day are defined to provide the model's environmental conditions using calibrated data loggers.

9.2.1. The chosen context:

The proposed evaporative cooling screen efficiency was tested through a systematic comparison of indoor conditions with outdoor ones, while simultaneously monitoring the system's water consumption to evaluate the observed results. Data loggers were installed in optimal indoor and outdoor locations to accurately record any changes in the internal environment compared to the external one, as shown in the following illustration.

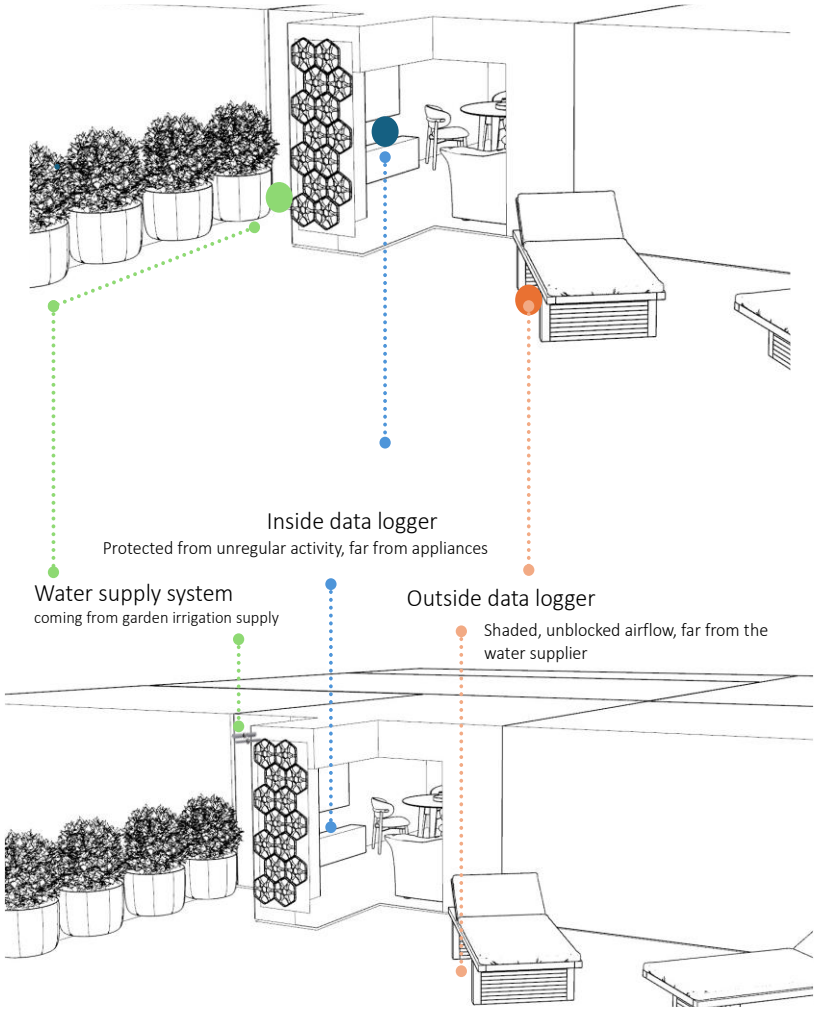


Figure 198 : Illustration showing the placement of data loggers for monitoring screen performance during the 1:1 application in Riyadh, KSA – source: Author

To evaluate the screen's impact on thermal conditions, the fieldwork was structured to continuously monitor indoor and outdoor relative humidity, air temperature, surface temperature, and air velocities before, during, and after installation. This approach allowed the research to document variations in indoor conditions and assess the screen's performance over time, providing a clearer understanding of its effectiveness in evaluating how indoor conditions were influenced by the screen application, as illustrated in the following diagram.

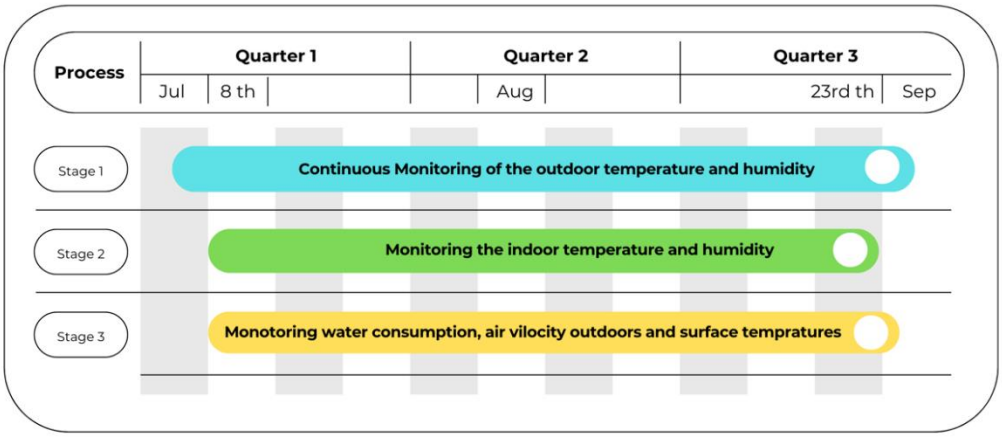


Figure 199 : Illustration showing the methodology for continuous monitoring of performance before and during the 1:1 application in Riyadh, KSA – source: Author

The proposed evaporative cooling screen's efficiency was tested by applying it directly to the aperture of a ground-floor room in an unoccupied domestic house in southeast Riyadh. The selected room, with an area of 9.30 m², had an aperture on the western façade, the most exposed to direct solar radiation, allowing the screen's performance to be evaluated under extreme conditions without obstruction from the surrounding context. Additionally, the aperture was chosen for its suitability in integrating a water supply system, utilising the house's garden irrigation system located nearby. This setup ensured that the evaporative screen was connected to the main water supply for continuous operation.

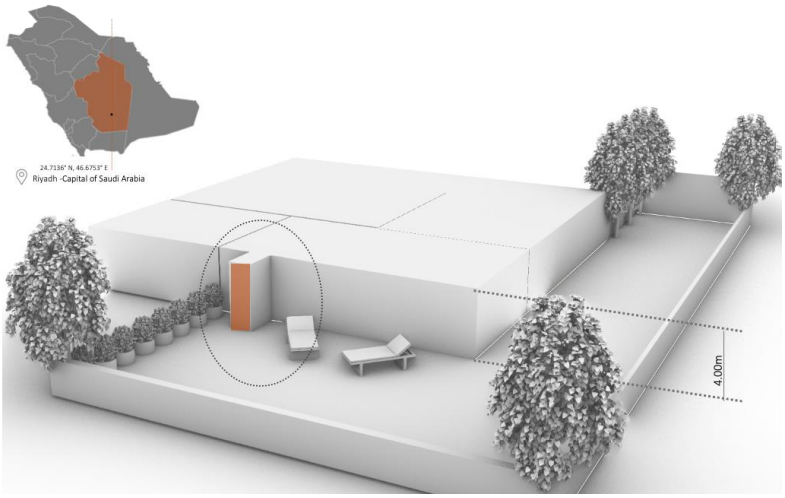


Figure 200 Illustration showing the contextual surroundings of the selected indoor space for the 1:1 application in Riyadh, KSA – source: Author

The aperture size was measured to determine the number of samples required to fully cover the area, as illustrated in the following diagram. The ceramic system, designed as a permeable vertical screen, was defined by its width and height in meters (m), which directly influenced the blockage ratio (BR), calculated at 65%. Based on these specifications, the samples were prepared and fired according to the required regulations to achieve the desired porosity, ensuring efficient system performance while preventing leaks or water drips (D.Jastrzebski, 1959). (*The process of preparing the samples and determining the optimal porosity is documented in the research journal*.)

Determining the number of the screens that were pre-prepared to cover the chosen aperture area is based on the following formula.

Equation 12 Equation used to define the number of components required to cover a given aperture — Source: Author

$$The\ number\ of\ elements\ needed\ to\ formulate\ the\ screen\ (N) = \frac{A(inlet)\ (\ cm^2)}{A(component)\ (784cm^2)}$$

Where:

- **A(Inlet):** The area of the aperture the screen is required to cover (cm^2); in this case, the area of the window was 0.93 m^2.
- **A(Component):** The area of the star component contained within the hexagonal form that constitutes the screen (i.e., the singular screen component) (cm^2), given as a fixed value based on the developed design (784cm^2 or 0.0784 m^2).

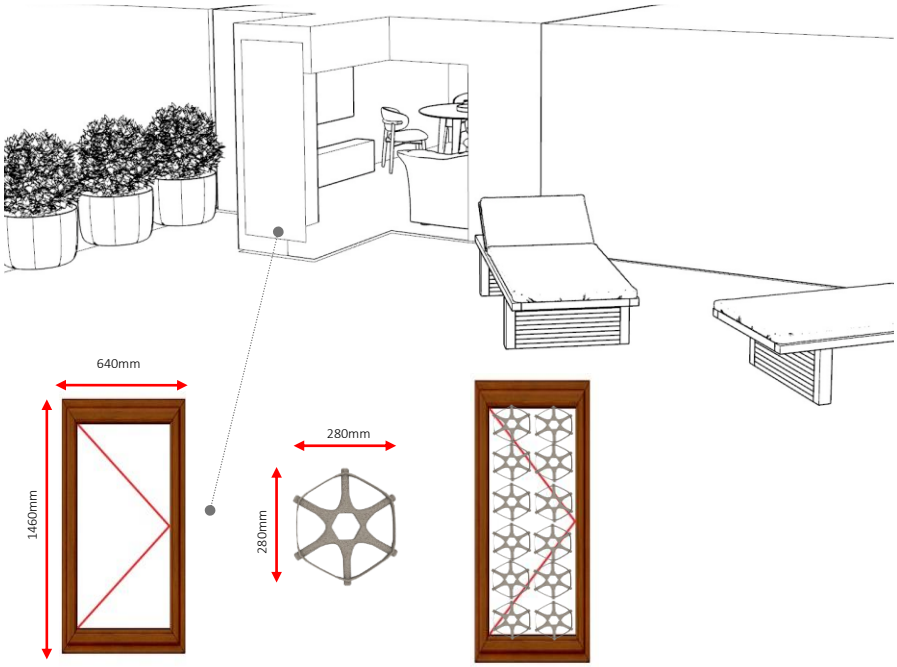


Figure 201 Illustration showing the dimensional details of the window aperture and the screen components - Source: Author

Based on the previous formula, the number of components required to form the screen is calculated as 11.8, which is approximately 12 components. This formula, which determines the number of

components needed to cover a given aperture area, can be further refined to accommodate specific design requirements. As shown in the following illustration, the same aperture area can be configured with varying numbers of components, resulting in different blockage ratios and a corresponding range of performance outcomes.

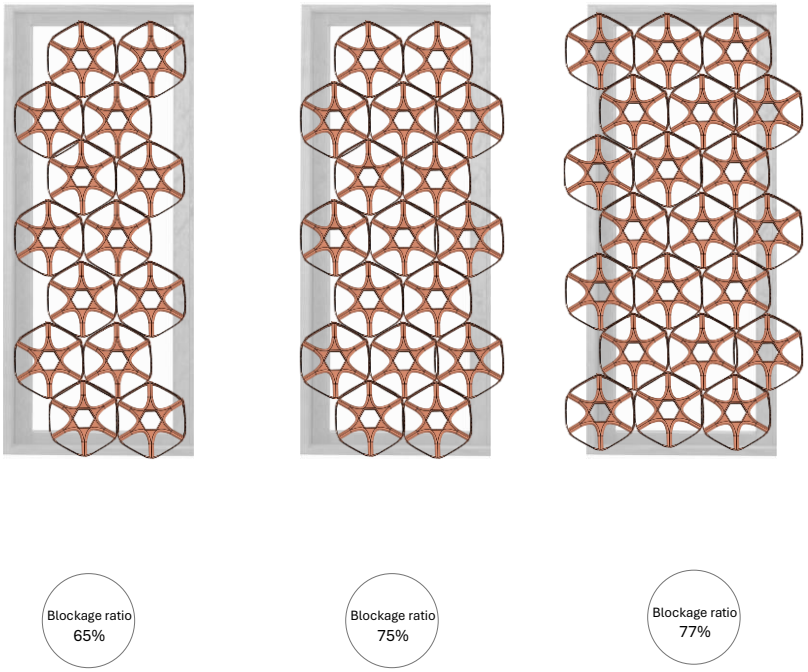


Figure 202 Illustration showing how the same aperture area can be configured with varying numbers of components, resulting in different blockage ratios - Source: Author

The screen was designed to provide flexibility, allowing for further exploration and adaptation based on specific design needs. By modifying the arrangement of its components, different blockage ratios can be achieved, directly affecting the number of elements required to construct the final screen.

The total number of components can be adjusted according to the selected blockage ratio, which may vary based on the façade orientation or the designer's preferences. Different orientations may require varying degrees of blockage, which can be achieved by adding or removing ceramic elements. Additionally, cost considerations can influence the chosen blockage ratio, as a lower blockage ratio requires fewer components, making it a more budget-friendly option.

To determine the number of components needed for a specific configuration, the following formula can be applied:

Equation 13 : Equation used to define the number of components required to cover a given aperture, depending on the desired blockage ratio - Source: Author

$$\textit{The number of elements needed to formulate the customed screen for a specfic Blockage Ratio } (N_{BR}) = \frac{A(inlet) - \Big(A(inlet) * \big(1 - BR(\%)\big)\Big)\,(cm^2)}{A(component)(784cm^2)}$$

Where:

BR is the blockage ratio of a screen, representing the percentage of the aperture area that is covered by the screen—effectively determining how much of the opening is obstructed. It is the inverse of the

perforation percentage, which refers to the open areas allowing airflow and light to pass through (Naguib, 2007). The blockage ratio is calculated by measuring the total area blocked by the screen, dividing it by the full aperture area, and multiplying by 100 to obtain a percentage. This percentage can be converted into a coefficient which, when subtracted from 1, yields the coefficient of the exposed area.

Choosing the appropriate blockage ratio allows for greater flexibility in designing the screen, as it directly influences airflow, shading, and cooling performance. By analysing different configurations of the screen's components and their corresponding blockage percentages, various design possibilities can be explored to achieve the desired balance between permeability and coverage. The available blockage ratios for the proposed screen are as follows:

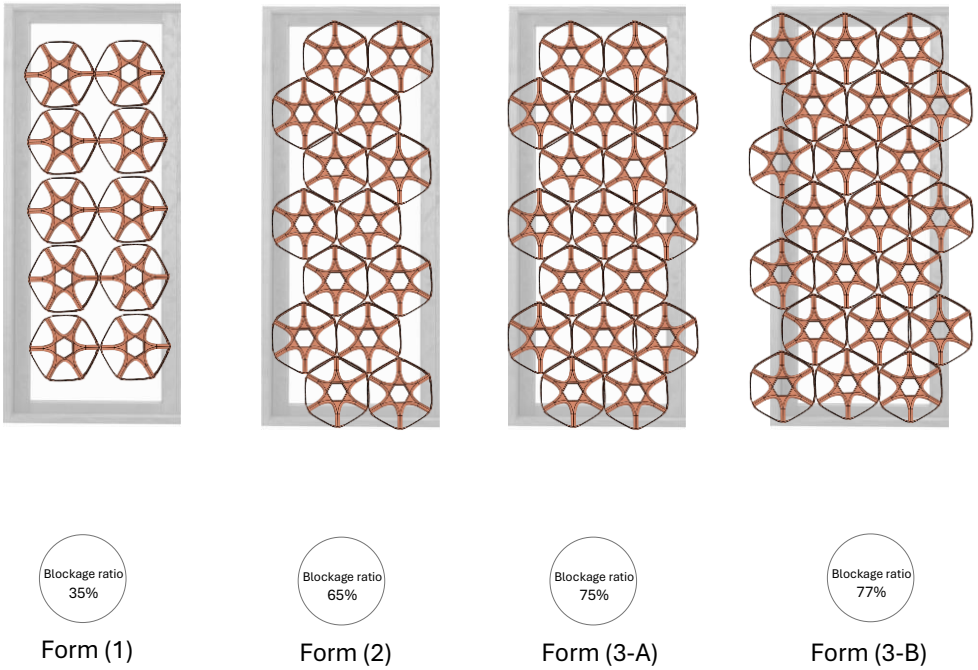


Figure 203 Illustration showing how the same aperture area can be configured with varying numbers of components, resulting in different blockage ratios - Source: Author

- Form 1: Blockage ratio of 35%. The components are arranged in columns, with each column having its own water supply inlet. No diagonal connections exist between the elements.
- Form 2: Blockage ratio of 65%. The component arrangement differs significantly from Form 1, as the elements in each column are connected diagonally. This configuration allows for a single water supply system serving all columns, offering greater flexibility in designing the water connection system. More components are required to account for the increased blocked area.
- Form 3-A: Blockage ratio of 75%. Similar in configuration to Form 2, but with a higher blockage ratio, requiring more components to cover a larger portion of the aperture.

- **Form 3-B:** Blockage ratio of 77%. Closely resembling Form 3-A, but with a slight increase in blockage ratio, necessitating additional components to further reduce the open area of the aperture.

9.2.2 The chosen indoor space:

The room, measuring 3m × 2.5m, with a single window of 0.93 m², was selected to simulate a compact, single-sided ventilated space. This setup reflects a common real-world application of evaporative cooling systems, particularly in resource-limited environments where efficient passive cooling solutions are essential.

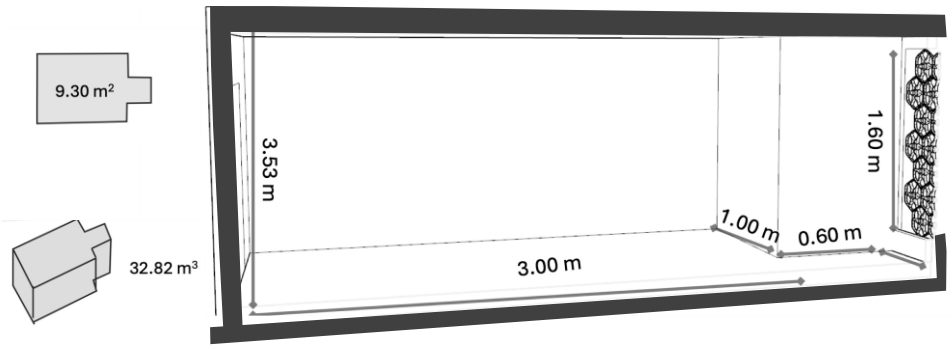


Figure 204: Illustration showing the contextual surroundings of the selected indoor space for the 1:1 application in Riyadh, KSA – source: Author

The primary reason for selecting this space was its unoccupied condition, ensuring that no external factors—such as internal heat gains or occupant activities—could interfere with the monitored data. This eliminated potential fluctuations in the results, allowing for a more precise assessment of the screen's range of performance.

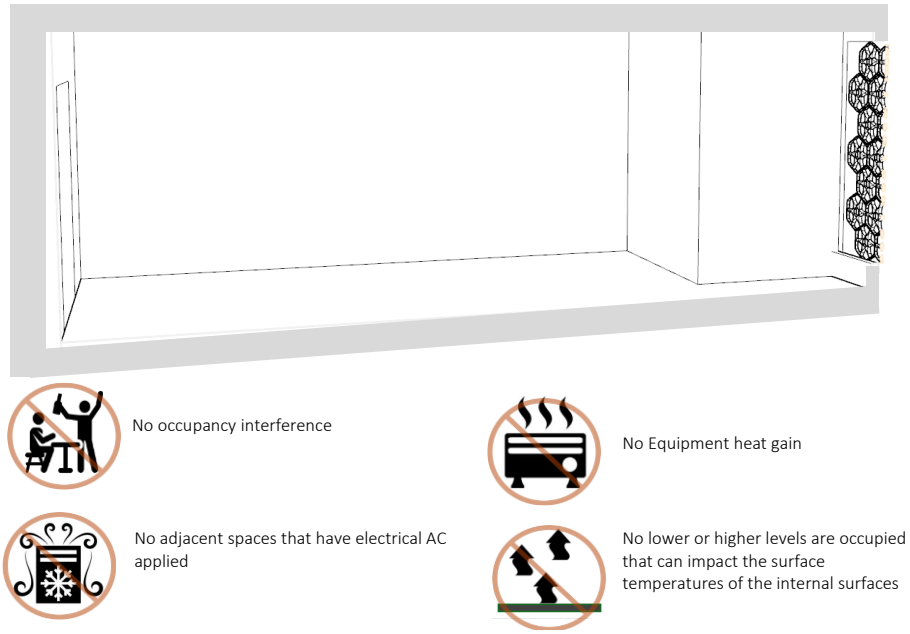


Figure 205 Illustration showing the spatial condition of the room, the window aperture, the screen, and the key architectural features within the space - Source: Author

The space is also not adjacent to any occupied areas, ensuring that no external cooling influences interfere with the monitored data. During this time of year, occupied spaces are typically cooled using electrical air conditioning systems set at low temperatures to counteract the extreme outdoor heat, as reported by the Electricity and Cogeneration Regulatory Authority (ECRA). This separation eliminates unintended cooling effects, allowing for an accurate evaluation of the screen's independent performance, as illustrated in the following bar chart (Alshahrani, 2019).

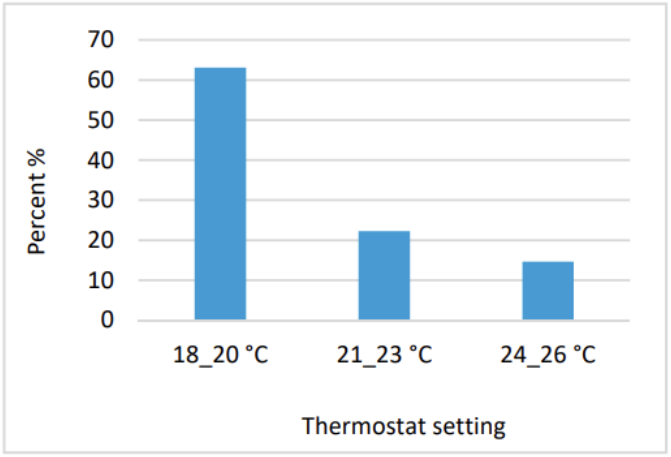


Figure 206 Bar chart illustrating household occupancy behaviour in KSA with regard to thermostat settings - Source: (Alshahrani, 2019)

Consequently, selecting the western façade for fieldwork was essential to test the screen's performance under extreme conditions, as this façade experiences the highest solar exposure during the operational period. This is demonstrated in the chart below, based on Meteonorm yearly weather data for Riyadh in 2024 (RCP2.6).

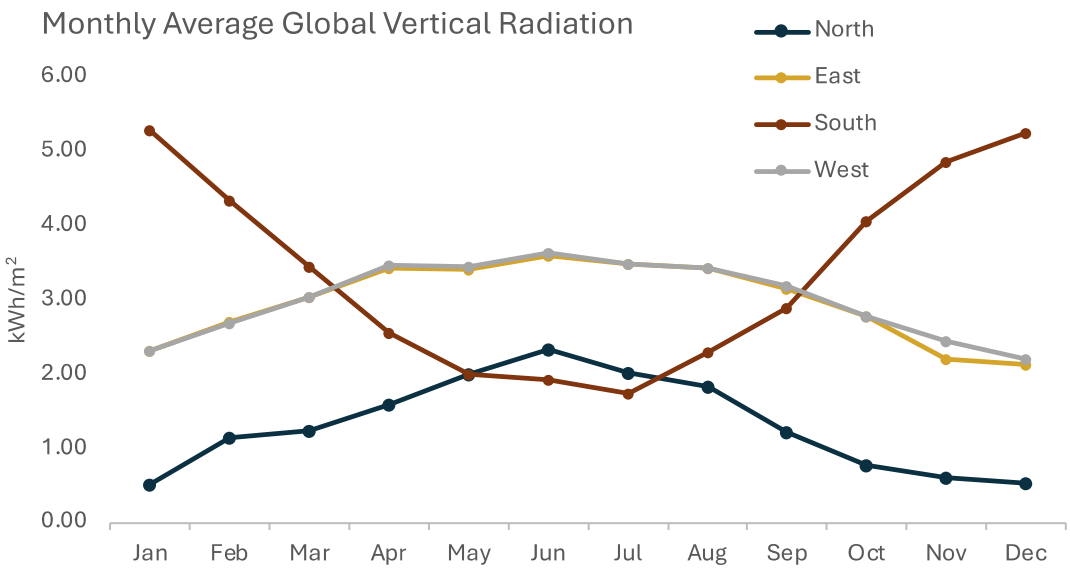


Figure 207 Monthly average global vertical solar radiation on each facade orientation - source: Meteonorm

9.2.3. The preparation of the components:

The samples were prepared according to the specified guidelines, with the necessary components fabricated based on detailed experimental protocols. Water absorption was measured at 13.6%, following the Standard Test Method ASTM C830-00 (2023) (C830-00, 2023).

The material properties of the clay, specifically the stoneware clay used in this study, were determined through a series of tests and experiments aimed at defining the desired porosity and durability. The process also identified the conditions required to achieve the target absorption characteristics. These experimental trials were documented in the research journal.

The samples were formed using slip casting in prefabricated moulds and were left to dry under controlled conditions. The drying process took place at a temperature of 50 °C, with humidity maintained between 50% and 70% to prevent rapid moisture loss, which could lead to cracking (D.Jastrzebski, 1959). The drying duration was carefully regulated, with the samples left for 48 to 50 hours to ensure a gradual reduction in moisture content.

Figure 208 Images documenting the drying and preparation process of the samples prior to firing, conducted at the Gorborn factory in Istanbul - Source: Author

The ceramic components were fired at a temperature range of 1050°C to 1070°C, with a controlled heating rate of 5°C/min to ensure uniform sintering. After the firing process, the components were allowed to cool gradually to prevent cracking, which could occur due to rapid changes in thermal conditions.

Once the drying, firing, and cooling processes were completed, the porosity of the samples was measured using the Standard Test Method ASTM C830-00 (2023). The following formula was applied to determine water absorption, which serves as an indicator of porosity (Simon, January 2023) (C830-00, 2023):

Equation 14 Water Absorption, (C830-00, 2023)

$$A_w(\%) = \frac{W_s - W_d}{W_d} \times 100$$

Where:

- A_w: The water absorption percentage, defined as the proportion of the sample's dry weight that can be absorbed as water (C830-00, 2023).
- W_s: The weight of the sample saturated with water after being soaked for 48 hours (C830-00, 2023).
- W_d: The weight of the dry sample after being placed in an oven at 105°C until a constant weight is achieved (C830-00, 2023).

Methodology:

Porosity is typically determined by measuring liquid penetration into the pores, with gravimetric techniques being the most commonly used (C830-00, 2023). This method involves weighing the material in both dry and saturated states. To begin, the dry weight of the samples was recorded prior to saturation. The samples were then placed in a gas oven at 105°C. After 24 hours of drying, their weights were measured again using an Adam ABW digital scale with an accuracy of 0.1g to ensure precise readings.

Next, the sample was immersed in a boiling water container with a capacity of 25 litres, at temperatures ranging between 85°C and 100°C. It was soaked for 2-3 hours, followed by 19 hours of continued soaking and gradual cooling in the same water. This procedure followed the guidelines of ASTM and ISO methods (Hansen, 2024), along with well-documented experimentation conducted by Tony Hansen (Hansen, 2024).

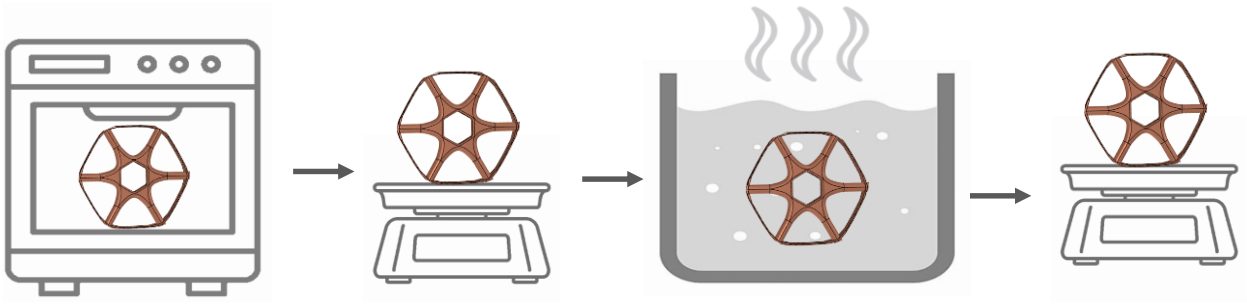


Figure 209 Steps followed for calculating porosity

The sample's porosity was determined after measuring its dry and saturated weights using the previously stated equation. Accordingly, the dry weight and the saturated weight were found to be:

Table 11 Weight and porosity of the tested ceramic sample

dry weight (g)	Saturated weight (g)	Porosity (%)
1372 g	1560 g	13.7%

According to experiments conducted to test different porosities achievable using the same Earth-Ware clay—and documented in the research journal—these values align with the porosity range of conventional terracotta products, which typically range from 6% to 25% (D.Jastrzebski, 1959). However, this range is known to influence the durability of the sample. Prior research has successively produced high-absorption ceramics with porosities reaching up to 50% (Vallejo, 2018), indicating potential for further development in this domain.

9.2.4. The Application of the samples:

As discussed in the previous chapters, the structural attachment system for mounting the screen onto the façade was designed with multiple configurations, as shown in the following illustration. However, configuration (b) was selected for implementation, as it proved to be the most practical and compatible option given the existing site conditions.

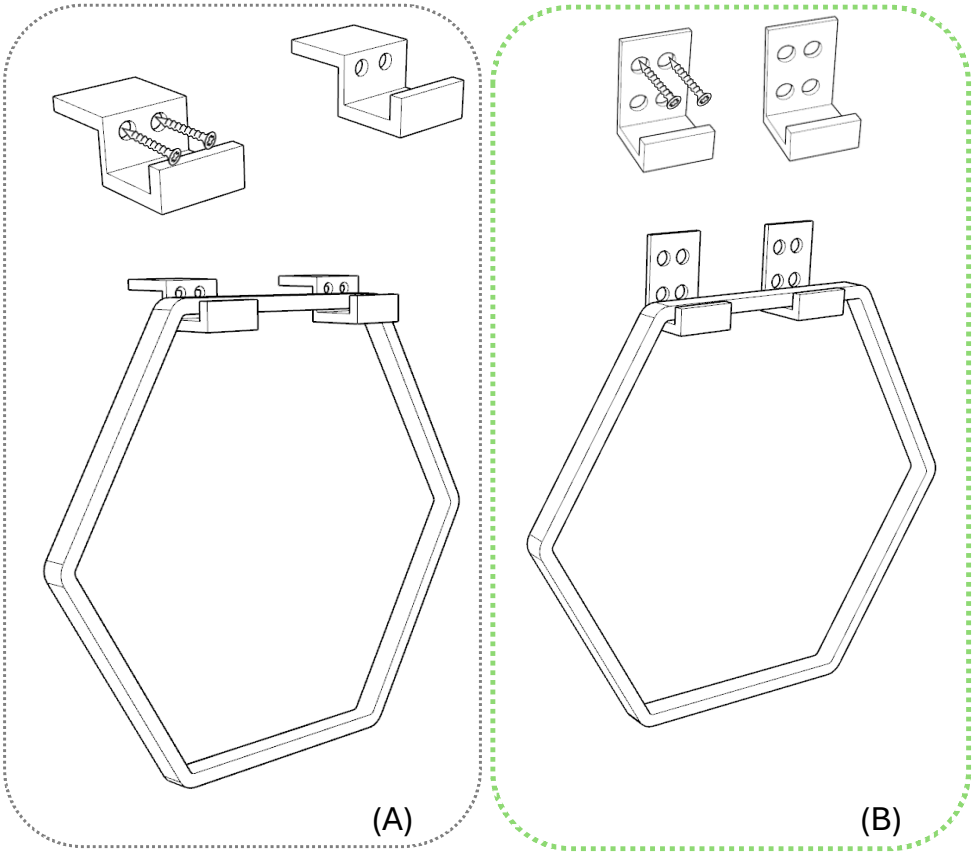


Figure 210 Proposed structural hanging system, where configuration (B) was selected - Source: Author

During the fieldwork, two different screen configurations were tested to assess the structural stability of varying blockage ratios achieved through different assembly methods. The results confirmed that both configurations demonstrated structural stability under the tested conditions.

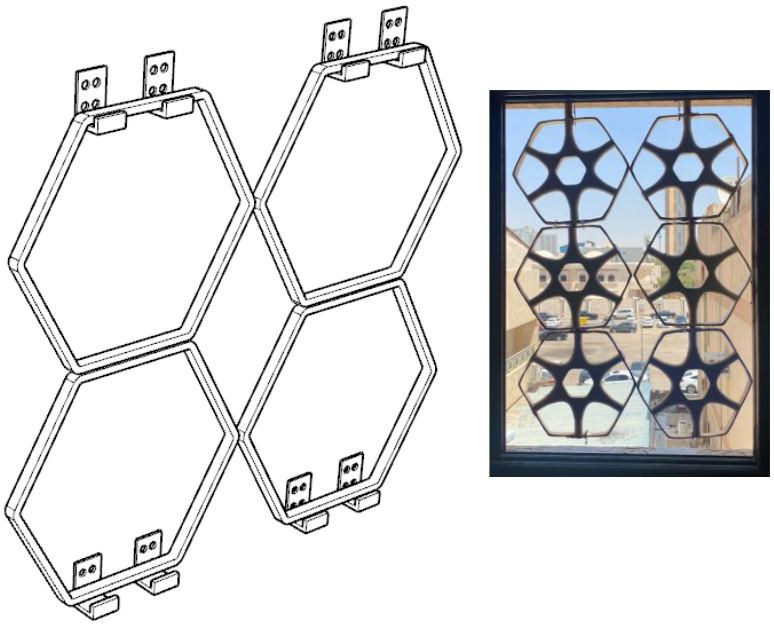


Figure 211 Aligned components used to formalise a lower blockage ratio through linear assembly - Source: Author

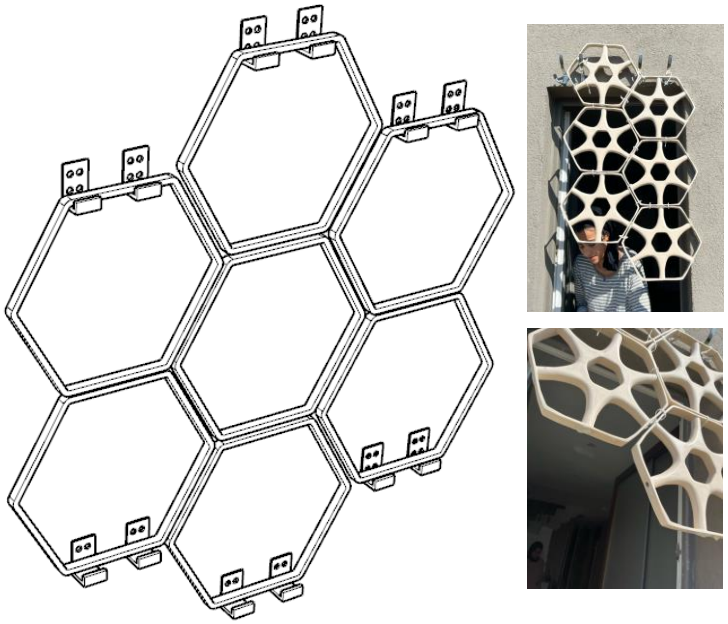


Figure 212 Shape 2: Diagonally assembled components used to formalise a higher blockage ratio through an interlinked configuration - Source: Author

Shape (1) refers to the placement of components in a parallel arrangement, intersecting at the edges. This design enhances structural alignment and results in a lower blockage ratio, which in turn produces a lower shading coefficient—allowing for increased airflow through the screen. In contrast, Shape (2) features a higher blockage ratio, leading to a higher shading coefficient and reduced permeability.

This approach was tested to highlight a key feature of the designed screen—its flexibility, inspired by the *Mashrabiya* concept. The design enables multiple configurations using the same components, emphasising functionality, adaptability, and modularity. It empowers architects and designers to implement flexible patterns that respond to cultural, environmental, and user-specific needs, as illustrated in the following bar chart. Given these considerations, Shape (2) was selected to better accommodate the extreme climatic conditions during the application period in Riyadh.

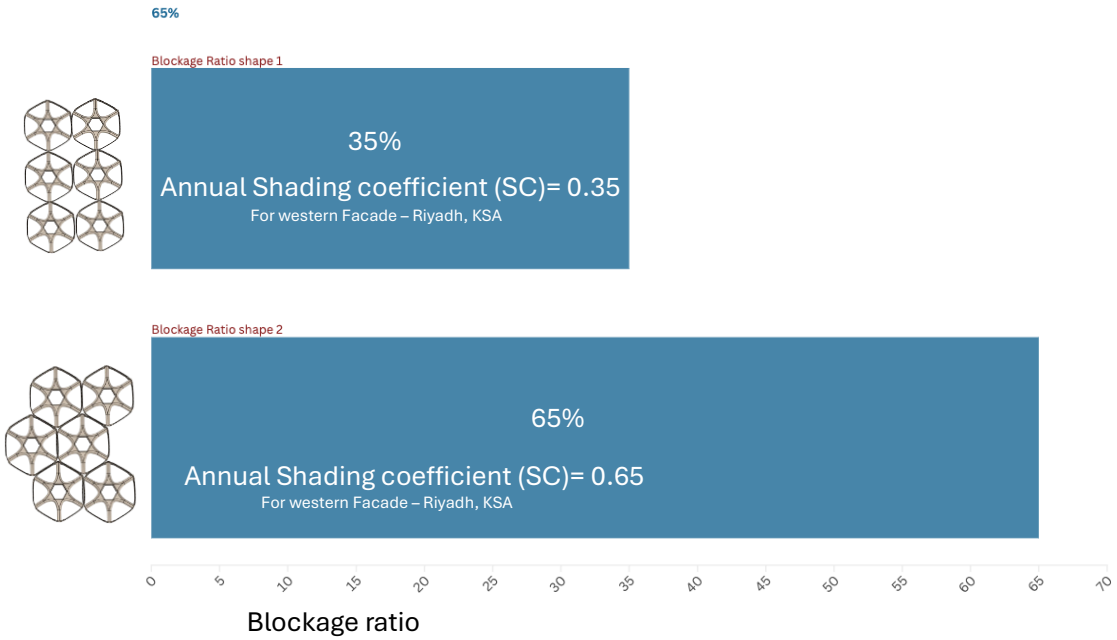


Figure 213 Chart visualising the variation in blockage ratio and shading coefficient between Shape (1) and Shape (2)- Source: Author.

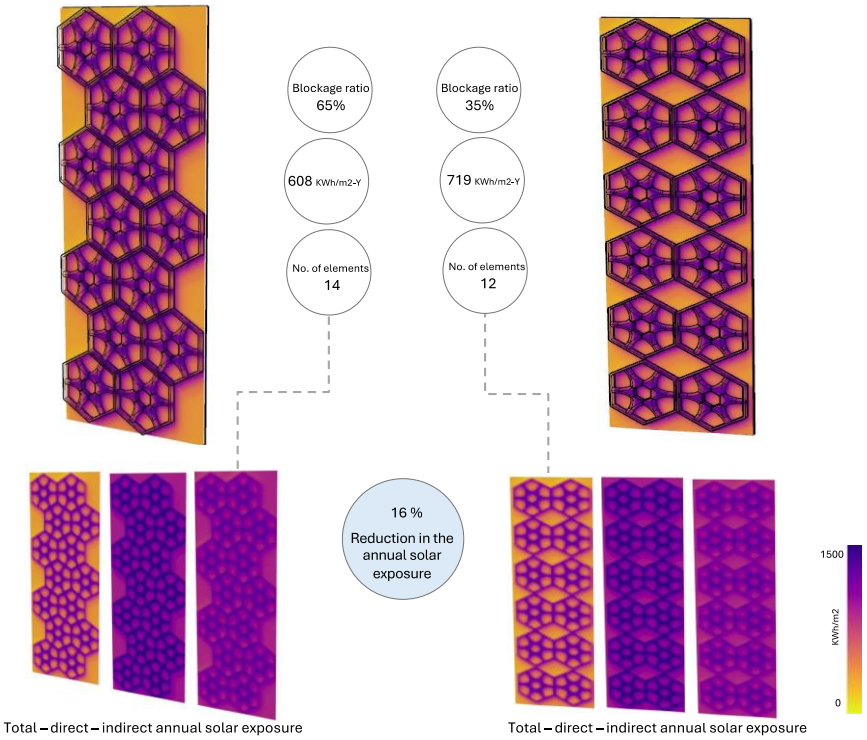


Figure 214 : Reduction in annual total solar exposure between Shape (2) and Shape (1) when applied to a west-facing facade in Riyadh, Saudi Arabia - Source: Author

On the other hand, the water supply system differs in both forms. In Shape (2), the channelled design enables more efficient water usage, as it requires only a single water output to supply the entire screen. This efficiency stems from the way each component is connected to the adjacent ones both diagonally and vertically. In contrast, Shape (1) permits only vertical connections, meaning each column operates independently and requires a separate water supply output, disconnected from adjacent columns.

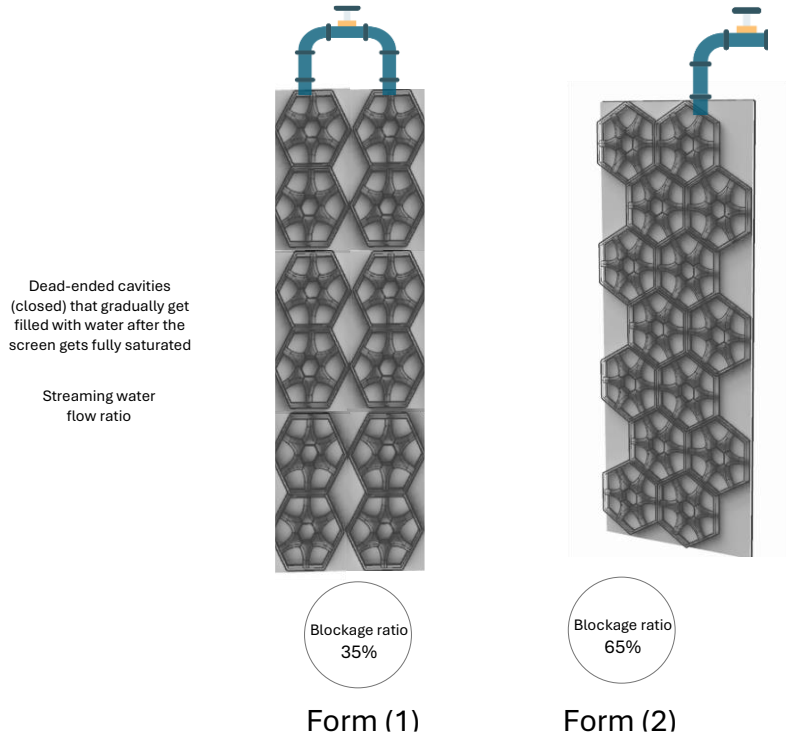


Figure 215 : Illustration showing the variation in the water supply system and flow dynamics within the component channels of shape (1) and Shape (2) - Source: Author.

9.3. The monitoring tools

1. **Data Loggers (Tiny tag Plus 2 — TGP-4500):** The calibrated data loggers, calibrated on 23rd May 2023, were used to measure air temperatures and relative humidity (RH) at both the intake and output points. Each probe is capable of recording temperatures ranging from 25°C to +85°C, with an accuracy of ±0.48°C at ambient conditions around 40°C. For relative humidity, the probes measure between 0% and 100%, with an accuracy of ±3.0% RH at 25°C. The data loggers offer strong resistance to moisture and condensation, ensuring reliable long-term data storage. Their precision and durability are essential for evaluating the system's performance under varying environmental conditions.
2. **Testo 410-1 Anemometer (Speed &Temperature):** This instrument was used to measure both air velocity and temperature. It is equipped with a 40 mm precision vane

for accurate air velocity readings and is designed to determine average values over time. Its ability to measure airflow dynamics and volume flow rate was essential for evaluating the effectiveness of evaporative cooling in dissipating heat.

3. Thermal Camera (UNI-T UTi160S Infrared Thermal Imager): Real-time thermal imaging was provided by the UNI-T camera, with an IR resolution of 160x120, response band of 8-14 µm, pixel spacing of 12µm, and a frame rate of ≤25Hz. The measuring range extended from -20°C to 550°C, with a temperature measurement accuracy of ±2°C /±2% and a resolution of 0.1°C. This enabled the visualisation of temperature gradients across the surface of the cooling screen, as well as the surrounding environment. It offered valuable insights into how the ceramic screen's surface temperature fluctuated throughout the day, contributing to the evaluation of its radiant cooling efficiency.

Altogether, these tools were instrumental in assessing the cooling system's performance by capturing both environmental and operational variables. They also facilitated the identification of spontaneous fluctuations or anomalies, allowing for timely diagnosis and resolution of emerging issues.

Accordingly, one of the data loggers was positioned indoors, at the centre of the room, mounted at a height of 1.1 meters and located 1.5 meters from the screen. This placement ensured continuous monitoring of indoor temperature and humidity variations. Meanwhile, a second logger was installed outdoors, sheltered from direct solar exposure, at a distance of 1.2 meters from the screen. This location represented the farthest feasible point within the site constraints and was sufficiently removed to avoid the cooling influence of the screen on the surrounding outdoor conditions.

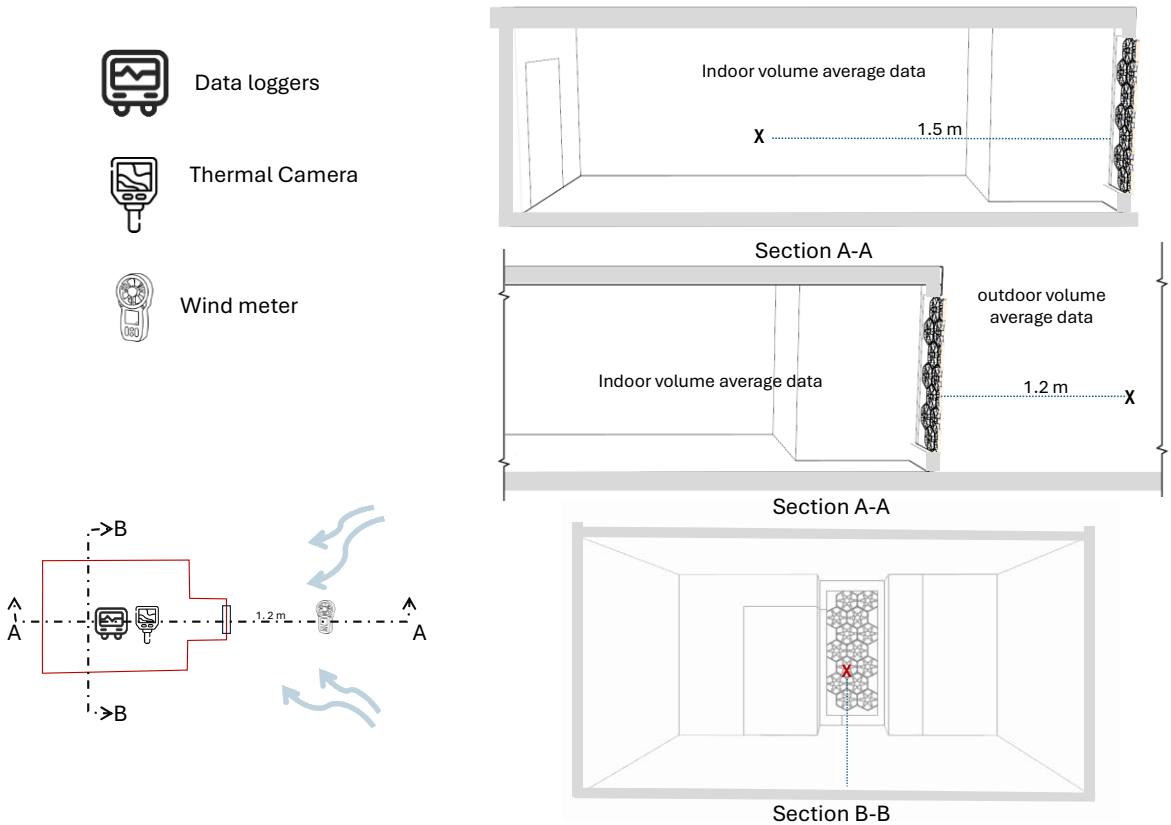


Figure 216: Methodology for positioning the monitoring instruments - Source: Author

The air velocity was monitored at various distances from the screen, all measured at a consistent height of 1.1 meters and from multiple directional angles. The dominant airflow direction was identified, and its readings were averaged to ensure representational accuracy. Measurements were taken at distances of 1.2 meters, 0.9 meters, 0.5 meters, and 0.3 meters to precisely capture the velocity of air entering the screen and assess its performance under varying proximity conditions.

9.5 Test Results and Performance Range:

To assess the effectiveness of the applied evaporative cooling system, experimental performance tests were conducted to monitor the screen's performance across different conditions. The following parameters were calculated to evaluate key performance metrics:

- Temperature reduction inside the room (Δ DBT).
- Boosted relative humidity.
- Evaporative cooling efficiency.
- Water hourly consumption by the system and Water Consumption Efficiency.

- Sensible Cooling Capacity.

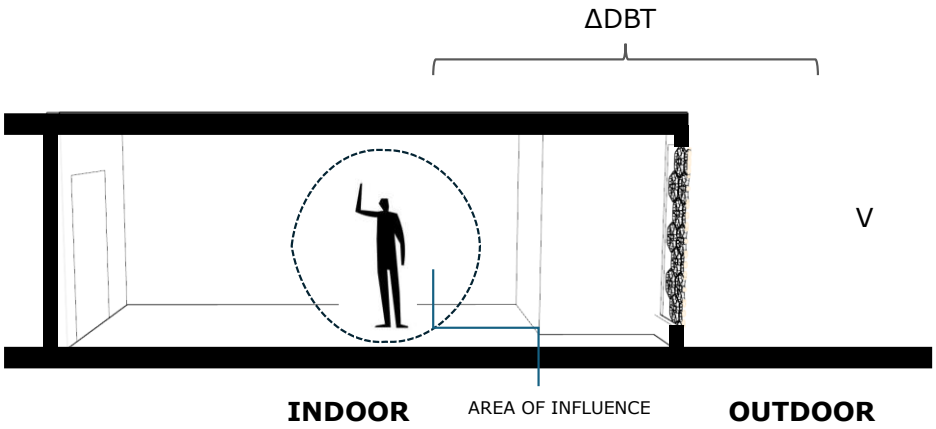


Figure 217 Illustration showing the variation in monitored parameters indoors and outdoors - Source: Author

9.5.1. Collected Key Performance Data:

- **Dry-Bulb outdoor $T_{db\text{-}out}$ (°C):** Measure the incoming air temperature and humidity before it passes through the screen, used to calculate theoretical maximum cooling.
- **Indoor Temperature $T_{db\text{-}in}$ (°C):** Continuously monitored dry-bulb temperature(°C) of the air in the middle of the room after passing through the cooling screen.
- **Humidity Levels (RH%):** Relative humidity is continuously measured before and after the air passes through the screen, helping to evaluate performance efficiency.
- **Wet-Bulb Temperature (°C):** Although not directly monitored, it was later calculated using the following formula, based on the recorded relative humidity and dry-bulb temperature for both indoor and outdoor conditions.

Equation 15 Formula used to calculate the wet-bulb Temperature (°C) - Source: (Kimball, 1977)

$$T_w = T\,\mathrm{atan}[0.151977(\mathrm{RH\%} + 8.313659)^{12}] + \mathrm{atan}(T_{Db} + RH\%) - \mathrm{atan}(RH\% - 1.676331) + 0.0039183(RH\%)^{\frac{3}{2}}\,\mathrm{atan}(0.023101\,RH\%) - 4686035.$$

The wet-bulb temperature was calculated separately for the outdoor and indoor data by inputting the respective relative humidity and dry-bulb temperature values.

- **Water Usage (m^3 or L):** Monitored water consumption rate to evaluate efficiency and determine how much water the system needs for each specific time. The process of hourly monitoring of the water consumption during the screen operating period is documented in the research journal.

- **Air Velocity and Flow Rate (m/s):** Measured air velocities across the screen and near the inlet, as they affect evaporation rate and cooling performance.

9.5.2. Calculated Key Performance Metrics:

- **Evaporative cooling efficiency (%):**

The evaporative cooling efficiency was calculated based on the collected key performance data using the following formula (Malli, 2011) (Kimball, 1977) :

Equation 16 Evaporative cooling Efficiency formula, Source: (Kimball, 1977)

$$\eta_{cooling}(\%) = \frac{T_{out} - T_{in}}{T_{out} - T_{wb(Out)}} \times 100$$

Where:

$\eta_{cooling}$ (%) is the Evaporative cooling efficiency percentage.

- **Sensible Cooling Capacity (w) and cooling capacity per m^2(W/ m^2):**

Calculating the cooling capacity is a vital tool to understand the efficiency of the screen, as it measures the evaporative cooling system's ability to remove the heat energy of the incoming air when it flows through the evaporative cooling screen, hence determining the system's energy efficiency. Accordingly, the following formula was used (Watt, 1997).

Equation 17 Sensible Cooling Capacity (w) - Source: (Watt, 1997)

$$Q_C = \dot{m} \times C_p \times (T_{out} - T_{in})$$

Where:

- **Q_c (W):** The cooling capacity of the evaporative cooling system in watts.
- **C_p (J/kg·°C):** The specific heat of the air, which is 1005 J/kg·°C in Riyadh during July and August (Elnaklah, August 2021) (Watt, 1997).
- **$\dot{m}$ (Kg/s):** The mass flow rate, calculated based on the volumetric flow rate derived from the airflow rate monitored during the screen application period, using the following formulas (Mata, 2013) (Watt, 1997):

Equation 18 Mass flow rate equation - Source: (Watt, 1997)

$$\dot{m} = \rho \times \dot{V}$$

$$\dot{V} = v \times A$$

Therefore:

$$\dot{m} = \rho \times v \times A$$

Where:

- **ρ (Kg/m³):** The air density, which can vary from one place to another based on changes in wind power, temperature and pressure recorded near the ground surface, was calculated and documented in some research projects to range between 1.09 Kg/m^3 to 1.26 Kg/m^3. The chosen value for this calculation was 1.18 Kg/m^3, based on the monthly average result in June in Riyadh (Rehman, 2007).
- **ν (m/s):** Monitored air velocities near the screen.
- **A (m²):** the sectional area of the window to which the screen is applied.
- **ΔT_{db} (°C):** The change in temperature that occurs due to the application of the screen and the building envelope. The change in air temperature is a valuable aspect of understanding the performance of the applied evaporative cooling screen, and the following formula was used (Kreider, 2019):

Equation 19 Difference in dry bulb temperature - source: (Kreider, 2019)

$$\Delta T_{db} = T_{db(outside)} - T_{db(inside)}$$

- **ΔRH (%):**

  Applying the evaporative cooling screen increases the absolute humidity inside compared to the absolute humidity outside. This increase helps understand the performance of the applied screen and correlate the monitored changes in the water consumed by the system, and the following formula was used (Kreider, 2019):

Equation 20 Increment in absolute Humidity - Source: (Vallejo, 2018)

$$\Delta AH = AH_{inside} - AH_{outside}$$

- **Water consumption efficiency:**

  The water-use efficiency (η_{water}) is determined by the ratio of the actual sensible cooling achieved to the maximum possible cooling that can be obtained from the evaporation of a given mass of water, which is equivalent to the heat of vaporisation (ΔHvap) (Pistochini, 2011):

Equation 21 Water consumption efficiency - source: (Pistochini, 2011)

$$\eta_{water} = \frac{1 \ / \ W(total)}{\Delta H\dot{v}ap}$$

Where:

- **η_{water}:** The water consumption efficiency of the screen.

- $\boldsymbol{\Delta Hvap}$ *(MJc/l):* Heat of vaporisation of water.
- $\boldsymbol{W(total)}$: The total water-use rate for evaporative cooling (Pistochini, 2011).
- m_{water}(kg): The total mass of water evaporated by the system, based on the hourly monitored water consumption. Accordingly, the following formula was used (Finch, 2001):

$$m_{water} = \rho_{water} \times V_{water}$$

Where:

- ρ_{water} (Kg/L): The density of the water (1 kg/L) (Finch, 2001).
- V_{water} (m^3 or L): The monitored volume of the water consumed by the screen during the operating time.

9.6 Analysed monitored data and calculated metricised range:

The performance range of the screen was monitored continuously for a month after installation, with data loggers tracking its effectiveness. Water consumption was recorded hourly during the screen's operating hours. Since the main water supply system relied on a tank positioned on the building's shaded roof, and the building was unoccupied during the testing period, any reduction in the tank's water level could be attributed primarily to the screen's consumption. The evaporative cooling screen was implemented from July 13th to August 23rd, though the process was briefly interrupted on the 17th due to hard water issues, requiring a three-day pause for cleaning and reinstallation. By July 25th, one week after deployment, the screen's performance had stabilised and remained consistent throughout the testing period.

The graph below illustrates temperature variations inside and outside the test environment from late July to August 23rd. The external temperature (blue curve) exhibits distinct daily fluctuation, with peaks exceeding 40°C and lows around 32°C. In contrast, the internal temperature (orange curve) follows a similar pattern but with a significantly reduced amplitude, demonstrating the cooling effect of the evaporative screen. This moderation of indoor temperature extremes highlights how the screen's efficiency in reducing peak heat, confirming its potential as a dependable passive cooling solution in this context.

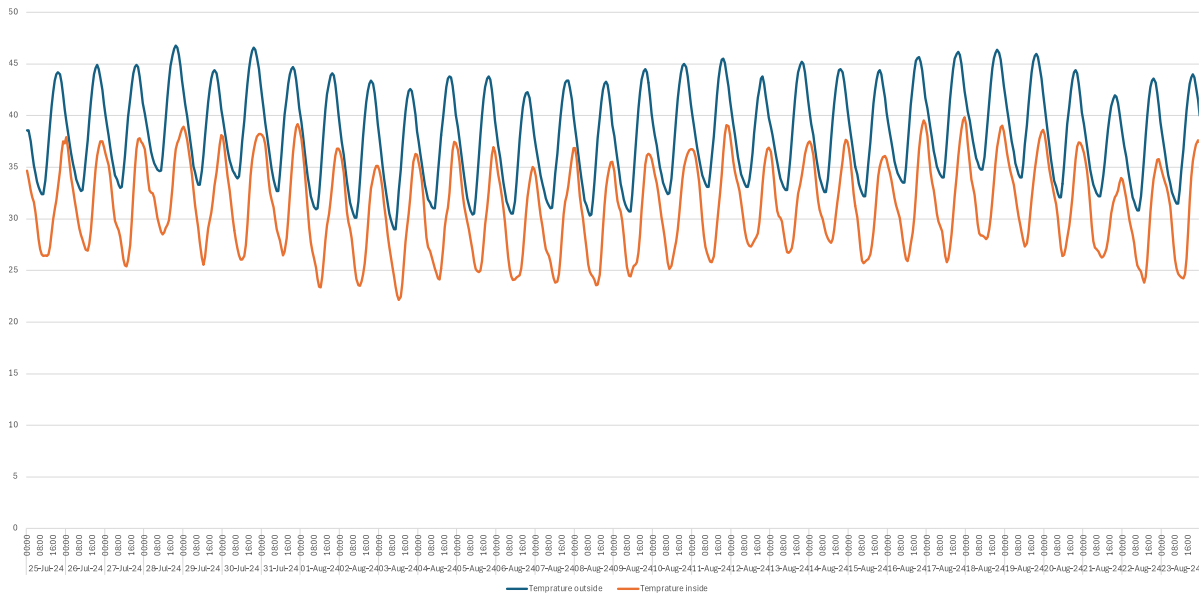


Figure 218 Monitored dry-bulb temperatures inside and outside the test environment from 25 July to 23 August 2024- Source: Author

9.6.1 Analysing the hourly performance range of the screen:

Zooming in on a specific day (25 July) provides valuable insights into the performance of the screen, allowing a clearer understanding of how its effectiveness changes during the daytime—when solar exposure reaches its peak on the façade—and how it gradually shifts throughout the nighttime.

The following line chart illustrates the variation between indoor and outdoor dry-bulb temperatures. The temperature difference gradually increases from 14:00, reaching its peak at 18:00, before showing a slight decline thereafter. This peak change is attributed to the variation in solar exposure, which plays a vital role in the rapid rise of air temperature during that period.

Even though the sun rises at 6:35 AM, its effect on air temperature and humidity outside is not instantaneous. The delay in air temperature change is due to the cumulative effect of solar radiation on the atmosphere and surfaces. In the early hours, the sun's low angle means that the incoming solar energy is less intense, resulting in a slower warming process. By 11:53 AM, as the sun's angle increases, the temperature of the air, the ground, and other surfaces rises, along with an increase in radiation intensity. This usually causes noticeable variations in temperature and humidity three to five hours after sunrise, although it also depends on factors such as local geology, wind, and cloud cover (Alshahrani, 2019).

On the other hand, the impact of the accumulated temperature increase on outdoor air continues for 3–4 hours after sunrise, with the peak occurring at 17:12. This is illustrated in the line chart, which displays indoor and outdoor dry-bulb temperatures during a typical day in the screen operating period. The chart highlights the performance range of the evaporative ceramic screen in moderating indoor temperatures throughout the day. As outdoor temperatures peak, the screened indoor environment remains significantly cooler, reflecting both the evaporative cooling and shading effects of the screen, as well as its impact on the west-facing façade.

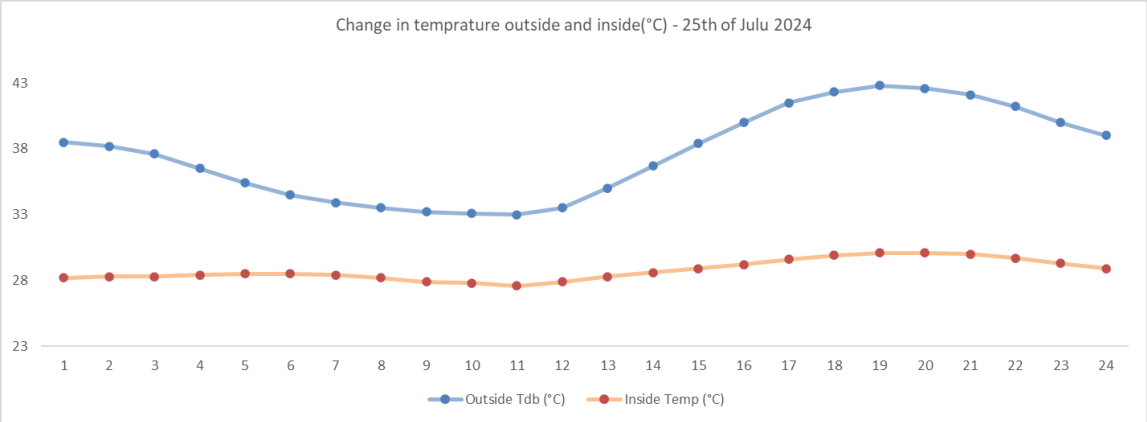


Figure 219 The fluctuation in air temperature (°C) inside and outside the room (air temperature of the incoming air), taken from the continuous monitoring data loggers over 24 hours (25TH Jul) – Source: Author

As the outdoor temperature decreases in the evening, the indoor temperature remains stable (yet lower than the outdoor temperature), demonstrating the continued influence of the building envelope along with the influence of the screen. The evaporative cooling screen plays a vital role in passive cooling, maintaining steady indoor conditions by minimising heat gain during the midday hours and contributing significantly to thermal comfort. A detailed analysis of the indoor-outdoor temperature differential (ΔDBT in °C) shows a gradual increase from 14:00, reaching a peak of 12.5°C at 18:00. Despite indoor temperatures remaining between 30–31°C, the pronounced gap is also driven by the shading performance of the evaporative cooling screen. When applied to a west-facing façade, which experiences the highest levels of solar radiation in the afternoon, the screen's shading effect becomes most effective from 14:00 to sunset (19:40). During this period, the screen significantly reduces direct solar gain, effectively reducing the impact of increasing external temperature on the indoor conditions and enhancing thermal comfort within the indoor space. This strengthens insulation against heat gain from the external conditions, highlighting the screen's potential to enhance performance.

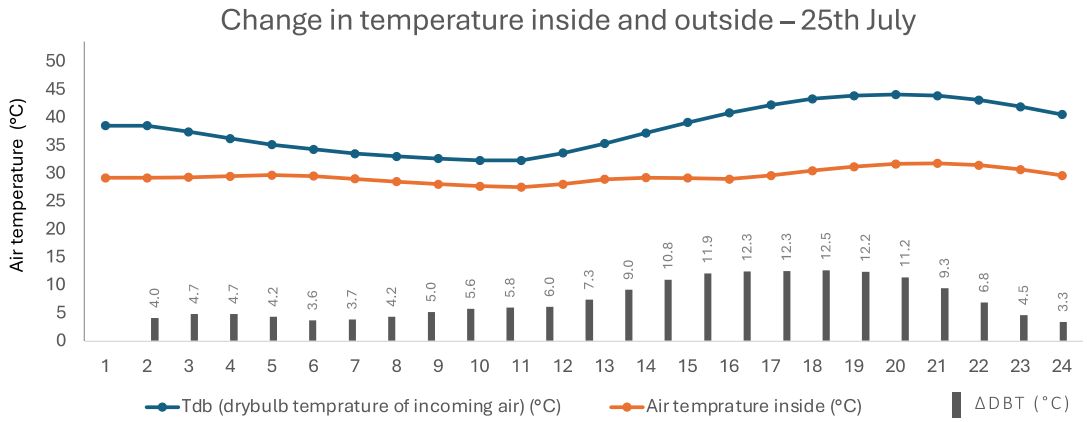


Figure 220 The fluctuation in air temperature (°C) inside and outside the room (air temperature of the incoming air) taken from the continuous monitoring data loggers over 24 hours (25th Jul) – Source: Author

The observed ΔDBT fluctuated considerably throughout the day, ranging from 3°C to 12.5°C. This differential between indoor and outdoor conditions is influenced not only by the performance of the evaporative cooling screen but also by the thermal properties of the building envelope.

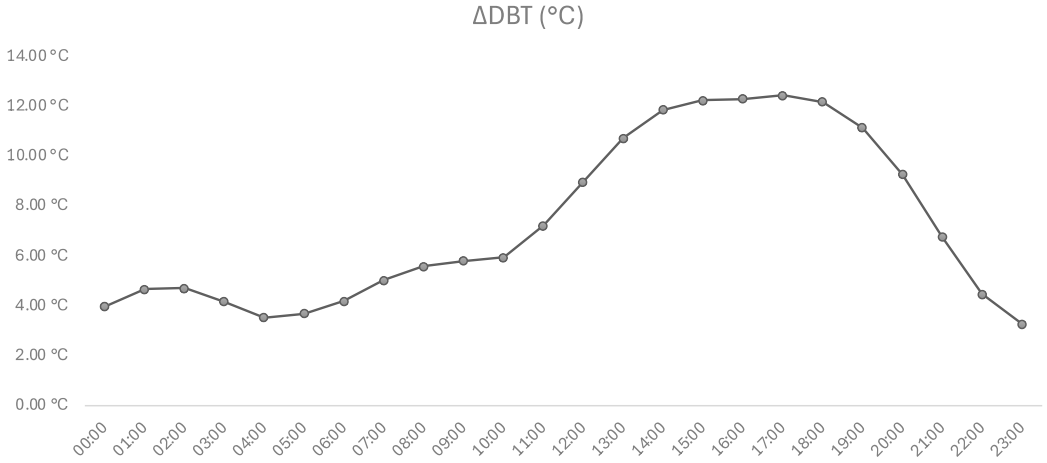


Figure 221 The fluctuation in dry-bulb temperature (ΔDBT in °C) throughout the day during the screen operating period, 25th of July 2024

To assess the screen's impact, the same space was monitored before its installation, selecting a day with outdoor conditions closely matching those observed after installation. Specifically, on July 18th, outdoor temperatures ranged between 37°C and 43°C, similar to the 36°C to 43°C range recorded on July 25th, when the screen was in place. Comparing the ΔDBT for both days provides a clear indication of the screen's contribution to moderating the indoor environment.

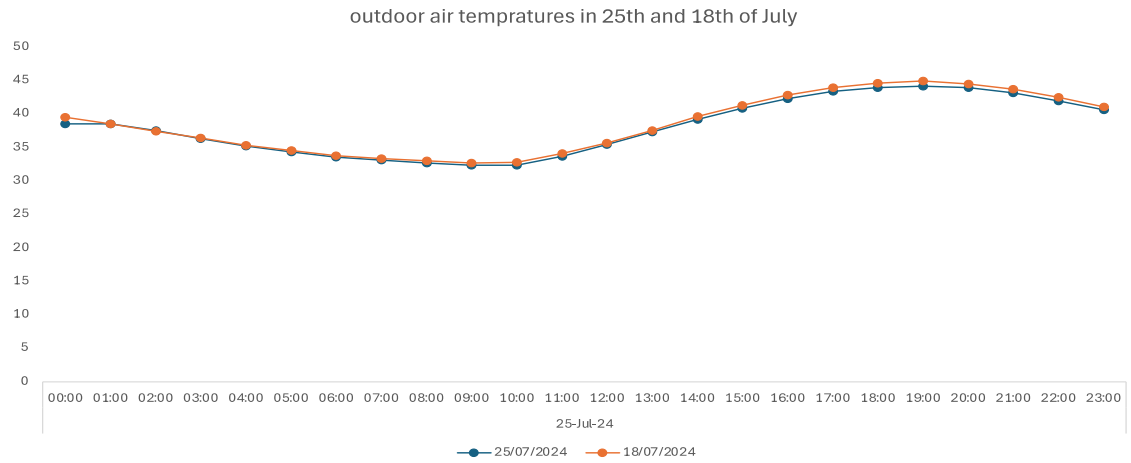


Figure 222 The outdoor temperature was monitored on the 18th and 25th of July 2024

Comparing the ΔDBT of the two days readily clarifies the distinct functions of the screen and the building envelope. On July 18th—the day before the screen was installed—the indoor temperature was lower than the outdoor conditions. The ΔDBT started at about 3°C early in the day and increased to around 7°C by the end of the day. This small differential shows the building envelope's insulation effect, which serves as a baseline for the enhanced performance expected after the screen's installation. On July 25th, the temperature difference between indoor and outdoor conditions was notably greater, particularly on the west-facing façade during peak hours. The ΔDBT gradually increased from midday, reaching 12°C by 18:00, as shown in the chart. This significant improvement reflects the combined benefits of the evaporative cooling and shading effects of the screen, which reduce direct solar heat gain while providing a more stable indoor environment.

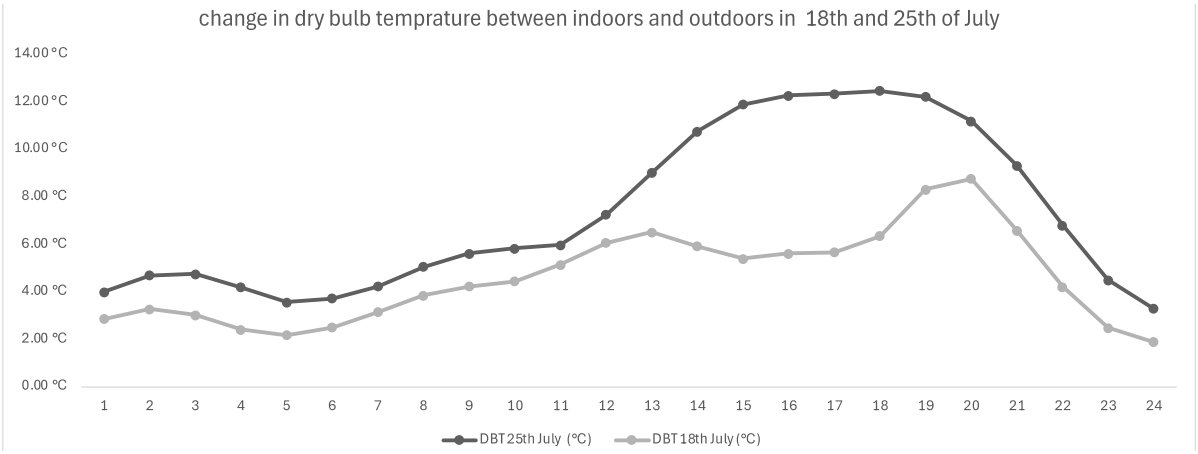


Figure 223 The fluctuation in dry-bulb temperature (ΔDBT in °C) throughout the day during the screen operating period, 25th and 18th of July 2024.

This day-long analysis from July 25 illustrates how the evaporative ceramic screen influences both temperature and relative humidity indoors compared to outdoors. Before and during sunrise, the air temperature remains relatively low, then gradually rises until it peaks between 15:00 and 18:00. While the outdoor temperature continues to increase through midday to late afternoon, the indoor

temperature begins to drop from around 15:00 until 19:00. This shift coincides with the sun's movement toward the west-facing façade, where the screen's extruded geometry effectively blocks direct solar radiation. Thermal imaging indicates that this shading action raises the screen's external ceramic surface temperature, accelerating water evaporation and leading to a slight uptick in indoor humidity after 19:00, likely linked to the accumulated evaporation and the higher water usage during peak morning hours. Meanwhile, the incoming air temperature still reaches its maximum in the late afternoon, but the indoor environment remains consistently cooler, underscoring the combined impact of shading and evaporative cooling. Outdoor humidity stays relatively low, whereas indoor humidity increases notably around midday when solar intensity is highest. This interplay of reduced temperature and rising humidity demonstrates the screen's dual benefit: it dampens heat gain while keeping indoor conditions more stable, affirming its effectiveness as a passive cooling strategy on a west-facing façade in a hot climate.

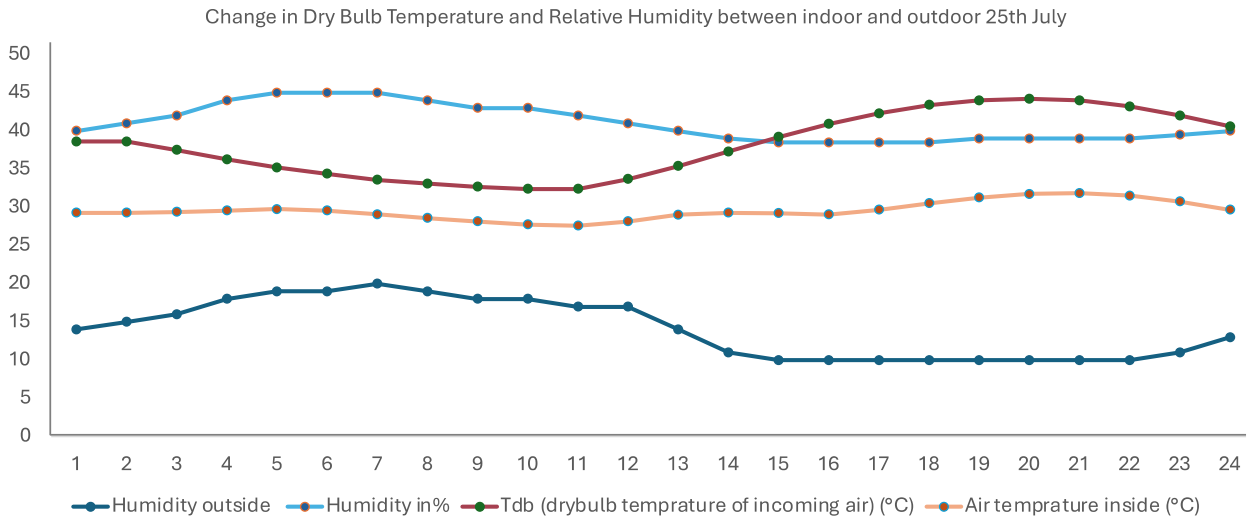


Figure 224 Change in Dry-Bulb Temperature and Relative Humidity between indoor and outdoor, 25th July.

During the night, external humidity gradually rises and reaches its highest level between 4:00 and 8:00, around sunrise. As the morning progresses, it decreases in tandem with outdoor conditions, only to climb again after sunset at approximately 17:12. Notably, despite a pronounced drop in outdoor relative humidity starting at midday, indoor humidity remains relatively stable, indicating that the screen's evaporative process continues to moderate the indoor environment beyond the hottest hours. This sustained elevation in indoor humidity, alongside the concurrent cooling observed in the temperature data, highlights the screen's overall effectiveness, as illustrated in the following chart.

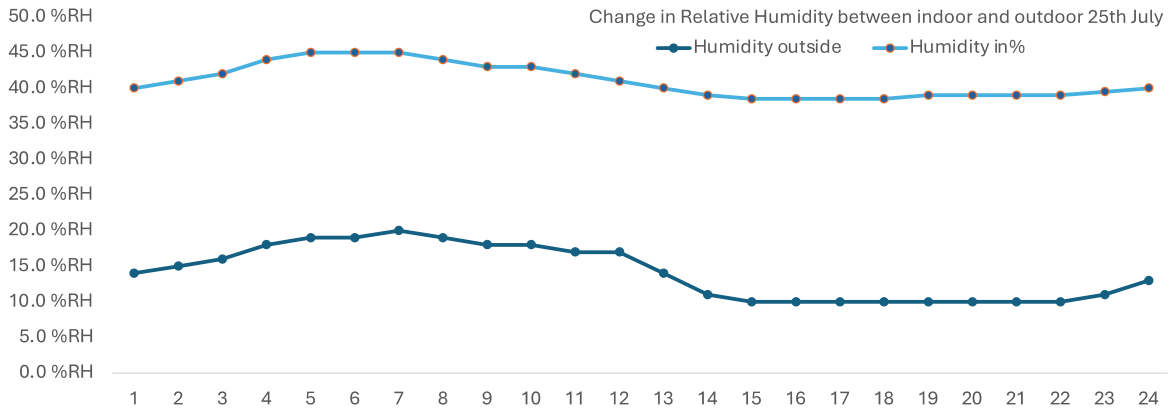


Figure 225 Change in the monitored relative humidity between indoor and outdoor conditions on 25th July 2024.

The given data illustrates the relationship between temperature and humidity inside the space. The stability in wet-bulb temperature indicates that, in the process of heating or cooling, there is a relative maintenance of evaporation potential. These factors are crucial when evaluating an evaporative cooling system, which mainly depends on the wet-bulb depression (the difference between air temperature and wet-bulb temperature) for its cooling effect.

In comparison to the outdoor environment, the indoor space remains noticeably cooler and more stable, especially during peak heating hours. By referencing a psychrometric chart, it becomes clear that indoor temperature and humidity levels often stay within or closer to typical comfort ranges, highlighting the system's effectiveness in providing a comfortable indoor conditions.

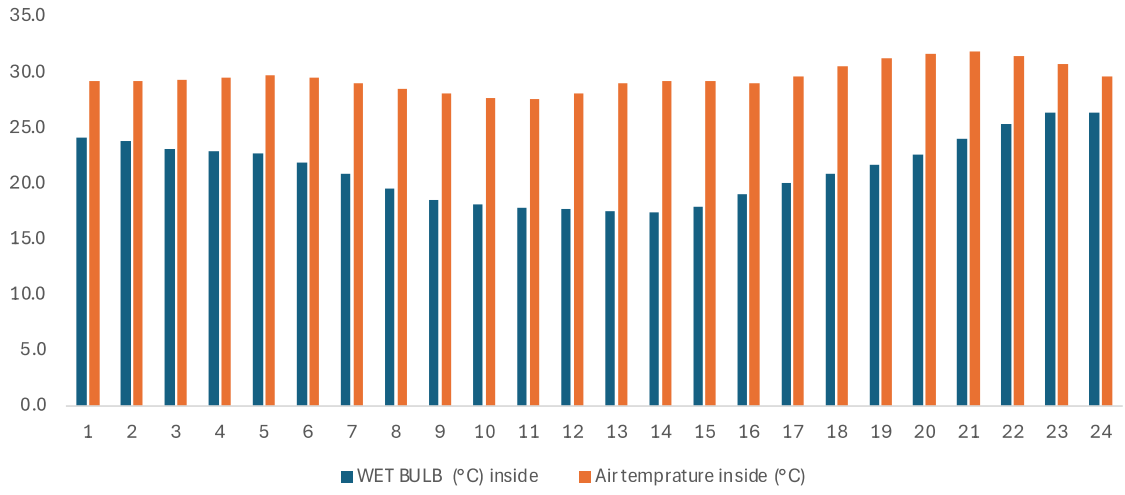


Figure 226 Relationship between dry-bulb temperature and wet-bulb temperature indoors on the 25th of July.

The following analysis uses psychrometric charts based on Riyadh's weather data at 1:00 PM and 3:00 PM on July 25th, 2024, to illustrate the effect of the evaporative cooling screen. At 1:00 PM, the chart shows the outdoor condition (red point) with high dry-bulb temperature and low humidity, while the indoor environment (orange point) is cooler and slightly more humid due to the screen's operation. The nearly constant enthalpy path between these points highlights how the screen effectively reduces indoor temperature while introducing a moderate increase in humidity. In the 3:00 PM chart, a more

significant temperature difference is observed (particularly since the results are for a context located on the west-facing façade), demonstrating the screen's ability to mitigate heat gain during peak solar hours.

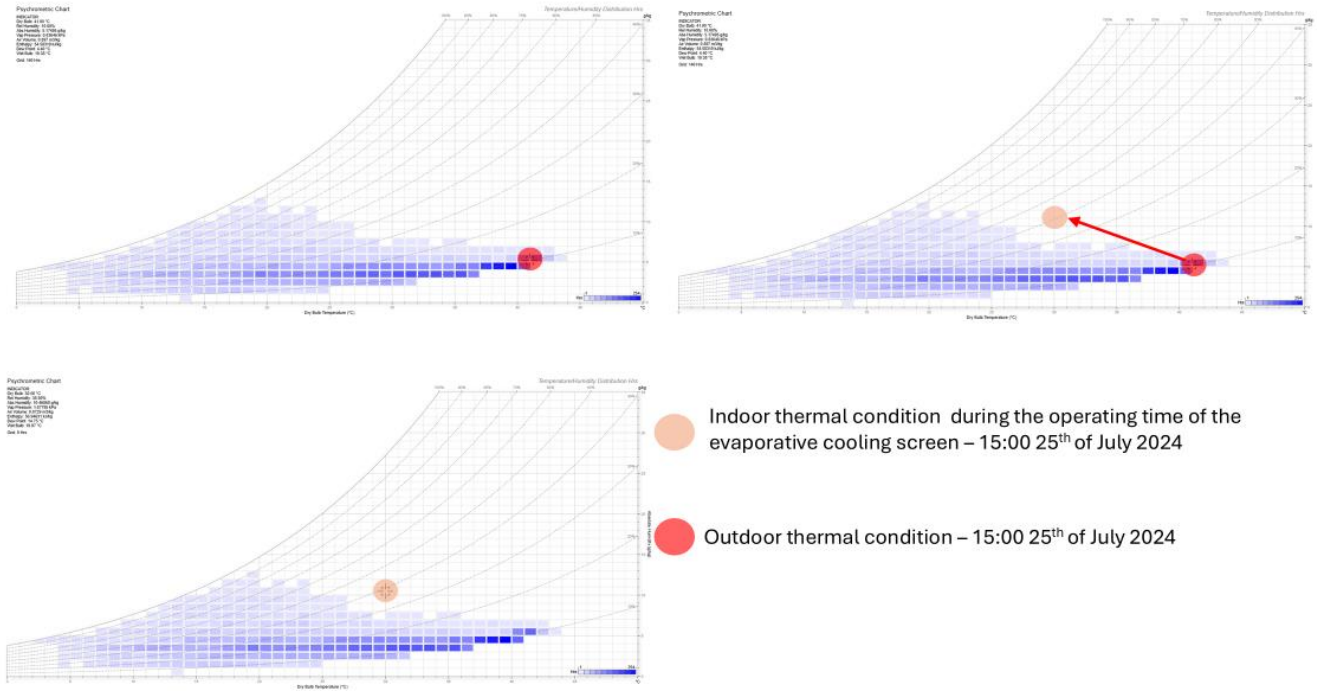


Figure 227 Change in thermal conditions in the outdoor and indoor monitored data on 25th July at 15:00, plotted on the psychrometric chart

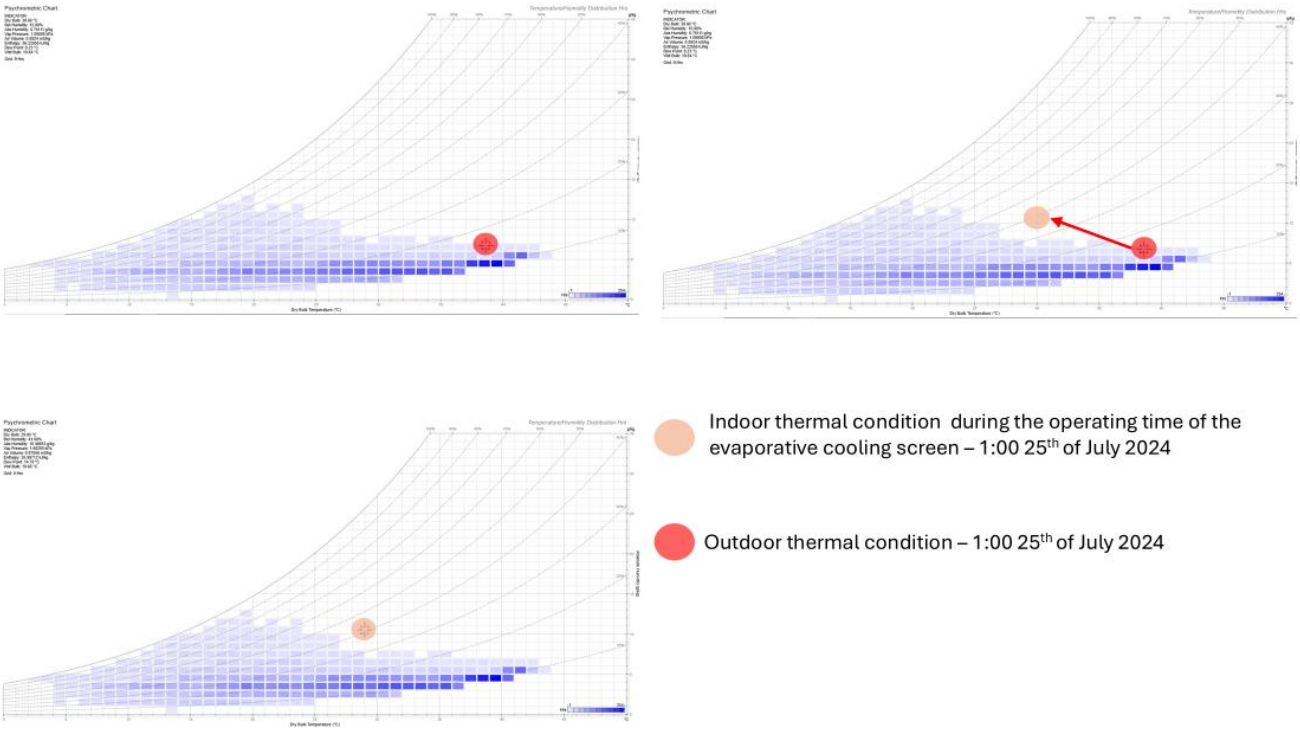


Figure 228 Change in thermal conditions in the outdoor and indoor monitored data on 25th July at 1:00 AM, plotted on the psychrometric chart.

The following chart illustrates how the gap between the dry-bulb temperature (DBT) and wet-bulb temperature (WBT) shifts throughout the day, indicating the potential for evaporative cooling. This gap typically widens in the mid-afternoon, aligning with peak outdoor conditions. When it is larger, more water can evaporate, leading to stronger cooling. By comparing the incoming air's dry-bulb temperature (the temperature of the air outside) with the outdoor WBT, it becomes clearer how effectively the screen lowers the incoming air temperature: the closer these two lines are, the more efficient the evaporative process becomes. This also reflects the theoretical cooling potential, as a larger DBT–WBT difference means the system can bring the incoming air closer to the wet-bulb temperature if operating efficiently.

In summary, the chart highlights the screen's ability to reduce air temperature, especially when the DBT–WBT difference is at its highest. As the gap narrows, humidity generally increases, which can influence both comfort and system performance.

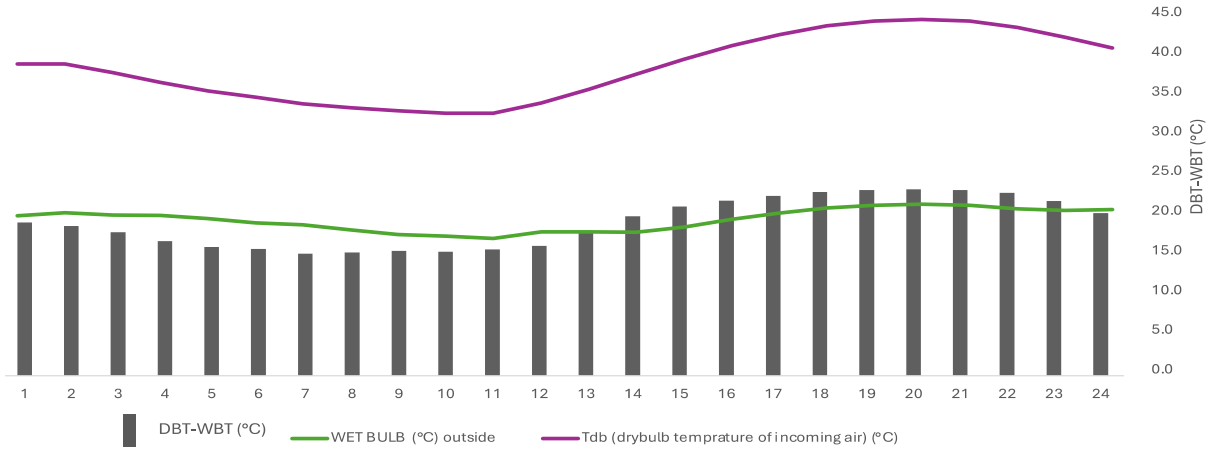


Figure 229 The change between outdoor dry-bulb temperature and wet-bulb temperature, based on monitored data on July 25th

Accordingly, the following graph illustrates the relationship between the outdoor DBT–WBT gap and the latent cooling measured indoors. A wider gap allows more water to evaporate, which in turn enhances the latent cooling effect. The graph also shows a slight delay between outdoor DBT–WBT conditions and the indoor latent cooling response. This lag is expected, given the building's thermal inertia and the time required for indoor air to adjust. Overall, the data confirms that latent cooling is directly linked to the DBT–WBT gap, with indoor performance reflecting these changes after a short delay.

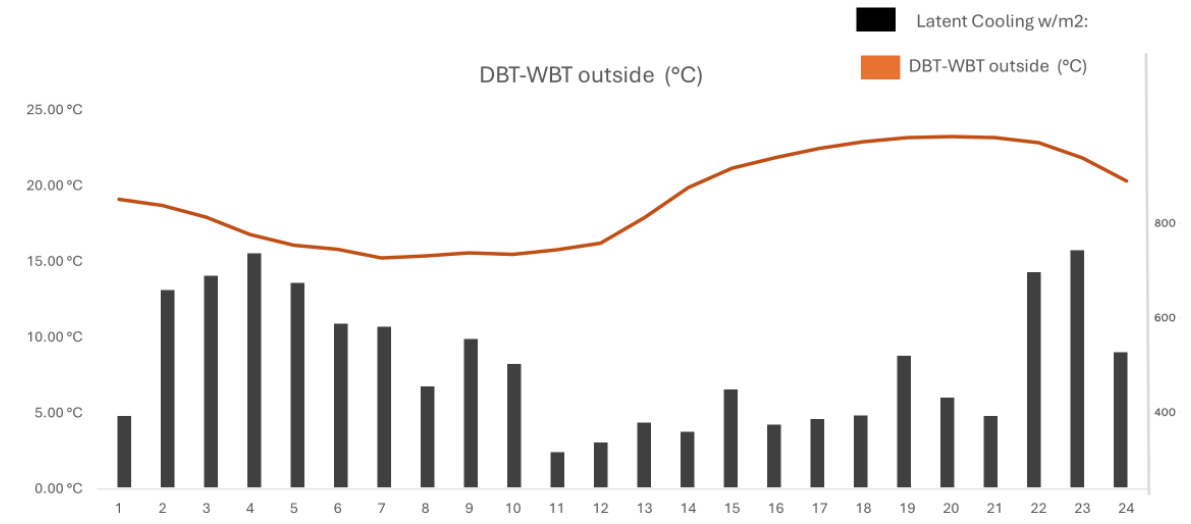


Figure 230 Change in outdoor dry-bulb and wet-bulb temperature in relation to the achieved latent cooling (W/m²).

Analysis of the hourly data confirms that the system delivers both sensible and latent cooling, measured in watts per unit area of the ceramic screen. Considering the operational hours, water consumption was monitored and found to be 21 litres per day, as shown in the following chart. The predicted reduction in air temperature at the specified user distance ranges from 2°C to 5.9°C after accounting for the effect of the building envelope. In addition, the cooling process increases the air moisture content by approximately 0.9 g/Kg to 1.4 g/Kg, as shown in the following detailed hourly metrics chart.

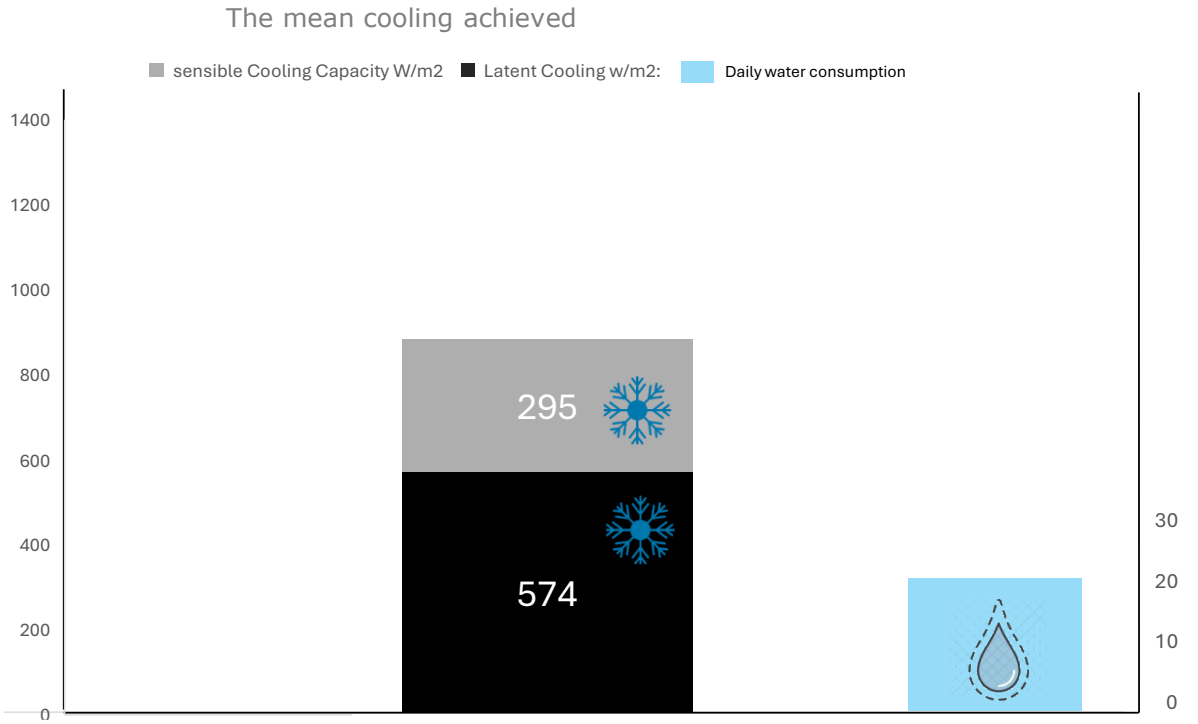


Figure 231 The cooling performance and daily water consumption of the ceramic screen during its operation on the 25th of July.

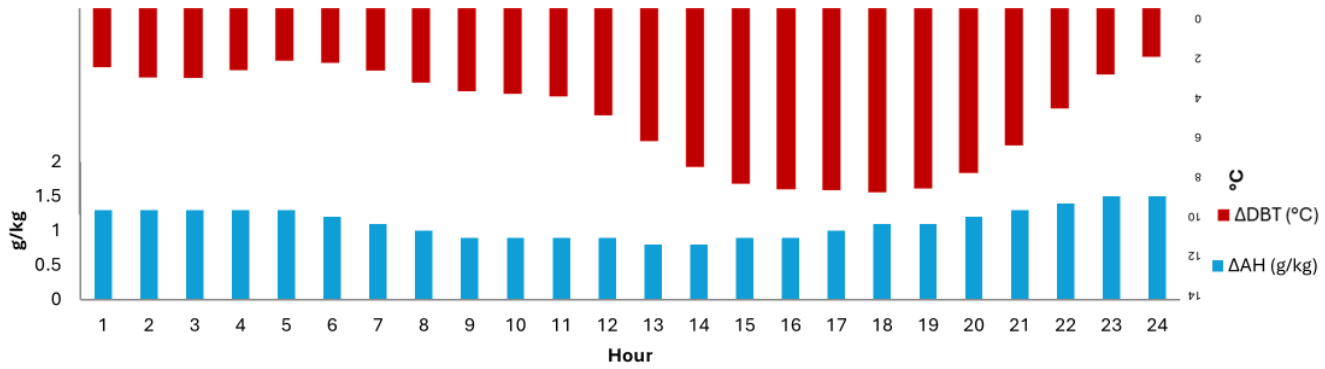


Figure 232 Hourly dry-bulb temperature reduction and absolute humidity increase on 25th July during the screen operation time in Riyadh

The following chart indicates that, under outdoor conditions allowing a larger difference between dry-bulb temperature (DBT) and wet-bulb temperature (WBT°C), the screen achieves more effective cooling, although humidity also increases (as shown by ΔAH). This highlights the principle of evaporative cooling, where reducing air temperature is balanced by a moderate rise in moisture levels.

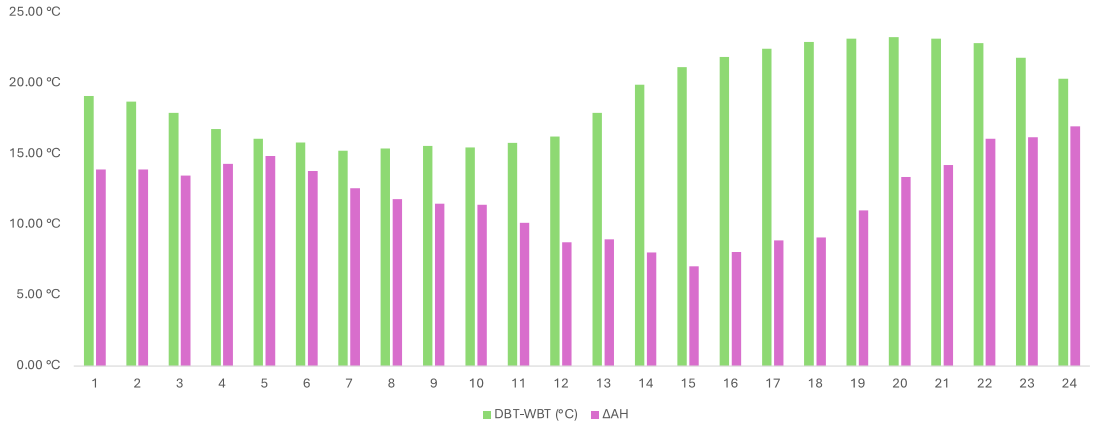


Figure 233 The fluctuation of dry-bulb temperature (DBT) and wet-bulb temperature in relation to the fluctuation in ΔAH

According to the thermal imaging taken during the screen's operation time, it is clear that the evaporative cooling screen plays an important role in reducing temperatures in areas where water flows evenly through the internal cavities of the screen. Cooler surfaces (darker patches) indicate where evaporation is efficiently removing heat from the incoming air, while warmer spots may point to sections with reduced water coverage or airflow. This contrast reveals how effectively the screen can lower temperatures by several degrees, contributing to a more comfortable environment. The variation in surface temperature also acts as a catalyst for maintaining the screen's performance over time, as shown in the following image, which collages thermal views taken from both inside and outside at different times of the day.

The ceramic surfaces show a clear fluctuation between absorbing heat and cooling off over time. In the early hours of the day, the ceramic surface temperature gradually rises, reaching its peak at 17:00. However, as the day progresses, the evaporative cooling effect becomes more apparent. There is a noticeable shift toward cooler surface temperatures on the screen and adjacent areas, indicating that

the ceramic screen does not retain heat indefinitely. This change illustrates how the ceramic not only absorbs heat under direct sunlight but also effectively releases it when conditions allow, resulting in a more balanced temperature profile by the end of the day. The thermal images also reveal the impact of the evaporative cooling screen, showing how the pattern draws heat away through evaporation and leaves cooler zones that stand out against the warmer background. These visuals clearly demonstrate how the screen reduces the intensity of extreme heat.

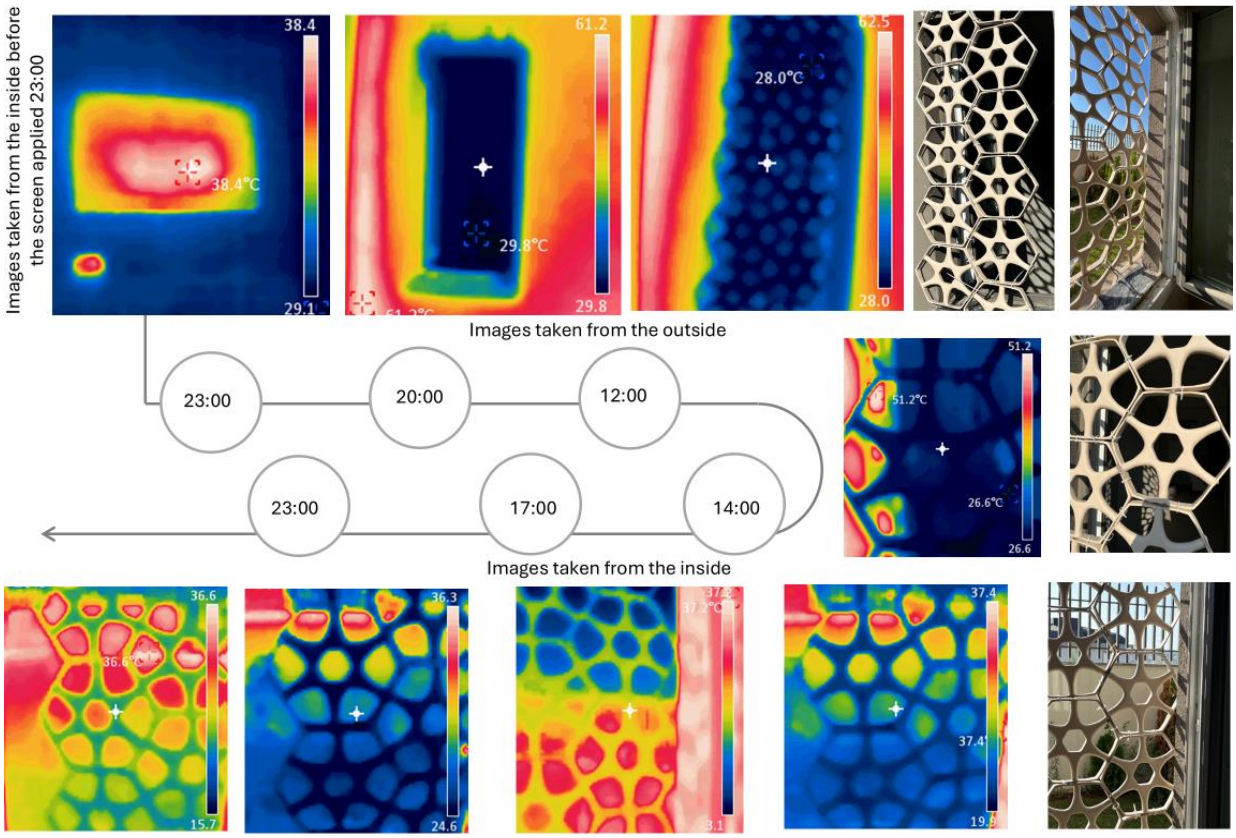


Figure 234 Thermal images taken of the screen during its operation throughout the day on 28th July.

9.6.2 Water consumption of the system:

Water consumption is a key performance metric for the ceramic evaporative cooling screen. Therefore, water usage was recorded hourly per square meter during system operation, providing insights into the system's response to varying environmental conditions. In addition, total daily water consumption was compared to a version without the extruded cap. This comparison illustrates how the mutual shading effect of the extruded cap influences water retention and evaporation rates, as shown in the accompanying chart.

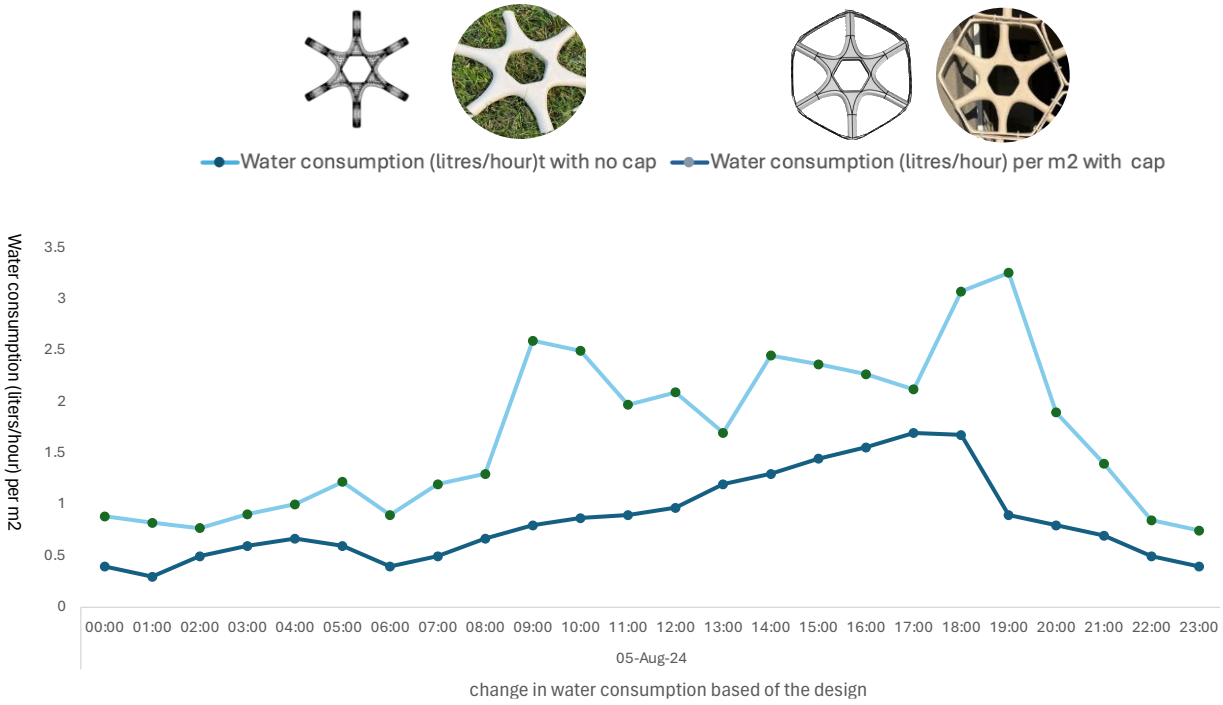


Figure 235 Hourly change in water consumption of the evaporative cooling ceramic screen component, based on the component's design.

Given the changes in water consumption, the average performance differs considerably, with the system without a cap showing higher consumption throughout the day, particularly around midday. In contrast, the version with the extruded cap maintains lower consumption levels, suggesting that the cap's shading effect helps reduce evaporation and overall water usage.

It was predicted that the component without a cap would consume more water than the one with the extruded cap, thanks to its design featuring mutual shading inspired by Mashrabiya. The design with the extruded cap indeed shows a reduction in water consumption by 17.2 litres. Notably, 87% of this reduction (15 litres) is achieved during the daytime due to the shading effect. However, during nighttime, when there is no solar exposure, the screen with the extruded cap still consumes 2.2 litres less, even though the exposed wet ceramic surface and external conditions remain the same, as shown in the following chart. This suggests that other factors may influence water consumption at night.

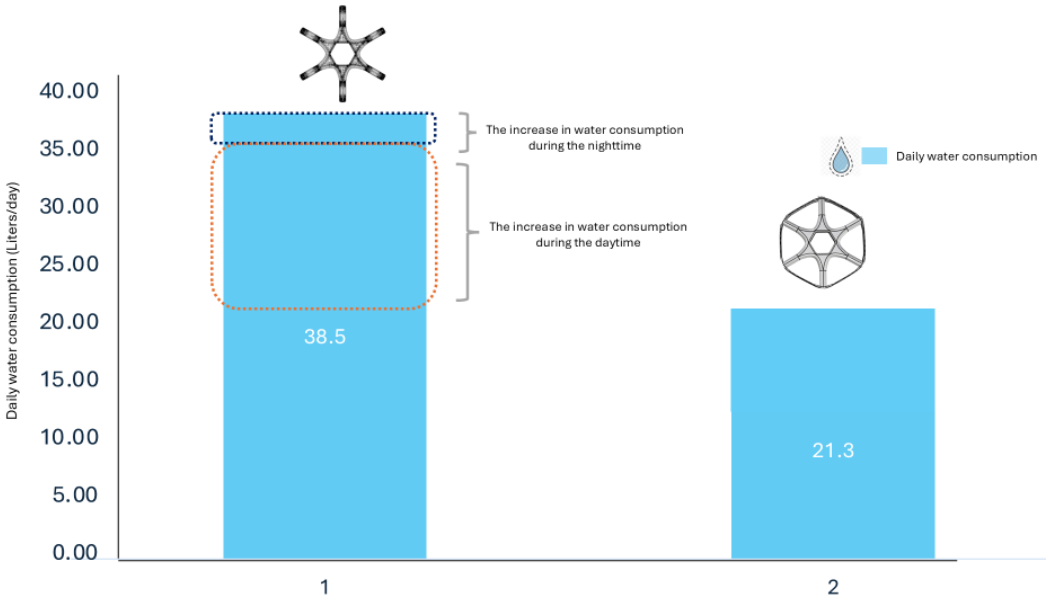


Figure 236 Total change in daily water consumption of the evaporative cooling ceramic screen component, based on the component's design.

The increase in night-time water consumption (an extra 2.2 litres) may be linked to the localised volumetric airflow near the screen. To investigate this, virtual CFD tests were conducted to assess how the extruded geometry affects airflow, using a monitored air velocity of 1.3 m/s, as shown in the following images.

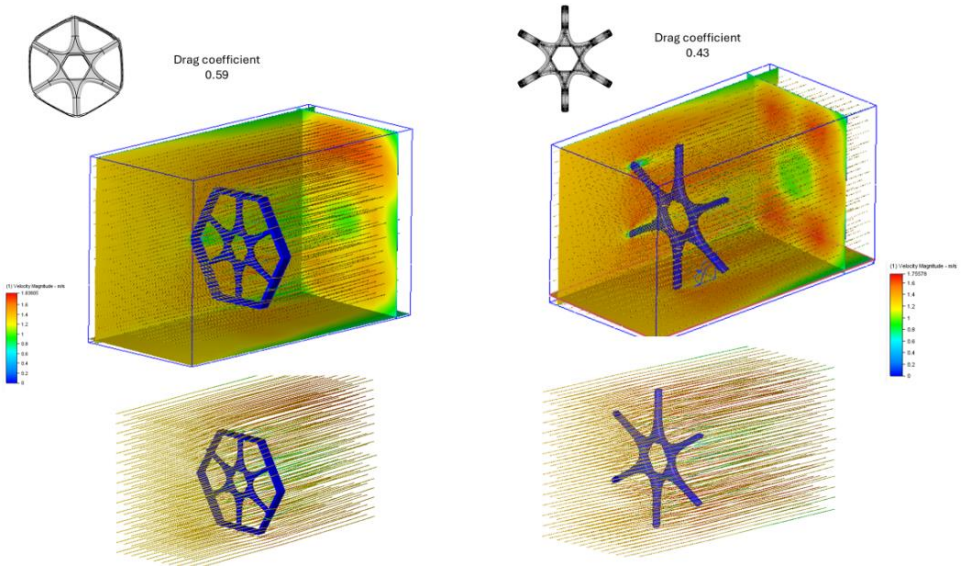


Figure 237 Change in the drag coefficient of the component based on the added extruded geometry in CFD simulations.

The simulations showed that the extruded cap slightly increases the component's drag, reducing airflow through the screen by a small margin. This minor obstruction accelerates the evaporation rate of the component without a cap, resulting in an average increase of approximately 0.2 litres per hour in water consumption during the night, as illustrated in the following chart.

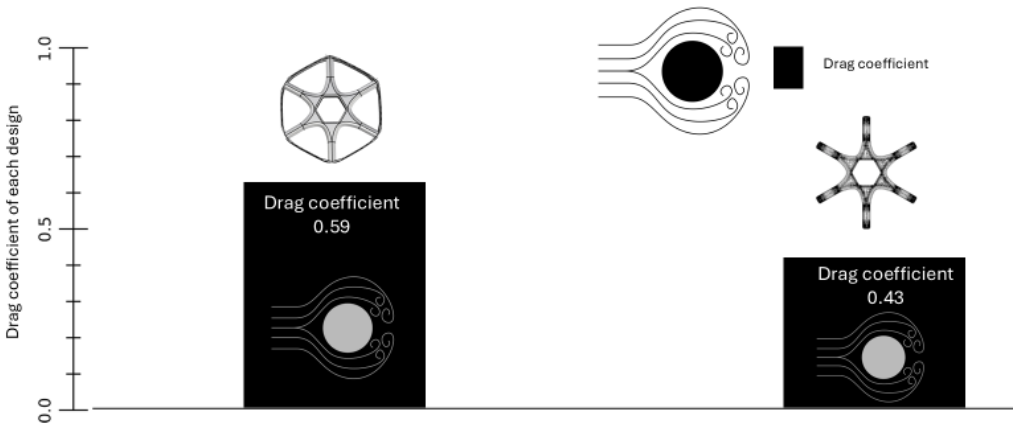


Figure 238 Change in the drag coefficient of the component based on the added extruded geometry.

The following chart shows how water consumption fluctuates in response to the average ΔDBT. As outdoor temperature increases throughout the day, peaking at 17:00, ΔDBT rises, and the screen uses more water to deliver stronger evaporative cooling. A similar peak is observed in both water consumption and ΔDBT around midday, indicating that the screen intensifies its cooling efforts when outdoor conditions are most extreme. Later in the day, as ΔDBT decreases, water usage also drops, suggesting that less cooling is required.

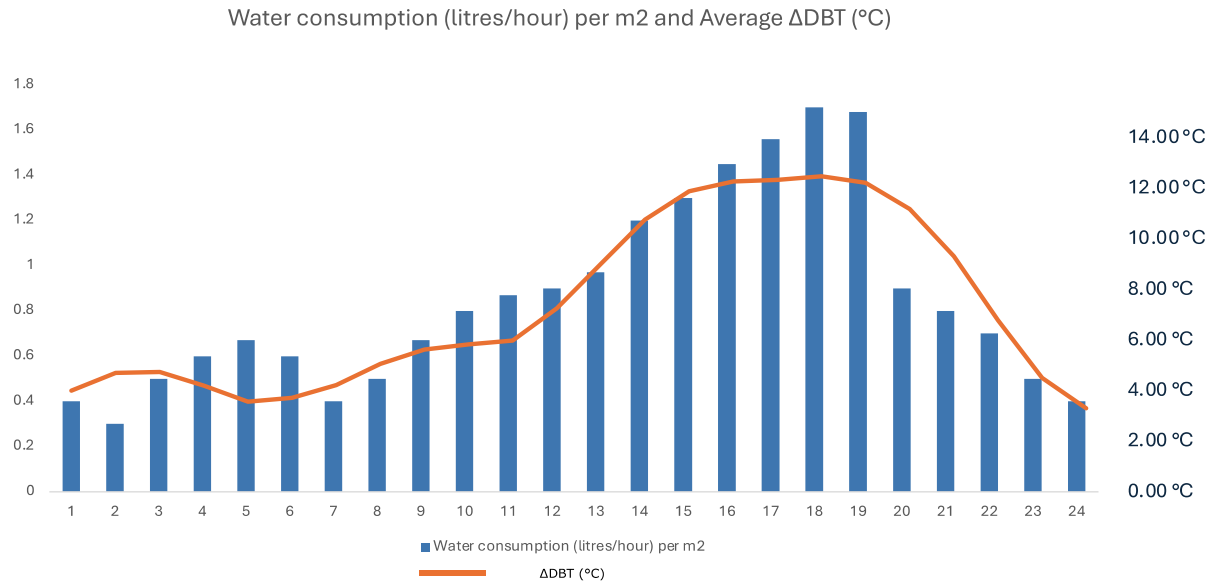


Figure 239 Monitored water consumption of the system in relation to the difference in dry-bulb temperature

Overall, the chart highlights the screen's water consumption fluctuation based on the temperature difference, achieving effective cooling during peak hours while reducing consumption when demand for cooling is lower.

The hourly fluctuation in water consumption responds to the indoor DBT–WBT difference, where, typically during the hotter parts of the day, the screen uses more water to boost evaporative cooling. Later, as this temperature gap decreases, water consumption gradually drops. This relationship indicates that the screen adjusts its cooling output (represented by water consumption) in response to

the indoor thermal load—using more water when conditions demand it and scaling back when they don't, as shown in the following chart.

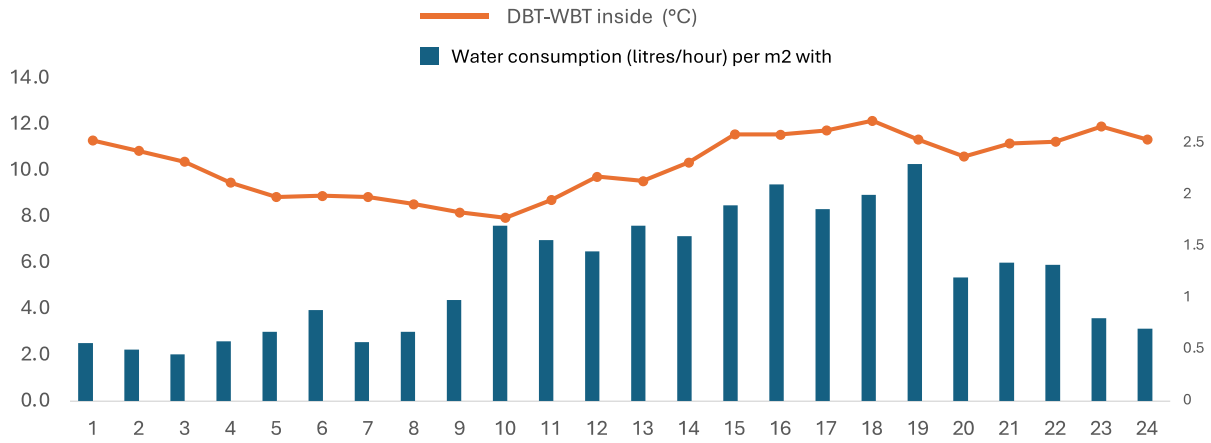


Figure 240 Hourly water consumption in relation to the DBT-WBT difference on 25th July 2024.

9.7 Conclusion

This chapter investigated the applicability of the proposed design under realistic conditions and how the screen would be applied to the façade, along with testing its performance range. Overall, the screen was successfully utilised and demonstrated high flexibility and adaptability across various contexts and spatial layouts. The design showcased several positive features related to the Mashrabiya concepts that initially inspired it.

The monitored and calculated performance range provided valuable insights into the effectiveness of the design and highlighted the benefits of initiating a hybrid design process—one that goes beyond merely mimicking Mashrabiya forms to understanding their functional value. This approach helped identify the core reasons behind the screen's strong performance and allowed for the evaluation of these inputs to inform future design improvements.

Ultimately, this underscores the value of the traditional Mashrabiya as a conceptual reference and emphasises the importance of a more innovative design methodology. By understanding the underlying principles of traditional designs, it becomes possible to develop environmentally responsive solutions that are both culturally grounded and technically effective.

The analysed data, as previously discussed, confirms that the 1:1 application of the proposed evaporative cooling screen performs effectively even under extreme conditions. The use of analytical tools to test the proposed design, based on continuous evaluation of environmental performance, enabled its development into a solution that is environmentally responsive to harsh climates. This was clearly observed during its application, where the realistic performance demonstrated a tangible impact in improving thermal conditions.

The results evaluated the performance of the screen applied to a 0.9 m² window in a 9.3 m² room using 12 elements. Here are the key findings:

- Temperature Reduction and Cooling Capacity:

With an outdoor dry-bulb temperature (DBT) of around 40°C and a wet-bulb temperature (WBT) near 19°C, the DBT–WBT gap is approximately 21 °C. The screen contributes to an 8°C indoor temperature reduction, lowering it from 40°C to approximately 31.9°C—representing roughly 57% of the total cooling potential. However, this temperature reduction is not solely attributed to the screen's evaporative cooling and shading performance; the building envelope also plays a role, reducing indoor temperatures by 3 to 4 °C on similar days without the screen.

This performance translates to a latent cooling capacity of 574 W per unit area under peak conditions and a sensible cooling capacity of 295 W per unit area. The relative humidity increased by up to 29% hourly, as the cooling process raised the air moisture content by approximately 0.9 g/kg to 1.4 g/kg.

- Response to DBT–WBT Variations

The cooling output closely follows the DBT–WBT gap. During midday, when this gap is at its widest, the system ramps up its latent cooling output. Indoor measurements, however, exhibit a slight lag compared to the outdoor DBT–WBT conditions—an expected outcome due to thermal inertia and the time required for indoor air to adjust.

- Water Consumption and Efficiency

Water usage is a crucial performance metric. When comparing the system with and without an extruded cap, the extruded cap—drawing inspiration from Mashrabiya—reduces overall water consumption by 17.2 litres per day. Most of this saving (approximately 15 litres) occurs during daytime hours, due to the mutual shading effect that limits solar heating and reduces evaporation. However, at night, the design with the extruded cap consumes 2.2 litres more. CFD analysis suggests that the cap increases drag, causing a minor airflow obstruction that accelerates evaporation during periods without solar exposure.

- Ceramic Surface Behaviour

Observations of the ceramic surfaces reveal a clear cycle: a gradual temperature increase during the early hours, peaking around 17:00, followed by noticeable cooling as the evaporative process takes effect. This dynamic behaviour demonstrates that the ceramic not only absorbs heat under direct sunlight but also releases it efficiently when conditions permit, contributing to a balanced temperature profile by the end of the day.

Accordingly, the 1:1 screen effectively utilises evaporative cooling as a strategy to moderate indoor temperatures, achieving a substantial cooling effect while optimising water usage. The design modifications, particularly the inclusion of the extruded cap, enhance performance by reducing daytime

water loss even though they introduce a slight increase in nighttime consumption. These results validate the design approach and demonstrate that the screen is well-suited to improving indoor comfort in extreme climates.

Through the process, the design has proved efficient as an evaporative cooling and shading system, performing effectively under one of the most extreme conditions—a west-facing façade for a single-sided naturally ventilated room in south Riyadh. The system's structural stability and feasibility make it suitable for application across various contexts.

Although limited inputs and a short testing timeframe restricted deeper exploration of the screen's potential, the proposed design nonetheless stands out as a promising foundation for future innovations. It is hoped that this design will inspire upcoming projects to adopt multifunctional screens capable of operating with a high degree of flexibility and adaptability across diverse environments.

Figure 241 Realistic vs. virtual testing of screen applicability -Source: Author

- Performance in Comparison to Other Innovative Approaches

This section aims to briefly compare the ceramic evaporative cooling screen proposed in this study—tested on a west-facing façade in Riyadh—with the Ecooler system, another evaporative cooling solution designed for dry climates, particularly in Egypt, as introduced in Chapter 1. While both approaches draw inspiration from the traditional Mashrabiya, each follows a distinct design philosophy and performance strategy. This comparison situates the proposed screen within the broader discourse on passive cooling technologies, especially in regions facing extreme temperatures, lower humidity and increasing water scarcity.

One of the key factors in comparing these two systems is their water consumption. Each Ecooler tile holds approximately 550 ml of water and, on average, releases half of that (roughly 275 ml) per day (Khan, 2025). Based on the data provided by the Ecooler developers, a full screen setup consumes around 9.5 litres daily, resulting in a total indoor water usage of approximately 28.5 litres per day. In contrast, the ceramic screen proposed and tested in this study used 21 Litres per day. However, with the addition of the Mashrabiya-inspired extruded cap, the system demonstrated a significant reduction in water usage during peak hours, saving up to 17 litres per day compared to the same setup without the cap.

Although the proposed ceramic screen requires more water overall, it also handles a greater cooling load. It delivers both sensible and latent cooling, with a noticeably higher temperature drop, especially when the difference between dry-bulb and wet-bulb temperatures peaks, such as the 21°C gap recorded in Riyadh. The screen's real advantage lies in its design refinements that reduce water loss and enhance adaptability to changing conditions. In contrast, the Ecooler system is geared towards slow, steady evaporation and serves a more decorative role, making it better suited for indoor or semi-transitional spaces.

Table 12 Ecooler cooling capabilities under varying environmental conditions, based on two variables: ambient temperature and humidity. source: (Khan, 2025).

Outside temperature	Relative humidity of surroundings								
	20%	25%	30%	35%	40%	45%	50%	55%	60%
Temperature in °C following cooling									
30	16.4	17.3	17.9	18.8	19.8	20.5	22.4	22.2	22.9
35	19.8	20.4	21.7	22.8	23.8	24.8	25.7	26.5	27.5
40	23.4	24.1	25.4	26.7	27.8	29.0	29.0	30.9	31.9

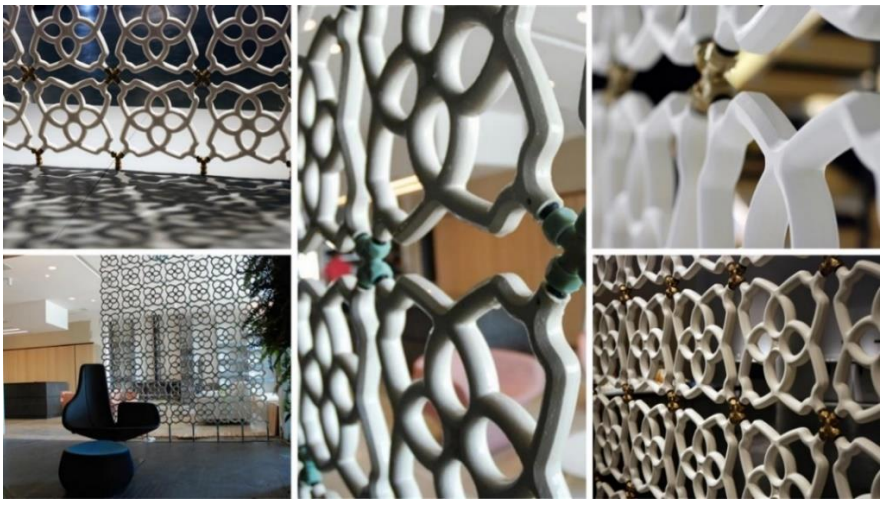


Figure 242 Ecooler documented application from the Ecooler Gallery – source: (Khan, 2025).

Another aspect of the comparison focuses on the role of solar exposure, as Ecooler was not tested in outdoor conditions where direct sunlight would affect its performance. To address this, a brief analysis was conducted to compare the variation in solar exposure between the proposed screen and Ecooler when both are placed in the same context and subjected to identical solar conditions. This comparison highlights how the proposed design addresses environmental factors that were not previously considered in Ecooler's testing, as illustrated in the following figure.

The following results illustrate the change in total annual solar exposure between Ecooler and the proposed ceramic screen when applied to a south-facing façade in Riyadh, Saudi Arabia, under the same context and environmental conditions.

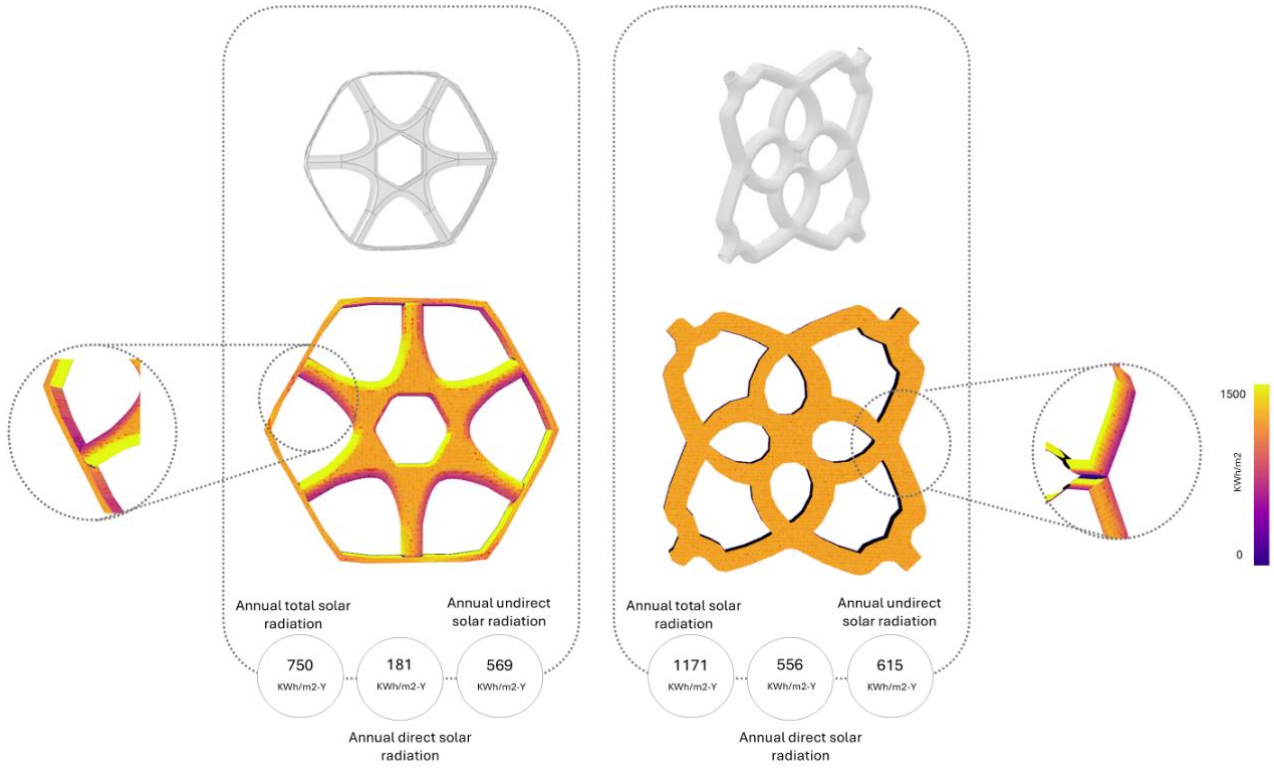


*Figure 243 Annual solar radiation analysis comparing the Ecooler and the proposed ceramic evaporative cooling screen on a south-facing façade in Riyadh. **Source**: Author, generated using Climate Studio.*

The results illustrate the difference in total annual solar exposure between the Ecooler and the proposed ceramic screen, both positioned on a west-facing façade in Riyadh under identical environmental conditions. The results demonstrate how each system responds to solar radiation in its own unique way, influenced by its form and geometry.

The proposed ceramic screen shows a lower annual total exposure of 750 kWh/m² compared to the Ecooler, which achieves a significantly higher total of 1171 kWh/m²/year. A closer look reveals that this difference results from how much direct solar radiation each system allows through due to the designed geometry. The ceramic screen blocks more of the direct solar exposure, allowing only 181 kWh/m², while the Ecooler geometry receives 615 kWh/m². That's over three times as much.

On the other hand, the proposed ceramic screen does allow more diffuse radiation of 569 kWh/m², which suggests that, while it shades well from harsh direct sunlight, it still enables softer ambient light to filter through. This aligns with its function as both a cooling and a daylighting device.
So, while Ecooler receives more solar exposure overall, the proposed ceramic screen's geometry does more to filter and control direct radiation, which is an essential factor in maintaining thermal comfort without completely blocking out natural light.

Another important point of comparison lies in the aerodynamic behaviour of each screen. This helps determine whether the proposed geometry—with its interlocking elements and mutual shading—is impeding airflow more than the Ecooler module. To assess this, both geometries were tested under the same wind conditions to observe differences in velocity distribution and drag performance.
The simulation shows that the proposed screen produces a slightly higher drag coefficient of 0.59, compared to 0.51 for the Ecooler. This suggests a marginally greater resistance to airflow. However, when looking at the velocity vectors and streamlines, it becomes clear that airflow through the proposed screen remains well distributed. The mutual shading elements do not fully block the passage of air; instead, the shading geometry redirects it while still allowing sufficient ventilation.

The Ecooler geometry, on the other hand, creates smoother, less turbulent airflow patterns with more consistent velocity magnitude throughout the section. While this supports a steady evaporative process, it may also mean reduced turbulence and mixing, which could limit the system's ability to adapt to fluctuating heat loads or maximise evaporative potential in peak conditions, especially when applied on a single-sided ventilation strategy.
In contrast, the proposed screen generates more complex airflow patterns with localised acceleration zones, especially near the shaded regions. This turbulence may contribute to enhanced heat exchange and moisture distribution within the system, supporting its dual role in both shading and cooling. So, although the drag is slightly higher, it seems to be a trade-off for more effective environmental modulation, which balances airflow, shading, and evaporation in a single form.

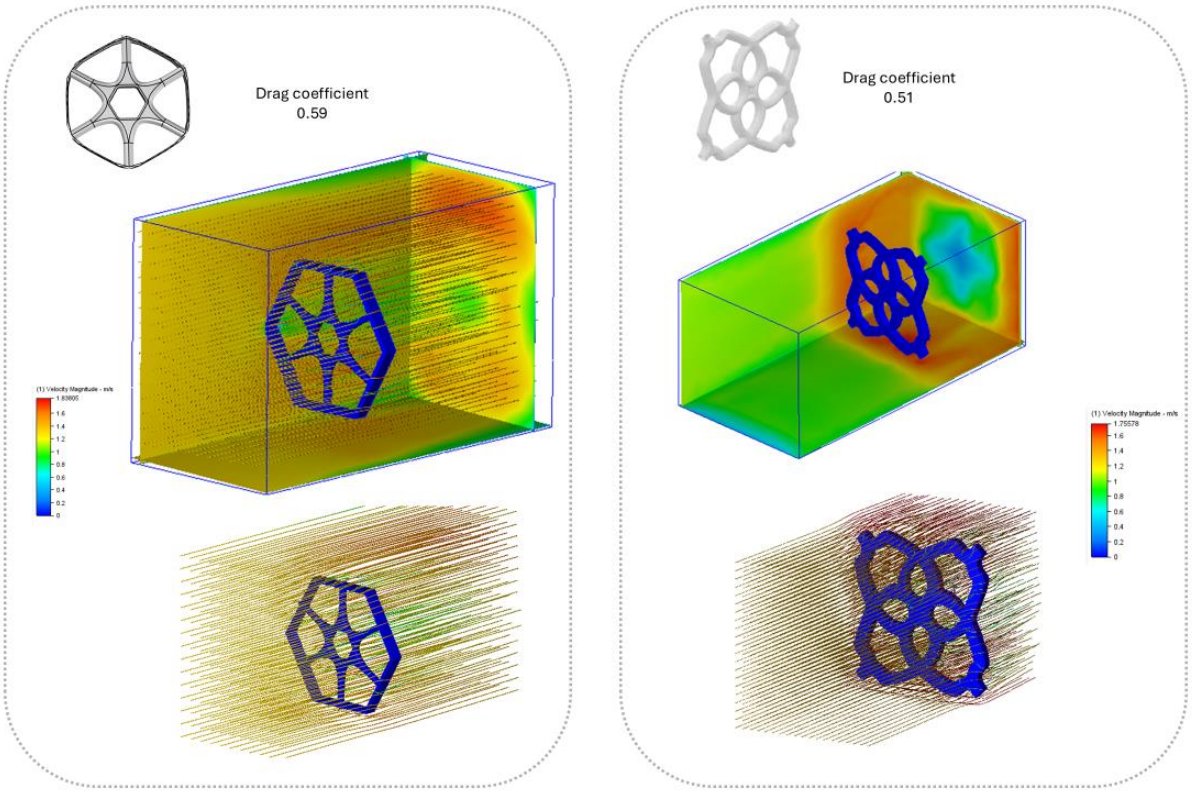


Figure 244 CFD analysis comparing aerodynamic performance and drag coefficients of the proposed ceramic screen and the Ecooler module. ***Source****: Author*

10. Conclusion

10. 1 Summary of Conclusion:

The primary aim of this study was to integrate passive evaporative cooling into contemporary architecture, not merely as a design solution, but as a culturally rooted environmental strategy. Drawing direct inspiration from the Mashrabiya, the study reinterprets this traditional Islamic façade element, which for centuries was meticulously crafted to respond to extreme climatic conditions long before the advent of mechanical cooling (Weber, 2014). The study presents an environmentally conscious alternative that merges climatic responsiveness with cultural identity by embedding traditional knowledge into a newly developed ceramic evaporative cooling screen.

Historically, the Mashrabiya served more than a mere shading function. Its intricate geometries and material composition were intentionally designed to regulate light, airflow, and privacy, while subtly contributing to passive cooling strategies. Moreover, in many traditional settings, a ceramic pot filled with water was placed behind the screen, allowing the air passing through to cool and absorb moisture, thus creating a naturally comfortable microclimate without isolating occupants from the outdoor environment (Weber, 2014).

Building on these principles, the main objective of the study was to develop a multifunctional ceramic screen that integrates both evaporative cooling and shading performance. This required investigating the environmental performance of the Mashrabiya, with a focus on understanding how its geometric configurations influence airflow modulation and mutual shading efficiency. The study also aimed to address the environmental challenges of hot and dry climates, particularly water scarcity, by optimising the screen's geometry and material selection to enhance cooling efficiency while minimising water consumption. To achieve these aims, the study set out to:

- Investigate and evaluate the environmental principles embedded in traditional Mashrabiya design, focusing on its geometric, material, and functional aspects as a foundation for developing a contemporary screen system.
- Design and develop ceramic prototypes that function both as wet media for direct evaporative cooling and as practical solar shading elements.
- Optimise the screen's performance by integrating mutual shading strategies with evaporative cooling to enhance thermal comfort while minimising water usage.

- Evaluate the environmental performance of the proposed screen through analytical testing and full-scale field application.

By doing so, this study aspires to contribute not only to improved environmental building performance, but also to the revival of culturally significant design approaches that respond meaningfully to both climate and place.

a) Unfolding the Principles of the Mashrabiya

The Mashrabiya is a complex architectural element that has evolved over time within Islamic architecture. It has undergone numerous changes in form and screen configurations (الزهراني, 2007). The literature review and the various resources discussed in chapters 2 and 3 provided valuable insights into the different types of Mashrabiya. Among these, Maymooni screens were identified as the most complex in terms of design, fabrication, and assembly. The intricate details of Maymooni screens offered significant insights into how the screen performs at a small scale.

b) The mutual shading and shading Performance Analysis

By zooming in on the details of Maymooni screens and understanding how the small, crafted elements impacted performance, it was found that these fine details influence not only the shading projected into the indoor space, but also the shading phenomenon that occurs on the screen surface itself, resulting in mutual shading. Analysing the Maymooni screen in Chapter 7 provided the main seed for the proposed ceramic screen. Through analysing the Mashrabiya screens using different analytical tools to assess shading performance through solar radiation analysis, it was found that it is possible to design a screen with simple extrusions capable of casting shadows onto itself. This design can achieve up to 61% self-shading, compared to a regular design without self-shading geometries. This means that the screen can shade itself without the need for an additional shading device to control solar accessibility to the system.

c) The Aerodynamic Behaviour and Micro-Scale Performance Analysis

The idea of incorporating shading devices as part of a ventilation strategy or an evaporative cooling system raises an issue of increased drag, which ultimately affects the aerodynamic performance around the screen. Accordingly, CFD analysis was conducted to examine the aerodynamic behaviour of the Maymooni screen. Various forms of the Maymooni screen were tested under fixed conditions to assess the impact of geometry on airflow, and the results provided insight into how air behaviour changes around these different configurations.

The analytical studies of the aerodynamics surrounding the screens revealed unique challenges in designing a perforated screen for effective passive cooling. At the micro-level, the aerodynamic behaviour around the small components of the screen was found to significantly influence the overall performance. This relationship was demonstrated through the calculation of the discharge coefficient for both the entire screen and its individual components. The results indicated a correlation resembling the butterfly effect, where minor changes in the screen's small parts contributed to broader outcomes in aerodynamics and shading performance.

The distinct shapes of these small components led to notable variations in performance across different cases. Aerodynamic testing and CFD simulations showed that small changes in cross-sectional geometry—such as the difference in drag between the NASA23015 aerofoil (drag coefficient of 0.07) and the circular cross-section (drag coefficient of 0.43)—were initially significant. However, when applied to the star-shaped component design, the drag results for both shapes became relatively close, demonstrating that subtle adjustments can substantially enhance overall drag efficiency. This underscores how micro-level refinements can optimise the system's aerodynamic behaviour, reinforcing the notion that iterative design has a meaningful impact on overall performance.

d) Translating Environmental Insights into Pattern Development

Analysing the different shapes and forms of the Maymooni screen provided valuable insights into what could be integrated into the new design of the proposed screen and what should be avoided, in order to develop a form that performs efficiently and achieves a balance between shading and ventilation—thereby facilitating the desired behaviour in evaporative cooling. In Chapter 8, the Mashrabiya served as a guiding reference in terms of environmental performance and as a foundation for developing a screen rooted in Islamic patterns. Understanding how Maymooni screens were designed and constructed shaped the methodology that led to the formulation of the base pattern. This base pattern was then deconstructed into layers to create a coherent composition that could serve both functional and conceptual purposes.

e) From Conceptual Inspiration to Modular Environmentally Efficient Screen Design

This dual action—cooling and shading—was profoundly inspiring for this study. The Mashrabiya pioneered an early vision of thermal comfort that relied not on mechanical isolation, but on adaptation and harmony with the surrounding environment. It introduced a fundamentally different dimension to the understanding of comfort—one that goes beyond merely blocking sunlight to actively transforming air quality.

Having valuable inputs from analysing the Maymooni screens of the Mashrabiya, the design of the proposed screen was structured to optimise environmental performance, while integrating knowledge derived from Islamic patterns. This resulted in an efficient approach to developing a multifunctional Islamic ceramic screen. The design evolved from a base pattern created by blending various Islamic screens, producing a generic Islamic motif that can be culturally adaptable across different regions due to its roots in multiple traditional typologies. The base pattern was deliberately kept simple and less complex, relying solely on two components—a star and a hexagon—merged and repeated modularly on a grid to form the final composition. This clear and minimal structure reflects one of the foundational layers common to a wide range of Islamic patterns, often serving as the underlying framework upon which more intricate designs are built.

f) Craft Knowledge Transfer and Ceramic Fabrication Process

The concept of working with craftsmen who understood the manufacturing aspects of the wooden Maymooni screens helped document many inherited details passed down through generations, overcome several limitations in analysing the screens, and better understand their design aspects—making Chapters 6 and 7 a key contribution to the study outcome in documenting both the screen details and the small crafted features in their wooden components. This same approach was followed during the fabrication of the proposed screen, where ceramic was chosen as the primary material. With knowledge gained from years of experience, clay artisans knew what to avoid in the design to prevent fabrication difficulties and make the screen more feasible to produce in any ceramic workshop using traditional fabrication methods and available slip-casting tools.

g) Design Flexibility and Architectural Adaptability

The ceramic cooling screen developed in this study was not merely a technical exercise, as explained in Chapter 8. It served as a means of exploring how far traditional architectural elements could be reinterpreted without losing the meaning and spirit they embody. The screen draws inspiration from the Mashrabiya—a traditional element that offers mutual shading strategies and a flexible structure. This reference to the Mashrabiya, which involves designing traditional screens, enables the new ceramic components to be modular and assembled in various patterns and densities, depending on environmental needs or architectural intentions. This flexibility allows the screen to adapt easily to different designs, be constructed and maintained without complex systems, and deliver varying performance depending on its arrangement.

h) Integrating Tradition and Innovation through Design Methodology

The study developed a methodology that combines cultural memory, material experimentation, and environmental testing at every stage to create a system responsive to both place and climate. It demonstrates that traditional Islamic screens need not be confined to museums; they can be reimagined, technically challenged, and revitalised through careful, evidence-based design. In this way, the real contribution of this study lies not only in the cooling screen itself, but also in the process of learning how to design in a manner that honours both the past and the future simultaneously.

I) Empirical Evaluation of the Proposed Screen in Real Climatic Conditions

The full-scale fieldwork was conducted during the hottest period of the year, as detailed in Chapter 9, to test the ceramic evaporative cooling screen under real environmental conditions. Installed on a west-facing façade in Riyadh, Saudi Arabia, the screen's 1:1 application was evaluated for its performance in evaporative cooling, shading, and airflow modification. Data loggers, thermal imaging, and air velocity measurements were employed to monitor the environmental changes generated by the screen over time.

The fieldwork revealed consistent cooling behind the screen, with air temperatures dropping by up to 8°C during peak hours, while relative humidity remained within a comfortable range. These findings confirmed that the screen functioned effectively as a shading device while supporting evaporative cooling. The results demonstrated that a single material system, when carefully designed, could achieve both functions without the need for mechanical assistance.

Beyond performance testing, the fieldwork revealed important nuances in how the screen interacts with natural elements—how the ceramic material absorbs and releases water, how porosity and pattern influence airflow, and how surface temperature responds to solar exposure. These findings highlighted the screen's ability to generate an indoor microclimate, reducing internal temperatures while maintaining both visual and cultural permeability. The testing also uncovered practical issues such as water distribution, uneven drying, and the influence of ambient wind—factors that were not apparent during the simulation stage. This layer of empirical work grounded the study, transitioning the screen from a conceptual prototype into a performative architectural element.

The study demonstrated that, when designed to expose a greater surface area, porous ceramic screens can effectively control solar exposure, facilitate airflow interaction, and absorb water efficiently—leading to significant improvements in thermal comfort in hot, dry environments. The screen's performance varied according to airflow velocity, aperture size, and ceramic saturation levels. These variables, monitored through Testo data loggers and FLIR thermal imaging, confirmed that the

orientation and geometry of the screen are critical factors influencing both cooling potential and shading effectiveness.

k) Comparative Performance and Adaptive Potential

Testing the proposed screen alongside an existing system like the Ecooler was never intended to prove superiority; rather, it aimed to understand the differences. While the proposed screen does consume more water, it also carries a significantly larger cooling load. It performs multiple functions—providing shading, sensible and latent cooling, and generating substantial temperature drops even under extreme conditions such as those in Riyadh. The study demonstrated that, through simple design adjustments like the extruded cap, it is possible to conserve considerable amounts of water during critical periods.

One concern from the outset was whether the complex geometry would interfere with airflow. The results indicate that it does not. While there is a slight increase in drag, air continues to flow through, and that movement is somewhat more dynamic—likely enhancing the cooling process rather than hindering it. In this sense, the screen is not merely passive but subtly reactive.

Compared to similar approaches in passive evaporative cooling screens, such as Ecooler, this study adopts a more adaptive strategy by integrating a modular screen into existing architectural apertures. The system demonstrated comparable evaporative trends and surface temperature dynamics, despite its smaller scale.

l) Passive Material Intelligence and Sustainable Cooling

The study underscores the importance of passive material intelligence in bio-integrated design. Rather than relying on mechanical intervention, the screen draws upon vernacular practices such as the Mashrabiya to establish an environmentally attuned architectural element. The ceramic material functions not only as a thermal mass and evaporative membrane, but also regulates solar accessibility, enhancing both spatial quality and privacy.

Notably, the findings suggest that effective passive cooling can be achieved even with relatively modest water consumption (17.2 l/day), addressing one of the primary critiques of evaporative technologies in arid regions.

Final Reflections

This study was never just solely about proposing a new evaporative cooling screen. It was a pioneering effort to establish a process that merges the profound environmental wisdom embedded in Islamic

architecture with the tools and evidence-based methods available today—sparking curiosity and interest through its innovative approach.

Every stage of the project, from early design thinking to testing and validation, was grounded in an approach that sought to respect and reinterpret traditional knowledge through real environmental performance testing. The final screen is the result of this dual commitment to cultural identity and environmental responsiveness. It draws directly from the Mashrabiya—not only by using mutual shading as a passive cooling strategy, but also by adopting the Mashrabiya's flexible structure, where individual units can create different patterns, densities, and performances depending on how they are arranged. The modularity of the design enables it to be easily built, adapted, and maintained, allowing it to respond to various architectural needs while preserving its core environmental function, thereby ensuring a sustainable future. In this sense, the contribution of the study lies as much in the methodology developed to shape the screen as in its physical form and in how it is designed to continue evolving with both tradition and climate in mind.

10.2 Contributions to Knowledge

This study contributes to the evolution of evaporative cooling screens as a façade component in the following ways:

- It demonstrates the feasibility of integrating porous ceramic evaporative screens into contemporary facades, specifically as shading devices for west-facing openings, while addressing the cultural and environmental aspects rooted in Mashrabiya.
- It provides field-tested data on thermal performance, bridging the gap between experimental laboratory studies and real-world architectural applications.
- It advances material understanding by linking micro-level properties (such as porosity and surface area) with macro-scale environmental effects (such as temperature reduction and shading efficacy).
- It frames ceramic cooling systems within a broader cultural and aesthetic context, reinforcing their responsiveness to heritage and climatic relevance.

In the end, a renewed engagement with the knowledge embedded in traditional architecture offers valuable insights into contemporary design challenges. Vernacular models often contain solutions that are more contextually attuned and ecologically responsive than those still being pursued by modern technology. Revisiting these architectural legacies is not merely a nostalgic gesture, but a critical act of reclaiming accumulated wisdom that can inform a more balanced and sustainable future

10.3 Limitations

Several limitations constrained the scope of this study. Field testing was conducted over a limited period during the peak summer month, focusing on a single façade orientation. While this ensured the screen was assessed under optimal evaporative conditions, it did not allow for a multi-seasonal evaluation, leaving many unknowns about its performance across complete seasonal cycles. The screen was also tested under fixed façade geometry and airflow conditions, which may not fully represent the variability found in real-world architectural contexts. Other limitations relate to material ageing, fouling, and long-term exposure, which were beyond the scope of this study. Water delivery was managed manually or through controlled irrigation, and future adaptations could benefit from automation or greywater integration to support scalability.

In addition to these known constraints, several other limitations emerged during the fieldwork itself. Highlighting how these issues were encountered and resolved is essential, as it offers valuable guidance for future development and helps similar projects navigate comparable challenges. The following section distinguishes between pre-existing limitations—primarily related to airflow dynamics and data collection strategies—and emerging ones, such as hard water mineral deposition, which affected performance mid-testing. The resolution of these issues contributes to improving the current prototype and enriches the broader conversation around maintaining the functionality of low-tech passive cooling systems under real environmental conditions.

- Limitations in the Fieldwork

The chosen site and indoor space for testing the screen under realistic conditions introduced several limitations. Some were anticipated, and pre-analytical approaches were taken to overcome them, while others emerged during the testing process.

- Pre-existing Limitations:

The site featured a single-sided ventilated area with an opening on the west-facing façade and no additional openings, which limited airflow and cooling efficiency. In single-sided natural ventilation, airflow depends on thermal drafts and wind currents, which influence the volume of fresh air entering the screen and reduce its cooling capacity. The temperature difference between indoor and outdoor environments affects airflow consistency; when outdoor temperatures drop at night, the reduced temperature difference leads to diminished airflow. Stagnation occurred as air moved downward around the screen and recirculated, decreasing evaporation and cooling efficiency. Additionally, the

effective aperture area of the window was reduced, further impacting the flow rate and cooling performance.

- Emerging Limitations:

A significant issue emerged with the use of hard water. On July 17th, a noticeable drop in performance was observed, as surface temperature monitoring, humidity levels, and water consumption all showed a decrease. The screen began dripping water from its connection points, indicating reduced evaporation. The problem was identified as mineral deposition caused by hard water, which clogged the internal pores of the screen and obstructed the evaporation process. The screen was disassembled, soaked in a vinegar and baking soda solution, and cleaned with high-pressure water to remove the deposits. After cleaning, it was reconnected to the water supply, and the issue was resolved.

10.4 Future Directions

While this study provides valuable insights, it also opens several avenues for deeper exploration:

- Seasonal and Orientation-Based Testing: Expanding performance evaluations across different climates and building orientations to capture temporal and directional variability.
- Passive Water Systems: Investigating lower-maintenance, passive water supply solutions—such as humidity harvesting or condensation-based systems—to enhance sustainability and reduce operational complexity.
- Dynamic CFD Modelling: Transitioning from static simulations to dynamic, time-sensitive CFD analyses that reflect real-world fluctuations throughout the day.
- Material Innovation: Exploring alternative ceramic compositions, including bio-based or low-fire options, to minimise embodied energy and environmental impact.
- Structural Integration: Assessing the feasibility of incorporating the screen into full building façades, with attention to wind and seismic resilience for architectural applications.

Ultimately, this study proposes that environmental performance and cultural meaning are not mutually exclusive—they can coexist within a single object, a unified system. The ceramic cooling screen represents one manifestation of this synthesis, but it is far from the only possibility. Countless other iterations remain to be imagined and realised. Future research should extend testing across diverse climatic zones and longer timeframes to assess year-round viability. Additionally, hybrid systems that integrate ceramic with bio-based materials—such as mycelium composites or alginate membranes

offer promising avenues for developing multifunctional architectural elements that combine ecological efficiency with symbolic depth.

Further refinement of design guidelines could be achieved through computational modelling of airflow across more complex interior layouts, allowing for greater responsiveness to spatial variability. Additionally, collaboration with local craftsmen and ceramicists offers an opportunity to embed cultural specificity into the system, yielding site-responsive adaptations that honour regional traditions while advancing environmental performance. These directions underscore the potential for architectural innovation that is both technically rigorous and deeply rooted in place.

"And maybe, if we look closely enough at what traditional architecture still knows, we'll find better answers to the questions that modern technology is still trying to solve." (Rofayda Salem,2021)

11. Journal

Documenting all the experimentation that has been carried during the research

11. Design Journal

Documenting all the experimentation that has been carried during the research

01. The Objectives

This journal is designed to provide valuable insights into the background and evolution of the research process, offering guidance for fellow researchers interested in this domain to help understand the successful trials and failures behind each stage and offer valuable insights regarding how the process was developed.

02. The Aims

This journal aims to document all the experimentations that were carried out in the research serving as a repository of all experimentations conducted throughout the research journey.

03. Documentation of the process

Design development

The design development of the screen was a process that contributed a lot to the research; it was the outcome of a long analytical journey that the research carried out. A hybrid approach was adopted, merging qualitative methods—such as interviews with artisans to capture traditional knowledge—with quantitative environmental analysis. Detailed examinations of Mashrabiya design details and laboratory scanning of Maymooni screens informed the design process. The ceramic evaporative cooling screen is fabricated using a traditional slip-casting method. Extensive testing ensured the chosen stoneware clay achieved the necessary porosity for efficient performance. Each screen component, measuring 28 x 28 cm, is designed to be easily installed on-site and connected to the main water supply through integrated cavities and channels, reducing water waste.

01 Understanding the fundamentals of Mashrabiya

Comes through testing samples and documenting which is also a useful approach to document the screens.

02 Developing the proposed screen

The development of the screen was a process that required parallel working with the material and fabrication methods to ensure the design applicability within the chosen material

a b c d

Finalizing the material

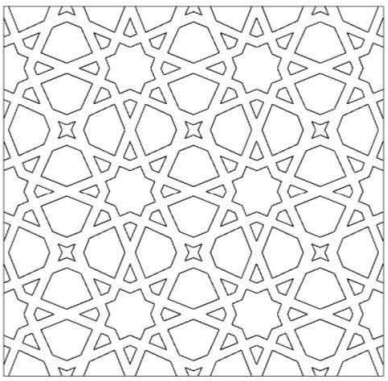
Testing the different colors of wood

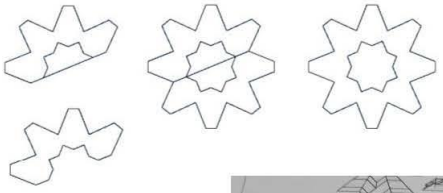
What color of clay ?

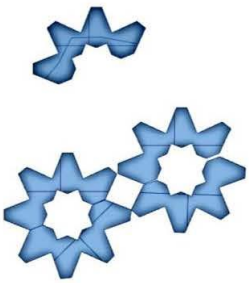
Accordingly, different colors of clay were used to fabricate the same geometry and were exposed to the same amount of solar incidence.

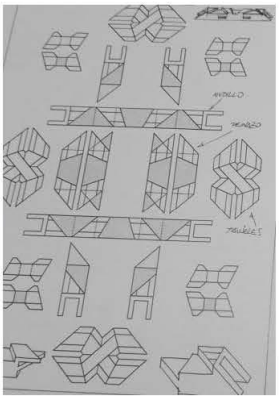
The Objective

Designing a pattern that is used for shading and evaporative cooling can be very tricky in terms of materiality. Although it is clear that clay will be used, clay comes with different colours and tones that might affect the probability of reflecting some of the solar incident rays within the screen itself.

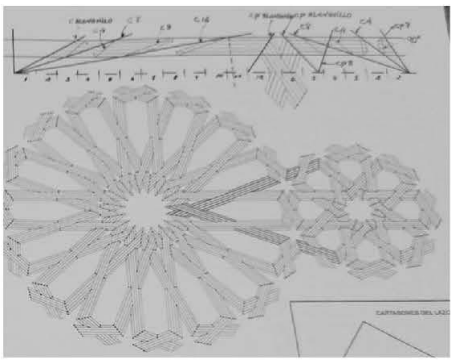








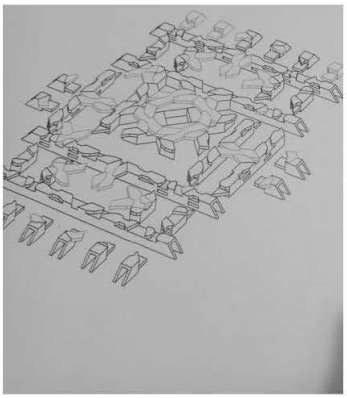
Source: Author



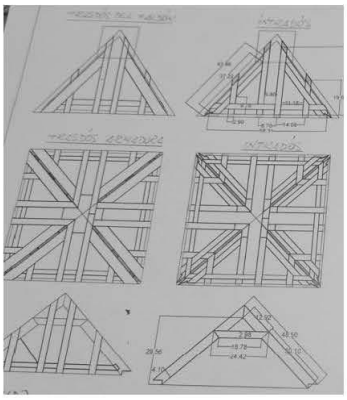
Source: Author

Developing a component from famous pattern

A component can be developed from a common Islamic pattern, but it won't be culturally responsive to all areas of application and the design will mimic the originality of the Islamic pattern



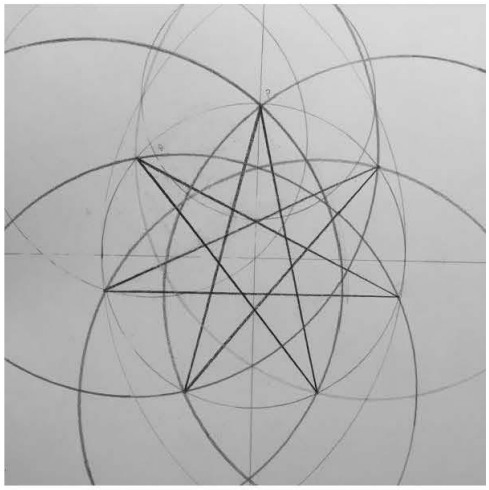
Source: Author

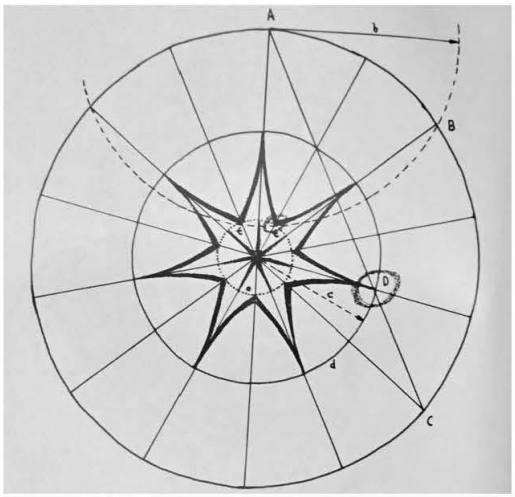


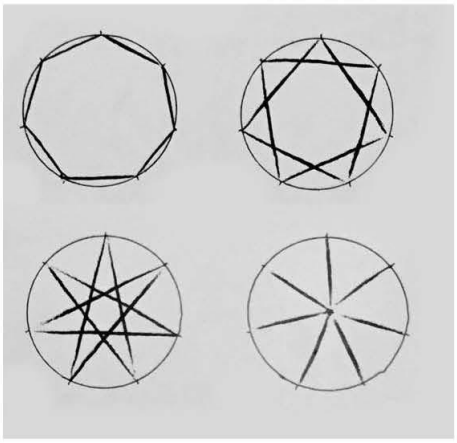
Source: Author

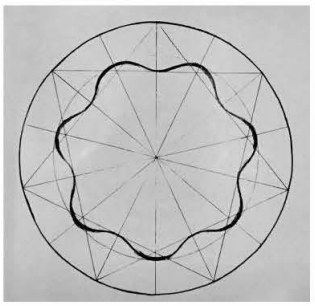
Studying the patterns.

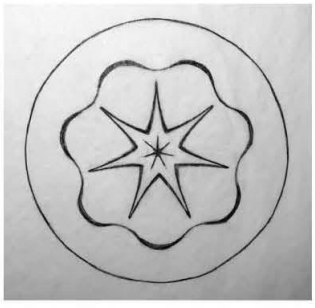
Understanding and defining the components that formalize each pattern.

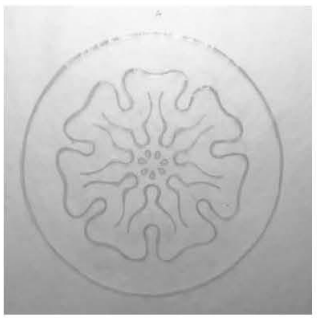


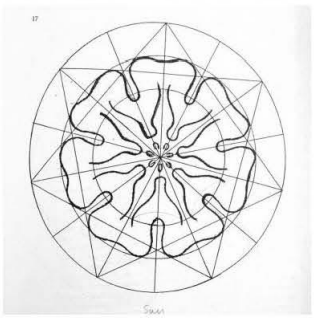












Geometrical sketches that were made to understand the pattern, and how the star shape can contribute to many forms found in Islamic screens. – Source:Author

Design development

Developing the organic forms that can come from the star component

Organic form with different levels of complexity

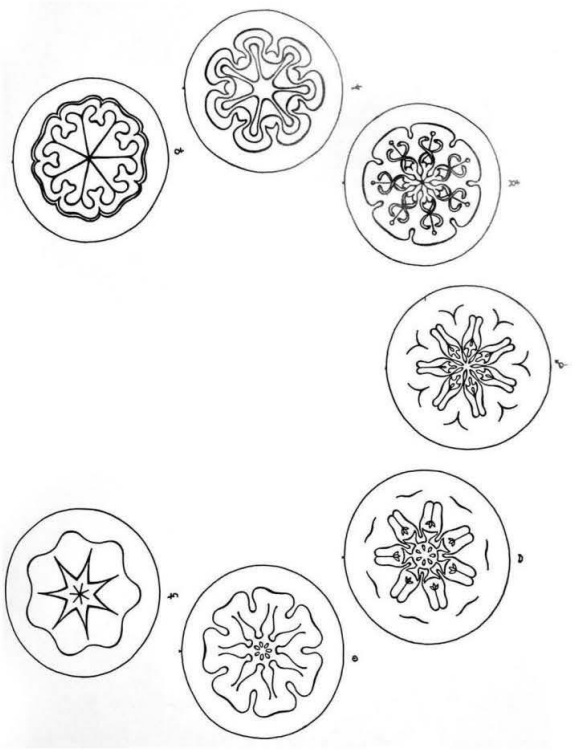
All shapes which how air, light, and shadow interact with the screen. The transitions between sketches show a process of evolving complexity: moving from basic symmetrical forms (like sketch a or b) toward more intricate and dynamic arrangements (like sketch f or g), which seem to create richer opportunities for controlled turbulence and diffused light penetration. The geometry in sketches like c and d introduces layered or nested elements, which could enhance surface area and therefore cooling potential. Meanwhile, designs like e and h appear more sculptural, evoking movement and possibly encouraging more active air engagement with the surface.

01 **Explorations of Geometry and Airflow**

These hand-drawn sketches represent an ea
and essential phase of design development
the ceramic cooling screen. The drawir
explore a variety of radial and rotatio
geometries, each rooted in traditional Islar
pattern logic while also responding
performance-driven needs — such as airfl
direction, surface exposure, and mut
shading.

02 **number of petals or arms, openness of form, and central density**

These hand-drawn sketches represent an early and essential phase of design development for the ceramic cooling screen. The drawings explore a variety of radial and rotational geometries, each rooted in traditional Islamic pattern logic while also responding to performance-driven needs — such as airflow direction, surface exposure, and mutual shading.

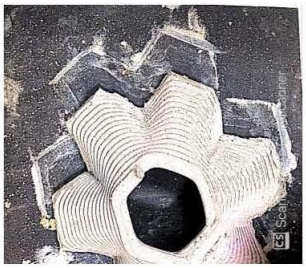


Source: Author

Experimenting with robotic ceramic fabrication

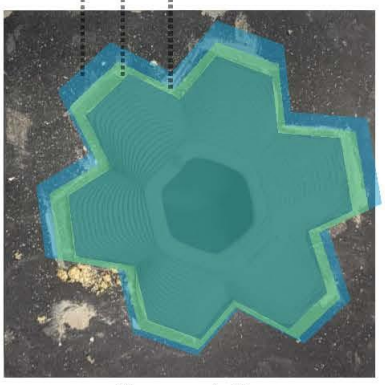
It was found the 3d robotic ceramic fabricator is a good new approach but also comes with limitations, not all the desired geometries can be made with this method, you can only formulate your design in RHINO through Loft command (only through curves).

Yet the process was helpful since the used clay was very liquid to avid the blockage of the nozzle making it close to the properties of the clay used in slip casting, it was a nice experiment to test the impact of the shrinkage factor on the geometry dismissions so this can be acknowledged latter in our modeling stage for slip casting.



Source: Author

Source: Author



Source: Author

Fabricating with Robotics

Avoiding complex geometries

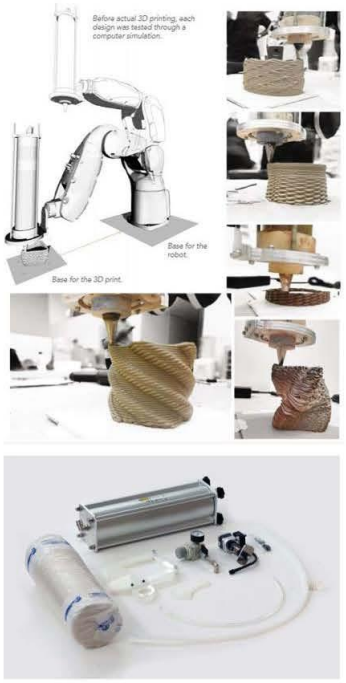
Very liquid ceramic

Construction support

Easy experimentation

If the pressure variation is documented, you can easily share a guide to ensure accuracy in mass production

Source: Author



Source: https://www.ibrahimibrahim.works/work/in-form-ed-clay

Testing initial geometries with similar clay using hand.

Source: Author

Testing different geometries with clay

Approaching the most suitable clay for initial modeling with hands .

Once the clay is chosen the draft modeling starts

The screen is printed in a 1:1 scale, and the chosen clay is applied on top of it with a transparent plastic film between them to avoid the clay sticking to the printed draft.

Source: Author

Source: Author

Source: Author

Source: Author

Source: Author

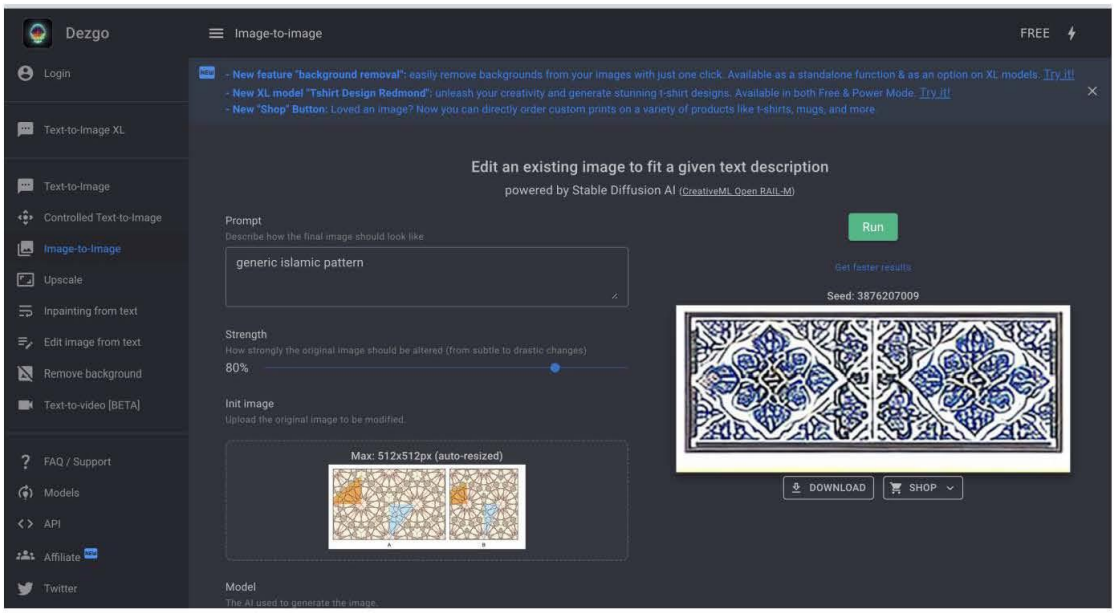
Proposed pattern design

Selecting all the documented Islamic screens

For defining the common feature between all these screen, the documented Islamic screens was collected in a JPEG format and the quality of each image was improves to 2000px

AI feature detection

After the screens was uploaded in the AI storage memory , an initial test was carried to say, " how would you image a ceramic element that is driven and relates to the screen that was uploaded" and the expected answer that came in sort of image was as expected – A STAR.



Source: Author & Midjourney

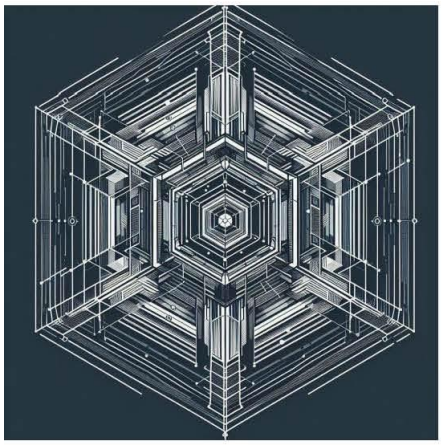
Source: Author & Midjourney

AI feature detection

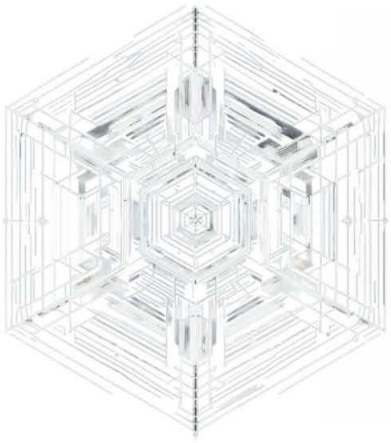
After the screens was uploaded in the AI storage memory , an initial test was carried to say, " how would you image a ceramic element that is driven and relates to the screen that was uploaded" and the expected answer that came in sort of image was as expected – A STAR.

More AI imagination

More testing into how the Islamic screens that uploaded can contribute to modern shapes and patterns that still has the same origin.

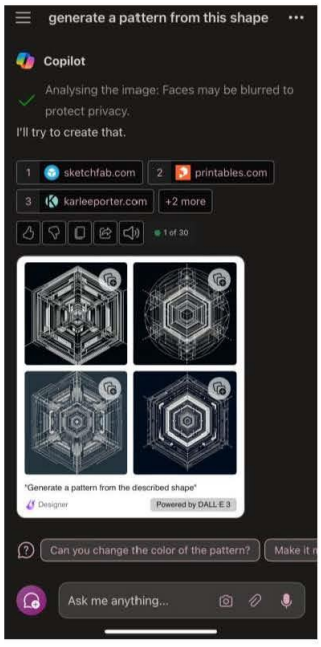


Source: Author &Copilot

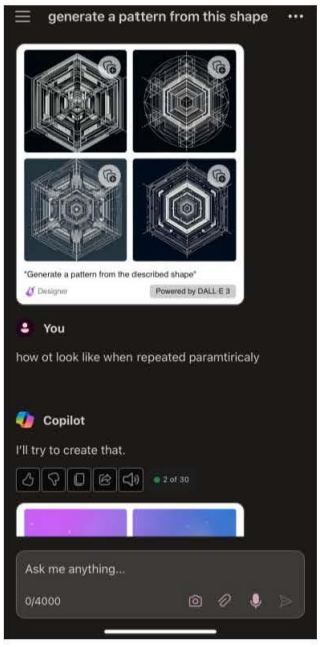


Source: Author &Copilot

Source: Author & Midjourney



Source: Author &Copilot



Source: Author &Copilot

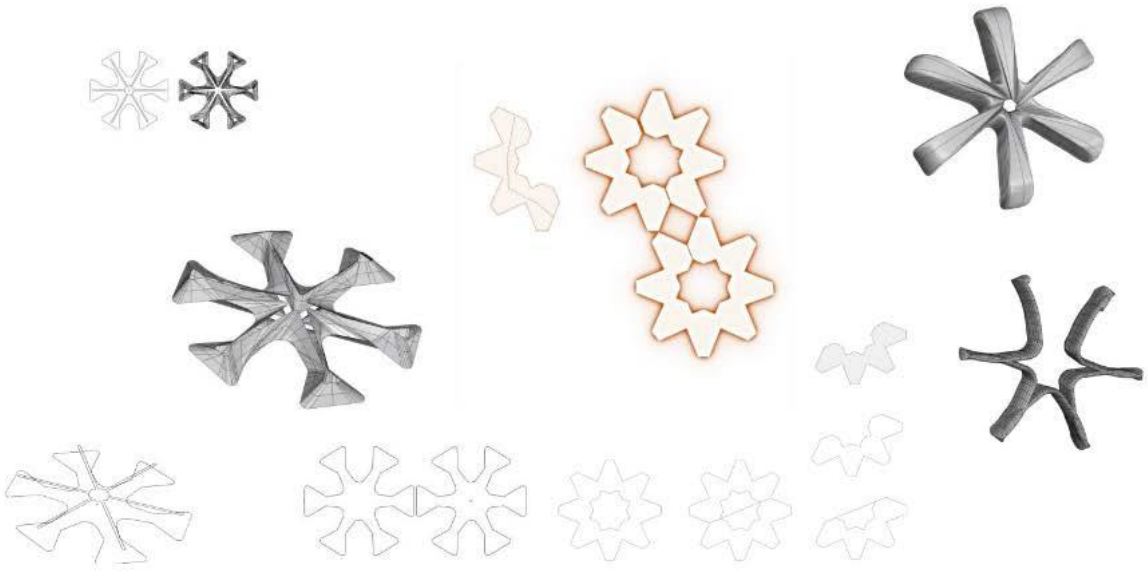


Figure - Documentation of the early stages for the modelling of the star geometry in Rhino - Source: Author

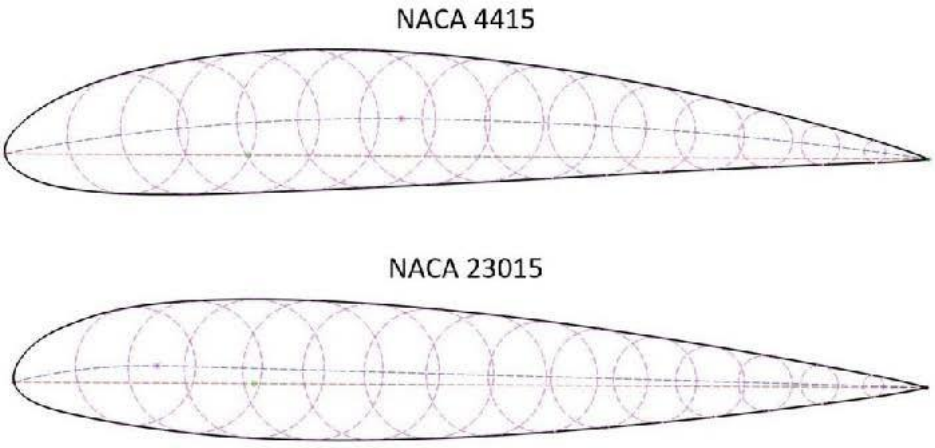
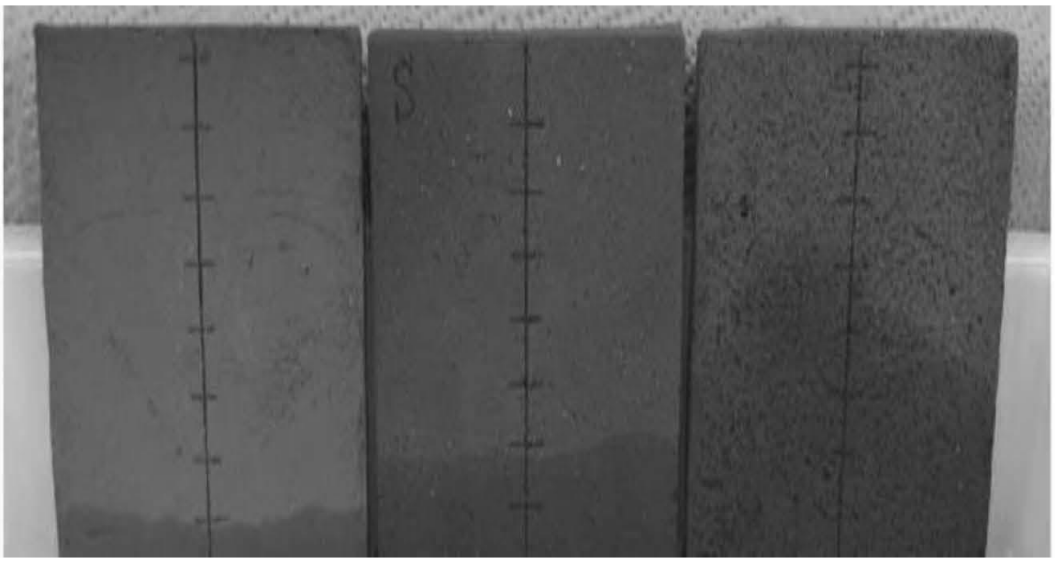


Figure – studding the best aerodynamic efficient cross sections - Source: Author

- **Water Retention:**

The clay should retain enough water to maintain a continuous evaporation process without excessive leakage.

- **Water absorption and porosity:**



What Porosity: Source: Author

Identify Key Parameters:

- **Water Retention:**

The clay should retain enough water to maintain a continuous evaporation process without excessive leakage.

- **Evaporation Rate:**

The rate at which water evaporates from the clay's surface.

- **Water Pressure:**

The pressure at which water is supplied to the clay screen.

Clay testing

Different clay tastings that were experimented by manipulating the time needed for drying and the firing temperature and time spent in the oven.

Test 1

Dry shrinkage in %:	3		
Firing temperature:	1070 °C	1140 °C	1240 °C
Firing colour:	white	light	light
Firing shrinkage %:	2	4	6
Water absorption %:	13	5	1
Coefficient of Thermal Expansion (CTE):			
20 °C - 400 °C	59	64	62
20° C - 500 °C	60	67	65
20 °C - 600 °C	73	80	74
CTE * 10^{-7}/K			
Moisture expansion in%:	0.185	0.08	

Test 2

Dry shrinkage in %:	3		
Firing temperature:	1070 °C	1140 °C	1240 °C
Firing colour:	white	light	light
Firing shrinkage %:	2	4	6
Water absorption %:	13	8	1
Coefficient of Thermal Expansion (CTE):			
20 °C - 400 °C	59	64	62
20° C - 500 °C	60	67	65
20 °C - 600 °C	73	80	74
CTE * 10^{-7}/K			
Moisture expansion in%:	0.185	0.08	

Test 5

Dry shrinkage in %:	3		
Firing temperature:	1070 °C	1140 °C	1240 °C
Firing colour:	white	light	light
Firing shrinkage %:	2	4	6
Water absorption %:	13	8	1
Coefficient of Thermal Expansion (CTE):			
20 °C - 400 °C	59	64	62
20° C - 500 °C	60	67	65
20 °C - 600 °C	73	80	74
CTE * 10^{-7}/K			
Moisture expansion in%:	0.185	0.08	

Test 6

Dry shrinkage in %:	3		
Firing temperature:	1070 °C	1140 °C	1240 °C
Firing colour:	white	light	light
Firing shrinkage %:	2	4	6
Water absorption %:	13	8	6
Coefficient of Thermal Expansion (CTE):			
20 °C - 400 °C	59	64	62
20° C - 500 °C	60	67	65
20 °C - 600 °C	73	80	74
CTE * 10^{-7}/K			
Moisture expansion in%:	0.185	0.08	

Test 9

Dry shrinkage in %:	3		
Firing temperature:	1070 °C	1140 °C	1240 °C
Firing colour:	white	light	light
Firing shrinkage %:	2	4	6
Water absorption %:	13	8	1
Coefficient of Thermal Expansion (CTE):			
20 °C - 400 °C	59	64	62
20° C - 500 °C	60	67	65
20 °C - 600 °C	73	80	74
CTE * 10^{-7}/K			
Moisture expansion in%:	0.185	0.08	

Test 10

Dry shrinkage in %:	3		
Firing temperature:	1070 °C	1140 °C	1240 °C
Firing colour:	white	light	light
Firing shrinkage %:	2	4	6
Water absorption %:	13	13	1
Coefficient of Thermal Expansion (CTE):			
20 °C - 400 °C	59	64	62
20° C - 500 °C	60	67	65
20 °C - 600 °C	73	80	74
CTE * 10^{-7}/K			
Moisture expansion in%:	0.185	0.08	

Body in powder form no. 209

Chemical analysis:

SiO_2	72.2 %
Al_2O_3	22.4 %
TiO_2	1.3 %
Fe_2O_3	0.9 %
CaO	0.0 %
MgO	0.3 %
K_2O	2.8 %
Na_2O	0.0 %

Ceramic data:

Dry shrinkage in %:	5	3	
Firing temperature:	1070 °C	1140 °C	1240 °C
Firing colour:	white	light	light
Firing shrinkage %:	2	4	6
Water absorption %:	13	1 2	1
Coefficient of Thermal Expansion (CTE):			
20 °C - 400 °C	59	64	62
20° C - 500 °C	60	67	65
20 °C - 600 °C	73	80	74
CTE * 10^{-7}/K			
Moisture expansion in%:	0.185	0.08	

Test 3

Dry shrinkage in %:	3		
Firing temperature:	1070 °C	1140 °C	1240 °C
Firing colour:	white	light	light
Firing shrinkage %:	2	4	6
Water absorption %:	13	7	1
Coefficient of Thermal Expansion (CTE):			
20 °C - 400 °C	59	64	62
20° C - 500 °C	60	67	65
20 °C - 600 °C	73	80	74
CTE * 10^{-7}/K			
Moisture expansion in%:	0.185	0.08	

Test 4

Dry shrinkage in %:	3		
Firing temperature:	1070 °C	1140 °C	1240 °C
Firing colour:	white	light	light
Firing shrinkage %:	2	4	6
Water absorption %:	13	11	1
Coefficient of Thermal Expansion (CTE):			
20 °C - 400 °C	59	64	62
20° C - 500 °C	60	67	65
20 °C - 600 °C	73	80	74
CTE * 10^{-7}/K			
Moisture expansion in%:	0.185	0.08	

Test 7

Dry shrinkage in %:	3		
Firing temperature:	1070 °C	1140 °C	1240 °C
Firing colour:	white	light	light
Firing shrinkage %:	2	4	6
Water absorption %:	13	8	5
Coefficient of Thermal Expansion (CTE):			
20 °C - 400 °C	59	64	62
20° C - 500 °C	60	67	65
20 °C - 600 °C	73	80	74
CTE * 10^{-7}/K			
Moisture expansion in%:	0.185	0.08	

Test 8

Dry shrinkage in %:	3		
Firing temperature:	1070 °C	1140 °C	1240 °C
Firing colour:	white	light	light
Firing shrinkage %:	2	4	6
Water absorption %:	13	8	8
Coefficient of Thermal Expansion (CTE):			
20 °C - 400 °C	59	64	62
20° C - 500 °C	60	67	65
20 °C - 600 °C	73	80	74
CTE * 10^{-7}/K			
Moisture expansion in%:	0.185	0.08	

Here are my 10minute sample porosity test results on stoneware casting slip

I will do a soak for 48 hours and test again on Saturday with an added 1050°c sample

All fired at 150 per hour

This is the formula I am using: P = ((wet - dry) / wet) * 100%

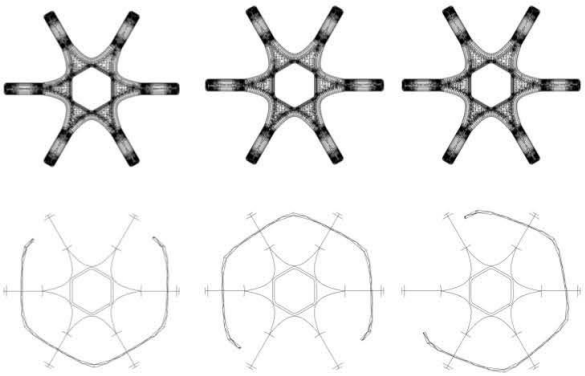
600°c (before quartz inversion, can be snapped easily, dark colour) = 11.32% absorption

800°c (about cone 017, just past quartz inversion I believe, cant be snapped easily, medium colour) = 11.88% absorption

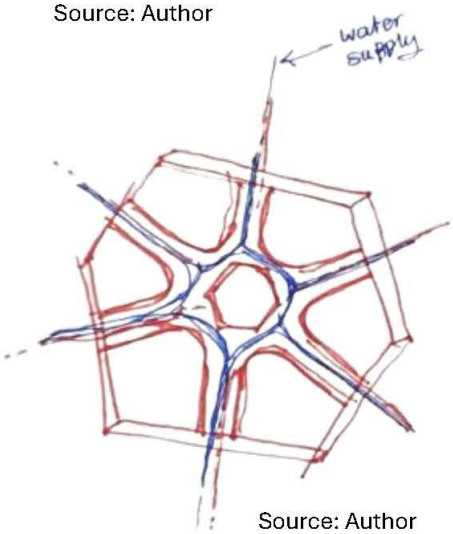
1000°c (about cone 09, more typical of bisque ware, light colour, more robust, most absorbent) = 12.54% absorption

These readings are interesting because generally the lower you fire the more absorption there is, but I theorise that because I've fired so low and so close to quartz inversion the pores in the clay haven't fully formed and in fact the clay is very brittle. So, please await the next set of tests and the 1050 test this weekend! I suspect the 1050 sample will go down in absorption.

The next step of the design was finding the possibility of using a small section of the hexagon and being able to rotate the form in response to the direct sunlight angles as explained in the following image. The objective is to check the practicability of this rotating element as an element of the methodical approach alongside conducting structural consequences examination and investigating how the element can be used functionally. However, upon execution, this technique raised concerns about its practical application at a 1:1 scale. A good operating system should have no impact on the scale of the service and the entire system's integrity. These problems underline that we should do the checking process via thorough analysis which will be done in the later part of this chapter.



Source: Author



Source: Author

Appendix fig – Documenting the early stages of the fabrication process of the sample and explaining to the craftsmen how the water cavities and channels should be Source: Author

Full-scale prototype options:

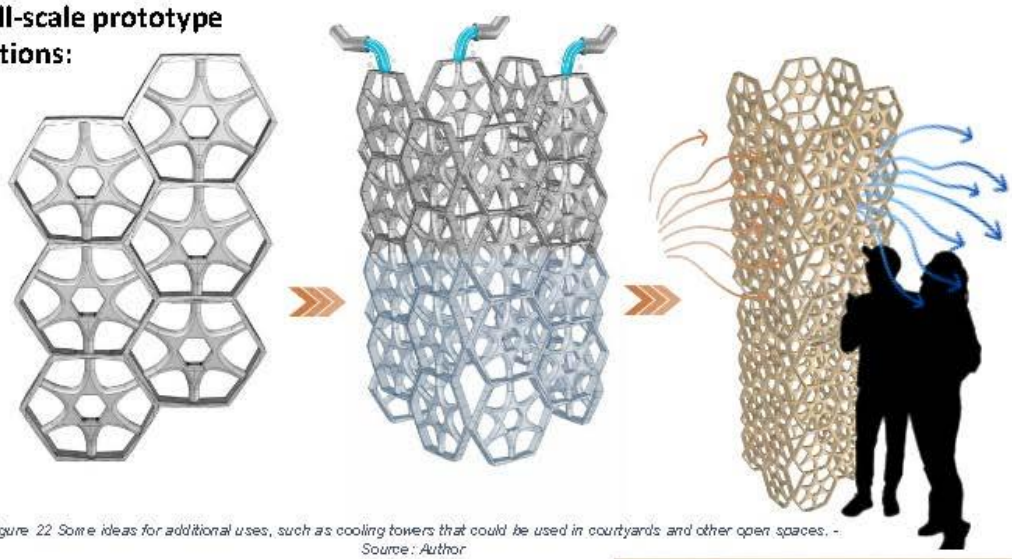


Figure 22 Some ideas for additional uses, such as cooling towers that could be used in courtyards and other open spaces. - Source: Author

- Wall.
- **Window**
- **Double façade**
- **Column**
- **Perforated Roof**
- **Cantilevered type**

The subsequent debates and analyses demonstrated a dedication to thorough design inspection to improve and optimise the suggested solution. Through such detailed examination, the design's feasibility and durability in real-world settings were thoroughly scrutinised, opening the path for informed decision-making and iterative design refinements.

The following image describes the experimenting process of drilling into the cork and the constraints encountered, which drove the process to iterate towards more flexible solutions in the manufacturing process.

Appendix fig – **the experimentation process of drilling into the cork and the constraints encountered, which drove the process to iterate towards more flexible solutions in the manufacturing process.**

Source: Author

Source: Author

Source: Author

Source: Author

Source: Author

Source: Author

Appendix fig – Documenting the fabrication process of the sample and how the craftsmen explored different approaches to control the thickness of the screen

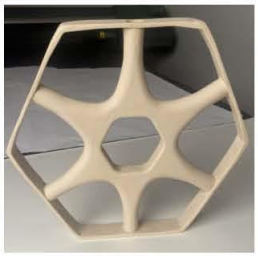
Experimenting with different connecting systems

Different experiments were caried using clay, plastic and heat guns, Cork and driller

Source: Author

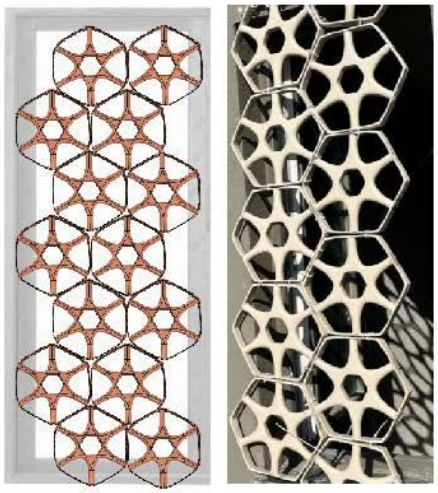
Source: Author

Source: Author



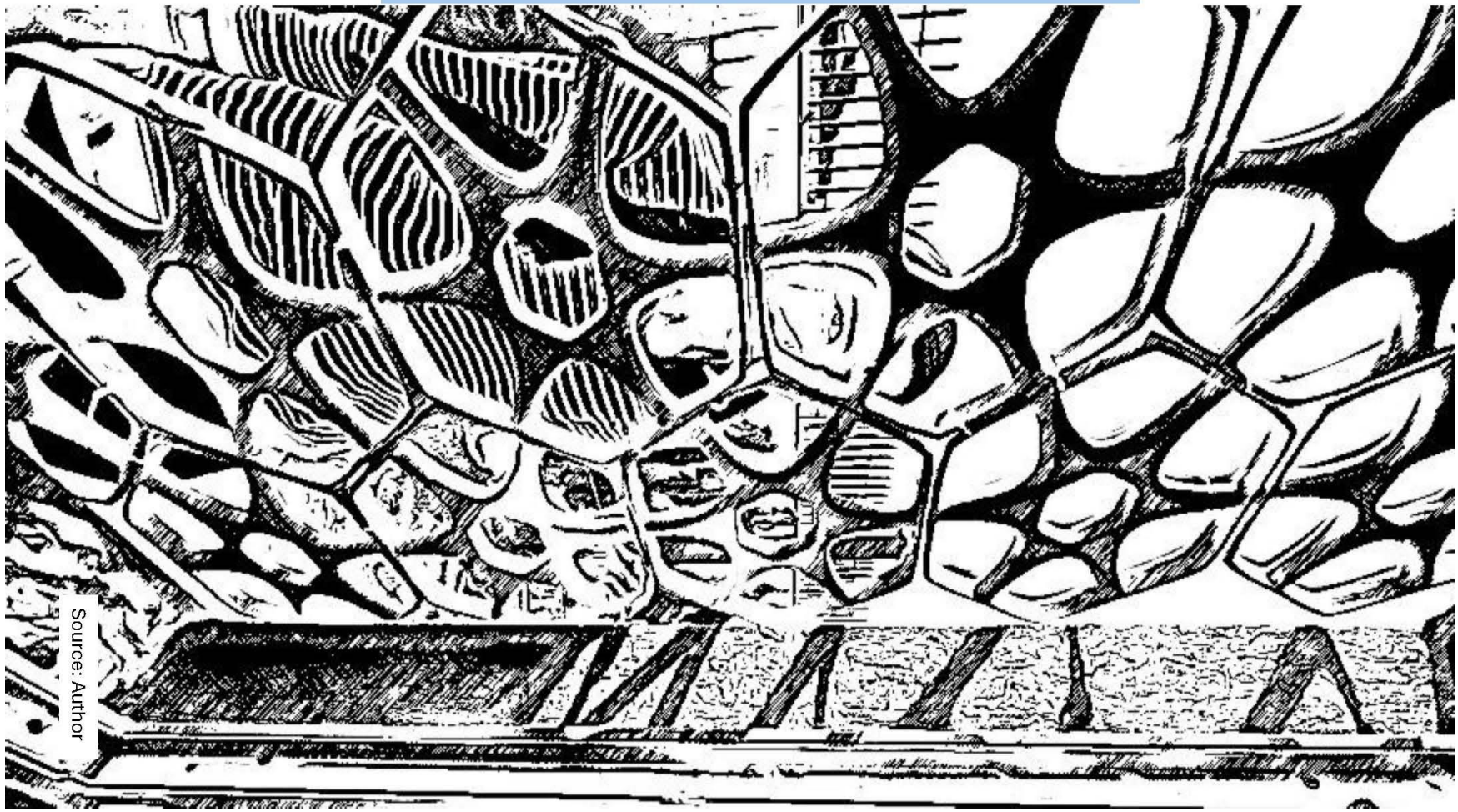
Appendix fig – experimenting with different connecting methods between the screen components, melted plastic was later excluded since it doesn't show applicability with extruded cap geometry that surrounded the stat geometry. Source: Author

Additionally, this design allowed for wider flexibility in how the components may be used in various structures rather than only on the apparatus; the graphic below offers some ideas for additional uses, such as cooling towers that could be used in courtyards and other open spaces.



Appendix fig – AI initial images that showed a realistic approach of how the screen will be applied

1:1 application of the screen



Source: Author

Water consumption

Calculate the Water Absorption:

•The amount of water absorbed by the sample is the difference between the wet and dry weights:

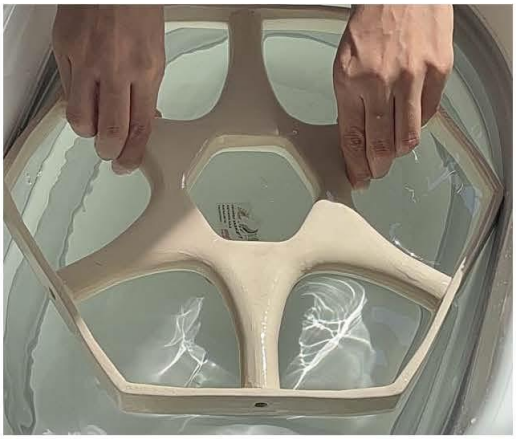
$$\text{Water Absorbed} = W_w - W_d$$

Calculate the Absorption Percentage:

•To express the water absorption capacity as a percentage of the dry weight, use this formula:

$$\text{Absorption Capacity (\%)} = \frac{W_w - W_d}{W_d} \times 100$$

Source: Author



Source: Author

Source: Author

Source: Author

Water consumption

Reservoir Method

•**What was followed**: A water reservoir (bucket, tank, or container) and a measuring stick or scale to check water levels.

•**Steps**:

- **Start with a Full System**: Begin by filling the ceramic cooling system completely, ensuring it's fully saturated with water.
- **Fill the Reservoir**: Fill the reservoir that supplies water to the ceramic cooling system to a known level (e.g., mark the initial water level).
- **Monitor Water Levels**: As the system operates, regularly check and record the water level in the reservoir (e.g., every hour or at the start and end of the day).
- **Calculate Usage**: Subtract the final water level from the initial water level to determine how much water was consumed.
- **Refill as Needed**: Track how much water you refill if the reservoir runs low.
- **Period of Operation**: Note the total time the system was running so you can calculate water consumption per hour/day.

Additionally, the semi transparent material in the water tank showed the clear level of water inside, so a scale was drawn on the tank to mark the water levels and a door camera was added to the tank enabling hourly monotiling of the reduction that happens in the water levels refereeing to the water consumed by the system during the evaporative cooling process (during the operation time)

Source: Author

Date	Time Period	Initial Water Level (L)	Final Water Level (L)	Water Consumed (L)	Refill Amount (L)	Total Consumption (L)
July 15	9:00–12:00	50	45	5	0	5
July 15	12:00–18:00	45	40	5	0	5

By the end of the monitoring period, It was possible to sum the total water used and track hourly consumption trends for one day of operating the system.
This manual monitoring approach is simple and effective, though it requires consistent attention to detail and regular checking.

The issue of Hard water:

•**What was followed**: the system was disassembled, and the screens was cleaned, and the tank was emptied (it only had 100 litters which was almost ¼ the tank capacity) and cleaned as well.

Source: Author

Source: Author

Rinse with fresh streaming water with high pressure 5000 PSI

Soak for 2 days

500 ml - white vinegar

300 grams Baking soda

Appendix fig documenting the hard water issue after the operation of the evaporative cooling screen

Source: Author

Hard water issue

Solutions to Hard Water

1.Water Softeners:

1. **Ion-Exchange Water Softeners**: The most common solution is to use ion-exchange water softeners, which replace calcium and magnesium ions with sodium or potassium ions, effectively softening the water.
2. **Reverse Osmosis (RO) Systems**: RO systems are effective at removing a wide range of impurities, including the minerals that cause water hardness. These systems are often used in homes and industries.

2.Desalination and Post-Treatment:

1. **Improved Desalination Processes**: Saudi Arabia is a world leader in desalination, but further improvement in desalination plants, along with post-treatment to remove residual minerals, can help reduce water hardness.

3.Precipitation Methods:

1. **Lime Softening**: Lime can be added to water to precipitate out calcium and magnesium, reducing hardness. This method is often used in large-scale water treatment plants.

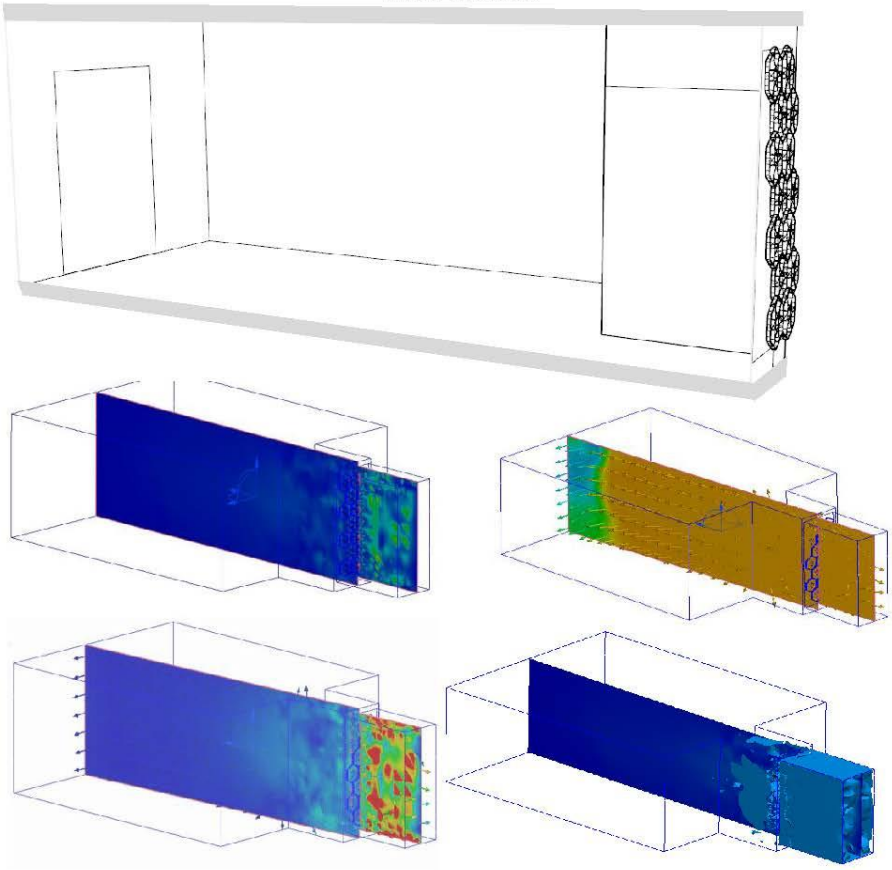
There are many substances that can dissolve limescale, including:
•Vinegar: An acidic solution that can dissolve limescale and other mineral deposits. You can soak the affected area in vinegar or spray a solution of equal parts vinegar and water on the area.
•Lemon juice: Can remove limescale deposits and freshen up machines.
•Baking soda: A household product that neutralizes acids and deep cleans.

CFD Spatial testing of the screen

Understanding virtually how the screen would perform in CFD and how the air will circulate inside the space to have a zoomed in look, what part of the screen applied to the aperture area will serve as an inlet and which part will serve as an outlet.

The test FAILD , CFD didn't understand the complex behavior around the screen and the changing in humidity and temperature since it is mostly used for steady behaviors. Yet this shows how important a realistic application of the screen is.

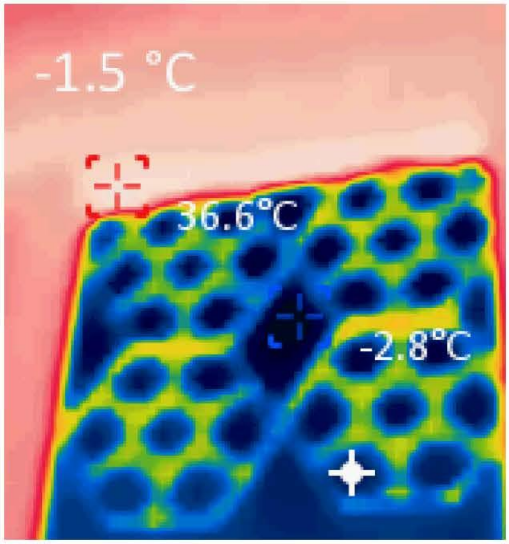
Source: Author

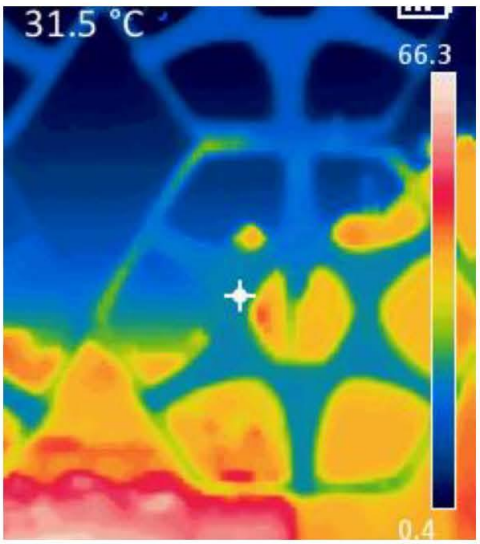


Appendix fig – Initial testing of the screen in another form where the components were applied perpendicular, evaluating how the water system can be connected to the screen in this form to ensure all the cavities get fully saturated Source: Author

Testing another design approach with the same components

The screens was applied first thin the cork was added to make the system watertight and avoid leakage and raw clay was used to close any gaps.

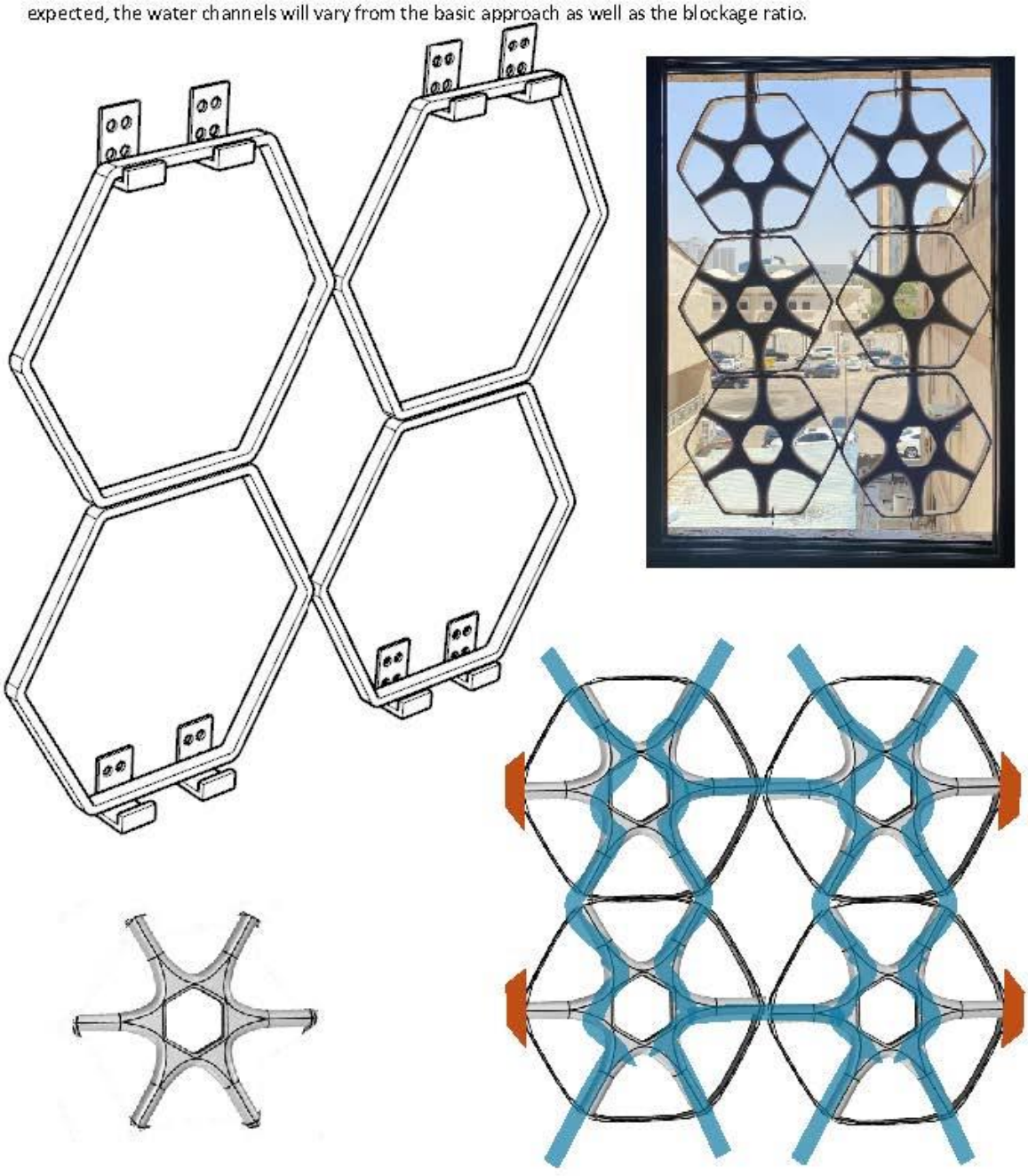




Appendix fig – Initial testing of the screen in another form where the components were applied perpendicular, at this stage, the water was not connected, but it was an initial test to see how the components would be placed, and since the components were already wet and saturated, thermal images showed a possible cooling effect Source: Author

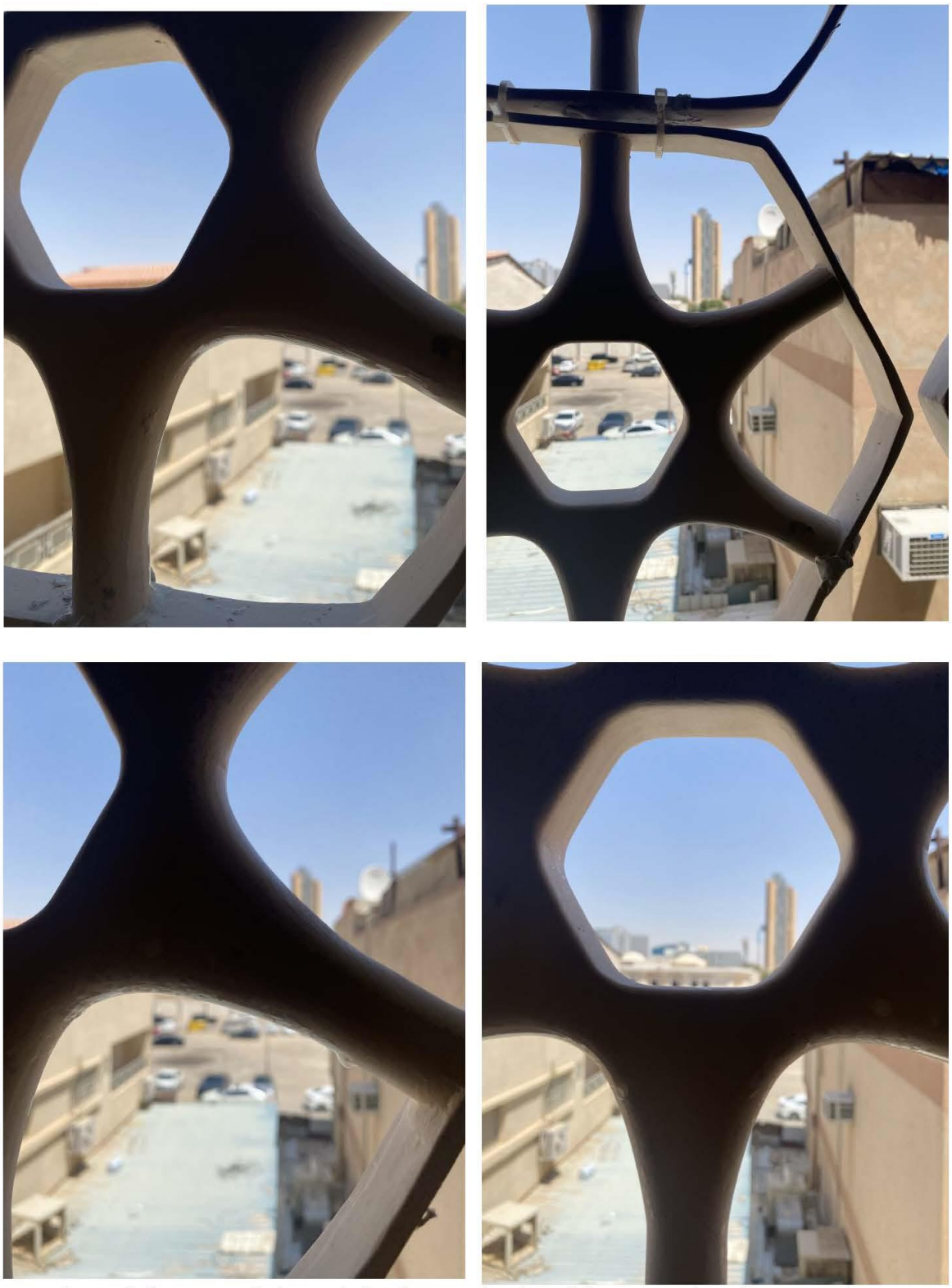
Testing another design approach with the same components

The design was structurally stable and safe to be applied, yet different performance was expected, the water channels will vary from the basic approach as well as the blockage ratio.



Appendix fig – Initial testing of the screen in another form where the components were applied perpendicular, evaluating how the water system can be connected to the screen in this form to ensure all the cavities get fully saturated, and at the same time block the undesired openings with raw clay to ensure no leak will happen.Source: Author

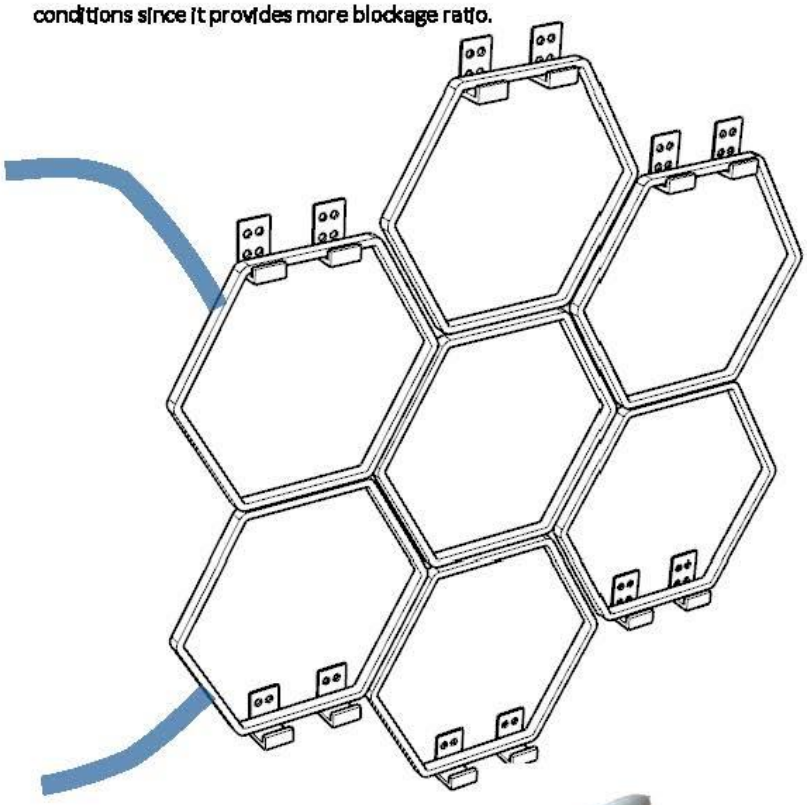
Appendix fig – Initial testing of the screen in another form where the components were applied perpendicular, evaluating how the water system can be connected to the screen in this form to ensure all the cavities get fully saturated, the regular water connection used was tilted. The water was blocked, and it was supplied to each column separately. Source: Author

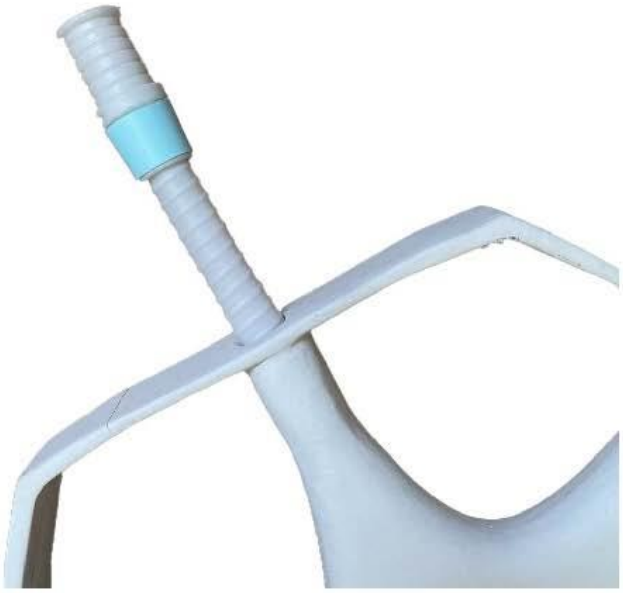


Appendix fig – zoomed in image during the operation time of the screen, where the evaporation rate reached the beak and the water droplets on the ceramic exterior surface were noticed.
Source: Author

Testing another design approach with the same components

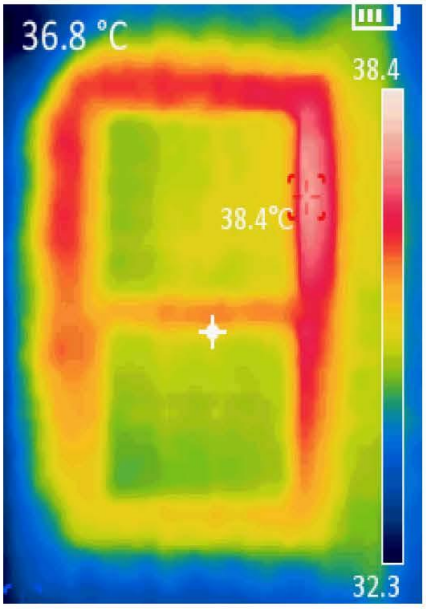
The design was structurally stable and safe to be applied, yet different performance was expected, the water channels will vary from the basic approach as well as the blockage ratio and this was the chosen approach since it suits more the extreme conditions since it provides more blockage ratio.

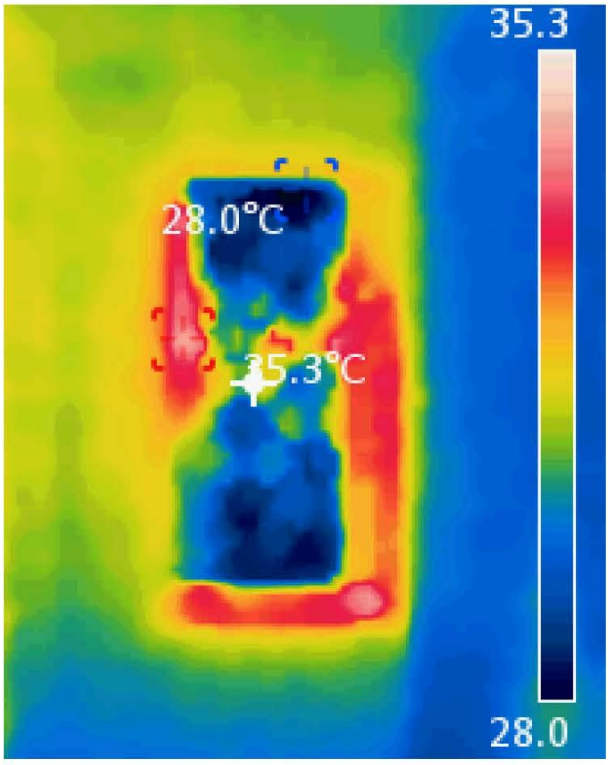


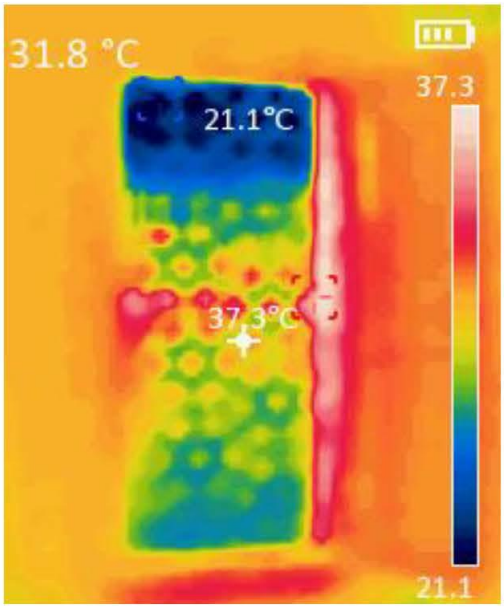


Appendix fig – testing alternative available water connection system during the fieldwork to experiment the most applicable method that minimise leak and water loss Source: Author

Appendix fig – broken sample that was used to break all the cap geometry parts to evaluate the difference in water consumption between the proposed design and the design without cap extrusion added







Appendix fig19- Thermal Images taken during the application of the screen in the 1:1 application process.

Appendix fig- Thermal Images taken during the application of the screen in the 1:1 application process.

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